The Igusa Square Root

The Borcherds determinant of $\phi_{0,1}$, its denominator algebra, and the $K3 \times E$ realization problem

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The square-root realization problem

Counting curves on a product of a K3 surface and an elliptic curve produces the reciprocal of a genus-two modular form. The geometric realization problem asks for more: a space of protected states, operations on those states, and an oriented square root of their determinant. These are different constructions. Their relation can be tested by asking what information survives a signed count.

Let S be a smooth complex projective K3 surface, so $K_S \simeq \mathcal{O}_S$ and $H^1(S, \mathcal{O}_S) = 0$. Let E be a smooth elliptic curve with an origin, and put $X = S \times E$. Choose a nonzero effective primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$. Here primitive means not an integer multiple greater than one of another integral class. For $d \geq 0$, the class (β_h, d) in $H_2(X, \mathbb{Z})$ is the sum of β_h in a K3 fiber and d copies of an elliptic fiber.

A one-dimensional subscheme $C \subset X$ has this curve class and an integer $n = \chi(\mathcal{O}_C)$. Translation in E moves C without changing its class. The reduced Donaldson–Thomas (DT) invariant counts translation orbits with the Behrend weight:

$$\mathrm{DT}_{n,(\beta_h,d)} = \int_{\mathrm{Hilb}^{(\beta_h,d),n}(X)/E} \nu \, de.$$

The Hilbert scheme parametrizes these subschemes, and ν is the integer-valued Behrend function of the quotient. The integral is the weighted Euler characteristic: each level set $\nu^{-1}(a)$ contributes $a\, e(\nu^{-1}(a))$, with the stack weighting for finite stabilizers.

The reduction removes the trivial obstruction direction induced by the holomorphic symplectic form of S. It is this reduced quotient theory that has a nonzero counting series.

To state its automorphic expression, write

$$\mathcal{H}_2 = \{Z = Z^t \in M_2(\mathbb{C}) : \mathrm{Im} Z > 0\}, \qquad Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix}, \qquad (q, r, s) = (e^{2\pi i z_1}, e^{2\pi i z_2}, e^{2\pi i z_3}).$$

Let χ_{10}^{OP} be the monic Igusa cusp form of weight ten, with coefficient $[qrs]\chi_{10}^{\mathrm{OP}} = 1$.

For fixed h, d, the counting series $\sum_n \mathrm{DT}_{n,(\beta_h,d)}\, y^n$ is a Laurent series at $y = 0$. Let $R_{h,d}(y) \in \mathbb{Q}(y)$ be its rational representative, whose existence follows from reduced DT/PT rationality [99, Theorem 3]. The primitive counting identity is an equality with rational coefficients in r:

$$Z_{\mathrm{OP}}^X(r, q, s) := \sum_{h,d \geq 0} R_{h,d}(-r) q^{h-1} s^{d-1} = -\frac{1}{\chi_{10}^{\mathrm{OP}}(Z)}.$$

Expansion of each rational coefficient at $r = 0$ recovers $\sum_n \mathrm{DT}_{n,(\beta_h,d)}(-r)^n$.

For each h, the invariant is computed on a projective K3 surface carrying β_h. Primitive deformation invariance makes this choice immaterial. The coefficient variable for Euler characteristic is $y = -r$. The variables q and s record the K3 square through $h - 1$ and elliptic degree through $d - 1$.

The equality uses reduced DT/stable-pair (Pandharipande–Thomas, PT) comparison, primitive Gromov–Witten (GW)/PT comparison, and the Oberdieck–Pixton theorem [99, Theorem 3(i)], [100, Proposition 5], [101, Theorem 1]. The respective comparisons and expansion domains are stated below.

The theta-normalized square root is the weight-five Igusa form Δ_5, fixed by

$$\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2, \qquad [q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64, \qquad 4096 = 64^2.$$

The subscript BPS refers to the Bogomolny–Prasad–Sommerfield protected-state interpretation being sought. The rescaled scalar $Z_{\mathrm{BPS}}^{K_3\times E} := -Z_{\mathrm{OP}}^{X}/4096$ is Δ_5^{-2}. This definition alone supplies neither a protected state space nor a trace on it. Such a trace requires the complex, cyclic pairing, completion, and comparison hypotheses specified in §12.

Squaring loses a character. Consider the flat complex line on a circle obtained from $\mathbb{R}\times\mathbb{C}$ by $(t, a) \sim (t+1, -a)$. Parallel transport around the circle multiplies a vector by -1. Its tensor square has transport $+1$, just like the trivial line. The two lines therefore have identical flat tensor squares but are not isomorphic as flat lines.

More generally a line may be twisted by any order-two flat line without changing its square. A determinant section consequently does not determine an oriented Pfaffian section: the square-root line and its transport remain additional data. This is the distinction used by Proposition 1.149.

For Δ_5, this lost datum is an order-two automorphic character. Recovering it geometrically is one part of the compact realization problem. A signed count gives the difference of two parity dimensions. Recovering the algebra of states also requires the individual dimensions and the multiplication. The first denominator calculation below shows where these separate questions occur.

Chapter I

The geometric realization problem

An infinite product can be realized as the character of an algebra, as an index of geometric complexes, or as the determinant of a family. To compare these realizations, one must compare what is multiplied. The root algebra has brackets and two parity dimensions at each root. Geometric complexes have extension maps, orientations and specialization. A determinant line has transport and a section.

The Igusa product gives a precise common target. Its root lattice fixes the charges, its Jacobi seed fixes the signed multiplicities, and its character fixes the automorphic line. Hilbert schemes provide independent geometric coefficient complexes. Their indices determine the scalar comparison, while their operations pose the stronger realization problem.

1 The lattice, roots and Weyl vector

A lattice is a finite free abelian group with an integral symmetric pairing. The denominator uses the Lorentzian lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus \langle 2 \rangle$, with basis (f_2, f_3, f_{-2}), pairings $(f_2, f_{-2}) = -1$, $(f_3, f_3) = 2$, and all other pairings zero. The two positive and one negative directions give signature $(2, 1)$. Here $U(k)$ is the rank-two lattice with Gram matrix $\left(\begin{smallmatrix} 0 & k \\ k & 0 \end{smallmatrix}\right)$. Changing the sign of one basis vector identifies the displayed hyperbolic plane with U.

The index-four sublattice $\Lambda^{2,1}_{II} = \mathbb{Z}(2f_2) \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}(2f_{-2})$ is abstractly $U(4) \oplus \langle 2 \rangle$, with determinant -32. For integral indices $(n, \ell, m) \in \mathbb{Z}^3$, define

$$\alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2}.$$

The three simple real roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ have Gram matrix

$$((\delta_i, \delta_j)) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix} = 4\mathbf{I}_3 - 2\mathbf{J}_3.$$

Here $\mathbf{J}_3$ is the all-ones matrix. The reflections $s_{\delta_i}(x) = x - (x, \delta_i)\delta_i$ preserve this form. They generate $W^{(2)}(\Lambda^{2,1}_{II})$.

Choose the chamber $\mathcal{P}_{II}$ containing the Weyl vector

$$\rho = \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \tfrac{1}{2}f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

Thus its wall inequalities in this convention are $(x, \delta_i) \le 0$ in the chosen negative cone.

2 Automorphic sections and the Jacobi seed

The square root transforms as a section of a line, rather than as an invariant function. Write $\mathbf{Sp}_4(\mathbb{Z})$ for the group preserving the standard alternating form on $\mathbb{Z}^4$. The notation $\mathbf{Sp}(2, \mathbb{Z})$ denotes the same group.

Its block matrix $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ acts by $gZ = (AZ + B)(CZ + D)^{-1}$. The weight-one automorphic line $\mathcal{L}$ on the quotient stack is defined by this factor $\det(CZ + D)$. A form of weight k and character ν satisfies

$$F(gZ) = \nu(g)\det(CZ + D)^k F(Z).$$

The product of the ten even genus-two theta constants is normalized so that its $q^{1/2}r^{1/2}s^{1/2}$ coefficient is $+64$. With this choice of sign it is Δ_5. Its character ν_{Δ_5} has order two [58, 84]. In particular

$$\mathcal{D}_X := \Delta_5 \in H^0(\mathbf{Sp}_4(\mathbb{Z})\backslash\mathcal{H}_2, \mathcal{L}^{\otimes 5} \otimes \nu_{\Delta_5}), \qquad \Delta_{10} := \Delta_5^2 \in M_{10}(\mathbf{Sp}_4(\mathbb{Z})).$$

The alternative names $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_{10}$ refer to this theta normalization. The monic form is $D_5 = 64^{-1}\Delta_5$, so $\chi_{10}^{\mathrm{OP}} = D_5^2$.

The product exponents come from one elliptic function with modular symmetry. A weak Jacobi form of weight zero and index one is a holomorphic function $\phi(\tau, z)$ on $\mathbb{H} \times \mathbb{C}$. It obeys the transformation laws

$$\phi\left(\frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d}\right) = e^{2\pi i c z^2/(c\tau+d)}\phi(\tau, z),$$
$$\phi(\tau, z + \lambda\tau + \mu) = e^{-2\pi i(\lambda^2\tau + 2\lambda z)}\phi(\tau, z) \quad (\lambda, \mu \in \mathbb{Z}),$$

for $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathbf{SL}_2(\mathbb{Z})$. Its Fourier expansion has nonnegative powers of $q_\tau = e^{2\pi i\tau}$.

The normalized seed is

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0, \ell \in \mathbb{Z}} f(n, \ell) q_\tau^n e^{2\pi i \ell z} = e^{-2\pi i z} + 10 + e^{2\pi i z} + O(q_\tau).$$

It is half the K3 elliptic genus. Its coefficients depend on $4n - \ell^2$, vanish below -1, and satisfy $f(1, 0) = 108$, $f(1, \pm 1) = -64$, $f(1, \pm 2) = 10$ [34, 39]. The elliptic law explains the discriminant: replacing ℓ by $\ell + 2\lambda$ replaces n by $n + \lambda\ell + \lambda^2$, preserving $4n - \ell^2$.

3 The automorphic product and denominator

To expand at a cusp, choose the positive indices

$$\Gamma_{\mathrm{eff}} = \{(n, \ell, m) : m > 0,\ n \geq 0,\ \ell \in \mathbb{Z}\} \cup \{(n, \ell, 0) : n > 0,\ \ell \in \mathbb{Z}\} \cup \{(0, \ell, 0) : \ell < 0\}.$$

Only $4nm - \ell^2 \geq -1$ contributes. Set $\Gamma_{\mathrm{act}} = \{\gamma \in \Gamma_{\mathrm{eff}} : f(nm, \ell) \neq 0\}$. Expand each factor in its positive monomial in a cusp chamber where $|q^n r^\ell s^m| < 1$ on this support. For example $\mathrm{Im} z_1 = \mathrm{Im} z_3 = t$, $\mathrm{Im} z_2 = -a$, with $0 < a < t$ and t sufficiently large, selects such a chamber.

The support condition bounds ℓ at fixed (n, m). The remaining degree-zero factor is $1 - r^{-1}$. These facts make coefficient extraction finite in the associated completion. The Borcherds product is

$$\Delta_5(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,\ell,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

A lattice-graded Lie superalgebra has even and odd root spaces. The signed multiplicity is their superdimension $\mathrm{smult}\,\mathfrak{g}_\alpha = \dim \mathfrak{g}_\alpha^{\bar{0}} - \dim \mathfrak{g}_\alpha^{\bar{1}}$. The Gritsenko–Nikulin Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\Delta_5}$ has $\mathrm{smult}\,\mathfrak{g}_{\alpha(n,\ell,m)} = f(nm, \ell)$ on the active support. Its denominator identity reads

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \sum_{w \in W^{(2)}} \mathrm{sgn}(w) w\left(e^\rho - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a) e^{\rho + a}\right) = 64^{-1}\Delta_5(2Z).$$

The symbols e^{α} are lattice characters. Doubling Z converts the half-exponent product convention to these denominator characters. The integers $m(a)$ are the imaginary-simple correction coefficients [52, Theorem 2.3(a)].

The first correction can be seen without reconstructing the whole algebra. Remove the leading monomial from the product and take s-degree zero. Up to q-degree one, its factors are

$$(1 - r^{-1})(1 - qr)(1 - q)^{10}(1 - qr^{-1}).$$

The coefficient of q is

$$-(1 - r^{-1})(r + 10 + r^{-1}) = -r - 9 + 9r^{-1} + r^{-2}.$$

Thus the constant term in r is -9.

In lattice notation it occurs at $\rho + 2f_2$. But

$$\rho^2 = -2 + \tfrac{1}{2} = -\tfrac{3}{2}, \qquad (\rho + 2f_2)^2 = -6 + \tfrac{1}{2} = -\tfrac{11}{2}.$$

Reflections preserve the norm, so this exponent is outside the Weyl orbit of ρ. The uncorrected alternating sum cannot give the product. The imaginary-root correction is necessary.

Signed multiplicities still leave a second question. At $\delta_{123} = \delta_1 + \delta_2 + \delta_3$, Proposition 4.6 computes $(\dim \mathfrak{g}^{\bar{0}}_{\delta_{123}}, \dim \mathfrak{g}^{\bar{1}}_{\delta_{123}}) = (29, 93)$. Their difference is $-64 = f(1, 1)$. The pair $(30, 94)$ has the same difference. The exponent -64 therefore cannot by itself recognize the parity-resolved root space, still less its brackets. This is why the compact primitive-algebra comparison requires more than the scalar denominator.

4 The regularized theta lift

The Jacobi transformation law also explains the product's modularity. Introduce the rank-five lattice $L = \Lambda^{3,2} = \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus \langle 2 \rangle$ of signature $(3, 2)$, and set $M = L(-1)$. The type-IV domain of negative oriented two-planes in L is the domain of positive oriented two-planes in M. The exterior-square representation identifies its chosen component with $\mathcal{H}_2$, and its arithmetic group with $\Gamma_{3,2} = \wedge^2\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$. This rank-five lattice carries the theta kernel. The rank-three cusp quotient carries the product indices.

The index-one elliptic theta functions are $\theta_{1,\mu}(\tau, z) = \sum_{\ell \equiv \mu \ (2)} q_\tau^{\ell^2/4} e^{2\pi i \ell z}$, for $\mu = 0, 1$. Writing $\phi_{0,1} = h_0\theta_{1,0} + h_1\theta_{1,1}$ gives $[q_\tau^{n-\ell^2/4}]h_{\ell \bmod 2} = f(n, \ell)$. The vector h has weight $-1/2$ and Weil representation $\rho_M = \rho_L^\vee$. Indeed the nontrivial theta component has eigenvalue i under $T : \tau \mapsto \tau + 1$, so its coefficient has eigenvalue $-i$. In particular $[q_\tau^0]h_0 = 10$.

For a positive plane P in M, decompose a lattice vector as $\lambda_+ + \lambda_-$, with $\lambda_+ \in P$ and $\lambda_- \in P^\perp$. The kernel and additive lift are

$$\Theta_M(\tau, P) = \sum_{\gamma \in M^*/M} e_\gamma \sum_{\lambda \in M + \gamma} e^{\pi i \tau \lambda_+^2 + \pi i \overline{\tau} \lambda_-^2},$$

$$\Phi_M(P, h) = \int^{\mathrm{reg}}_{\mathbf{SL}_2(\mathbb{Z})\backslash\mathbb{H}} v \sum_\gamma h_\gamma(\tau) \overline{\Theta_{M,\gamma}(\tau, P)} \, \frac{du\, dv}{v^2}.$$

Here $\tau = u + iv$, squares use the form of M, and e_γ is the standard basis of the group algebra $\mathbb{C}[M^*/M]$. The Gaussian has absolute value $e^{-\pi v(\lambda_+^2 - \lambda_-^2)}$, which decays.

The Hermitian pairing has weight $(1,1)$. Multiplication by v makes the integrand invariant. Regularization means the constant term of the meromorphic continuation obtained by inserting v^{-s} in the truncated integral, as specified in §4.

Borcherds's theorem constructs a multiplicative form Ψ_M from this additive function. A homogeneous null coordinate $w = X_M + iY_M$ satisfies $(w,w)_M = 0$ and $(w,\overline{w})_M > 0$. Its oriented plane is $P(w) = \operatorname{span}_{\mathbb{R}}(X_M, Y_M)$. With $k = [q_\tau^0]h_0/2 = 5$ and $C_B = \Gamma'(1)/2 + \log\sqrt{2\pi}$, the precise relation is

$$\Phi_M = -4\log\|\Psi_M\| - 4kC_B, \qquad \|\Psi_M\| = |\Psi_M(w)|\,|Y_M|^k, \qquad |Y_M| = \sqrt{(Y_M,Y_M)_M}.$$

The theorem applies to weight $-1/2$ inputs of type ρ_M with integral coefficients at nonpositive indices [10, Theorem 13.3]. Its general output weight varies with the input. For this chosen seed, the primitive-cusp product with leading coefficient $+1$ is D_5.

Choose the ordered basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ of L, with nonzero pairings $(f_1, f_{-1})_L = (f_2, f_{-2})_L = -1$ and $f_3^2 = 2$. In this basis the cusp frame is $w = (z_2^2 - z_1z_3, z_3, z_2, z_1, 1)$. Substitution gives $w^2 = 0$ and $|Y_M|^2 = 2\det(\operatorname{Im}Z)$. Therefore

$$\Phi_M(P,h) = -4\log\Big(|\Delta_5(Z)|\,[2\det(\operatorname{Im}Z)]^{5/2}\Big) + 4\log 64 - 20C_B.$$

The norm determines no phase.

The positive leading coefficient 64 fixes the theta normalization separately. Thus the multiplicative lift, denoted $\Theta^{\mathrm{Bor}}_{\mathrm{sing}}(h) = 64D_5$, is Δ_5, and its denominator is $64^{-1}\Delta_5(2Z)$. These are the automorphic, denominator, and theta descriptions of one form.

5 GEOMETRIC COEFFICIENTS AND THE NORMALIZED HALF-INDEX

The seed already has a geometric coefficient theory. For a smooth projective complex variety Y of dimension d, put

$$\mathcal{E}_{q,r}(Y) = r^{-d/2}\bigotimes_{j\ge 1}\Big(\Lambda_{-rq^{j-1}}T_Y^*\otimes\Lambda_{-r^{-1}q^j}T_Y\otimes\operatorname{Sym}_{q^j}T_Y^*\otimes\operatorname{Sym}_{q^j}T_Y\Big). \tag{1.1}$$

Here $\Lambda_t V = \sum_i t^i\Lambda^i V$ and $\operatorname{Sym}_t V = \sum_i t^i\operatorname{Sym}^i V$. An exterior power $\Lambda^i V$ has internal parity i modulo two. The minus signs record this parity in the displayed class. Thus each coefficient is an actual finite direct sum of vector bundles with parity, rather than only a virtual rank.

Apply derived global sections coefficient by coefficient. The resulting complex has total parity given by internal parity plus cohomological degree. Its signed Euler characteristic is

$$\operatorname{Ell}(Y;q,r) = \chi\big(Y,\mathcal{E}_{q,r}(Y)\big).$$

Completion is in nonnegative powers of q. At fixed q-degree the r-support is finite, and projectivity makes every coefficient cohomology finite-dimensional. This fixes both the carrier and the meaning of its index.

For a surface Y, let $Y^{[N]}$ be the Hilbert scheme of length-N zero-dimensional subschemes and define

$$F_Y(s,q,r) = \sum_{N\ge 0} s^N\operatorname{Ell}(Y^{[N]};q,r).$$

The coefficient complexes are $R\Gamma(Y^{[N]},\mathcal{E}_{q,r}(Y^{[N]}))$.

The Hilbert-scheme elliptic-genus theorem gives

$$F_Y = \prod_{a>0,\, b\geq 0,\, \ell\in\mathbb{Z}} (1 - s^a q^b r^\ell)^{-c_Y(ab,\ell)}, \qquad \mathrm{Ell}(Y;q,r) = \sum_{b,\ell} c_Y(b,\ell) q^b r^\ell. \tag{1.2}$$

This is the numerical McKay comparison for the crepant Hilbert–Chow resolution.[1] It does not identify the extension or chiral operations on the coefficient complexes.

Separate the factors with zero s-degree by setting

$$b(q,r) = q^{1/2} r^{1/2} (1 - r^{-1}) \prod_{j\geq 1} (1 - q^j)^{10} (1 - rq^j)(1 - r^{-1} q^j), \qquad \mathcal{B}(q,r) = b(q,r)^2.$$

The square is

$$\mathcal{B} = q(r + r^{-1} - 2) \prod_{j\geq 1} (1 - q^j)^{20} (1 - rq^j)^2 (1 - r^{-1} q^j)^2.$$

For the monic form $D_5 = \Delta_5/64$ and $\Xi = D_5^2 = \chi_{10}^{\mathrm{OP}}$, cancellation gives

$$F_S = \frac{s\mathcal{B}}{\Xi} = \prod_{a>0, b\geq 0, \ell\in\mathbb{Z}} (1 - s^a q^b r^\ell)^{-2f(ab,\ell)}. \tag{1.3}$$

The numerator is part of the normalization. It removes the boundary pole and restores the length-zero coefficient 1. The cancelled expression lies in $\mathbb{Z}[r^{\pm 1}][[s,q]]$.

Theorem 5.1 *(The Enriques half-index).* Let T be a smooth projective complex Enriques surface. Then

$$F_T = \frac{s^{1/2} b}{D_5} = \prod_{a>0, b\geq 0, \ell\in\mathbb{Z}} (1 - s^a q^b r^\ell)^{-f(ab,\ell)}, \qquad F_T^2 = F_S. \tag{1.4}$$

These are identities of normalized elliptic-index series.

Proof. The canonical cover $\pi : \widetilde{S} \to T$ is an étale K3 double cover, so $T_{\widetilde{S}} = \pi^* T_T$.[2] The elliptic characteristic class therefore pulls back, and integration over the cover multiplies it by two. Thus $\mathrm{Ell}(T) = \phi_{0,1}$.

Equation (1.2) gives the product. Separating its zero-s factors in D_5 gives the quotient, and squaring doubles every exponent. □

The surface T is a separate geometric input. No Enriques involution is required on the originally chosen S. The equality $F_T^2 = F_S$ compares indices of independently defined complexes. It supplies no equivalence between their states or their products.

The canonical line K_T defines a character on the surface T. The automorphic character belongs to a line over Siegel moduli. A geometric square-root comparison must relate these bases and lines.

The first nontrivial length calculation makes the distinction visible. At $q = 0$, the Enriques seed has coefficients $r^{-1} + 10 + r$. The length-two term comes from two length-one factors or one length-two factor, giving

$$[s^2 q^0] F_T = r^2 + 11r + 66 + 11r^{-1} + r^{-2},$$
$$[s^2 q^0] F_S = 3r^2 + 42r + 234 + 42r^{-1} + 3r^{-2}.$$

[1] L. Borisov and A. Libgober, *McKay correspondence for elliptic genera*, arXiv:math/0206241, Corollary 6.7. The coefficient bundle convention is their equation (3). The hypotheses here are smoothness and projectivity.

[2] I. Dolgachev, *A brief introduction to Enriques surfaces*, arXiv:1412.7744v2, Section 3, pp. 7–8. Only the complex case is used.

For example the constant coefficient of the first line is $\binom{11}{2} + 1 + 10 = 66$. The summands count two weight-zero length-one contributions, the opposite-weight pair, and the weight-zero length-two term.

Only after cancelling the boundary factors may one set $r = 1$. All positive-energy exterior and symmetric factors in (1.1) then cancel. The remaining index is the topological Euler characteristic. Consequently

$$F_T(s,q,1) = \prod_{a\geq 1}(1-s^a)^{-12} = 1 + 12s + 90s^2 + 520s^3 + \cdots,$$
$$F_S(s,q,1) = \prod_{a\geq 1}(1-s^a)^{-24} = 1 + 24s + 324s^2 + 3200s^3 + \cdots.$$

The raw quotients in (1.3) and (1.4) have vanishing numerator and denominator at $r = 1$. Their cancelled series specify the value.

5.1 A coefficient before taking its index

The two parities can be calculated directly at positive energy. Write $\Omega_T = T_T^*$ and let $K_T = \Lambda^2\Omega_T$. Since $K_T^{\otimes 2} \simeq \mathcal{O}_T$, the qr coefficient of (1.1) is the internally even bundle

$$B_{11} = \operatorname{Sym}^2 \Omega_T \oplus K_T \oplus \Omega_T \oplus T_T. \tag{1.5}$$

Indeed the expression through q-degree one is

$$r^{-1}(1 - r\Omega_T + r^2 K_T)\big(1 + q((1-r)\Omega_T + (1-r^{-1})T_T)\big) + O(q^2).$$

Extracting qr gives $\Omega_T^{\otimes 2} + K_T \otimes \Omega_T + K_T \otimes T_T$. The rank-two identities $K_T \otimes T_T \simeq \Omega_T$ and $K_T \otimes \Omega_T \simeq T_T$ give (1.5).

On an Enriques surface, $c_2(T) = 12$, $\chi(\mathcal{O}_T) = 1$, and $c_1(T) = 0$ rationally. Its Hodge numbers give $h^1(\Omega_T) = 10$.

The K3 cover has no nonzero holomorphic symmetric cotangent tensors of positive degree.[3]

Pullback and Serre duality therefore make H^0 and H^2 vanish for T_T, Ω_T, and $\operatorname{Sym}^2 \Omega_T$.

For the last bundle, Riemann–Roch gives

$$\operatorname{ch}(\operatorname{Sym}^2 \Omega_T) = 3 - 4c_2(T), \qquad \chi(\operatorname{Sym}^2 \Omega_T) = 3 - 48 = -45.$$

The same formula for T_T gives $\chi(T_T) = -10$. The canonical line has cohomology dimensions $(0,0,1)$.

Adding the four summands yields

$$(h^0, h^1, h^2)(T, B_{11}) = (0, 65, 1), \qquad \operatorname{sdim} H^*(T, B_{11}) = 1 - 65 = -64.$$

Thus the actual coefficient has parity dimensions $(1|65)$.

The denominator root space has $(29|93)$ at the corresponding interior root, as computed above. Their indices agree, but the two parity spaces are not isomorphic. A geometric comparison must account for the differential, the operations and their possible kernels. It may instead be an action of the root algebra on the state complex.

[3] S. Kobayashi, *The first Chern class and holomorphic symmetric tensor fields*. See Corollary 8 in J. Math. Soc. Japan **32** (1980), 325–329.

6 Reduced counting and the scalar normalization

The Hilbert-scheme count in the opening can be compared with maps and sheaves. A stable pair on X is a map $\mathcal{O}_X \to F$, with F pure of dimension one and zero-dimensional cokernel. The reduced invariant $\mathrm{PT}_{n,(\beta,d)}$ is the corresponding Behrend-weighted quotient count by E. For nonzero K3 class, Oberdieck proves

$$\sum_n \mathrm{DT}_{n,(\beta,d)}\, y^n = \sum_n \mathrm{PT}_{n,(\beta,d)}\, y^n.$$

There is no degree-zero correction factor: $\chi_{\mathrm{top}}(X) = \chi_{\mathrm{top}}(S)\chi_{\mathrm{top}}(E) = 24 \cdot 0 = 0$ [99, §0.3, Theorem 3(i)].

For maps, let $\overline{M}^{\bullet}_{g,1}(X, (\beta_h, d))$ be the moduli of possibly disconnected stable maps, with no contracted connected components, one marked point, and arithmetic genus g. Choose $\beta_h^\vee \in H^2(S, \mathbb{Q})$ with $\langle \beta_h, \beta_h^\vee \rangle = 1$. If π_S, π_E are the projections, the reduced Gromov–Witten invariant is

$$N_{g,h,d} = \int_{[\overline{M}^{\bullet}_{g,1}(X,(\beta_h,d))]^{\mathrm{red}}} \mathrm{ev}_1^*(\pi_S^*\beta_h^\vee \cup \pi_E^*[\mathrm{o}_E]).$$

The insertion removes the translation orbit, and the value is independent of the dual class. Primitive deformation invariance makes it depend on h, rather than the chosen primitive class.

A nonzero disconnected contribution has one component nontrivial over S. Its other components have genus one. Thus its arithmetic genus is nonnegative [100, §1.2].

Write

$$Z^{\mathrm{rat}}_{\mathrm{PT}}(y, q, s) = \sum_{h,d\geq 0} R_{h,d}(y) q^{h-1} s^{d-1}.$$

The GW/PT comparison of Oberdieck–Pandharipande gives

$$Z_{\mathrm{GW}}(u, q, s) := \sum_{g,h,d\geq 0} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1} = Z^{\mathrm{rat}}_{\mathrm{PT}}(-e^{iu}, q, s),$$

where the right side means its Laurent expansion at $u = 0$ [100, Proposition 5]. Oberdieck–Pixton prove $Z_{\mathrm{GW}} = -1/\chi^{\mathrm{OP}}_{10}(r, q, s)$ under $r = e^{iu}$ [101, Theorem 1]. Their theorem proves the primitive Igusa formula conjectured in [100].

Combining the comparisons gives precisely

$$Z^X_{\mathrm{OP}} = Z^{\mathrm{rat}}_{\mathrm{PT}}(-r, q, s) = -4096\Delta_5^{-2}, \qquad Z^{K3\times E}_{\mathrm{BPS}} = \Delta_{10}^{-1} = -\tfrac{1}{4096} Z^X_{\mathrm{OP}}.$$

These are identities of coefficientwise rational functions before choosing Laurent expansions. The DT/PT expansion is at $r = 0$. The chosen product chamber instead gives the expansion at $r = \infty$, and the GW substitution gives the expansion at $u = 0$. For example the coefficient of $q^{-1}s^{-1}$ is

$$-\frac{r}{(1-r)^2} = \begin{cases} -\sum_{n\geq 1} n r^n, & |r| < 1, \\ -\sum_{n\geq 1} n r^{-n}, & |r| > 1. \end{cases}$$

Thus the two Laurent series have the same rational representative.

Under the GW substitution, $r - 2 + r^{-1} = 2\cos u - 2 = -u^2 + O(u^4)$, so $-1/(r - 2 + r^{-1}) = u^{-2} + O(1)$, consistent with genus zero. The sign $y = -r$ is independent of the orientation character. The formula stated here does not include imprimitive K3 classes.

7 Five distinct numerical invariants

The counting and algebraic constructions use several Euler characteristics and ranks. The group $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$ of triples (n, ℓ, m) is called the *Gram-index lattice*. Its exponent map α identifies it with $\Lambda_{II}^{2,1}$. Write $r_{\mathrm{Gram}} = 3$ for its rank.

The lattice $\widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$, with pairing $(r,c,a)\cdot(r',c',a') = c\cdot c' - ra' - r'a$, is called the *Mukai lattice*. Its algebraic sublattice replaces $H^2(S,\mathbb{Z})$ by $\mathrm{NS}(S)$. Here $\mathrm{NS}(S)$ is the image of the first Chern class of line bundles in integral cohomology. The five numerical values are

$$(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, r_{\mathrm{Gram}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24).$$

The first is $\chi(\mathcal{O}_X) = 2\cdot 0 = 0$. The Hodge numbers $h^{0,q}(X)$, for $q = 0, 1, 2, 3$, are $(1, 1, 1, 1)$, giving the second invariant $1 - 1 + 1 - 1 = 0$.

The third counts the three Gram coordinates of the type-II Heisenberg–Cartan algebra. It is not the rank of the algebraic Mukai lattice $H^0(S,\mathbb{Z}) \oplus \mathrm{NS}(S) \oplus H^4(S,\mathbb{Z})$, which has rank five when $\mathrm{NS}(S) \simeq U \oplus \langle -2\rangle$. The fourth is the Borcherds weight $f(0,0)/2 = 10/2 = 5$. The last is $\chi_{\mathrm{top}}(S) = 1 + 22 + 1 = 24$.

A genus-one Heisenberg bar coefficient is a different quantity. Its definition requires the chosen bar complex, a trace functional, and a normalized genus-one class. Denote this coefficient by $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$ when these data are fixed. The comparison $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = r_{\mathrm{Gram}} = 3$ requires a map identifying that normalized coefficient with the Gram rank. Rescaling the trace rescales the coefficient and leaves the rank unchanged. The five-entry tuple with $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$ in its third position therefore assumes this additional comparison.

In particular $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_S)$ would give $5 = 0 + 2$, which is false. The values $\chi(\mathcal{O}_S) = 2$ and $\chi_{\mathrm{top}}(S) = 24$ are different invariants.

For any admissible Borcherds input Φ_N, the weight formula instead reads $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$, with $c_N(0)$ the zero-coset constant coefficient of that input. The $(t, N; k) = (1, 1; 5)$ row of the Cléry–Gritsenko catalog uses $\phi_{0,1}$ [50, 52]. The Fake-Monster product Φ_{12} uses constant coefficient 24, giving weight 12, with its denominator on the Leech-extension lattice $\mathrm{II}_{25,1}$ [8, 9]. The two weights belong to different inputs and lattices.

For the cyclic Chaudhuri–Hockney–Lykken (CHL) orbifold inputs with orders $N = 1, 2, 3, 4, 6$, the eta-product cycle shapes are 1^{24}, $1^8 2^8$, $1^6 3^6$, $1^4 2^2 4^4$, $1^2 2^2 3^2 6^2$. Their eta weights are 12, 8, 6, 5, 4. Subtracting two gives the squared-product weights 10, 6, 4, 3, 2, hence square-root weights $5, 3, 2, 3/2, 1$ [48, Table 1, §3].

These rows have different orbifold inputs. Even a comparison with the unchanged number $0 + 2$ would have equality at $N = 3$. The counterexample at $N = 1$ already disproves a universal identification of the Borcherds weight with the Hodge and holomorphic Euler invariants.

8 Charges and extension operations

The root indices must be compared with geometric charges before extension operations can be transported. For two Mukai vectors (Q, P), the quadratic function

$$\Pi_X(Q,P) = \big((Q,Q)/2, (Q,P), (P,P)/2\big)$$

is called the *Gram map*. Sheaf charges lie in the algebraic Mukai sublattice. The following calculation also applies to the full lattice.

The first coordinate of $\Pi_X(Q+Q', P+P')$ differs from the sum of the first coordinates by (Q, Q'). The middle differs by $(Q, P') + (Q', P)$, and the last by (P, P'). Thus the Gram map is quadratic, rather than additive.

Its polarization does give an additive refinement. Let $C = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z})$ and set

$$B(c, c') = \Pi_X(c + c') - \Pi_X(c) - \Pi_X(c').$$

On $C \oplus \mathbb{Z}^3$ define

$$(c, \lambda) \star (c', \lambda') = (c + c', \lambda + \lambda' + B(c, c')).$$

The map $(c, \lambda) \mapsto (c, \lambda - \Pi_X(c))$ identifies this law with componentwise addition. In particular it is associative and commutative.

Its group-homomorphic sections over C are $c \mapsto (c, \Pi_X(c) + L(c))$ for additive maps $L : C \to \mathbb{Z}^3$. This constructs the charge-group refinement. A Hall comparison must still transport the coefficient complexes and their extension maps through that refinement.

There is a second geometric issue before choosing coefficients. Write $\mathcal{M}_\alpha$ for translation-invariant moduli stacks of coherent sheaves on $X = S \times E$ in an extension-closed class. Their charges add in extensions.

Let $\mathcal{H}^{\mathrm{ext}}_{\alpha,\beta}$ parametrize short exact sequences with endpoint charges α, β. Set

$$\mathcal{Q}_\alpha = [\mathcal{M}_\alpha / E], \qquad \mathcal{H}_{\alpha,\beta} = [\mathcal{H}^{\mathrm{ext}}_{\alpha,\beta} / E].$$

The quotient uses simultaneous translation of the whole sequence. Every quotient has a map to the classifying stack BE of principal E-bundles.

Proposition 8.1. The reduced extension correspondence has endpoints over BE:

$$\mathcal{Q}_\alpha \times_{BE} \mathcal{Q}_\beta \longleftarrow \mathcal{H}_{\alpha,\beta} \longrightarrow \mathcal{Q}_{\alpha+\beta}. \tag{1.6}$$

Its left-hand space is canonically $[(\mathcal{M}_\alpha \times \mathcal{M}_\beta)/E_{\mathrm{diag}}]$. Iterated correspondences are the corresponding flag stacks on one common torsor. The zero object gives the relative unit BE.

Proof. Over a test scheme, an object in the fiber product is a pair of E-torsors, equivariant families, and an isomorphism of torsors. Transport across that isomorphism places both families on one torsor. Repeating a single torsor gives the inverse construction, on both objects and arrows. An extension on this torsor gives the span.

Either bracketing of three extensions gives a three-step flag. Taking preimages and quotients recovers the intermediate objects, and further flags give all refinement compatibilities. The zero sheaf has every E-translation as a stabilizer, so its quotient is BE. □

Quotienting the endpoints independently would replace the fiber product over BE by a product over a point and discard this torsor identification. To define Hall operations from (1.6), choose a coefficient theory with pullback and pushforward. Their composites require orientation and support conditions on the correspondence. Coherent complexes and vanishing-cycle sheaves have different such operations. The correspondence alone does not identify them.

8.1 A reduced coefficient as a matrix element

Reduction can change which functional on an algebra represents the counting invariant. This has an elementary model. Let H and R be unital complex algebras and suppose

$$I : H \longrightarrow R[\epsilon]/(\epsilon^2), \qquad I(a) = I_0(a) + \epsilon I_1(a)$$

is a unital algebra homomorphism, with ϵ central. Comparing its two coefficients gives

$$I_1(ab) = I_0(a)I_1(b) + I_1(a)I_0(b), \qquad I_1(1) = 0, \tag{1.7}$$

while I_0 is unital and multiplicative. The reduced coefficient I_1 therefore need not be a character.

Nevertheless it is a matrix element of a unital action:

$$\rho(a) = \begin{pmatrix} I_0(a) & I_1(a) \\ 0 & I_0(a) \end{pmatrix} \quad \text{on the right } R\text{-module } R \oplus R.$$

Equation (1.7) is precisely the off-diagonal equation in $\rho(ab) = \rho(a)\rho(b)$. The first summand is invariant, and the quotient is another copy of the module defined by I_0. The coefficient from the second summand to the first is I_1, including its vanishing on the unit. This model distinguishes a state action from an algebra character. To use it in reduced curve counting requires an independently constructed geometric map I and a comparison with the actual state.

9 COLLISION COEFFICIENTS BEFORE GLOBAL INTEGRATION

The coefficient must retain how supports meet. The displacement parameter in the following model belongs to a spectral elliptic curve. It is distinct from the geometric factor E of $S \times E$ and from the Jacobi parameter curve. An identification of those roles requires a map between the corresponding families.

In a formal coordinate near zero displacement, the three equations are

$$x(x + h) = 0, \qquad y(y + h) = 0, \qquad xy = 0.$$

For $h \neq 0$ their solutions are $(0, 0)$, $(-h, 0)$ and $(0, -h)$. At $h = 0$ the three points meet. The coherent derived zero locus retains the equations through

$$K_h = \mathbb{C}[h, x, y] \otimes \Lambda(\epsilon_1, \epsilon_2, \epsilon_3), \qquad (d\epsilon_1, d\epsilon_2, d\epsilon_3) = (x(x + h), y(y + h), xy), \qquad |\epsilon_i| = -1.$$

Its negative cohomology records relations among the equations.

Adding dual multiplier coordinates gives a different construction:

$$W_h = Px(x + h) + Qy(y + h) + Txy.$$

Its critical equations include derivatives in P, Q, T. The critical completion, its constructible coefficients and its matrix factorizations are developed in Chapters 13 and 14. The coherent and constructible comparisons specify different recipients, gradings and support functors.

One composition already explains why the full coefficient matters. A matrix factorization of W_h is a parity-graded free module with an odd operator D satisfying $D^2 = W_h \,\mathrm{id}$.

For $R = \mathbb{C}[h, P, Q, T]$, take the two branch factorizations D_0, D_1 in (14.4). Their morphism complexes have odd closed maps $f : D_0 \to D_1$ and $g : D_1 \to D_0$ represented by a Clifford generator γ_1 with $\gamma_1^2 = -P$. Their return is

$$[gf] = -P[1_{D_0}] \neq 0.$$

It is killed by h, as proved in Proposition 4.1. Inverting h contracts the off-diagonal morphisms and loses this return. A comparison made only after that localization therefore does not construct the comparison at the collision.

The central critical coefficient also has a nonzero monodromy summand with zero global cohomology, by Theorem 7.3 and Proposition 9.1. Its absence from the global periodic invariant does not make it the zero sheaf. Thus the coefficient functor and its operations must be constructed before taking global sections or an index.

10 The compact Pfaffian realization

A compact construction must produce a square-root line and its section before comparing it with Δ_5. The target $X = S \times E$ has a holomorphic volume form after one is chosen on each factor. The construction requires reduced self-Ext complexes on its moduli of objects, describing infinitesimal deformations and obstructions. Their determinant lines are candidates for the square of a Pfaffian line. The reduced complexes, orientations, sections, and their gluing are the geometric data to be constructed.

A morphism from the obstruction sheaf to the structure sheaf is called a *cosection*. Its kernel alone does not define a reduced perfect obstruction theory. The reduction here requires the actual perfect complex and compatible maps on extension families.

An E_3-prefactorization datum assigns complexes to open sets and compatible operations for disjoint insertions, with the three-dimensional little-disks operations where specified. The notation $\mathfrak{A}_X \in E_3\text{-HolFA}(X)$ means a chosen such holomorphic datum. A Calabi–Yau category alone is not an asserted construction of this datum [24, 25, 72, 74, 108].

Choose a stability condition with Harder–Narasimhan filtrations: finite filtrations whose semistable quotients have strictly decreasing phases. Let R select a finite range of their types. The construction needs a hybrid configuration carrier on E. It contains ordinary finite configurations and the wrapped strata for objects supported over the whole elliptic factor.

Charges lie in a lattice $\Gamma_{X,R}$ with quadratic Gram map Π_X. Vanishing-cycle sheaves Φ_R record the local critical geometry, and an orientation o_R is needed to glue them. Hall correspondences H_R encode extensions and their composition. The source coalgebra $C_{X,R}$ and its Koszul cobar $\Theta_{\mathrm{Kos},R}$ retain the differentials and higher operations needed to compare that source with the denominator. The primitive part after Gram descent is P_R^{Π}.

These objects, a Pfaffian line $\mathcal{L}_{\mathrm{Pf},R}$ and section $\mathrm{pf}_{X,R}$, its orientation character $\epsilon_{o,R}$, and a recognition map Rec_R form compatible finite data. Their inverse limit is

$$\mathfrak{D}_X^{DI} = \varprojlim_R (\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathcal{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}, \mathrm{Rec}_R).$$

Every entry requires compatible transition maps. This inverse system specifies the target of the construction. The compact geometric entries remain to be constructed.

The orientation hypotheses have separate roles. Condition (Do) requires an orientation through the degeneration, agreeing with the generic reduced orientation. Condition $(O1)$ requires a square root of the reduced determinant line on the quotient by E, extending over the cusp boundary. Condition $(O1)^+$ requires Weyl-equivariant root lifts with a trivialized sign cocycle on the actual action groupoid. A reduction to constant-group cohomology $H^2(W^{(2)}; \mathbb{F}_2)$ is additional data.

Condition $(O2_{\mathrm{atlas}})$ requires type-II wall charts with a rank-one normal Pfaffian factor, a section-compatible root lift and reflection-invariant units. The reflection changes the normal coordinate's sign, and the chosen lift has residual character -1 on each simple reflection. The formal set of twelve signs

has $2^{12} = 4096$ elements; a geometric orbit of that size needs its own bijection. The equality with 64^2 is numerical.

The finite geometric Pfaffian datum (P_{fin}) further requires the following data at each finite stage, on the reduced compact quotient $\mathfrak{Q}_R$.

(i) Reduced perfect self-Ext complexes K_R with odd skew self-duality $K_R \simeq K_R^\vee[1]$, compatible with reduction and the elliptic quotient.

(ii) Orientations and Pfaffian lines with specified isomorphisms $\mathcal{L}_{\mathrm{Pf},R}^{\otimes 2} \simeq \mathcal{L}_{\det,R}$.

(iii) Finite skew blocks and Pfaffian sections, with the prescribed active support, signed exponents, and full even and odd dimensions.

(iv) Type-II wall factorizations $\mathrm{pf}_R = \upsilon_R u_R \mathrm{pf}_R^{\tan}$, with υ_R an invariant unit and normal divisor order one.

(v) Line comparisons with $\mathcal{L}^5 \otimes \nu_{\Delta_5}$, compatible with determinant squares and cusp frames, with leading coefficient 64.

(vi) Strict section-preserving transitions with closed images and vanishing $\lim^1$ for the line and section systems.

These hypotheses retain the finite geometric complexes and maps. Equality of scalar products supplies none of these data.

The orientation character acts on lattice reflections. To compare it with automorphy, extend the Maass character to the homogeneous null cone over $\mathcal{H}_2$. At the cusp frame $w(Z)$, put

$$\widehat{\Delta}_5(tw(Z)) = t^{-5}\Delta_5(Z), \qquad t \in \mathbb{C}^\times.$$

Write $\sigma : \mathbf{Sp}_4(\mathbb{Z}) \to \Gamma_{3,2}$ for the exterior-square homomorphism. Its frame relation $\sigma(g)w(Z) = \det(CZ + D)w(gZ)$ gives $\widehat{\Delta}_5(\sigma(g)w) = \nu_{\Delta_5}(g)\widehat{\Delta}_5(w)$. The character descends through the kernel $\{\pm\mathbf{I}_4\}$, where it is trivial. Since $\Gamma_{3,2} \subset \mathrm{SO}(L)$ and $\det(-\mathbf{I}_L) = -1$, it has a unique extension

$$\widetilde{\nu}_{\Delta_5} : \Gamma_{3,2} \times \{\pm\mathbf{I}_L\} \longrightarrow \{\pm 1\}, \qquad \widetilde{\nu}_{\Delta_5}(-\mathbf{I}_L) = (-1)^5 = -1.$$

The central value is forced by the homogeneous degree.

Extend each s_{δ_i} by the identity on $f_{\pm 1}$. Choose $g_i = \mathrm{diag}(V_i, V_i^{-t})$, where

$$V_1 = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad V_2 = \begin{pmatrix} -1 & -2 \\ 0 & 1 \end{pmatrix}, \qquad V_3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

These matrices are symplectic, with $\sigma(g_i) = -s_{\delta_i}$ and $\det(V_i^{-t}) = -1$. Maass's formula gives $\nu_{\Delta_5}(g_i) = 1$ [52, (1.5)]. Thus $\Delta_5(s_{\delta_i}Z) = -\Delta_5(Z)$, while $\widetilde{\nu}_{\Delta_5}(s_{\delta_i}) = -1$. The extended character restricts to the determinant on $W^{(2)}(\Lambda_{II}^{2,1})$, since the three reflections generate it. These are the linear reflection signs of [52, Propositions 2.1–2.2].

THEOREM 10.1 *(Pfaffian–Dirac).* Assume $(D0)$, $(O1)$, $(O1)^+$, $(O2_{\mathrm{atlas}})$, and (P_{fin}) on the cosection-reduced critical charts of $K3 \times E$, with the compatible inverse system just described. Then

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI}) = \Delta_5, \qquad \epsilon_o = \widetilde{\nu}_{\Delta_5} \quad \text{on } W^{(2)}(\Lambda_{II}^{2,1}).$$

The comparison uses the specified automorphic line maps and cusp frames.

The orientation comparison checks the three generators: $\epsilon_o(s_{\partial_i}) = -1 = \widetilde{\nu}_{\Delta_5}(s_{\partial_i})$. The section comparison uses the finite Fourier data and strict transitions in (P_{fin}). The detailed recognition statement is Theorem 2.1. Construction of its compact hypotheses remains a geometric problem.

Remark 10.2 *(Primitive recognition).* The protected primitive Lie superalgebra is

$$\mathcal{P}_X = P_X^{\Pi,\mathrm{red}} = P_X := H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X)/\overline{\operatorname{Rad}\langle\,,\,\rangle_X}.$$

Here $\mathcal{F}_X$ is the hybrid factorization sheaf, $\overline{\Pi}^{\Theta}_{X,*}$ is its completed Gram pushforward, and the closed radical is the part annihilated by its Hall pairing. The assertion $\mathcal{P}_X \cong \mathfrak{g}_{\Delta_5}$ requires the source representatives, relations, pairings, radical quotient, Poincaré–Birkhoff–Witt comparison, and full parity dimensions of $(S1)$–$(S10)$ on cofinal relation-closed windows. The finite Koszul comparisons are additional chain-level requirements. They are distinct from Pfaffian recognition and are not implied by equality of signed exponents.

11 Mirror periods and geometric comparison objects

The rank-seventeen lattice-polarized K3 family of Chapter 15 supplies a classical Siegel period model. The compact mirror problem asks for a separate three-dimensional period construction $\mathcal{P}^{(3)}_{K3\times E}$, with a map to $\mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z})$, a geometric meromorphic section and a comparison with the automorphic line identifying that section with Δ_5^{-2}. The stronger assertion of a period quotient requires its quotient map and an inverse period comparison as well. On a common period cover where the period map is normally unramified, the predicted polar divisor is $2\mathfrak{D}_{\mathrm{mir}}$, with reduced support the type-II Humbert–Heegner divisor. Its generic coarse reflection quotient has order one instead. These base and line conditions belong to Conjecture 4.2; the compact comparison is not used as a theorem hypothesis.

Definition 11.1 *(The five comparison objects).* With the preceding domains and hypotheses, the five objects are

$$(\mathcal{D}_X, \operatorname{den}(\mathfrak{g}_{\Delta_5}), \operatorname{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI}), \operatorname{Disc}(\mathcal{P} \to \mathbb{D}), \Theta^{\mathrm{Bor}}_{\mathrm{sing}}(h)).$$

The automorphic section, denominator, and multiplicative theta lift agree after the factor 64 and substitution $Z \mapsto 2Z$. The Pfaffian agrees under all five geometric hypotheses of Theorem 10.1. The inverse-square mirror comparison is conjectural. The primitive-algebra comparison has the separate hypotheses of Remark 10.2.

12 Protected traces and the physical realization

A physical interpretation must specify which states are counted and how observables act on them. A signed partition function alone specifies neither operation. If a boundary theory is described by a factorization differential graded category C, a chosen boundary vacuum b has an endomorphism algebra $A_b = \operatorname{End}_C(b)$. Composition in C gives the boundary product. The bar construction records twisting and coupling data.

Bulk observables are compared with the derived chiral center $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$, whose action must respect boundary products and locality. A trace on that center is a further operation.

The compact comparison requires a source coalgebra C_X and a cobar comparison with the denominator chiral object $\mathsf{Den}_{\Delta_5,E}$. It must preserve Weyl transport, orientation, Hall pairings, radical quotients and the Poincaré–Birkhoff–Witt filtration. These are the chiral Koszul source datum, finite Koszul comparison datum, and strict Mittag–Leffler conditions of Definitions 1.153 and 1.156 and Proposition 1.159. They retain chain-level information that the denominator forgets.

For a protected trace, choose a chart algebra A_b in a specified completed chiral bar–cobar domain. Its comparison with the coalgebra has the type

$$\mathrm{Bar}^{\mathrm{ch}}(A_b) \simeq C_X, \qquad \text{or equivalently in that domain} \qquad A_b \simeq \Omega^{\mathrm{ch}} C_X.$$

The first map is a coalgebra map and the second is an algebra map. Require their filtrations and completions to agree with the finite inverse system.

The trace further requires concentration and stratification comparisons on this actual domain, and a degree -3 cyclic pairing

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \otimes Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \longrightarrow \mathbb{C}[-3].$$

The completion must support the stated chiral-Koszul comparisons, and the pairing must be compatible with the volume form, Pfaffian line, and transitions. Finally its scalar normalization must agree with the product and the reduced enumerative convention.

Let $t_{h,d,n}$ be the normalized supertrace on the (h, d, n)-graded summand of the supplied protected complex. The index n is Euler characteristic, while h and d record the K3 square and elliptic degree as in the counting series. Write the full graded trace as

$$\mathrm{Tr}^{\mathrm{bulk}}_{\mathrm{gr}}((-1)^F \mathrm{id}) := \sum_{h,d \geq 0} \sum_{n \in \mathbb{Z}} t_{h,d,n} r^n q^{h-1} s^{d-1},$$

using the specified completion and the DT expansion at $r = 0$. Here F denotes parity.

Suppose the specified concentration and stratification comparisons identify this trace with the normalized reduced state. Then the protected trace identity is

$$\mathrm{Tr}^{\mathrm{bulk}}_{\mathrm{gr}}((-1)^F \mathrm{id}) = \Delta_s^{-2},$$

as an identity after coefficientwise rationalization in r. A trace at a fixed Euler degree n_0 is the coefficient $[r^{n_0}]$ in the $r = 0$ expansion, rather than this full series. The OP scalar is -4096 times the full graded trace. Construction of this compact trace datum and construction of a gravity-line algebra acting on boundary states remain distinct problems.

Remark 12.1 *(Information retained by the comparisons).* Choosing a vacuum, passing to cohomology, taking a radical quotient, completing, and taking a trace each require their own maps. The scalar identity does not reverse these operations.

The Pfaffian comparison retains a line and its character. Primitive recognition retains brackets and parity. The trace comparison retains a cyclic pairing. A proposed gravitational realization must preserve all operations used to compose and glue boundary observables.

A formal local model illustrates the algebraic data without providing compact gluing. Fix a worldline $\mathbb{R}_t \times \{0_s\} \times \{p\}$ in $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$, with a formal skyscraper boundary condition at p and multiplicity space $\mathbb{C}^N$. In the formal Darboux chart $\widehat{\mathcal{O}}_{\mathbb{C}^2,p} = \mathbb{C}[[z_1, z_2]]$, the transverse coordinates act by matrices $\phi_1, \phi_2 \in \mathfrak{gl}_N$.

Work in the formal completion at $\phi_1 = \phi_2 = 0$, the length-N skyscraper with trivial transverse action. Let $\mathrm{Kosz}([\phi_1, \phi_2])$ be the power-series algebra in the matrix entries, tensored with odd generators η_{ij} in degree -1, with differential $d\eta_{ij} = [\phi_1, \phi_2]_{ij}$. It describes the completed derived commuting locus. Infinitesimal simultaneous conjugation is imposed by the continuous Chevalley–Eilenberg cochain complex

$$A^{\mathrm{cl}}_{\partial,N} = C^{\bullet}_{\mathrm{CE}}(\mathfrak{gl}_N, \mathrm{Kosz}([\phi_1, \phi_2])).$$

The ghosts have degree one, and the total differential combines the Koszul differential with the Lie-algebra action and bracket. This is a classical formal-local boundary model in the mixed holomorphic–topological setting with local geometry $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ [26, 27]. Its use as a compact chiral-Hochschild chart requires a comparison map and descent, rather than only equality of local expressions.

The simplest gluing obstruction has a precise surface carrier. For a holomorphic-symplectic surface (Y, ω) and a symplectic vector field v, the one-form $\alpha_v = -\iota_v \omega$ is closed. Choose local holomorphic functions h_i with $dh_i = \alpha_v$. Their differences on overlaps are locally constant and define

$$\mathrm{per}_Y(v) = [h_i - h_j] \in H^1(Y, \mathbb{C}_Y).$$

This class vanishes exactly when the primitives can be changed by constants to glue to one holomorphic Hamiltonian. Equivalently it is the connecting image for $0 \to \mathbb{C}_Y \to \mathcal{O}_Y \xrightarrow{d} \Omega^1_{Y,\mathrm{cl}} \to 0$.

For the threefold $X = S \times E$, the notation $\mathrm{ob}^{\mathrm{hol}}_{\mathrm{dR}}(X)$ names the proposed compact descent obstruction. Defining its target complex and the map transporting the local surface Hamiltonians to it remains part of the compact construction. The surface period class alone supplies neither. A global theory must also establish locality, anomaly cancellation, and the quantum master equation. A three-dimensional gravitational path integral with internal factor $K3 \times E$ requires both that global field theory and the gravity-line algebra acting on boundary states. The enumerative identity alone supplies neither.

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Part I

Lattices and automorphic algebras

Chapter 2

K3 lattices and spherical objects

The three real simple roots of the Igusa denominator have matrix

$$G = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

We first realize this matrix as the Euler form of three line bundles on a projective K3 surface. Their extension spaces then give a doubled quiver with two arrows between each pair of vertices. A second choice of the line bundles makes their Hilbert polynomials equal, so that polystability supplies a formal differential graded algebra.

These constructions take place on the surface. Pushing the bundles onto a fiber of $S \times E$ introduces a normal deformation and cancels the Euler form. To compare the resulting threefold charges with Igusa exponents, we must therefore distinguish the surface pairing, the threefold Euler form, and the quadratic Gram map. We compute all three before considering their compatibility with Hall operations.

1 Cohomological and algebraic Mukai lattices

Let S be a smooth complex projective K3 surface. Write $\omega_S \in H^4(S, \mathbb{Z})$ for the class of a point, normalized by $\int_S \omega_S = 1$. The Néron–Severi group $\mathrm{NS}(S)$ is the image of the first Chern class map $\mathrm{Pic}(S) \to H^2(S, \mathbb{Z})$. Its rank is $\rho(S)$.

Definition 1.1 *(Mukai lattice).* The cohomological Mukai lattice is

$$\Lambda_S = \widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}).$$

We write its elements as (r, D, s), with the last coordinate denoting $s\omega_S$, and use the pairing

$$\langle (r, D, s), (r', D', s') \rangle = D \cdot D' - rs' - r's.$$

The algebraic Mukai lattice is its sublattice

$$\Lambda_S^{\mathrm{alg}} = \mathbb{Z} \oplus \mathrm{NS}(S) \oplus \mathbb{Z}.$$

For $F \in D^b \operatorname{Coh}(S)$, its Mukai vector is

$$v(F) = \operatorname{ch}(F)\sqrt{\operatorname{td}(S)} = \big(\operatorname{rk} F, c_1(F), \operatorname{rk} F + \operatorname{ch}_2(F)\big).$$

Here Chern characters and ranks of a complex are alternating sums, and $\operatorname{ch}_2(F)$ denotes its integral over S.

The formula uses $\mathrm{td}(S) = 1 + 2\omega_S$ and $\sqrt{\mathrm{td}(S)} = 1 + \omega_S$. The last coordinate is integral because D^2 is even. The two lattices have different ranks: Λ_S has rank 24, whereas Λ_S^{alg} has rank $\rho(S) + 2$.

The Euler pairing is $\chi_S(F, G) = \sum_i (-1)^i \dim_{\mathbb{C}} \mathrm{Ext}^i_S(F, G)$. The numerical Grothendieck group is $K_0(S)$ modulo the radical of this pairing.

PROPOSITION 1.2 *(Riemann–Roch and numerical classes).* For $F, G \in D^b \mathrm{Coh}(S)$,

$$\chi_S(F, G) = -\langle v(F), v(G) \rangle.$$

The Mukai vector induces an isomorphism from the numerical Grothendieck group of S to Λ_S^{alg}.

Proof. The involution $(r, D, s)^\vee = (r, -D, s)$ satisfies

$$\int_S (r, D, s)^\vee (r', D', s') = rs' + r's - D \cdot D'.$$

Hirzebruch–Riemann–Roch identifies this expression with $\chi_S(F, G)$ when the vectors are $v(F)$ and $v(G)$.

The vectors of $\mathcal{O}_S$, a point sheaf $\mathcal{O}_x$, and a line bundle $\mathcal{O}_S(D)$ are

$$(1, 0, 1), \qquad (0, 0, 1), \qquad (1, D, 1 + D^2/2).$$

Their integer combinations generate Λ_S^{alg}. Its pairing is nondegenerate by the Hodge index theorem. Thus the kernel of v is precisely the radical of the Euler form, which proves the assertion about numerical classes. □

For a zero-dimensional subscheme $Z \subset S$ of length a, the exact sequence $0 \to I_Z \to \mathcal{O}_S \to \mathcal{O}_Z \to 0$ gives

$$v(I_Z) = (1, 0, 1 - a), \qquad \langle v(I_Z), v(I_Z) \rangle = 2a - 2.$$

In particular, a point sheaf and an ideal of a point have different Mukai vectors. The numerical quotient is essential in Proposition 1.2: quotienting the algebraic Grothendieck group merely by torsion does not produce the full cohomological lattice. The construction and Riemann–Roch convention are those of [57, 91, 92].

Example 1.3 *(Points, ideals and divisors).* For a point p, a zero-dimensional subscheme Z of length a, and a smooth curve C of genus g, the vectors are

F	$v_S(F)$
$\mathcal{O}_S$	$(1, 0, 1)$
$\mathcal{O}_p$	$(0, 0, 1)$
I_Z	$(1, 0, 1 - a)$
$\mathcal{O}_S(D)$	$(1, D, 1 + D^2/2)$
$I_C = \mathcal{O}_S(-C)$	$(1, -C, g)$
$\mathcal{O}_C$	$(0, C, 1 - g)$.

The ideal sequences give the second, third and last rows. The exponential Chern character gives the line-bundle rows, and adjunction gives $C^2 = 2g - 2$. In particular,

$$v_S(\mathcal{O}_p)^2 = 0, \quad v_S(I_Z)^2 = 2a - 2, \quad \langle v_S(\mathcal{O}_S), v_S(\mathcal{O}_p) \rangle = -1.$$

The last equality agrees with $\chi_S(\mathcal{O}_S, \mathcal{O}_p) = 1$.

Remark 1.4 *(The hyperbolic sign).* The K3 lattice $H^2(S,\mathbb{Z})$ is $U^{\oplus 3}\oplus E_8(-1)^{\oplus 2}$, where U has basis e,f with $e^2=f^2=0$ and $e\cdot f=1$. The vectors $(1,0,0)$ and $(0,0,1)$ have pairing -1. Replacing the second by its negative identifies their span with U. Hence

$$\Lambda_S\cong U^{\oplus 4}\oplus E_8(-1)^{\oplus 2}$$

is even and unimodular of signature $(4,20)$. Changing only the sign of the hyperbolic off-diagonal entries gives an isometric integral lattice. Riemann–Roch fixes which sign belongs to the chosen Mukai vector.

Lemma 1.5 *(Two hyperbolic planes).* The full lattice Λ_S contains an orthogonal primitive copy of $U\oplus U$. Such a copy need not lie in Λ_S^{alg}.

Proof. Take two summands in the displayed decomposition. For a projective surface of Picard rank one, the algebraic lattice has rank three and cannot contain a rank-four sublattice. □

2 Three spherical line bundles

An object $L\in D^b\operatorname{Coh}(S)$ is spherical if $\operatorname{Ext}^0(L,L)=\operatorname{Ext}^2(L,L)=\mathbb{C}$ and its other self-extension groups vanish. Every line bundle on a K3 surface is spherical, since its self-extensions are $H^*(S,\mathcal{O}_S)$. The problem is to arrange the extensions between three different line bundles.

We use the following consequence of the K3 period theorem. A primitive even sublattice of signature $(1,r-1)$ in the K3 lattice is the Néron–Severi lattice of some projective K3 surface. If a positive vector in that lattice is perpendicular to no vector of square -2, a marking can put it in the ample cone. Indeed, a generic period makes the prescribed lattice exactly algebraic, and reflections in square--2 classes move positive chambers to the ample chamber [57, Chapters 7, 8 and 14].

Proposition 2.1 *(A triple with the required Euler form).* For every positive integer d, there is a polarized K3 surface (S,H) with $H^2=2d$ and line bundles L_0,L_1,L_2 such that

$$\dim\operatorname{Ext}^k(L_i,L_j)=\begin{cases}1, & i=j,\ k=0,2,\\ 2, & i\neq j,\ k=1,\\ 0, & \text{otherwise.}\end{cases}$$

Thus $(\chi_S(L_i,L_j))_{i,j}=G$. The bundles can be chosen with $L_0=\mathcal{O}_S$ and

$$\chi(L_0(mH))=dm^2+2,\qquad \chi(L_i(mH))=dm^2-2\quad(i=1,2).$$

Proof. In three orthogonal hyperbolic planes choose bases e_i,f_i and set

$$D_1=e_1-4f_1,\qquad D_2=-4f_1+e_2-4f_2,\qquad H=e_3+df_3.$$

The lattice $N_d=\langle D_1,D_2,H\rangle$ is primitive: the coefficient minor in rows e_1,e_2,e_3 is the identity. Its Gram matrix is

$$\begin{pmatrix}-8 & -4 & 0\\ -4 & -8 & 0\\ 0 & 0 & 2d\end{pmatrix}.$$

It has signature $(1,2)$. Every vector perpendicular to H has square $-8(a^2+ab+b^2)$, so none has square -2. The period theorem therefore gives a marked surface with $\mathrm{NS}(S)=N_d$ and H ample.

Take $L_i = \mathcal{O}_S(D_i)$ for $i = 1, 2$. Each of D_1, D_2, $D_1 - D_2$ has square -8 and is perpendicular to H. A nonzero effective divisor has positive intersection with H. Consequently neither sign of any of these three classes is effective.

For each such class D, we obtain

$$h^0(\mathcal{O}_S(D)) = h^2(\mathcal{O}_S(D)) = 0, \qquad \chi(\mathcal{O}_S(D)) = 2 + D^2/2 = -2.$$

Hence $h^1(\mathcal{O}_S(D)) = 2$, giving all the off-diagonal extensions. The diagonal groups follow from sphericality. Riemann–Roch applied to $D_i + mH$ gives the Hilbert polynomials. □

All three bundles have slope zero with respect to H, but their reduced Hilbert polynomials are different. A direct sum containing L_0 and one of L_1, L_2 is therefore not Gieseker polystable [56, Definitions 1.2.3–1.2.4]. To apply the formality theorem for polystable sheaves, we construct a triple with one common polynomial.

THEOREM 2.2 *(A common-polynomial spherical triple).* There is a polarized K3 surface (S, H) with $H^2 = 8$ and three line bundles L_1, L_2, L_3 having the extension groups of Proposition 2.1. Each is H-Gieseker stable, and each has Hilbert polynomial $4m^2 - 2$.

Proof. In $U^{\oplus 3}$ put

$$D_1 = e_1 - 4f_1, \qquad D_2 = -4f_1 + e_2 - 4f_2,$$
$$D_3 = -4f_1 - 4f_2 + e_3 - 4f_3, \qquad V = f_1 + f_2 + 2f_3.$$

The sublattice $N = \langle D_1, D_2, D_3, V \rangle$ is primitive, since its coefficient minor in rows e_1, e_2, e_3, f_1 has determinant one. In the stated basis its matrix is

$$N = \begin{pmatrix} -8 & -4 & -4 & 1 \\ -4 & -8 & -4 & 1 \\ -4 & -4 & -8 & 2 \\ 1 & 1 & 2 & 0 \end{pmatrix}.$$

Set $H = D_3 + 4V = e_3 + 4f_3$. Then $H^2 = 8$, $H \cdot D_i = 0$, and $H \cdot V = 2$. The span of the D_i is negative definite, with

$$\left(\sum_i a_i D_i\right)^2 = -4\left(\sum_i a_i^2 + \left(\sum_i a_i\right)^2\right).$$

Thus N has signature $(1, 3)$. A vector in N perpendicular to H has zero V coordinate, and its square is divisible by eight. In particular, $H^\perp \cap N$ contains no square--2 class.

Choose a surface with $\mathrm{NS}(S) = N$ and H ample. Set $L_i = \mathcal{O}_S(D_i)$. For $i \neq j$, the class $D_j - D_i$ is nonzero, has square -8, and is perpendicular to H. The previous cohomology argument gives $\mathrm{Ext}^1(L_i, L_j) = \mathbb{C}^2$ and no other off-diagonal extensions. A line bundle is Gieseker stable: a proper subsheaf of the same rank has a nonzero torsion quotient and therefore a smaller reduced Hilbert polynomial. Finally,

$$\chi(L_i(mH)) = 2 + (D_i + mH)^2/2 = 4m^2 - 2.$$

□

3 Integral gluing and the root sublattice

The primitive embedding above specifies more than a rational intersection form. The ample class and the negative classes do not split the Picard lattice integrally. We compute the missing classes before using the spherical vectors as a root lattice.

For an integral nondegenerate lattice K, let $K^* = \{x \in K \otimes \mathbb{Q} : (x, K) \subset \mathbb{Z}\}$ and $A_K = K^*/K$. If K is even, its discriminant quadratic form is $q_K(x + K) = x^2 \bmod 2\mathbb{Z}$.

Proposition 3.1 *(The Picard gluing).* In Theorem 2.2, let $L = \langle D_1, D_2, D_3, H\rangle$. Then

$$[N : L] = 4, \qquad N/L \cong \mathbb{Z}/4, \qquad \det L = -2048, \qquad \det N = -128.$$

The overlattice is obtained by adjoining the isotropic discriminant class of $(H - D_3)/4$. Moreover,

$$A_N \cong \mathbb{Z}/8 \oplus \mathbb{Z}/16.$$

Proof. The change of basis $H = D_3 + 4V$ has determinant four. It identifies N/L with the cyclic group generated by $V + L$. In $L \otimes \mathbb{Q}$,

$$V = (H - D_3)/4, \qquad V^2 = 0, \qquad \big((V, D_1), (V, D_2), (V, D_3), (V, H)\big) = (1, 1, 2, 2),$$

where the last expression lists the four pairings. Thus $V \in L^*$ and its class is isotropic of exact order four.

The D_i block has eigenvalues $-16, -4, -4$, and the orthogonal H block is (8). Hence $\det L = -2048$ and $\det N = \det L/4^2 = -128$. The greatest common divisors of the one-, two-, and three-row minors of the displayed matrix of N are $1, 1, 8$. Its Smith invariant factors are therefore $1, 1, 8, 16$, which give the stated discriminant group. □

The corresponding Mukai lattice is $M = \mathbb{Z} \oplus N \oplus \mathbb{Z}$. Write a vector as (r, a, b, c, t, s), meaning $(r, aD_1 + bD_2 + cD_3 + tV, s)$. The three spherical vectors are

$$v_i = v(L_i) = (1, D_i, -3).$$

Let $F \subset M$ be their integer span.

Proposition 3.2 *(The spherical lattice and its complement).* The lattice F is saturated in M, and its Gram matrix is $-G$. Its orthogonal complement consists exactly of

$$C(r, a, c) = (r, a, a, c, 4(c - a), 3r - 16a), \qquad r, a, c \in \mathbb{Z}.$$

In this parametrization the complement has matrix

$$C = \begin{pmatrix} -6 & 16 & 0 \\ 16 & -40 & -8 \\ 0 & -8 & 8 \end{pmatrix}, \qquad \det C = 256, \qquad A_C \cong \mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/16.$$

The inclusion $F \oplus C \subset M$ has quotient $\mathbb{Z}/2 \oplus \mathbb{Z}/4$ and index eight.

Proof. The coefficient matrix of the v_i has an identity minor in rows a, b, c, proving saturation. Their pairings are

$$\langle v_i, v_j\rangle = D_i \cdot D_j + 6 = \begin{cases} -2 & i = j, \\ 2 & i \neq j. \end{cases}$$

For $x = (r, a, b, c, t, s)$, the differences of the equations $\langle x, v_i \rangle = 0$ first give $b = a$ and $t = 4(c - a)$. The remaining equation gives $s = 3r - 16a$. These equations have no further divisibility condition, so the displayed parametrization exhausts the integral complement.

Substitution into the Mukai pairing gives the matrix C. Its entries have greatest common divisor two, and its two-row minors have greatest common divisor sixteen. Together with its determinant, these give Smith factors 2, 8, 16.

To compute the quotient, impose the three relations $v_i = 0$ in the free group with coordinate generators R, A, B, C_0, T, S_0. They identify $A = B = C_0 = -R + 3S_0$. The three complement generators then impose

$$R + 3S_0 = 0, \qquad -2R - 10S_0 - 4T = 0, \qquad -R + 3S_0 + 4T = 0.$$

Eliminating R leaves $4S_0 + 4T = 0$ and $6S_0 + 4T = 0$. Their relation matrix has Smith factors 2, 4. As a check on the index, $\det M = 128$ and $\det F \det C = 32 \cdot 256 = 8^2 \det M$. □

The negative of the Mukai pairing on F is the Euler matrix G. This constructs an integral copy of the real-root lattice. It does not identify all algebraic charges with that copy. For example, $(0, 0, 1)$ lies outside $F \otimes \mathbb{Q}$: a combination of the v_i with zero divisor part has every coefficient zero. Point charges therefore require additional classes even rationally.

4 The extension quiver and its formal deformation equation

Choose finite-dimensional vector spaces W_i, not all zero, and put $F_W = \bigoplus_{i=1}^{3} L_i \otimes W_i$, using the common-polynomial triple. This sheaf is H-polystable. Its self-extension spaces in degrees zero and two are

$$\operatorname{Ext}^0(F_W, F_W) = \bigoplus_i \operatorname{End}(W_i), \qquad \operatorname{Ext}^2(F_W, F_W) = \bigoplus_i \operatorname{End}(W_i)^*.$$

The trace pairing identifies each dual matrix space with its original. The degree-one space is

$$\operatorname{Ext}^1(F_W, F_W) = \bigoplus_{i \neq j} \operatorname{Ext}^1(L_i, L_j) \otimes \operatorname{Hom}(W_i, W_j).$$

Choose two arrows $a : i \to j$ for each $i < j$. A basis of $\operatorname{Ext}^1(L_i, L_j)$ assigns a map $X_a : W_i \to W_j$ to each arrow. Use a Serre-dual basis of $\operatorname{Ext}^1(L_j, L_i)$ to assign $Y_a : W_j \to W_i$.

Degree-one Serre duality is alternating. Choose the dual bases so that the quadratic Yoneda product is the moment map

$$\mu_i(X, Y) = \sum_{a:\, \mathrm{head}(a) = i} X_a Y_a - \sum_{a:\, \mathrm{tail}(a) = i} Y_a X_a.$$

This also fixes the sign of the duality convention. The trace of $\sum_i \mu_i$ vanishes because every arrow contributes one commutator.

THEOREM 4.1 *(Pointed formal deformation model).* The algebra $R \operatorname{Hom}_S(F_W, F_W)$ is formal. Let A be a local Artin $\mathbb{C}$-algebra with maximal ideal $\mathfrak{m}_A$. Choose a formality comparison. Deformations with an identified special fiber are then described by

$$(X, Y) \in \operatorname{Ext}^1(F_W, F_W) \otimes \mathfrak{m}_A, \qquad \mu_i(X, Y) = 0,$$

modulo the gauge group $\exp(\operatorname{Ext}^0(F_W, F_W) \otimes \mathfrak{m}_A)$.

Proof. Theorem 1.2 of Budur–Zhang applies to every Gieseker-polystable sheaf on a projective K3 surface, with no genericity condition on its ample polarization [16]. It gives a quasi-isomorphism zigzag from $R\operatorname{Hom}_S(F_W, F_W)$ to its cohomology algebra.

Passing to the commutator differential graded Lie algebra preserves this comparison. Over the nilpotent ideal $\mathfrak{m}_A$, its Maurer–Cartan groupoid is invariant under quasi-isomorphism. In the cohomology algebra the differential is zero, so the equation is $u^2 = 0$ for u of degree one. Products of off-diagonal extensions can land only in the diagonal degree-two spaces. Serre duality identifies those products with the displayed μ_i, giving the quadratic equations and their gauge action. □

The quotient here fixes an identification of the special fiber. An equivariant comparison with the action of $\prod_i \mathbf{GL}(W_i)$ is additional structure on the formality map. Likewise, a cyclic comparison must preserve the trace up to the specified homotopies. Formality for one sheaf supplies neither compatibility with extension flags nor an orientation on a global moduli stack. These requirements will enter the Hall construction separately.

5 Passage to the threefold

Definition 5.1 *(The product threefold).* Let E be a smooth complex elliptic curve with origin 0_E and put $X = S \times E$. Write p_S, p_E for the projections. Choose nonzero holomorphic volume forms on S and E and use their product to trivialize K_X.

This is a Calabi–Yau threefold in the trivial-canonical sense. Its irregularity is $h^{1,0}(X) = 1$, and

$$\chi_{\mathrm{top}}(X) = 24 \cdot 0 = 0, \qquad \chi(\mathcal{O}_X) = 2 \cdot 0 = 0.$$

The chosen volume form determines the trace used by the Calabi–Yau structure.

Proposition 5.2 *(Extensions on a surface fiber).* For $t \in E$, let $i_t : S \hookrightarrow X$ be the fiber inclusion. For perfect complexes A, B on S, there is a natural decomposition

$$\operatorname{Ext}^k_X(i_{t*}A, i_{t*}B) = \operatorname{Ext}^k_S(A, B) \oplus \big(\operatorname{Ext}^{k-1}_S(A, B) \otimes T_tE\big).$$

In particular, the self-extension dimensions of $i_{t*}L_i$ are $(1, 1, 1, 1)$ in degrees $0, 1, 2, 3$. For $i \neq j$, the dimensions are $(0, 2, 2, 0)$. All their threefold Euler pairings vanish.

Proof. We have $i_{t*}A = A \boxtimes \mathcal{O}_t$. The derived Künneth formula for Hom gives

$$R\operatorname{Hom}_X(i_{t*}A, i_{t*}B) \simeq R\operatorname{Hom}_S(A, B) \otimes R\operatorname{Hom}_E(\mathcal{O}_t, \mathcal{O}_t).$$

The second factor has cohomology $\mathbb{C}$ in degree zero and T_tE in degree one. This follows from a local resolution by the equation of the point. Tensoring gives the asserted decomposition. The two summands have opposite Euler signs and the same dimensions. □

The normal line T_tE is part of this formula. A local parameter identifies it with $\mathbb{C}$ and gives an exterior-generator model for the normal extension algebra. Changing the parameter rescales that generator. Moreover, $i_{t*}L_i$ has two-dimensional support. A category consisting only of sheaves supported on curves in X does not contain these objects.

Remark 5.3 *(The two K3 Euler characteristics).* The integer $\chi(\mathcal{O}_S) = 2$ appears in the line-bundle Riemann–Roch formula. The integer $\chi_{\mathrm{top}}(S) = 24$ appears in the elliptic genus. With $\phi_{0,1} = r^{-1} + 10 + r + O(q)$, the K3 elliptic genus is $2\phi_{0,1}$, whose specialization at the elliptic variable zero is 24. The Borcherds product of $\phi_{0,1}$ instead has weight $10/2 = 5$ [9, 39, 52]. These are different operations on the input coefficients.

The distinction persists in the CHL examples. Write k_N for the weight of the relevant genus-two form and $w_N = k_N/2$ for its square-root weight. The eta products entering the additive seeds have weights 12, 8, 6, 5, 4 at $N = 1, 2, 3, 4, 5$, respectively. The common factor $\vartheta_1(\tau, z)^2/\eta(\tau)^6$ has weight -2, so the weights are

N	1	2	3	4	5
k_N	10	6	4	3	2
w_N	5	3	2	3/2	1.

The prime-level formula is $k_N = 24/(N+1) - 2$ for $N = 2, 3, 5$ [47, 60]. At $N = 4$, the eta product is $\eta(\tau)^4\eta(2\tau)^2\eta(4\tau)^4$, giving weight five before the Jacobi factor [48, Table 1 and (3.17)]. Thus the numerical equality $3 + \chi(\mathcal{O}_S) = 5$ at $N = 1$ is not a formula for this sequence of weights. It supplies no comparison between a chiral coefficient and an automorphic weight.

6 Paired charges and the Gram map

Let $\omega_E \in H^2(E, \mathbb{Z})$ satisfy $\int_E \omega_E = 1$. Since $H^{\mathrm{odd}}(S, \mathbb{Z}) = 0$, every even cohomology class on X has unique components in $\widetilde{H}(S, \mathbb{Z}) \otimes (\mathbb{Z} \oplus \mathbb{Z}\omega_E)$.

Definition 6.1 *(Paired formal charge).* The formal charge lattice and its algebraic sublattice are

$$\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S, \qquad \Gamma_X^{\mathrm{alg}} = \Lambda_S^{\mathrm{alg}} \oplus \Lambda_S^{\mathrm{alg}}.$$

We order a charge as $c = (Q, P)$ by writing

$$v_X = P \otimes 1_E + Q \otimes \omega_E.$$

The notation Γ_X^{phys} will mean this same formal lattice when used without a specified stability condition. For a perfect complex $\mathcal{F}$ on X, its actual charge is $v_X(\mathcal{F}) = \mathrm{ch}(\mathcal{F})\sqrt{\mathrm{td}(X)}$.

Both components are invariant under translation in E. For an actual complex, $P = v(Li_t^*\mathcal{F})$ and $Q = v(Rp_{S*}\mathcal{F})$, so both are algebraic integral Mukai vectors. Conversely, external products with $\mathcal{O}_E$ and $\mathcal{O}_t$ generate both summands numerically. These formulas also explain why Q and P cannot be assigned support dimensions merely from their order in the pair.

Riemann–Roch on X gives the skew form

$$\chi_X(\mathcal{F}, \mathcal{G}) = \langle Q, P' \rangle - \langle P, Q' \rangle.$$

Indeed, duality changes the sign of the ω_E component, and integration on S turns the Mukai pairing into the negative Euler pairing. A symmetric cross-pairing on the same group is an auxiliary form and is not this threefold Euler form.

Definition 6.2 *(Gram degrees).* The Gram-index group is $\Gamma_{\mathrm{gram}} = \mathbb{Z}^3$. The Gram map is

$$\Pi_X(Q, P) = \left(\frac{\langle Q, Q \rangle}{2}, \langle Q, P \rangle, \frac{\langle P, P \rangle}{2} \right).$$

For $\gamma = (n, \ell, m)$ set $T(\gamma) = \left(\begin{smallmatrix} n & \ell/2 \\ \ell/2 & m \end{smallmatrix}\right)$.

Proposition 6.3 *(Gram matrix and Fourier character).* For $\Pi_X(Q, P) = (n, \ell, m)$, the Gram matrix of Q, P is $2T(\gamma)$, and its determinant is $4nm - \ell^2$. If $Z = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right)$, $q = e^{2\pi i \tau}$, $r = e^{2\pi i z}$ and $s = e^{2\pi i \sigma}$, then

$$e^{2\pi i \operatorname{tr}(T(\gamma)Z)} = q^n r^\ell s^m.$$

Proof. Evenness of Λ_S makes n, m integral. The matrix and exponential identities follow by multiplying their two-by-two entries. □

At the standard cusp, we use the positive additive semigroup

$$\begin{aligned}\Gamma_{\text{eff}} = {} & \{(n, \ell, m) : m > 0,\ n \geq 0\} \\ & \cup \{(n, \ell, 0) : n > 0\} \cup \{(0, \ell, 0) : \ell < 0\}.\end{aligned}$$

The omitted coordinate ℓ ranges over all integers in the first two sets. Its Jacobi-active subset Γ_{act} consists of the triples with $f(nm, \ell) \neq 0$, where $\phi_{0,1} = \sum f(a, b) q^a r^b$. Positivity of an exponent is an arithmetic condition. It does not assert existence of a sheaf with that charge.

Proposition 6.4 *(The ideal-sheaf charge).* Let $Y \subset X$ have dimension at most one, with fundamental one-cycle

$$[Y] = \beta \otimes \omega_E + d\omega_S \otimes 1_E$$

and holomorphic Euler characteristic $n_Y = \chi(\mathcal{O}_Y)$. Then

$$P(I_Y) = (1, 0, 1 - d), \qquad Q(I_Y) = (0, -\beta, -n_Y),$$

and

$$\Pi_X(I_Y) = (\beta^2/2, n_Y, d - 1).$$

Proof. The ideal sequence gives $\operatorname{ch}(I_Y) = 1 - [Y] - n_Y \omega_S \omega_E$. Here the degree-six coefficient is $-n_Y$ because $c_1(X) = 0$ and Y has dimension at most one. Multiplication by $\sqrt{\operatorname{td}(X)} = 1 + \omega_S$ gives the two displayed components. Their Mukai pairings give the three Gram coordinates. □

In this sector d is the degree of projection of the fundamental cycle to E. Its Gram exponent is $d - 1$. The distinction persists outside this sector, where the Gram map remains quadratic and a support-degree function requires a specified class of objects.

6.1 The quadratic additivity defect

For charges $c = (Q, P)$ and $c' = (Q', P')$, define

$$B(c, c') = \big(\langle Q, Q' \rangle, \langle Q, P' \rangle + \langle Q', P \rangle, \langle P, P' \rangle\big).$$

This symmetric bilinear map measures the failure of the Gram degrees to add.

Lemma 6.5 *(Polarization).* For all formal charges,

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c'), \qquad B(c, c) = 2\Pi_X(c).$$

If $(\delta q)(c, c') = q(c) + q(c') - q(c + c')$, then $B = -\delta \Pi_X$. In particular,

$$B(b, c) - B(a + b, c) + B(a, b + c) - B(a, b) = 0.$$

Proof. Expanding $(Q+Q')^2/2$ gives the first coordinate and its cross term $\langle Q, Q'\rangle$. Expanding $(Q+Q', P+P')$ gives the two middle cross terms. The third coordinate is the same calculation with P. Putting $c' = c$ gives the diagonal identity. Bilinearity cancels the four terms of the final equation. □

Remark 6.6 *(The primitive is quadratic).* An additive map q has $\delta q = 0$. Since B is nonzero, an additive change of coordinates cannot have coboundary $-B$. The identity $B = -\delta\Pi_X$ instead uses the quadratic map Π_X. The diagonal value $2\Pi_X(c)$ is its polarization identity.

6.2 *Primitive formal lifts*

LEMMA 6.7 *(A saturated lift of every Gram triple).* Choose two orthogonal hyperbolic planes in Λ_S, with bases e_1, f_1, e_2, f_2. For every $(n, \ell, m) \in \mathbb{Z}^3$, the vectors

$$Q = e_1 + nf_1, \qquad P = \ell f_1 + e_2 + mf_2$$

have Gram degree (n, ℓ, m) and span a primitive rank-two sublattice. The greatest common divisor of the coordinates of $Q \wedge P$ is one.

Proof. The hyperbolic pairings give $Q^2 = 2n$, $P^2 = 2m$ and $Q \cdot P = \ell$. The coefficient minor in rows e_1, e_2 is the identity matrix. It proves saturation and gives coefficient one on $e_1 \wedge e_2$ in $Q \wedge P$. □

Remark 6.8 *(Formal and geometric lifts).* The lemma constructs an integral cohomological pair for each prescribed Gram matrix. Algebraic pairs must lie in Γ_X^{alg}. A stable sheaf with such a pair requires further geometric conditions and a nonempty moduli space. Even when individual pairs are realized, a comparison with Hall multiplication requires compatibility with addition of classes in exact triangles.

7 AN ADDITIVE EXTENSION OF GRAM DEGREES

A Hall operation adds classes of objects. The Gram map is quadratic, so it cannot itself serve as an additive grading map. Introduce a second coordinate to record its cross terms.

Definition 7.1 *(Normal-ordered extension).* On the set $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \times \Gamma_{\mathrm{gram}}$ define

$$(c, T) \star (c', T') = (c + c', T + T' + B(c, c')).$$

The *normal-ordered* Gram map and the raw placement are

$$\overline{\Pi}_X(c, T) = \Pi_X(c) - T, \qquad i_0(c) = (c, 0).$$

THEOREM 7.2 *(The extension splits).* The operation $\star$ makes $\widehat{\Gamma}_X$ an abelian group. Its identity is $(0, 0)$, its inverse formula is

$$(c, T)^{-1} = (-c, -T + 2\Pi_X(c)),$$

and $\overline{\Pi}_X$ is a homomorphism. The map

$$\Psi : \widehat{\Gamma}_X \longrightarrow \Gamma_X^{\mathrm{form}} \oplus \Gamma_{\mathrm{gram}}, \qquad (c, T) \longmapsto (c, T - \Pi_X(c))$$

is an isomorphism to the ordinary direct sum. In particular, $c \mapsto (c, \Pi_X(c))$ is an additive section of the charge projection.

Proof. The cocycle identity proves associativity, and symmetry of B proves commutativity. Since $B(c,-c) = -2\Pi_X(c)$, the displayed inverse multiplies with (c,T) to $(0,0)$. Polarization gives

$$\Pi_X(c+c') - T - T' - B(c,c') = \Pi_X(c) - T + \Pi_X(c') - T',$$

which proves additivity of $\overline{\Pi}_X$. The same identity shows that Υ is additive, with inverse $(c,S) \mapsto (c, S + \Pi_X(c))$. □

The extension is thus split as an abelian group. Its role is to retain the correction coordinate in a proposed graded construction. The raw placement i_0 is a set map, not a homomorphism, and the additive section has zero normal-ordered Gram degree. Neither map provides an additive lift of arbitrary raw Gram degrees.

7.1 *Products of raw placements*

Lemma 7.3 *(The flag formula).* For $c_1, \dots, c_k \in \Gamma_X^{\mathrm{form}}$,

$$i_0(c_1) \star \cdots \star i_0(c_k) = \left(\sum_i c_i, \sum_{i<j} B(c_i, c_j) \right),$$
$$\overline{\Pi}_X\big(i_0(c_1) \star \cdots \star i_0(c_k)\big) = \sum_i \Pi_X(c_i).$$

Proof. Adding one factor introduces $B(\sum_{i<k} c_i, c_k) = \sum_{i<k} B(c_i, c_k)$. Induction gives the first identity. Iterated polarization gives $\Pi_X(\sum_i c_i) = \sum_i \Pi_X(c_i) + \sum_{i<j} B(c_i, c_j)$, which proves the second. □

This is an identity of charge groups. To use it on moduli of flags, one must construct the additional coordinate and its transition maps on those moduli. A group law alone does not give that geometric data.

7.2 *An obstruction along a root string*

Definition 7.4 (Raw regrading). For a vector space $V = \bigoplus_c V_c$ graded by formal charges, define

$$(\Pi_{X,*}^{\mathrm{raw}} V)_\gamma = \bigoplus_{\Pi_X(c)=\gamma} V_c.$$

A fixed-lift choice instead retains only $V_{L(\gamma)}$, where $\Pi_X(L(\gamma)) = \gamma$ for the selected degrees.

Vector-space regrading is always defined when the indicated sums exist. It need not preserve the degree of a multiplication or a Lie bracket.

Theorem 7.5 *(The fixed-lift root-string obstruction).* Let $\mathfrak{p} = \bigoplus_c \mathfrak{p}_c$ be a charge-graded Lie superalgebra. Suppose $x \in \mathfrak{p}_c$ and $y \in \mathfrak{p}_{c'}$ satisfy

$$[x,y] \neq 0, \qquad [x,[x,y]] \neq 0, \qquad \Pi_X(c) \neq 0.$$

The original bracket cannot be homogeneous for raw Gram degree on both of these channels. Consequently, fixed raw lifts of the real simple generators cannot preserve the nonzero second root strings of the matrix G.

Proof. Homogeneity of the first bracket requires

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c'), \quad \text{hence } B(c, c') = 0.$$

Homogeneity of the second requires $B(c, c + c') = 0$. But

$$B(c, c + c') = B(c, c) + B(c, c') = 2\Pi_X(c) \neq 0.$$

This is a contradiction. For the real simple generators attached to G, the off-diagonal Cartan integer is -2. The corresponding $\mathfrak{sl}_2$ root string has nonzero first and second brackets, so it meets the stated hypotheses [52]. □

Remark 7.6 (Scope of the obstruction). The theorem concerns the original bracket on homogeneous chosen charges. It does not exclude a different bracket, a different grading, or a comparison that mixes source lifts. The additive map $\overline{\Pi}_X$ removes this degree obstruction for an algebra already graded by $\widehat{\Gamma}_X$. Constructing such an algebra and comparing its primitive bracket remain separate problems.

7.3 Operations preserved by additive regrading

For a homomorphism $f : A \to B$ of abelian groups, let

$$(f_*V)_b = \bigoplus_{a : f(a)=b} V_a.$$

We use direct sums throughout this construction. Cohomological degree and super parity are retained as separate gradings.

PROPOSITION 7.7 *(Additive regrading).* On graded $\mathbb{Q}$-complexes, f_* is exact and symmetric monoidal. It preserves homogeneous products, Lie superbrackets, coalgebra maps, and their identities. It commutes with the direct-sum bar construction of an augmented differential graded algebra, including every finite length truncation.

Proof. A degree component of f_* is a direct sum of complexes, so exactness holds over $\mathbb{Q}$. The tensor comparison is the identification

$$\begin{aligned}(f_*V \otimes f_*W)_b &= \bigoplus_{f(a)+f(a')=b} V_a \otimes W_{a'} \\ &= (f_*(V \otimes W))_b.\end{aligned}$$

The equality uses additivity of f. This identification preserves the symmetry and its Koszul signs, since it changes neither cohomological degree nor parity. It therefore preserves all identities expressed by the stated operations.

For an augmented algebra $A_\bullet$, write $\bar{A}_\bullet$ for the augmentation ideal. The bar coalgebra has underlying complex $\bigoplus_{j\geq 0}(\bar{A}_\bullet[1])^{\otimes j}$, where $[1]$ lowers cohomological degree by one. Its differential comes from the internal differential and multiplication. Exactness, the tensor comparison, and preservation of these maps identify the two bar constructions. The same argument applies after restricting to length at most any fixed integer. □

Taking $f = \overline{\Pi}_X$ gives the appropriate additive regrading once the extended grading exists. Pairings of degree zero, bracket filtrations and finite Poincaré–Birkhoff–Witt constructions are preserved for the same reason. Completed products require an additional finiteness or continuity argument. The finite versions, including separate even and odd dimensions and zero-degree radicals, are developed in Appendix 8.

Remark 7.8 *(The root comparison).* The root coordinates used later identify a Gram triple with

$$\alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2} \in \Lambda^{2,1}_{II}.$$

Their form has $f_3^2 = 2$, $(f_2, f_{-2}) = -1$, and all other pairings zero. Thus $\alpha(n, \ell, m)^2 = 2\ell^2 - 8nm$. The three triples $(1, 1, 0)$, $(0, 1, 1)$ and $(0, -1, 0)$ give matrix G. This arithmetic identification is independent of the saturated surface embedding $F \subset M$. A geometric realization must compare the corresponding operations, as well as these lattices.

Proposition 7.9 *(Sheaves representing the three simple Gram degrees).* Choose $p \in S$ and a length-two subscheme $Z \subset S$. The three pairs of coherent sheaves

$$x_{\delta_1} = (I_Z, I_p), \qquad x_{\delta_2} = (I_p, I_Z), \qquad x_{\delta_3} = (I_p, \mathcal{O}_p)$$

have Gram degrees $(1, 1, 0)$, $(0, 1, 1)$ and $(0, -1, 0)$, respectively, when their vectors are ordered as (Q, P). These degrees are also represented by coherent sheaves on X: for a fixed $t \in E$, put

$$\begin{aligned} \mathcal{F}_1 &= i_{t*} I_Z \oplus p_S^* I_p, \\ \mathcal{F}_2 &= i_{t*} I_p \oplus p_S^* I_Z, \\ \mathcal{F}_3 &= i_{t*} I_p \oplus p_S^* \mathcal{O}_p. \end{aligned}$$

Proof. The vectors $v_S(I_Z) = (1, 0, -1)$, $v_S(I_p) = (1, 0, 0)$ and $v_S(\mathcal{O}_p) = (0, 0, 1)$ give

	$Q^2/2$	$Q \cdot P$	$P^2/2$
x_{δ_1}	1	1	0
x_{δ_2}	0	1	1
x_{δ_3}	0	-1	0.

For any sheaf A on S, the external-product formulas give

$$v_X(i_{t*}A) = v_S(A) \otimes \omega_E, \qquad v_X(p_S^* A) = v_S(A) \otimes 1_E.$$

They identify the charges of $\mathcal{F}_i$ with the displayed pairs. All supports are proper because X is projective. □

The sheaves $\mathcal{F}_1$ and $\mathcal{F}_2$ have rank one and a surface-supported direct summand. The sheaf $\mathcal{F}_3$ has surface and curve summands. Thus this construction gives actual coherent charges, but not pure sheaves of one fixed dimension. A realization by stable objects in a specified heart, with the required extension operations and oriented coefficients, is a further problem. The spherical line bundles of §2 answer a different question: they realize the additive Euler matrix of the real roots.

8 GEOMETRIC SUPPORT OVER THE ELLIPTIC CURVE

The support of an object gives another constraint on a proposed factorization construction. A finite configuration on E can record objects whose projection is finite. A curve projecting dominantly to E requires a different parameter space.

LEMMA 8.1 *(Projection of an effective curve).* Let $Z = \sum_i m_i[C_i]$ be an effective one-cycle in $S \times E$. If $(p_E)_* Z = b[E]$ with $b > 0$, then $p_E(|Z|) = E$.

Proof. The proper image of an irreducible curve C_i in E is a point or all of E. A dominant component contributes $m_i \deg(C_i/E) > 0$ to b, whereas a component mapping to a point contributes zero. If $b > 0$, some component is dominant. □

Remark 8.2 *(Support degree and Gram exponent).* For ideal sheaves in Proposition 6.4, the support degree is d and the third Gram coordinate is $d - 1$. For a sheaf on one surface fiber, the projection is a point even though its support has dimension two. Thus projection-finiteness and support dimension are distinct conditions. A moduli problem must specify both before it can be assigned to a finite-configuration or dominant-support parameter space.

The finite-configuration space is the Ran prestack of E: its T-points are nonempty finite subsets of $\mathrm{Hom}(T, E)$. We call objects projecting onto E wrapped objects. Combining the two support types requires moduli for wrapped objects and correspondences for mixed extensions. Their existence and compatibility do not follow from adjoining a second set of points to the Ran prestack. Appendix 8 states the stability, finiteness, and equivariance conditions for such a construction.

Definition 8.3 *(Finite Harder–Narasimhan type).* Fix a stability condition and a collection of factor labels. A Harder–Narasimhan filtration is a finite filtration with semistable quotients of strictly decreasing phase. When every quotient has a chosen factor label, its *retained HN type* is the ordered word of factor and phase labels.

A finite set of such words can bound the types of objects used in an extension correspondence. The finite-window proof and the further conditions needed for moduli of these objects are given in Appendix 8.

The surface calculation supplies concrete spherical objects, a root sublattice, and a formal deformation equation. The threefold calculation supplies their normal extensions and actual paired charges. A global Hall comparison must connect these constructions through extension correspondences with a specified coefficient theory, trace and orientation. The following chapters develop the arithmetic target before returning to those geometric operations.

Chapter 3

The automorphic denominator

1 The section to be realized

A geometric realization of the Igusa denominator must recover a section of a specified line, with its character and boundary behavior. These data can be determined before constructing the geometric source. The Jacobi coefficients give a convergent product. Its theta normalization then fixes a weight-five section, and its square gives the monic weight-ten form in the primitive curve-counting theorem.

The three constructions involved have different inputs. The automorphic product starts from a weak Jacobi form. The Borcherds–Kac–Moody denominator starts from a Lie superalgebra with specified generators and relations. A compact Pfaffian would start from a family of complexes with a pairing, reduction and orientation. This chapter identifies the first section and its relation to the second denominator. Its boundary calculations also give necessary conditions on the third construction.

2 Siegel space and the coefficient cone

For the classical theory of Siegel modular forms, see [41, Chapter I] and [114, §§2–3]. Write

$$\mathbb{H}_2 = \{Z = Z^t \in M_2(\mathbb{C}) : \operatorname{Im} Z > 0\}, \qquad Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix}.$$

The action of $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \in \mathbf{Sp}_4(\mathbb{Z})$ is $gZ = (AZ + B)(CZ + D)^{-1}$. Put $j(g, Z) = \det(CZ + D)$. A modular form of integral weight k and character ν satisfies

$$F(gZ) = \nu(g)\, j(g, Z)^k F(Z).$$

Thus it is a section of $\mathcal{L}^k \otimes \nu$ on the quotient stack $[\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2]$, where $\mathcal{L}$ is the determinant Hodge line and ν denotes the character line. All line-valued identities below use this quotient.

At the standard rational cusp $\mathfrak{c}_\infty$, set

$$q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$

The three elementary root monomials are

$$x_1 = qr, \qquad x_2 = sr, \qquad x_3 = r^{-1}.$$

They correspond to the charges $(1, 1, 0)$, $(0, 1, 1)$, $(0, -1, 0)$, in the order (n, ℓ, m).

Their nonnegative span is the monoid

$$C_+ = \{(n, \ell, m) \in \mathbb{Z}^3 : n \geq 0,\ m \geq 0, \quad n + m - \ell \geq 0\}.$$

Indeed

$$q^n r^\ell s^m = x_1^n x_2^m x_3^{n+m-\ell}.$$

The total root height and its completion are therefore

$$h(n,\ell,m) = 2(n+m) - \ell, \qquad \widehat{\mathbb{Z}[C_+]} = \mathbb{Z}[[x_1, x_2, x_3]].$$

At height h, this monoid has $\binom{h+2}{2}$ elements. Consequently multiplication and products of factors of increasing positive height are defined coefficientwise.

The usual ordering set for the Borcherds product is

$$\begin{aligned}\Gamma_{\mathrm{eff}} = {} & \{(n,\ell,m) : m > 0,\ n \geq 0,\ \ell \in \mathbb{Z}\} \\ & \cup \{(n,\ell,0) : n > 0,\ \ell \in \mathbb{Z}\} \cup \{(0,\ell,0) : \ell < 0\}.\end{aligned}$$

This larger semigroup records the conventional order of the indices. The coefficient cone C_+ supplies the completion. In particular, the entire ordering semigroup does not have finite decomposition sets: $(2,0,0) = (1,a,0) + (1,-a,0)$ for every integer a. The Jacobi support proved below places every nontrivial product factor in $C_+ \setminus \{0\}$.

In analytic coordinates $y_j = \mathrm{Im} z_j$, the corresponding chamber is

$$-y_2 > 0, \qquad y_1 + y_2 > 0, \qquad y_3 + y_2 > 0.$$

These inequalities imply $\mathrm{Im} Z > 0$ and $|x_i| < 1$. A useful ray is $(y_1, y_2, y_3) = (T, -T/2, T)$, on which $|x_i| = e^{-\pi T}$. The product theorem will give absolute convergence sufficiently far along this chamber. The more restrictive ordering $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ is a possible cusp subregion. Formal identities use total height and require no limit of complex variables.

3 The Jacobi input

For the Jacobi variables $(\tau, z) \in \mathbb{H}_1 \times \mathbb{C}$, use $q = e^{2\pi i \tau}$ and $r = e^{2\pi i z}$. The weight-zero, index-one weak Jacobi form is

$$\phi_{0,1}(\tau, z) = 4 \sum_{a=2}^{4} \frac{\theta_a(\tau,z)^2}{\theta_a(\tau,0)^2} = \sum_{n \geq 0,\ \ell \in \mathbb{Z}} f(n,\ell) q^n r^\ell.$$

Here

$$\begin{aligned}\theta_2(\tau,z) &= \sum_{a \in \mathbb{Z}} q^{(a+1/2)^2/2} r^{a+1/2}, \\ \theta_3(\tau,z) &= \sum_{a \in \mathbb{Z}} q^{a^2/2} r^a, \qquad \theta_4(\tau,z) = \sum_{a \in \mathbb{Z}} (-1)^a q^{a^2/2} r^a.\end{aligned}$$

The denominators are nonzero on $\mathbb{H}_1$.

The theta transformation formulas give the modular law

$$\phi_{0,1}\left(\frac{a\tau+b}{c\tau+d}, \frac{z}{c\tau+d}\right) = e^{2\pi i c z^2/(c\tau+d)} \phi_{0,1}(\tau, z)$$

for $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathbf{SL}_2(\mathbb{Z})$, and the elliptic law

$$\phi_{0,1}(\tau, z + a\tau + b) = q^{-a^2} r^{-2a} \phi_{0,1}(\tau, z), \qquad a, b \in \mathbb{Z}.$$

Comparing coefficients gives

$$f(n,\ell) = f(n + a\ell + a^2, \ell + 2a).$$

Together with $z \mapsto -z$, this implies that $f(n,\ell)$ depends only on $4n-\ell^2$. Reduction of ℓ to 0 or 1, followed by weak holomorphy, gives

$$f(n,\ell)=0 \quad \text{if } 4n-\ell^2<-1.$$

The normalization is $\phi_{0,1}=r^{-1}+10+r+O(q)$. It generates the one-dimensional space $J^{\mathrm{weak}}_{0,1}$ [39]. For a K3 surface S, the elliptic genus has constant term $2r^{-1}+20+2r$, so $\mathrm{Ell}(S)=2\phi_{0,1}$. This is the normalization used in the Hilbert-scheme formulas of Chapter 1 [34].

Expansion of the three theta quotients gives

	$\ell=-3$	-2	-1	0	1	2	3
$n=0$	0	0	1	10	1	0	0
$n=1$	0	10	-64	108	-64	10	0
$n=2$	1	108	-513	808	-513	108	1

(3.1)

Appendix 2 gives the coefficient calculation. The constant coefficient 10 determines the weight. The coefficient at discriminant -1 determines the simple reflective divisor [49]. The negative coefficients will be signed root multiplicities.

Define the active set by

$$\Gamma_{\mathrm{act}}=\{(n,\ell,m)\in\Gamma_{\mathrm{eff}} : f(nm,\ell)\neq 0\}.$$

For $n+m>0$, its elements satisfy $\ell^2\le 4nm+1$. If $\ell>n+m$, integrality would give $\ell^2\ge(n+m+1)^2>4nm+1$, a contradiction. For $n=m=0$, the only active index is $(0,-1,0)$. It follows that

$$\Gamma_{\mathrm{act}}\subset C_+\setminus\{0\}.$$

This proves the local finiteness needed for the formal product and for each of its inverses.

4 The theta normalization

The theta phase matters for a square root. Choose representatives $a,b\in\{0,1\}^2$, and define the genus-two theta constants by

$$\nu_{a,b}(Z)=\sum_{u\in\mathbb{Z}^2}\exp\!\big(\pi i\big((u+a/2)^tZ(u+a/2)+b^tu\big)\big).$$

Use the ten characteristics with a^tb even, and put

$$\Delta_5(Z)=\prod_{a^tb\ \text{even}}\nu_{a,b}(Z).$$

This is the convention of Gritsenko–Nikulin [52, (1.2)].

In the convention with phase $b^t(u+a/2)$, each constant acquires $e^{\pi i a^tb/2}$. The product of those phases is -1, coming from $a=b=(1,1)$. Specifying the series removes this otherwise hidden sign.

Proposition 4.1 *(Theta-constant leading coefficient).* In the chosen convention,

$$[q^{1/2}s^{1/2}]\Delta_5=64(r^{1/2}-r^{-1/2}),\qquad [q^{1/2}r^{1/2}s^{1/2}]\Delta_5=64.$$

Proof. The four constants with $a = (0, 0)$ start with 1. The two with $a = (1, 0)$ start with $2q^{1/8}$, and the two with $a = (0, 1)$ start with $2s^{1/8}$. For $a = (1, 1)$, the two even characteristics give

$$2q^{1/8}s^{1/8}(r^{1/4} + r^{-1/4}), \qquad 2q^{1/8}s^{1/8}(r^{1/4} - r^{-1/4}).$$

Their product retains both elliptic monomials. Multiplication by the other eight leading terms gives the stated coefficient. □

Definition 4.2 *(Monic Borcherds form).* Set $D_5 = 64^{-1}\Delta_5$. Its coefficient at $q^{1/2}r^{1/2}s^{1/2}$ is 1. The half powers mean the exponentials $e^{\pi i z_1}$, $e^{\pi i z_2}$, $e^{\pi i z_3}$ on $\mathbb{H}_2$.

The factor $r^{1/2} - r^{-1/2}$ will give the separating divisor. Calling the form monic fixes one coefficient. It does not discard the other term in that coefficient of $q^{1/2}s^{1/2}$.

5 The Borcherds product

The Jacobi form has integral coefficients, weight zero and index one. The Borcherds product theorem in this case gives a global modular form from precisely these data [52, Theorem 4.1], [54, Theorem 2.1 and Example 2.4].

THEOREM 5.1 *(Borcherds–Gritsenko–Nikulin product).* The monic form has the expansion

$$D_5(Z) = (qrs)^{1/2} \prod_{\gamma=(n,\ell,m)\in\Gamma_{\mathrm{act}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

After division by $(qrs)^{1/2}$, this is an element of $1 + (x_1, x_2, x_3)\mathbb{Z}[[x_1, x_2, x_3]]$. Analytically, the product converges absolutely sufficiently near $\mathfrak{c}_\infty$ in the specified chamber. It continues to a holomorphic Siegel cusp form of weight five, with a character of order two on $\mathbf{Sp}_4(\mathbb{Z})$.

Proof. The active-cone calculation proves the formal assertion. The analytic product theorem supplies the continuation, its weight $f(0, 0)/2 = 5$, and its rational quadratic divisors. Gritsenko–Nikulin identify this product with their theta product divided by 64 [52, Theorem 4.1]. Their later convention writes the monic form itself as Δ_5 [54, (2.16)]. Proposition 4.1 fixes the conversion to the convention used here. □

For example, convergence holds for sufficiently large T on the ray $(T, -T/2, T)$, and on a neighborhood of this ray far enough toward the cusp. On smaller compact subsets the convergence is normal. The general convergence statement and its continuation follow from the same product theorem. A formal height completion alone would not supply analytic convergence.

COROLLARY 5.2 *(Theta-normalized section).* With zero exponents omitted or retained, respectively,

$$\Delta_5 = 64(qrs)^{1/2} \prod_{\Gamma_{\mathrm{act}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)} = 64(qrs)^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

The composite index nm is the same one that occurs in the Hilbert-scheme elliptic-genus product. There the K3 coefficients are $2f(nm, \ell)$, and the positive-m product is an inverse. The boundary factors and the factor of two distinguish that index from the present square root.

5.1 The automorphic line

Definition 5.3 *(The Borcherds–Gritsenko–Nikulin section).* For $X = S \times E$, write $\mathcal{D}_X = \Delta_5$ for the section of $\mathcal{L}^5 \otimes \nu_5$ specified above. Here ν_5 is the Maass character, also denoted ν_{Δ_5}, whose values are given below.

Remark 5.4 *(The meaning of $\mathcal{D}_X$).* The subscript names the geometric realization problem. The section itself is defined on Siegel space. A geometric interpretation requires a parameter space B, a map $\sigma : B \to [\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2]$, and an isomorphism from an independently constructed line on B to $\sigma^*(\mathcal{L}^5 \otimes \nu_5)$. The introduction of the three Fourier variables does not define this map or line isomorphism.

6 Weight, character and divisor

6.1 The weight

Proposition 6.1 *(The Borcherds weight).* For the fixed Jacobi input, put $c_1(0) = f(0,0) = 10$. Then

$$\operatorname{wt}(\mathcal{D}_X) = \frac{f(0,0)}{2} = 5 = \frac{c_1(0)}{2} = \kappa_{\mathrm{BKM}}(\Delta_5).$$

Here κ_{BKM} denotes the automorphic weight.

Proof. This is the weight formula in Theorem 5.1. □

Integral weight makes $j(g,Z)^5$ single-valued on the Siegel group. A nontrivial character remains. The input metaplectic representation in Chapter 5 concerns a different group.

The same weight formula applies to the other seven diagonal-divisor forms in Appendix 2. For a congruence input its constant term is the weighted contribution of all cusps. The parameter pair (t, N), consisting of the Jacobi index and congruence level, must be retained.

6.2 The Maass character

The character $\nu_5 : \mathbf{Sp}_4(\mathbb{Z}) \to \{\pm 1\}$ is determined on the usual generators by

$$\nu_5\begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix} = 1, \qquad \nu_5\begin{pmatrix} I_2 & B \\ 0 & I_2 \end{pmatrix} = (-1)^{b_1+b_2+b_3}, \quad B = \begin{pmatrix} b_1 & b_2 \\ b_2 & b_3 \end{pmatrix},$$

and, for $U = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$,

$$\nu_5\begin{pmatrix} U^{-t} & 0 \\ 0 & U \end{pmatrix} = (-1)^{(1+a+d)(1+b+c)+ad}.$$

These are Maass's formulas [84], also reproduced in [52, (1.3)–(1.5)]. In particular $\nu_5^2 = 1$. The character is nontrivial, since translation by $B = \operatorname{diag}(1,0)$ has value -1. Thus Δ_5 is a section of the character-twisted Hodge line, whereas Δ_5^2 is a section of $\mathcal{L}^{10}$.

6.3 The diagonal divisor

Let

$$\mathcal{H}_{\mathrm{diag}} = \left\{ \begin{pmatrix} \tau & 0 \\ 0 & \sigma \end{pmatrix} : \tau, \sigma \in \mathbb{H}_1 \right\}.$$

The divisor of Δ_5 on $\mathbb{H}_2$ is

$$\operatorname{div}(\Delta_5) = \sum_{\mathcal{H}\in \mathbf{Sp}_4(\mathbb{Z})\cdot\mathcal{H}_{\mathrm{diag}}} \mathcal{H},$$

where each distinct component occurs once [50, Theorem 1.4]. It descends to the Humbert divisor of discriminant one. The order here is measured upstairs, or on the quotient stack with its line. Coarse quotient ramification should not be inserted into this multiplicity.

In the rank-three lattice

$$M_0 = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}, \qquad (f_2, f_{-2}) = -1, \quad f_3^2 = 2,$$

all other basis pairings vanish. The three simple roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

They span the type-II lattice $M_{II} = \mathbb{Z}(2f_2) \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}(2f_{-2})$.

Their Gram matrix and product degrees are

$$((\delta_i, \delta_j)) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}, \qquad (1,1,0),\ (0,1,1),\ (0,-1,0).$$

Each corresponding factor has exponent 1. Its wall is a component of the simple Humbert divisor. The reciprocal Δ_5^{-2} consequently has order -2 there. The separating calculation below also proves order one directly at $z_2 = 0$.

6.4 Linear reflections and the slash action

The action on the tube coordinates has a different character from the weight-five slash action. To specify both, put

$$\zeta = z_3 f_2 + z_2 f_3 + z_1 f_{-2}, \qquad s_\beta(v) = v - (v,\beta)\beta \quad (\beta^2 = 2),$$

and let $s_\beta Z$ denote the period corresponding to $s_\beta \zeta$. The three reflections permuting the simple roots have normals

$$\alpha_1 = f_{-2} - f_2, \qquad \alpha_2 = f_2 - f_3, \qquad \alpha_3 = f_{-2} - f_3.$$

The orthogonal realization separates the two signs intrinsically. Let L and $\sigma : \mathbf{Sp}_4(\mathbb{Z}) \to \Gamma_{3,2} \subset \mathbf{SO}(L)$ be the exterior-square lattice and homomorphism of Theorem 2.2. Its cusp frame satisfies $\sigma(g)w(Z) = j(g,Z)w(gZ)$.

Definition 6.2 (Homogeneous orthogonal character). On the nonzero null cone over $\mathbb{H}_2$, set

$$\widehat{\Delta}_5(tw(Z)) = t^{-5}\Delta_5(Z).$$

The character of its orthogonal transformation law is

$$\widetilde{\nu}_5 : \Gamma_{3,2} \times \{\pm \mathbf{I}_L\} \longrightarrow \{\pm 1\}, \qquad \widetilde{\nu}_5(\eta\sigma(g)) = \eta^{-5}\nu_5(g) \quad (\eta \in \{\pm 1\}).$$

We also write $\widetilde{\nu}_{\Delta_5} = \widetilde{\nu}_5$.

The definition is independent of the lift g, since $\ker\sigma = \{\pm\mathbf{I}_4\}$ and $\nu_5(-\mathbf{I}_4) = 1$. The two orthogonal factors intersect trivially: $\dim L = 5$ gives $\det(-\mathbf{I}_L) = -1$. Homogeneity forces $\widetilde{\nu}_5(-\mathbf{I}_L) = -1$, and the frame relation gives $\widehat{\Delta}_5(\sigma(g)w) = \nu_5(g)\widehat{\Delta}_5(w)$. Extend each rank-three reflection by the identity on the complementary hyperbolic plane in L.

Lemma 6.3 *(Reflection signs and symplectic lifts).* The linear tube action satisfies

$$\Delta_5(s_{\delta_i}Z) = -\Delta_5(Z), \qquad \Delta_5(s_{\alpha_i}Z) = \Delta_5(Z).$$

For the block-diagonal symplectic lifts g_β below, the Maass and slash-action values are instead

$$\nu_5(g_{\delta_i}) = 1, \qquad \nu_5(g_{\alpha_i}) = -1, \qquad \Delta_5|_5 g_\beta = \nu_5(g_\beta)\Delta_5.$$

The homogeneous orthogonal values are

$$\widetilde{\nu}_5(s_{\delta_i}) = -1, \qquad \widetilde{\nu}_5(s_{\alpha_i}) = 1.$$

Proof. Take $g_\beta = \operatorname{diag}(U_\beta^{-t}, U_\beta)$, with

β	δ_1	δ_2	δ_3
U_β	$\left(\begin{smallmatrix} 1 & -2 \\ 0 & -1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & 0 \\ 2 & -1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$
β	α_1	α_2	α_3
U_β	$\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & -1 \\ 0 & -1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & 0 \\ 1 & -1 \end{smallmatrix}\right)$

Each determinant is -1. The action $Z \mapsto U_\beta^{-t} Z U_\beta^{-1}$ is s_β on the tube coordinates. For $U = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, its matrix in the basis (f_2, f_3, f_{-2}) is

$$\begin{pmatrix} a^2 & -2ab & b^2 \\ -ac & ad+bc & -bd \\ c^2 & -2cd & d^2 \end{pmatrix}.$$

Substitution verifies all six reflections.

Maass's formula gives $+1$ in the first three cases and -1 in the last three. Since $j(g_\beta, Z) = \det U_\beta = -1$, the functional equation gives $\Delta_5(s_\beta Z) = -\nu_5(g_\beta)\Delta_5(Z)$. The cusp frame obeys $w(s_\beta Z) = s_\beta w(Z)$, so the frame relation also gives $\sigma(g_\beta) = -s_\beta$. Consequently $\widetilde{\nu}_5(s_\beta) = -\nu_5(g_\beta)$. These equations prove all three assertions. □

Write ε_5 for the character of this linear tube action. It is the restriction of $\widetilde{\nu}_5$ to the reflections fixing the complementary hyperbolic plane. For the reflection group $W^{(2)}(M_{II})$ and the chamber symmetry group S_3, the lemma gives

$$\varepsilon_5(w) = \det(w), \qquad \varepsilon_5(a) = 1 \quad (a \in S_3).$$

These are the signs used in the alternating denominator identity [52, Propositions 2.1–2.2].

The divisibility of $\beta \in M_0$ is the positive generator of the ideal $(\beta, M_0) \subset \mathbb{Z}$. The type-I walls carry no divisor of Δ_5. Their primitive normals have divisibility one in M_0, whereas a type-II normal has divisibility two. The latter admits the half-normal in the dual lattice that supports the negative Fourier index of the lift. This divisor criterion is made explicit in Chapter 5.

7 The coefficientwise determinant

For each $\gamma = (n, \ell, m) \in \Gamma_{\mathrm{act}}$, choose a finite super vector space $V_\gamma = V_\gamma^{\bar{0}} \oplus V_\gamma^{\bar{1}}$ with

$$\operatorname{sdim} V_\gamma = \dim V_\gamma^{\bar{0}} - \dim V_\gamma^{\bar{1}} = f(nm, \ell).$$

Such spaces exist by placing $|f(nm, \ell)|$ basis vectors in the parity given by its sign. This realizes the integer without specifying a bracket or geometric coefficient complex. We use the split Grothendieck group of finite super vector spaces with the additional relation $[\Pi V] = -[V]$, where Π reverses parity. Its classes are determined by superdimension.

The graded sum $V = \bigoplus_\gamma V_\gamma$ has finite-dimensional total-height components.

For $x^\gamma = q^n r^\ell s^m$, define the completed Berezinian by

$$\operatorname{Ber}_V(1 - x) = \prod_\gamma \frac{\det(1 - x^\gamma \mid V_\gamma^{\bar{0}})}{\det(1 - x^\gamma \mid V_\gamma^{\bar{1}})}.$$

Total height makes every truncation finite.

LEMMA 7.1 *(The K_0 superdeterminant).* The Berezinian depends only on the super-Grothendieck classes and is

$$\operatorname{sdet}_V(1 - x) := \operatorname{Ber}_V(1 - x) = \prod_{\Gamma_{\mathrm{act}}} (1 - x^\gamma)^{f(nm,\ell)}.$$

The Euler supercharacter of the super-symmetric algebra is its inverse.

Proof. An even basis vector contributes $1 - x^\gamma$ to the determinant. An odd vector contributes its reciprocal. This proves the first identity on each finite height truncation and hence on their inverse limit. In a super-symmetric algebra, an even generator gives a polynomial algebra with supercharacter $(1 - x^\gamma)^{-1}$. An odd generator gives an exterior algebra with supercharacter $1 - x^\gamma$. Multiplication gives the inverse product. □

PROPOSITION 7.2 *(The automorphic continuation of the determinant).* Put $v_0(Z) = 64(qrs)^{1/2}$. Then

$$v_0(Z)\ \operatorname{sdet}_V(1 - x) = \mathcal{D}_X(Z) = \Delta_5(Z)$$

in the height-completed cusp expansion. The left side has the analytic continuation given by Theorem 5.1.

Proof. The lemma identifies every factor of the product theorem. The theta coefficient fixes v_0. On the convergence region this is an equality of holomorphic functions. Analytic continuation gives the stated section. □

Remark 7.3 (Scope of the determinant identity). An Euler class can also come from an actual derived supercomplex. Its Berezinian still remembers only the signed dimensions used above. A comparison with a Hall algebra requires compatible extension maps. A comparison with a Pfaffian requires a family, a skew pairing, an orientation and modular descent. None of these operations is defined by the product of Euler classes.

8 The complete boundary factors

The zero-s factors give the first Fourier–Jacobi coefficient. Define

$$b(q,r) = q^{1/2}(r^{1/2} - r^{-1/2}) \prod_{a\geq 1}(1-q^a)^{10}(1-rq^a)(1-r^{-1}q^a).$$

Separating the factors with $m = 0$ gives the exact identity

$$D_5 = s^{1/2} b(q,r) \prod_{\substack{m>0,\ n\geq 0\\ \ell\in\mathbb{Z}}} (1-q^n r^\ell s^m)^{f(nm,\ell)}.$$

Thus $[s^{1/2}]D_5 = b$.

Writing

$$\eta(\tau) = q^{1/24}\prod_{a\geq 1}(1-q^a), \qquad \vartheta(\tau,z) = q^{1/8}(r^{1/2}-r^{-1/2})\prod_{a\geq 1}(1-q^a)(1-rq^a)(1-r^{-1}q^a),$$

one has $b = \eta^9\vartheta$. This is the weight-five, index-$\frac{1}{2}$ Jacobi form whose additive lift gives D_5 [50, (13)]. Its multiplier on the Jacobi parabolic is $v_\eta^{12}\times v_H$, where v_η is the Dedekind eta multiplier and v_H is the odd-theta Heisenberg character.

For a K3 surface S and a chosen Enriques surface T, the normalized elliptic-index series of Chapter 1 are therefore

$$F_S = \frac{s\,b^2}{D_5^2}, \qquad F_T = \frac{s^{1/2}b}{D_5}, \qquad F_T^2 = F_S.$$

These quotients have their boundary factors cancelled before specialization at $r = 1$. Raw reciprocal cusp forms have poles there.

Proposition 8.1 *(The separating leading section).* For $z = z_2$ near zero and $z_1, z_3 \in \mathbb{H}_1$,

$$D_5(Z) = 2\pi i z\,\eta(z_1)^{12}\eta(z_3)^{12} + O(z^3).$$

Consequently

$$D_5(Z)^2 = (2\pi i z)^2\eta(z_1)^{24}\eta(z_3)^{24} + O(z^4).$$

Proof. At $r = 1$, the theta quotient gives $\phi_{0,1}(\tau,0) = 12$. Hence

$$\sum_\ell f(n,\ell) = \begin{cases} 12, & n = 0,\\ 0, & n > 0.\end{cases}$$

After removing $r^{1/2} - r^{-1/2}$ from D_5, the factors with $n, m > 0$ cancel at $r = 1$. The two boundary directions give

$$\left.\frac{D_5}{r^{1/2}-r^{-1/2}}\right|_{r=1} = (qs)^{1/2}\prod_{a\geq 1}(1-q^a)^{12}(1-s^a)^{12} = \eta(z_1)^{12}\eta(z_3)^{12}.$$

This calculation first holds on a product convergence region and then continues along the diagonal. The quotient is even in z, since the remaining factors pair ℓ and $-\ell$.

Finally $r^{1/2} - r^{-1/2} = 2\pi i z + O(z^3)$. □

The eta factors never vanish on $\mathbb{H}_1$. This proves the simple order on the diagonal and fixes its leading section. An eventual geometric line comparison must reproduce both the factor $2\pi i$ and the two eta powers.

9 THE SQUARE AND THE COUNTING NORMALIZATION

Set

$$\Xi = D_5^2 = \chi_{10}^{\mathrm{OP}}, \qquad \Delta_{10} = \Delta_5^2 = 4096\,\Xi.$$

The character of Δ_5 disappears on squaring. Both Ξ and Δ_{10} are weight-ten forms, with different leading coefficients.

PROPOSITION 9.1 *(The weight-ten normalization).* Both counting papers [100, 101] denote Ξ by χ_{10}. Its product is

$$\Xi = qrs \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{2f(nm,\ell)}.$$

In the unscaled notation of this book,

$$\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_{10} = \Delta_5^2.$$

Thus this last bare χ_{10} is 4096 times the form bearing that name in the two cited counting papers.

Proof. Squaring the product doubles its exponents and squares its prefactor. The product in [100, (12)] and [101, (3)] has prefactor qrs and coefficients $c(4nm - \ell^2) = 2f(nm, \ell)$. It is therefore Ξ, coefficient by coefficient. The unscaled equalities are the stated notation. □

For primitive nonzero K3 curve classes $\beta_h^2 = 2h - 2$, the reduced disconnected Gromov–Witten theorem of Oberdieck–Pixton [101, Theorem 1] is

$$\sum_{g,h,d \geq 0} N_{g,h,d}\, u^{2g-2} q^{h-1} s^{d-1} = -\frac{1}{\Xi(q, e^{iu}, s)}.$$

The right side means its Laurent expansion at $u = 0$. The DT/PT and GW/PT comparisons in Chapter 1 give the rational-coefficient form of the primitive DT series,

$$Z_{\mathrm{OP}}^X = -\Xi^{-1} = -4096\,\Delta_5^{-2}.$$

The DT Euler variable is $-r$. Its expansion at $r = 0$, a product expansion in the chamber $|r| > 1$, and a genus expansion at $r = e^{iu}$ are different expansions of the same rational coefficients. Arbitrary imprimitive classes require additional statements.

The convention $Z_{\mathrm{BPS}}^{K3\times E} := -Z_{\mathrm{OP}}^X/4096 = \Delta_5^{-2}$ names a rescaled scalar function. Constructing a protected state space and a trace with this value is a further problem. Squaring also loses the order-two character of the line and the sign of its section.

10 THE ROOT DENOMINATOR

For the Gritsenko–Nikulin Lie superalgebra $\mathfrak{g}_{\Delta_5}$, let $\mathrm{den}(\mathfrak{g}_{\Delta_5})$ denote its Weyl–Kac–Borcherds denominator in the character convention $e^{-2\pi i(\alpha,\zeta)}$. The change from product degrees to roots is

$$\alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2} = n\delta_1 + m\delta_2 + (n + m - \ell)\delta_3.$$

It identifies the charge lattice with M_{II}, an index-four sublattice of M_0. Direct pairing gives

$$q^n r^\ell s^m = e^{-\pi i(\alpha(n,\ell,m),\zeta)}.$$

Proposition 10.1 *(The doubled denominator).* In the specified character convention,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5})(Z) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

Proof. Gritsenko–Nikulin's denominator theorem identifies the signed root multiplicities with $f(nm, \ell)$, and its Weyl vector is $\rho = f_2 - \frac{1}{2}f_3 + f_{-2}$ [52, §3 and Theorem 4.1]. Replacing Z by $2Z$ changes each $e^{-\pi i(\alpha,\zeta)}$ to $e^{-2\pi i(\alpha,\zeta)}$, including the Weyl prefactor. The scalar is the theta normalization already fixed above. □

The simple imaginary generators and their parities are part of the Lie superalgebra construction in Chapter 4. A graded super vector space with the same Euler dimensions would give the same product but would not define this denominator algebra.

II The orthogonal lift and the geometric comparison

The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ identifies Siegel space with a type-IV domain. Chapter 5 constructs the regularized theta lift on that domain, starting from the weight-$-\frac{1}{2}$ vector-valued theta components of $\phi_{0,1}$. Its multiplicative product is D_5 in the monic cusp normalization.

Theorem 11.1 *(The automorphic target).* The section $\mathcal{D}_X = \Delta_5$ has weight five, Maass character ν_5, and simple Humbert divisor of discriminant one. Its cusp expansion is the height-completed Borcherds product, its first Fourier–Jacobi coefficient is $64b$, and its separating leading section is

$$128\pi i z_2\, \eta(z_1)^{12}\eta(z_3)^{12}.$$

Its square is $4096\,\Xi$, and its root denominator is $64^{-1}\Delta_5(2Z)$.

Proof. The assertions follow from the product theorem, Maass's formulas, the diagonal-divisor theorem, and the two boundary calculations. □

The retained hypotheses in Appendix 7 specify the Do-HN datum and quotient orientation. They also specify Weyl lifts, the wall datum ($O2_{\mathrm{atlas}}$), and the finite Pfaffian datum (P_{fin}). The relative Pfaffian–Dirac statement is treated in Chapter 7 under these data.

A geometric realization must define $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ on its own line and compare it with $\sigma^*\mathcal{D}_X$. The line isomorphism must preserve the character and both boundary sections computed above. An identification of Hall or chiral operations has additional content beyond this line comparison. The mirror-discriminant interpretation likewise requires an independently defined period map and divisor comparison.

Chapter 4

The denominator algebra

The product expansion of the Igusa form assigns an integer to every positive lattice degree. To obtain a Lie superalgebra, one must also specify generators and relations. At the first interior degree, the product exponent is -64. The algebra constructed below has 29 even and 93 odd vectors there. Their difference explains the exponent. Their brackets require the presentation.

Gritsenko and Nikulin construct this presentation by adjoining imaginary simple generators to a rank-three hyperbolic Kac–Moody algebra [52, Sections 2–3 and 6]. We first describe its lattice and chamber, then extract the imaginary generators from the chamber Fourier expansion. The resulting superdenominator is $64^{-1}\Delta_5(2Z)$. Its ordinary denominator supplies the additional information needed to recover both parity dimensions at every degree.

Finite root spaces then provide concrete comparison spaces for the compact $K3 \times E$ problem. The comparison itself must preserve brackets, pairings and coefficient objects. Equality of denominators supplies only its numerical consequence.

1 The lattice and its chamber

Write U for the integral hyperbolic plane, with isotropic generators whose pairing is -1. The lattice relevant to the chosen cusp is

$$\Lambda^{2,1} = U \oplus \langle 2 \rangle = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2},$$

with bilinear form

$$(f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2, \qquad (f_2, f_2) = (f_{-2}, f_{-2}) = (f_2, f_3) = (f_{-2}, f_3) = 0. \tag{4.1}$$

It is even of signature $(2, 1)$. Adjoining a second hyperbolic plane gives the orthogonal lattice $\Lambda^{3,2}$ used in Section 2.1.

Definition 1.1 *(Type-II sublattice).* The type-II lattice is

$$L := \Lambda^{2,1}_{II} = 2\mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus 2\mathbb{Z}f_{-2}.$$

Its dual and discriminant group are

$$L^* = \tfrac{1}{2}\Lambda^{2,1}, \qquad L^*/L \simeq \mathbb{Z}/4 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/4.$$

Thus $[\Lambda^{2,1} : L] = 4$, $[L^* : \Lambda^{2,1}] = 8$ and $L \simeq U(4) \oplus \langle 2 \rangle$, with determinant -32.

The three vectors

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3$$

form an integral basis of L. They have square two and divisibility two in $\Lambda^{2,1}$: the ideal $(\delta_i, \Lambda^{2,1})$ is $2\mathbb{Z}$.

LEMMA 1.2 *(Cartan matrix).* The Gram matrix of this basis is

$$A = ((\delta_i, \delta_j)) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix} = 4I_3 - 2J_3, \tag{4.2}$$

where J_3 is the all-ones matrix. Its eigenvalues are -2, 4, 4.

Proof. For example, $(2f_2 - f_3, 2f_{-2} - f_3) = -4 + 2 = -2$. The other entries follow from (4.1). The line spanned by $(1,1,1)$ has eigenvalue -2. The plane of coordinate sum zero has eigenvalue 4. □

The Gram-index lattice of Chapter 2 is $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$. Its identification with L is

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2} = n\delta_1 + m\delta_2 + (n+m-l)\delta_3. \tag{4.3}$$

If $\gamma = (n,l,m)$ and $\gamma' = (n',l',m')$, then

$$\langle \gamma, \gamma' \rangle_{\mathrm{ind}} = 2(nm' + n'm) - ll',$$
$$(\alpha(\gamma), \alpha(\gamma')) = -2\langle \gamma, \gamma' \rangle_{\mathrm{ind}}, \qquad \alpha(\gamma)^2 = -2(4nm - l^2).$$

This integral map is fixed throughout the chapter. In particular, changing an exponential coordinate does not change its image lattice.

1.1 *Reflections and the positive cone*

For a positive-norm vector β, its orthogonal reflection is

$$s_\beta(x) = x - \frac{2(x,\beta)}{(\beta,\beta)}\beta.$$

It preserves an integral lattice when (β,β) divides $2(\beta, L)$. For a square-two vector the formula is

$$s_\beta(x) = x - (x,\beta)\beta. \tag{4.4}$$

Let V_+ be the component of $\{x \in L_{\mathbb{R}} : x^2 < 0\}$ containing $\delta_1 + \delta_2 + \delta_3$. We also write $C(\Lambda^{2,1})_+ = V_+$.

Definition 1.3 *(Type-II Weyl chamber).* The chamber and its closed cone are

$$\mathcal{P}_{II} = \{\mathbb{R}_{>0}x : x \in V_+,\ (x,\delta_i) \le 0\ (1 \le i \le 3)\}, \qquad C = \{x \in L_{\mathbb{R}} : (x,\delta_i) \le 0\ (1 \le i \le 3)\}.$$

The primitive wall normals are $\mathcal{P}_{II,\mathrm{prim}} = \{\delta_1, \delta_2, \delta_3\}$. The notation $\mathbb{R}_{>0}\mathcal{P}_{II}$ will include the nonzero ideal boundary of C.

LEMMA 1.4 *(Chamber and dual cone).* Put $Q_+ = \mathbb{N}\delta_1 + \mathbb{N}\delta_2 + \mathbb{N}\delta_3$ and $\Delta(L)_+ = \mathbb{R}_{\ge 0}Q_+$. With the nonpositive pairing convention for the dual cone, one has

$$\Delta(L)_+^* = C \subset \overline{V}_+ \subset \Delta(L)_+.$$

Proof. Write $x = \sum_i b_i \delta_i$ and $t_i = -(x, \delta_i)$. Since $A^{-1} = (I_3 - J_3)/4$, the chamber inequalities give

$$b_i = \frac{\sum_{j \neq i} t_j}{4} \geq 0, \qquad x^2 = -\frac{1}{2} \sum_{i<j} t_i t_j \leq 0.$$

A nonzero such vector belongs to the future component. This proves the first inclusion.

For the second, let $s = \sum_i b_i \geq 0$ on $\overline{V}_+$. The norm inequality reads $2 \sum_i b_i^2 \leq s^2$. If $b_1 < 0$, then

$$\sum_i b_i^2 \geq b_1^2 + \tfrac{1}{2}(s - b_1)^2 > \tfrac{1}{2} s^2,$$

a contradiction. The other coordinates are nonnegative by symmetry. □

Remark 1.5 *(Divisibility classes).* A primitive square-two vector in $\Lambda^{2,1}$ has divisibility one or two, called type I or type II. To see the parity distinction, write it as $af_2 + bf_3 + cf_{-2}$. Its norm equation is $b^2 - ac = 1$. The divisibility is $\gcd(a, 2b, c)$. It is two precisely when a, c are even, in which case b is odd.

Otherwise $a + b + c$ is even. Thus type I vectors lie in

$$\Lambda_I^{2,1} = \{af_2 + bf_3 + cf_{-2} : a + b + c \equiv 0 \pmod 2\},$$

and type II vectors lie in L. The vectors $f_{-2} - f_2, f_2 - f_3, f_{-2} - f_3$ are type I. The δ_i are type II. The divisor of Δ_5 uses the latter orbit, as proved in Chapter 3.

Lemma 1.6 *(Reflection groups).* Let $W^{(2)}(\Lambda^{2,1})$ be generated by its square-two reflections, and let W_I, W_{II} be generated by the type I and type II reflections. Then

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}), \qquad [W^{(2)} : W_I] = 2, \qquad [W^{(2)} : W_{II}] = 6,$$

and

$$W^{(2)}(\Lambda^{2,1}) = W_{II} \rtimes \mathrm{Aut}(\mathcal{P}_{II}), \qquad \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3.$$

The group W_{II} is the Coxeter group on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ with relations $s_{\delta_i}^2 = 1$ only.

Proof. These are the reflection decompositions for $U \oplus \langle 2 \rangle$ in [52, Section 2, equations (2.1)–(2.3)], using Nikulin's classification [97]. The type-II chamber is an ideal triangle: each pair of its wall normals has Gram matrix $\left(\begin{smallmatrix} 2 & -2 \\ -2 & 2 \end{smallmatrix}\right)$. Consequently distinct simple reflections have no finite braid relation. The chamber stabilizer permutes its three ideal vertices, and the three reflections in the following lemma realize all these permutations. Its order gives the index six. The type-I reflection subgroup has the separate order-two chamber symmetry described in the cited decomposition. □

Lemma 1.7 *(S_3 chamber action).* The reflections in $f_{-2} - f_2, f_2 - f_3$ and $f_{-2} - f_3$ act on $(\delta_1, \delta_2, \delta_3)$ as (12), (13) and (23), respectively.

Proof. For the first reflection, f_2 and f_{-2} are exchanged and f_3 is fixed. For the second, substitution in (4.4) exchanges $2f_2 - f_3$ and f_3 and fixes $2f_{-2} - f_3$. The third follows by exchanging f_2 and f_{-2}. Each permutation preserves A. □

Proposition 1.8 *(Weyl vector).* The equations

$$(\rho, \delta_i) = -\tfrac{1}{2}\delta_i^2 = -1 \tag{4.5}$$

have the unique solution

$$\rho = \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \tfrac{1}{2} f_3 + f_{-2} \in L^* \cap V_+, \qquad \rho^2 = -\tfrac{3}{2}.$$

Proof. The matrix A is invertible and $A(1,1,1)^T = (-2,-2,-2)^T$. This determines the coefficients of ρ. Its norm follows from (4.1). □

Remark 1.9 *(The fixed line).* The chamber action fixes ρ. Its fixed subspace in $L_{\mathbb{Q}}$ is exactly $\mathbb{Q}\rho$, since it permutes the three basis vectors δ_i. In particular, the factor $1/2$ in the Weyl vector is fixed by (4.5). The sum of the three roots has pairing -2 with each wall.

LEMMA 1.10 *(Isotropic chamber rays).* The only primitive isotropic rays of $C \cap L$ are

$$a_{13} = \delta_1 + \delta_3 = 2f_2, \qquad a_{23} = \delta_2 + \delta_3 = 2f_{-2}, \qquad a_{12} = \delta_1 + \delta_2 = 2f_2 - 2f_3 + 2f_{-2}.$$

The chamber group permutes them transitively.

Proof. In the proof of Lemma 1.4, a chamber vector is isotropic precisely when $\sum_{i<j} t_i t_j = 0$. For nonnegative t_i, this means that at most one is nonzero. The resulting three rays have the displayed primitive integral generators. The S_3 action permutes the pairs of indices. □

2 From Fourier coefficients to simple generators

The Fourier coefficients of the weak Jacobi form are denoted by

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0,\ l \in \mathbb{Z}} f(n,l) q^n r^l = r^{-1} + 10 + r + O(q).$$

Elliptic translation and evenness make $f(n,l)$ a function of $4n - l^2$. It vanishes for $4n - l^2 < -1$. Let Γ_{eff} be the cusp support

$$\{m > 0, n \geq 0, l \in \mathbb{Z}\} \cup \{m = 0, n > 0, l \in \mathbb{Z}\} \cup \{m = n = 0, l < 0\} \subset \mathbb{Z}^3.$$

Definition 2.1 *(Active support).* Set

$$\Gamma_{\text{act}} = \{(n,l,m) \in \Gamma_{\text{eff}} : f(nm, l) \neq 0\}, \qquad C_{\text{Borch}} = \mathbb{R}_{\geq 0}\alpha(\Gamma_{\text{act}}).$$

LEMMA 2.2 *(Positive root coordinates).* Every active degree has

$$\alpha(n,l,m) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3 \in Q_+ \setminus \{0\}.$$

Moreover $C_{\text{Borch}} = \Delta(L)_+$.

Proof. The first two coefficients are nonnegative. Activity implies $l^2 \leq 4nm + 1$. If $n, m > 0$ and $l \geq n + m + 1$, then $l^2 \geq 4nm + (n-m)^2 + 2(n+m) + 1 > 4nm + 1$. The cases with $nm = 0$ follow from $|l| \leq 1$ and the cusp convention. Thus $n + m - l \geq 0$. The active triples $(1,1,0)$, $(0,1,1)$, $(0,-1,0)$ give the three simple vectors, proving equality of the cones. □

For $\beta = \sum b_i \delta_i \in Q_+$ define its height by $\operatorname{ht}\beta = b_1 + b_2 + b_3$. There are finitely many lattice degrees below any fixed height. We use the corresponding completion $\mathbb{Z}[[Q_+]]$: its elements are formal sums $\sum_{\beta \in Q_+} a_\beta X^\beta$, and multiplication is defined by the finite sums over $\beta = \beta_1 + \beta_2$. Product factors below always have positive height. This specifies a common formal domain before any infinite product is formed.

Lemma 2.3 *(Leading coefficient).*

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64.$$

Proof. The ten-theta product gives the leading term $64q^{1/2}s^{1/2}(r^{1/2} - r^{-1/2})$ by Proposition 4.1. □

Theorem 2.4 *(Borcherds product).* In the fixed cusp expansion,

$$D_5(Z) := 64^{-1}\Delta_5(Z) = q^{1/2}r^{1/2}s^{1/2}\prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1 - q^n r^l s^m)^{f(nm,l)}. \tag{4.6}$$

Here $q^n r^l s^m = e^{-\pi i(\alpha(n,l,m),z)}$ and $q^{1/2}r^{1/2}s^{1/2} = e^{-\pi i(\rho,z)}$ in the tube coordinates of Chapter 3.

Proof. This is the multiplicative lift of $\phi_{0,1}$ [54, Theorem 2.1 and Example 2.4], equivalently the orthogonal Borcherds lift on $\Lambda^{3,2}$ [10, Theorem 13.3]. Lemma 2.2 places its nontrivial factors in the height completion. The boundary factor $(1 - r^{-1})$ has degree δ_3. The leading coefficient fixes the normalization. □

The chamber expansion of this product specifies the imaginary simple generators. It is convenient to use $W = W_{II}$ and to include the isotropic boundary in $C \setminus \{0\}$.

Theorem 2.5 *(Chamber Fourier correction).* There are unique integers $m(a)$, indexed by $a \in (C \cap L) \setminus \{0\}$, such that

$$D_5(Z) = \sum_{w\in W}\det(w)\left(e^{-\pi i(w\rho,z)} - \sum_{a\in(C\cap L)\setminus\{0\}} m(a)e^{-\pi i(w(\rho+a),z)}\right).$$

Proof. Write $D_5 = \sum_\lambda c_\lambda e^{-\pi i(\lambda,z)}$ with $\lambda \in \rho + L$. Its nonzero Fourier coefficients lie in V_+ by the cusp condition. The linear reflection law from Lemma 6.3 gives $c_{w\lambda} = \det(w)c_\lambda$. A coefficient on a reflecting wall vanishes.

Each surviving orbit therefore has a unique representative in the chamber interior. For this representative $\lambda = \rho + a$, the pairings (a, δ_i) are even integers, so $(\lambda, \delta_i) < 0$ implies $(a, \delta_i) \le 0$. Lemma 1.4 puts a in C.

The coefficient at ρ is one. Define $m(a) = -c_{\rho+a}$ for $a \neq 0$ and sum the alternating orbits. This is [52, Theorem 2.3(a)]. The chamber condition concerns the orbit representative: $s_{\delta_i}\rho = \rho + \delta_i$ is another nonzero Fourier term, although $\delta_i^2 = 2$. □

Lemma 2.6 *(Null-ray correction).* For any of the three primitive null vectors $a = a_{ij}$,

$$1 - \sum_{n\ge1} m(na)t^n = \prod_{n\ge1}(1 - t^n)^9, \qquad \tau(na) := 9. \tag{4.7}$$

For $a = 2f_2$, the left side is also $1 + 64^{-1}\sum_{n\ge1} c_\Delta(1 + 2n, 1, 1)t^n$, where $c_\Delta(u, v, w)$ denotes the coefficient of $q^{u/2}r^{v/2}s^{w/2}$.

Proof. The first Fourier–Jacobi coefficient of Δ_5 is the lift input $\eta^9\vartheta$. Its restriction along this ray gives (4.7) [52, Section 3]. The chamber permutations carry it to the other two rays. □

Definition 2.7 *(Imaginary simple generators).* At a timelike $a \in C \cap L$, prescribe

$$\mu_{\bar{0}}(a) = \max(m(a), 0), \qquad \mu_{\bar{1}}(a) = \max(-m(a), 0).$$

At every nonzero multiple of a primitive chamber null vector, prescribe $\mu_{\bar{0}}(na_{ij}) = 9$ and $\mu_{\bar{1}}(na_{ij}) = 0$. All other imaginary simple fibers are zero. Denote this multiset by $\Delta^{im}_{\text{simp}}$.

Remark 2.8 *(Simple parity and root parity).* The signs of $m(a)$ determine the parity of the timelike *simple* generators in this presentation. A full root space also contains brackets of lower degrees. Its product exponent is the difference of its even and odd dimensions. The sign of that exponent does not determine the parity of every vector. At a_{ij}, nine simple vectors and one real-real bracket give ten even vectors. At $\delta_1 + \delta_2 + \delta_3$, both parities occur.

3 The Borcherds presentation

LEMMA 3.1 *(Cartan inequalities).* Distinct simple vectors have nonpositive pairing. Real simple vectors have square two, imaginary simple vectors have nonpositive square, and $2(\delta_i, a)/\delta_i^2$ is integral for every simple vector a.

Proof. The real pairings are the off-diagonal entries of A. The chamber inequalities give $(a, \delta_i) \le 0$ for imaginary a. Two vectors in the same closed future cone have nonpositive pairing. Integrality follows from the integral form on L. Repeated copies of a null vector have pairing zero. Repeated copies of a timelike vector have negative pairing. □

Definition 3.2 *(Borcherds–Cartan superdatum).* Let I index one even generator at each δ_i, and $\mu_{\bar{p}}(a)$ generators of parity $\bar{p}$ at every imaginary simple degree a. Denote their degree and parity maps by $i \mapsto \alpha_i \in L$ and $i \mapsto p(i) \in \mathbb{Z}/2$. Together with the even Cartan space $\mathfrak{h} = L \otimes \mathbb{C}$ and its form, these data define $\mathscr{B}_{\Delta_s}$.

Definition 3.3 *(Denominator algebra).* The Lie superalgebra $\mathfrak{g}_{\Delta_s} = \mathfrak{g}$ is generated by $\mathfrak{h}$ and e_i, f_i for $i \in I$, with $|e_i| = |f_i| = p(i)$ and degrees $\alpha_i, -\alpha_i$. Identify $h_i = \alpha_i$ in $\mathfrak{h}$. Impose

$$[h, h'] = 0, \qquad [h, e_j] = (h, \alpha_j)e_j, \qquad [h, f_j] = -(h, \alpha_j)f_j,$$
$$[e_i, f_j] = \delta_{ij} h_i.$$

For a real simple index i and $j \neq i$, impose

$$(\operatorname{ad} e_i)^{1-2(\alpha_i,\alpha_j)/\alpha_i^2} e_j = 0, \qquad (\operatorname{ad} f_i)^{1-2(\alpha_i,\alpha_j)/\alpha_i^2} f_j = 0.$$

Finally, impose

$$[e_i, e_j] = [f_i, f_j] = 0 \quad \text{if } (\alpha_i, \alpha_j) = 0.$$

All brackets satisfy super-antisymmetry and the super Jacobi identity. For homogeneous vectors,

$$[x, y] = -(-1)^{|x||y|}[y, x].$$

These are the symmetrizable generalized Kac–Moody relations without odd real simple roots [52, Statement 6.4 and equations (6.26)–(6.30)]. They can be obtained from the derived contragredient algebra with one independent Cartan generator for each $i \in I$ by quotienting its central Cartan kernel under $h_i \mapsto \alpha_i$. The specified Cartan space is nondegenerate and spanned by the three real vectors.

Remark 3.4 *(Real Serre exponent).* For distinct real indices, $1 - 2(\delta_i, \delta_j)/\delta_i^2 = 3$. Thus $(\operatorname{ad} e_i)^3 e_j = 0$, while $[e_i, e_j]$ survives. The matrix $A/2$ is a rescaled bilinear form, not the generalized Cartan matrix in this normalization: real diagonal entries of that matrix are two.

Lemma 3.5 *(Triangular decomposition and local finiteness).* There is a decomposition

$$\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+, \qquad \mathfrak{n}_\pm = \bigoplus_{\beta \in Q_+ \setminus \{0\}} \mathfrak{g}_{\pm\beta},$$

with $\mathfrak{g}_0 = \mathfrak{h}$ and $[\mathfrak{g}_\beta, \mathfrak{g}_\gamma] \subset \mathfrak{g}_{\beta+\gamma}$. Every homogeneous root space is finite-dimensional. More strongly, the sum of its dimensions over $0 < \operatorname{ht}\beta \le N$ is finite for every N.

Proof. The triangular decomposition follows from the symmetrizable presentation theorem just cited. For its grading, first use the free abelian group $\widetilde{Q} = \bigoplus_{i \in I} \mathbb{Z} r_i$, then send r_i to α_i. Every α_i has positive height. Only finitely many simple degrees have height at most N, and each has finite multiplicity. A positive word of total height at most N has length at most N and uses only this finite set of generators. There are therefore finitely many such words.

This proves the finite-fiber condition of [52, (6.29)–(6.30)] and the asserted finiteness after passing to relations. No nonempty positive word has degree zero. □

Definition 3.6 *(Root sets and signed multiplicity).* The positive roots are $\mathcal{R}_+ = \{\beta \in Q_+ \setminus \{0\} : \mathfrak{g}_\beta \neq 0\}$. The negative roots are $-\mathcal{R}_+$. The real roots form $\Delta^{re} = W\{\delta_1, \delta_2, \delta_3\}$, and all remaining roots are imaginary. The imaginary root spaces are the homogeneous pieces of the presentation, including all brackets that survive its relations. They are not just Weyl translates of the simple fibers. For every lattice degree set

$$e_\beta = \dim \mathfrak{g}_{\beta,\bar{0}}, \qquad o_\beta = \dim \mathfrak{g}_{\beta,\bar{1}}, \qquad \operatorname{smult}(\beta) = e_\beta - o_\beta.$$

We also use $\operatorname{mult}_{\bar{p}}(\beta)$ for the two dimensions.

For a real root β, write $\mathbb{H}_\beta = \{\mathbb{R}_{>0} x : x \in V_+,\ (x, \beta) = 0\}$. The three simple mirrors bound $\mathcal{P}_{II}$. Their Weyl translates form $\bigcup_{\beta \in \Delta^{re}} \mathbb{H}_\beta$. The type-I chamber reflections permute the three simple mirrors. A physical wall interpretation additionally requires a comparison with the geometric parameter space.

The chosen cusp also splits the positive half. For $\beta = \alpha(n, l, m)$, let $c(\beta) = m$, the second simple-root coordinate. Then $\mathfrak{n}_+ = \mathfrak{n}_+^{>0} \oplus \mathfrak{n}_+^0$, where

$$\mathfrak{n}_+^{>0} = \bigoplus_{\beta \in \mathcal{R}_+,\ c(\beta) > 0} \mathfrak{g}_\beta,$$

$$\mathfrak{n}_+^0 = \bigoplus_{\beta \in \mathcal{R}_+,\ c(\beta) = 0} \mathfrak{g}_\beta.$$

The second summand is supported on the cusp boundary. They do not split the Cartan or negative half. Additivity of c makes the first an ideal of $\mathfrak{n}_+$ and the second a subalgebra. This cusp degree is distinct from the chamber multiplicity $m(a)$.

Lemma 3.7 *(Real subalgebra).* The real generators and $\mathfrak{h}$ generate the Kac–Moody algebra $\mathfrak{g}(A)$. Its inclusion in $\mathfrak{g}$ is injective.

Proof. In the positive presentation, every relation involving an imaginary index has positive degree in that index for the finer $\widetilde{Q}$-grading. Hence it imposes no relation on the purely real words. The remaining positive relations are exactly the real Serre relations. The same argument applies to the negative half.

Together with the injective Cartan identification and triangular decomposition, this proves injectivity of the full real subalgebra. □

Remark 3.8 *(Automorphic correction).* Every connected rank-two principal submatrix of A is affine $A_1^{(1)}$, while A has Lorentzian signature. Thus $\mathfrak{g}(A)$ is hyperbolic. The imaginary simple generators enlarge it to $\mathfrak{g}$. This is the automorphic correction of [52, Section 3]. Both algebras have real-root multiplicity one and real Serre exponent three. Their imaginary root spaces differ.

The correction provides the explicit modular denominator below.

TABLE 4.1: Real generators and the first interior root.

	$\mathfrak{g}(A)$	$\mathfrak{g}$
Real simple vectors	$\delta_1, \delta_2, \delta_3$	$\delta_1, \delta_2, \delta_3$
Imaginary simple fibers	0	$\mu_{\bar{p}}(a)$
First interior parity dimensions	2|0	29|93

LEMMA 3.9 *(Invariant bilinear form).* There is a unique even nondegenerate invariant supersymmetric form $B_{\mathfrak{g}}$ extending the form on $\mathfrak{h}$. It satisfies

$$B_{\mathfrak{g}}(e_i, f_j) = \delta_{ij}, \qquad B_{\mathfrak{g}}([x, y], z) = B_{\mathfrak{g}}(x, [y, z]).$$

Root spaces pair only in opposite degrees, and their opposite-degree pairings are perfect.

Proof. Use the invariant-form theorem for symmetrizable contragredient superalgebras [66, Chapter 2], with the super modification and central quotient of [52, Section 6, Type 1 and (6.28)]. The Cartan form here is nondegenerate, so after that quotient no Cartan radical remains. The theorem gives nondegeneracy and invariance.

Its normalization can also be read directly:

$$B_{\mathfrak{g}}([e_i, f_j], h) = B_{\mathfrak{g}}(e_i, [f_j, h]) = (\alpha_j, h) B_{\mathfrak{g}}(e_i, f_j).$$

The left side is $\delta_{ij}(\alpha_i, h)$. Since $\alpha_j \neq 0$ and the Cartan form is nondegenerate, this forces the stated value. Invariance recursively determines the pairings of bracket words.

For weights β, γ, invariance under h gives $(h, \beta + \gamma) B_{\mathfrak{g}}(x_\beta, y_\gamma) = 0$. Thus only opposite weights pair. Local finiteness then turns the nondegenerate opposite pairing into a perfect pairing. □

LEMMA 3.10 *(Even Weyl transport).* The operators $\exp(\operatorname{ad} e_i) \exp(-\operatorname{ad} f_i) \exp(\operatorname{ad} e_i)$ at real simple indices induce even isomorphisms $\mathfrak{g}_{\beta,\bar{p}} \simeq \mathfrak{g}_{s_i\beta,\bar{p}}$.

Proof. The Serre relations give local nilpotence of the real adjoint actions. These exponentials therefore define the usual Weyl operators [66, Chapter 3]. The generators are even, so the operators preserve parity [52, Section 6]. □

LEMMA 3.11 *(Real roots and simple imaginary fibers).* Every real root has parity dimensions 1|0. At an imaginary simple degree a, let E_a^{simp} be the span of its specified generators. Its signed dimension is $\mu_{\bar{0}}(a) - \mu_{\bar{1}}(a)$. The full root space may contain further bracket vectors.

Proof. The assertions hold at the simple generators by the positive presentation. Weyl transport proves the real-root assertion on the entire real orbit. In the fine positive grading, simple generators and decomposable words have distinct degrees, even when their images in L coincide. □

4 Two denominator identities

For the following formulas let $X^\beta = e^{-2\pi i(\beta,z)}$. $X^\rho\mathbb{Z}[[Q_+]]$ denotes the shifted height-completed module. Define the chamber correction series

$$S_s = 1 - \sum_{a\in(C\cap L)\setminus\{0\}} m(a)X^a,$$

$$S_o = 1 - \sum_{\substack{a\in C\cap L\\ a^2<0}} |m(a)|X^a + \sum_{i<j}\left(\prod_{n\geq 1}(1-X^{na_{ij}})^9 - 1\right).$$

The null-ray terms in S_o retain their eta-product coefficients. Only the timelike coefficients undergo the absolute-value operation.

Theorem 4.1 *(Superdenominator and ordinary denominator).* The algebra $\mathfrak{g}$ satisfies

$$X^\rho D_s := \sum_{w\in W}\det(w)X^{w\rho}w(S_s) = 64^{-1}\Delta_5(2Z),$$
$$D_s = \prod_{\beta\in Q_+\setminus\{0\}}(1-X^\beta)^{e_\beta - o_\beta},$$
$$D_o := X^{-\rho}\sum_{w\in W}\det(w)X^{w\rho}w(S_o) = \prod_{\beta\in Q_+\setminus\{0\}}(1-X^\beta)^{e_\beta}(1+X^\beta)^{-o_\beta}.$$

In particular,

$$\mathrm{den}(\mathfrak{g}) := X^\rho D_s = 64^{-1}\Delta_5(2Z),$$

and, for every $\beta = \alpha(n,l,m)\in Q_+\setminus\{0\}$,

$$e_\beta - o_\beta = f(nm,l). \tag{4.8}$$

Proof. Apply the two superalgebra denominator identities [52, Statements 6.7′–6.8′] to the presentation. In the super identity, the imaginary correction is S_s. For the ordinary identity, each timelike simple generator contributes minus its ordinary multiplicity. Two timelike future vectors cannot be orthogonal, nor can a timelike vector be orthogonal to a nonzero future null vector. Orthogonal collections of imaginary generators therefore consist either of one timelike generator or of generators on one null ray.

All null generators are even, so these collections give the three eta products in S_o. This proves both product formulas.

The Weyl sums are locally finite in height. For a reduced word w, $w\rho-\rho$ is the sum of its positive inversion roots, hence has height at least the word length. Also $wa - a\in Q_+$ for $a\in C$. Only finitely many Weyl words and chamber degrees can therefore contribute to a fixed height. The positive root spaces are finite by Lemma 3.5.

The chamber Fourier formula with z replaced by $2z$ gives the first equality with $\Delta_5(2Z)$. Compare its product with (4.6). Euler factors in $\mathbb{Z}[[Q_+]]$ are unique: remove them successively in increasing height, where the first nonconstant term of $(1-X^\beta)^a$ is $-aX^\beta$. Thus the exponents agree, including zero exponents. □

Remark 4.2 *(Exponential normalization).* The substitution $Z\mapsto 2Z$ sends q,r,s to q^2,r^2,s^2 and $e^{-\pi i(\beta,z)}$ to $e^{-2\pi i(\beta,z)}$. It leaves the integral lattice L and the degree map α unchanged. In particular $L\neq 2\Lambda^{2,1}$: its f_3 direction is primitive. The Jacobi form $\phi_{0,1}$ is the multiplicative input, whereas the

additive lift uses $\eta^9\vartheta$. The constant $64 = 2^6$ is the leading theta-product coefficient. The number 2^{12} in a stipulated half-Hilbert orientation model counts a different finite orbit. It does not normalize this denominator.

PROPOSITION 4.3 *(Recovery of both parities).* Write the ordinary denominator uniquely as $D_o = \prod_{\beta>0}(1 - X^\beta)^{b_\beta}$, and put $f_\beta = f(nm, l)$ when $\beta = \alpha(n, l, m)$. The parity dimensions of the fixed presentation are determined recursively by

$$o_\beta = \frac{b_\beta - f_\beta + o_{\beta/2}}{2}, \qquad e_\beta = f_\beta + o_\beta, \tag{4.9}$$

where $o_{\beta/2} = 0$ if $\beta/2 \notin Q_+ \setminus \{0\}$.

Proof. The identity $(1 + X^\beta)^{-o_\beta} = (1 - X^\beta)^{o_\beta}(1 - X^{2\beta})^{-o_\beta}$ gives $b_\beta = e_\beta + o_\beta - o_{\beta/2}$. Combine it with $f_\beta = e_\beta - o_\beta$. The half-degree has smaller height, so the formula is recursive. Existence of the presented algebra and its two denominator identities ensure nonnegative integral results. This argument applies at every height. It does not infer an infinite assertion from a finite table. □

4.1 *The first root spaces*

PROPOSITION 4.4 *(Jacobi coefficients through degree four).* The nonzero coefficients for $0 \le n \le 4$ are as follows. $f(n, -l) = f(n, l)$ and omitted entries vanish.

$n\backslash\lvert l\rvert$	0	1	2	3	4
0	10	1	0	0	0
1	108	−64	10	0	0
2	808	−513	108	1	0
3	4016	−2752	808	−64	0
4	16524	−11775	4016	−513	10

Proof. Expand the three theta quotients defining $\phi_{0,1}$ in Chapter 3, or its theta decomposition in Lemma 3.1 [39]. Elliptic translation supplies the repeated discriminant coefficients. The value $\phi_{0,1}(\tau, 0) = 12$ checks each positive-n row sum. In particular $2\phi_{0,1}$ has value 24, the K3 Euler number. Squaring (4.6) gives

$$\Delta_5^2 = 64^2 qrs \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{2f(nm,l)}.$$

□

PROPOSITION 4.5 *(Low-degree brackets).* The positive roots of height at most three are

$\mathrm{ht}\,\beta$	β	β^2	$e_\beta\lvert o_\beta$
1	δ_i	2	1|0
2	$a_{ij} = \delta_i + \delta_j$	0	10|0
3	$2\delta_i + \delta_j$ $(i \neq j)$	2	1|0
3	$\delta_{123} := \delta_1 + \delta_2 + \delta_3$	−6	29|93

(4.10)

The inverse Gram coordinate is $\gamma(\sum b_i\delta_i) = (b_1, b_1 + b_2 - b_3, b_2)$. For $i \neq j$, set $E_{ij} = [e_i, e_j]$ and $E_{iij} = [e_i, E_{ij}]$. For $\{i, j, k\} = \{1, 2, 3\}$ one has

$$E_{ji} = -E_{ij}, \qquad [h_i, E_{ij}] = [h_j, E_{ij}] = 0, \qquad [h_k, E_{ij}] = -4E_{ij},$$
$$[f_i, E_{ij}] = 2e_j, \qquad [f_j, E_{ij}] = -2e_i, \qquad [f_k, E_{ij}] = 0,$$
$$[e_i, E_{iij}] = 0, \qquad [f_i, E_{iij}] = 2E_{ij}, \qquad [f_j, E_{iij}] = 0.$$

For $i < j$, a basis in degree a_{ij} consists of E_{ij} and nine even simple vectors $u_{ij,r}$, $1 \le r \le 9$. They satisfy

$$[e_i, u_{ij,r}] = [e_j, u_{ij,r}] = [f_i, u_{ij,r}] = [f_j, u_{ij,r}] = 0, \qquad (\operatorname{ad} e_k)^5 u_{ij,r} = 0.$$

Proof. The inverse degree map and the norm formula give the table's degrees and norms. Purely real positive words of height two give one bracket in each pair degree. Null-ray simple generators add nine, with orthogonality giving the displayed zeros. Chevalley and Jacobi give, for example, $[f_i, [e_i, e_j]] = [-h_i, e_j] = 2e_j$ and $[f_i, [e_i, E_{ij}]] = [-h_i, E_{ij}] + [e_i, 2e_j] = 2E_{ij}$. The real Serre exponents are three on e_j and five on $u_{ij,r}$ because $(\delta_k, a_{ij}) = -4$. The interior count is proved next. □

Proposition 4.6 *(The first interior degree).* One has $m(\delta_{123}) = -93$ and $e_{\delta_{123}}|o_{\delta_{123}} = 29|93$. The even space contains the two-dimensional real span

$$T_1 = [e_1, E_{23}], \qquad T_2 = [e_2, E_{31}], \qquad T_3 = [e_3, E_{12}], \qquad T_1 + T_2 + T_3 = 0,$$

and the 27 vectors $[e_k, u_{ij,r}]$. The odd space consists of 93 imaginary simple vectors, denoted $c^1, \dots, c^{93}$.

Proof. The multilinear part of the free Lie algebra on three real generators has dimension two. No Serre relation has this fine degree. Each mixed real-null word is also free of a relation at this degree. The fine generator grading separates the 27 choices from each other and from the real words. Thus the even contribution is $2 + 27 = 29$.

The chamber Fourier coefficient is $m(\delta_{123}) = -64^{-1}c_\Delta(3, 3, 3) = -93$: the coefficient of qrs in the normalized product is $64 + 29 = 93$. It prescribes 93 odd simple generators. No bracket involving an odd generator can yet occur at smaller total height. Hence the signed root multiplicity is $29 - 93 = -64 = f(1, 1)$. □

Proposition 4.7 *(Further interior parity dimensions).* Up to permutation of simple-root coordinates,

(b_1, b_2, b_3)	$m(\beta)$	f_β	e_β	o_β
(1, 1, 1)	−93	−64	29	93
(1, 1, 2)	90	108	201	93
(2, 2, 1)	54	−513	510	1023
(2, 2, 2)	−540	4016	10136	6120

Proof. The chamber coefficient definition computes the displayed m values from the finite product expansion in these degrees. Formula (4.9) gives the two root dimensions.

There is also a direct word count. At (1, 1, 2), there are three purely real words and $9 + 9 + 9$ words containing one null simple generator. Brackets between generators of degrees a_{13} and a_{23} add 81 vectors. The remaining 90 even vectors are simple generators. Their sum is 201. The odd vectors are $[e_3, c^u]$, $1 \le u \le 93$.

At $(2,2,1)$, an odd word contains one c^u. Its remaining degree $(1,1,0)$ is represented by the two ordered real actions or one of the nine generators in degree a_{12}. This gives $93(2+9) = 1023$.

At $(2,2,2)$, there are 540 new odd simple generators. Each c^u permits six ordered actions by three distinct real generators. It also permits both orders of a null generator and the complementary real generator, with $3 \cdot 9$ choices for the null generator. Thus

$$o_{(2,2,2)} = 540 + 93(6 + 3 \cdot 2 \cdot 9) = 6120.$$

Fine generator degrees separate these summands. The real Serre relations involving c^u require three repetitions of one real index and do not occur in the listed action degrees. The null orthogonality relations give precisely the reductions just used. Finally $e_\beta = f_\beta + o_\beta$ gives 510 and 10136. □

Remark 4.8 *(Real strings).* For distinct i, j, k, the real $\mathfrak{sl}_2$ at k acts on the string through a_{ij} with lowest weight -4. The degrees are $a_{ij} + s\delta_k$, $0 \le s \le 4$. $a_{ij} - \delta_k$ and $a_{ij} + 5\delta_k$ are not roots. This is the real-root string theorem [66, Proposition 5.1(c)], with the superalgebra convention of [52, Section 6]. The affine subalgebra at i, j contributes one even vector at every na_{ij}, and the null simple fibers contribute nine. Thus $\mathfrak{g}_{na_{ij}}$ has dimensions 10|0. If $C_{k,s} = a_{ij} + s\delta_k$ and $D_k = \delta_k + 2a_{ij}$, the corresponding dimensions are

$$2a_{ij} : 10|0, \quad C_{k,2} : 201|93, \quad C_{k,3} : 29|93, \quad C_{k,4} : 10|0, \quad C_{k,5} : 0|0, \qquad D_k : 510|1023.$$

The two transported rows follow from even Weyl symmetry. These are abstract root dimensions. Comparison maps still require actual vectors and operations in the compact source.

COROLLARY 4.9 *(First signed multiplicities).* The values at δ_i, a_{ij}, δ_{123} are respectively 1, 10, -64. Their parity dimensions are 1|0, 10|0, 29|93.

Proof. Subtract the parity dimensions in (4.10). □

5 A DIFFERENTIAL AND ITS FIRST SERRE OBSTRUCTION

The two parities at $\gamma := \delta_{123}$ permit a complex with smaller cohomology but the same signed dimension. Whether such a complex respects the root operations is a further question. Consider first the genuine positive-height quotient

$$L_{\le 3} = \mathfrak{n}_+ / \bigoplus_{\operatorname{ht}\beta > 3} \mathfrak{g}_\beta.$$

Its dimensions by height are 3, 30, and 35|93, respectively, so its total dimension is 161. Give its even simple generators cohomological degree zero and its 93 odd generators c^u degree -1. Cochain parity supplies the super sign.

PROPOSITION 5.1 *(Cancellation in the height-three quotient).* Choose a basis $w_1, \dots, w_{29}$ of $W_\gamma := \mathfrak{g}_{\gamma,\bar{0}}$. The assignments

$$dc^u = w_u \quad (1 \le u \le 28), \qquad dc^u = 0 \quad (29 \le u \le 93), \qquad de_i = du_{ij,r} = 0$$

define a dg Lie structure on $L_{\le 3}$. At γ its cohomology is $H^{-1} = \mathbb{C}^{65}$ and $H^0 = \mathbb{C}$.

Proof. Every height-three vector is central in this positive quotient. The assigned derivation therefore respects the relations and squares to zero. Its degree-γ complex is $\mathbb{C}^{93} \longrightarrow \mathbb{C}^{29}$ with rank 28. The pairs (c^u, w_u) for $u \le 28$ are contractible, with homotopy $w_u \mapsto c^u$. The remaining 65 odd and one even vector give the stated cohomology. □

This calculation explains how $29 - 93 = -64$ can coexist with cohomology dimensions $1|65$. It makes no choice of a geometric differential. Such a choice would have to arise from the coefficient complexes and preserve their operations. Even abstractly, the unchanged Borcherds relations constrain the extension.

Lemma 5.2 *(Lowering detects the interior even space).* Put $E_i = \operatorname{ad} e_i$, $F_i = \operatorname{ad} f_i$ and $H_i = \operatorname{ad} h_i$. The map

$$W_\gamma \longrightarrow \bigoplus_i \mathfrak{g}_{\gamma-\delta_i}, \qquad w \longmapsto (F_1 w, F_2 w, F_3 w)$$

is injective, and $F_i^2 W_\gamma = 0$.

Proof. For the real basis $u = [e_1, [e_2, e_3]]$ and $v = [e_2, [e_1, e_3]]$, Chevalley and Jacobi give

	u	v
F_1	$4E_{23}$	$2E_{23}$
F_2	$2E_{13}$	$4E_{13}$
F_3	$-2E_{12}$	$2E_{12}$

The first two coefficient rows have determinant 12.

For a null simple vector $b \in \operatorname{span}\{u_{ij,r}\}$,

$$F_k[e_k, b] = 4b, \qquad F_i[e_k, b] = F_j[e_k, b] = 0 \quad (\{i, j, k\} = \{1, 2, 3\}).$$

The nine simple directions at a_{ij} are independent of E_{ij}. These blocks therefore detect all 27 mixed coefficients. The real block detects the remaining two. A second lowering has a negative simple-root coordinate and vanishes. □

Theorem 5.3 *(First obstruction to extending the differential).* A root-preserving odd derivation of the unchanged positive Borcherds algebra vanishes on all c^u. Set the rank-28 differential above to zero on all other simple generators. It descends to every positive-height quotient of height at most five, and fails at height six.

Proof. The purely even real-root spaces force $de_i = 0$. For $w = dc^u \in W_\gamma$, differentiation of the defining relation $E_i^3 c^u = 0$ gives $E_i^3 w = 0$. On W_γ we have $H_i w = -2w$ and $F_i^2 w = 0$. The $\mathfrak{sl}_2$ commutation relations give

$$F_i E_i^a F_i^b w = E_i^a F_i^{b+1} w - a(-2 - 2b + a - 1) E_i^{a-1} F_i^b w.$$

Four successive lowerings of $E_i^3 w$ are $E_i^3 F_i w$, $6E_i^2 F_i w$, $36E_i F_i w$, and $144 F_i w$. Hence $E_i^3 w = 0$ implies $F_i w = 0$ for every i. Lemma 5.2 forces $w = 0$. The negative generators only detect the positive relations. No extension of d to them is assumed.

The first relations involving c^u are the real Serre relations at $\gamma + 3\delta_i$, of height six. Orthogonality gives none at lower height: $(\gamma, \delta_i) = -2$ and $(\gamma, a_{ij}) = -4$. Thus through height five every surviving relation either contains no c^u and is closed, or is absent by height. The differential preserves height and squares to zero on generators, so it descends through those quotients. At height six the injective detector rules out its nonzero image. □

There is a separate restriction on retaining the real algebra in cohomology. If R_γ is its two-dimensional real multilinear part and $W' \subset W_\gamma$ has dimension 28, then

$$\dim(W' \cap R_\gamma) \geq 28 + 2 - 29 = 1.$$

Thus making W' exact while keeping the real generators closed kills at least one nonzero real bracket in cohomology.

PROPOSITION 5.4 *(Free attachment of the desired boundaries).* Let $\mathfrak{a}$ be the ordinary Lie algebra generated by the real and null simple vectors with their prescribed relations. Choose $w_1, \dots, w_{28} \in \mathfrak{a}_\gamma$ and form

$$\mathcal{L} = \mathfrak{a} * \mathrm{Lie}(c^1, \dots, c^{93}), \qquad dc^u = \begin{cases} w_u, & u \le 28, \\ 0, & u > 28, \end{cases} \qquad d\mathfrak{a} = 0,$$

where the c^u have degree -1 and root γ. This is a dg Lie algebra with $H^0(\mathcal{L}) = \mathfrak{a}/\langle w_1, \dots, w_{28}\rangle_{\mathrm{Lie}}$. The differential extends to its positive-height completion.

Proof. The derivation extends freely and its square is zero on generators. Degree zero is $\mathfrak{a}$. Every degree--1 word has one c^u, so its differential is an iterated bracket of w_u with $\mathfrak{a}$. These images generate exactly the stated ideal. Root degree is preserved, giving continuity in positive height. □

This attachment imposes no new Serre relation on the c^u. For example, $d[c^i, c^j] = -[c^j, w_i] - [c^i, w_j]$ records relations among its boundaries in negative cohomological degrees. For a homotopy-invariant derived quotient, make the attachment to a semifree resolution of $\mathfrak{a}$ using chosen cycle lifts of the w_u. Imposing an additional differential $dz = E_i^3 c^u$ would require $d^2 z = E_i^3 w_u = 0$. The theorem shows why that requirement fails in the original root algebra. The construction provides a different derived algebra, whose comparison with geometric elliptic coefficients remains to be made.

6 FINITE OPERATIONS AND COMPLETIONS

A finite set of roots gives finite-dimensional vector spaces. It gives a Lie algebra only when the bracket has a suitable quotient or closure property. The positive-height quotients just used have this property because positive heights add.

PROPOSITION 6.1 *(Positive completion).* The subspace

$$\mathfrak{n}_{+,>N} = \bigoplus_{\mathrm{ht}\,\beta > N} \mathfrak{g}_\beta$$

is a Lie ideal of $\mathfrak{n}_+$. Its finite-dimensional nilpotent quotients define the topological Lie superalgebra

$$\widehat{\mathfrak{n}}_+ = \varprojlim_N \mathfrak{n}_+/\mathfrak{n}_{+,>N} = \prod_{\beta>0} \mathfrak{g}_\beta.$$

The bracket projected to $\mathfrak{h} \oplus \bigoplus_i (\mathfrak{g}_{\delta_i} \oplus \mathfrak{g}_{-\delta_i})$ fails Jacobi.

Proof. A bracket of positive degrees has their sum as degree, so the specified subspace is an ideal. An iterated bracket of more than N positive inputs has height greater than N. The quotient is therefore nilpotent and finite-dimensional. Every coefficient of the completed bracket is a finite sum, giving continuity.

For the projected full bracket, take $i \neq j$. The term $[e_i, [e_j, f_j]] = [e_i, h_j] = 2e_i$ survives, whereas the intermediate $[e_i, e_j]$ is projected to zero and $[e_i, f_j] = 0$. Thus the derivation form of Jacobi has defect $2e_i$. □

Definition 6.2 *(Finite homogeneous operation datum).* A finite set $W \subset \mathcal{R}_+$ is *downward saturated* if it contains every positive root $\beta' \le \beta \in W$, where $\beta - \beta' \in Q_+$. Put $S_W = W \cup (-W) \cup \{0\}$. The target datum contains the spaces $\mathfrak{g}_{\beta,\bar{p}}$ for $\beta \in S_W$ and $B_{\mathfrak{g}}$ on opposite degrees. It retains each homogeneous bracket map whose input and output degrees belong to S_W. It also specifies the simple generators, their defining relations, and bases of the finite enveloping pieces defined below. Every iterated identity is required in a larger set containing all its intermediate degrees.

Such a datum is a fragment of the graded algebra. It is not assigned a bracket that sets omitted outputs to zero. In particular, its finite matrices describe operations inherited from $\mathfrak{g}$, not an enveloping algebra of the fragment itself.

Proposition 6.3 *(Finite filtered enveloping spaces).* Choose homogeneous root bases and a basis of $\mathfrak{h}$. Give a root basis vector weight $|\operatorname{ht}\beta|$ and a Cartan basis vector weight one. Ordered super-PBW monomials of total weight at most N span a finite-dimensional space $F_N U(\mathfrak{g})$. These spaces exhaust $U(\mathfrak{g})$ and satisfy

$$F_M F_N \subset F_{M+N}, \qquad \Delta(F_N) \subset \sum_{a+b\le N} F_a \otimes F_b, \qquad S(F_N) \subset F_N.$$

Here $\Delta(x) = x \otimes 1 + 1 \otimes x$ and $S(x) = -x$ on $\mathfrak{g}$. The ordinary PBW length filtration is an additional filtration. Its intersection with each F_N is finite-dimensional.

Proof. Only finitely many root and Cartan basis vectors have weight at most N, and each has positive weight. There are finitely many ordered monomials of bounded total weight, with an odd vector occurring at most once. Reordering uses $xy - (-1)^{|x||y|} yx = [x, y]$. The bracket weight is at most the sum of the input weights, including opposite roots whose bracket lies in the Cartan. Thus multiplication and the Hopf operations respect the stated bounds. The super PBW theorem proves that these monomials form bases. □

A finite filtered piece is generally not an algebra. Already in $U(\mathfrak{sl}_2)$, order the basis by $f < h < e$ and project each product to PBW length at most one. The projected product satisfies $e * f = h, h * f = -2f$ and $f * f = 0$, so

$$(e * f) * f - e * (f * f) = -2f.$$

The associative multiplication is the map into the larger filtered piece specified by the proposition.

Theorem 6.4 *(Reconstruction from finite operations).* Let $P = \bigoplus_{\beta\in L} P_\beta$ be a graded Lie superalgebra with finite-dimensional homogeneous spaces and no support outside $\mathcal{R}_+ \cup (-\mathcal{R}_+) \cup \{0\}$. Suppose compatible even isomorphisms $A_\beta : P_\beta \to \mathfrak{g}_\beta$ are given on an exhaustion of all root degrees and the Cartan. If every homogeneous bracket square commutes in some finite set containing its inputs and output, then $\bigoplus_\beta A_\beta$ is a graded Lie isomorphism. If the opposite-degree pairing squares commute, it also preserves the specified invariant forms. It induces a filtered Hopf isomorphism of the ordinary enveloping algebras and an isomorphism of the positive-height Lie completions.

Proof. Compatibility defines A on every homogeneous space. Algebraic elements have finite support. Every pair of such elements is contained in a finite set of homogeneous bracket squares, so A preserves brackets. The inverses have the same property.

Pairing compatibility is checked on the finitely many opposite-degree terms. The enveloping assertion follows by its universal property and the super PBW theorem. On the positive half, the maps commute with the genuine height quotients. The compatible inverses induce the inverse on their limit. □

PROPOSITION 6.5 *(Obstruction to the full root-product topology).* The bracket of $\mathfrak{g}$ has no jointly continuous extension to $\mathfrak{h} \oplus \prod_{\beta \neq 0} \mathfrak{g}_\beta$ with its coordinatewise linear topology and discrete Cartan factor.

Proof. Fix a primitive null ray $a = a_{ij}$ and choose one pair of simple generators e_n, f_n in degrees $na, -na$ for each $n \geq 1$. Chevalley gives $[e_n, f_m] = \delta_{nm} na$. The two partial sums $\sum_{n=1}^{N} e_n$ and $\sum_{n=1}^{N} f_n$ converge coordinatewise, but their brackets have Cartan component $N(N+1)a/2$, which is not eventually constant. This contradicts continuity. □

One valid enlargement is $\mathfrak{n}_- \oplus \mathfrak{h} \oplus \widehat{\mathfrak{n}}_+$, with the negative and Cartan summands discrete. A negative element has finite support, so every mixed bracket coefficient is a finite sum. Its negative output is again finite. Positive-positive brackets are continuous by height. This defines a topological Lie algebra. A completion allowing both infinite signs needs a different summation domain and a new continuity argument.

6.1 *Inverse limits of maps and complexes*

A compatible family of literal isomorphisms of inverse systems induces an isomorphism on inverse limits: the compatible inverse maps provide its inverse. No exactness hypothesis is needed for that assertion. Exactness becomes relevant when forming quotient objects or passing from quasi-isomorphisms of complexes to ordinary limits.

PROPOSITION 6.6 *(Finite towers).* Every countable inverse system of finite-dimensional vector spaces is Mittag–Leffler. Consequently, for a compatible short exact sequence $0 \to K_N \to V_N \to Q_N \to 0$ with each K_N finite-dimensional, its inverse limit is exact.

Proof. At a fixed index N, the images of later terms in K_N form a descending chain of subspaces. Their dimensions stabilize, hence so do the subspaces. This is the Mittag–Leffler condition and implies $R^1 \varprojlim K_N = 0$ for a countable tower [117, Section 3.5]. Apply the inverse-limit exact sequence. □

The assertion applies to primitive spaces, pairing kernels and finite PBW pieces only after those spaces and their transition maps exist. Kernels themselves commute with limits of an actual commuting inverse-system map.

Dualization reverses arrows: if $T_N : V_{N+1} \to V_N$, then $T_N^* : V_N^* \to V_{N+1}^*$. Thus finite perfect pairings do not, by themselves, define an inverse tower of dual spaces. For a surjective inverse tower of finite-dimensional spaces, the continuous dual of its limit is $\varinjlim V_N^*$. Comparison with a second inverse limit requires specified adjoint transition maps. Over a general coefficient ring, finite generation alone also does not imply the descending-chain argument used above.

PROPOSITION 6.7 *(Ordinary limits of complexes).* For a countable tower of cochain complexes, degreewise Mittag–Leffler chain groups suffice to identify its ordinary inverse limit with its homotopy inverse limit. Without such a condition, ordinary inverse limits need not preserve quasi-isomorphisms, even for towers of acyclic complexes.

Proof. Let T shift a product of the chain groups using the transition maps. The homotopy inverse limit is $\mathrm{Cone}(1 - T : \prod C_N \to \prod C_N)[-1]$. Degreewise Mittag–Leffler makes $1 - T$ degreewise surjective. Its kernel complex, the ordinary limit, is then quasi-isomorphic to that homotopy fiber. Homotopy limits preserve equivalences.

For failure without this hypothesis, take the acyclic complexes

$$C_N = \left(t^N \mathbb{C}[t] \hookrightarrow \mathbb{C}[t] \longrightarrow \mathbb{C}[t]/(t^N)\right)$$

in degrees $-1, 0, 1$. The transition maps are inclusion, identity and reduction, respectively. Their ordinary limit is $0 \to \mathbb{C}[t] \to \mathbb{C}[[t]]$, whose degree-one cohomology is $\mathbb{C}[[t]]/\mathbb{C}[t]$. The compatible residues $1 + t + \cdots + t^{N-1}$ have no polynomial lift: such a lift would have coefficient one in every degree. The infinite-dimensional tail spaces here are not covered by the preceding proposition. □

7 Information in the superdenominator

Lemma 7.1 *(Visible and zero exponents).* On Γ_{act}, the signed root dimension is $\mathrm{smult}(\alpha(n,l,m)) = f(nm,l)$. A zero exponent can arise from an absent root space or from equal, nonzero parity dimensions. The superdenominator alone does not distinguish these possibilities.

Proof. The equality is (4.8). A factor with exponent zero is one for either of the stated parity possibilities. The ordinary denominator of the fixed presentation resolves that ambiguity by Proposition 4.3. □

Proposition 7.2 *(Information in a superdenominator).* A product $\prod_{\beta>0}(1 - X^{\beta})^{f_\beta}$ in the positive height completion determines every signed exponent f_β. Taken without a presentation or ordinary denominator, it determines neither the two parity dimensions nor a bracket or invariant form.

Proof. Taking logarithms over $\mathbb{Q}$ gives

$$\log D_s = -\sum_{\beta>0}\sum_{r\geq 1}\frac{f_\beta}{r}X^{r\beta}.$$

At each height the already known smaller degrees can be subtracted, leaving f_β. But adjoining an even-odd pair in any degree does not change f_β. On fixed graded spaces, the abelian bracket also gives the same determinant. It differs from the Borcherds bracket, for which $[e_i, f_i] = h_i \neq 0$. □

Proposition 7.3 *(The super Grothendieck quotient).* Let $\mathcal{B}$ be a locally finite super vector space graded by the positive monoid, with $\mathrm{sdim}\,\mathcal{B}_{\alpha(n,l,m)} = f(nm,l)$. Its completed superdeterminant gives

$$64q^{1/2}r^{1/2}s^{1/2}\prod_{\Gamma_{\mathrm{eff}}}(1-q^n r^l s^m)^{\mathrm{sdim}\,\mathcal{B}_{\alpha(n,l,m)}} = \Delta_5(Z).$$

This invariant factors through the quotient of the completed Grothendieck group by $[\Pi V] = -[V]$, where Π reverses parity. The ordinary Grothendieck group of super vector spaces retains both parity dimensions.

Proof. In each degree, ordinary K_0 is $\mathbb{Z}[\mathbb{C}_{\bar{0}}] \oplus \mathbb{Z}[\mathbb{C}_{\bar{1}}]$. Superdimension is the map $(e,o) \mapsto e - o$, whose kernel is generated by $(1,1)$. The superdeterminant is $(1 - X^{\beta})^{e-o}$, so it factors through that quotient. Local finiteness makes the product meaningful in height. The specified signed dimensions identify it with (4.6). □

Remark 7.4 *(The extra datum in parity recovery).* The pair 29|93 uses the minimal simple-generator prescription and the Borcherds relations. The superdenominator by itself admits all pairs $(a, a+64)$. Formula (4.9) recovers the actual pair because its second input, D_o, is constructed from that prescription. This does not reconstruct a compact Hall object from a numerical product.

Remark 7.5 *(Cusp and root coordinates).* The cusp notation uses (n, l, m) and the root notation uses $(b_1, b_2, b_3) = (n, m, n + m - l)$. Their common active monomials lie in Q_+ by Lemma 2.2. The cusp order and the chosen Weyl chamber specify expansions of the same lattice characters through α. This coordinate identification is an integral map. It does not supply a duality of coefficient objects.

For a compact source, the normal-ordered charge homomorphism $\overline{\Pi}_X$ must still intertwine its operations with this grading. The quadratic defect of an uncorrected charge assignment is not an extra term in the target Cartan form.

8 THE INVARIANT FORM AND PRIMITIVE RECOGNITION

If an associative super-bialgebra H is given, its primitives are $P(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}$. Compatibility of product and coproduct makes them a Lie superalgebra under the supercommutator. Recognition of $\mathfrak{g}$ in such a source requires a Lie form on the primitives. That form must be distinguished from a pairing adjoint to the ordinary Hopf product and coproduct.

PROPOSITION 8.1 *(Hopf adjointness on primitives).* Let B be an even Hopf pairing on a super-bialgebra over $\mathbb{C}$, with $B(xy, z) = B(x \otimes y, \Delta z)$ and the unit-counit identities. Then $B([P(H), P(H)], P(H)) = 0$. Its restriction cannot be the nondegenerate invariant form of $\mathfrak{g}$. The contradiction already occurs on a real $\mathfrak{sl}_2$ triple.

Proof. For primitive z, the right side of the adjointness identity has one unit factor. If x, y are also primitive, the unit-counit identities give $B(xy, z) = 0$, and similarly for the reversed product, with the Koszul tensor sign. Thus the primitive commutator pairs to zero with every primitive. On an even real triple in $\mathfrak{g}$,

$$[h, e] = 2e, \qquad B_{\mathfrak{g}}(e, f) = 1, \qquad B_{\mathfrak{g}}([h, e], f) = 2.$$

This contradicts the asserted Hopf adjointness. Even without a unit normalization, for even primitives the expression $B(x, z)B(y, 1) + B(x, 1)B(y, z)$ is symmetric in x, y, giving the same contradiction after subtraction. □

The obstruction fixes the recipient: ordinary primitives and their ordinary coproduct. It does not exclude a Hall pairing between different positive and negative objects, a braided coproduct, or an independently constructed invariant Lie form. Each such choice has its own compatibility equations.

PROPOSITION 8.2 *(Radical quotient from a Lie form).* Let P be a Lie superalgebra with an even invariant form B_P, and let R be its radical. Then R is a Lie ideal. The two-sided ideal $\langle R \rangle$ in $U(P)$ is a Hopf ideal, and

$$U(P)/\langle R \rangle \simeq U(P/R)$$

as Hopf superalgebras.

Proof. Invariance implies $B_P([x, r], y) = 0$ for every $r \in R$ and all homogeneous x, y, so $[P, R] \subset R$. In $U(P)$ the elements of R are primitive. Therefore $\Delta\langle R \rangle \subset \langle R \rangle \otimes U(P) + U(P) \otimes \langle R \rangle$, the counit kills this ideal, and the antipode preserves it. The enveloping universal property gives the displayed quotient isomorphism. □

For a compact source, first construct P_X and its invariant form B_X. Then construct a root-graded homomorphism

$$\mathfrak{g} \longrightarrow P_X/\operatorname{Rad} B_X$$

from specified source representatives of the real and imaginary simple generators. The defining relations, generation and absence of extra relations must be proved. On finite homogeneous spaces, these statements can be expressed by bracket, pairing and comparison matrices as in Theorem 6.4. An arbitrary full-rank integral matrix is insufficient over a coefficient ring: multiplication by two on $\mathbb{Z}$ has full rank and is not invertible. Over $\mathbb{C}$, equal dimensions and invertible comparison matrices do give the required vector-space isomorphisms.

Proposition 8.3 *(Recognition on all primitives).* Let H be a connected cocommutative Hopf superalgebra over $\mathbb{C}$. Here connected means that the reduced coproduct is elementwise conilpotent. Then the natural map $U(P(H)) \to H$ is an isomorphism. Consequently a morphism between such Hopf superalgebras is an isomorphism if it is an isomorphism on all primitives.

Proof. In the convolution algebra of endomorphisms of H, put $J = \mathrm{id} - \eta\varepsilon$ and

$$e = \log^*(\mathrm{id}) = \sum_{n\geq 1} \frac{(-1)^{n+1}}{n} J^{*n}.$$

Every sum is finite on an element by conilpotence. In $\mathrm{Hom}(H, H \otimes H)$, the maps $x \mapsto x \otimes 1$ and $x \mapsto 1 \otimes x$ commute for convolution by cocommutativity, and their product is Δ. Taking logarithms therefore gives $\Delta e(x) = e(x) \otimes 1 + 1 \otimes e(x)$. Thus e takes values in $P(H)$. The identity $\exp^*(e) = \mathrm{id}$ expresses each element as a finite sum of products of these primitives, proving surjectivity.

For injectivity, suppose the kernel has a nonzero element of least PBW filtration degree d. Its counit is zero, and its reduced coproduct lies in the tensor square of filtration degree at most $d - 1$. The map is injective on that smaller filtration, so the reduced coproduct must vanish.

By super PBW in characteristic zero, the primitives of $U(P(H))$ are exactly $P(H)$. Indeed, in the associated graded symmetric coalgebra the coproduct component of lengths $(1, d-1)$, followed by multiplication, is d times the identity in length d. Thus a primitive has length at most one. This contradicts injectivity on $P(H)$. The last assertion follows by functoriality of the enveloping map. □

Agreement on a chosen list of root generators has no such force without these recipient and generation hypotheses. An isomorphism of global primitive spaces also does not identify a sheaf of chiral coefficients without a comparison of its local operations.

9 Current algebras and their coefficients

The root algebra determines a chiral algebra on a smooth complex curve C. This construction supplies a concrete comparison object. Its local state spaces include current descendants absent from the root denominator.

Definition 9.1 *(Current chiral envelope).* Let $\mathcal{D}_C$ be the sheaf of differential operators on C. The induced right differential module

$$\mathrm{Cur}_C(\mathfrak{g}) = \mathfrak{g} \otimes_{\mathbb{C}} \mathcal{D}_C$$

has the current Lie-star bracket induced by $[a, b]_{\mathfrak{g}}$. Here a Lie-star bracket is a differential-module map to the small-diagonal direct image, with antisymmetry and Jacobi. Its level-zero Beilinson–Drinfeld chiral envelope is

$$V_C = \mathsf{Den}_{\Delta_s, C} := U_C^{\mathrm{ch}} \mathrm{Cur}_C(\mathfrak{g}).$$

A chiral bracket extends the Lie-star bracket to inputs on the complement of the diagonal, with the corresponding differential-module and Jacobi conditions. The envelope has the local vacuum map $\omega_C \to V_C$.

PROPOSITION 9.2 *(Local current realization).* The uncompleted ind differential-module envelope V_C exists. On a coordinate disk its vacuum module is the level-zero induced module

$$U(\mathfrak{g}[t,t^{-1}]) \otimes_{U(\mathfrak{g}[t])} \mathbb{C},$$

and its PBW vector space is $\mathrm{Sym}_{\mathrm{super}}(\mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}])$. The zero product of the current states recovers $[a,b]_{\mathfrak{g}}$. Its root superdenominator is $64^{-1}\Delta_5(2Z)$.

Proof. The induced current Lie-star module is [7, Section 2.5.6(b)(ii)]. The chiral enveloping construction and its PBW theorem are [7, Sections 3.7.1 and 3.7.14]. The induced underlying $\mathcal{O}_C$-module is flat. Locally, the current operator product is $a(z)b(w) \sim [a,b](w)/(z-w)$, with no central double pole at level zero.

PBW orders the negative loop modes and gives the displayed vacuum module [65]. Thus the first current states recover the original bracket by their zero product. Apply Theorem 4.1 to that root algebra. □

The last assertion concerns the current bracket, not a character of the entire vacuum module. Give a mode $a \otimes t^{-r}$ energy r, with formal energy variable u. For the positive-root current envelope, PBW gives the supercharacter

$$\prod_{\beta>0} \prod_{r\geq 1} (1-u^r X^{\beta})^{-\operatorname{smult}(\beta)}.$$

Each fixed root height and energy has finite dimension.

The energy is the loop-rotation grading. This formula does not require a Sugawara conformal vector for the infinite-dimensional algebra. The product includes every descendant mode, so it differs from the root superdeterminant even before a compact state space is considered.

PROPOSITION 9.3 *(An infinite current coefficient).* For the full two-sided root algebra, the energy-two, root-charge-zero part of the universal current vacuum module is infinite-dimensional.

Proof. At each $n a_{ij}$ choose an even null simple generator and its negative partner. The PBW monomials $(e_n \otimes t^{-1})(f_n \otimes t^{-1})|0\rangle$ have energy two and charge zero. They are independent for all $n \geq 1$. Thus the unpolarized root-and-energy character is not defined by finite coefficient dimensions. A different convergence prescription or a separately constructed state is needed to define such a trace. □

9.1 An associative envelope on Ran space

An associative bar construction needs an associative recipient, a tensor product and a unit. We describe one recipient in which the current chiral algebra has such an envelope. Let $\mathsf{FinSurj}_+$ be the category of nonempty finite sets and surjections. Set

$$\mathcal{R}_C = \varprojlim_{\mathsf{FinSurj}_+} \mathrm{DMod}(C^I), \qquad \mathcal{R}_C^+ = D(\mathbb{C}) \times \mathcal{R}_C,$$

where a surjection $I \twoheadrightarrow J$ acts by exceptional pullback along its diagonal $C^J \to C^I$. An object is therefore a compatible family of right differential modules, together with an isolated empty component.

For an ordered partition $I = I_1 \sqcup I_2$, let j_{I_1,I_2} be the open locus where points in different blocks are distinct. Chiral convolution is

$$(F \star G)_I = \bigoplus_{I=I_1\sqcup I_2} j_{I_1,I_2,*}\, j^!_{I_1,I_2}(F_{I_1} \boxtimes G_{I_2}).$$

The empty-block summands use the empty coefficients. The unit K has $K_\varnothing = \mathbb{C}$ and $K_I = 0$ for $I \neq \varnothing$. Call a family positive if its empty component is zero.

For a right differential module N on C, its positive main-diagonal family is $(\iota_* N)_I = \delta_{I,*} N$. The chiral bracket of V_C makes $\mathcal{L}_C := \iota_* V_C$ a Lie object for $\star$. Define the augmented associative envelope

$$\mathcal{A}_{\Delta,C} = U_\star(\mathcal{L}_C), \qquad (\mathcal{A}_{\Delta,C})_\varnothing = \mathbb{C}.$$

This algebra and the BD envelope V_C have different carriers.

Proposition 9.4 *(Bar of the associative envelope).* In $(\mathcal{R}^+_C, \star, K)$ there are equivalences

$$B\mathcal{A}_{\Delta,C} \simeq \mathrm{CE}_\star(\mathcal{L}_C), \qquad \Omega\mathrm{CE}_\star(\mathcal{L}_C) \simeq \mathcal{A}_{\Delta,C}.$$

The first coalgebra satisfies factorization. The bar is coaugmented and counital, its reduced coproduct is coassociative, and the cobar recovers an augmented associative algebra with empty component $\mathbb{C}$.

Proof. The Ran convolution is symmetric and exact. Its positive r-fold words vanish on a configuration of cardinality less than r. This finite configuration bound makes the normalized bar–cobar comparison finite at every configuration.

Flattening a bar–cobar word with r original inputs gives $r-1$ gaps, each with the two states "inside a block" and "between blocks". On the associated graded the gap differential is a tensor product of two-term identity complexes. It is contractible for $r > 1$. The empty and one-input terms survive. The finite filtration proves the unit and counit equivalences, including their coherent versions.

For the envelope, the comparison from the Chevalley–Eilenberg coalgebra to the bar is suspended symmetrization

$$sx_1 \cdots sx_r \longmapsto \sum_{\sigma\in S_r} \varepsilon(\sigma; sx_1, \ldots, sx_r)[x_{\sigma(1)}|\cdots|x_{\sigma(r)}].$$

It has no factor $1/r!$: the unshuffle coproduct on the source corresponds to deconcatenation of the permutation sum. Its binary differential is the envelope commutator relation. The PBW filtration reduces it to the abelian Koszul comparison, contracted in positive polynomial weight by the Euler homotopy divided by that weight. Characteristic zero and the finite configuration bound give an equivalence. These constructions in the Ran coefficient category are described in [81].

Finally, the Chevalley–Eilenberg coalgebra of a chiral Lie object satisfies factorization precisely when that Lie object is supported on the main diagonal [40, Theorems 5.1.1–5.2.1]. Here it is $\iota_* V_C$, so the hypothesis holds. The associative cobar assertion follows from the first paragraph. □

The Francis–Gaitsgory equivalence in the last step is between Lie and commutative coalgebra objects. The associative statement uses the envelope and its associative bar comparison in addition. It does not construct a second independently defined chiral field algebra.

Likewise, restriction to ordered words, pullback to a collision atlas, and transposition of coefficient modules are different operations. A transposed algebra needs a duality domain and the correct exchanged extensions, such as j_* and $j_!$. The generally infinite current coefficients are not automatically bounded holonomic objects. None of these operations turns a scalar supercharacter into a map of coefficient systems.

10 The character and the weight

The automorphic character is part of the denominator normalization. For $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \in \mathbf{Sp}_4(\mathbb{Z})$ and $j(g, Z) = \det(CZ + D)$, one has

$$\Delta_5(gZ) = \nu_{\Delta_5}(g)\, j(g, Z)^5 \Delta_5(Z), \qquad \nu_{\Delta_5}^2 = 1.$$

Consequently Δ_5^2 has weight ten and trivial character. The homogeneous orthogonal character $\widetilde{\nu}_{\Delta_5}$ is the one in Definition 6.2. Its central value at $-\mathbf{I}$ is -1.

LEMMA 10.1 *(Reflection values).* The character is -1 on each s_{δ_i} and $+1$ on each of the three type-I chamber reflections.

Proof. Their symplectic lifts g_β satisfy $\sigma(g_\beta) = -s_\beta$ and $j(g_\beta, Z) = -1$. Maass's character [84], in the explicit form [52, (1.5), Propositions 2.1–2.2], has values $+1$ on the type-II lifts and -1 on the type-I lifts. Multiplication by the homogeneous central value -1 gives the claimed reflection values. □

LEMMA 10.2 *(Linear action and slash action).* For $h = wa \in W \rtimes \mathrm{Aut}(\mathcal{P}_{II})$, one has

$$\widetilde{\nu}_{\Delta_5}|_W = \det, \qquad \widetilde{\nu}_{\Delta_5}|_{\mathrm{Aut}(\mathcal{P}_{II})} = 1, \qquad \Delta_5(hZ) = \det(w)\Delta_5(Z).$$

For a lift g_h obtained by multiplying the six reflection lifts, $j(g_h, Z) = \det(w)\det(a)$ and

$$(\Delta_5|_5 g_h)(Z) = \nu_{\Delta_5}(g_h)\Delta_5(Z) = \det(a)\Delta_5(Z).$$

Proof. The generator values prove the restrictions and linear action. The cocycle rule multiplies the constant factors -1 of the lifts, giving $j(g_h, Z) = \det(h)$. Dividing the linear action by its fifth power gives the slash action. Changing a lift by $-\mathbf{I}_4$ leaves these factors unchanged. □

Remark 10.3 *(Eight Coxeter characters).* Since the only Coxeter relations are $s_i^2 = 1$, a character $W \to \{\pm 1\}$ can choose each of its three signs independently. There are eight such characters. The reflection calculation selects $(-1, -1, -1)$, hence the determinant. Rescaling Δ_5 cannot alter these values. Identifying them with transport on a compact orientation line still requires that line and its equivariant comparison with the automorphic line.

10.1 Borcherds weight

The notation κ_{BKM} will mean the modular weight of a specified Borcherds product. It is distinct from a categorical trace, a current-algebra parameter or an Euler characteristic.

THEOREM 10.4 *(Weight of the denominator).* For an admissible Borcherds input with constant coefficient $c(0, 0)$, the product has weight $c(0, 0)/2$. In the present case,

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \frac{f(0, 0)}{2} = \frac{10}{2} = 5.$$

Proof. Apply [10, Theorem 13.3] to the weight-$-1/2$ Weil input obtained by theta decomposition of $\phi_{0,1}$ on $\Lambda^{3,2}$. Its constant coefficient is ten. The exceptional orthogonal identification transports this weight to the Siegel form, as in Chapter 5. □

For comparison, the cyclic CHL eta products at orders 1, 2, 3, 4, 6 have weights 12, 8, 6, 5, 4:

N	g_N	$\mathrm{wt}(g_N)$	$(\mathrm{wt}(g_N)-2)/2$
1	η^{24}	12	5
2	$\eta^8\eta(2\tau)^8$	8	3
3	$\eta^6\eta(3\tau)^6$	6	2
4	$\eta^4\eta(2\tau)^2\eta(4\tau)^4$	5	3/2
6	$\eta^2\eta(2\tau)^2\eta(3\tau)^2\eta(6\tau)^2$	4	1

Here unlabeled eta factors have argument τ. Multiplication by ϑ^2/η^6 subtracts two from the seed weight. The last column is its half-weight.

The lifting prescriptions and their characters are specified in [47, 48, 60]. For the corresponding normalized square-root Borcherds inputs, the weight formula reads $c_N(0) = (10, 6, 4, 3, 2)$ and gives precisely that column. Half-integral weight requires the appropriate cover.

The frame shapes also describe the permutation action on twenty-four coordinates [20]. The number of fixed coordinates is not this constant coefficient.

These cyclic orders do not index the eight pairs of the Gritsenko–Cléry catalog. In its input-pair order

$$(1,1), (2,1), (1,2), (3,1), (1,3), (4,1), (1,4), (2,2),$$

the weights are $(5, 2, 3, 1, 2, \frac{1}{2}, \frac{3}{2}, 1)$ and the Borcherds constants are $(10, 4, 6, 2, 4, 1, 3, 2)$ [50, Theorem 1.4]. Appendix 2 gives the exact cusp and parameter dictionary. The common $(1,1)$ member is the present Δ_5.

Remark 10.5 *(Different characteristics).* The numerical tuple $(0, 0, 3, 5, 24)$ can record proposed categorical and chiral characteristics together with the Cartan rank, Borcherds weight and K3 Euler number. The last three numbers have the present interpretations $\mathrm{rk}\, L = 3$, $\mathrm{wt}\,\Delta_5 = 5$ and $\chi_{\mathrm{top}}(K3) = 24$. A categorical or chiral trace in either first slot requires its own object and trace definition. The Hodge Euler characteristic of $K3 \times E$ is zero because $\chi(\mathcal{O}_E) = 0$, while $\chi(\mathcal{O}_{K3}) = 2$. Reading a proposed additive formula through these Euler numbers gives

$$5 \neq 0 + 2.$$

The different numerical equality $3 + 2 = 5$ uses the Cartan rank. It cannot supply the varying CHL weights 5, 3, 2, 3/2, 1 from a fixed rank-three lattice and the same K3 Euler number. Thus neither equality constructs an additive identity among these independently defined characteristics.

Remark 10.6 *(Two inputs to the weight formula).* The present input has constant ten and weight five. The fake-monster denominator on the Lorentzian lattice $\mathrm{II}_{25,1}$ comes from η^{-24}, whose constant coefficient is twenty-four, and has weight twelve [8, 9]. Its orthogonal automorphic product lives on the corresponding signature-$(26, 2)$ extension. The values five and twelve are evaluations of the same weight rule on different lattices and different inputs.

11 The compact comparison

Theorem 11.1 *(Denominator algebra).* The specified Borcherds presentation has Cartan space $L \otimes \mathbb{C}$, locally finite positive and negative root spaces, an invariant nondegenerate form and a triangular decomposition. Its superdenominator and weight are

$$\mathrm{den}(\mathfrak{g}) = 64^{-1}\Delta_5(2Z), \qquad \kappa_{\mathrm{BKM}}(\Delta_5) = 5,$$

and its signed root dimensions are $f(nm, l)$ under α. Its two parity dimensions are determined by (4.9).

Proof. The presentation and invariant-form results give the first assertions. The two denominator identities and the weight theorem give the remaining formulas. □

COROLLARY 11.2 *(Automorphic and theta realizations).* The root superdenominator is the normalized doubled Igusa form. That form is also obtained from the multiplicative singular theta lift of the Weil component of $\phi_{0,1}$ on $\Lambda^{3,2} \simeq U \oplus U \oplus \langle 2 \rangle$, with the normalization of Theorem 2.4.

Proof. Combine the denominator theorem with the singular-lift description [10, Theorem 13.3] and the exceptional exterior-square map of Section 2.1. □

Remark 11.3 *(Arithmetic consequences).* The polar coefficient $f(0, 1) = 1$ determines the real reflective divisor and $f(0, 0) = 10$ determines the weight. The remaining Jacobi coefficients give the signed root multiplicities and the theta-lift product. These arithmetic identities hold at every degree by the lift theorems. A finite coefficient table illustrates them. It cannot establish a compact protected index or identify its operations.

A compact protected Pfaffian asks for additional geometric data. Chapter 6 specifies a finite Dirac family, its reduced orientation and orientation-compatible transition maps. It also requires a coherent wall atlas and a finite geometric Pfaffian datum. The relevant hypotheses are (Do), $(O1)$, $(O1)^+$, $(O2_{\mathrm{atlas}})$ and (P_{fin}) there. Their comparison aims at

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5, \qquad \epsilon_o|_W = \widetilde{\nu}_{\Delta_5}|_W.$$

These are conditional geometric identities with specified line trivializations and normalization. They are separate from the unconditional abstract denominator theorem. A mirror discriminant comparison remains conjectural as formulated in Chapter 15.

Remark 11.4 *(Boundary algebra and center).* Suppose a factorization category $\mathcal{C}$ and a chosen boundary object b have been constructed. Its boundary algebra is $A_b = R\,\mathrm{End}_{\mathcal{C}}(b)$ in the specified coefficient recipient. A proposed bulk description by its derived center requires the bulk-to-center map, its operation compatibility and its reconstruction hypotheses. An identification with Hochschild cochains uses the corresponding enrichment and representability conditions [81, 82]. Where a chiral Swiss-cheese brace action is defined, the comparison must intertwine its mixed bulk-boundary operations. A gravitational interpretation additionally requires line observables and their operator product.

The root algebra $\mathfrak{g}$ is not, by its denominator identity, that center. Nor may a bar coalgebra C_X be substituted for the associative boundary algebra in a center expression without a proved comparison in the relevant category.

For a supplied compact coalgebra in $\mathcal{R}_E^+$, the current construction provides a precise possible comparison target,

$$C_X \longrightarrow B\mathcal{A}_{\Delta,E}, \qquad \Omega C_X \longrightarrow \mathcal{A}_{\Delta,E}.$$

A constructible Hall system must first be realized as such a right differential-module family. The morphism must preserve the collision operations, the coaugmentation and the transition maps. Its cone must be acyclic before it is called an equivalence. A word-cut coalgebra gives only its specified ordered restriction. The missing comparison with full collision diagrams remains part of the compact problem.

The local current vacuum $\omega_E \to V_E$, empty convolution unit K, augmentation to K, and prepared physical state have four different roles. A protected operator trace requires a state or module, its action and its trace functional. Neither the current construction nor the scalar product chooses that state. The compact recognition problem requires these coefficient and operation maps together with the invariant Lie comparison. Their completion must support all the stated operations. The algebraic constructions of this chapter fix their codomains and their first nontrivial root tests.

Chapter 5

The singular theta lift

1 From Jacobi coefficients to an automorphic section

The product in Chapter 3 gives a holomorphic section from the coefficients of $\phi_{0,1}$. The singular theta lift explains why the product is automorphic. Its integral first produces a real function on an orthogonal symmetric domain. This function determines the norm of a meromorphic section. A monic cusp normalization fixes its phase and identifies the section with D_5.

Three normalizations will enter. The positive leading coefficient fixes D_5. The specified theta constants give $\Delta_5 = 64D_5$. The root-denominator character convention gives $\mathrm{den}(\mathfrak{g}_{\Delta_5})(Z) = D_5(2Z)$. We derive the lattice maps and metric factors that relate these expressions.

1.1 The Jacobi form

Definition 1.1 *(Weak Jacobi form).* Fix integers k and $t > 0$. A weak Jacobi form of weight k and index t is a holomorphic function $\phi : \mathbb{H}_1 \times \mathbb{C} \to \mathbb{C}$ satisfying

$$\phi\left(\frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d}\right) = (c\tau + d)^k e^{2\pi i t c z^2/(c\tau+d)} \phi(\tau, z)$$

for $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathbf{SL}_2(\mathbb{Z})$, and

$$\phi(\tau, z + \lambda\tau + \mu) = e^{-2\pi i t(\lambda^2\tau + 2\lambda z)} \phi(\tau, z) \quad (\lambda, \mu \in \mathbb{Z}).$$

At the cusp its Fourier expansion has the form $\sum_{n \geq 0, \ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell$.

The elliptic law identifies coefficients with the same $(4tn - \ell^2, \ell \bmod 2t)$. At index one the residue is determined by the discriminant. The normalization $r^{-1} + 10 + r + O(q)$ singles out $\phi_{0,1}$.

Lemma 1.2 *(Theta-quotient presentation).* With the genus-one theta functions fixed in Chapter 3,

$$\phi_{0,1}(\tau, z) = 4\left(\frac{\theta_2(\tau, z)^2}{\theta_2(\tau, 0)^2} + \frac{\theta_3(\tau, z)^2}{\theta_3(\tau, 0)^2} + \frac{\theta_4(\tau, z)^2}{\theta_4(\tau, 0)^2}\right).$$

Its coefficients are those in (3.1), and $f(n, \ell) = 0$ for $4n - \ell^2 < -1$. Moreover $\phi_{0,1}(\tau, 0) = 12$.

Proof. The modular transformations permute the three even theta quotients and give their common index-one exponential factor. The elliptic transformations give the second law in the definition. Their q^0 sum is $r^{-1} + 10 + r$. The one-dimensionality of $J^{\mathrm{weak}}_{0,1}$ therefore identifies the sum [39]. Coefficient recurrence gives the support bound, and setting $z = 0$ gives $4(1 + 1 + 1) = 12$. The expansion in Appendix 2 computes the displayed Fourier table directly. □

The classes of discriminants 0, 4, −1, respectively, give

$$f(0, 0) = f(1, \pm 2) = 10, \quad f(1, 0) = f(2, \pm 2) = 108, \quad f(0, \pm 1) = f(2, \pm 3) = 1.$$

They are coefficient identities of the Jacobi form. The factor 64 in the genus-two theta product is a separate normalization calculation.

2 The orthogonal domain

The lift uses a rank-five lattice. Its explicit construction also determines the relation between the symplectic and orthogonal automorphic lines.

Definition 2.1 *(The exterior-square lattice).* Let $\Lambda^4 = \bigoplus_{i=1}^4 \mathbb{Z} e_i$, with volume $\mathrm{vol} = e_1 \wedge e_2 \wedge e_3 \wedge e_4$. On $\wedge^2 \Lambda^4$ define $u \wedge v = (u, v)\mathrm{vol}$. Put

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4, \qquad L = \Lambda^{3,2} = q_0^{\perp}.$$

The pairing on the exterior square is even and unimodular of signature $(3, 3)$, and $q_0^2 = -2$. An integral basis of its orthogonal complement is

$$\begin{aligned} f_1 &= e_1 \wedge e_2, \quad f_2 = e_2 \wedge e_3, \quad f_3 = e_1 \wedge e_3 - e_2 \wedge e_4, \\ f_{-2} &= e_4 \wedge e_1, \quad f_{-1} = e_4 \wedge e_3. \end{aligned}$$

Indeed the orthogonality equation sets the coefficients of $e_1 \wedge e_3$ and $e_2 \wedge e_4$ to opposite values. The remaining four coefficients are arbitrary integers.

In this basis the Gram matrix is

$$G_L = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus

$$L = \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2], \qquad \Lambda^{(1,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad \det G_L = 2.$$

Its discriminant group is $\mathbb{Z}/2\mathbb{Z}$, generated by $f_3/2 + L$, with quadratic value $1/4 \bmod \mathbb{Z}$.

The type-IV domain consists of the negative null lines

$$\mathbb{H}^{\mathrm{IV}} = \{[w] \in \mathbb{P}(L \otimes \mathbb{C}) : (w, w) = 0, \quad (w, \overline{w}) < 0\}.$$

Choose the component containing the vectors

$$w(Z) = (z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1), \qquad Z \in \mathbb{H}_2,$$

written in the f-basis. Direct pairing gives

$$(w(Z), w(Z))_L = 0, \qquad (w(Z), \overline{w(Z)})_L = -4 \det(\mathrm{Im} Z).$$

The negative real plane spanned by the real and imaginary parts of $w(Z)$, with this order, determines its orientation.

2.1 *The exceptional isomorphism*

The stabilizer of q_0 in $\mathbf{SL}_4$ is the symplectic group for

$$J_4(x, y)\,\mathrm{vol} = -x \wedge y \wedge q_0.$$

It preserves L. Let $\Gamma_{3,2}$ be the projective exterior-square image of $\mathbf{Sp}_4(\mathbb{Z})$ in $\mathrm{PO}^+(L)$, the group preserving the chosen domain component. Since L has odd rank, each projective isometry has a unique representative of determinant one. We use that representative when acting on the cone. The integral determinant-one isometries preserving this component form $\mathrm{SO}^+(L)$.

Theorem 2.2 *(Exterior-square exceptional isomorphism).* The exterior-square action induces

$$\mathbf{Sp}_4(\mathbb{Z})/\{\pm I_4\} \simeq \Gamma_{3,2} \simeq \mathrm{SO}^+(L).$$

The map $Z \mapsto [w(Z)]$ identifies $\mathbb{H}_2$ with the chosen component $\mathbb{H}_+^{\mathrm{IV}}$. For $g \in \mathbf{Sp}_4(\mathbb{R})$, its homogeneous lifts satisfy

$$(\wedge^2 g)w(Z) = j(g, Z)\, w(gZ), \qquad j(g, Z) = \det(CZ + D).$$

Proof. The exterior-square pairing is $\mathbf{SL}_4$-invariant. The stabilizer description therefore gives the homomorphism. Its kernel on $q_0^{\perp}$ is $\{\pm I_4\}$. The integral image is $\mathrm{SO}^+(L)$ by Gritsenko–Nikulin's integral identification [52, Lemma 1.1].

For the action on the domain, let $Q_1(Z), Q_2(Z)$ be the columns of $\left(\begin{smallmatrix} Z \\ I_2 \end{smallmatrix}\right)$. Their exterior product satisfies $w(Z) = -Q_1(Z) \wedge Q_2(Z)$. The matrix identity

$$g\begin{pmatrix} Z \\ I_2 \end{pmatrix} = \begin{pmatrix} gZ \\ I_2 \end{pmatrix}(CZ + D)$$

proves the homogeneous transformation formula.

The null equation and negativity give $\det(\mathrm{Im} Z) > 0$. The selected component imposes $\mathrm{Im} z_1 > 0$, hence $\mathrm{Im} Z > 0$. Conversely each negative null line in this component has the displayed affine representative. Its last coordinate cannot vanish: the orthogonal complement of an isotropic real vector has negative index one and contains no negative definite two-plane. This proves the domain identification. □

The corresponding real-group isomorphisms are [69, Chapter VI, Problems 25–27]

$$\mathbf{PSp}_4(\mathbb{R}) \simeq \mathrm{SO}_0(3, 2), \qquad \mathbf{Sp}_4(\mathbb{R}) \simeq \mathbf{Spin}(3, 2).$$

Both groups give the same symmetric domain, with maximal compact groups $\mathbf{U}(2)$ and $\mathrm{SO}(3) \times \mathrm{SO}(2)$ related by the spin covering. This covering concerns the output group. The metaplectic group $\mathbf{Mp}_2(\mathbb{Z})$ below acts on the one-dimensional input upper half-plane.

A homogeneous function Ψ of degree $-k$ on the null cone defines $F(Z) = \Psi(w(Z))$. If $\Psi((\wedge^2 g)w) = \chi(g)\Psi(w)$, the theorem gives

$$F(gZ) = \chi(g)\, j(g, Z)^k F(Z).$$

This proves the identification of the two automorphic lines, including their characters.

3 The Weil input

Borcherds's integral uses a lattice of signature $(2,3)$. Set $M = L(-1)$: its underlying abelian group is L, and its pairing is the negative of the one above. Write

$$Q_M(v) = \frac{(v,v)_M}{2}, \qquad A_M = M^*/M = \{0,\gamma\}, \qquad Q_M(\gamma) = -\frac{1}{4} \bmod \mathbb{Z}.$$

The standard basis of $\mathbb{C}[A_M]$ is e_0, e_γ.

For $e(t) = e^{2\pi i t}$, the Weil representation ρ_M of $\mathbf{Mp}_2(\mathbb{Z})$ is given on the standard lifts by

$$\rho_M(T) = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}, \qquad \rho_M(S) = \frac{e(1/8)}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Here

$$T = \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, 1 \right), \qquad S = \left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \sqrt{\tau} \right), \quad 0 < \arg\sqrt{\tau} < \frac{\pi}{2}.$$

The hyperbolic summands are unimodular, so these are the discriminant matrices of the rank-one lattice $[-2]$. In particular $\rho_M = \rho_L^\vee$.

LEMMA 3.1 *(Theta decomposition).* For

$$\theta_{1,\mu}(\tau,z) = \sum_{\ell \equiv \mu \ (2)} q^{\ell^2/4} r^\ell, \qquad \mu = 0,1,$$

there are unique functions h_0, h_1 such that

$$\phi_{0,1} = h_0\theta_{1,0} + h_1\theta_{1,1}, \qquad [q^{n-\ell^2/4}]h_{\ell \bmod 2} = f(n,\ell).$$

The vector $h = h_0 e_0 + h_1 e_\gamma$, also denoted $\phi_{0,1}^{\mathrm{Weil}}$, is a weakly holomorphic modular form of weight $-\frac{1}{2}$ and representation ρ_M. Its first terms are

$$h_0 = 10 + 108q + 808q^2 + O(q^3), \qquad h_1 = q^{-1/4}(1 - 64q - 513q^2 + O(q^3)).$$

Proof. The elliptic coefficient recurrence identifies all exponents with the same discriminant and residue. Grouping the even and odd powers gives the decomposition and uniqueness. Poisson summation gives the theta vector weight $\frac{1}{2}$ and Weil representation ρ_L. Since its contraction with h has weight zero and trivial representation, h has the dual representation and weight $-\frac{1}{2}$ [39, Theorem 5.1]. For example, $\theta_{1,1}(\tau+1,z) = i\theta_{1,1}(\tau,z)$, forcing $h_1(\tau+1) = -ih_1(\tau)$. The coefficient table proves the displayed expansions. □

The support bound has become a simple principal part: the only negative-index coefficient of h is $q^{-1/4}e_\gamma$, with coefficient 1. The constant coefficient of the zero component is 10. These two numbers will determine the divisor and the weight.

4 The regularized integral

An oriented negative two-plane of L is an oriented positive two-plane of M. For such a plane P, decompose $\lambda = \lambda_+ + \lambda_-$ by orthogonal projection to P and $P^\perp$. The second summand has negative definite pairing.

Definition 4.1 *(Siegel theta series).* For $\tau = u + iv \in \mathbb{H}_1$, define

$$\Theta_M(\tau, P) = \sum_{\delta \in A_M} e_\delta \sum_{\lambda \in M + \delta} \exp\bigl(\pi i \tau (\lambda_+, \lambda_+)_M + \pi i \overline{\tau} (\lambda_-, \lambda_-)_M\bigr).$$

Each summand has absolute value

$$\exp\bigl[-\pi v \bigl((\lambda_+, \lambda_+)_M - (\lambda_-, \lambda_-)_M\bigr)\bigr].$$

The expression in parentheses is positive definite. This proves convergence of the kernel in every lattice direction. Its modular weight is $(1, \frac{3}{2})$ and its representation is ρ_M.

Let $F = \sum_\delta F_\delta e_\delta$ be weakly holomorphic of weight $-\frac{1}{2}$ for ρ_M. Write

$$F_\delta(\tau) = \sum_{a \in \mathbb{Q}} c_\delta(a) q^a, \qquad c_\delta(a) = 0 \text{ for } a \ll 0,$$

and suppose $c_\delta(a) \in \mathbb{Z}$ for $a \le 0$. Use the Hermitian contraction

$$\langle F, \Theta_M \rangle = \sum_\delta F_\delta \, \overline{\Theta_{M,\delta}}.$$

Its weight is $(1, 1)$, since $(-\frac{1}{2}, 0) + (\frac{3}{2}, 1) = (1, 1)$. Multiplication by v makes it invariant.

For the usual modular fundamental domain truncated at $v = T$, denoted $\mathcal{F}_T$, set

$$\Phi_M(P, F) = \mathrm{CT}_{s=0}\left[\lim_{T \to \infty} \int_{\mathcal{F}_T} v \langle F(\tau), \Theta_M(\tau, P) \rangle v^{-s} \frac{du\, dv}{v^2}\right].$$

The limit is taken first in the convergent half-plane and then continued meromorphically in s. The constant Laurent coefficient is Borcherds's regularization [10, §6]. The resulting function is evaluated away from its rational quadratic singularities. In particular, the conjugated kernel and the power of v are part of the definition.

To state the norm identity, write $w = X_M + iY_M$ on the positive null cone of M, and put

$$P(w) = \mathrm{span}_{\mathbb{R}}(X_M, Y_M), \qquad |Y_M| = \sqrt{(Y_M, Y_M)_M}, \qquad C_B = \frac{\Gamma'(1)}{2} + \log \sqrt{2\pi}.$$

The null equation says that X_M and Y_M are orthogonal with equal positive norm.

Theorem 4.2 *(Singular theta lift* [10, Theorem 13.3]*).* For the input F above, put $k = c_0(0)/2$. There is a meromorphic homogeneous function $\Psi_M(w, F)$ of degree $-k$, with the following properties.

(1) It is an automorphic form of weight k, with a unitary character, for the component-preserving isometries of M that preserve F.

(2) Its order along the divisor orthogonal to a rational negative line $\mathbb{R}\lambda$ is

$$\mathrm{ord}_{\lambda^\perp} \Psi_M = \sum_{\substack{x > 0 \\ x\lambda \in M^*}} c_{x\lambda + M}\bigl(x^2 Q_M(\lambda)\bigr).$$

Each divisor occurs once in this formula, and all contributing positive multiples are included. Non-negative orders imply holomorphy in this rank-five case.

(3) Its norm satisfies

$$\log|\Psi_M(w,F)| = -\frac{1}{4}\Phi_M(P(w),F) - k\big(\log|Y_M| + C_B\big).$$

Equivalently, for $\|\Psi_M\| = |\Psi_M(w,F)|\,|Y_M|^k$,

$$\Phi_M(P,F) = -4\log\|\Psi_M\| - 4kC_B.$$

(4) Choose a primitive isotropic z in a unimodular hyperbolic summand and an isotropic $z' \in M$ with $(z,z')_M = 1$. Let $K = (M \cap z^\perp)/\mathbb{Z}z$, identified with the orthogonal complement of their span. Thus K has signature $(1,2)$ and $A_K = A_M$. In the cusp section $w = z' + \zeta - Q_M(\zeta)z$, the product is

$$\Psi_M(w,F) = C\,e\big((\rho_{\mathrm{Bor}},\zeta)_M\big) \prod_{\substack{\lambda\in K^*\\(\lambda,W)_M>0}} \big(1 - e\big((\lambda,\zeta)_M\big)\big)^{c_{\lambda+K}(Q_M(\lambda))}.$$

Here W is a Weyl chamber, ρ_{Bor} its Weyl vector, and $|C| = 1$. The product converges sufficiently near the cusp with $\mathrm{Im}\zeta \in W$.

The assertion about $|C|$ uses the primitive unimodular hyperbolic summand in the statement. For a general cusp, the product constant also contains the finite factors in [10, Theorem 13.3(5)]. The integral fixes an absolute norm. At the present cusp we choose the remaining phase so that $C = 1$, and denote the resulting product by $\mathcal{B}_M(F)$.

THEOREM 4.3 *(Dependence on the input).* In a common cusp chamber, the monic products satisfy

$$\mathcal{B}_M(F_1 + F_2) = \mathcal{B}_M(F_1)\mathcal{B}_M(F_2).$$

Their weights and divisor orders add, and their characters multiply. The zero input gives the constant function 1.

Proof. The integral is linear in F. In the cusp product each exponent is a Fourier coefficient of F, and the Weyl vector depends linearly on this input. The normalized products therefore multiply factor by factor. Continuation proves the equality of meromorphic sections. The weight and divisor formulas give the remaining assertions. □

The target thus varies with F. A general multiplicative lift takes values in meromorphic sections of different weights and characters. The form of weight five arises from the particular vector h.

5 The Igusa specialization

Choose $z = f_1$, $z' = f_{-1}$ in M. Then

$$K = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}, \qquad \zeta = z_3 f_2 + z_2 f_3 + z_1 f_{-2}.$$

For $(n,\ell,m) \in \mathbb{Z}^3$, define

$$\lambda(n,\ell,m) = nf_2 - \frac{\ell}{2}f_3 + mf_{-2} \in K^*.$$

The coordinate and quadratic identities are

$$(\lambda,\zeta)_M = nz_1 + \ell z_2 + mz_3, \qquad Q_M(\lambda) = nm - \frac{\ell^2}{4}.$$

The class of λ in A_K is $\ell \bmod 2$. The theta decomposition therefore identifies the exponent:

$$c_{\lambda+K}(Q_M(\lambda)) = f(nm,\ell).$$

In this formula the zero coefficients allow n, m to range over all integers. A nonzero coefficient has $4nm - \ell^2 \geq -1$. In the chamber $y_1 + y_2 > 0$, $y_3 + y_2 > 0$, $y_2 < 0$, the positive covectors on this support are precisely the $\lambda(n,\ell,m)$ with $(n,\ell,m) \in \Gamma_{\mathrm{act}}$. Indeed $nm < 0$ is excluded by the support bound. For $n, m > 0$, the positive-cone calculation of Chapter 3 gives the claimed sign. The case $n, m < 0$ has the opposite sign.

If $nm = 0$, the allowed elliptic exponents are $0, \pm 1$, and the same inequalities decide the sign directly.

5.1 *The two Weyl-vector coordinates*

Theorem 5.1 *(Weyl vector of the lift).* The cusp exponents of $\mathcal{B}_M(h)$ are $(A,B,C) = (\frac{1}{2},\frac{1}{2},\frac{1}{2})$. In the $e((\lambda,\zeta)_M)$ convention of the integral,

$$\rho_{\mathrm{Bor}} = \frac{1}{2}f_2 - \frac{1}{4}f_3 + \frac{1}{2}f_{-2}.$$

In the root convention $e^{-\pi i(\alpha,\zeta)_L}$, the vector is

$$\rho = 2\rho_{\mathrm{Bor}} = f_2 - \frac{1}{2}f_3 + f_{-2} = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3).$$

It satisfies $(\rho,\delta_i)_L = -1$ for $i = 1, 2, 3$.

Proof. The Jacobi product theorem [54, Theorem 2.1] gives

$$A = \frac{1}{24}\sum_{\ell} f(0,\ell), \qquad B = \frac{1}{2}\sum_{\ell>0} \ell f(0,\ell), \qquad C = \frac{1}{4}\sum_{\ell} \ell^2 f(0,\ell).$$

The row $r^{-1} + 10 + r$ makes all three numbers $1/2$. The vector $\lambda(A,B,C)$ is ρ_{Bor}. Since $\alpha(n,\ell,m) = 2\lambda(n,\ell,m)$ and $(\ ,\ )_M = -(\ ,\ )_L$, the root vector is twice this covector.

The simple-root Gram matrix has diagonal entries 2 and off-diagonal entries -2. Its row sums are -2, proving $(\rho,\delta_i)_L = -1$. Its eigenvalues are $-2, 4, 4$, so its determinant is -32. The simple roots form an integral basis of M_{II}, and these integral pairings show $\rho \in M_{II}^*$. □

The same half-integral exponent vector is therefore represented by different vectors in the two exponential conventions. The factor of two belongs to the pairing formula, before the separate substitution $Z \mapsto 2Z$ in the denominator.

Theorem 5.2 *(The Igusa form as multiplicative lift).* The phase-normalized multiplicative lift of h is D_5:

$$\mathcal{B}_M(h)(w(Z)) = D_5(Z).$$

In the fixed theta convention define $\Theta(h) = \Theta^{\mathrm{Bor}}_{\mathrm{sing}}(h) := 64\,\mathcal{B}_M(h)$. Then

$$\Theta(h)(w(Z)) = \Delta_5(Z).$$

The notation $\Theta(h)$ denotes this multiplicative section, whereas $\Phi_M(P,h)$ denotes the regularized additive function.

Proof. The cusp covector formula identifies each product exponent. The Weyl-vector theorem identifies the prefactor $(qrs)^{1/2}$. Both products have leading coefficient 1, so Theorem 5.1 identifies the monic lift with D_5 on the convergence region. Continuation gives the global equality. The theta coefficient then gives the second equality. □

The divisor formula can also be read directly from the input. Its only negative coefficient is $c_\gamma(-1/4) = 1$. Thus a divisor is supported on a vector $\lambda \in M^*$ with $Q_M(\lambda) = -1/4$. Every smaller positive multiple has quadratic value strictly between $-1/4$ and 0, impossible in $Q_M(M^*) \subset \frac{1}{4}\mathbb{Z}$. Larger multiples have zero negative coefficient. Each supported divisor has order one.

Equivalently, $\delta = 2\lambda \in L$ has norm 2 and divisibility two. A type-I norm-two vector has divisibility one, so its half does not lie in the dual lattice and its divisor has order zero. The supported divisors are the translates of the diagonal identified by the Igusa product. The absence of poles, the weight 5, and the cusp form identity agree with the general lift theorem.

5.2 *The Petersson norm in Siegel coordinates*

The cusp section in the theorem is exactly $w(Z)$, since

$$Q_M(\zeta) = z_1 z_3 - z_2^2, \qquad (w(Z), f_1)_M = 1.$$

Its imaginary part has last coordinate zero, and hence

$$(\mathrm{Im} w(Z), \mathrm{Im} w(Z))_M = 2y_1y_3 - 2y_2^2 = 2\det(\mathrm{Im} Z).$$

The norm formula becomes

$$\Phi_M(P(Z), h) = -4\log\Big(|D_5(Z)|[2\det(\mathrm{Im} Z)]^{5/2}\Big) - 20C_B.$$

For the theta-normalized form it is

$$\Phi_M(P(Z), h) = -4\log\Big(|\Delta_5(Z)|[2\det(\mathrm{Im} Z)]^{5/2}\Big) + 4\log 64 - 20C_B.$$

The additive constant therefore records both the chosen regularization and the theta normalization. A formula for the norm does not determine the phase of the section.

5.3 *The character*

THEOREM 5.3 *(Multiplier system of the lift).* On $\mathbf{Sp}_4(\mathbb{Z})$, the lift has Maass character ν_5. Its Jacobi-parabolic restriction is $v_\eta^{12} \times v_H$. For the linear tube action, the character is $\varepsilon_5|_{W^{(2)}(M_{II})} = \det$ and $\varepsilon_5|_{S_3} = 1$. It is the restriction of the homogeneous orthogonal character $\widetilde{\nu}_5$ of Definition 6.2. These characters are related on the chosen symplectic reflection lifts by

$$\varepsilon_5(s_\beta) = \widetilde{\nu}_5(s_\beta) = -\nu_5(g_\beta).$$

Proof. Equality with D_5 gives the Maass character. The first Fourier–Jacobi coefficient is $b = \eta^9\vartheta$. The odd theta function has multiplier $v_\eta^3 \times v_H$, giving the asserted parabolic restriction [50, (13)]. The six reflection matrices and their weight factors are computed in Lemma 6.3. □

The finite Weil representation and the automorphic character enter at different stages. The former acts on the two input components. The latter describes the output line. Borcherds's theorem supplies a unitary character for each admissible input. Here its identification with the Maass character follows from the product equality, including its phase.

The rank-three lattice $M_{\mathrm{o}} = \Lambda^{(1,1)} \oplus [2]$ is a primitive summand of L, with orthogonal complement the second hyperbolic plane. The BKM grading lattice M_{II} has index four in M_{o}. Thus L has rank five and discriminant two, while M_{II} has rank three and determinant -32. The decomposition $L \simeq \mathrm{II}_{2,2} \oplus [2]$ is another way to write the two hyperbolic planes and the rank-one summand. It does not identify L with a unimodular lattice.

6 The denominator and the weight

Theorem 6.1 *(Lift–denominator compatibility).* For the Gritsenko–Nikulin algebra $\mathfrak{g}_{\Delta_5}$ on M_{II},

$$\operatorname{den}(\mathfrak{g}_{\Delta_5})(Z) = D_5(2Z) = e^{-2\pi i(\rho,\zeta)_L} \prod_{\alpha \in \mathcal{R}_+} (1 - e^{-2\pi i(\alpha,\zeta)_L})^{\operatorname{smult}(\alpha)}.$$

Here $\mathcal{R}_+$ is the positive root set, and

$$\operatorname{smult}(\alpha(n,\ell,m)) = f(nm,\ell) \quad \text{on } \Gamma_{\mathrm{act}}.$$

Its negative-norm simple-root coefficients $m(a)$ and isotropic simple-root coefficients $\tau(a)$ are those in the Gritsenko–Nikulin presentation.

Proof. The algebra and its denominator theorem are constructed in [52, §§2–4 and §6]. The equality of exponents follows from the cusp covector calculation. The root convention has $q^n r^\ell s^m = e^{-\pi i(\alpha,\zeta)_L}$. Doubling Z changes this to the denominator character and doubles the Weyl exponent in the same way. This gives $D_5(2Z)$. □

6.1 Agreement of the three constructions

Theorem 6.2 *(Automorphic, root and theta identities).* After the exceptional domain identification,

$$\mathcal{D}_X(Z) = \Theta(h)(w(Z)) = 64 \operatorname{den}(\mathfrak{g}_{\Delta_5})(Z/2) = \Delta_5(Z).$$

All three expressions define the same section of $\mathcal{L}^5 \otimes \nu_5$.

Proof. The first equality uses the explicitly normalized lift. The second is the denominator rescaling just proved. The automorphic line identification follows from $(\wedge^2 g)w(Z) = j(g,Z)w(gZ)$. □

The equality $\mathcal{D}_X = \Delta_5$ is the notation for the automorphic target. The substantive comparisons are the multiplicative-lift theorem and the denominator theorem. Neither changes the input category of the other construction.

6.2 The weight coordinate

Theorem 6.3 *(The weight identity at index one).* For $h = \phi^{\mathrm{Weil}}_{0,1}$,

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \operatorname{wt} \mathcal{B}_M(h) = \frac{c_0(0)}{2} = \frac{f(0,0)}{2} = 5.$$

Proof. The zero component of the theta decomposition has constant term 10. Apply the weight formula of Theorem 4.2. □

The eight-form catalog uses the same weight principle. At congruence level, its vector-valued zero coefficient is assembled from the Jacobi expansions at every cusp, as specified in Appendix 2. The pair (t, N) must therefore accompany such a constant. It cannot be replaced by a CHL order without identifying the actual input.

The number 5 is independent of $e(S) = 24$, $\chi(\mathcal{O}_S) = 2$, and the rank 3 of the root-degree lattice. An equality $3 + 2 = 5$ alone constructs no comparison of the associated operations. Replacing the lattice rank by a chiral or Heisenberg invariant requires the normalized bar, trace and comparison data of Chapter 1.

7 GEOMETRIC COMPARISON DATA

The primitive input of this construction is h, together with the lattice, Weil representation and cusp. The regularized integral determines a norm. The multiplicative lift supplies a section and its automorphy, and the cusp choice fixes the phase. These passages have distinct mathematical content.

COROLLARY 7.1 *(The automorphic part of the realization problem).* The automorphic section, the rescaled BKM denominator and the normalized multiplicative theta lift coincide as in Theorem 6.2. Their comparison with a compact Pfaffian additionally requires a geometric parameter map, a descended line isomorphism and a compatible section. The mirror-discriminant comparison requires its period map and divisor identification.

Proof. The identity is the preceding theorem. The additional comparisons have their source lines and bases in the compact and mirror constructions, respectively. They require the displayed maps before equality of sections can be formulated. □

In the sequence of Chapter 1, the theta integral is an operation on primitive inputs. The section Δ_5 is its normalized multiplicative value on h. The BKM algebra retains its bracket and root spaces, which its denominator supercharacter forgets. The relative Pfaffian assertion of Appendix 7 retains its Do-HN, quotient-orientation, Weyl-transport, wall and finite Pfaffian hypotheses. A scalar trace comparison at the level of Definition 7.20 does not supply an operation-preserving Hall–Borcherds map. Such a map, with its state and reduction data, is still needed for the proposed compact operator or gravitational realization.

The output weight is integral in the present case. Half-integral weights in other members of the catalog require a square-root automorphy factor with its multiplier. Equivalently, one uses a suitable cover of the output group. This output choice is separate from the $\mathbf{Mp}_2(\mathbb{Z})$ action on the Weil input. The distinction preserves the character information needed for any subsequent geometric line comparison.

Part II

Formal geometry and Hall operations

Chapter 6

Formal geometry and the compact comparison

The denominator algebra supplies a target for a geometric Hall construction. To reach it, one must first construct coefficients on moduli of objects, maps on extension flags and compatible orientation data. The product of their signed dimensions is a consequence to compare, not a definition of those operations. This chapter begins with the formal deformation model, then constructs or specifies the maps needed by the compact problem.

Let S be a smooth projective K3 surface, E a smooth elliptic curve and $X = S \times E$. Fix the polarization and stability data, the elliptic origin and the support conditions of Appendix 8. Singular K3 limits, singular elliptic curves, imprimitive classes, the compact Hall cusp and multiple-component degenerations need their own coefficient and specialization maps. Results below are used on the stated smooth and bounded recipients.

1 The source and its finite comparisons

Three established reference constructions will recur: the normalized Borcherds product $D_5 = \Delta_5/64$, the denominator algebra $\mathfrak{g}_{\Delta_5}$ with denominator $D_5(2Z)$, and the regularized theta lift. A geometric Pfaffian, source coalgebra or compact state must be constructed independently before being compared with these targets.

At a stage R, the proposed geometric datum is

$$\mathfrak{D}^{\mathrm{DI}}_{X,R} = (\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P^{\Pi}_R, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}, \mathrm{Rec}_R).$$

The entries denote, respectively, a field complex when that refinement is required, the local and wrapped carrier, charges and their Gram map, critical coefficients and orientation, Hall operations and primitives, a source coalgebra and its associative comparison, a Pfaffian line and section, Weyl transport and primitive recognition. Their precise definitions and morphisms are developed below. A compatible cofiltered system of such stages is the Dirac–Igusa pro-object $\{\mathfrak{D}^{\mathrm{DI}}_{X,R}\}_R$. An effective geometric inverse limit is an additional construction.

Formal charges and geometric supports

The Mukai–Gram map Π_X is quadratic. Its polarization B defines an additive normal-ordered central lift $\overline{\Pi}_X$, as in Appendix 3. Pointwise lattice lifts of Gram triples and the root assignment

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$$

are explicit. They supply no effectivity or semistability statement for a compact sheaf. The raw zero fibre $K_{\Pi,0} = \Pi_X^{-1}(0, 0, 0)$ is not an additive subgroup, so it cannot be used as a quotient charge group.

After an actual additive regrading has been fixed, summing finite vector spaces over its fibres is exact and retains even and odd parts separately. It also preserves a supplied filtration and triangular decomposition. The tensor-coalgebra comparison uses that additive grading; the raw quadratic Gram map does not commute with tensor degrees. These linear constructions retain their precise scope in Proposition 1.173. They do not prove that a geometric Hall bracket, pairing radical or coefficient functor descends to the regraded source.

A finite target test window therefore starts from actual bounded geometric charges, their normal-ordering translates and the image identity in Definition 1.189. A geometric exhaustion must represent every required relation degree and its intermediate outputs. The first simple target triples are

$$\gamma(\delta_1) = (1, 1, 0), \qquad \gamma(\delta_2) = (0, 1, 1), \qquad \gamma(\delta_3) = (0, -1, 0).$$

The sheaf representatives of Proposition 7.9 realize these Gram values. Their wall semistability, reduced wall charts and operation comparisons are further geometric data.

The moduli and coefficient hypotheses

The Abramovich–Polishchuk heart and its Liu tilt retain the polynomial charge, weak stability function, torsion pair and central charge together. A finite subobject- and quotient-closed collection with finitely many isomorphism classes and a strict integer length has the noetherian and HN properties of Propositions 8.4 and 8.6. The maximal-phase, maximal-length construction bounds the number of HN factors by the initial length. If length is additive, their lengths sum to that length. This finite collection need not be additive or closed under all extensions.

Finite geometric stages require additional bounded class data, finite-type loci and the object and flag maps actually used. Semistability need not be closed under every specialization. The ambient CY3 moduli stack has a shifted symplectic structure; its scalar directions prevent an automatic two-term reduction on all substacks. The universal complex may require a scalar gerbe or a twisted recipient, even when its weight-zero endomorphism complex descends. Scalar rigidification does not force finite residual inertia. Closed strata, retained extensions, HN factors and geometric duality must be supplied at the support and heart where they are asserted.

These requirements constitute the finite-moduli and reduced-atlas hypotheses. They are not consequences of a finite charge list. The reduced critical object is denoted Φ_o: its orientation enters through the sign-torsor construction, not an ordinary tensor of a constructible complex with a coherent line. The common translation quotient, proper or compact-support maps, cosections, roots and full flag coherences belong to the same coefficient system.

If an E_3-field refinement is sought, its formal target charts, field complexes and operadic maps are additional input. A BV realization requires its quantum master equation, anomaly and framing conditions. The proposed $\mathrm{Sp}^{\mathrm{ch}}_{K_3,E}$ must be constructed on these complexes and preserve compact supports, factorization and stage transitions. A current-envelope target does not define this specialization functor.

1.1 *Formal sheaf models and elliptic support*

The spherical collection gives a local algebra of sheaf deformations. To obtain Hall operations, this algebra must describe extensions of several objects at once and retain their traces. We first construct that formal comparison. We then distinguish sheaves supported over finitely many points of E from sheaves whose support dominates E. Their extension diagrams can be formed together, and elliptic translation acts on the entire diagram.

One cyclic model for the spherical collection

Use the common-polynomial line bundles L_1, L_2, L_3 of Theorem 2.2. Fix a nonzero holomorphic form Ω_S. Their Dolbeault dg category has morphisms

$$\mathcal{D}(i,j) = \mathcal{A}^{0,*}(S, L_i^\vee \otimes L_j), \qquad d = \bar\partial, \qquad \langle a, b\rangle = \int_S \Omega_S \wedge \operatorname{tr}(ba).$$

The pairing has degree -2 and induces Serre duality. Set $A(i,j) = \operatorname{Ext}^*_S(L_i, L_j)$ and $\ell = \bigoplus_i \mathbb{C}1_i$. Thus $A^0 = \ell$, A is concentrated in degrees 0, 1, 2, and A^2 is dual to ℓ.

Proposition 1.1 *(Cyclic formality of the collection).* There is a strictly unital cyclic minimal model on A with only its Yoneda multiplication m_2 nonzero. One cyclic A_∞ quasi-equivalence $F : A \to \mathcal{D}$ can be chosen for the whole collection, before choosing multiplicities or flags.

Proof. The cyclic minimal-model and strictification results of [31, Lemmas 4.4–4.5, Proposition 4.8 and Corollary 4.9] apply to this simple-minded 2-Calabi–Yau collection. The nonnegative-degree hypothesis for strictification holds. They give the cyclic model with strict vertex units.

For $n \geq 3$, an input in $A^0 = \ell$ makes m_n vanish by strict unitality. If all inputs have positive degree and one has degree two, the output has degree at least three, since $|m_n| = 2 - n$. The remaining possibility has every input of degree one and output of degree two. Pair that output with a vertex unit. Cyclicity moves the unit into an input, making the value zero. Perfection of the pairing $A^2 \otimes A^0 \to \mathbb{C}$ forces the output to vanish. □

For multiplicity spaces V_i, write $F_{\mathbf{n}} = \bigoplus_i L_i \otimes V_i$ and $n_i = \dim V_i$. Matrix amplification replaces $A(i,j)$ by $A(i,j) \otimes \operatorname{Hom}(V_i, V_j)$, composes morphisms by matrix multiplication, and composes the cyclic trace with matrix trace. Write $A_{\mathbf{n}}$ for this matrix amplification. The group $G_{\mathbf{n}} = \prod_i GL(V_i)$ acts by changing bases.

Choose a quiver Q with two arrows from i to j for each $i < j$. Its double has dual variables $X_a : V_{t(a)} \to V_{h(a)}$ and $Y_a : V_{h(a)} \to V_{t(a)}$. With dual arrow bases chosen according to the Serre pairing, the surface Maurer–Cartan equation is

$$\mu_i(X,Y) = \sum_{h(a)=i} X_a Y_a - \sum_{t(a)=i} Y_a X_a = 0. \tag{6.1}$$

For a local Artinian base with maximal ideal $\mathfrak{m}$, the pointed deformation problem uses $\mathfrak{m}$-valued arrows and the formal gauge group $\exp(\mathfrak{g}_{\mathbf{n}} \otimes \mathfrak{m})$, where $\mathfrak{g}_{\mathbf{n}} = \bigoplus_i \operatorname{End}(V_i)$. The full affine quotient by $G_{\mathbf{n}}$ is a separate algebraic quiver model. Its comparison with sheaf moduli here is formal near the specified polystable object.

The normal operator and the derived critical quotient

Embed S as the fibre over $e \in E$ and choose a normal coordinate at e. The normal Koszul resolution adds an exterior generator θ of degree one. The formal threefold algebra is $A_{\mathbf{n}} \otimes \Lambda(\theta)$. A degree-one element $x + z\theta$ has square

$$(x + z\theta)^2 = x^2 + (xz - zx)\theta.$$

The first equation is (6.1); the second says that the normal endomorphisms commute with the arrows. Both are critical equations of

$$\begin{aligned}\mathcal{W}_{\mathbf{n}}(X,Y,Z) &= \sum_i \operatorname{Tr}(Z_i\mu_i)\\ &= \sum_{a\in Q_1} \operatorname{Tr}\big(Y_a(Z_{h(a)}X_a - X_aZ_{t(a)})\big).\end{aligned}$$

The potential is defined on $\mathcal{V}_{\mathbf{n}} = \operatorname{Rep}(\overline{Q},\mathbf{n}) \oplus \mathfrak{g}_{\mathbf{n}}$. Its equations are

$$\mu_i = 0, \qquad Z_{h(a)}X_a = X_aZ_{t(a)}, \qquad Y_aZ_{h(a)} = Z_{t(a)}Y_a.$$

Taking the derived critical locus on the smooth quotient $[\mathcal{V}_{\mathbf{n}}/G_{\mathbf{n}}]$ retains gauge symmetry. At a critical point its endomorphism-degree complex is

$$\mathfrak{g}_{\mathbf{n}} \xrightarrow{\rho} \mathcal{V}_{\mathbf{n}} \xrightarrow{\operatorname{Hess}\mathcal{W}_{\mathbf{n}}} \mathcal{V}_{\mathbf{n}}^* \xrightarrow{-\rho^*} \mathfrak{g}_{\mathbf{n}}^*. \tag{6.2}$$

The degrees are 0, 1, 2, 3; the tangent complex is its shift by [1]. Invariance of $\mathcal{W}$ gives $(\operatorname{Hess}\mathcal{W})\rho = 0$ and its dual identity. The outer terms record automorphisms and their duals. Omitting them changes the derived critical quotient.

For $N = \sum_i n_i > 0$, set

$$c = N^{-1}\sum_i \operatorname{Tr} Z_i, \qquad Z_i^{\circ} = Z_i - cI_{V_i}.$$

Since $\sum_i \operatorname{Tr}\mu_i = 0$, the potential is independent of c. Therefore

$$\operatorname{Crit}_{[\mathcal{V}_{\mathbf{n}}/G_{\mathbf{n}}]}(\mathcal{W}) \simeq T^*[-1]\mathbb{A}^1_c \times \operatorname{Crit}_{[\mathcal{V}_{\mathbf{n}}^{\circ}/G_{\mathbf{n}}]}(\mathcal{W}^{\circ}).$$

Quotienting translations of c removes its tangent direction; the centred critical model removes its dual obstruction direction as well. These are different operations. On a flag with block dimensions N_i, removing the total centre imposes $\sum_i N_ic_i = 0$ and leaves the relative centres. The zero object has $N = 0$ and requires its separate unit construction.

The ambient quotient has canonical line $\det(\mathcal{V}_{\mathbf{n}}^*) \otimes \det\mathfrak{g}_{\mathbf{n}}$. The opposite arrow spaces cancel in its determinant character, as do the loop and gauge terms. An ordering of the hyperbolic pairs gives a local trivialization. Matching this orientation with a prescribed compact orientation is a further comparison, even when the formal cyclic algebras have been identified.

Dimensional reduction and the point sector

Write $\varphi_f = \phi_f[-1]$ for shifted vanishing cycles, so that $\varphi_0\mathbb{Q} = \mathbb{Q}$. In the mixed Hodge version, $(-r)$ denotes a Tate twist. The potential above is linear in the $r = \dim\mathfrak{g}_{\mathbf{n}}$ loop coordinates. This gives a coefficient comparison after compactly supported pushforward.

PROPOSITION 1.2 *(Dimensional reduction of the tripled potential).* There is a natural comparison

$$R\Gamma_c(\mathcal{V}_{\mathbf{n}}, \varphi_{\mathcal{W}_{\mathbf{n}}}\mathbb{Q}) \simeq R\Gamma_c(\mu^{-1}(0),\mathbb{Q})[-2r](-r).$$

The comparison is $G_{\mathbf{n}}$-equivariant, interpreted through finite-dimensional approximations. On the centred model replace r by $r-1$ when $\sum_i n_i > 0$; the zero locus on the right remains $\mu^{-1}(0)$. The zero-dimensional representation is left unchanged.

Proof. For $f(x,z) = \sum_{i=1}^{r} z_i f_i(x)$, the dimensional-reduction theorem identifies $\pi_! \varphi_f \pi^* K$ with $\pi_! \pi^* i_* i^* K$, where i includes the common zero locus of the f_i and π is the rank-r vector-bundle projection [28, Theorem A.1 and Corollary A.9]. Apply it to the moment-map coordinates and $K = \mathbb{Q}$. Compactly supported cohomology of the affine fibre is $\mathbb{Q}[-2r](-r)$, giving the formula. On the trace-free loop space the potential pairs with the projection of μ to its dual. That projection has the same zero locus: μ already has total trace zero. The fibre rank is now $r-1$. □

For the preprojective stack $[\mu^{-1}(0)/G_{\mathbf{n}}]$, the dual compactly supported cohomology is pure of Tate type [29, Theorem A]. This controls its mixed Hodge weights; it does not identify a compact reduced coefficient system with the preprojective one.

This is a pushforward comparison. It does not identify the vanishing-cycle sheaf at each critical point with a constant sheaf, or construct reduced compact Hall operations.

For point sheaves the cyclic model is also explicit. Fix $x \in X$ and a volume-normalized tangent basis e_1, e_2, e_3, with e_3 in the elliptic direction. The cyclic analytic minimal model of $\mathcal{O}_x$ is $\Lambda^\bullet T_x X$ with exterior multiplication and the volume pairing [115, Theorem 4.23 and Proposition 5.18]. Matrix amplification for $\mathcal{O}_x^{\oplus n}$ gives

$$a = X_1 e_1 + X_2 e_2 + Z e_3, \qquad \frac{1}{3}\operatorname{vol}_x \operatorname{Tr}(a^3) = \operatorname{Tr}(Z[X_1, X_2]).$$

Its three critical equations are the pairwise commutators of X_1, X_2, Z. In the formal neighbourhood of the zero triple they describe the deformations of $\mathcal{O}_x^{\oplus n}$ modulo GL_n. Over a fixed closed point, support at x requires the coordinate operators to be jointly nilpotent. The full affine commuting-triple variety also contains other supports. Removing $\operatorname{Tr} Z/n$ centres this local model in the elliptic direction. The gauge terms and relative centres remain as in (6.2).

Finite flags use the same comparison

We use the suspended bar convention: the bar differential has degree one and F has degree zero. Its Taylor coefficients below act on the suspended morphism spaces. Matrix amplification of these coefficients is the path sum

$$(F_n^{\mathrm{mat}}(a_n, \dots, a_1))_{i_n i_0} = \sum_{i_1, \dots, i_{n-1}} F_n((a_n)_{i_n i_{n-1}}, \dots, (a_1)_{i_1 i_0}).$$

An ordered twisted flag is a finite upper-triangular matrix of objects with a strictly upper-triangular degree-one twisting element δ satisfying the Maurer–Cartan equation. Its successive quotients are the diagonal objects. Write $\overline{\delta} = s\delta$, of degree zero in the suspended space, and regard the a_i below as suspended morphisms.

Theorem 1.3 *(Cyclic comparison on finite formal flags).* The chosen F extends to finite twisted flags by

$$F_*(\overline{\delta}) = \sum_{n \geq 1} F_n^{\mathrm{mat}}(\overline{\delta}, \dots, \overline{\delta}),$$

$$F_k^{\mathrm{tw}}(a_k, \dots, a_1) = \sum_{j_0, \dots, j_k \geq 0} F^{\mathrm{mat}}_{k + j_0 + \cdots + j_k}(\overline{\delta}_k^{j_k}, a_k, \dots, a_1, \overline{\delta}_0^{j_0}).$$

A power here denotes repeated inputs. These formulas give a cyclic quasi-equivalence, equivariant under changes of multiplicity bases and compatible with every regrouping of a finite flag. They also apply with nilpotent Artinian parameters and in the compatible inverse-limit formal completion.

Proof. A strictly upper-triangular insertion advances along the finite flag, so the sums terminate between any two displayed morphisms. With Artinian parameters, use flag distance and the maximal-ideal filtration together; each fixed precision still has finitely many terms. The identity $b_{\mathcal{D}}F = Fb_{A}$, evaluated on repeated $\overline{\partial}$ and then on distinguished morphism inputs, gives the Maurer–Cartan equation and all twisted functor equations.

On the associated graded of these finite filtrations, the linear map is a direct sum of the original quasi-isomorphisms F_1. It is therefore a quasi-isomorphism on each morphism complex. Lift a flag inductively by lifting its next degree-one extension class and taking its cone. For a small Artinian extension, the obstruction lies in degree two; injectivity of $H^2(F_1)$ removes it, surjectivity of $H^1(F_1)$ matches the extension class, and a degree-zero gauge correction matches representatives.

The trace is the original cyclic trace followed by matrix trace. Rotating a closed matrix path preserves it with the Koszul sign. Regrouping a flag partitions the same paths into consecutive subpaths. Thus the comparisons preserve the cyclic class and agree under further subdivision. The formal completion is the inverse limit of these compatible finite-precision formulas. □

This theorem constructs the formal comparison on all finite flags. An analytic realization requires convergence. A comparison of constructible critical coefficients on compact moduli further requires algebraic or analytic charts, their orientation descent and the coefficient morphisms on extension correspondences. Those requirements remain even though the formal category comparison is available.

CRITICAL COEFFICIENTS ON SPECTRAL CONFIGURATIONS

The polynomial quiver model also gives coefficients on its actual spectral configuration spaces. Work in rational constructible derived categories with smooth descent on quotient stacks, allowing the unbounded complexes produced by classifying stacks. For this construction keep the common centre. Put

$$\mathcal{Y}_{\mathbf{n}} = [\mathcal{V}_{\mathbf{n}}/G_{\mathbf{n}}], \qquad d_{\mathbf{n}} = \dim \mathcal{V}_{\mathbf{n}} - \dim G_{\mathbf{n}} = 2\sum_{a\in Q_1} n_{t(a)}n_{h(a)}, \qquad \Phi_{\mathbf{n}} = \varphi_{\mathcal{W}_{\mathbf{n}}}\mathbb{Q}_{\mathcal{Y}_{\mathbf{n}}}[d_{\mathbf{n}}].$$

This normalization uses the local quotient orientation already specified. In mixed Hodge coefficients include the twist $(d_{\mathbf{n}}/2)$, which is integral.

PROPOSITION 1.4 *(Spectral factorization).* On the open where the spectrum of Z splits into two ordered disjoint parts, the critical stack and its coefficient are the products for those two parts. These comparisons are compatible with every refinement into finitely many disjoint spectral blocks.

Proof. Coprime factors of the characteristic polynomial give polynomial idempotents in Z by Bezout's identity. They split the vertex spaces in families on the resultant open. The equations commuting the arrows with Z make every arrow preserve these subspaces. The cross-block normal variables have paired potentials

$$\operatorname{Tr}(Y\mathcal{L}(X)), \qquad \mathcal{L}(X) = Z_{\beta}X - XZ_{\alpha}.$$

The eigenvalues of $\mathcal{L}$ are differences of eigenvalues in the two blocks. Its determinant is invertible on this open, so the polarized change $Y' = \mathcal{L}^*(Y)$ turns the potential into $\operatorname{Tr}(Y'X)$.

There are $r(\alpha,\beta) = \sum_{a\in Q_1}(\alpha_{t(a)}\beta_{h(a)} + \beta_{t(a)}\alpha_{h(a)})$ hyperbolic pairs. Their vanishing-cycle factor is $\mathbb{Q}[-2r](-r)$ by dimensional reduction. The difference $d_{\alpha+\beta} - d_{\alpha} - d_{\beta} = 2r$ cancels this shift, and the Hodge normalization cancels its Tate twist. Thom–Sebastiani and stabilization give the coefficient comparison [12, Theorems 2.13 and 5.4].

The chosen polarization fixes its orientation generator: dimensional reduction identifies it with the complex orientation of the X-space. The change in Y fixes that space. Under refinement, the same hyperbolic pairs are regrouped; Künneth multiplies their even-degree orientation generators. The resulting normalized generators have degree zero, so all reassociations agree. These constructions are equivariant and descend to the quotient. □

Let $\mathfrak{M}_{\mathbf{n}}$ be the classical critical stack and let $\pi_{\mathbf{n}}$ record the characteristic divisors of its normal operators in $\prod_i \mathrm{Sym}^{n_i} \mathbb{A}^1$. Define $C_{\mathbf{n}} = R\pi_{\mathbf{n}!}\Phi_{\mathbf{n}}$. For a word $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_r)$ of nonzero dimension vectors, let $\theta_{\boldsymbol{\gamma}}$ place multiplicity $(\gamma_j)_i$ at the jth point in colour i. The word coefficient is

$$C_{\boldsymbol{\gamma}} = \theta_{\boldsymbol{\gamma}}^{!} C_{\sum_j \gamma_j} \quad \text{on } (\mathbb{A}^1)^r.$$

Exceptional pullback retains the collision coefficients and their shifts. The empty coefficient is $\mathbb{Q}$. For a consecutive cut $A|B$, denote by $j_{A|B}$ the open inclusion where every point in A differs from every point in B. The cut tensor has $A|B$ component $R\,j_{A|B,*}\,j_{A|B}^{*}(C_A \boxtimes C_B)$.

Theorem 1.5 *(The spectral cut coalgebra).* The word coefficients have coassociative maps

$$C_{\boldsymbol{\gamma}} \longrightarrow R\,j_{A|B,*}\,j_{A|B}^{*}(C_A \boxtimes C_B)$$

that are equivalences after restriction to the cut open. Their empty coefficient gives the counit and coaugmentation. With this cut tensor, their cobar is the unital dg algebra

$$\Omega_{\mathrm{cut}} C = \widehat{T}_{\mathrm{cut}}(\overline{C}[-1]), \qquad dq_c = -q_{dc} + \sum (-1)^{|c_1|+1} q_{c_1} q_{c_2}, \quad \overline{\Delta} c = \sum c_1 \otimes c_2.$$

Here $|q_c| = |c| + 1$, and completion is the product over total dimension vectors.

Proof. Spectral factorization, base change for $R\pi_!$ and Künneth identify the restricted coefficient with the external product. The unit of the adjunction $j^* \dashv R\,j_*$ gives the displayed map. Two successive cuts use the same three-block factorization and the same open direct image, proving coassociativity. Empty cuts give the unit comparisons.

Extend the displayed differential as a derivation. Its linear and quadratic parts anticommute because Δ is a chain map; its two successive splitting terms cancel by coassociativity. Hence $d^2 = 0$. At a fixed total dimension $\mathbf{n}$, a word of nonempty factors has length at most $\sum_i n_i$. Thus the completion has only finitely many word lengths in each dimension, even when its stack cohomology is unbounded. □

This is an associative construction on charged ordered configurations. Its tensor uses consecutive cuts and their separation opens. A Hall multiplication instead uses extensions. Their comparison must preserve these operations on their stated coefficient categories.

The support left by failed cuts

For a word of length r, let U_i be its prefix cut open after position i, and put $Z_r = (\mathbb{A}^1)^r \setminus \bigcup_{i=1}^{r-1} U_i$. Write $R\underline{\Gamma}_{Z_r}$ for the coefficient functor of sections supported in this closed set. Expanding the cut cobar by the subsets of the $r-1$ cuts gives the augmented Čech complex of these opens, shifted by $[-1]$. Therefore

$$(\overline{\Omega_{\mathrm{cut}} C})_{\boldsymbol{\gamma}} \simeq R\underline{\Gamma}_{Z_r}(C_{\boldsymbol{\gamma}})[-1]. \tag{6.3}$$

Indeed the term with k cuts has the $k+1$ generator shifts, and the cobar splitting signs are the alternating restriction signs. The augmented Čech complex is the fibre of restriction to the union of the cut opens. For two positions this says

$$R\underline{\Gamma}_{Z_2}(C)[-1] = \mathrm{Cone}(C \longrightarrow Rj_*j^*C)[-2],$$

using $\mathrm{Cone}(f)^n = D^n \oplus C^{n+1}$ for $f : C \to D$.

The first supports already distinguish cuts from the full diagonal. If Δ_{ij} denotes equality of positions i, j, then

$$Z_2 = \Delta_{12}, \qquad Z_3 = \Delta_{13}, \qquad Z_4 = \Delta_{14} \cup (\Delta_{13} \cap \Delta_{24}).$$

To verify the last formula, assume positions one and four differ. Failure of the first and third cuts forces an equality joining one to an interior position and another joining four to an interior position. The two interiors cannot be equal. Pairing one with two and three with four leaves the middle cut open; pairing one with three and two with four closes all three cuts. The remaining possibilities force one to equal four. This proves the formula. In particular, the crossed-pair component must be retained even away from the total diagonal.

The coefficient on a collision is also larger than its restriction off that collision. For two one-dimensional colours with r paired arrows, the centred potential is $f = sQ(x, y)$, where $Q = \sum_{i=1}^{r} x_i y_i$. The effective relative group $\mathbb{G}_m$ acts with weights $1, -1, 0$ on x, y, s; the full quotient retains a separate diagonal scalar stabilizer. Its critical ideal is

$$(Q, sx_1, \dots, sx_r, sy_1, \dots, sy_r) = (x, y) \cap (s, Q).$$

For the intersection identity, reduce $sa + Qb$ modulo (x, y); since s is a nonzerodivisor there, membership forces $a \in (x, y)$. For $r = 2$, the invariant matrix $(x_i y_j)$ on the collision component has trace zero and rank at most one. Writing it as $\left(\begin{smallmatrix} u & v \\ w & -u \end{smallmatrix}\right)$ identifies its affine quotient with $u^2 + vw = 0$. On $x \neq 0$ the quotient is $T^*\mathbb{P}^1$.

PROPOSITION 1.6 *(The collision stalk).* For $r \geq 2$, the stalk of $\varphi_f \mathbb{Q}[2r+1]$ at the origin has rank one in degrees $1 - 2r$, -1, 0 and vanishes otherwise. At a nonzero point of the spectral axis only degree -1 remains. At a smooth nonzero point of the collision quadric only degree $1 - 2r$ remains.

Proof. The map $(s, x, y) \mapsto (s, x, sy)$ identifies $f^{-1}(1)$ with $\mathbb{C}^* \times Q^{-1}(1)$. Projection of $Q^{-1}(1)$ to $x \neq 0$ is an affine bundle with fibre $\mathbb{C}^{r-1}$. Thus $f^{-1}(1)$ has the homotopy type of $S^1 \times S^{2r-1}$. Homogeneity rescales a small Milnor fibre into a large ball sublevel of $f^{-1}(1)$. The squared norm is a proper semialgebraic function on this smooth affine fibre; its critical set has finitely many connected components, on each of which its value is constant. Above those values the sublevels are deformation retracts of the whole fibre. Thus the reduced Milnor cohomology has rank one in degrees 1, $2r-1$, $2r$. The shift gives $H^j(\varphi_f \mathbb{Q}[2r+1])_0 = \widetilde{H}^{j+2r}(F_0, \mathbb{Q})$. Off the origin, the transverse functions are respectively the nondegenerate quadratic form Q and the node sQ with Q as a local coordinate. Their Milnor spheres give the two remaining assertions. □

These are atlas coefficients; the equivariant quotient also retains its stabilizer directions. The gauge complex has the relative infinitesimal action of $\mathbb{G}_m$ in addition to the displayed critical equations.

Supported comparison equations

Fix a dg model of the supported coefficient category and two uncurved A_∞ algebras A, B in it. Put $sA = A[1]$ and $sB = B[1]$, and write b_A, b_B for their degree-one bar coderivations. A comparison has Taylor components F_n of degree zero satisfying $b_B F = F b_A$. With the components below arity n fixed, set

$$\mathcal{H}_n = \mathrm{Hom}_{\mathrm{supp}}((sA)^{\otimes n}, sB), \qquad R_n = (b_B F_{<n} - F_{<n} b_A)_n.$$

The tensor and Hom here retain the specified support and diagram conditions. The Hom differential is $\partial H = b_{B,1} H - (-1)^{|H|} H b_{A,1}$, where the source has its tensor differential.

Proposition 1.7 *(The next comparison component).* If all lower arity equations hold, then $\partial R_n = 0$. The next component exists exactly when $[R_n] = 0 \in H^1(\mathcal{H}_n)$, and it solves $\partial F_n = -R_n$. If $\partial h + h\partial = 1 - P$ is a contraction onto closed cohomology representatives and $P R_n = 0$, one solution is $F_n = -h R_n$. Solutions modulo boundaries, with the lower components fixed, form a torsor under $H^0(\mathcal{H}_n)$.

Proof. For $D = b_B F - F b_A$, the square-zero identities give $b_B D + D b_A = 0$. Its arity-n part is $\partial R_n = 0$ because all lower defects vanish. Adding F_n changes the defect by ∂F_n, giving the stated obstruction and solution. Differences of solutions are degree-zero cycles; homotopies fixing the lower components add boundaries. □

For the compact comparison, these equations must hold on the whole collision coefficients. Already the complex $\mathbb{C}[s]e \to \mathbb{C}[s]$, with $|e| = -1$ and $de = s$, has $H^0 = \mathbb{C}[s]/(s)$ although it becomes contractible after inverting s. A primitive containing s^{-1} therefore does not solve the equation on the collision. At four inputs, (6.3) also retains the crossed-pair support.

Local and horizontal support

For a coherent sheaf on X, call its support *local over* E when its image under $p_E : X \to E$ is finite, and *wrapped* when that image is all of E. Properness and irreducibility of E give these alternatives for a nonempty support. The fibre sheaves $i_* L_i$ above are local. In a short exact sequence, the middle support is the union of the two endpoint supports. Hence an extension is local precisely when both endpoints are local.

Proposition 1.8 *(Effective elliptic degree).* Let $Z = \sum_i m_i [C_i]$ be an effective one-cycle on X, with $m_i > 0$. Write $(p_E)_* Z = b[E]$. Then $b > 0$ if and only if some C_i dominates E, in which case $p_E(|Z|) = E$. For a coherent sheaf of support dimension at most one, its fundamental one-cycle has $b = 0$ precisely when its support is local over E.

Proof. Each proper irreducible curve maps either to a point, contributing zero, or onto E, contributing its positive degree. Positivity of the multiplicities prevents cancellation. Zero-dimensional support adds only finitely many image points. □

Corollary 1.9. A sheaf with positive effective elliptic degree is not supported over any finite configuration in E. Finite-support sheaf moduli alone therefore do not contain that geometric sector.

This is a statement about support of sheaves. A factorization algebra on E can carry other labels or act on additional modules; the proposition does not prohibit such constructions. It also does not identify b with an exponent of the Igusa product.

Definition 1.10 *(Geometric degree at a bounded stage).* Choose a bounded family of coherent sheaves, a finite set of charges Γ_R, and the extension stacks to be used at that stage. On the sector of support dimension at most one set $b_R^{\mathrm{geom}}(A) = \deg(p_E)_* \operatorname{cyc}_1(A)$. On a larger coherent-sheaf sector one may instead fix an ample H on S and an integer n for which every generic-fibre sheaf $A|_{S_{\mathbb{C}(E)}}(nH)$ in the chosen family is generated by sections and has no higher cohomology. Set

$$b_{R,n}^{\mathrm{geom}}(A) = \operatorname{rk} R p_{E*}(A \otimes p_S^* \mathcal{O}_S(nH)).$$

The regularity condition and a common choice of n are part of this larger-domain construction. We write b_R^{geom} for the chosen degree when its domain is fixed.

The existence of sufficiently large twists is the relative Serre and global-generation statement; a common bound must hold on the chosen families [109, Tags 0B5S and 01PR]. The degree is additive on their Grothendieck classes. On the larger domain it is the generic-fibre Euler characteristic, which by the regularity hypothesis equals the dimension of global sections. It is nonnegative, and it is zero precisely when that generic-fibre sheaf is zero. Since the sheaf is generated by its sections, this is precisely the local-support condition. On the one-dimensional support sector the generic fibre has finite length, so the two definitions agree and do not depend on n.

Definition 1.11 *(Additive support data).* The charges of every permitted short exact sequence must satisfy $\gamma_C = \gamma_A + \gamma_B$. A degree on these charges is additive support data when it represents the chosen geometric degree of each object and satisfies

$$b_R^{\mathrm{geom}}(\gamma_A + \gamma_B) = b_R^{\mathrm{geom}}(\gamma_A) + b_R^{\mathrm{geom}}(\gamma_B), \qquad b_R^{\mathrm{geom}}(0) = 0.$$

Only operations whose endpoint, intermediate and total charges are present are included at a bounded stage.

PROPOSITION 1.12 *(Support and addition).* Both constructions in Definition 1.10 are additive on the stated coherent-sheaf domains. Consequently the support types obey LL = L, whereas a product involving a wrapped input has type W.

Proof. Fundamental cycles are additive for short exact sequences of sheaves of dimension at most one. For the second construction, derived pushforward is additive in the Grothendieck group and rank is additive. The support characterization just proved gives the type rule. The direct support-union argument gives the same rule without choosing a numerical degree. □

The regularity hypothesis matters. For a line bundle L_i and the polarization H of the common-polynomial triple, $\chi(S, L_i(nH)) = 4n^2 - 2$. Thus the horizontal sheaf $L_i \boxtimes \mathcal{O}_E$ has that generic Euler characteristic, which is negative at $n = 0$. It becomes a nonnegative rank only on the acyclic, generated range. A virtual perfect-complex class can also have cancellations, so effectiveness must not be inferred from its numerical degree alone.

THE CONFIGURATION AND FAMILY SPACES

For a finite set I, a labelled configuration is a tuple $\mathbf{e} \in E^I$. Let $Z_I \subset E \times E^I$ be the union of the graphs of its coordinates. A closed local incidence stack at fixed thickness N parametrizes a flat family A with

$$\mathcal{I}_{S \times Z_I}^N A = 0.$$

We denote it by $\mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,I,N}$. Keeping N retains infinitesimal support at collisions. Set-theoretic support in Z_I does not impose thickness one. A bounded choice of N, or the specified filtered system in N, is included whenever these incidence stacks are used.

Thickness also enters extension closure. If the same ideal kills the two endpoints to powers N_A and N_B, it kills the middle sheaf to power $N_A + N_B$: its N_A-th power sends the middle sheaf into the subobject, and the next N_B powers kill that subobject. Thus collision and extension maps must use compatible thickness bounds, rather than assume that one fixed bound is closed under all extensions.

Definition 1.13 *(Wrapped families and scalar rigidification).* Let $\mathcal{M}^{\mathrm{wr}}_{\eta,R}$ be a chosen algebraic stack of wrapped sheaves. When its scalar subgroup can be rigidified, write

$$\rho_{\eta,R} : \mathcal{M}^{\mathrm{wr}}_{\eta,R} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}.$$

The map removes the central scalar automorphisms. It is directed from the original stack to the rigidified stack. Elliptic translation has not been divided out. A coefficient or universal object on the rigidification requires descent along ρ; the universal sheaf on the original stack is not assumed to descend.

Lemma 1.14 *(Translation on families).* Support translation acts by $a \cdot A = (\mathrm{id}_S \times \tau_a)_* A$, where $\tau_a(e) = e + a$, and gives a coherent E-action on the sheaf and flag stacks. It descends through a scalar rigidification compatible with this central subgroup.

Proof. The equality $\tau_a \tau_b = \tau_{a+b}$ gives the composition isomorphism on sheaves and all their morphisms. Its associativity is the associativity of composition of translations. Scalar multiplication commutes with this action, so the universal property of rigidification gives the descended action. □

Proposition 1.15 *(Finite type after rigidification).* Suppose the chosen wrapped stack is finite type and its scalar rigidification is an algebraic $\mathbb{G}_m$-gerbe. Then the rigidified stack is finite type. No universal sheaf on it is implied. In particular, the finite-type premise holds for a finite open cover by specified finite-type semistable substacks. Finiteness of the numerical index set and boundedness of each substack are separate premises of this construction.

Proof. The gerbe is smooth and surjective, and finite type over the base can be checked on such a cover. Descent of a particular sheaf is a separate condition on its scalar action and descent isomorphisms. The finite disjoint union of the open substacks is a smooth surjective finite-type cover, proving the additional assertion. □

Definition 1.16 *(A carrier for local and wrapped families).* Choose the charge and moduli domains above. A presentation of the hybrid carrier consists of a finite labelled local configuration $(I, \alpha, \mathbf{e})$ and a finite list of wrapped families $(W_j)_{j \in J}$ with charges η_j. Form the prestack colimit of

$$E^I \times \prod_{j \in J} \mathcal{M}^{\mathrm{wr}}_{\eta_j,R}$$

over relabellings and local collision maps. Under a surjection $I \twoheadrightarrow I'$, the diagonal repeats coordinates and the new local colour is the sum of the colours in each fibre. Only defined charge sums occur. Denote this prestack by $\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$. Compatible scalar rigidifications can be used in this diagram when the required family and coefficient descent is supplied.

The indexing category contains all specified finite presentations; it is not generally a finite category. A topology τ gives its stackification $a_\tau \operatorname{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$. An operation or coefficient object on that stackification must satisfy descent. Neither taking the colimit nor stackifying defines a Hall coefficient system.

Definition 1.17 *(Local and wrapped subcarriers).* The local subcarrier has $J = \varnothing$; the purely wrapped subcarrier has $I = \varnothing$. Mixed presentations have both. When the geometric degree is defined on the chosen domain, its charge types are

$$\Gamma_R^{\mathrm{loc}} = \{\gamma : b_R^{\mathrm{geom}}(\gamma) = 0\}, \qquad \Gamma_R^{\mathrm{wr}} = \{\gamma : b_R^{\mathrm{geom}}(\gamma) > 0\}.$$

The local incidence stacks and the critical sheaves placed over these carriers are additional objects, not part of their points.

A wrapped family is retained as a family, with its automorphisms. Replacing it by its degree or by a single point of E would discard that information. The ordered extension diagrams below retain still more information: which object is a subobject and which is a quotient.

Ordered extensions and proper targets

We use quotient-first input order. Thus $\mathfrak{E}_{\alpha,\beta}$ parametrizes short exact sequences

$$0 \longrightarrow B_\beta \longrightarrow C_{\alpha+\beta} \longrightarrow A_\alpha \longrightarrow 0,$$

with endpoint map $p(C) = (A, B)$ and middle-object map $q(C) = C$. All sheaves, maps and configuration parameters belong to the chosen domains. The charge set at height R is only a window: a nonzero positive charge cannot lie in a finite set closed under all of its positive multiples. Operations at this stage are therefore partial, unless a separate quotient construction makes them total.

Definition 1.18 *(The four binary correspondence types).* Restrict the ordered extension stack to the indicated support types:

type	endpoints (A, B)	middle object
LL	$(A_{\mathrm{L}}, B_{\mathrm{L}})$	C_{L}
LW	$(A_{\mathrm{L}}, B_{\mathrm{W}})$	C_{W}
WL	$(A_{\mathrm{W}}, B_{\mathrm{L}})$	C_{W}
WW	$(A_{\mathrm{W}}, B_{\mathrm{W}})$	C_{W}.

Local endpoints retain their closed configuration presentations and thicknesses. Wrapped endpoints retain their sheaf families. These are correspondences before elliptic quotient. Scalar rigidification is applied to a compatible whole diagram when such a diagram and its coefficient descent have been specified.

PROPOSITION 1.19 *(A proper coherent-sheaf domain).* Let C be a flat family of coherent sheaves on the projective threefold X, over a base T. Fix the Hilbert polynomial of the quotient A. If all additional endpoint and support conditions cut out a closed subfunctor of $\operatorname{Quot}_T(C)$, then the corresponding middle-object map q is proper over T.

Proof. The relative Quot scheme with that polynomial is projective over T [56, Theorem 2.2.4]. Its universal quotient is flat, and its kernel is flat because C is flat. The required sequences therefore form the stated closed subfunctor of a projective T-scheme, which is proper. Fixed annihilator conditions defining the local incidence spaces are imposed inside this Quot problem. □

Proposition 1.20 *(Mixed and wrapped target maps).* The same properness criterion holds for the mixed and wrapped coherent-sheaf correspondences, and for their finite flag versions. If instead the chosen conditions define only an open substack, the restriction of q need not be proper. A product with ordinary critical cohomology is then available by this route only when the specific coefficient complex has proper support over the target, or another appropriate integration morphism has been constructed.

Proof. Use the same relative Quot scheme, or its projective iterated flag version, and impose the closed conditions. An open subspace of a proper scheme need not be proper: $\mathbb{A}^1 \subset \mathbb{P}^1$ is an example. Thus mere openness of a stability condition gives no properness conclusion. The support alternative is exactly the condition identifying the relevant exceptional and ordinary pushforwards. □

In particular, ordinary Quot properness does not establish this criterion for all perfect complexes in a tilted heart. Such a heart needs its actual extension moduli and the corresponding properness or integration theorem. A bound on the number of numerical charges also does not prove that their moduli are finite type.

Flags and their eight support words

A three-step ordered flag is written in descending order,

$$C = F^0 \supset F^1 \supset F^2 \supset F^3 = 0, \qquad A_i = F^{i-1}/F^i.$$

This convention matches quotient-first multiplication. Every intermediate sheaf and its configuration belongs to the specified domain. For a partial charge window, both regroupings are compared only when all their intermediate charges are present.

Definition 1.21 *(Binary support words).* The support word of this flag is $(\epsilon_1, \epsilon_2, \epsilon_3)$, with $\epsilon_i = \mathrm{L}$ or W according to A_i. The eight possibilities are

$$\mathrm{LLL}, \quad \mathrm{LLW}, \quad \mathrm{LWL}, \quad \mathrm{WLL}, \quad \mathrm{LWW}, \quad \mathrm{WLW}, \quad \mathrm{WWL}, \quad \mathrm{WWW}.$$

Each block has type L exactly when all its factors do. The word specifies filtration order, not physical time order.

Proposition 1.22 *(The two extension compositions).* For each of the eight words, the two iterated extension correspondences are naturally equivalent to the stack of flags with those successive quotients. These equivalences commute with base change in the parameter scheme.

Proof. For the first composition, take a sequence with quotient C_{12} and subobject A_3, together with a sequence expressing C_{12} as an extension of A_1 by A_2. The inverse image of A_2 defines F^1, and $F^2 = A_3$. For the other composition, take F^1 as the extension of A_2 by A_3 and then extend A_1 by F^1. These constructions are inverse on families and their isomorphisms. Kernels, inverse images and the indicated quotients commute with base change for these flat exact sequences. The support-union rule gives precisely the displayed word types. □

This is a geometric comparison of correspondences. To deduce an equality of operations, the two coefficient morphisms on that flag stack must agree. We state the needed coefficient framework before making that deduction.

COEFFICIENT MAPS AND SIX OPERATIONS

Fix a characteristic-zero coefficient field k; the rational constructions above use $k = \mathbb{Q}$.

Definition 1.23 *(An admissible coefficient theory).* Choose constructible derived categories $D_c(Y)$ on all spaces and stacks in the correspondence diagram. Require the six operations f^*, Rf_*, $Rf_!$, $f^!$, tensor product and internal Hom on the maps being used, with coherent composition, the needed Cartesian base-change isomorphisms and the projection formula. Each chosen coefficient and its image must remain in the stated categories. On quotient stacks this includes the full smooth or equivariant descent diagrams. Boundedness of cohomology or perfection is an additional condition whenever a finite dual or ordinary tensor product requires it. For a compactification $f = \overline{f}\, j$, with j open and $\overline{f}$ proper, the exceptional image is $Rf_!K = R\overline{f}_* j_! K$. If f is proper, take $j = 1$ and $\overline{f} = f$. Independence of this presentation is proved in Proposition 1.105.

Write $\mathcal{M}_\gamma$ for the chosen object stack and choose $K_\gamma \in D_c(\mathcal{M}_\gamma)$. For ordinary critical cohomology, a binary operation uses an integer source shift $\kappa(\alpha, \beta)$ and a coefficient morphism

$$u_{\alpha,\beta} : p^*(K_\alpha \boxtimes K_\beta)[\kappa(\alpha,\beta)] \longrightarrow q^! K_{\alpha+\beta}, \tag{6.4}$$

together with properness of the source coefficient's support over the target of q. This identifies its ordinary and exceptional pushforwards. Pullback on ordinary cohomology, then $Rq_! u$ and the trace, gives the product. A Thom–Sebastiani isomorphism on a product of critical charts contributes to u only after its comparison with the extension diagram has been constructed. Appendix 4.2 gives a theorem that constructs these morphisms on its unreduced moduli domain. More explicitly, a comparison $p^*(K_\alpha \boxtimes K_\beta)[\kappa] \simeq K_{\mathfrak{E}}$ can be used when it is supplied together with $K_{\mathfrak{E}} \to q^! K_{\alpha+\beta}$. Their composite is $u_{\alpha,\beta}$. The first isomorphism alone does not supply the second arrow.

PROPOSITION 1.24 *(Base change and projection formula).* In the chosen admissible coefficient theory, a Cartesian square with horizontal maps f, f' and vertical maps q, q' gives

$$f^* Rq_! K \simeq Rq'_! f'^* K, \qquad Rq_!(K \otimes q^* L) \simeq Rq_! K \otimes L.$$

Consequently the pull–push functors of composable extension correspondences identify with the functor of their fibre-product correspondence. The coefficient morphisms themselves must be compatible with these identifications.

Proof. Apply the base-change and projection-formula isomorphisms in the definition, then the composition isomorphism for exceptional pushforward. This compares the functors on their common domain; it makes no assertion that two independently chosen coefficient morphisms are equal. □

PROPOSITION 1.25 *(Critical coefficient transport).* Suppose oriented critical charts and their coefficient morphisms are compatible on the extension and flag diagrams. Their Thom–Sebastiani and hyperbolic comparison maps then define the morphisms (6.4) compatibly with base change and projection formula. This implication holds for local, mixed and wrapped correspondences on that domain.

Proof. The chartwise morphisms are objects of the descent diagram for morphisms of constructible complexes. Their specified overlap homotopies glue them. Naturality of the six operations transports the base-change and projection-formula comparisons through this descent. The critical Hall theorem supplies one such coherent system under its own hypotheses; an arbitrary orientation line alone does not supply the morphisms. □

Proposition 1.26 *(Associativity for the eight words).* Assume that the two composites of the maps u agree on each three-step flag stack, including their shifts and orientation identifications. Then every defined triple product satisfies

$$m_{\alpha+\beta,\gamma}(m_{\alpha,\beta}\otimes 1) = m_{\alpha,\beta+\gamma}(1\otimes m_{\beta,\gamma})$$

with the tensor associativity constraint determined by κ.

Proof. Proposition 1.22 identifies the two geometric compositions with one flag stack. Base change and projection formula identify the induced pull–push functors there. The assumed equality of coefficient morphisms makes their traces equal. This proves the displayed equation for all eight words. □

Proposition 1.27 *(Four-input coherence).* The geometric associators obtained by forgetting steps of a four-step flag satisfy the pentagon. If the coefficient comparisons are coherent on that flag diagram, their associators satisfy the same pentagon. Writing $a_{\alpha,\beta,\gamma}$ for reassociation from left to right, the identity is

$$a_{\alpha,\beta,\gamma+\delta}a_{\alpha+\beta,\gamma,\delta} = (1\otimes a_{\beta,\gamma,\delta})a_{\alpha,\beta+\gamma,\delta}(a_{\alpha,\beta,\gamma}\otimes 1).$$

Proof. Both paths around the pentagon identify a filtration with the same successive quotients and forget the same intermediate steps. The inverse-image and quotient constructions in the preceding proof agree under repeated subdivision. Applying a coherent coefficient functor preserves this equality. □

A pentagon for categorical associators controls reassociation in that category. It does not by itself provide all higher chain homotopies of an A_∞ operation or all collision coherences of a factorization object.

Definition 1.28 *(The coloured flag diagram).* An ordered profile consists of finitely many local or wrapped inputs with their configuration and family data. A planar rooted tree records an iterated extension of that profile; each edge carries the sum of the charges below it and the corresponding support type. Vertices carry the indicated extension or flag correspondence. Grafting gives composition, and contracting an internal edge regroups a flag. A coherent coefficient system on this diagram specifies the operations and comparison homotopies on all finite trees, together with units, local relabellings, configuration refinements and overlap descent. Every requested coherence is on its actual configuration space, including collisions. The notation $\mathfrak{o}_R^{\mathrm{col}} = 0$ abbreviates these tree, unit, symmetry, refinement, descent and overlap conditions. Each condition belongs to its own diagram; the notation does not assert that they form one cohomology class.

Proposition 1.29 *(From coherent flags to operations).* A coherent coefficient system on the coloured flag diagram induces the corresponding associative or homotopy-coherent operations, with its specified units and local symmetries. The conclusion is restricted to the charge and support domain of that system.

Proof. Compose the coefficient morphisms and their pull–push functors along each tree. Edge contraction gives the comparison between composites. The prescribed higher compatibilities are exactly the relations among these comparisons. The six-functor composition constraints transport them to the resulting operations. □

UNITS, DISJOINT SUPPORT AND INSERTION ORDER

Definition 1.30 *(The zero-object correspondence).* Before elliptic quotient, the distinguished zero object has moduli stack pt, empty support, and charge zero. Its two unit correspondences are

$$0 \longrightarrow A \xrightarrow{1} A \longrightarrow 0 \longrightarrow 0, \qquad 0 \longrightarrow 0 \longrightarrow A \xrightarrow{1} A \longrightarrow 0.$$

Choose the coefficient at zero and the comparison morphisms so that these correspondences induce the identity. Scalar rigidification of nonzero objects does not rigidify the zero object by a nonexistent scalar subgroup of its automorphisms.

PROPOSITION 1.31 *(Local and wrapped units).* With these unit coefficient identifications, both zero-endpoint operations are the identity on every local or wrapped coefficient object.

Proof. Each displayed sequence is uniquely determined by A, including its morphisms. Both legs identify with the identity map on its moduli stack. The coefficient comparison and the trace are then the stipulated identity maps. □

PROPOSITION 1.32 *(Local symmetry and configuration descent).* Sheaves with disjoint supports over E have vanishing mutual Ext groups. Their extension correspondence is the direct-sum correspondence. If its oriented coefficient comparison is the external-product comparison, the local operation on disjoint configurations is symmetric with the coefficient category's Koszul signs. A coefficient object on the unlabelled carrier requires compatible equivariant descent under relabellings and under the specified collision refinements. For a colour-preserving permutation, this is equivariant descent under $\mathfrak{S}_\alpha = \{\sigma \in \mathrm{Aut}(I) : \alpha_{\sigma(i)} = \alpha_i\}$. The corresponding quotient is $[\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I,N}/\mathfrak{S}_\alpha]$; its coefficient includes the full equivariant identifications.

Proof. All local Ext sheaves between disjoint closed supports vanish, so their global Ext groups vanish. The extensions split uniquely relative to their endpoints. Permuting the summands is compatible with their product charts and external tensor products, giving the stated symmetry. On labelled charts the descent is the full equivariant diagram; collision maps require its additional refinement comparisons. Explicitly, a relabelling $\sigma : I \to I'$ sends (α_i, e_i) to $(\alpha_{\sigma^{-1}(i')}, e_{\sigma^{-1}(i')})$. It identifies the graph unions $Z_I, Z_{I'}$, preserves the annihilator condition at thickness N, and leaves the sheaf itself unchanged. Its induced coefficient isomorphisms must satisfy the permutation composition law before quotient descent. □

In particular, summing independently over every E^I would count the same collision configurations repeatedly. A coefficient system on the carrier uses the actual colimit and its compatible maps, or an equivalent descent construction, rather than an unrelated direct sum over all finite labels. On a permitted collision, the one-label presentation and the two-coincident-label presentation are identified by descent. A direct sum over their label sets keeps two independent copies.

PROPOSITION 1.33 *(Ordered wrapped insertions).* For overlapping supports, exchanging two inputs changes which one is the subobject. The corresponding extension stacks need not be identified. Planar order must therefore be retained unless a specified symmetry or braiding compares the two operations.

Proof. The exchanged correspondence parametrizes $0 \to A \to C' \to B \to 0$, whereas the original parametrizes $0 \to B \to C \to A \to 0$. They involve opposite Ext groups and have no natural identification from the definition of an extension. The disjoint-support splitting argument does not apply when those groups are nonzero. □

Definition 1.34 *(Two-sided mixed insertions).* A two-sided mixed correspondence is the flag stack with ordered successive quotients (A_-, W, A_+), where the first and last objects have local support and W is wrapped. Collections of local inputs may replace A_- and A_+, with their original orders and configurations retained. Forgetting the left or right steps gives the one-sided mixed correspondences.

Elliptic quotient over a common torsor

Let BE denote the stack of E-torsors, and let Y be a stack with an E-action. A point of $[Y/E]$ over T is an E-torsor $P \to T$ together with an equivariant map $P \to Y$. The map $[Y/E] \to BE$ remembers P. For the unrigidified sheaf stacks put $\mathcal{Q}_\gamma = [\mathcal{M}_\gamma/E]$ and $\mathcal{H}_{\alpha,\beta} = [\mathfrak{E}_{\alpha,\beta}/E]$, using simultaneous translation of the exact sequence.

Proposition 1.35 *(Common-torsor Hall spans).* There is a natural equivalence

$$\mathcal{Q}_\alpha \times_{BE} \mathcal{Q}_\beta \simeq [(\mathcal{M}_\alpha \times \mathcal{M}_\beta)/E_{\mathrm{diag}}].$$

The extension diagram descends to a relative span

$$\mathcal{Q}_\alpha \times_{BE} \mathcal{Q}_\beta \longleftarrow \mathcal{H}_{\alpha,\beta} \longrightarrow \mathcal{Q}_{\alpha+\beta}.$$

Its iterated compositions are identified by the same flag constructions, and its relative zero-object unit is BE.

Proof. An object of the fibre product consists of two torsors, their families, and an isomorphism between the torsors. Transport one family across this isomorphism. Both families now live on one torsor; this is an object of the diagonal quotient. Repeating that torsor in the two factors gives the inverse equivalence on objects and arrows, compatibly with base change.

An equivariant extension on the common torsor has equivariant endpoints and middle object. Thus the extension span descends. The inverse-image and quotient constructions for flags are translation equivariant, so they identify its two compositions. Regroupings of longer flags forget the same steps. The zero object is fixed by translation and has quotient $[\mathrm{pt}/E] = BE$. Extensions with zero give the relative unit identities. □

Definition 1.36 *(Quotients of correspondence diagrams).* For an E-equivariant correspondence $Y_0 \leftarrow Z \to Y_1$, assign the span $[Y_0/E] \leftarrow [Z/E] \to [Y_1/E]$ over BE. Composition uses relative fibre products over the middle quotient. Denote this assignment on the chosen diagrams by $\mathcal{Q}^{\mathrm{corr}}_{E,R}$. It is a quotient of diagrams, not a cosection reduction of obstruction complexes. For a coefficient realization of this assignment, $\mathfrak{o}^{Q}_{R} = 0$ will mean compatibility with composition, base change, the critical coefficient maps, flag refinement, the chosen cohomology realization, operation comparison and transitions. Its components are defined only after the maps in question have been supplied.

Proposition 1.37 *(Composition and Cartesian squares).* For equivariant maps $Z \to Y \leftarrow W$,

$$[(Z \times_Y W)/E] \simeq [Z/E] \times_{[Y/E]} [W/E].$$

These equivalences preserve the composition and base-change squares of the correspondence diagrams and all their flag refinements.

Proof. A point on the right gives a common torsor after transporting across its specified isomorphism. The equality of the two maps to Y is then exactly an equivariant point of $Z \times_Y W$. This construction and its inverse commute with pullback and with forgetting a stage of a longer diagram. The resulting associativity identifications are therefore coherent. □

PROPOSITION 1.38 *(What independent quotients forget).* Replacing $\mathcal{Q}_\alpha \times_{BE} \mathcal{Q}_\beta$ by $\mathcal{Q}_\alpha \times \mathcal{Q}_\beta$ discards the identification of their torsors. It is not generally an equivalent input stack. A construction directly on quotient stacks is valid when it retains this relative input and its coefficient maps.

Proof. Take two copies of E with the translation action. Their diagonal quotient is E, represented by the difference of the two coordinates. Each independent quotient $[E/E]$ is a point, so the product of independent quotients is a point. The difference coordinate is lost. The common-torsor fibre product retains it by Proposition 1.35. □

Thus the obstruction is to forgetting the common torsor, not to working with quotient stacks. The relative construction already gives the correct quotient correspondence. Obtaining a reduced critical operation on it still requires reduced coefficients and their comparison maps.

COEFFICIENT DESCENT AND ITS VARIANCE

For an E-stack Y, a descended coefficient consists of $\overline{K}_Y \in D_c([Y/E])$ and its coherent pullback identification with K_Y on Y. Equivalently it is an object of the full descent diagram on the action groupoid $E^\bullet \times Y$. This includes the higher compatibility maps; an isomorphism on $E \times Y$ without its cocycle data is insufficient. If critical coefficients are normalized perversely, the smooth-atlas dimension shifts are included in the stated identification.

PROPOSITION 1.39 *(Descent of the critical comparisons).* Suppose all coefficient objects, binary morphisms and their flag homotopies form a compatible equivariant descent diagram. Then they descend to the quotient correspondence diagram, with its relative input over BE. Their base-change, projection and Thom–Sebastiani comparisons descend with them.

Proof. The constructible category on the quotient is the totalization of its equivariant descent categories. The supplied morphisms and homotopies define a morphism in this totalization. The functorial comparison diagrams are equivariant by hypothesis, so they define the corresponding diagrams after totalization. This uses the full descent data; forgetting to the underlying complex on Y is not a fully faithful test for all equivariant derived morphisms. □

The translation quotient is not automatically within the critical Hall theorem's moduli hypotheses. For example, its relative zero has stabilizer E, which is not affine, whereas Theorem 4.2 assumes affine stabilizers. Moreover, removing an orbit and its dual obstruction direction requires the reduced critical construction, not just the quotient of the classical stack.

Fix a Verdier duality functor $\mathbb{D}_Y$ in the chosen coefficient theory. A useful dual compact-support convention is

$$\mathbb{B}(Y;K) := R\,\mathrm{Hom}(R\Gamma_c(Y,K),k).$$

When duality applies it is $R\Gamma(Y,\mathbb{D}_Y K)$. For $K = k_Y$, this is the usual dualizing-complex description of Borel–Moore chains, with its indicated cohomological indexing. Crucially, it is contravariant in K.

PROPOSITION 1.40 *(Dual coefficients on the quotient).* When the indicated functors exist, the descended coefficient gives $\mathbb{B}([Y/E];\overline{K}_Y)$. A morphism $f : K \to L$ induces

$$\mathbb{B}(Y;L) \longrightarrow \mathbb{B}(Y;K),$$

and duality reverses composition. An operation obtained by dualizing a compact-support operation must therefore reverse its coefficient arrows. Identification of dual tensor products additionally requires the stated finiteness or completed-tensor hypothesis.

Proof. Apply $R\,\mathrm{Hom}(-,k)$ to the morphism $R\Gamma_c(Y,K) \to R\Gamma_c(Y,L)$. For $K \xrightarrow{f} L \xrightarrow{g} M$, the resulting map is $f^\vee g^\vee = (gf)^\vee$. The tensor-duality identification is valid for perfect complexes; outside that domain its completion or replacement must be specified. Descent gives the quotient coefficient object before this duality is taken. □

This does not produce a natural map from nonequivariant chains on Y to chains on $[Y/E]$, nor an equivalence between them. Their coefficient and stabilizer information differs. In particular, finite type of a stack does not imply finite-dimensional cohomology: already $H^*(B\mathbb{G}_m,\mathbb{Q}) = \mathbb{Q}[c]$ is unbounded.

Proposition 1.41 *(Comparison of operations).* Suppose a coefficient realization $Q_{E,R}$ of the equivariant correspondence diagram is supplied, together with natural comparisons

$$\theta_{\mu,R} : Q_{E,R}\mu_R \xrightarrow{\sim} \overline{\mu}_R Q_{E,R}^{\otimes n}$$

for its n-input operations. The tensor on the quotient side is the specified relative tensor. If these comparisons commute with flag refinement, they preserve associativity. If they also commute with the separate collision diagrams, they preserve the specified coproduct and its compatibilities.

Proof. Use the comparison to replace each operation in either composite. Naturality and the assumed flag compatibility identify the two replacements with the comparison for the common longer flag. The same argument on the collision diagram proves the coproduct assertion. A coproduct comparison is used only after that coproduct has been defined. □

Proposition 1.42 *(Primitives and transitions under comparison).* An operation-preserving comparison of coaugmented counital coalgebras maps primitive elements to primitive elements. To identify their primitive objects, an equivalence or a suitable kernel-preservation property is required. Comparisons at two bounded stages commute with transition only when the coefficient maps and their natural transformations satisfy that transition square.

Proof. Apply the comparison to $\Delta x = 1 \otimes x + x \otimes 1$ and use preservation of the coaugmentation and coproduct. This proves preservation of primitive elements. Equality with the whole target kernel needs more than an inclusion, hence the stated additional hypothesis. The transition assertion follows by composing the supplied commuting squares. None is implied by the existence of the maps on isolated stages. □

Anchors retain less than families

An anchor is auxiliary data for recording a family's position or spectral invariants. It is not needed to define the common-torsor quotient. For a perfect family A, normalize its determinant pushforward to degree zero:

$$\lambda^{\det}(A) = \det R p_{E*} A \otimes \mathcal{O}_E(-\chi(A)\mathrm{o}_E) \quad \text{in } \mathrm{Pic}^{\mathrm{o}}(E) \simeq E.$$

Proposition 8.44 gives $\lambda^{\det}(a \cdot A) = \lambda^{\det}(A) + \chi(A)a$. For $\chi \neq 0$, base change along $[\chi] : E \to E$ gives a weight-one anchor and a finite residual $E[|\chi|]$-quotient, as constructed in Proposition 8.45. For $\chi = 0$, this determinant is translation invariant.

Definition 1.43 *(Anchor data on a correspondence).* Given chosen anchor maps on the object stacks, retain their pullbacks from every endpoint, intermediate object and middle object of an extension or flag. If a proposed anchor comparison identifies or composes these values, specify that morphism and its

compatibility with flag refinement before passing to a quotient. The original families remain part of the correspondence.

For semistable degree-zero vector bundles on E, the Jordan–Hölder divisor $\Lambda(V) = \sum_i [L_i] \in \mathrm{Sym}^r \mathrm{Pic}^0(E)$ has Abel sum $\det V$. It distinguishes $\mathcal{O}_E^{\oplus 2}$ from $L \oplus L^{-1}$ for nontrivial L. But translations preserve each degree-zero line's isomorphism class, hence preserve $\Lambda(V)$. This spectral invariant does not repair orbit separation on a free translation orbit.

Definition 1.44 *(Requirements on a chosen anchor comparison).* An anchor comparison specifies its actual target, its definition on families, its translation weight, and any relation to the determinant. To replace a family parameter by that target one must additionally identify the resulting kernel equivalence relation. Compatibility with extensions and transitions is imposed as a commutative diagram. Divisor union applies to extensions of semistable degree-zero elliptic bundles, where their Jordan–Hölder factors concatenate; it is not imposed on arbitrary perfect pushforwards. Write $\mathfrak{v}_R^\lambda = 0$ for existence of these maps, their stated translation weight, their prescribed kernel equivalence relation, extension compatibility, and transition compatibility. These are conditions on the chosen maps, with the zero-weight obstruction below still in force.

PROPOSITION 1.45 *(The zero-weight obstruction).* On a nontrivial translation orbit, a translation-invariant spectral datum cannot determine an E-valued anchor of weight one. In particular this applies to the degree-zero Jordan–Hölder divisor. On the nonzero-χ locus the determinant cover instead gives the weight-one construction, with its finite stabilizer descent still to be specified.

Proof. A function of an invariant datum has the same value on A and $a \cdot A$. Weight one requires those values to differ by a. For $a \neq 0$ these statements contradict one another. The nonzero-weight assertion is the explicit isogeny construction already cited. This rules out the stated invariant replacement, not every possible anchor construction. □

GEOMETRIC DEGREE AND AUTOMORPHIC DEGREE

The normal-ordered charge map $\overline{\Pi}_X : \widehat{\Gamma} \to \Lambda^{2,1}_{II}$ is additive on its specified charge carrier. Write $\overline{\Pi}_X(\gamma) = (n, \ell, m)$ and $m^{\mathrm{Bch}} = \mathrm{pr}_3 \overline{\Pi}_X$. The integer m^{Bch} is a root or trace grading. The geometric degree b_R^{geom} measures support over E.

Definition 1.46 *(A comparison of degree maps).* A comparison specifies a domain of geometric charges and the maps from that domain to the two grading groups. Any equality between the two degrees is stated on that domain and checked against addition on its permitted extensions. An affine shift in a scalar generating series is recorded separately from an additive charge map.

PROPOSITION 1.47 *(An affine shift is not additive).* If b and m are additive on two charges and their sum, then $b = m + 1$ cannot hold on all three. It may still describe the absolute indexing of selected coefficients in a generating series.

Proof. If the equality holds on α and β, then $b(\alpha + \beta) = m(\alpha) + m(\beta) + 2 = m(\alpha + \beta) + 2$, rather than $m(\alpha + \beta) + 1$. □

For example, a scalar series written in s^{d-1} has an exponent $m = d - 1$. The shift accounts for its prefactor. It cannot be used unchanged as an additive degree law for Hall extensions.

Proposition 1.48 *(The three simple wall labels).* The simple type-II wall images are

$$(1,1,0), \qquad (0,1,1), \qquad (0,-1,0).$$

Suppose three actual sheaf classes have these images under $\overline{\Pi}_X$, and suppose their geometric elliptic degrees are respectively 1, 2, 1. Then all three classes have wrapped support. Local insertions against them use mixed correspondences, and insertions on both sides use the ordered two-sided flag.

Proof. The wall images are Proposition 2.1. The assumed positive geometric degrees give wrapped support by Proposition 1.8, or by the specified larger-domain degree construction. The definitions of the mixed correspondences then give the stated profiles. The degrees 1, 2, 1 agree with $d = m + 1$ on these three selected labels; the previous proposition prevents extending that affine rule to all their sums as an additive identification. □

Proposition 1.49 *(Wall labels after quotient).* Translation preserves the numerical charges, so the common-torsor quotient preserves the three selected type-II labels and their mixed profiles. It introduces no further divisor wall. The type-I reflections act by the chamber permutations described in Lemma 1.7. Their reflecting vectors are $f_{-2} - f_2$, $f_2 - f_3$ and $f_{-2} - f_3$; they do not become type-II divisor roots under this quotient.

Proof. Translation is connected to the identity and preserves numerical classes. Quotienting the same object and flag stacks consequently retains their labels. The two root types have the different ambient divisibilities and divisor orders stated in Remark 1.5. An elliptic quotient changes neither the reference lattice nor these divisor orders. This conclusion concerns the selected wall classes; it neither constructs their sheaf representatives nor exhausts every possible geometric charge. □

Protected integration is a further map

The coefficient object is not determined by its Euler characteristic. A scalar invariant requires a specified functional on that object, and compatibility with Hall multiplication requires an equation for the functional on extension correspondences.

Definition 1.50 *(Protected integration at a bounded stage).* Let $\mathcal{H}_R = \bigoplus_{\gamma \in \Gamma_R} \mathcal{H}_{\gamma,R}$ be a chosen reduced coefficient realization, with the operations already constructed on its charge domain. Choose a target complex $T_R = \bigoplus_\gamma (T_R)_\gamma$ and degree-zero chain maps

$$I^{\mathrm{prot}}_{\gamma,R} : \mathcal{H}_{\gamma,R} \longrightarrow (T_R)_\gamma.$$

Their sum is I^{prot}_R. A scalar-labelled target has $(T_R)_\gamma = ke_{\omega_R(\gamma)}$ in degree zero; the label map ω_R and its relation to $\overline{\Pi}_X$ must be specified. A finite charge window carries only those target products that are defined on the window, unless a separate algebra quotient has been proved to exist.

Proposition 1.51 *(The integration equations).* The finite sum above is a chain map. It preserves the specified multiplication precisely when, on every permitted pair,

$$I^{\mathrm{prot}}_R m_R = m_{T,R}(I^{\mathrm{prot}}_R \otimes I^{\mathrm{prot}}_R).$$

If source and target also have specified coproducts, preservation of them is the additional equation

$$\Delta_{T,R} I^{\mathrm{prot}}_R = (I^{\mathrm{prot}}_R \otimes I^{\mathrm{prot}}_R)\Delta_R.$$

The relevant shifts, tensor products and units are included in these equations. Neither equation follows from the existence of the individual functionals.

Proof. A finite direct sum of chain maps is a chain map. The displayed identities are the algebra and coalgebra morphism conditions on each homogeneous component. Summing them gives the corresponding condition on every element in the allowed domain. Conversely, restricting an operation-preserving map to a homogeneous component gives those identities. □

PROPOSITION 1.52 *(Primitive compatibility and scalar targets).* A coaugmentation- and coproduct-preserving map sends primitives to primitives. If the target is a monoid algebra with its group-like coproduct $\Delta e_t = e_t \otimes e_t$ and unit e_1, its primitive space is zero. Such a map therefore kills every source primitive.

Proof. Apply the map to the primitive equation. For the second claim, write a primitive as $x = \sum_t c_t e_t$. For $t \neq 1$, the coefficient of $e_t \otimes e_t$ in $\Delta x = e_1 \otimes x + x \otimes e_1$ gives $c_t = 0$. The coefficient of $e_1 \otimes e_1$ then gives $c_1 = 2c_1$, hence $c_1 = 0$ as well. □

This is a restriction on that target coproduct. A numerical integration map may still be useful as an algebra map or trace, without preserving a primitive Lie algebra. To compare operator primitives one needs a target with the appropriate primitive structure and an actual intertwining map.

Another scalar coalgebra can retain a primitive. Let e_n have charge $n \geq 0$, with

$$\Delta e_n = \sum_{a+b=n} e_a \otimes e_b, \qquad \varepsilon(e_n) = \delta_{n0}, \qquad e_a e_b = \binom{a+b}{a} e_{a+b}.$$

The map $e_n \mapsto x^n/n!$ identifies this Hopf algebra with the additive polynomial Hopf algebra, so its primitive space is ke_1. The split components of its coproduct have coefficient one, but its product has binomial coefficients. Using instead the monoid product $e_a e_b = e_{a+b}$ breaks bialgebra compatibility: the coefficient of $e_1 \otimes e_1$ is one in Δe_2 and two in $(\Delta e_1)^2$. Thus a scalar coproduct and a scalar product must be compared together.

If source and target primitive projections π_P, π_T are chosen, their compatibility with integration is the additional equation $I_R^{\mathrm{prot}} \pi_P = \pi_T I_R^{\mathrm{prot}}$. Preservation of the primitive subspace alone does not give this equation for arbitrary projections.

PROPOSITION 1.53 *(Homogeneous integration).* Suppose the target has monomials $q^n r^\ell s^m$ graded by $\overline{\Pi}_X(\gamma) = (n, \ell, m)$. Then an integration map preserves this grading exactly when it maps each source component to the corresponding monomial component. The geometric degree b_R^{geom} does not determine m without the comparison of Definition 1.46.

Proof. The grading condition is the statement that each homogeneous component has the displayed image. Additivity of $\overline{\Pi}_X$ makes target multiplication homogeneous. The geometric degree has a different definition and can be substituted only after the asserted relation between the maps is proved on this domain. □

When the numerical charge index Γ_R is not already the normal-ordered extension, choose its lift $\widehat{\omega}_R : \Gamma_R \to \widehat{\Gamma}$ explicitly. For a target degree map g_R, the homogeneous comparison is

$$g_R \omega_R = \overline{\Pi}_X \widehat{\omega}_R.$$

All three maps respect the permitted additions. The preceding formula uses $\widehat{\omega}_R = 1$ when the source was already indexed by extended charges.

For a transition $\rho_{R'R} : \mathcal{H}_R \to \mathcal{H}_{R'}$ and $t_{R'R} : T_R \to T_{R'}$, the integration compatibility is $t_{R'R} I_R^{\mathrm{prot}} = I_{R'}^{\mathrm{prot}} \rho_{R'R}$. It must hold before forming a limiting map. A finite-type bound alone gives neither this equation nor exactness of the inverse limit.

The compact factorization problem

The K3 stability conditions and their projective moduli [5, 13] provide the surface input. A choice of moduli on $S \times E$ must additionally satisfy the product-stability and support conditions of Appendix 8.

Definition 1.54 *(A local and wrapped factorization datum).* A compact hybrid datum on the chosen charge domain consists of local incidence stacks and wrapped family stacks, their extension and collision diagrams, and an admissible coefficient theory on those diagrams. Its coefficient objects are equipped with the required critical and reduced orientation structures. Its maps include the binary comparisons, all finite flag coherences, local configuration descent, and the separate collision comparisons. Elliptic quotient is taken over a common torsor, with coherent coefficient descent and relative unit BE. Optional anchors and protected integration maps carry the specific comparison diagrams just defined.

At bounded stages all charge sums and intermediate objects must belong to the permitted domain. A family of such data has specified transition maps preserving its objects, operations, units and coherences. The completion, when needed, is formed only with a stated topology and the corresponding inverse-limit or derived inverse-limit construction. We abbreviate this collection of existence and compatibility requirements by $\mathfrak{O}_{\mathrm{hyb},R} = 0$. The notation includes the anchor and integration conditions only when those maps are requested. It is not an assertion that a coefficient object exists because its prospective defects have been named.

Proposition 1.55 *(Scope of the constructed data).* The formal cyclic comparison and its finite-flag extension identify the specified formal sheaf deformations with the corresponding completions of the quiver models. The common-torsor construction gives the relative elliptic quotient of the sheaf extension diagrams. The critical Hall theorem supplies coefficient operations on its unreduced moduli domain. A compact reduced hybrid factorization object follows when these constructions are joined by the coefficient, orientation and collision comparisons in the preceding definition.

Proof. The first assertion is Proposition 1.1 and Theorem 1.3; the second is Proposition 1.35; the third is Theorem 4.2. With the additional comparisons, all prescribed operations and coherences are maps on one coefficient diagram. Proposition 1.29 then gives the claimed factorization operations on that domain. □

Corollary 1.56 *(The compact comparison remains a coefficient problem).* A degree label, a determinant anchor, a product of independent quotients or the scalar Igusa product does not construct the compact reduced coefficient comparisons. Working relatively over BE resolves the common-torsor part of the quotient problem; it leaves the reduced critical maps and their collision compatibilities to be constructed.

The distinction is visible already at a collision. A formal comparison of deformation algebras acts on the completed neighbourhood, whereas a constructible coefficient comparison must act on the actual extension and collision spaces with the required supports. A homotopy defined only after inverting a separation parameter does not solve an equation on its zero locus. These supported maps, together with their reduced orientations, are the next inputs needed for the compact Hall structure.

A holomorphic factorization theory on X would provide another possible source. Its pushforward along p_E and its comparison with the Hall coefficients require specified functors preserving the relevant

supports, tensor products, units and collision operations. The right-D-module chiral construction of [7] and the smooth factorization framework of [25] use different coefficient categories; an equivalence between their realizations is an additional map. Neither the local deformation model nor a common scalar trace supplies that map.

1.2 *Reduced complexes and coherent orientations*

An orientation is a square root together with its squaring map. Its compatibility with an extension is another isomorphism, and compatibility with a flag is an equation between such isomorphisms. These distinctions matter on $X = S \times E$: removing the elliptic translation direction, changing a formal coordinate on E, and descending through a stabilizer all act on the deformation complex. We first construct the complexes and local comparisons. We then formulate the global orientation they must preserve.

THE DEFORMATION COMPLEX AND THE REDUCED RECIPIENT

Fix a smooth projective K3 surface S, an elliptic curve E, and nonzero forms $\sigma_S \in H^0(S, \Omega_S^2)$ and $\omega_E \in H^0(E, \Omega_E^1)$. Their product trivializes K_X. K3 stability conditions and projective moduli of stable objects [5, 13] provide the surface input. On X, we additionally fix the product-stability and support conditions of Appendix 8. For a perfect family $\mathcal{F}$ on $X \times B$, put

$$K_{\mathcal{F}} = R p_* R\mathcal{H}om(\mathcal{F}, \mathcal{F}), \qquad p : X \times B \longrightarrow B.$$

We work on bases for which this complex is perfect and its formation commutes with the base changes in use. Proper smooth projection and a perfect family give this property. At a point represented by F, the cohomology of K_F is $\mathrm{Ext}^*(F, F)$.

PROPOSITION 1.57 *(Restriction of the shifted symplectic structure).* The derived stack of perfect complexes on X carries the (-1)-shifted symplectic structure associated with the chosen volume form. A derived open substack inherits it. On a different derived substack, the pulled-back closed two-form is symplectic only if its induced map from tangent to shifted cotangent complexes is an equivalence. Finiteness of a stability window does not imply this condition. At F, the tangent complex of the unrigidified stack is $K_F[1]$; in particular,

$$H^{-1}(T_F) = \mathrm{Hom}(F, F).$$

For nonzero simple F, its cotangent complex therefore has a nonzero H^1, and is not of amplitude $[-1, 0]$.

Proof. The shifted form on the derived moduli stack is the trace construction of [104]. An open immersion identifies tangent and cotangent complexes, so it preserves nondegeneracy. For a general morphism, pulling back a closed form preserves closedness but gives no such identification. The tangent formula follows from the endomorphism deformation complex. Duality gives $H^1(L_F) = \mathrm{Hom}(F, F)^\vee$. The identity of a nonzero simple object spans a copy of $\mathbb{C}$ in this group. □

PROPOSITION 1.58 *(Critical truncation and orientation coefficients).* Let $\mathfrak{U}$ be a derived scheme locally of finite presentation carrying a (-1)-shifted symplectic form, and let $U = t_0\mathfrak{U}$. Then U has its associated d-critical structure, with canonical line

$$K_{U,s} \simeq \det(L_{\mathfrak{U}})|_{U^{\mathrm{red}}}.$$

A square-root pair for this line defines the oriented vanishing-cycle sheaf on U. On a critical chart $i : R \hookrightarrow V$, with function $f : V \to \mathbb{A}^1$, write ϕ_f for the vanishing-cycle functor shifted by $[-1]$. Its expression is

$$\mathcal{P}_{U,s}|_R \simeq i^*\phi_f \mathbb{Q}_V[\dim V] \otimes_{\mu_2} Q_{R,V,f,i},$$

with the convention for shifted vanishing cycles fixed as in [12, Theorem 6.9]. Here $Q_{R,V,f,i}$ is the principal μ_2-bundle of square-root identifications with i^*K_V.

Proof. The d-critical structure and canonical-line comparison are [12, Theorem 6.6]. The oriented coefficient sheaf and the comparison on embeddings of charts are Theorem 6.9 of that paper. The tensor in the displayed formula uses the sign representation of μ_2. It is a tensor of constructible coefficients; it is not a tensor with the coherent orientation line. On an Artin stack one must also supply compatible smooth-atlas descent. A reduced or rigidified moduli problem must first satisfy the hypotheses on its own derived recipient. □

Choose the translation vector v_E dual to ω_E. Contraction of the Atiyah class gives its infinitesimal action

$$u_F = \iota_{v_E} \mathrm{At}(F) \in \mathrm{Ext}^1(F, F).$$

Let $\mathrm{Tr}_X : \mathrm{Ext}^3(F, F) \to \mathbb{C}$ be the Serre trace with the chosen volume form.

Proposition 1.59 *(Semiregularity and the translation direction).* The functional

$$\sigma_F : \mathrm{Ext}^2(F, F) \longrightarrow \mathbb{C}, \qquad \xi \longmapsto \mathrm{Tr}_X(\xi \circ u_F)$$

is dual to the infinitesimal translation class. It is nonzero, hence surjective, precisely when $u_F \neq 0$. With compatible relative Atiyah and trace maps it is induced by a morphism

$$K_{\mathcal{F}} \longrightarrow \mathcal{O}_B[-2].$$

Proof. Serre duality gives a perfect pairing between $\mathrm{Ext}^2(F, F)$ and $\mathrm{Ext}^1(F, F)$. Evaluation against u_F proves the pointwise assertion. The relative Atiyah class, contraction, composition and trace give the displayed morphism in families. Its target has cohomology in degree two, as the functional requires. Equivalently, the Atiyah–trace expression can be paired with the pulled-back K3 two-form; the volume convention fixes the sign of that expression. A statement of surjectivity on a family needs nonvanishing at every relevant point, not only at its generic point. □

Definition 1.60 *(A cosection fibre).* Let K_0 be a specified perfect deformation complex with a morphism $\sigma : K_0 \to \mathcal{O}_B[-2]$. Its cosection fibre is

$$K^{\mathrm{fib}} = \mathrm{fib}(\sigma), \qquad K^{\mathrm{fib}} \longrightarrow K_0 \xrightarrow{\sigma} \mathcal{O}_B[-2].$$

One possible K_0 is the fibre of the trace map $K_{\mathcal{F}} \to R p_* \mathcal{O}_{X \times B}$; using it requires restricting the cosection to that fibre. We use the notation K^{red} only after specifying which scalar, orbit and dual obstruction directions the reduced deformation problem removes.

Proposition 1.61 *(Perfectness and scalar descent).* The fibre of a morphism of perfect complexes is perfect. The endomorphism complex of a family on a scalar $\mathbb{G}_m$-gerbe has weight zero and descends to the rigidified base, whenever the stated perfect-complex descent applies. The trace, relative Atiyah class along X, contraction and cosection maps descend with it. These statements do not assert that the universal family, of scalar weight one, descends.

Proof. Perfect complexes form a stable category. Scalar conjugation acts trivially on $R\mathcal{H}om(\mathcal{F},\mathcal{F})$, since the two scalar weights cancel. Locally trivialize the gerbe and apply descent to this weight-zero complex and its natural maps. The same cancellation proves compatibility on overlaps. It does not cancel the weight of $\mathcal{F}$ itself. □

The trace-free fibre need not remove scalar automorphisms. Indeed, $\operatorname{tr}(\mathrm{id}_F) = \operatorname{rk}(F)$; for a rank-zero sheaf, the identity survives this trace. Likewise, deleting one obstruction quotient is not by itself a symplectic reduction. At a point where $u_F \neq 0$, Serre duality induces a perfect pairing between $\operatorname{Ext}^1(F,F)/\mathbb{C}u_F$ and $\ker \sigma_F \subset \operatorname{Ext}^2(F,F)$. This is the linear reduction of the orbit direction and its dual obstruction. Constructing the corresponding family, its derived critical charts and their gluing remains necessary. A perfect obstruction theory, a cosection-localized cycle [67], and such an oriented critical family are different constructions.

There are positive reduced geometric constructions in more specific moduli problems. For stable pairs on $S \times E$ with nonzero K3 curve class, the quotient and reduced virtual classes of [99, Theorems 1–2] satisfy the comparison stated in Theorem 4.7. That result retains its precise stable-pair domain. Extending it to the reduced critical coefficient and complete-flag structures above is a further problem.

Square-root pairs and their obstruction

For a perfect complex K, use the determinant convention

$$\det K = \bigotimes_j (\det K^j)^{(-1)^j}.$$

Thus $\det(K[1]) = (\det K)^{-1}$ and $\det(K^\vee) = (\det K)^{-1}$. Determinants of exact triangles carry the usual functorial identifications [33, 70].

Definition 1.62 *(The reduced determinant line).* A specified perfect reduced complex determines $\mathscr{D}^{\mathrm{red}} = \det K^{\mathrm{red}}$. An orientation is a pair

$$(o,\rho), \qquad \rho : o^{\otimes 2} \xrightarrow{\sim} \mathscr{D}^{\mathrm{red}}.$$

When this orientation is used for critical coefficients, a comparison between $\mathscr{D}^{\mathrm{red}}|_{B^{\mathrm{red}}}$ and the canonical d-critical line is part of the datum. A shift or dualization of the deformation complex changes that comparison.

PROPOSITION 1.63 *(Kummer obstruction and choices).* Over a complex scheme or stack on its relevant descent site, let $\delta : \operatorname{Pic}(B) \to H^2(B,\mu_2)$ be the Kummer boundary. A line $\mathscr{D}$ has a square-root pair if and only if $\alpha = \delta(\mathscr{D})$ vanishes. If it vanishes, the isomorphism classes of square-root pairs form a torsor under $H^1(B,\mu_2)$; the automorphisms of a pair form $H^0(B,\mu_2)$. The underlying root lines, after forgetting their squaring maps, form a torsor under $\operatorname{Pic}(B)[2]$.

Proof. Apply cohomology to

$$1 \longrightarrow \mu_2 \longrightarrow \mathbb{G}_m \xrightarrow{z\mapsto z^2} \mathbb{G}_m \longrightarrow 1;$$

see [109, Tags 03PK, 03PL]. Local roots of transition functions have a μ_2-valued defect on triple overlaps. This is $\delta(\mathscr{D})$, and a trivializing cochain glues a root pair. The ratio of two pairs is a line with a trivialization of its square, which is a μ_2-torsor. A root-preserving automorphism is a section of μ_2. Forgetting the trivialization gives the last assertion. In general $H^1(B,\mu_2)$ is larger than $\operatorname{Pic}(B)[2]$, because units modulo squares also occur. □

Proposition 1.64 *(Integral and rational characteristic classes).* The image of the Kummer class of a complex line in analytic cohomology is its integral first Chern class reduced modulo two. It is also the second Stiefel–Whitney class of the underlying oriented real plane. A rational Grothendieck–Riemann–Roch formula alone does not determine this class.

Proof. The first assertion follows by comparing the square map on transition functions with the integral exponential sequence. For a complex line, $w_2 = c_1 \bmod 2$, as in [90]. The map from integral to rational cohomology kills torsion, so its value cannot recover arbitrary mod-two information. For example, the sign line on $B\mu_2$ has nonzero Kummer obstruction although its rational first Chern class vanishes. Its character has no square root character of μ_2. □

For a proper smooth morphism $p : Y \to B$ of smooth schemes and a perfect complex $\mathcal{K}$, the rational calculation retains its usual form:

$$c_1(\det R p_* \mathcal{K})_{\mathbb{Q}} = \left[p_* \big(\operatorname{ch}(\mathcal{K}) \operatorname{td}(T_p) \big) \right]_1 .$$

It is the degree-one part of Grothendieck–Riemann–Roch [109, Tag 02UO]. The Deligne–Mumford stack version uses the appropriate representation-valued theory [111]. Neither rational calculation removes the need for the integral Kummer class.

A cosection is a morphism of complexes on B, not a map of bases along which a line can be pulled back. If a reduction constructs an additional perfect complex C_{cs}, one may instead define

$$\mathscr{D}_{\mathrm{tw}} = \mathscr{D}^{\mathrm{red}} \otimes \det C_{\mathrm{cs}}, \qquad \alpha_{\mathrm{red}} = \delta(\mathscr{D}_{\mathrm{tw}}).$$

This notation requires the construction of C_{cs}. The cokernel of a cosection has a determinant only when supplied with the requisite perfect model. The root criterion applies to the actual line $\mathscr{D}_{\mathrm{tw}}$, including its torsion information.

Formal coordinates and the full determinant unit

Consider sheaves near one fibre of $S \times E$, annihilated by a locally uniform power of a chosen normal coordinate t. Such a sheaf is equivalently a coherent sheaf F on S with a nilpotent endomorphism N_F: multiplication by t is N_F. Restriction and extension of this module structure preserve exact sequences and finite flags. With strict endomorphisms on chosen resolutions, put

$$C(F,G) = R \operatorname{Hom}_S(F,G), \qquad \mathcal{K}(F,G) = C(F,G) \otimes \Lambda(\theta), \quad |\theta| = 1.$$

For a homogeneous coefficient a, the differential and composition are

$$D(a\theta^{\epsilon}) = d_C a\, \theta^{\epsilon} + (-1)^{|a|}(N_G a - a N_F)\theta^{\epsilon+1},$$
$$(a\theta^{\epsilon})(b\theta^{\delta}) = (-1)^{\epsilon|b|} ab\, \theta^{\epsilon+\delta}, \qquad \theta^2 = 0.$$

The relative resolution

$$0 \longrightarrow \mathcal{O}_S[t] \otimes F \xrightarrow{\,t - N_F\,} \mathcal{O}_S[t] \otimes F \longrightarrow F \longrightarrow 0$$

combined with a resolution on S computes this Hom complex. The commutator with N commutes with d_C and is a derivation, so these formulas define a dg category. This construction uses the actual endomorphisms; it does not require formality of $C(F,G)$.

Let $f(t) = \sum_{r\geq 1} c_r t^r$, with c_1 invertible. For composable maps $a_i : F_{i-1} \to F_i$, define

$$D_n^f(a_n, \dots, a_1) = \sum_{r\geq n} c_r \sum_{j_0+\cdots+j_n=r-n} N_{F_n}^{j_n} a_n N_{F_{n-1}}^{j_{n-1}} \cdots a_1 N_{F_0}^{j_0}.$$

At fixed arity every sum is finite under the chosen nilpotence bounds. The coefficient maps need not commute with N. The first divided difference satisfies

$$D_1^f(N_G a - a N_F) = f(N_G)a - a f(N_F).$$

THEOREM 1.65 *(Coordinate functors at every arity).* Use suspension s of degree -1, with bar operations

$$b_1(sa) = -s(Da), \qquad b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

There is a strictly unital A_∞ functor $\mathcal{F}_f$ from the category with endomorphisms N_F to that with endomorphisms $f(N_F)$. Its first component is the identity on unmarked coefficients and D_1^f on the θ coefficient. For $n \geq 2$, its suspended components are

$$(\mathcal{F}_f)_n\big(s(a_n\theta), \dots, s(a_1\theta)\big) = s\big(D_n^f(a_n, \dots, a_1)\theta\big),$$

and vanish if any input is unmarked. For another invertible coordinate g, these coalgebra maps satisfy

$$\mathcal{F}_g \mathcal{F}_f = \mathcal{F}_{g\circ f}, \qquad \mathcal{F}_{f^{-1}} \mathcal{F}_f = \mathrm{id}.$$

Proof. On marked inputs the suspended degree is the coefficient degree. Consequently the internal differential obeys exactly the graded Leibniz rule for D_n^f; the closed endomorphisms N_F have degree zero. Products of two marked inputs vanish. In the remaining higher identity, exactly one input is unmarked. Replacing that input by its commutator with N makes the inner sum telescope over the adjacent exponent. For the monomial $f(t) = t^r$, its endpoint terms are the two adjacent compositions, with the tensor signs dictated by b_1, b_2. Linearity and finite nilpotence give the identity for f. With two or more unmarked inputs only the ordinary binary composition identity can survive. The unit is unmarked, so all higher components containing it vanish.

For composition, place the N_{F_i} on the diagonal of an upper triangular block matrix and the a_i on its first off-diagonal. The relevant n-step block of f of this matrix is D_n^f. Applying g and summing over intervening blocks is precisely the partition formula for composition of the suspended functors. The identity $g(f(M)) = (g \circ f)(M)$ proves the claimed equality. The same calculation with the inverse formal series proves the inverse identity at each nilpotence bound. □

Already in arity two, the higher map is needed:

$$D_1^f(ab) - a D_1^f(b) = D_2^f(N_H a - a N_G, b).$$

It is this identity, with its graded counterparts, that corrects the failure of the first component to preserve composition.

The K3 Serre trace $\tau_S : C(F,F) \to \mathbb{C}[-2]$ gives a CY3 trace on the θ coefficient. For complementary powers of θ, the pairing is

$$\langle a\theta^\epsilon, b\theta^\delta \rangle = (-1)^{\epsilon|b|} \tau_S(ab), \qquad \epsilon + \delta = 1.$$

If the normal volume is $h(t)\,dt$, its coefficient after changing to $f(t)$ is $h(t)/f'(t)$. Cyclicity gives the exact trace identity

$$\tau_S\Big(h(N)f'(N)^{-1}D_1^f(b)\Big) = \tau_S(h(N)b).$$

Indeed, for any coefficient $k(N)$, cyclicity gives $\tau_S(k(N)D_1^f(b)) = \tau_S(k(N)f'(N)b)$. Take $k(N) = h(N)f'(N)^{-1}$. This proves transport of the displayed trace functional. A comparison of negative cyclic classes or a global critical coefficient functor requires its own construction.

Proposition 1.66 *(The determinant and its nonreduced square root).* Let the coefficient base be a local Artin $\mathbb{C}$-algebra R with residue field $\mathbb{C}$, and let $C(F,G)$ have a bounded finite free model. The determinant of the coordinate map on $\mathcal{K}(F,G)$ is

$$d_f(F,G) = \prod_j \det\Big(D_1^f|_{C^j(F,G)}\Big)^{(-1)^{j+1}}.$$

On a self-Hom complex put $\chi = \chi(C(F,F))$. Suppose its reduction has determinant $\bar{c}_1^{-\chi}$, and χ is even. Then $\rho_f = c_1^{\chi}\,d_f(F,F)$ lies in $1+\mathfrak{m}_R$, and the unique root with reduction $\bar{c}_1^{-\chi/2}$ is

$$j_f(F) = c_1^{-\chi/2}\exp\Big(\tfrac{1}{2}\log\rho_f\Big), \qquad j_f(F)^2 = d_f(F,F).$$

Proof. The unmarked part of the complex is unchanged, whereas the marked copy of C^j is in degree $j+1$. This gives the determinant exponent. Both logarithm and exponential are finite series on the nilpotent maximal ideal. Squaring is an automorphism of $1+\mathfrak{m}_R$, proving existence and uniqueness. On the reduced K3 self-Hom complex, evenness of the Mukai lattice gives $\chi(F,F)\in 2\mathbb{Z}$. The determinant of a polynomial in the commuting nilpotent left and right actions has constant diagonal c_1, which gives the stated reduced determinant. □

For example, take $R=\mathbb{C}[\varepsilon]/(\varepsilon^2)$, $F=\mathcal{O}_S$, $N=\varepsilon\,\mathrm{id}$, and $f(t)=t+at^2$. The two self-Ext groups in degrees zero and two each see $D_1^f = 1+2a\varepsilon$. Hence

$$d_f(F,F) = 1-4a\varepsilon, \qquad j_f(F) = 1-2a\varepsilon.$$

Their reduced values are both one, but the actual coordinate comparison is not. Nilpotent units cannot be discarded from an orientation map.

Equivariant square roots and the Borel filtration

Let M be connected and carry an E-action and an equivariant determinant line $\mathcal{D}$. An orientation on $[M/E]$ is an equivariant square-root pair on M. Equivariance includes its cocycle on $E\times E\times M$. We write $M_E = EE\times_E M$ for the analytic homotopy quotient when discussing mod-two characteristic classes.

Proposition 1.67 *(The base part of an equivariant obstruction).* The degree-two base term in the Borel spectral sequence is

$$E_\infty^{2,0} = H^2(BE,\mathbb{F}_2)/\mathrm{im}\big(d_2 : H^1(M,\mathbb{F}_2)\to H^2(BE,\mathbb{F}_2)\big).$$

It identifies with the image of $H^2(BE,\mathbb{F}_2)$ in $H^2(M_E,\mathbb{F}_2)$. There is generally no canonical projection from the latter group to $H^2(BE,\mathbb{F}_2)$. If a class α_E restricts to zero on M, then it is the pullback of a base class a, determined modulo $\mathrm{im}\,d_2$. It vanishes precisely when that representative lies in $\mathrm{im}\,d_2$.

Proof. Topologically E is a real two-torus, so BE is $K(\mathbb{Z}^2, 2)$, with $H^1(BE, \mathbb{F}_2) = 0$. In total degree two, the only incoming differential to $(2, 0)$ is the stated transgression, and no outgoing differential survives. The filtration has no $(1, 1)$ term. Hence a class vanishing on the fibre lies in its $(2, 0)$ piece. The edge map is a pullback from the base, not a projection onto it. Choosing generators u_1, u_2 of $H^2(BE, \mathbb{F}_2)$ writes a representative as $a_1u_1 + a_2u_2$, but does not make that representative unique. □

For the translation action of E on itself, the homotopy quotient is contractible. Both base generators are transgressions. This free action gives a concrete reason that two coefficients in $H^2(BE)$ cannot be assigned canonically to an arbitrary equivariant class. On a free locus admitting a geometric quotient, the equivariant root problem is the ordinary root problem on that quotient.

PROPOSITION 1.68 *(Equivariant nulltrivializations).* The vanishing of $\delta(\mathscr{D})$ on $[M/E]$ is equivalent to the existence of an equivariant square-root pair. Its choices form a torsor under $H^1([M/E], \mu_2)$, with automorphism group $H^0([M/E], \mu_2)$. A nonequivariant root and a vanishing ordinary class on M do not specify this equivariant nulltrivialization.

Proof. Apply the Kummer argument on the quotient stack, or apply it on the simplicial action presentation. The equations on double and triple overlaps include the action and its composition. Forgetting those equations gives only the nonequivariant root. □

FINITE STABILIZERS AND CHARACTER LINES

For an object x with finite stabilizer $H \subset E$, restriction along $BH \to [M/E]$ is always defined. It requires no choice of a Borel edge splitting. Such restrictions are necessary local tests; they need not detect the whole equivariant obstruction. Write $\beta_x \in H^2(BH, \mu_2)$ for the restricted gerbe class.

PROPOSITION 1.69 *(Finite root gerbes).* The root gerbe at x is neutral precisely when $\beta_x = 0$. In that case root pairs form an $H^1(BH, \mu_2)$-torsor. For the root gerbe of an actual line, write its character as $\chi \in \mathrm{Hom}(H, \mathbb{C}^*)$, additively. Its root characters are exactly the solutions of

$$2\lambda = \chi.$$

Proof. A line on BH is a one-dimensional representation. Its tensor square doubles its character. The Kummer sequence therefore gives both the obstruction and the stated torsor. More generally, a μ_2-gerbe has a normalized group two-cocycle; a trivialization is a cochain whose coboundary is that cocycle. □

PROPOSITION 1.70 *(Stack descent and inertia).* A root character λ need not vanish for an orientation on BH or on the quotient stack. Vanishing is an additional condition for descent to the coarse space, or for a specified inertia-trivial normalization. Existence of a root and this additional condition are distinct requirements.

Proof. Take $H = \mu_2$ and the trivial determinant line. Both the trivial line and the sign line, with their squaring maps, are roots on $B\mu_2$. The sign line does not descend to the point, because its inertia acts nontrivially. Conversely the determinant sign character itself has no root character: doubling either character of μ_2 gives zero. Thus a gerbe obstruction precedes the choice of root linearization, while coarse descent further restricts that choice. □

The finite subgroups of a complex elliptic curve are abelian of rank at most two. Their odd-primary part contributes no positive-degree cohomology with $\mathbb{F}_2$ coefficients. The following formulas include cyclic subgroups and unequal two-primary factors.

Proposition 1.71 *(Two-primary classes).* Let $H_{(2)} = C_{2^a} \times C_{2^b}$, omitting a factor when it is trivial. A nontrivial cyclic factor has generators x in degree one and u in degree two, with

$$H^*(BC_{2^a}, \mathbb{F}_2) = \begin{cases} \mathbb{F}_2[x], & a = 1, \quad u = x^2, \\ \Lambda(x) \otimes \mathbb{F}_2[u], & a > 1. \end{cases}$$

For two nontrivial factors,

$$H^2(BH, \mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2(x_1 x_2) \oplus \mathbb{F}_2 u_2.$$

If a character line has exponents k_1, k_2 on these factors, its Kummer class is

$$\delta(\chi) = (k_1 \bmod 2)u_1 + (k_2 \bmod 2)u_2.$$

In particular the mixed term of an arbitrary gerbe is available in H^2, but it cannot occur in the root gerbe of a character line. When $a = b \geq 2$, the nonzero class $x_1 x_2$ restricts to zero on every cyclic subgroup of $H_{(2)}$. Thus cyclic restrictions do not determine an arbitrary degree-two gerbe class.

Proof. The periodic resolution of a cyclic group computes the stated ring: the square of the degree-one generator is nonzero for order two and zero for higher powers of two. The Künneth formula gives the three degree-two summands. A character of the product is a tensor product of characters pulled back from the two factors. The Kummer boundary is additive and commutes with pullback. On a cyclic factor, doubling characters has cokernel $\mathbb{F}_2$, detected by the parity of its exponent; this defines u. There is therefore no mixed term. Restriction and transfer for the odd-primary subgroup show that its positive-degree $\mathbb{F}_2$-cohomology vanishes. To check the cyclic restrictions when $a = b \geq 2$, let $C \subset H_{(2)}$ be cyclic. If C has order two, then $C \subset 2H_{(2)}$, so both mod-two characters x_1, x_2 restrict to zero. If $|C| \geq 4$, its degree-one cohomology is spanned by a class ξ with $\xi^2 = 0$. Both restricted characters are multiples of ξ, so their product is zero. The trivial subgroup also has zero degree-two cohomology. The Künneth summand above shows that $x_1 x_2$ itself is nonzero. □

Proposition 1.72 *(The Klein four calculation).* For $H = C_2 \times C_2$, every degree-two class has a unique expression

$$\beta = b_{20} x_{\mathrm{I}}^2 + b_{11} x_1 x_2 + b_{02} x_2^2.$$

Its restrictions to the first factor, second factor and diagonal have coefficients b_{20}, b_{02} and $b_{20} + b_{11} + b_{02}$, respectively. These three restrictions detect β. For a line-root gerbe one already has $b_{11} = 0$.

Proof. Pull back the polynomial ring $\mathbb{F}_2[x_1, x_2]$ along the three inclusions. The diagonal sends both generators to the same class x. The resulting three linear equations recover all three coefficients. The final assertion is the character-line calculation above. For higher cyclic orders, the degree-one restriction to the order-two subgroup may vanish; the same three-test argument cannot be transferred without computing those restriction maps. □

Proposition 1.73 *(Scope of stabilizer detection).* An equivariant root restricts to a root at every stabilizer. The converse requires a descent statement for the global gerbe, including its variation on the moduli space and the action overlaps. It does not follow from the pointwise classes β_x alone.

Proof. Restriction preserves square-root pairs. Pointwise stabilizer restriction forgets the remaining overlap maps. Already for a free action all stabilizers are trivial, whereas the quotient may carry a line without a square root. Thus stabilizer tests cannot in general recover its Kummer class. □

CROSS DETERMINANTS AND COMPLETE FLAGS

For a perfect pair on a proper CY3, put

$$L(F,G) = \det R\operatorname{Hom}_X(F,G), \qquad \mathscr{D}(F) = \det R\operatorname{Hom}_X(F,F).$$

Serre duality identifies $R\operatorname{Hom}(G,F) \simeq R\operatorname{Hom}(F,G)^\vee[-3]$. Both the dual and the odd shift invert the determinant, so their composite preserves it.

PROPOSITION 1.74 *(The cross-determinant square).* For a direct sum, and for an exact triangle represented by a strict filtered perfect model with subquotients F, G, determinant functoriality gives

$$\mathscr{D}(E) \simeq \mathscr{D}(F) \otimes \mathscr{D}(G) \otimes L(F,G)^{\otimes 2}.$$

For a strict finite flag with subquotients $F_1, \dots, F_r$,

$$\mathscr{D}(E) \simeq \bigotimes_i \mathscr{D}(F_i) \otimes \bigotimes_{i<j} L(F_i,F_j)^{\otimes 2}.$$

These determinant identifications are compatible with regrouping the flag, with the determinant signs fixed by its order.

Proof. The direct-sum endomorphism complex has its four Hom blocks. The two cross blocks have equal determinants by CY3 duality. For a filtered model, filter the endomorphism complex by the difference of the flag indices. Its associated graded consists of all the Hom blocks. Multiplicativity of determinant on this filtration gives the same formula without splitting the extension. Refining or regrouping the filtration gives the canonical determinant comparison for the same filtered complex. Pair the off-diagonal blocks using Serre duality in their fixed order. □

A trace-free or reduced construction changes this formula. For specified reduced determinant lines define the excess line

$$A(F,G) = \mathscr{D}^{\mathrm{red}}(E) \otimes \big(\mathscr{D}^{\mathrm{red}}(F) \otimes \mathscr{D}^{\mathrm{red}}(G) \otimes L(F,G)^2\big)^{-1}.$$

One must compute $A(F,G)$ and supply its root if a square-root multiplication uses it. Reducing the two endpoints independently need not give the reduction of their extension.

For a concrete trace calculation take

$$H = \mathbb{C}[u]/(u^2) \otimes \Lambda(\eta), \quad |u| = 2, \quad |\eta| = 1, \quad \tau(u\eta) = 1.$$

In the trace-free block of sizes $n, m > 0$, the common scalar complement is

$$t \longmapsto \operatorname{diag}(tI_n/n, -tI_m/m).$$

Its trace pairing is $(n+m)\tau(tt')/(nm)$. Deleting each block trace separately would delete this nonzero paired summand. For a flag of block sizes n_i, the same complement is $\{(t_i) : \sum_i t_i = 0\}$, with pairing $\sum_i \tau(t_i t_i')/n_i$. Regrouping a flag retains this one common trace condition.

This calculation also has nonsplit families. Over $\mathbb{C}[t]$, take the upper triangular trace-free matrices spanned by $h = \operatorname{diag}(1,-1)$ and $e = e_{12}$, tensored with H, and differential $[t\eta e, -]$. Then

$$dh = -2t\eta e, \qquad de = 0.$$

The cohomology contains $\mathbb{C}[t]/(t)$ classes represented by ηe and $u\eta e$, in degrees one and three. Replacing the family by a split fibre loses this torsion and its derived specialization. Determinants of the actual filtered complex remain well defined throughout the family.

Definition 1.75 *(Strong reduced orientation).* On a specified reduced moduli problem, a *strong reduced orientation* consists of a root pair (o, ρ), compatible equivariant descent, and extension isomorphisms

$$\gamma_{F,G} : o_E \xrightarrow{\sim} o_F \otimes o_G \otimes L(F,G) \otimes a(F,G), \qquad a(F,G)^2 \simeq A(F,G).$$

Their squares are the prescribed determinant comparisons. On every three-step flag, the two composites of γ, including the cross and excess lines, agree. They agree with the chosen unit on the zero object and commute with the stated transitions and base changes. These are equations of line isomorphisms over the full bases, including their nilpotents.

Proposition 1.76 *(Coordinate roots on strict flags).* For a strict flag in the K3 formal endomorphism category, with the Serre-compatible determinant comparisons above, the coordinate factors satisfy

$$d_f(E,E) = d_f(F,F)d_f(G,G)d_f(F,G)^2, \qquad j_f(E) = j_f(F)\,j_f(G)d_f(F,G).$$

The second formula uses the specified Artin roots and extends to all finite flags by multiplication over their ordered pairs.

Proof. The upper triangular endomorphism acts on the filtered Hom complex; its associated graded has the four Hom blocks. Determinants are products on that filtration. The divided-difference operators on opposite cross blocks are adjoint for the K3 Serre pairing, so their determinants agree. This proves the first formula. The Euler characteristics obey $\chi(E,E) = \chi(F,F) + \chi(G,G) + 2\chi(F,G)$. Consequently both sides of the root formula have the same square and the same reduced value $\bar{c}_1^{-\chi(E,E)/2}$. Uniqueness of the root in $1 + \mathfrak{m}_R$ proves equality on the full Artin base. Refinement of the same filtration proves the all-flag formula. □

Even squares and units do not force the three-step equation. Starting with compatible maps indexed by positive masses, multiply $\gamma_{2,1}$ by -1 and leave the other maps unchanged, including the unit maps. The two composites at masses $(1,1,1)$ then differ by -1, while all squares are unchanged. There are nevertheless two useful positive corrections. A normalized mass-only sign three-cocycle $c(a,b,c)$ on $\mathbb{N}$ is the coboundary of

$$\beta(a,b) = \prod_{j=0}^{b-1} c(a,j,1).$$

The cocycle equation telescopes to $\partial\beta = c$, so rescaling by β removes this mass-only associator. For a normalized constant sign two-cocycle α on a finite group G of odd order, $b(g) = \prod_{h\in G} \alpha(g,h)$ similarly satisfies $\partial b = \alpha^{|G|} = \alpha$. Neither calculation trivializes a sign cocycle varying over the moduli space. For example, $\alpha((a,b),(c,d)) = (-1)^{ad}$ on C_2^2 is not a coboundary: it is nonsymmetric, whereas a coboundary on an abelian group with trivial coefficients is symmetric.

Oriented coefficients and Hall operations

On an oriented critical moduli scheme, the pointwise Euler characteristic of the vanishing-cycle sheaf is the Behrend function ν; the sign local system has rank one and does not change this Euler characteristic. This is the numerical comparison of [12, Remark 6.14], underlying the weighted-Euler theories of [6, 62]. Passing to ν forgets monodromy and extension morphisms. Those data are still needed for a product on critical cohomology.

PROPOSITION 1.77 *(Thom–Sebastiani and quotient descent).* Compatible strong orientation data identify the sign torsors in Thom–Sebastiani comparisons and descend the oriented coefficient sheaves to the quotient stack. These comparisons use the actual critical charts and their embeddings. They do not follow from an isomorphism between determinant classes in a Picard group alone.

Proof. On a critical chart the coefficient is twisted by the principal bundle of square-root identifications. An orientation isomorphism identifies these bundles; equality on triple overlaps gives equality of their descent maps. Apply the Thom–Sebastiani isomorphism to the vanishing-cycle factor and tensor its sign-bundle comparison. Equivariance supplies the same equations on the action presentation. This constructs descent to the stack, where stabilizer characters are retained. Coarse descent would impose the additional inertia-trivial condition discussed above. □

PROPOSITION 1.78 *(Operations require their coefficient morphisms).* For an E-equivariant moduli space M, put $\mathcal{M} = [M/E]$. Suppose an extension correspondence

$$\mathcal{M} \times_{BE} \mathcal{M} \xleftarrow{p} \mathcal{E} \xrightarrow{q} \mathcal{M}$$

has a specified critical coefficient comparison, with its shifts, support conditions and functors, giving $q_! p^*(\mathcal{P} \boxtimes_{BE} \mathcal{P}) \to \mathcal{P}$. Fix a cohomology theory in which the specified pull–push and Künneth maps exist, so that this comparison induces the Hall product. Assume the two resulting comparisons agree on three-step flags and satisfy the relevant base-change identities. Then the induced Hall product is associative. A collision coproduct requires a separately specified correspondence and coefficient morphism, with its own coassociativity and support identities. The orientation and the product alone do not supply it.

Proof. Both parenthesizations of the product pull back to the complete three-step flag correspondence. Proper or supported base change, projection and the assumed coefficient equality identify their pushforwards. The same argument proves coassociativity only after the collision coefficient morphism and its complete-flag equality have been supplied. Inverting Thom–Sebastiani on a local product chart does not construct this global morphism on collision support. □

The tensor is over BE because both endpoints of an extension use one elliptic torsor. Quotient the whole correspondence with its diagonal action before separating its maps. The zero object retains its map to BE; it is not replaced by a point before checking the unit. The common-torsor equivalence of Section 1.3 fixes this convention independently of any reduction of the deformation complex.

PROPOSITION 1.79 *(Primitive projections after descent).* Let a descended coefficient theory already carry the specified connected conilpotent cocommutative Hopf structure over a characteristic-zero field. The convolution logarithm of the identity is then its projection to the kernel of the reduced coproduct on the domain proved in Section 1.6. If transition functors preserve the Hopf maps and their finite convolution sums, they preserve this projection. Without these hypotheses, a chosen splitting of the primitive kernel is additional data.

Proof. The formula for the projection is a finite sum on every conilpotence stage and is natural for the given Hopf maps. Apply the transition functor term by term. An orientation comparison by itself asserts none of the coalgebra, connectedness or exactness hypotheses needed for that formula. □

Weyl transport as a root-groupoid problem

Let $W = W^{(2)}(\Lambda_{II}^{2,1})$. Its three simple reflections have the presentation described in Lemma 1.7. To transport an orientation, one first needs geometric maps or correspondences between the relevant strata; the lattice reflection alone gives no pullback of a line. For clarity, write g^* for the specified pullback functor, with its composition identifications.

Definition 1.80 *(The groupoid of defined transports).* Choose strata and specified invertible geometric transports between them. Let $\mathcal{G}$ be their groupoid, with all defined composites and their relations. A finite set of strata or generating arrows need not make $\mathcal{G}$ finite: there may be infinitely many distinct loops. A finite cochain calculation applies only after a finite model for the cochains in the required degrees has been specified.

Definition 1.81 *(Root transport and its sign cocycle).* Suppose each stratum U has a root pair (o_U, ρ_U), and each arrow $g : U \to V$ has a determinant comparison $d_g : \mathscr{D}_U \to g^*\mathscr{D}_V$. A lift is an isomorphism

$$\tau_g : o_U \longrightarrow g^* o_V, \qquad (g^*\rho_V)\tau_g^{\otimes 2} = d_g \rho_U.$$

When the determinant comparisons are coherent, two composites of such lifts differ by a section of μ_2. In particular define

$$c(g,h) = \tau_{hg}^{-1}\,(g^*\tau_h)\,\tau_g \in H^0(U, \mu_2).$$

This is a two-cocycle on $\mathcal{G}$ with the pullback coefficient system $U \mapsto H^0(U, \mu_2)$. Strong Weyl transport, denoted $(O1)^+$, is a coherent lift preserving the full root, extension, unit and equivariant descent data.

Proposition 1.82 *(Existence, torsor defects and strictification).* Given d_g, the obstruction to identifying the two root pairs is the class of their ratio in $H^1(U, \mu_2)$. If it vanishes, lifts form a torsor under $H^0(U, \mu_2)$. Once lifts exist, they can be made coherent precisely when $[c] = 0$. Multiplying lifts by a one-cochain b changes c by δb. For an involution s, coherent normalized transport satisfies

$$(s^*\tau_s)\tau_s = \mathrm{id}.$$

The determinant square equation alone does not impose this identity.

Proof. Use d_g to regard g^*o_V and o_U as roots of one line. Their ratio is a μ_2-torsor. A trivialization is the required lift, and its automorphisms give the stated freedom. Associativity of three lifted arrows proves the cocycle equation. Substitution of $b(g)\tau_g$ gives the coboundary formula, so strictification is equivalent to $[c] = 0$. The pair of arrows (s, s) gives the involutive relation. Before strictification their composite may instead be the nontrivial automorphism of the root. □

For constant coefficients $\mathbb{F}_2$, a specified finite cochain model makes the equation $\delta b = c$ a linear system. It is soluble if and only if adjoining c to the matrix of $\delta : C^1 \to C^2$ does not increase its rank. One must use cochains for the actual groupoid. A transitive action groupoid is equivalent to the stabilizer group of one object. Its cohomology is therefore stabilizer cohomology, not automatically $H^*(W)$. Only an action on a single fixed object gives BW.

Proposition 1.83 *(Naturality of the obstruction data).* A determinant comparison transports the Kummer class by pullback. A root lift transports its chosen nulltrivialization. At a stabilizer isomorphism $\varphi : H_U \to H_V$, the equations are

$$\beta_U = \varphi^*\beta_V, \qquad \lambda_U = \varphi^*\lambda_V.$$

If inertia-trivial normalization is imposed, it is preserved by the same equation with $\lambda_U = \lambda_V = 0$. The base-filtration part of a Borel class is preserved as a filtration class, without choosing representatives in $H^2(BE)$.

Proof. Kummer boundaries, restrictions and character pullbacks commute with the geometric pullback. A root isomorphism gives the same comparison before passing to cohomology, hence transports the nulltrivialization as well. In chosen cohomology bases, if A_φ represents φ^*, the defect is $\beta_U - A_\varphi \beta_V$, and similarly in degree one. This fixes the variance of the finite matrix equation. Naturality of the spectral sequence preserves its filtration and quotient terms; it does not create a splitting. □

PROPOSITION 1.84 *(Compatibility of strictifications).* Suppose transition pullbacks r identify two systems of projective lifts with cocycles $c_R, c_{R'}$. Cochains killing these cocycles give compatible strictifications when their restrictions differ by the specified change of trivialization. In constant additive coefficients this condition is

$$r(b_R) - b_{R'} = \delta a_{RR'},$$

with compatible zero-cochains $a_{RR'}$ on successive transitions.

Proof. Conjugating the root lift at each object by its sign a changes the arrow cochain by δa. Thus the displayed equation is exactly a natural isomorphism of the two strict transports. Composition of these natural isomorphisms is the additional compatibility equation on a triple of transition levels. □

REFLECTION CHARACTERS AND DIVISOR MONODROMY

Definition 1.85 *(A character extracted from root transport).* To obtain a scalar character from $(O1)^+$, choose compatible reference identifications of the transported lines, or a fixed one-dimensional fibre representation. Suppose the residual signs $\sigma_i \in \{\pm 1\}$ of the three simple lifts are constant on the relevant components and compatible with transitions. They define

$$\epsilon_o(s_{i_1} \cdots s_{i_k}) = \prod_{j=1}^{k} \sigma_{i_j}.$$

For the present type-II group, the only Coxeter relations are $s_i^2 = 1$; hence this is a character once the residual signs have been constructed. An action on moving lines, without the reference identifications, is not itself such a scalar character.

PROPOSITION 1.86 *(A simple Pfaffian wall).* Suppose a local Pfaffian section is $p = vue$, where e is a line frame and v is invertible. Fix a reference comparison of source and target frames. Let the base reflection send $u \mapsto -u$ and preserve v. Relative to that reference comparison the pulled-back section has ratio -1. If the actual root lift satisfies

$$\tau_s(p) = s^* p,$$

its residual sign relative to the reference comparison is -1. Under the component and transition compatibility in the preceding definition, this gives $\epsilon_o(s_i) = -1$.

Proof. In the reference frames, write $\tau_s(e) = \lambda s^* e$. The displayed compatibility gives $\lambda vu = -vu$ on the wall complement, hence $\lambda = -1$. Squaring loses the sign. Preserving a reference frame alone would not identify its action with this section-compatible root lift. The calculation uses the stipulated reflection, not parallel transport along a real path through a removed wall. □

Proposition 1.87 *(The determinant and automorphic characters).* If the source character exists and takes value -1 on all three simple type-II reflections, then

$$\epsilon_o = \det_W = \widetilde{\nu}_{\Delta_5}|_W .$$

Proof. The determinant of a real reflection is -1, so equality on the generators gives the first equality. The second is the homogeneous orthogonal character calculation in Lemma 10.2. The homogeneous character is used here: its value on an orthogonal reflection is not the bare symplectic Maass character of a chosen lift. This automorphic identity does not construct the source character. □

Proposition 1.88 *(Chamber symmetries).* The homogeneous character $\widetilde{\nu}_{\Delta_5}$ is trivial on the type-I chamber symmetry group $\Sigma \simeq S_3$. A character ϵ_o taking value -1 on all three simple reflections is Σ-invariant. Its abstract extensions to $W \rtimes \Sigma$ are exactly

$$\widetilde{\epsilon}_\chi(w, \sigma) = \epsilon_o(w)\chi(\sigma), \qquad \chi \in \{1, \mathrm{sgn}\}.$$

The automorphically compatible extension has $\chi = 1$.

Proof. The type-I transpositions have homogeneous character $+1$, by the calculation in Chapter 3; they generate Σ. Chamber symmetry permutes the three equal generator values of ϵ_o, proving invariance. The displayed formula then respects the semidirect product. Two extensions differ by a character of S_3, whose only sign characters are the two listed. □

Proposition 1.89 *(Geometric extension to chamber symmetries).* An abstract extension is realized on an oriented Hall theory only after supplying chamber transports of its actual correspondences, coefficient maps and root pairs. These must preserve the specified product, coproduct, unit, pairing and transitions, and satisfy the S_3 relations and

$$\theta_\sigma \tau_i \theta_\sigma^{-1} = \tau_{\sigma(i)}$$

with the appropriate pullbacks. Such coherent transports define an action of $W \rtimes S_3$.

Proof. The relations present the semidirect product. An action on the geometric objects and all their comparison maps satisfying these relations factors through that presentation. The character criterion for a line does not assert the existence of these correspondences. □

Definition 1.90 *(A local divisor monodromy datum).* Let $\mathcal{U}$ be a complex analytic family near a smooth point of a divisor, with a specified transverse disk and punctured disk Δ^*. Choose its reduced critical recipient and orientation. Suppose the relevant supported cohomology of the oriented vanishing-cycle sheaf forms a finite-rank local system $\mathbb{V}$ on Δ^*. A base point and the positive meridian then define

$$T_D : \mathbb{V}_{z_o} \xrightarrow{\sim} \mathbb{V}_{z_o}$$

up to conjugacy. For a rank-one sign local system its value lies in $\{\pm 1\}$. The required local constancy, supports and finite-dimensionality are hypotheses on the actual family.

Proposition 1.91 *(Divisor and Weyl monodromy have different domains).* An imaginary simple root contributes a multiplicity to the BKM presentation but no simple Weyl reflection. A meridian around a geometric divisor labelled by that root therefore has no prescribed value under $\epsilon_o : W \to \{\pm 1\}$. Its local monodromy requires the preceding analytic datum; a Maass character or a root multiplicity alone does not supply it.

Proof. The Weyl group is generated by reflections in the real simple roots. The meridian instead belongs to the fundamental group of a specified punctured analytic neighbourhood. Identifying their actions would require a map of these geometric actions and their coefficient systems, not merely a common root label. □

PARITY, DUALITY AND CYCLIC TRACES

PROPOSITION 1.92 *(Local root profiles and orientation lifts).* The target root profiles are $(1 \mid 0)$ for each real simple root, $(10 \mid 0)$ for each isotropic a_{ij}, and $(29 \mid 93)$ for $\delta_1 + \delta_2 + \delta_3$, with the parity convention of Chapter 4. A compact chart realizes these profiles only if an actual graded comparison identifies its relevant cohomology with those spaces. Their parity does not alone determine a lift to a determinant square root.

Proof. The profiles are the target calculations in the BKM chapter. A graded linear isomorphism preserves them, so it gives the conditional chart assertion. For the last statement, consider one odd line. Parity acts by -1 on its Berezinian. A scalar lift λ to a square root would have to satisfy $\lambda^2 = -1$, hence $\lambda = \pm i$ over $\mathbb{C}$. Such a lift is not an involutive sign action. More generally the Berezinian parity is $(-1)^q$ on a space of dimension $(p \mid q)$; a Pfaffian-half comparison is additional structure. In particular $q = 93$ cannot be replaced by an assumed orientation sign without that comparison. □

PROPOSITION 1.93 *(Opposite degrees under CY3 duality).* If a specified comparison identifies $K_{-\gamma} \simeq K_\gamma^\vee[-3]$, then

$$\det K_{-\gamma} \simeq \det K_\gamma.$$

An unshifted dual would instead have inverse determinant. The corresponding square-root identification must square to the actual shifted determinant comparison.

Proof. Dualization and an odd cohomological shift each invert determinant. Their composite gives the displayed equality. The distinction is visible for $K_\gamma = \mathcal{O}(1)$ in degree zero on $\mathbb{P}^1$: the shifted dual has determinant $\mathcal{O}(1)$, whereas the unshifted dual has determinant $\mathcal{O}(-1)$. □

PROPOSITION 1.94 *(Opposite products and Chevalley transport).* Suppose geometric duality defines an involution ϑ on a specified graded associative Hall theory, with

$$\vartheta(xy) = (-1)^{|x||y|}\vartheta(y)\vartheta(x), \qquad \vartheta^2 = 1,$$

and compatible orientation, coproduct and transition comparisons. It then induces the corresponding opposite-product involution on its primitive Lie superalgebra. Agreement with a target Chevalley involution is an additional equality of these maps under the primitive comparison.

Proof. Substitute the displayed formula into the supercommutator. One obtains $\vartheta([x, y]) = -[\vartheta(x), \vartheta(y)]$. Equivalently it is the indicated Lie anti-involution, with a minus sign converting it to a Lie automorphism if that convention is used. The coproduct comparison preserves primitives, and the determinant square comparison preserves their orientation data. None of these maps is determined by dimensions of opposite root spaces. □

PROPOSITION 1.95 *(Associative traces and primitive forms).* Let A be a graded associative algebra or dg algebra, ℓ a line, and $\mathrm{tr} : A \to \ell[-3]$ a closed cyclic trace. Then

$$B(x, y) = \mathrm{tr}(xy), \qquad B(x, y) = (-1)^{|x||y|}B(y, x), \qquad B(xy, z) = B(x, yz).$$

Its nonzero homogeneous values have $|x| + |y| = 3$, when ℓ is in degree zero. On the commutator Lie superalgebra the restricted form is invariant. This is a different condition from Hopf adjointness on a full bialgebra.

Proof. These are respectively the definition, graded cyclicity, associativity and degree of the trace. Expanding the commutators gives Lie invariance with the same Koszul signs. In contrast, for an ordinary Hopf pairing and primitive elements x, y, z, Hopf adjointness gives

$$\langle xy, z\rangle = \langle x, z\rangle\epsilon(y) + \epsilon(x)\langle y, z\rangle = 0.$$

Thus it also annihilates $[x, y]$ against z. An invariant Lie form on a nonabelian primitive algebra need not do this. A product and a trace must be defined on A, not on a primitive subspace which is generally not closed under that product. □

Proposition 1.96 *(The induced quotient pairing).* Let a bilinear pairing of finite-dimensional spaces have matrix G, with convention $B(x, y) = x^t G y$. Its left and right radicals are $\ker G^t$ and $\ker G$. It induces a nondegenerate pairing on the two radical quotients. If Q_L, Q_R are quotient maps and s_L, s_R are sections, its matrix is

$$\overline{G} = s_L^t G s_R, \qquad Q_L s_L = 1, \quad Q_R s_R = 1.$$

The expression is independent of the sections. It is generally not $Q_L G Q_R^t$.

Proof. Changing a section adds a map into the appropriate radical, which pairs to zero. If a quotient class pairs to zero with every opposite class, a representative lies in the radical and its quotient is zero. This proves nondegeneracy over a field. Over a coefficient ring one must separately require finite projective quotients and that the resulting map to the dual is an isomorphism; a nonzero determinant or full generic rank is insufficient. Compatibility with a Lie bracket, a Hopf operation or transitions further requires the corresponding ideal, coideal or kernel identities. □

Compatible towers and effective orientation lines

Definition 1.97 *(Orientation-preserving transitions).* For specified transition maps $t_{RR'}$, an orientation transition is a root isomorphism

$$\theta^o_{RR'} : t^*_{RR'} o_{R'} \xrightarrow{\sim} o_R.$$

Its square is the determinant comparison, and it preserves the chosen equivariant, extension, unit and Weyl identifications. Identity and composition are equations of these isomorphisms on all successive transitions.

Proposition 1.98 *(Formal orientation towers).* A compatible family of orientation objects and transition isomorphisms defines an object in the homotopy limit of their groupoids. Every finite projection is the specified root pair, with its comparisons. This formal statement does not require the orientation line itself to have trivial Picard class.

Proof. An object of the homotopy limit is precisely such a compatible system, including its composition coherence. Applying any finite projection recovers its component. Vanishing of the Kummer obstruction says that the determinant has a root; it does not say that the root is a trivial line. □

PROPOSITION 1.99 *(Classes in a tower of root groupoids).* Consider a countable tower and a compatible family of isomorphism classes of roots. Choices of compatible representatives and edge isomorphisms have, when nonempty, ambiguity measured by $\varprojlim{}^{1} A_R$, where A_R is the automorphism group of the chosen root at level R. Here $A_R = H^0(B_R, \mu_2)$, with the induced transition maps. If this automorphism tower is Mittag–Leffler, the ambiguity vanishes. To obtain a line on a geometric limit, one additionally needs an effective comparison between its root groupoid and the homotopy limit. An equivalence of those groupoids, together with the stated vanishing, identifies its isomorphism classes with the compatible finite classes.

Proof. Fix representatives. Changing each adjacent transition isomorphism is an element of $\prod_R A_R$; changing the representative identifications acts by $(a_R) \mapsto (a_R - t_R a_{R+1})$ in additive notation. The quotient is the usual $\varprojlim{}^{1} A_R$. The Mittag–Leffler condition makes this quotient zero. This is an automorphism calculation; there is no presumed abelian group of roots of a fixed line. Finally an effective comparison transfers the formal objects and isomorphisms to the geometric limit. Without it, a formal compatible system is all that has been constructed. □

Definition 1.100 *(Coefficient-preserving transitions).* A transition of oriented critical coefficients is a specified isomorphism between the chosen pullback of $\mathcal{P}_{R'}$ and $\mathcal{P}_R$, with the required shift and support convention. It commutes with the orientation sign torsors, critical-chart comparisons, duality and the coefficient maps defining the retained Hall operations. These isomorphisms satisfy the transition composition equation.

PROPOSITION 1.101 *(Preservation of critical coefficients).* Coefficient-preserving transitions identify the induced finite operations and their orientation comparisons. Passing to a chosen completed operation further requires the limit and completed-tensor hypotheses for that coefficient category.

Proof. Apply the specified transition isomorphisms to the defining pull–push diagram. Their commutation with its coefficient maps identifies the two operations. A limiting diagram follows only if the relevant functors, supports and tensor products commute with that limit. Properness of the underlying stack map alone does not assert this coefficient comparison. □

The orientation theorems for sheaf moduli and real elliptic operators [63, 64] provide substantial orientation constructions in their own settings. Applying them to the compact reduced problem requires the comparison with its actual reduced complex and extension data. The formal coordinate functors, cross determinant squares and finite character calculations above supply explicit parts of that problem. The remaining global question is the construction of one reduced oriented coefficient system on the common elliptic quotient, compatible with complete flags, geometric transitions and the stipulated reflection transports.

1.3 Critical Hall operations and primitive Lie algebras

The formal deformation model of §1.1 supplies actual extension categories and polynomial critical models. Their Hall algebra requires a coefficient theory on the extension stacks. The global critical Hall theorem gives such a multiplication on ordinary critical cohomology under its strong orientation and properness hypotheses. The compact reduced $S \times E$ problem asks for additional comparisons with the reduced quotient coefficients. A collision coproduct and a primitive Lie form are further structures, with separate equations.

We work over a characteristic-zero field k, using $k = \mathbb{Q}$ for constructible coefficients before extension of scalars. All cohomological shifts and orientation parities are part of the coefficient objects. When an ordinary superalgebra is used below, its multiplication is even and its tensor symmetry is $\tau(x \otimes y) = (-1)^{|x||y|} y \otimes x$. Passing from an index-shifted critical Hall multiplication to this category requires the stated regrading or comparison.

The coefficient object and its multiplication

Definition 1.102 *(A bounded geometric coefficient object).* Fix chosen moduli components $\mathcal{M}_{R,\gamma}$ and critical coefficients $K_{R,\gamma}$, including any reduction and orientation data. Choose one realization, for example

$$\mathcal{H}_{R,\gamma} = R\Gamma(\mathcal{M}_{R,\gamma}, K_{R,\gamma}), \qquad \mathcal{H}_R = \bigoplus_{\gamma \in \Gamma_R} \mathcal{H}_{R,\gamma}.$$

If instead compact-support or dual compact-support realization is used, name it explicitly and use its coefficient variance. In particular $\mathbb{B}(M; K) = R\,\mathrm{Hom}(R\Gamma_c(M, K), k)$ is contravariant in K. The set Γ_R is finite when a finite charge window is specified. Finite-dimensional cohomology or a perfect chain model is an additional hypothesis, not a consequence of this finite indexing set.

For a bounded ordinary-cohomology model, let $\mathcal{E}_{\alpha,\beta}$ be the ordered extension stack with maps p to its endpoints and q to its middle object. The required coefficient morphism is

$$u_{\alpha,\beta} : p^*(K_\alpha \boxtimes K_\beta)[\kappa(\alpha,\beta)] \longrightarrow q^! K_{\alpha+\beta}.$$

The source shift κ is fixed by this formula, as in (1). On a translation quotient, the endpoint stack is the fibre product over BE and the coefficient tensor is the specified relative tensor.

Definition 1.103 *(A geometric Hall product).* Put $L = p^*(K_\alpha \boxtimes K_\beta)[\kappa(\alpha,\beta)]$. Suppose $u : L \to q^! K_{\alpha+\beta}$ is defined, the support of L is proper over the target, and Künneth holds in the chosen realization. The Hall product is the composite

$$(\mathcal{H}_\alpha \otimes \mathcal{H}_\beta)[\kappa(\alpha,\beta)] \longrightarrow R\Gamma(\mathcal{M}_{\alpha+\beta}, Rq_! L) \longrightarrow \mathcal{H}_{\alpha+\beta}.$$

The first arrow is pullback followed by $Rq_* L \simeq Rq_! L$. The second applies $Rq_! u$ and the counit $Rq_! q^! \to 1$. Only permitted charge sums are included at a bounded stage.

Proposition 1.104 *(The pullback and integration legs).* The product is defined under the stated coefficient, Künneth and proper-support hypotheses. Existence of $Rq_!$ for a nonproper map alone does not provide the last arrow on ordinary cohomology. For compactly supported cohomology, pullback along p needs its corresponding support condition instead.

Proof. Ordinary cohomology admits pullback through the unit $K \to Rp_* p^* K$. Proper support identifies $Rq_* L$ with $Rq_! L$, after which $Rq_! u$ and the trace give the last arrow. Without that support condition an arbitrary global section on the extension stack need not have proper support over the target. Dually, pulling a compactly supported section back along a map with nonproper fibres need not leave compact support. □

Proposition 1.105 *(Independence of an exceptional-pushforward model).* Suppose a map in the coefficient theory is compactifiable as $q = \overline{q}\, j$, with j open and $\overline{q}$ proper, and exceptional pushforward is computed by $Rq_! = R\overline{q}_*\, j_!$. Two such compactifications give the same functor when the chosen six-functor theory supplies its proper base-change and composition identifications.

Proof. Compare both compactifications with the closure of the graph of the identity on their common open domain inside their product. The proper comparison maps restrict to the identity on that open. Proper base change and extension by zero identify the two pushforwards with the one through this common refinement. The composition constraints give the cocycle identification for three choices. This concerns $q_!$; it does not remove a proper-support requirement elsewhere in the Hall product. □

PROPOSITION 1.106 *(Associativity of the coefficient product).* If the two composites of u agree on the three-step extension flag, including their shifts and orientation identifications, then the Hall multiplication is associative on all permitted triples. Coherent zero-endpoint comparisons give its unit.

Proof. The two geometric compositions are the same flag stack by Proposition 1.22. Base change, projection formula and composition identify their pull–push functors. The coefficient equality gives equality of the resulting maps. The two zero-endpoint sequences identify with the identity correspondence and give the unit equations. □

Theorem 4.2 constructs these morphisms and their coherence for the specified unreduced 3-Calabi–Yau moduli with strong orientation data. This applies to the Ginzburg categories of the tripled-quiver models. On a compact projective threefold with trivial canonical bundle, orientation data exists by [64, Theorem 4.4]; the strong associativity condition remains additional. A formal cyclic equivalence between deformation categories does not alone identify their constructible coefficient maps on an open algebraic moduli stack.

AN ACTUAL CONSTRUCTIBLE HALL CALCULATION

There is a concrete algebraic result before critical coefficients are compared. Consider the Euler-characteristic Hall algebra of finite-dimensional nilpotent representations of the preprojective algebra of the quiver in (6.1). Functions are constructible, and convolution integrates over invariant subspaces. Use the same quotient-first convention as above. Let e_i be the characteristic function of the vertex simple. The divided power $e_i^{(b)} = e_i^{*b}/b!$ is the characteristic function of $S_i^{\oplus b}$: a complete flag in $\mathbb{C}^b$ has Euler characteristic $b!$.

THEOREM 1.107 *(The cubic real-root Serre relation).* For distinct vertices i, j,

$$(\operatorname{ad} e_i)^3(e_j) = 0.$$

The assignment of the three Chevalley generators to e_i therefore defines a homomorphism from the positive real-root Kac–Moody Lie algebra with Euler matrix G into this constructible Hall algebra.

Proof. Only dimension vector $(3,1)$ on i, j contributes. Combine the two forward arrows into $X : \mathbb{C}^3 \to \mathbb{C}^2$ and the reverse arrows into $Y : \mathbb{C}^2 \to \mathbb{C}^3$. The preprojective relation at i gives $YX = 0$, hence $\operatorname{rk} X + \operatorname{rk} Y \le 2$. Put $I = \operatorname{im} Y$ and $K = \ker X$.

For $e_i^{(b)} * e_j * e_i^{(3-b)}$, a descending flag has its last i-subspace U of dimension $3-b$. Invariance requires $I \subset U \subset K$. If $I \not\subset K$, all these flag spaces are empty. Otherwise, writing $r = \dim I$ and $s = \dim K$, their Euler characteristics are $\binom{s-r}{3-b-r}$. We have $s - r = 3 - \operatorname{rk} X - \operatorname{rk} Y \ge 1$. Consequently their alternating sum is

$$\sum_{b=0}^{3} (-1)^b \binom{s-r}{3-b-r} = (-1)^{3-r}(1-1)^{s-r} = 0.$$

Multiplying by $3!$ converts the divided-power relation into the cubic commutator, up to its irrelevant overall sign. The same argument with the two vertices exchanged proves the opposite relation. □

This is a relation for actual convolution operators with constructible-function coefficients. It proves neither injectivity nor an identification of all imaginary generators. Transporting it to critical cohomology or to compact reduced sheaves needs a comparison preserving convolution. In particular, matching the Euler matrix G alone is insufficient.

A spectral coproduct on critical quiver cohomology

The quiver theory also has a coproduct constructed from direct sum and the ambient Euler class. Let Q' be a finite quiver, T a torus acting on its arrows, and W an invariant potential. This includes the tripled quiver with its canonical potential. For a dimension vector d, put $G_d = \prod_i GL_{d_i}$ and

$$R = H^*(BT;\mathbb{Q}), \qquad M_d = [\mathrm{Rep}_d(Q')/G_d], \qquad C_d = H^*_T(M_d, \varphi_{\mathrm{Tr}\,W}\mathbb{Q}), \qquad C = \bigoplus_d C_d.$$

Assume the relative Künneth maps over BT are isomorphisms for the finite products used below. The direct sum has finite dimension-vector support. Every fixed dimension vector has finitely many decompositions.

Write $E_{i,d}$ for the tautological bundles and L_a for the line bundle on BT associated to the character of arrow a. The ambient two-term Hom complex, in degrees zero and one, is

$$\Theta_{d,e} = \left[\bigoplus_i E_{i,d}^\vee \boxtimes E_{i,e} \longrightarrow \bigoplus_{a:i\to j} E_{i,d}^\vee \boxtimes E_{j,e} \otimes L_a\right].$$

Its virtual rank is the quiver Euler form $\chi(d,e) = \sum_i d_i e_i - \sum_{a:i\to j} d_i e_j$. This complex belongs to the smooth representation stack; it is not the threefold self-Ext complex of the critical locus. Set $\Psi(V,z) = z^{\mathrm{rk}\,V} c(V, z^{-1})$, expanding the Chern polynomial of a virtual bundle at $z = \infty$.

Direct sum induces a critical-cohomology pullback $\delta_{d,e} : C_{d+e} \to C_d \otimes_R C_e$. Tensoring the vertex spaces with a line gives a coaction $\mathrm{act}_z^* = e^{z\mathsf{T}}$, polynomial on each class. Thus T is locally nilpotent, $|z| = 2$ and $|\mathsf{T}| = -2$.

Theorem 1.108 *(Critical vertex coproduct).* The maps

$$\Delta_{d,e}(z) = \Psi(\Theta_{d,e}, -z) e^{z\mathsf{T}_1} \delta_{d,e} : C_{d+e} \longrightarrow (C_d \otimes_R C_e)((z^{-1}))$$

satisfy

$$(\Delta_{d,e}(z)\otimes 1)\Delta_{d+e,f}(w) = (1\otimes \Delta_{e,f}(w))\Delta_{d,e+f}(z+w),$$
$$\Delta(z)\mathsf{T} = \partial_z \Delta(z) + (1\otimes \mathsf{T})\Delta(z),$$
$$(\varepsilon\otimes 1)\Delta(z) = 1, \qquad (1\otimes\varepsilon)\Delta(z) = e^{z\mathsf{T}}.$$

Here ε is restriction to the zero representation. Rational factors in the first identity are compared with $z, w, z+w$ invertible and expanded in the same iterated Laurent region. The (d,e) component has ordinary cohomological degree $2\chi(d,e)$, including the degree of z.

Proof. The unit of direct-sum adjunction, natural pullback for vanishing cycles, Thom–Sebastiani and Künneth define δ. Associativity of direct sum makes it coassociative. Line scaling commutes with direct sum, so T is a coderivation. Splitting either variable of Θ gives a direct sum of the corresponding Hom complexes. Translating its first variable by u tensors it with L_{-u}, and translating the second tensors it with L_u.

Consequently both iterated coproducts have factor

$$\Psi(\Theta_{12}, -z)\Psi(\Theta_{13}, -z-w)\Psi(\Theta_{23}, -w)$$

on $e^{(z+w)\mathsf{T}_1 + w\mathsf{T}_2}\delta^{(3)}$. This proves coassociativity. Simultaneous translation fixes Θ, and $\partial_z\Psi(\Theta, -z) = \mathsf{T}_1\Psi(\Theta, -z)$, which prove the translation identity. A zero-dimensional factor has $\Theta = 0$, giving the two counit equations. Finally $z^{\chi-r}c_r(\Theta)$ has degree 2χ. This is the construction of [61, Theorem 2.3.1] in ordinary cohomological grading. □

For the Hall regrading $A_d = C_d[-\chi(d,d)]$, the component degree becomes $\chi(d,e) - \chi(e,d)$. Tensor symmetries must be transported with this regrading. With the compatible sign twist of the Hall product, the coproduct satisfies the vertex bialgebra identity in the meromorphically braided coefficient category [61, Theorems 4.5.2 and 4.5.5]. For a symmetric quiver that category consists of modules over the tautological cup-product ring; for a nonsymmetric quiver it uses the separate cohomology rings in each dimension vector. These are concrete algebraic recipients, with their prescribed braiding and spectral completion.

The spectral parameter has a geometric role. A localized Euler factor $y - x + \lambda$ becomes $y - x + \lambda - z$ under first-factor line scaling. Its leading coefficient is -1, so it is invertible as a Laurent series even if the unshifted class is a zero divisor. For a single arrow $1 \to 2$, dimensions $d = (1,0)$ and $e = (0,1)$, and zero arrow weight,

$$\Delta_{d,e}(z)(1) = \frac{1}{y - x - z} = -\sum_{r \geq 0}(y-x)^r z^{-r-1}.$$

The general localized-to-spectral comparison is [61, Theorem 3.3.2]. It does not extend a localized coefficient across its collision divisor. Nor does it identify this coproduct with the cut coproduct of Theorem 1.5: one uses line scaling on cohomology, the other uses separation opens in spectral configuration spaces. A comparison with the compact reduced theory must specify its own tensor, support, coefficient maps and relation to these two constructions.

There is also a localized bialgebra on the dual compactly supported critical cohomology of semistable quiver representations. For a fixed slope and a stability condition generic at that slope, with the locally closed Serre-subcategory restriction of [30, §6.1], its tensor product inverts the prescribed cross-factor equivariant Euler classes. The perverse associated graded has a Hopf structure without localization in the specified symmetric category and is free commutative there; after the stated sign twist it is free supercommutative [30, Theorem C]. The PBW comparison with the original Hall object is a comparison of graded coefficient objects and need not be an algebra isomorphism. Thus localization and passage to the associated graded each change the recipient of the coproduct assertion.

A collision coproduct is separate data

Definition 1.109 *(A geometric collision coproduct).* For each decomposition $\gamma = \alpha + \beta$, specify a collision span

$$\mathcal{M}_\gamma \xleftarrow{r} C_{\alpha,\beta} \xrightarrow{s} \mathcal{M}_\alpha \times \mathcal{M}_\beta$$

and a coefficient morphism that induces a map in the chosen realization. For ordinary cohomology, one possible form is

$$v_{\alpha,\beta} : r^*K_\gamma \longrightarrow s^!(K_\alpha \boxtimes K_\beta)[d_{\alpha,\beta}],$$

with the support of r^*K_γ proper over the endpoint stack. The resulting component is

$$\mathcal{H}_\gamma \longrightarrow (\mathcal{H}_\alpha \otimes \mathcal{H}_\beta)[d_{\alpha,\beta}].$$

A coproduct on a direct sum requires finitely many nonzero components on every element; otherwise the target and its completion must be specified. The analogous construction on quotients uses the relative endpoint product over BE.

Proposition 1.110 *(The coproduct integration leg).* Under the coefficient and support hypotheses in this definition, pullback along r, then v, then the proper trace along s gives the stated coproduct component.

Proof. Apply the same ordinary-cohomology pull–push construction as for multiplication, with the collision span and its specified coefficient map. Proper support is used at the trace, and Künneth identifies the endpoint cohomology with the chosen tensor product. □

Proposition 1.111 *(Coassociativity and compatibility).* Suppose the two successive collision diagrams are identified, and their coefficient composites agree. Then the coproduct is coassociative. Suppose in addition that the diagram comparing collision after extension with extension after two collisions carries the corresponding coefficient equality. In an ordinary super realization it gives

$$\Delta m = (m \otimes m)(1 \otimes \tau \otimes 1)(\Delta \otimes \Delta).$$

Unit and counit compatibility are additional equations.

Proof. The common triple-collision diagram gives the first equality by base change and composition of the coefficient maps. In the second diagram, the middle two factors must be interchanged before the two products are formed. This is τ, so its Koszul sign is part of the displayed equality. These arguments compare the given maps; they do not construct a missing collision coefficient morphism from the extension product. □

The ordinary associative bar coalgebra has another coproduct: cutting a word into two consecutive subwords. Its tensor, support and differential are those of the bar construction. Identifying it with a geometric collision coalgebra requires an operation-preserving comparison. Neither a shared notation Δ nor equality of Euler characteristics supplies that comparison; see §1.5.

The complete bialgebra equations

Assume now that H is a complex of super vector spaces with even chain maps $m, \eta, \Delta, \varepsilon$, with the ordinary tensor product. All subsequent assertions in this subsection have this carrier. If a geometrically constructed object is only an algebra in a shifted or relative category, the following criterion applies after its specified comparison to this carrier.

Definition 1.112 *(Bialgebra defects).* The bialgebra defects are the differences between the two sides of the following equations:

$$\begin{aligned} m(m \otimes 1) &= m(1 \otimes m), & m(\eta \otimes 1) &= 1 = m(1 \otimes \eta), \\ (\Delta \otimes 1)\Delta &= (1 \otimes \Delta)\Delta, & (\varepsilon \otimes 1)\Delta &= 1 = (1 \otimes \varepsilon)\Delta, \\ \Delta m &= (m \otimes m)(1 \otimes \tau \otimes 1)(\Delta \otimes \Delta), & \Delta\eta &= \eta \otimes \eta, \\ \varepsilon m &= \varepsilon \otimes \varepsilon, & \varepsilon\eta &= 1_k. \end{aligned}$$

Call their collection $\mathfrak{O}_{\mathrm{Hbialg},R}$ on a bounded realization. Vanishing means that every displayed map identity holds on its defined domain. In finite bases these are matrix and structure-constant equations. A differential is required to commute with all four structure maps.

The last three equations cannot simply be omitted. For example, let $H = ke_1 \oplus ke_2$ with $e_i e_j = \delta_{ij} e_i$, unit $e_1 + e_2$, $\Delta e_i = e_i \otimes e_i$, and $\varepsilon(e_i) = 1$. Multiplication is associative and unital, comultiplication is coassociative and counital, and Δ is multiplicative. Nevertheless

$$\Delta(1) = e_1 \otimes e_1 + e_2 \otimes e_2 \neq 1 \otimes 1, \qquad \varepsilon(e_1 e_2) = 0 \neq 1, \qquad \varepsilon(1) = 2.$$

This is the algebra of the groupoid with two identity objects, not a bialgebra with the given structure maps.

Definition 1.113 *(Compact geometric bialgebra data).* A compact geometric bialgebra datum consists of a coefficient object, the extension and collision coefficient maps, zero-object unit and counit, and the comparison to the stated tensor category. The geometry must supply the proper-support or exceptional integration maps it uses. Coherent flag, collision and mixed diagrams must induce all equations of Definition 1.112. If finite matrices are used, their bases and finite-dimensional models are part of the datum.

PROPOSITION 1.114 *(Geometric data and algebraic identities).* Such a datum gives a bialgebra on its specified domain. Its primitive Lie algebra is obtained from the reduced coproduct below. The conclusion is conditional on the geometric and coefficient comparisons, not just on formal names for the four structure maps.

Proof. The flag and mixed correspondence comparisons give the displayed identities by the preceding propositions. The zero-object and counit diagrams give the unit equations, counit equations and the last three scalar compatibilities. These are exactly the bialgebra axioms. □

PRIMITIVES, PARITY AND AN ANTIPODE

Definition 1.115 *(The primitive subspace).* Let H be a counital coalgebra with coaugmentation η, $\varepsilon\eta = 1$ and $\Delta\eta = \eta \otimes \eta$. Put $I = \ker \varepsilon$ and $\pi = 1 - \eta\varepsilon : H \to I$. Its reduced coproduct and primitive space are

$$\overline{\Delta} = (\pi \otimes \pi)\Delta|_I : I \longrightarrow I \otimes I, \qquad P(H) = \ker \overline{\Delta}.$$

For $x \in I$, this is $\overline{\Delta}x = \Delta x - 1 \otimes x - x \otimes 1$. A linear projection onto $P(H)$, if desired, is an extra choice unless another theorem constructs one naturally.

Definition 1.116 *(Primitive Hall bracket).* For an ordinary super bialgebra H, the primitive bracket is

$$[x, y] = m(x, y) - (-1)^{|x||y|} m(y, x), \qquad x, y \in P(H).$$

It is the restriction of the supercommutator in H.

PROPOSITION 1.117 *(Primitive closure and Jacobi).* The bracket takes values in $P(H)$ and makes it a Lie superalgebra. It satisfies

$$[x, [y, z]] = [[x, y], z] + (-1)^{|x||y|} [y, [x, z]].$$

A compatible differential restricts to its dg Lie differential.

Proof. Multiplicativity of the coproduct gives

$$\Delta(xy) = xy \otimes 1 + x \otimes y + (-1)^{|x||y|} y \otimes x + 1 \otimes xy$$

for primitives. The mixed terms cancel in the supercommutator, so the bracket is primitive. Expanding the three commutators using associative multiplication gives the displayed Jacobi identity. Compatibility of the differential with the coaugmentation, counit, multiplication and coproduct proves that it restricts and is a derivation. □

Proposition 1.118 *(What multiplication alone determines).* An associative superalgebra has a Lie superalgebra under its supercommutator, but its primitive subspace is not determined without a coproduct. On $k[t]$, the two bialgebra structures

$$\Delta_a t = t \otimes 1 + 1 \otimes t, \quad \varepsilon_a(t) = 0, \qquad \Delta_m t = t \otimes t, \quad \varepsilon_m(t) = 1$$

have the same multiplication and different primitive spaces.

Proof. For the additive structure, $P(k[t]) = kt$: the mixed terms in $\Delta_a(t^n)$ exclude every highest degree $n \geq 2$. For the multiplicative structure, every basis element t^n is group-like. Comparing diagonal coefficients as in Proposition 1.52 shows that its primitive space is zero. □

The ambiguity also occurs among connected positively graded Hopf structures. Give $k[x, y]$ degrees 1, 2. One coproduct makes both x and y primitive; another has

$$\Delta x = x \otimes 1 + 1 \otimes x, \qquad \Delta y = y \otimes 1 + 1 \otimes y + x \otimes x.$$

For the second, $y - x^2/2$ is primitive. This change of polynomial coordinates identifies it with the additive Hopf algebra on two generators and proves coassociativity. The two primitive planes inside the fixed polynomial algebra are $\operatorname{span}\{x, y\}$ and $\operatorname{span}\{x, y - x^2/2\}$.

Proposition 1.119 *(Charge and parity).* Suppose the product preserves an additive charge grading $\widehat{\Gamma}$ and cohomological parity. If the coproduct and coaugmentation preserve the grading, its primitive space is graded, and

$$[P_{\widehat{\alpha},\bar{p}}, P_{\widehat{\beta},\bar{q}}] \subset P_{\widehat{\alpha}+\widehat{\beta},\bar{p}+\bar{q}}.$$

Regrading by the additive map $\overline{\Pi}_X$ preserves this law. A raw quadratic charge expression can be used only through the normal-ordered extension on which it becomes additive.

Proof. Both terms in the bracket have the same summed charge and parity. Primitive closure was proved above. The composite with an additive grading map sends their common charge to $\overline{\Pi}_X(\widehat{\alpha}) + \overline{\Pi}_X(\widehat{\beta})$. The construction and group law of the normal-ordered charge extension are given in Chapter 2. □

The primitive projection is not part of this proof. In the connected cocommutative characteristic-zero domain, the convolution logarithm of the identity gives a natural primitive projection; Proposition 8.3 proves that result with its local-nilpotence hypotheses. It must not be applied to a general noncocommutative Hall bialgebra without those hypotheses.

Definition 1.120 *(Antipode).* An antipode on a bialgebra is a chain map $S : H \to H$ satisfying

$$m(S \otimes 1)\Delta = \eta\varepsilon = m(1 \otimes S)\Delta.$$

A bialgebra admitting this convolution inverse of its identity is a Hopf algebra. On a connected positively graded bialgebra, the antipode is defined recursively by

$$S(1) = 1, \qquad S(x) = -x - \sum S(x')x'', \qquad \overline{\Delta}x = \sum x' \otimes x''.$$

Each reduced factor has lower positive degree.

PROPOSITION 1.121 *(Existence from conilpotence).* If the augmentation projection $\pi = 1 - \eta\varepsilon$ is locally nilpotent under convolution, then

$$S = \sum_{n\geq 0} (-1)^n \pi^{\star n}$$

is an antipode. In particular this holds for a connected positively graded bialgebra. It preserves additive charge. A bialgebra need not admit an antipode: the multiplicative structure on $k[t]$ above is an example.

Proof. The convolution unit is $\eta\varepsilon$, and $1_H = \eta\varepsilon + \pi$. Local nilpotence makes the geometric series finite on each element. Multiplying the series by $\eta\varepsilon + \pi$ on either side gives $\eta\varepsilon$, proving both identities. In a connected positive grading, iterated reduced coproducts strictly split the total degree, so they eventually vanish. Every term has the original total charge. For multiplicative $k[t]$, an antipode would satisfy $S(t)t = 1$, which has no polynomial solution. □

An antipode therefore preserves an additive charge; it does not create the opposite charge sector. A two-sided comparison or a Drinfeld double has to construct that opposite sector separately.

FINITE POSITIVE WINDOWS

Let $H = \bigoplus_{n\geq 0} H_n$ have additive positive grading. The direct sum $H_{\leq R}$ carries every product with total degree at most R. If Δ preserves total degree, this window is a subcoalgebra. Declaring all higher products to be zero instead gives the algebra quotient $H/H_{>R}$, but that ideal need not be a coideal.

For the additive Hopf algebra $k[x]$, take $N \geq 2$. Then

$$\Delta(x^N) = x^N \otimes 1 + 1 \otimes x^N + \sum_{i=1}^{N-1} \binom{N}{i} x^i \otimes x^{N-i}.$$

The coefficient of $x \otimes x^{N-1}$ is $N \neq 0$. Thus (x^N) is not a coideal, and the algebra $k[x]/(x^N)$ does not inherit the additive coproduct in the ordinary tensor category. It still has the partial operations obtained by restricting total degree. This elementary example prevents a bounded Hall charge window from being called a finite Hopf algebra merely by setting operations outside it to zero.

A PRIMITIVE LIE FORM AND A HOPF PAIRING

A compact effective Hall support generally contains no opposite charge. To pair γ with $-\gamma$, choose an enlarged carrier or independently defined opposite object and a comparison realizing that pairing. An antipode on a positively graded Hall algebra does not supply it.

Definition 1.122 *(A positive–negative primitive form).* Let $L = \bigoplus_{\gamma,\bar{p}} L_{\gamma,\bar{p}}$ be a Lie superalgebra containing the required positive and negative sectors. An even degree-zero primitive form is a bilinear map $B : L \otimes L \to k$ pairing only opposite charges and equal parities. It is supersymmetric when $B(x, y) = (-1)^{|x||y|} B(y, x)$, and invariant when

$$B([x, y], z) = B(x, [y, z]).$$

Its invariance defect is $\mathfrak{o}_F(x, y, z) = B([x, y], z) + (-1)^{|x||y|} B(y, [x, z])$. For an even supersymmetric form, its vanishing for all homogeneous triples is equivalent to the displayed invariance identity, by super skew-symmetry of the bracket. One possible geometric construction starts with an even supertrace tr_o on a two-sided graded associative realization and sets $B(x, y) = \mathrm{tr}_o(xy)$ on its chosen Lie subspace. The existence and descent of that trace are part of the data.

Proposition 1.123 *(Trace forms).* If the trace is even, vanishes in nonzero charge, and satisfies $\mathrm{tr}_0(xy) = (-1)^{|x||y|}\,\mathrm{tr}_0(yx)$, then its product form has the stated homogeneity, supersymmetry and Lie invariance.

Proof. Charge preservation of multiplication and the degree-zero trace give homogeneity. Its parity gives zero between unequal parities, and its cyclic identity gives supersymmetry. Finally,

$$\begin{aligned} B([x,y],z) &= \mathrm{tr}_0(xyz) - (-1)^{|x||y|}\,\mathrm{tr}_0(yxz) \\ &= \mathrm{tr}_0(xyz) - (-1)^{|y||z|}\,\mathrm{tr}_0(xzy) = B(x,[y,z]). \end{aligned}$$ □

Definition 1.124 *(Pairing identities on their carriers).* For a primitive Lie form, its identities are evenness, homogeneity, supersymmetry and Lie invariance. Its radical and quotient pairing are then formed on L. A Hopf pairing instead is a bilinear map $\beta : H \otimes K \to k$ between full Hopf superalgebras such that

$$\beta(xy,z) = \beta(x \otimes y, \Delta_K z), \qquad \beta(x,yz) = \beta(\Delta_H x, y \otimes z),$$

with $\beta(1,z) = \varepsilon_K(z)$ and $\beta(x,1) = \varepsilon_H(x)$. The tensor pairing includes the super sign

$$\beta(x \otimes y, u \otimes v) = (-1)^{|y||u|}\beta(x,u)\beta(y,v).$$

Antipode compatibility is imposed with the convention of the chosen pairing. These equations concern the full algebra products, not a product asserted to close on primitives. The notation $\mathfrak{O}_{\mathrm{Hpair},R} = 0$ records the identities for the pairing on its specified carrier. A Lie-form comparison uses the first set; a double uses the full Hopf pairing equations. It does not require the same form to serve both roles.

Proposition 1.125 *(Hopf adjointness on primitives).* If x, y are primitives in H and z is primitive in K, then $\beta(xy,z) = 0$. Hence the restriction of a Hopf pairing to primitives annihilates $[P(H),P(H)]$ in its first argument. If that restricted pairing is nondegenerate there, then $P(H)$ is abelian.

Proof. Insert $\Delta_K z = z \otimes 1 + 1 \otimes z$ in the first adjointness equation. Each term contains either $\varepsilon_H(x)$ or $\varepsilon_H(y)$, both zero. Subtract the two products to obtain the commutator assertion. Nondegeneracy forces every such commutator to vanish. □

Thus a nondegenerate invariant form on a nonabelian primitive Lie superalgebra cannot simultaneously be the restriction of this Hopf pairing. The primitive-recognition problem uses the Lie form. A Drinfeld double uses an appropriate Hopf pairing on its full positive and negative halves. Their comparison must retain this distinction.

Radicals and quotient forms

Proposition 1.126 *(Finite quotient pairing).* Let $G : V \otimes W \to k$ be a bilinear pairing of finite-dimensional spaces. Quotienting V by its left radical and W by its right radical gives a perfect pairing. If Q_V, Q_W are matrices of chosen quotient lifts, its matrix is

$$\overline{G} = Q_V^{t} G Q_W.$$

It has rank equal to both quotient dimensions. To apply this criterion to geometric primitives, the actual matrix G and its two radicals must first be identified.

Proof. Both radicals vanish against the opposite space, so the pairing descends. A nonzero class in either quotient has nonzero pairing with some vector in the opposite space by the definition of its radical. Thus both adjoint maps are injective. Their dimensions are both rank G, so they are isomorphisms. Evaluating on the chosen lifts gives the displayed matrix. □

Definition 1.127 *(Geometric pairing data).* For the compact primitive comparison, geometric pairing data specify the positive and negative carrier, the pairing correspondence or trace, its orientation and degree shifts, and the cyclic three-input comparison proving Lie invariance. The form and its radical are then computed on the resulting primitive Lie algebra. If a Hopf pairing for a double is also required, its two full Hopf algebras, product–coproduct adjointness, unit–counit normalization and antipode compatibility are additional data. They are not imposed as substitute axioms on the primitive Lie form.

PROPOSITION 1.128 *(Lie radicals and enveloping quotients).* Let L have an even supersymmetric invariant form B, and let $R = \operatorname{rad} B$. Then R is a Lie ideal, and L/R has the descended nondegenerate invariant form. The ideal generated by R in $U(L)$ is a Hopf ideal, and

$$U(L)/(U(L)R) \simeq U(L/R).$$

This construction does not require B to be a Hopf pairing.

Proof. For homogeneous $x \in L$, $r \in R$ and $z \in L$, invariance and supersymmetry give

$$B([x, r], z) = -(-1)^{|x||r|} B(r, [x, z]) = 0.$$

Thus R is an ideal. The form descends and is nondegenerate by the definition of the radical. In $U(L)$ each r is primitive, so $\Delta r = r \otimes 1 + 1 \otimes r$, $\varepsilon(r) = 0$ and $S(r) = -r$. The generated ideal is therefore also a coideal and is stable under the antipode. The universal property of the enveloping algebra identifies its quotient with $U(L/R)$. □

PROPOSITION 1.129 *(Radicals of a full Hopf pairing).* Let $\beta : H \otimes K \to k$ be a Hopf pairing. Suppose its induced quotient tensor pairings separate points, as they do in finite-dimensional quotients. Then its left and right radicals are Hopf ideals in their respective full algebras.

Proof. If r is in the left radical, product adjointness gives $\beta(hr, z) = \beta(h \otimes r, \Delta z) = 0$, and similarly for rh. Pairing r against the unit gives $\varepsilon_H(r) = 0$. The other adjointness equation says $\beta(\Delta r, y \otimes z) = \beta(r, yz) = 0$. Separation by the quotient tensor pairing forces the image of Δr in the tensor of the two quotient spaces to vanish. Thus $\Delta r \in R \otimes H + H \otimes R$. Antipode compatibility gives stability under S_H. The argument for the right radical is identical with the roles exchanged. □

PROPOSITION 1.130 *(What the pairing matrix proves).* A finite matrix and its kernel determine a bilinear radical and its perfect quotient pairing. Lie-ideal stability requires invariance of the primitive form. Hopf-ideal stability in a full algebra requires the full pairing hypotheses just stated, or the enveloping-algebra construction from a Lie ideal. These are distinct ways to construct a quotient.

Proof. The linear-algebra assertion is Proposition 1.126. The other assertions follow from the two preceding propositions, whose proofs use the respective invariance and adjointness identities. A kernel computation alone contains neither identity. □

The resulting quotient Lie algebra is still a geometric source for the recognition problem. A comparison with the BKM target must prove its generators, brackets, form and absence of extra relations as stated in §1.6. A matrix rank or a signed dimension does not give that map.

Transition maps

Let $R' \le R$. Write $T_{R'R} : H_R \to H_{R'}$ for a proposed inverse transition. It must first be defined by a morphism of coefficient diagrams or by another specified chain construction. The direction of a dual transition is reversed and is not silently identified with this one.

Definition 1.131 *(Multiplicative transitions).* A multiplicative transition commutes with the differential, unit and multiplication:

$$T_{R'R} m_R = m_{R'}(T_{R'R} \otimes T_{R'R}), \qquad T_{R'R}\eta_R = \eta_{R'}.$$

For partial charge windows the first equation is imposed only where both sides are defined. The transition comparisons must also respect shifts, orientation and the relative tensor when those are present.

Proposition 1.132 *(Multiplication under transition).* A coefficient-diagram transition preserving the extension maps and their trace comparisons satisfies these equations.

Proof. Move the transition through each leg of the defining composite for the product, using its supplied pullback, coefficient and trace naturality squares. Their composition is the displayed multiplicative square. The zero-object square gives the unit. □

Definition 1.133 *(Coalgebra transitions).* A coalgebra transition satisfies

$$\Delta_{R'} T_{R'R} = (T_{R'R} \otimes T_{R'R})\Delta_R, \qquad \varepsilon_{R'} T_{R'R} = \varepsilon_R, \qquad T_{R'R}\eta_R = \eta_{R'}.$$

The maps for successive stages compose and preserve the specified degree and parity.

Proposition 1.134 *(Coproduct under transition).* Transitions of collision coefficient diagrams preserving all their comparison maps satisfy these equations.

Proof. Compose the naturality squares for the collision pullback, coefficient morphism and trace. They give the coproduct square. The empty-output and zero-object squares give the counit and coaugmentation identities. □

Definition 1.135 *(Primitive transitions).* A primitive transition is the restriction $T^{\mathrm{Prim}}_{R'R} : P(H_R) \to P(H_{R'})$ of a coaugmentation- and coproduct-preserving transition. If an independently chosen primitive projection is used, its commutation with $T_{R'R}$ is an additional equation.

Proposition 1.136 *(Preservation of the primitive kernel).* Every coalgebra transition above preserves primitives, and multiplicative transitions also preserve their Lie bracket.

Proof. The reduced-coproduct square commutes, so its source kernel maps into its target kernel. Multiplicativity then preserves the two terms of the supercommutator. □

Definition 1.137 *(Radical transitions).* Let L_R and $L_{R'}$ have the actual primitive forms B_R and $B_{R'}$, and let $T : L_R \to L_{R'}$ preserve the Lie bracket. A sufficient pairing condition for radical descent is: for every opposite homogeneous $y' \in L_{R'}$, there exists $a(y') \in L_R$ such that

$$B_{R'}(Tx, y') = B_R(x, a(y')) \quad \text{for all } x \in L_R.$$

An adjoint lift satisfying $Ta(y') = y'$ is one possible stronger choice. In finite coordinates the condition is $T_\gamma^t G_{R',\gamma} = G_{R,\gamma} A_{-\gamma}$.

PROPOSITION 1.138 *(Radical descent).* Under this condition $T(\operatorname{rad} B_R) \subset \operatorname{rad} B_{R'}$. Hence T descends to the quotient Lie algebras. Compatible radical-preserving transitions give a quotient inverse system.

Proof. For x in the source radical, the right side of the pairing identity vanishes for every y'. Thus Tx lies in the target radical. The quotient map is well defined, and it is a Lie map because T is one. Its composition laws are inherited from the original transitions. □

COALGEBRAS AND THEIR COMPLETIONS

Definition 1.139 *(A geometric conilpotent coalgebra).* A geometric coalgebra $C_{X,R}$ is a coefficient object with a specified tensor, coaugmentation, counit and geometric collision coproduct. It is conilpotent if each element is killed by a sufficiently long iterated reduced coproduct. A sufficient filtration is $F_0 = k1 \subset F_1 \subset \cdots$, exhaustive, with

$$\overline{\Delta} F_n \subset \sum_{i,j>0,\ i+j\le n} F_i \otimes F_j.$$

If this is a chiral coalgebra on E or on its Ran space, the coefficient category, chiral tensor, supports and filtration are part of the definition; ordinary vector-space data alone do not define it. Transitions preserve all these structures.

PROPOSITION 1.140 *(Conilpotence and a completed limit).* The displayed positive filtration implies conilpotence of each algebraic element. An inverse limit of such coalgebras need not be conilpotent as an ordinary coalgebra. It carries the corresponding completed coproduct when that coproduct is continuous for the specified inverse-limit topology.

Proof. Every positive factor in an iterated reduced coproduct consumes at least one unit of filtration degree, so more than n factors kill F_n. For the limit, consider the tensor coalgebra truncated by word length and take its inverse limit. An element with nonzero components in all lengths has nonzero reduced coproducts of arbitrarily high length. The compatible finite coproducts land instead in the completed tensor product, where the induced limit map is continuous. □

PROPOSITION 1.141 *(Exactness of finite towers).* For a countable inverse tower of actual finite-dimensional vector spaces in each fixed degree, the Mittag–Leffler condition holds and $R^1 \varprojlim$ vanishes. Thus a short exact sequence of such towers remains exact under inverse limit. In particular, once compatible finite primitive and radical towers have been constructed,

$$\varprojlim_R (L_R/\operatorname{rad} B_R) \simeq (\varprojlim_R L_R)/(\varprojlim_R \operatorname{rad} B_R).$$

The kernel of an actual commuting map of inverse systems commutes with inverse limit without a Mittag–Leffler hypothesis. Operations on the limits still require their stated continuous tensor products.

Proof. The images in a fixed finite-dimensional term form a descending chain, which stabilizes. This is Mittag–Leffler and gives $R^1 \varprojlim = 0$ for the countable tower, as proved in Proposition 6.6. Apply the derived inverse-limit exact sequence to the radical short exact sequence. Kernels commute with inverse limits because both are limits. The continuous operation statement concerns the definition of their target topology, not exactness alone. □

The finiteness premise is substantial for moduli coefficients. For unbounded or infinite-dimensional systems one must prove an appropriate Mittag–Leffler property or retain the derived inverse-limit correction. Fibrewise K3 boundedness alone also does not construct a finite-type stability moduli space on $S \times E$. On the product hearts and stability conditions of [1, 75], the required input is boundedness in every selected class, together with a support bound making the permitted charge window finite. Threefold boundedness results such as [105] apply only after their stability hypotheses have been verified. Appendix 8 states these product-stability and support requirements.

Integral forms and the compact comparison

Definition 1.142 *(An integral-form quantum-group comparison).* An integral-form comparison starts with a specified graded Hopf algebra and a lattice stable under its operations. It includes a separated complete topology, an appropriate full Hopf pairing for the double, its cross-relations and Cartan part, and a comparison preserving these structures with the proposed quantum group. A stable-envelope realization also specifies the geometric correspondence, chamber, polarization, support and normalization conditions, and its chamber composition maps. For the Igusa problem, one proposed target is denoted $Y_\hbar(\widehat{\mathfrak{g}}_{\Delta_3})$. Here the chosen central extension, its quantum-group presentation and its coproduct have to be defined independently. The desired comparison is then

$$D_\hbar(H^+)^{\mathrm{int}}_{\mathrm{compl}} \longrightarrow Y_\hbar(\widehat{\mathfrak{g}}_{\Delta_3}).$$

An isomorphism would require an inverse preserving these operations. The notation for the target does not itself construct a Yangian for the Borcherds Lie superalgebra. The compact Hall–Drinfeld proposal of [79] is subject to the same coefficient, pairing and completion requirements.

Proposition 1.143 *(Scope of stable-envelope transport).* The stable-envelope theorem and the Schiffmann–Vasserot comparisons construct quantum-group objects on their specified symplectic and instanton moduli. Applying them to the compact reduced Hall construction requires the preceding comparison data. The formal tripled-quiver chart alone does not extend those objects or their operators to compact moduli.

Proof. Appendix 4.6 states the geometric hypotheses and the actual algebras of those constructions. A formal chart identifies a completed local deformation problem. It does not furnish a proper symplectic resolution, a torus action and stable envelopes on the whole compact moduli space, or the required coefficient and operator comparison. When those data are supplied, the induced map can be tested on the products, coproducts and double cross-relations stated in the definition. □

The normal-ordered charge group is independent of a stability chamber, but the selected sheaf moduli and their Hall operations can change with the chamber. A wall-crossing comparison must preserve the actual coefficient operations and match the primitive curve-counting chamber before its scalar series is identified with the normalized Igusa denominator. This is additional to a charge-lattice identification.

The original compact problem therefore has several concrete next maps. First construct the reduced oriented coefficient morphism on extensions and its equality on three-step flags. Then construct the collision coproduct and the mixed product–coproduct comparison. An operator realization of the BKM algebra further requires the primitive Lie form and the recognition map; a geometric Pfaffian or protected trace requires its own family, state and modular-line comparison. The local constructions above supply domains and equations for these tasks without replacing them by the scalar product. The collective notation $\mathfrak{O}_{\mathrm{Hall},R} = 0$ means that these geometric data and their relevant bialgebra, pairing and transition

equations have actually been supplied. Exactness of an inverse limit is imposed at the chosen completion, not inferred from the name of a finite stage.

1.4 Finite skew blocks and the protected Pfaffian

A square root of a determinant line need not come with a section. A skew operator supplies one: its Pfaffian. This subsection first constructs that finite algebraic object, then states the geometric comparisons needed to regard it as the protected section on the compact reduced moduli space.

The Pfaffian of a skew two-term complex

Let B be a complex base, V a vector bundle of even rank $2n$, and $D : V \to V^\vee$ a skew map. Regard

$$K_D = [V \xrightarrow{D} V^\vee]$$

as a complex in degrees $[-1, 0]$. In a frame its skew matrix has a Pfaffian, and under a frame change A it satisfies

$$\mathrm{Pf}(A^t D A) = \det(A)\,\mathrm{Pf}(D).$$

Thus these local functions define a section of $\det V^\vee$.

Definition 1.144 *(Pfaffian line and section).* For this skew complex define

$$\mathscr{L}_{\mathrm{Pf}}(K_D) = \det V^\vee, \qquad \mathrm{pf}(D) \in H^0(B, \mathscr{L}_{\mathrm{Pf}}(K_D)).$$

The determinant identity gives

$$\mathscr{L}_{\mathrm{Pf}}(K_D)^{\otimes 2} \simeq \det K_D, \qquad \mathrm{pf}(D)^2 = \det(D).$$

A geometric Pfaffian datum specifies such local skew models and compatible Pfaffian-line and section comparisons on their overlaps, or another precise construction of the same line and section. An orientation of a critical locus is a root pair; it does not by itself specify these operators or their Pfaffian sections.

The section is also $\omega_D^n/n!$, where $\omega_D \in \Lambda^2 V^\vee$ is the alternating form of D. In a hyperbolic basis its coefficient is the product of the skew-block entries and its square is the determinant. The identity holds on the dense set of invertible skew matrices and hence, being polynomial, on all skew matrices over $\mathbb{C}$. It therefore holds in families by the same universal polynomial identity.

The degree convention matters. For the unshifted CY3 self-Ext complex $K = R\,\mathrm{Hom}(F, F)$, Serre duality has the form $K \simeq K^\vee[-3]$. If a specified reduced complex has only Ext degrees one and two, shifting it by $[2]$ places those terms in degrees $[-1, 0]$, and changes the self-duality to a map to its dual shifted by $[1]$. A comparison with K_D must include this shift, the skew symmetry and the determinant-square map. One cannot impose the latter self-duality on the unshifted complex without changing its meaning. The shifted degree condition alone also does not provide a skew realization or a Pfaffian section.

Hyperbolic factors and the super ratio

Choose a positive charge monoid Γ^+ and a height function h which is positive on its nonzero elements and locally finite on the active degrees. Work in its completed monoid algebra, or at a fixed height jet. Then $1 - x^\gamma$ is a unit for $\gamma \neq 0$. Let $P_\gamma = P_{\gamma,\bar{0}} \oplus P_{\gamma,\bar{1}}$ have finite dimension $(p_\gamma \mid q_\gamma)$.

Definition 1.145 *(A finite skew block).* For one basis vector of either parity use the ordered hyperbolic block

$$J_\gamma = \begin{pmatrix} 0 & 1 - x^\gamma \\ -(1 - x^\gamma) & 0 \end{pmatrix}.$$

Take p_γ such blocks in the even part and q_γ in the odd part. Their super-Pfaffian means the ratio of the two ordinary Pfaffians:

$$\mathrm{sPf}(J_{\gamma,P}) = \frac{\mathrm{Pf}(J_\gamma^{\oplus p_\gamma})}{\mathrm{Pf}(J_\gamma^{\oplus q_\gamma})} = (1 - x^\gamma)^{p_\gamma - q_\gamma}.$$

The ordinary Pfaffian of all the blocks together has exponent $p_\gamma + q_\gamma$, not $p_\gamma - q_\gamma$. The ratio is defined on the formal unit domain just specified, or as a meromorphic section where its denominator is allowed.

The matrix is linear in the scalar $1 - x^\gamma$. This does not make it the symbol of the Hall bracket or an analytic Dirac operator. An abelian three-dimensional Lie algebra and the three-dimensional Heisenberg algebra give identical such blocks when assigned the same graded vector space, but have different brackets. In a Chevalley–Eilenberg model the bracket is a binary operation in its differential; in an enveloping algebra it appears in the Poisson bracket of the PBW associated graded. Relating either construction to a differential operator requires a separate map.

A finite geometric comparison

Definition 1.146 *(Finite geometric Pfaffian data).* A finite geometric Pfaffian datum (P_{fin}) consists of a specified reduced quotient base $\mathfrak{Q}_R$, its perfect reduced complex, oriented critical coefficients and skew realization with Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$ and section $\mathrm{pf}_{X,R}$. The skew realization is compatible with the shift and square map above.

On a specified cusp chart, the datum further gives an oriented block comparison with the finite family $J_{\gamma,P}$, including its determinant of frame change and every residual unit. The scalar normalization is fixed as $64x^\rho$, where $x^\rho = q^{1/2}r^{1/2}s^{1/2}$. Thus the block comparison identifies the section itself, rather than only its rank or squared determinant. Each P_γ is compared as a graded space with the target root space V_γ. In particular the source is zero where the target graded space is zero; vanishing of its signed dimension alone is not this condition.

At a simple type-II wall there is a compatible local factorization $\mathrm{pf}_{X,R} = vu\ \mathrm{pf}^{\tan}_{X,R}$, with simple normal zero, invertible v and a stipulated reflection lift. For the sign conclusion the reference comparison preserves the tangential factor and unit, the base map sends u to $-u$, and the actual root lift carries the section to its pullback.

Finally the datum specifies its parameter map, Pfaffian-to-automorphic line comparison and transition isomorphisms. Transitions preserve the square maps, the sections, the root comparisons and the critical coefficient data. An effective geometric limit is required when these are claimed to define a line and section on that limit.

This definition retains the distinction between a positive height jet and an actual finite moduli space. On the former, adding roots of higher height does not change a lower jet. It need not leave an ordinary section on the latter unchanged. A transition of the moduli problem must therefore be supplied separately. Likewise a root with signed multiplicity zero can still have equal nonzero even and odd parts; the full graded comparison detects them.

PROPOSITION 1.147 *(The finite Pfaffian product).* Suppose (P_{fin}) supplies the stated block and normalization comparison on a finite active set A_R. Then

$$\mathrm{pf}_{X,R}\,|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma \in A_R} (1 - x^\gamma)^{p_\gamma - q_\gamma}.$$

If $P_\gamma \simeq V_\gamma$ with both parities preserved, its exponents equal the target Borcherds exponents $a_\Delta(\gamma) = \operatorname{sdim} V_\gamma$.

Proof. The Pfaffian is multiplicative on the ordered orthogonal sum of skew blocks. Divide the odd Pfaffian into the even Pfaffian and use the specified frame and normalization to obtain the formula. The graded root comparison gives the exponent equality. Merely knowing the ranks and the leading coefficient would not prove this formula: multiplying a section by a nonconstant unit with constant term one preserves those data. The actual block comparison excludes that additional unit. □

Write $J(z) = \left(\begin{smallmatrix} 0 & z \\ -z & 0 \end{smallmatrix}\right)$. Then $J(1 - x)$ and $J(1 - x)(1 + x^2)$ have the same rank, leading coefficient and simple divisor at $x = 1$ wherever $1 + x^2$ is invertible. Their Pfaffian sections differ. The calculation exhibits the need to identify every unit in the block comparison.

Definition 1.148 *(The automorphic line comparison).* Let $\kappa : B \to \mathcal{A}_2$ be the specified parameter map, on the chosen stack or cover carrying the weight-five character line. A Pfaffian-to-automorphic comparison is an isomorphism

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \kappa^*(\mathcal{L}^5 \otimes \nu_{\Delta_5}),$$

compatible with the square map, the cusp frame and the stated boundary extension. A section identity means $\iota_{\mathrm{aut}}(\mathrm{pf}_X) = \kappa^* \Delta_5$. It includes the map κ; the two lines do not initially live on the same base.

PROPOSITION 1.149 *(What scalar squaring forgets).* Local scalar products, their divisors, their leading coefficients and their squares do not determine the descent of a square-root section. In particular they do not construct ι_{aut}.

Proof. Let T_χ be a nontrivial sign local system on a punctured quotient chart. Tensoring a local square-root section by a flat frame of T_χ leaves its local expression and its square unchanged. Around a loop with $\chi = -1$, that frame changes sign. The two local sections therefore have different descent data, even though their squares agree. This example concerns the flat or equivariant line with its specified descent; it makes no claim that a given scalar expression has two distinct global holomorphic descents. Those descents are precisely the missing information. □

TYPE-II CROSSINGS AND LOCAL NORMALIZATION

Definition 1.150 *(A rank-one Pfaffian crossing).* A normal Pfaffian crossing is a rank-two skew block

$$D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad \mathrm{Pf}(D_u) = u,$$

with a specified reference frame comparison and base reflection $u \mapsto -u$. Its tangential Pfaffian is an invariant unit in that reference comparison. The orientation lift is required to carry the section to its pullback. The phrase ''rank one'' counts one hyperbolic Pfaffian factor; the skew matrix itself has size two.

Proposition 1.151 *(The normal sign).* A rank-one crossing changes its Pfaffian section by -1 and has determinant u^2. If all three simple walls have these comparisons and they assemble to the coherent character ϵ_o, then $\epsilon_o = \det_W$ on the type-II Weyl group. Its automorphic target is the homogeneous orthogonal character $\widetilde{\nu}_{\Delta_5}|_W$.

Proof. The two-by-two Pfaffian is u, so its pulled-back expression is $-u$ and its square is u^2. Section compatibility forces the root lift to have residual sign -1 in the reference frames. The three generator values determine the character, as proved in the preceding subsection. A real path crossing $u = 0$ is not a path in the wall complement; the stipulated lift is part of the datum. □

The abstract sign set

$$\Omega_{12} = \{\pm 1\}^{12}$$

has cardinality $2^{12} = 4096$. To interpret this number as a count of geometric wall configurations, one needs a bijection from a complete geometric atlas to that set, including stabilizer weights and identifications on overlaps. The finite sign sum alone does not construct such an atlas.

From the finite product to the compact section

Local finiteness in positive height makes the products compatible as jets: only finitely many factors contribute to any fixed height. This gives a formal product. To obtain the holomorphic automorphic section one also needs convergence on a cusp domain, the parameter and line comparisons, and analytic continuation with the stated boundary behaviour. Under these assumptions the target product is Δ_5, with normalization

$$\Delta_5 = 64 q^{1/2} r^{1/2} s^{1/2} (1 + \cdots), \qquad \Delta_5^2 = \chi_{10},$$

where the complete boundary factors are those of Chapter 3.

If an independently defined protected trace satisfies the geometric sewing identity

$$Z_X = -4096\, \iota_{\mathrm{aut}}(\mathrm{pf}_X)^{-2},$$

then the section comparison gives $Z_X = -4096\, \kappa^*(\Delta_5^{-2})$. The trace and its sewing identity remain additional geometric data. The scalar branch sign in this formula does not determine ϵ_o, and the finite algebraic skew blocks do not alone construct the compact state space or its operator.

1.5 *The source bar coalgebra and the compact comparison*

The hybrid correspondences must first define an algebra in a specified coefficient category. Only then does its associative bar have a meaning. We use the unitalized Ran category $(\mathcal{R}_E^+, \star, K)$ of Section 9.1, with right differential-module coefficients, cross-block chiral convolution and empty unit K. A realization of constructible Hall coefficients in this category is additional geometric data. It must carry the products, orientation transport and higher collision coherences, including wrapped and mixed configurations.

Write $V_E = \mathsf{Den}_{\Delta_5, E}$ for the BD current envelope and $\mathcal{A}_{\Delta,E} = U_\star(\iota_* V_E)$ for its associative Ran envelope. The associative comparison in this subsection has $\mathcal{A}_{\Delta,E}$ as its target. A comparison directly with V_E as a BD chiral algebra is a different statement and needs the corresponding chiral functor. The bar–cobar equivalence in Proposition 9.4 does not identify these two recipients.

Lemma 1.152 *(Source and target bars).* For an augmented associative target T with $T_\varnothing = \mathbb{C}$, the counit $\Omega BT \to T$ is an internal equivalence in the specified Ran category. It constructs neither a compact-source algebra nor a morphism from such an algebra to T.

Proof. The counit depends functorially on the multiplication and augmentation of T. Its construction has no input consisting of compact correspondences, a realization of their coefficients, orientations or a source-to-target morphism. In particular, $C_X := BT$ would define a coalgebra from T, without identifying it with a coalgebra obtained from the compact geometry. □

Coefficient realization and the source coalgebra

At a stage R, denote the realized hybrid algebra by $A_{X,R}$. The index may bound geometric complexity or configuration arity. It does not by itself specify a root-height quotient. Whenever a target stage T_R is used, it must be an actual augmented associative object with supplied transition maps. A projection to symmetric positive and negative root degrees is not such an object, by Proposition 6.1. Identification of the limit of the T_R with $\mathcal{A}_{\Delta,E}$ is a separate comparison.

Definition 1.153 *(Chiral Koszul source datum).* For fixed stability, support and stage data (σ, S, R), a source datum $\mathfrak{K}^{\mathrm{Kos}}_{X,R}$ consists of the following objects and morphisms in $\mathcal{R}^{+}_{E}$.

(i) A realization of $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ as an augmented associative algebra $A_{X,R}$ with empty component $\mathbb{C}$. Its multiplication and all coherent collision operations come from the supplied hybrid correspondences and orientations.

(ii) A coaugmented counital coassociative coalgebra $C_{X,R}$ with empty component $\mathbb{C}$, differential d_C, coproduct Δ^{ch}_R and an exhaustive filtration $F^{\mathrm{co}}_{R,\bullet}$. The reduced coproduct is locally nilpotent on each finite configuration and retained filtration piece. If a global finite bound on that stage is used, it is included in the datum.

(iii) A morphism of coalgebras $b_{X,R} : C_{X,R} \to BA_{X,R}$ compatible with charge descent, coaugmentation, counit and the supplied geometric operations. Its cone is acyclic. The geometric construction of $C_{X,R}$ and this comparison are specified on local, wrapped and mixed inputs, with all higher-colored collision coherences.

(iv) Compatible transition morphisms on these objects. Their coherent diagrams include the reduced E-quotient and the normal-ordered charge homomorphism $\overline{\Pi}_X$.

A source-to-target comparison additionally supplies an augmented associative morphism $\vartheta_R : A_{X,R} \to T_R$ and proves its cone acyclic. Define

$$\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} := \varepsilon_{A_{X,R}} \circ \Omega b_{X,R} : \Omega C_{X,R} \longrightarrow A_{X,R}, \qquad \Theta_{\mathrm{Kos},R} := \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}.$$

The symbols B, Ω in this datum always refer to $\star$ and K.

The finite binary flag atlas supplies particular associativity and coproduct diagrams. It does not replace the higher-colored, unit, refinement and quotient coherences required by this definition. Nor does the projection-finite local sector provide the coefficient realization of wrapped inputs. A restriction to ordered words records the word-cut part of $C_{X,R}$. Comparison with the full Ran coalgebra must preserve every collision operation.

The obstruction notation can retain these separate requirements:

$$\mathfrak{O}_{\mathrm{Kos},R} = (\mathfrak{o}^{C}_{R}, \mathfrak{o}^{\mathrm{fil}}_{R}, \mathfrak{o}^{\mathrm{conil}}_{R}, \mathfrak{o}^{\Delta,\mathrm{ch}}_{R}, \{\mathfrak{o}^{C,\mathrm{tr}}_{R'R}\}, H^{\bullet}\,\mathrm{Cone}\, b_{X,R}, \mathfrak{o}^{\Omega}_{R}, \mathfrak{o}^{\mathrm{wrKos}}_{R}, \mathfrak{o}^{\mathrm{hyb}}_{R}, \mathfrak{o}^{\Pi}_{R}, \mathfrak{o}^{\mathrm{sep}}_{R}, \mathfrak{o}^{\mathrm{Prim}}_{R}).$$

Here the first entries mean the stated coalgebra, filtration and transition identities. $\mathfrak{o}^{\mathrm{wrKos}}_{R}$ includes the wrapped coefficient realization and its bar domain. The last entries require compatible charge descent,

separation of source from target, and any specified ordinary primitive realization. An absent object or map is an unresolved construction, rather than a numerical value of this tuple.

Proposition 1.154 *(Finite-length source bar).* Suppose the realized augmented algebra $A_{X,R}$ is supplied and its positive augmentation ideal $a_{X,R}$ has no nonzero retained $\star$-word of length greater than N_R. Then its normalized bar coalgebra is

$$C^{\mathrm{bar}}_{X,R} = BA_{X,R} = K \oplus \bigoplus_{1 \le p \le N_R} (sa_{X,R})^{\star p},$$

with the internal and multiplication differentials. The suspension s is the bar suspension. Multiplication has degree one after suspension. The coproduct is deconcatenation:

$$\Delta[x_1|\cdots|x_p] = 1 \otimes [x_1|\cdots|x_p] + [x_1|\cdots|x_p] \otimes 1 + \sum_{i=1}^{p-1} [x_1|\cdots|x_i] \otimes [x_{i+1}|\cdots|x_p].$$

It is conilpotent and supplies a source-bar model with $b_{X,R}$ the identity.

Without a uniform N_R, the same construction is finite at each configuration by the positive arity bound.

Proof. The normalized associative bar construction in this tensor category has the displayed terms. Associativity with its coherent form gives the square-zero bar differential. Cutting words gives a counital coassociative coproduct compatible with it. Each reduced cut lowers the available length, so more than N_R nonempty blocks cannot occur.

On a configuration I, positivity gives the bound $|I|$ instead. These are coefficient-category statements. To identify the coproduct with geometric cutting correspondences requires the realization and coherence maps stipulated in the source datum. □

If all nonempty geometric colors have positive HN height, the bound on total HN height can imply $N_R \le R$. That is a hypothesis on the retained geometric algebra, not a conclusion about the full two-sided root algebra. Additivity of $\overline{\Pi}_X$ ensures that a bar word, each cut and each multiplication term have the same descended Gram degree. This proves compatibility of gradings. It constructs neither the realization nor the comparison ϑ_R.

Corollary 1.155 *(The remaining source-to-target map).* For a source datum in the specified Ran category, $\varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$ is an equivalence. Hence $\Theta_{\mathrm{Kos},R}$ is an equivalence exactly when ϑ_R is. A bar construction in another coefficient category requires its own counit theorem before this conclusion applies.

Proof. The coalgebra map $b_{X,R}$ is an equivalence by assumption. Apply the associative cobar and the bar–cobar counit of Proposition 9.4. The two-out-of-three property proves the assertion. A target-internal counit supplies no part of ϑ_R. □

For a strict representative, the comparison equations include

$$d_T\vartheta_R = \vartheta_R d_X, \quad \vartheta_R\mu_X = \mu_T(\vartheta_R \star \vartheta_R), \quad \vartheta_R\eta_X = \eta_T, \quad \vartheta_R\rho^X_{R'R} = \rho^T_{R'R}\vartheta_{R'}.$$

For coherent representatives, these equations are replaced by the specified coherent homotopies. In either case the full coefficient cone $\mathrm{Cone}\,\vartheta_R$ must be acyclic. The Lie–commutative chiral theorem of [40, Theorems 5.1.1–5.2.1] does not supply this compact associative comparison. It enters the envelope construction at precisely the scope proved in Proposition 9.4.

COMPATIBILITY WITH THE GEOMETRIC STRUCTURES

Definition 1.156 *(Finite Koszul comparison datum).* A comparison datum $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$ supplements the source datum by the following maps and commuting diagrams.

(K0) The source counit and ϑ_R have acyclic cones. If finite filtrations are used to prove this, the associated-graded cone comparisons are acyclic at every retained coefficient and parity.

(K1) Every specified Weyl transport is intertwined by $\Theta_{\mathrm{Kos},R}$. A reflection may map between chambers or require a larger finite operation domain. Both domain and codomain are specified. In chain representatives the equation is $\Theta\tau_i^X - \tau_i^T\Theta = dH_i^W + H_i^W d$. The H_i^W carry the higher coherences for their defined compositions. No finite symmetric root set is assumed invariant under the whole Weyl group.

(K2) On the perfect complexes defining the geometric Pfaffian and determinant lines, a supplied line isomorphism intertwines orientation transport. Its type-II reflection sign is $\epsilon_o(s_{\delta_i}) = \widetilde{\nu}_{\Delta_\vee}(s_{\delta_i}) = -1$. The wall and chamber isomorphisms satisfy the cocycle identities of $(O1)^+$ and $(O2_{\mathrm{atlas}})$.

(K3) The associative map preserves products and units, and its bar preserves coproducts, coaugmentations and counits. If an ordinary Hall/Hopf realization is used to extract primitives, it is separately supplied and its operation diagrams commute. The invariant Lie form on those primitives is intertwined with $B_{\mathfrak{g}}$. It is not identified with a pairing adjoint to the ordinary primitive Hopf coproduct. Such an identification is excluded by Proposition 8.1.

(K4) The primitive comparison carries the invariant source Lie radical onto the target radical. It induces the comparison on their quotients. When the target is $\mathfrak{g}$, its invariant form has zero radical. Kernel and quotient squares are required in the actual coefficient realization.

(K5) This primitive comparison is the same map ι_R^{Prim} used in recognition. The induced maps of ordinary enveloping algebras preserve PBW length and the finite weighted pieces of Proposition 6.3.

(K6) Stage transitions and all the preceding structures form coherent diagrams. In a strict chain model, $\Theta_R\Omega(\rho^C_{R'R}) - \rho^T_{R'R}\Theta_{R'} = dH^{\mathrm{tr}}_{R'R} + H^{\mathrm{tr}}_{R'R}d$, with the composition homotopies also specified.

(K7) Completion uses the homotopy inverse limit of these diagrams. To use ordinary inverse limits, require degreewise Mittag–Leffler chain groups for the objects and comparison cones. Alternatively, supply another explicit comparison between ordinary and homotopy limits. Bar and cobar commute with the chosen completed construction only when that additional comparison has been proved.

The notation $\mathfrak{O}_{\mathrm{Kcmp},R}$ denotes the two cone cohomologies together with the Weyl, line, operation, radical, PBW and transition defects in these diagrams. Their vanishing includes the supplied coherent nullhomotopies, not just equal matrix ranks.

PROPOSITION 1.157 *(Compatible finite comparison).* Under (K0)–(K6), the map $\Theta_{\mathrm{Kos},R} : \Omega C_{X,R} \to T_R$ is an augmented associative equivalence compatible with the specified Weyl, orientation, primitive, invariant-form and enveloping diagrams.

Proof. Clause (K0) and the source counit give the equivalence. Each remaining compatibility is a supplied commuting square or coherent homotopy, and composition with the counit preserves it. The radical comparison descends to quotient spaces. The ordinary enveloping universal property gives the PBW comparisons. This conclusion uses the stated realizations of these structures, not an automatic extraction of a Hall object from the target. □

Proposition 1.158 *(Agreement on selected generators).* Agreement with a prescribed map on a selected family of root vectors does not imply an equivalence of associative coefficient objects. For a supplied comparison ϑ_R, the remaining condition is acyclicity of its full coefficient cone.

Proof. The failure already occurs for ordinary augmented dg algebras. Let B be such an algebra and let M be a dg B-bimodule with nonzero cohomology. The square-zero extension $A = B \oplus M$ has

$$(b, m)(b', m') = (bb', bm' + mb'), \qquad M^2 = 0.$$

Its augmented projection $p : A \to B$ fixes every selected vector in the copy of B, but $\mathrm{Cone}(p) \simeq M[1]$ is not acyclic. This example asserts neither a connected Hopf structure on A nor agreement on all its primitives. Under those additional hypotheses, Proposition 8.3 gives the positive ordinary Hopf recognition theorem. □

To establish the analogous conclusion in a chiral coefficient category one must inspect the local comparison cone, not merely the ordinary example. Additional wrapped, mixed or descendant coefficients can lie outside the selected root-vector family. The invariant form and the ordinary primitive map likewise need the specified realizations in (K3)–(K5).

Proposition 1.159 *(Passage to completion).* Suppose source and comparison data are supplied on a cofinal sequence of stages, with the coherent transitions of (K6). Then their homotopy inverse limits give equivalences

$$\operatorname*{holim}_R C_{X,R} \simeq \operatorname*{holim}_R BA_{X,R}, \qquad \operatorname*{holim}_R \Omega C_{X,R} \simeq \operatorname*{holim}_R T_R,$$

with all specified compatible operations. Under (K7), the same assertions hold for the ordinary inverse limits of the chosen chain models. Writing these limits as BA_X, ΩC_X and $\mathcal{A}_{\Delta,E}$ additionally requires the corresponding bar, cobar and target-limit comparisons.

Proof. A homotopy limit is a functor on the coherent diagram category and preserves equivalences. Apply it to $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ and their compatibility diagrams. The ordinary-limit assertion follows from Proposition 6.7. For actual degreewise finite-dimensional towers, the required Mittag–Leffler property is automatic once their transition maps exist. In general it is not automatic, as the polynomial-tail example there shows. Nothing in preservation of equivalences by limits alone says that bar or cobar commutes with that limit. □

Twisting cochains and the boundary algebra

A twisting cochain from $C_{X,R}$ to an augmented algebra has unsuspended degree one. It satisfies

$$d\tau + \tau d + \mu(\tau \star \tau)\overline{\Delta} = 0,$$

with the suspension signs fixed by the bar convention. Equivalently it is a morphism from the cobar. For a bar source it includes the canonical twisting cochain. This equation describes coupling data at the specified carrier and tensor. It does not identify the bar coalgebra with the bulk algebra.

A proposed bulk center belongs to the associative boundary algebra $A_b = R\,\mathrm{End}(b)$ and requires its separate map and hypotheses, as in Remark 11.4. The empty coefficient K, the BD vacuum, the augmentation of $A_{X,R}$ and a prepared state are not interchangeable. In particular, obtaining a completed coefficient equivalence to $\mathcal{A}_{\Delta,E}$ still leaves the state comparison and its protected trace to be constructed. The geometric primitive recognition in the next subsection is another operation-preserving comparison, with its own explicit source data.

1.6 Primitive recognition

The compact recognition problem asks for a graded Lie superalgebra isomorphism from geometric primitives to the denominator algebra $\mathfrak{g} = \mathfrak{g}_{\Delta_5}$ of Chapter 4. Its first interior comparison must identify 29 even and 93 odd directions and their brackets. The scalar -64 specifies only their difference.

We first work with algebraic direct sums over root degrees. Positive height completion is available by Proposition 6.1. A completion of both signs must be separately defined. In particular, the full product of positive and negative root spaces is not a topological Lie algebra with the coordinatewise topology. This distinction applies equally to the proposed compact source and to the target.

The source and its presentation map

Assume the hybrid geometry supplies an ordinary Hall super-bialgebra H_X in a specified coefficient realization. Its associative product m_X, unit and counital coproduct Δ_X satisfy

$$\Delta_X m_X = (m_X \otimes m_X)(1 \otimes \tau \otimes 1)(\Delta_X \otimes \Delta_X),$$

where τ is the Koszul braiding. Its actual primitives form a Lie superalgebra under the supercommutator. If a cochain realization is used, all differentials and the passage to protected cohomology must preserve the operations needed for this assertion.

Denote the descended primitive algebra by $P_X^{\Pi,\mathrm{desc}}$. A formula such as

$$P_X^{\Pi,\mathrm{desc}} = H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X)$$

means that the coefficient realization, primitive functor and charge descent have been constructed and shown to commute in that formula. It is not a definition of those maps from the charge labels alone.

Let B_X be an independent even invariant Lie form on this algebra, and set

$$R_X = \operatorname{Rad} B_X, \qquad P_X^{\Pi,\mathrm{red}} = P_X^{\Pi,\mathrm{desc}}/R_X.$$

Invariance makes R_X a Lie ideal. Its generated ideal in the ordinary enveloping algebra is a Hopf ideal by Proposition 8.2. The invariant form is not required to be adjoint to the ordinary primitive Hopf coproduct. That requirement would contradict the real $\mathfrak{sl}_2$ sector.

Let $\mathfrak{F}$ be the free Lie superalgebra on a basis of $\mathfrak{h} = L \otimes \mathbb{C}$ and the symbols e_i, f_i with their prescribed parities. Let J_{BK} be the ideal generated by Definition 3.3, including the linear identification $h_i = \alpha_i \in \mathfrak{h}$. Thus $\mathfrak{F}/J_{\mathrm{BK}} = \mathfrak{g}$. This is the Gritsenko–Nikulin presentation [52, Section 3 and Statement 6.4], with its product and lattice forms in [8, 53, 54, 66]. No further radical quotient of $\mathfrak{g}$ is needed: its invariant form is already nondegenerate.

THEOREM 1.160 *(Recognition from geometric representatives).* Suppose the following data are supplied over $\mathbb{C}$.

(S1) The source primitive algebra and its normal-ordered descent are constructed. Their bracket is graded by Γ_{gram}, and each descended homogeneous parity space is finite-dimensional.

(S2) An even identification $P_{X,0}^{\Pi,\mathrm{red}} \simeq \mathfrak{h}$ matches the source degrees with L through $\alpha(n, l, m)$.

(S3) For every simple index i, the source supplies e_i^X, f_i^X in degrees $\pm\alpha^{-1}(\alpha_i)$ and parity $p(i)$. It also supplies h_i^X corresponding to α_i in the Cartan. The imaginary fibers have the prescribed $\mu_{\tilde{p}}(a)$ generators.

(S4) These representatives satisfy the Cartan, Chevalley, real Serre, orthogonality and super sign relations of Definition 3.3 in the source bracket.

(S5) The invariant source Lie form descends to $P_X^{\Pi,\mathrm{red}}$, and the presentation map preserves it and $B_{\mathfrak{g}}$. The radical and quotient maps are actual maps of the supplied source objects.

(S6) For the induced map $\pi : \mathfrak{F} \to P_X^{\Pi,\mathrm{red}}$, one has $\ker \pi = J_{\mathrm{BK}}$.

(S7) The Cartan and simple representatives generate $P_X^{\Pi,\mathrm{red}}$ by finite bracket words.

(S8) The source and target ordinary enveloping algebras carry their PBW length filtrations and the finite weighted pieces of Proposition 6.3. The comparison uses these actual algebras, not envelopes of symmetric root fragments.

(S9) Whenever a topological conclusion is desired, a completion domain and continuous operations are specified on both sides. The comparison and its inverse extend continuously. For positive height this uses the genuine positive Lie quotients. For another domain it is a separate hypothesis.

(S10) The source parity dimensions agree with e_β, o_β in every positive and negative root degree, and vanish in non-root degrees. At δ_{123} they are $29|93$, not a choice inferred from the signed value -64.

Then $\mathfrak{g} \to P_X^{\Pi,\mathrm{red}}$ is a graded Lie isomorphism preserving the invariant form. It induces a filtered Hopf isomorphism of ordinary enveloping algebras. Under (S9), it also identifies the specified completions. Its superdenominator is $64^{-1}\Delta_5(2Z)$.

Proof. The representatives and (S4) factor π through $\mathfrak{g}$. Clause (S6) gives injectivity and (S7) gives surjectivity. The degree, parity and form conditions make this the claimed structured isomorphism.

The ordinary enveloping universal property and PBW give the filtered Hopf assertion. Continuity of both maps proves the completed statement in (S9). The denominator follows from Theorem 4.1.

In particular, (S10) is a consequence of the established isomorphism, while its finite parity tests can be used in proving injectivity or surjectivity. It is not a substitute for them. □

For a real pair the required Serre exponent is three. At $a_{ij} = \delta_i + \delta_j$, the source must supply nine even null simple representatives in addition to $[e_i^X, e_j^X]$. For the complementary real index k, their exponent is five. The relations at δ_{123} also restrict possible source differentials: Theorem 5.3 forbids the rank-28 cancellation on the unchanged presentation beyond height five. A derived replacement must therefore identify its changed relations and its comparison map.

If the independent coefficient data of Definition 1.156 are supplied, their primitive comparison can be required to be the inverse of the isomorphism above. The resulting statement concerns the associative Ran envelope $\mathcal{A}_{\Delta,E}$ in its specified category. It does not follow solely from the ordinary primitive isomorphism.

Finite root sets and geometric support

Definition 1.161 *(Cofinal finite recognition data).* For $\nu \geq 1$ set

$$W_\nu = \{\beta \in \mathcal{R}_+ : \mathrm{ht}\,\beta \leq \nu\}, \qquad S_{W_\nu} = W_\nu \cup (-W_\nu) \cup \{0\}, \qquad \Gamma(W_\nu) = \alpha^{-1}(W_\nu).$$

Finite recognition data on an exhaustion consist of the following compatible homogeneous spaces and maps.

(R0) Geometric source spaces with specified compact supports, coefficient summands, orientation data and charge maps. Their basis vectors are classes of these objects.

(R1) The simple representatives and Cartan maps in each retained degree, with the prescribed parities.

(R2) The homogeneous bracket maps and the defining relation identities. Each iterated identity is evaluated in a finite enlargement retaining all intermediate degrees.

(R3) Even bijections $A_{\beta,\bar{p}} : P^{\Pi,\mathrm{red}}_{X,\alpha^{-1}(\beta),\bar{p}} \to \mathfrak{g}_{\beta,\bar{p}}$ for all retained root degrees in both signs and in degree zero. Require the source spaces to vanish in every non-root degree. Include these zero-space tests in the finite degree exhaustion.

(R4) Opposite-degree invariant Lie pairings, their radicals and quotient maps. The comparison intertwines the pairings and brackets. Where ordinary Hall products and coproducts are used to define primitives, their bialgebra identities hold in finite operation diagrams with the necessary output spaces.

(R5) No extra relation in each bounded word space: $\ker(\pi|_{F_N\mathfrak{F}}) = J_{\mathrm{BK}} \cap F_N\mathfrak{F}$. Here F_N bounds total generator weight, assigning weight one to Cartan generators and $\mathrm{ht}\,\alpha_i$ to e_i, f_i. This finite-dimensional word bound is different from fixing only the total root degree.

(R6) Every retained source vector is a finite bracket-word combination of the supplied representatives, in some finite enlargement containing its intermediate degrees.

(R7) The ordinary enveloping comparison preserves PBW length and the finite weighted pieces. Its maps are the ones induced by the primitive comparison.

(R8) Restriction and enlargement maps agree on common homogeneous operations. For a completed statement, actual quotient or inverse-system maps are supplied in a completion domain with continuous operations, as in (S9).

The maps $A_{\beta,\bar{p}}$ must commute with every retained bracket and pairing square. Root sets carry partial operations as in Definition 6.2. No zero extension of omitted brackets is imposed.

A bounded root degree alone does not make the free presentation finite: positive and negative generators can cancel in arbitrarily long words. The total generator-weight bound in (R5) does give finite-dimensional word spaces. The intersection $J_{\mathrm{BK}} \cap F_N\mathfrak{F}$ is the actual ideal intersection, not merely the span of defining relators whose written degrees happen to lie in a root set.

PROPOSITION 1.162 *(Height sets and their inverse degree map).* The W_ν are finite, downward saturated and cofinal in $\mathcal{R}_+$. Their Gram preimages are

$$\Gamma(W_\nu) = \{(b_1, b_1 + b_2 - b_3, b_2) : \textstyle\sum_i b_i\delta_i \in W_\nu\}.$$

They lie in the product cusp support.

Proof. There are finitely many nonnegative triples with sum at most ν. A proper positive subdegree has no greater height. Every positive root has finite height.

Solving the three coefficients in (4.3) gives the displayed inverse. If $b_2 > 0$ then $m > 0$. If $b_2 = 0, b_1 > 0$ then $m = 0, n > 0$. If $b_1 = b_2 = 0, b_3 > 0$ then $n = m = 0, l < 0$. □

COROLLARY 1.163 *(Exhaustion of positive roots).* Every positive root β lies in $W_{\mathrm{ht}\,\beta}$, and $\bigcup_\nu W_\nu = \mathcal{R}_+$.

Proof. Finiteness of height proves the assertion. A root of zero superdimension still occurs, since roots were defined by their spaces. □

Definition 1.164 *(Geometric support exhaustion).* A discrete geometric support exhaustion uses a set $\mathscr{S}^{\mathrm{cpt}}_X$ of compact support types and a height $h_X : \mathscr{S}^{\mathrm{cpt}}_X \to \mathbb{N}$ with finite fibers. Its stages are $\mathscr{S}^{\mathrm{cpt}}_{X,\le R} = \{s :$

$h_X(s) \le R\}$. Each type carries its actual source stratum, proper support, stabilizer and orientation data. Geometric transition maps between the associated moduli or coefficient objects are also specified. Inclusions of the index sets do not alone give these maps.

Proposition 1.165 *(Support exhaustion).* Such an exhaustion contains every retained support type in a finite stage. It does not assert that the union of the corresponding moduli has finite dimension or finitely many points.

Proof. The type s occurs at $R = h_X(s)$. Finiteness concerns the index set of types, not the geometric objects they index. □

Definition 1.166 *(Compatibility of source and root degrees).* For each geometric stage, specify its normal-ordered degree map $d_R : \mathscr{S}^{\mathrm{cpt}}_{X,\le R} \to \Gamma_{\mathrm{gram}}$ and an index $\nu(R)$ such that $\alpha(d_R(s)) \in S_{W_{\nu(R)}}$ whenever that support contributes to the primitive comparison. The maps agree on shared support types as R increases, and $\nu(R)$ can be chosen nondecreasing.

Proposition 1.167 *(Degree-compatible finite comparison).* These data place each retained geometric primitive space in a specified target degree and preserve that assignment under transition.

Proof. Apply the defining equality with α at each support type. Compatibility of d_R makes the same equality hold in every later stage. Negative degrees and the Cartan use $-W_\nu$ and $\{0\}$. □

The existence of a geometric support exhaustion and its operation maps is part of the compact construction. The target height exhaustion supplies no such geometric maps. In a proposed source whose support cannot be indexed with finite fibers, another finite coefficient construction must replace this discrete formulation. That limitation concerns this exhaustion method, not existence of all possible compact sources.

Recognition in finite homogeneous coordinates

Proposition 1.168 *(Target operation spaces).* A finite comparison with $\mathfrak{g}$ requires its two parity spaces, bracket maps, opposite-degree invariant pairings and ordinary enveloping pieces on the retained operation domains. The signed dimensions f_β or simple coefficients $m(\beta)$ alone do not define these maps.

Proof. A bracket square has a vector space at each input and at its output, and a pairing square has the specified opposite-degree spaces. A scalar dimension cannot supply their structure maps. The parity recursion constructs dimensions for the target presentation. Bases and operations still come from that presentation. □

Proposition 1.169 *(Finite homogeneous recognition).* Finite recognition data determine an isomorphism of the retained homogeneous operation diagrams

$$A_W : P_X^{\Pi,\mathrm{red}}|_{\alpha^{-1}(S_W)} \xrightarrow{\sim} \mathfrak{g}|_{S_W}.$$

It preserves the Cartan, simple representatives, retained brackets and invariant pairings. Cofinal compatible data identify the full algebraic graded Lie algebras.

Proof. Each parity block of A_W is bijective by (R3), and the bracket and pairing squares commute by (R2)–(R4). Their inverses preserve the same maps. Every homogeneous operation occurs in some finite enlargement, so the compatible family identifies all operations on the direct sums. Equivalently, (R5)–(R6) prove injectivity and surjectivity of the source presentation map. This is Theorem 6.4. No enveloping algebra of a symmetric root fragment is used. □

Definition 1.170 *(Finite source matrices).* Choose homogeneous bases of the geometric coefficient and primitive spaces. Denote the ordinary Hall product, coproduct, bracket and invariant Lie form by matrices M, D, B, G, and the unit and counit by η, ϵ. These matrices act on their specified input and output spaces, with enlargement when a product increases the retained weight. Kernel inclusions, quotient maps and source-to-target maps are denoted K, Q, A.

The coefficient basis is obtained from the supplied compact support, vanishing-cycle or intersection-cohomology summand, reduced and quotient orientations, finite-stabilizer action and protected integration. Thom–Sebastiani and transition isomorphisms specify how these coefficients enter the Hall operations. These geometric objects and maps precede their coordinate matrices.

Require associativity of M, coassociativity of D, and the left and right unit and counit laws. Require also

$$DM = (M \otimes M)(1 \otimes \tau \otimes 1)(D \otimes D), \qquad B = M - M\tau \quad \text{on primitives,}$$

and the invariant-form identities

$$G([x, y], z) = G(x, [y, z]), \qquad G(x, y) = (-1)^{|x||y|} G(y, x).$$

The comparison squares include

$$A_{\beta+\gamma} B^{X}_{\beta,\gamma} = B^{\mathfrak{g}}_{\beta,\gamma} (A_\beta \otimes A_\gamma), \qquad G^{X}_{\beta} = A^{\mathrm{t}}_{\beta} G^{\mathfrak{g}}_{\beta} A_{-\beta}$$

in homogeneous bases, with the super sign convention on reversed arguments.

Radical inclusion and quotient matrices satisfy $\operatorname{im} K = \ker G$, $QK = 0$, and induce a nondegenerate form on the quotient. The relation matrices express the defining Cartan, Chevalley, Serre and orthogonality identities. Their actual ideal intersections enter (R5). The bracket-word spans enter (R6).

The induced enveloping matrices use the super PBW relations and finite weighted pieces. They need not satisfy Hopf adjointness with the same matrix G. Proposition 8.1 proves that the latter requirement is incompatible with this target.

PROPOSITION 1.171 *(Coordinate form of recognition).* Over $\mathbb{C}$, the finite source spaces and maps of (R0)–(R8) are equivalently specified by the matrices above. Their geometric construction, exact kernel equalities and commuting operation diagrams remain part of the specification. This equivalence is independent of the chosen homogeneous bases.

Proof. A choice of basis turns every given linear map into a matrix. Composition, kernel and intertwining identities are unchanged by conjugating the source and target bases. Conversely, matrices on the specified geometric spaces define those maps when they satisfy the stated identities. Equality of a proposed relation subspace with a kernel requires inclusion and equality of dimensions, not only two unrelated ranks. Invertibility of each comparison block gives its actual inverse. Bare matrices of the right sizes do not identify their underlying spaces with geometric coefficients. □

Proposition 1.172 *(Cofinal recognition).* Suppose the algebraic source and its presentation map are constructed. Conditions (S1)–(S8) and (S10) can be checked on cofinal finite homogeneous operation domains and bounded word spaces. A completed recognition statement additionally requires (S9).

Proof. Every algebraic vector and relation is a finite linear combination of words, and hence occurs at some total generator-weight bound. Every bracket or pairing square involves finitely many degrees. Thus the finite data exhaust the defining maps and identities. Conversely, a global graded isomorphism and its specified geometric source restrict to these diagrams and preserve their kernels and parities. A topology and convergence of its operations are not consequences of an algebraic finite-word assertion, so the completed conclusion retains (S9). □

Transitions, parity and PBW

Charge descent changes the grading before any source-to-target comparison is made. Its formal effect on supplied algebraic data can be stated separately.

Proposition 1.173 *(Additive regrading).* Let $q : \Gamma \to \Lambda$ be a homomorphism of charge groups and $V = \bigoplus_{\gamma\in\Gamma} V_\gamma$ a graded super vector space. Define

$$(q_*V)_{\lambda,\bar{p}} = \bigoplus_{q(\gamma)=\lambda} V_{\gamma,\bar{p}}.$$

Regrading preserves both parity summands and any supplied homogeneous filtration. For a graded Lie superalgebra L, it gives

$$q_*U(L) \simeq U(q_*L), \qquad q_*\,\mathrm{gr}_{\mathrm{PBW}}\,U(L) \simeq \mathrm{Sym}_{\mathrm{super}}(q_*L).$$

A supplied positive/Cartan/negative direct-sum decomposition survives as a labeled decomposition. To recover those labels from the new charge alone, require q to preserve the corresponding positive, zero and negative supports.

Proof. Regrading retains every vector, its parity and its filtration membership. Additivity gives $q(\gamma + \gamma') = q(\gamma) + q(\gamma')$, so each homogeneous product, coproduct or bracket has the new required degree. It changes neither the tensor algebra nor its supercommutator relations. Hence it preserves the enveloping algebra, its PBW length and its associated graded.

The three labeled summands remain direct because their underlying subspaces have not changed. If their new charge supports overlap, charge alone cannot distinguish them. Finite retained vector spaces remain finite, but arbitrary charge fibers need not have finite dimension. A completed regrading therefore requires its own summation and continuity hypotheses. □

Apply this proposition to the supplied normal-ordered homomorphism $\overline{\Pi}_X$. It preserves constructed parity, PBW and triangular data. It does not supply source dimensions, representatives, relations or the geometric maps in the recognition theorem.

An inclusion of finite root sets is an enlargement of an operation diagram. It is not automatically a quotient morphism of Lie algebras. In the following statements, L_ν and their transitions are actual Lie superalgebras and morphisms, for example the positive-height quotients. This supplies the domain needed by the enveloping functor.

Definition 1.174 *(PBW transition data).* For an even Lie morphism $T^{Q}_{\nu+1,\nu} : L_{\nu+1} \to L_{\nu}$, its PBW transition is the induced map $U(T^{Q}_{\nu+1,\nu}) : U(L_{\nu+1}) \to U(L_{\nu})$. When weighted finite pieces are retained, assume T^{Q} respects that weight filtration as well. The comparison maps and transitions are required to form commuting diagrams.

PROPOSITION 1.175 *(Functorial PBW filtration).* The induced map preserves PBW length. Its associated-graded map is $\mathrm{Sym}_{\mathrm{super}}(T^{Q})$, and these maps compose functorially. Under the additional weight condition, they preserve the specified finite weighted pieces.

Proof. A generator maps to a generator in L_{ν}. A word of length at most d maps to such a word. Preservation of the Lie bracket carries the source supercommutator relation to the target relation, so the tensor-algebra map descends. Super PBW identifies its symbol with the symmetric map. Both construction and composition are functorial. The weight assertion follows by adding the weights of generators in each word. □

Definition 1.176 *(Parity transition data).* Let J_{ν} be $+1$ on $V_{\nu,\bar{0}}$ and -1 on $V_{\nu,\bar{1}}$. A parity-preserving transition satisfies

$$J_{\nu}T_{\nu+1,\nu} = T_{\nu+1,\nu}J_{\nu+1}.$$

The same equation is required in each retained positive, negative and Cartan degree, compatibly with its comparison map.

PROPOSITION 1.177 *(Parity blocks).* The equation above is equivalent to zero off-diagonal parity blocks. It restricts the inverse system separately to both parities.

Proof. If $J_{\nu+1}x = (-1)^{p}x$, then $J_{\nu}Tx = TJ_{\nu+1}x = (-1)^{p}Tx$. Conversely, a map preserving both eigenspaces commutes with J. The restricted maps inherit identity and composition from T. □

Definition 1.178 *(Finite primitive towers).* A finite primitive tower consists of actual finite-dimensional spaces $P_{\nu,\gamma,\bar{p}}$ and compatible inverse-system maps $t_{\nu'\nu}$ preserving degree and parity. The notation does not assert that these are projections of a symmetric root fragment as Lie algebras.

PROPOSITION 1.179 *(Primitive inverse limits).* Every such countable tower is Mittag–Leffler and satisfies $R^{1}\varprojlim_{\nu} P_{\nu,\gamma,\bar{p}} = 0$.

Proof. Images in a fixed finite-dimensional term form a descending chain and stabilize. Apply Proposition 6.6. No additional stabilization hypothesis is needed after these finite spaces and their actual transition maps have been constructed. □

Definition 1.180 *(Pairing-kernel towers).* At each degree and parity, the invariant form defines

$$\lambda_{\nu,\gamma} : P_{\nu,\gamma} \to P^{*}_{\nu,-\gamma}, \qquad \rho_{\nu,\gamma} : P_{\nu,-\gamma} \to P^{*}_{\nu,\gamma},$$

with kernels $K^{\ell}_{\nu,\gamma}$ and $K^{r}_{\nu,\gamma}$. To form a kernel tower, require that the specified transitions carry each kernel into the corresponding kernel and compose. For a map of inverse systems with these dual codomains, specify the codomain transitions separately. Dualization reverses the direction of an inverse transition.

PROPOSITION 1.181 *(Kernel inverse limits).* The countable towers of these finite kernels have $R^{1}\varprojlim K^{\ell}_{\nu,\gamma} = R^{1}\varprojlim K^{r}_{\nu,\gamma} = 0$. For an actual commuting map of inverse systems, its kernel commutes with inverse limit without a Mittag–Leffler assumption.

Proof. Finite-dimensionality gives the first assertion. The second follows because the kernel and the inverse limit are limits, which commute. It makes no identification between an inverse limit of dual spaces and a continuous dual unless the necessary variance and topology have been supplied. □

Definition 1.182 *(Finite PBW associated-graded towers).* Suppose actual enveloping transitions preserve finite weighted pieces $F_N U(L_\nu)$ and PBW length $P_d U(L_\nu)$. Their finite associated-graded pieces are

$$G_{\nu,N,d,\chi,\bar{p}} = \left(\frac{F_N U(L_\nu) \cap P_d U(L_\nu)}{F_N U(L_\nu) \cap P_{d-1} U(L_\nu)} \right)_{\chi,\bar{p}}.$$

The induced quotient maps define the corresponding inverse towers.

Proposition 1.183 *(PBW inverse limits).* These towers satisfy $R^1 \varprojlim G_{\nu,N,d,\chi,\bar{p}} = 0$. For the limit filtration defined by the finite filtered inverse systems, taking the indicated associated-graded quotient commutes with inverse limit.

Proof. The G_ν are finite-dimensional. The short exact sequence of the two filtered pieces and their quotient also has finite-dimensional kernel. Proposition 6.6 therefore proves exactness after inverse limit and gives the associated-graded quotient. This does not identify $U(\varprojlim L_\nu)$ with $\varprojlim U(L_\nu)$. Such a comparison has its own completion requirements. □

Proposition 1.184 *(Compatible limits and radical quotients).* Compatible finite-level isomorphisms $A_\nu : V_\nu \to W_\nu'$ induce an isomorphism of ordinary inverse limits without an additional exactness hypothesis. For a compatible short exact sequence of finite-dimensional spaces $0 \to R_\nu \to P_\nu \to Q_\nu \to 0$, one has

$$\varprojlim Q_\nu \simeq (\varprojlim P_\nu)/(\varprojlim R_\nu).$$

If the towers are given the inverse-limit linear topology, $\varprojlim R_\nu$ is closed. These assertions preserve operations only when those operations are defined continuously on the limits.

Proof. The inverses A_ν^{-1} are compatible and induce the inverse map on limits. The quotient assertion follows from $R^1 \varprojlim R_\nu = 0$. Its kernel is closed because it is the intersection of the inverse images of the finite-stage kernels. Continuous operation squares pass to limits. The obstruction in Proposition 6.5 shows why a vector-space limit of symmetric root fragments is not, by these linear facts alone, a completed Lie algebra. □

The first enlarged degree set

The height-three root set contains thirteen degrees. To evaluate the first real Serre relations, doubled null degrees, complementary strings and odd-odd products, consider the finite degree set

$$\begin{aligned} W_{\mathrm{rel}}^{+} = W_3 \cup \{3\delta_i + \delta_j : i \neq j\} \cup \{2a_{ij} : i < j\} \\ \cup \{a_{ij} + s\delta_k : \{i,j,k\} = \{1,2,3\},\ 2 \le s \le 5\} \cup \{2\delta_{123}\}. \end{aligned}$$

It has $13 + 6 + 3 + 12 + 1 = 35$ degrees, including nine zero root spaces at the real and complementary terminal degrees. Its nonzero proper subdegrees add exactly $D_k = \delta_k + 2a_{ij}$, giving a 38-degree set. Thus a degree set that includes zero codomains is different from a subset of $\mathcal{R}_+$.

PROPOSITION 1.185 *(Comparison spaces in the enlarged set).* The full parity dimensions at $C_{k,2} = a_{ij} + 2\delta_k$, D_k and $2\delta_{123}$ are respectively

$$201|93, \qquad 510|1023, \qquad 10136|6120.$$

Every downward-saturated root set containing $2\delta_{123}$ contains the D_k. Recognition there additionally requires the source coefficient spaces, invariant pairings and the bracket and enveloping comparison maps in those degrees.

Proof. The dimensions are Proposition 4.7. The inequalities $0 < D_k < 2\delta_{123}$ force their inclusion. The operation requirements are those of Proposition 1.168. The signed values 108, −513, 4016 and simple coefficients 90, 54, −540 describe different quantities and do not supply the geometric maps. □

A further degree-enlargement rule illustrates why one should retain finite operation diagrams instead of requiring a single finite set closed under all such enlargements. For a degree β with $(\delta_i, \beta) < 0$, define

$$T_i(\beta) = \beta + \big(1 - (\delta_i, \beta)\big)\delta_i.$$

On an imaginary simple generator this is its real Serre output degree. On a composite root, this formula only specifies a test degree. It does not assert that the corresponding power of $\operatorname{ad} e_i$ kills the entire root space. The distinction is detected by Theorem 5.3.

For example, apply T_i to the ten degrees $2a_{ij}$, $C_{k,2}$, D_k and $2\delta_{123}$ inside the 38-degree set. There are twenty-one distinct new output degrees. Eighteen have nonzero signed coefficient. Adjoining their nonzero proper subdegrees creates 126 incidences across thirty distinct new degrees, so the enlarged set has $38 + 21 + 30 = 89$ degrees.

These finite counts use the full Jacobi coefficients after elliptic reduction. In particular,

β	(nm, l)	$f(nm, l)$
$(2, 6, 1)$	$(12, 7)$	1
$(8, 2, 2)$	$(16, 8)$	10
$(5, 2, 1)$	$(10, 6)$	108
$(7, 2, 2)$	$(14, 7)$	−513

and the same values hold after exchanging the first two root coordinates. Elliptic translation reduces these coefficients to $f(0, 1)$, $f(0, 0)$, $f(1, 0)$ and $f(2, 1)$, respectively.

A finite Fourier expansion cannot assign zero to an uncomputed coefficient beyond its degree bound. Such a substitution would already miscount this first enlargement.

PROPOSITION 1.186 *(An unbounded affine staircase).* Suppose a set of nonnegative lattice degrees is closed under T_i whenever its pairing is negative. Suppose it contains all proper subdegrees with nonzero Jacobi coefficient. If it contains the three real simple degrees, it is infinite.

Proof. For $N \geq 2$, define

$$\begin{aligned} A_N = \{&(0, N, N-1), (0, N, N), (0, N+1, N), \\ &(N, 0, N-1), (N, 0, N), (N+1, 0, N)\}, \\ B_N = \{&(0, N, N+1), (0, N+1, N+1), (0, N+1, N+2), \\ &(N, 0, N+1), (N+1, 0, N+1), (N+1, 0, N+2)\}. \end{aligned}$$

For $\beta = (0, N+1, N)$, the pairing with δ_3 is -2, so $T_3(\beta) = (0, N+1, N+3)$. Its three indicated proper subdegrees in the first half of B_N have Gram q-degree zero and elliptic exponents $-1, 0, -1$, hence coefficients 1, 10, 1. Exchanging the first two coordinates gives the other half. Therefore A_N forces B_N.

For $\beta = (0, N+1, N+2)$, the pairing with δ_2 is -2, so $T_2(\beta) = (0, N+4, N+2)$. Its proper subdegrees include the first half of A_{N+2}, again with coefficients 1, 10, 1. The coordinate exchange gives $B_N \Rightarrow A_{N+2}$. Both displayed output degrees have zero coefficient, since their elliptic exponent has absolute value two at q-degree zero.

Finally, $T_3(\delta_2) = (0, 1, 3)$ forces the root $(0, 1, 2)$. Its T_2 output is $(0, 4, 2)$, whose nonzero proper subdegrees include $(0, 2, 1)$, $(0, 2, 2)$, $(0, 3, 2)$. Starting with δ_1 gives their coordinate exchanges, so A_2 is present. Induction yields A_{2+2k} and B_{2+2k} for all $k \geq 0$. Their coordinates are unbounded. In particular this includes the six-degree families at every even $N \geq 14$. □

This is an obstruction to that finite closure rule. It does not obstruct the positive Lie quotients or the cofinal finite operation diagrams used above. A compact construction need only provide each specified operation in a sufficiently large valid domain, with compatible maps. It need not make an infinite root algebra into a single finite algebra.

The remaining compact construction

Lemma 1.187 *(The same determinant with an abelian bracket).* Equip the graded super vector space underlying $\mathfrak{g}$ with the zero bracket. Its root superdeterminant is unchanged, but it is not isomorphic to $\mathfrak{g}$ as a Lie superalgebra.

Proof. The determinant uses only the signed graded dimensions. The new bracket is abelian, while $[e_i, f_i] = h_i \neq 0$ in $\mathfrak{g}$. An abelian Lie superalgebra is a valid comparison object here. No counital Hopf structure with zero coproduct is being asserted. □

Lemma 1.188 *(Scalar data and geometric recognition).* A scalar denominator, Pfaffian product or protected index does not by itself determine the geometric primitive recognition datum. Target dimensions or target PBW spaces likewise do not construct source coefficient spaces or an intertwining map.

Proof. The preceding lemma keeps the entire signed determinant while changing the bracket. Adding an even-odd pair also keeps it while changing both parity dimensions. The matrices in Definition 1.170 require actual source spaces, operations and geometric coefficient maps before their comparison with the target can be formed. Those objects are absent from the scalar formulas. □

The unresolved compact data include admissible correspondences, coefficient realizations, the invariant Lie form and source representatives. Generation, kernel equality and the preceding coefficient comparison must also be constructed. They can be expressed as the relation and operation defects

$$\mathfrak{O}_{\mathrm{Rec},\nu} = \left(\mathfrak{o}_\nu^{(R1)}, \ldots, \mathfrak{o}_\nu^{(R8)}, \mathfrak{o}_{\nu+1,\nu}^{\mathrm{tr}}\right),$$

with each entry referring to the corresponding map and identity. An unresolved geometric map cannot be replaced by a vanishing numerical defect. The orientation and Pfaffian data of Appendix 7 do not alone construct these source maps.

The target denominator is established by [52, Sections 3–4], [54, Theorem 2.1 and Section 5.1.1] and the Borcherds lift [9, 10]. An additional Drinfeld double, a derived center or a vacuum-evaluated algebra is a further construction with its own tensor and pairing data. A local toric Hall algebra does not identify

the compact $K3 \times E$ primitive algebra without a map between the actual coefficient objects. The first compact comparison must construct the source operations on the explicit real, null and interior spaces above. These finite comparisons must then extend to global recognition.

1.7 *Finite geometric systems and their limits*

The preceding constructions have different meanings of finiteness. A bounded set of charges is combinatorial. A moduli stack of finite type is geometric. A finite-dimensional coefficient space is a property of a specified cohomology theory. None of these conditions alone supplies the other two. For example, $H^*(B\mathbb{G}_m;\mathbb{Q}) = \mathbb{Q}[c_1]$ is unbounded even though $B\mathbb{G}_m$ is an Artin stack of finite type.

We organize the compact comparison by finite operation diagrams and then by inverse systems. A finite stage must contain the input and output spaces of the operations being compared. It need not be closed under every extension or bracket. When a calculation leaves the retained range, its larger output stage must be specified. This permits finite calculations without asserting that a bounded root fragment is a Lie algebra or that a bounded nonzero charge set is closed under arbitrary addition.

CHARGES AND TARGET WINDOWS

Fix the stability and support data of Appendix 8. Let $F_{\sigma,S}(R) \subset \Gamma_X^{\mathrm{form}}$ be a supplied finite set of charges with positive height bounded by R, and let $\Gamma_R^{HN} \subset F_{\sigma,S}(R)$ be the charges of the geometric strata to be used. The product-stability construction has central charge

$$L_F(n) = Z_S\big(Rp_{S,*}(F \otimes p_E^* H^n)\big),$$

where H has degree one on E and the K3 stability datum is rational [75]. Its numerical formula does not prove boundedness of the chosen moduli problem. The relevant class bounds, bounded families and finite-type substacks must be supplied at their actual support and stability scope.

Let Π_X and B be the Gram map and normal-ordering cocycle of Section 3.3. The cocycle equation is

$$B(c,c') + B(c+c',c'') = B(c',c'') + B(c,c'+c'').$$

It makes the central lift associative on its declared charge monoid. Restrictions to finite sets retain only the products whose outputs are present.

Definition 1.189 *(Finite target test window).* For $c \in \Gamma_R^{HN}$, set

$$\mathcal{T}_R(c) = \left\{ \sum_{i<j} B(c_i,c_j) : c = \sum_i c_i,\ c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \le R \right\}.$$

Assume these translate sets are finite, and define

$$\widehat{\Gamma}_R = \{(c,T) : c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\},$$
$$\Gamma_R^{\mathrm{test}} = \{\Pi_X(c) + T : (c,T) \in \widehat{\Gamma}_R\}.$$

This is a finite target-degree set. Compatible inclusions of the charge and translate sets give inclusions of the target sets as R increases.

The image formula constructs a set of Gram degrees after its inputs have been supplied. It does not construct sheaves representing those degrees, nor does it make the test window exhaustive.

Definition 1.190 *(Exhaustion of a specified target set).* Let $\mathcal{D}_{\mathrm{rel}}$ be a specified set of degrees needed by the target relations. A finite-stage realization of this set assigns to each $\gamma \in \mathcal{D}_{\mathrm{rel}}$ a finite height $R(\gamma)$, a charge $c_\gamma \in \Gamma^{HN}_{R(\gamma)}$, and a translate $T_\gamma \in \mathcal{T}_{R(\gamma)}(c_\gamma)$, such that

$$\Pi_X(c_\gamma) + T_\gamma = \gamma.$$

The assignments are compatible with the stage inclusions. If the target set has a finite prefix and symbolic tail families γ_N, the tail requires formulas R_N, c_N, T_N valid for every allowed N, with the same image identity. All intermediate degrees of each asserted relation must be included in an appropriate finite enlargement.

Proposition 1.191 *(Coverage of target degrees).* Under this datum, every degree of $\mathcal{D}_{\mathrm{rel}}$ belongs to Γ^{test}_R for some finite R.

Proof. The pair (c_γ, T_γ) belongs to $\widehat{\Gamma}_{R(\gamma)}$ and has image γ. The same argument applies uniformly to each symbolic family. □

A relation cascade, including a family of the form $A_N \Rightarrow B_N \Rightarrow A_{N+2}$, can describe target degrees that must be retained. It supplies neither the geometric charges c_N nor their normal-ordering translates. Conversely, a finite list of represented degrees does not prove coverage of an infinite cascade. The target calculations and the geometric exhaustion are distinct parts of recognition.

Geometric stages and transitions

A geometric stage consists of the chosen finite-type object stacks, extension stacks and flag stacks, with the support and stability conditions needed for their maps. The full CY3 deformation problem has its shifted symplectic structure on its stated derived stack [104]. Restricting or reducing that problem requires a comparison preserving the relevant structure; a finite-type substack is not automatically a two-term critical chart. The reduced constructible coefficients and orientations are those constructed on the actual reduced critical problem [12, 64].

A universal complex may live on a scalar gerbe. Its weight-zero endomorphism complex and natural maps can descend without descent of the weight-one universal object. Scalar rigidification likewise does not imply finite residual inertia: direct sums can retain positive-dimensional automorphism groups. Both facts affect the coefficient theory and its quotient maps.

Definition 1.192 *(Finite geometric transition datum).* For $R' \le R$, specify object, extension and flag morphisms

$$t^a_{RR'} : \mathfrak{M}^a_R \longrightarrow \mathfrak{M}^a_{R'}, \qquad a \in \{\mathrm{obj}, \mathrm{ext}, \mathrm{fl}\},$$

and closed retained substacks $\mathfrak{Z}^a_R \subset \mathfrak{M}^a_R$ carried into $\mathfrak{Z}^a_{R'}$. Require the restricted morphisms to be proper; a closed immersion is one allowed special case. For every coefficient operation, include the appropriate base-change square and its comparison morphism. Include the maps of deformation complexes, cosections, constructible coefficients and root pairs, with their overlap and composition coherences.

Proposition 1.193 *(Restricted geometric transitions).* These data give proper transition maps on the retained substacks. When their coefficient and base-change comparisons are equivalences, they transport the specified coefficient operations and orientation pairs compatibly with composition.

Proof. The retained-locus condition restricts each supplied morphism to the chosen source and target. Properness is a hypothesis on those maps. The coefficient assertion follows by composing their specified pullback or pushforward comparisons with the base-change isomorphisms. Their coherence gives equality of the composite transports. □

A closed graph alone does not make a morphism a closed immersion. Nor do inclusions of bounded HN substacks naturally give morphisms in the opposite direction. The direction of a transition and the variance of its coefficient functor are part of the datum. A direct system obtained from open inclusions is not the inverse system above without an additional construction.

The eight kinds of structure

The following structures specify the operations that a finite comparison must retain.

(i) *Charge and support.* The geometric classes, Gram map, normal-ordering cocycle and finite operation degrees are those above. The elliptic support degree is defined by $(p_E)_* \operatorname{cyc}_1(F) = b^{\mathrm{geom}}[E]$. It is additive on the relevant exact sequences. It is not the third Borcherds coordinate m^{Bch} without a proved comparison. A positive effective elliptic degree projects onto all of E; at degree zero the image of the curve support in E is a finite set.

(ii) *Moduli and coefficients.* Specify the derived object and extension stacks, full flag correspondences, reduced deformation theory, critical coefficient system and supports. A proper target map can be obtained from a closed relative Quot locus in the sheaf recipient when its boundedness and closedness hypotheses hold. It does not imply properness of the source map, and the argument does not apply to an arbitrary tilted-heart extension problem without another construction.

(iii) *Hybrid operations.* The local and wrapped sectors of Section 1.1 use their actual configuration spaces, wrapped prequotients and anchors. The LL, LW, WL, WW correspondences retain the source and target anchor data before quotient. Their three-input comparisons include all eight words in $\{L, W\}^3$, and the full higher-flag coherences are further structure. The wrapped product of quotient stacks uses a common BE fibre, not two independently forgotten translation torsors. The quotient-after-correspondence pseudofunctor and protected integration maps must satisfy the relevant base-change, projection and Thom–Sebastiani identities.

(iv) *Orientation and transport.* Specify root pairs on the actual quotient, cross-determinant factors on extensions and coherent flag comparisons. On a finite-stabilizer locus, a root obstruction and the character of a chosen root are separate conditions. Weyl lifts act on this same datum and preserve its square maps, critical coefficients and transition comparisons.

(v) *Hall and primitive operations.* Specify product, coproduct, unit, counit and the tensor in which the bialgebra equation is asserted. The ordinary Koszul and meromorphic braided tensors give different equations. A product alone gives neither cooperation nor a primitive projection. In a connected cocommutative ordinary realization over characteristic zero, the Eulerian projection has the hypotheses of Section 1.6. The invariant Lie form, its radical and its generated Hopf ideal are distinct from any separately supplied full Hopf pairing.

(vi) *Pfaffians.* Supply the finite skew complexes, their shifts and square maps, the Pfaffian lines and sections, and the geometric wall charts. The actual parameter and automorphic-line comparisons fix the target and its normalization. Positive-height product convergence and analytic continuation are separate from geometric effectivity of the family.

(vii) *Associative coefficients.* Supply the counital, coaugmented source coalgebra with its conilpotence or the precise replacement filtration, and the associative cobar recipient of Section 1.5. The finite comparison maps preserve its differential, support, tensor and operations. Lie–commutative chiral duality is not a substitute for this associative comparison.

(viii) *Recognition.* Supply geometric simple and Cartan representatives, relation maps, kernel equalities, generation, both root parities and PBW comparisons. A finite calculation must retain all intermediate outputs. Compatible comparisons in every bounded weighted word space give the all-level algebraic assertion; a topological conclusion also requires continuous inverse maps on a specified completion.

For wrapped sheaves, a determinant anchor can record translation when its Euler characteristic is nonzero. At Euler characteristic zero that information can vanish. Jordan–Hölder data may then separate additional objects but do not automatically determine an elliptic translation orbit. Additional anchors must be chosen and their separating property proved. Similarly, a change of origin can change a line's Picard class without determining an equivariant stabilizer character. The constructions and examples of Section 1.1 keep these distinctions.

On the formal nilpotent support of the local deformation problem, the cyclic local model and its full flag extensions provide genuine local operations. Their comparison with a global reduced compact coefficient functor, its proper correspondence maps and quotient descent remains a separate geometric construction. The finite organization above preserves this positive local result without treating a list of required objects as proof of global existence.

Morphisms of finite comparison data

Definition 1.194 *(Finite-stage Dirac–Igusa datum and morphism).* In a fixed coefficient setting, a stage is a tuple

$$\mathfrak{D}^{\mathrm{DI}}_{X,R} = (\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P^{\Pi}_R, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}, \mathrm{Rec}_R),$$

whose components carry the specified structures above. In particular, an E_3-source is actual additional operadic data when that component is required; it is not inferred from its scalar series. A morphism consists of the following compatible maps.

(M1) Maps of charge and support diagrams preserving Π_X, the cocycle B and the retained products.

(M2) Geometric and coefficient transition maps as in Definition 1.192, with the required proper support and variance.

(M3) Maps of the hybrid carriers and all specified coloured-flag operations, compatible with the quotient pseudofunctor and protected integration.

(M4) Isomorphisms of the pulled-back orientation pairs, compatible Weyl lifts and wall charts, preserving their square maps, normal coordinates and section comparisons.

(M5) Maps of the chosen Hall data preserving product, coproduct, units and counits in their declared tensor. Preserve primitives, the invariant Lie form and its radical; require a further Hopf-pairing comparison only where that pairing is independently supplied.

(M6) Isomorphisms of pulled-back Pfaffian lines carrying the specified sections to one another and commuting with their square and automorphic-line comparisons.

(M7) Coalgebra maps preserving the differential, counit, coaugmentation, filtration and the associative comparison Θ_{Kos}.

(M8) Maps of the recognition diagrams preserving the representatives, relations, kernels, parity pieces, invariant forms, radicals and PBW filtrations, with all required intermediate operation spaces.

Composition is componentwise, with the coherence required by the chosen coefficient category. Fix strict models when an ordinary category C_X is used below. Alternatively retain an enhanced category and all its mapping spaces. A finite-stage isomorphism is an invertible morphism of these structured data, not merely an isomorphism of their underlying vector spaces.

The preservation results in the preceding subsections apply to these maps at their individual scopes: orientation pairs and critical coefficients, Hall products and coproducts, primitive subspaces, invariant-form radicals, PBW filtrations and parity. They do not follow from a scalar equality. In particular, maps between nonorthogonal matrix quotients must be checked before a radical or pairing is transported.

PRO-OBJECTS AND ACTUAL INVERSE LIMITS

Definition 1.195 *(The Dirac–Igusa pro-object).* A compatible cofiltered diagram $R \mapsto \mathfrak{D}^{\mathrm{DI}}_{X,R}$ in C_X defines the pro-object $\mathfrak{D}^{\mathrm{DI}}_X = \{\mathfrak{D}^{\mathrm{DI}}_{X,R}\}_R$. A pro-isomorphism is an invertible morphism in $\mathrm{Pro}(\mathsf{C}_X)$. No componentwise inverse limit is implicit in this notation.

PROPOSITION 1.196 *(Mapping formula and cofinal invariance).* For two such systems $\mathfrak{E} = (\mathfrak{E}_S)_S$ and $\mathfrak{D} = (\mathfrak{D}_R)_R$,

$$\mathrm{Hom}_{\mathrm{Pro}(\mathsf{C}_X)}(\mathfrak{E}, \mathfrak{D}) = \varprojlim_R \varinjlim_S \mathrm{Hom}_{\mathsf{C}_X}(\mathfrak{E}_S, \mathfrak{D}_R).$$

For a constant source T, this reduces to

$$\mathrm{Hom}_{\mathrm{Pro}(\mathsf{C}_X)}(T, \mathfrak{D}) = \varprojlim_R \mathrm{Hom}_{\mathsf{C}_X}(T, \mathfrak{D}_R).$$

Cofinal restriction gives an isomorphic pro-object. Compatible finite-stage isomorphisms on common cofinal systems are a sufficient construction of a pro-isomorphism.

Proof. The first formula defines the morphisms of the pro-category. A constant source removes the inner colimit. Cofinal restriction preserves these mapping functors, hence the represented pro-object. Compatible finite-stage inverses give inverse elements in the two mapping sets, proving the last assertion. In an enhanced pro-category, the corresponding statement uses mapping spaces and homotopy limits and colimits. □

The sufficient condition is not necessary in a general pro-category. For example, the vector-space tower

$$\mathbb{C} \xleftarrow{0} \mathbb{C} \xleftarrow{0} \mathbb{C} \xleftarrow{0} \cdots$$

is pro-zero: for every target stage, the next transition kills the identity class in the defining colimit. Every stage is nevertheless nonzero, so no cofinal set of stages is levelwise isomorphic to the zero tower. Replacing the zero transitions by identities gives the constant nonzero pro-object. Thus the transition maps are decisive.

When an actual inverse limit exists, it belongs to a specified complete category. For modules its projection topology has basic neighbourhoods given by kernels of the stage projections. For line or root-pair components the appropriate construction is a limit of Picard groupoids or stacks, with its descent

data. Neither existence of a pro-object nor Mittag–Leffler vanishing for some cohomology groups proves geometric effectivity of all its components.

Definition 1.197 *(Degreewise Mittag–Leffler condition).* Choose a countable cofinal sequence and coefficient complexes $K_{R,a}$ on the retained object, extension, wrapped or flag stacks. Let

$$C^q_{R,a} = H^q_c(\mathfrak{M}_{R,a}, K_{R,a})$$

when compactly supported cohomology is the selected theory, and use the specified dualizing-complex model in a Borel–Moore theory. The actual six-functor maps define a tower only after their variance and identity and composition laws have been fixed. It is Mittag–Leffler in degree q if, for every R_0, the images of $C^q_{R,a} \to C^q_{R_0,a}$ are constant for all sufficiently large R.

Proposition 1.198 *(Degreewise exactness of a countable limit).* For a countable tower of modules satisfying the preceding condition, $R^1 \varprojlim_R C^q_{R,a} = 0$. For a tower of complexes C_R, there is a short exact sequence

$$0 \longrightarrow \varprojlim{}^1 H^{q-1}(C_R) \longrightarrow H^q(R\varprojlim C_R) \longrightarrow \varprojlim H^q(C_R) \longrightarrow 0.$$

In particular, degreewise Mittag–Leffler cohomology gives the right-hand identification.

Proof. The derived limit of a countable module tower is represented by the two-term product complex with map

$$1 - T : \prod_n A_n \longrightarrow \prod_n A_n, \qquad (1-T)(x)_n = x_n - u_n x_{n+1}.$$

For a surjective tower this map is surjective by successive lifting. For a Mittag–Leffler tower, its stabilized images form a surjective subtower. The quotient is pro-zero. On a pro-zero tower, $\sum_{j\geq 0} T^j$ is finite in each fixed output coordinate and inverts $1 - T$. The exact sequence of these towers therefore gives surjectivity of $1 - T$ on the original tower, proving the vanishing of its cokernel $\varprojlim{}^1$. Applying the same product-cone model to a tower of complexes gives the displayed cohomology sequence. □

Every tower of finite-dimensional vector spaces over a fixed field is Mittag–Leffler: its images in a fixed stage form a descending chain of subspaces. The analogous assertion for finite-rank integral modules is false, as the tower $\mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{2} \cdots$ shows. The images $2^n\mathbb{Z}$ never stabilize. Finite type of a stack, a proper transition map or an Euler characteristic does not establish the finite-dimensional or Mittag–Leffler hypotheses for a chosen coefficient theory.

Finite and limiting comparison statements

Theorem 1.199 *(Formation of the finite comparison system).* Suppose the eight kinds of structure above are constructed, with morphisms (M1)–(M8) satisfying their identity and composition laws on a cofiltered stage system. They define a Dirac–Igusa pro-object. The primitive, pairing-kernel, PBW and coefficient cohomology limits agree with the cohomology of their derived limits whenever each of those towers satisfies the Mittag–Leffler hypotheses just stated. No geometric effectivity or analytic continuation is implied by this conclusion.

Proof. The component maps and their coherence define the diagram in C_X, hence its pro-object. Apply the preceding proposition to each of the indicated coefficient towers. The last sentence distinguishes these limits from the further geometric and analytic realizations. □

THEOREM 1.200 *(Finite Pfaffian comparison).* At a finite stage carrying the geometric skew and orientation data, the Pfaffian product and local reflection identities hold under their specified line comparisons. A primitive recognition or associative cobar conclusion holds only with the corresponding finite source maps and relations. If these comparisons and their inverses commute with every stage transition, they give the stated comparison of pro-objects and, when defined, of their compatible limits.

Proof. The finite skew identity is $\mathrm{pf}(D)^2 = \det D$, and orthogonal block sums multiply Pfaffians. Section-compatible wall lifts give the local sign of Proposition 1.151. The primitive and coefficient conclusions are the separate recognition and associative comparison theorems of the preceding subsections. Compatible inverse maps induce a pro-isomorphism by Proposition 1.196; applying an available limit functor preserves that equivalence. □

For $K3 \times E$, the finite moduli, reduced coefficients, orientation transport and compact comparison maps remain the geometric inputs. The local cyclic charts, formal charge formulas, target relations and finite skew identities provide concrete parts of the construction. A global realization requires their compatible geometric extensions, rather than only their scalar shadows.

1.8 The four geometric comparison conditions

The compact comparison needs a reduced coefficient theory, compatible orientation pairs, Weyl transport and geometric wall charts. We denote these requirements by $(D0)$, $(O1)$, $(O1)^+$ and $(O2)$. They are existence and compatibility conditions; they are not four classes in a common cohomology group. The Kummer and group-cocycle classes below detect particular parts of them.

The tuple of Definition 7.18 keeps these inputs together with finite geometric Pfaffians:

$$(D0_{\mathrm{HN}}, O1_{\mathrm{quot}}, \mathcal{T}_W, O2_{\mathrm{atlas}}, P_{\mathrm{fin}}).$$

Primitive recognition, an associative coefficient comparison and a protected state require additional data. In particular, constructing this tuple would not identify the compact primitive algebra from its superdimension alone.

DEGENERATION AND REDUCED COEFFICIENTS

At each finite geometric stage R, condition $(D0)$ asks for a specified degeneration or specialization to a zero-dimensional sector. Its domain must be stated before using flatness. On a fixed projective target, a flat family of coherent sheaves has constant Hilbert polynomial. A sheaf with nonzero effective curve class has a positive linear coefficient, whereas a zero-dimensional sheaf has a constant polynomial. Such a family cannot specialize the former to the latter. Likewise a family of perfect complexes preserves its Chern character in a fixed cohomological marking.

A proposed comparison changing the target, support theory or charge therefore has to construct that change. The datum $D0_{\mathrm{HN}}$ specifies its geometric recipient, coefficient functor and transition maps, together with the reduced deformation and orientation comparisons. It must also provide specialization of the constructible critical coefficients, Pfaffian lines and protected integration maps on every required correspondence. The boundary $b_R^{\mathrm{geom}} \to 0$ in a hybrid Ran carrier is not by itself such a geometric specialization.

For extension maps q, p_1, p_2, a compatible semiregularity cosection satisfies

$$q^* \mathrm{CS}_C = p_1^* \mathrm{CS}_A + p_2^* \mathrm{CS}_B$$

on the common deformation complex, with the stated reductions and shifts. This equality is required throughout the specialization and under the stage transitions. The nonzero-K3-class surjectivity criterion of Section 1.2 does not assert surjectivity in curve class zero.

A perfect reduced deformation complex, a cosection-localized virtual cycle and an oriented critical coefficient are different objects. The orientation constructions of [62, 64] do not identify all three or supply every specialization above. Theorem 7.2 describes a consequence of a supplied compatible system. Failure of this proposed degeneration route does not rule out an independently constructed reduced Hall theory.

Quotient orientation

Let K^{red} be the specified perfect reduced complex on a geometric coefficient base B, and write $\mathscr{D}^{\mathrm{red}} = \det K^{\mathrm{red}}$. Condition $(O1)$ asks for a root pair

$$(o, \rho), \qquad \rho : o^{\otimes 2} \xrightarrow{\sim} \mathscr{D}^{\mathrm{red}},$$

on the required quotient stack, with coherent extension-flag comparisons and stage transitions. If a cosection correction is included, it must have a perfect model C_{cs}. The actual line to be rooted is then

$$\mathscr{D}_{\mathrm{tw}} = \mathscr{D}^{\mathrm{red}} \otimes \det C_{\mathrm{cs}}.$$

Its obstruction is the Kummer boundary

$$\alpha^{\mathrm{red}} = \delta(\mathscr{D}_{\mathrm{tw}}) \in H^2(B, \mu_2).$$

In analytic cohomology it is the mod-two reduction of the integral first Chern class. Rational Grothendieck–Riemann–Roch does not recover its possible two-torsion. The vanishing of α^{red} permits a root pair; a choice of pair and its extension-flag coherence remain necessary.

For an E-action, the equivariant root problem is on $[B/E]$. On a free-action locus this is the quotient itself. There is no universal obstruction obtained merely by writing

$$a_1 u_1 + a_2 u_2 \in H^2(BE; \mathbb{F}_2).$$

Such a class contributes only through the actual pullback to the chosen quotient problem; that pullback need not be injective.

On a finite-stabilizer stratum with stabilizer H, one must solve the equivariant Kummer problem $\widetilde{\beta}^H \in H^2_H(B; \mu_2)$. For coarse descent, every stabilizer must also act trivially on the chosen root fibre. Thus a zero gerbe obstruction does not remove a nonzero degree-one sign character $\lambda^H \in H^1(BH; \mathbb{F}_2)$. For $H = E[2]$, the three order-two restrictions detect the three coefficients of the degree-two class. For $E[N]$, with $4 \mid N$, the mixed class $e_1 e_2$ in the two-primary cohomology is invisible on every cyclic subgroup. Its coefficient $A_{12}^{(N)}$ must be checked independently. These are the finite-group calculations underlying Theorem 7.9.

Orientation enters the critical coefficient through its sign torsor and the corresponding constructible local system. An ordinary tensor of a coherent root line with a constructible complex is not the coefficient construction. Nor does descent of the weight-zero endomorphism complex imply descent of the weight-one universal sheaf.

COHERENT WEYL TRANSPORT

Set $W = W^{(2)}(\Lambda_{II}^{2,1})$. Condition $(O1)^+$ supplies geometric maps between the required strata and lifts of their action to the root pairs. At a finite stage these maps may define only a partial action groupoid $\mathcal{G}_R$. Its composition defect belongs to

$$c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mu_2}),$$

with the actual action on the sign coefficients. A one-cochain η_R with $\delta\eta_R = c_{o,R}$ rectifies the lifts. One uses $H^2(W; \mathbb{F}_2)$ only after identifying the relevant complete action and constant coefficient system with that group cohomology problem.

The corrected lifts must satisfy $\tau_i^2 = 1$, preserve the quotient root and its square map, and commute with stage transitions. The torsor defect $\omega_{i,C}$ measuring compatibility with the quotient orientation must vanish with a specified trivialization. The free-product Coxeter presentation has no finite braid relations between distinct generators. It therefore simplifies the relations to verify, but it does not construct the lifts or their scalar signs. The sign comparison in Theorem 7.10 additionally uses the local sections of $(O2)$.

PFAFFIAN WALL CHARTS

Condition $(O2)$ supplies the wall atlas of Definition 5.4. At each required simple wall the geometric skew complex is compared with a tangential factor and the normal rank-two block

$$[O^2 \xrightarrow{D_u} (O^2)^\vee], \qquad D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}.$$

The normal Pfaffian is u; its determinant is u^2. An invariant tangential Pfaffian unit and a section-compatible lift of $u \mapsto -u$ give the local residual sign -1. The comparison includes the required shift of the geometric self-Ext complex. It is not inferred from the scalar complex $[O \xrightarrow{u} O]$.

The charts must cover the geometric locus, with compatible splittings, roots, sections and integration maps on components, boundaries and overlaps. Their transitions must agree with the Weyl lifts. Theorem 7.16 transports such a supplied system. The local matrix computation supplies none of its global covering or gluing maps.

The set $\{\pm 1\}^{12}$ has 4096 elements. To regard it as a geometric atlas count requires a separate bijection, including stabilizer weights and overlap identifications. Its cardinality does not determine a reflection character. The equality $4096 = 64^2$ in the scalar normalization $Z_{\mathrm{OP}}^X = -4096\Delta_5^{-2}$ is independent of that geometric counting problem.

THE JOINT CONDITIONAL COMPARISON

THEOREM 1.201 *(Pfaffian and primitive comparison from geometric data).* Suppose the four geometric conditions above are supplied on a common cofinal system, together with the following additional data.

(i) The finite geometric Pfaffian datum of Definition 1.146, including the parameter and line maps, section normalization and the effective formal or analytic limit required for the conclusion.

(ii) The primitive recognition data (S1)–(S10) of Theorem 1.160, or their compatible finite realization with every intermediate operation degree retained and the stated convergence hypotheses.

(iii) The associative coefficient source and the finite comparison of Definition 1.156, on its declared tensor and support, with compatible inverse maps and the derived-limit hypotheses needed to pass to the chosen completion.

Then the geometric Pfaffian is identified with Δ_5 by its specified line comparison, and

$$\epsilon_o = \widetilde{\nu}_{\Delta_5}|_W, \qquad P_X^{\Pi,\mathrm{red}} \simeq \mathfrak{g}_{\Delta_5}.$$

The third datum additionally identifies the source cobar with the associative Ran target $\mathcal{A}_{\Delta,E}$ of Section 1.5. Each conclusion is at the coefficient, topology and operation scope of its own hypotheses.

Proof. The wall models and coherent lifts give the generator signs -1, hence the character comparison by Theorem 5.3 and the Weyl action. The finite skew Pfaffians multiply by the finite product formula. The stipulated parameter, line and limit comparisons identify that section with the normalized automorphic product.

For primitives, the geometric representatives define the presentation map. Equality of its relation kernel and the defining ideal gives injectivity; generation gives surjectivity, as in Theorem 1.160. The independent coefficient maps give the cobar comparison and preserve the required Weyl, orientation, pairing and filtration data by Proposition 1.157. Passing to the specified limit uses its derived-limit hypotheses, not just the existence of the finite objects. □

The cofinal collection is first a pro-object $\{\mathfrak{D}^{\mathrm{DI}}_{X,R}\}_R$ in the category where its maps and operations have been defined. An inverse limit is an additional construction in a specified complete category. The four geometric conditions, primitive recognition and the associative comparison are separate parts of that construction. Automorphic catalogue entries fix target products and characters; they do not supply the compact quotient orientation, Weyl lifts, Pfaffian atlas or geometric representatives.

2 Assembly of the geometric comparison

The constructions above have supplied formal local models and finite algebraic comparisons. Their compact extension can now be stated as a conditional assembly theorem, with each operation in its own recipient.

Theorem 2.1 *(The structured comparison system).* Suppose a common cofiltered family of geometric stages carries the following compatible structures.

(L1) The local, wrapped and mixed carriers of Section 1.1, with actual anchors, common-torsor quotients and all required coefficient and flag maps. When an E_3-source and chiral specialization are asserted, their field, operadic, anomaly, support and descent hypotheses are supplied on that same system.

(L2) Reduced orientation pairs, their quotient and cross-determinant comparisons, coherent Weyl lifts and stage maps, as in Section 1.2. Trivial stabilizer characters are required wherever coarse descent is part of the comparison.

(L3) The Hall product, coproduct, unit and counit in their declared tensor, with the coefficient maps of Section 1.3. Primitive projection and a full Hopf pairing are additional structures with their own hypotheses. The invariant Lie form on primitives has its separate radical quotient. A double comparison also requires an opposite half, cross relations, integral form and completion; stable-envelope transport is used only when its geometric maps have been supplied.

(L4) The actual skew complexes, Pfaffian lines, sections and parameter and automorphic-line comparisons of Section 1.4, including the boundary normalization and the stated geometric or analytic effectivity.

(L5) The source coalgebra and the associative comparison of Section 1.5, with the finite comparison maps and their inverses. A bar of the full factorization algebra is used only after its higher operations and coherences exist. An ordinary bar and a current envelope remain separate recipients.

(L6) The compact source representatives, relation kernel, generation, both parities, invariant form and PBW data of (S1)–(S10), or their exhaustive finite realization in Definition 1.161.

(L7) Morphisms (M1)–(M8) satisfying the identity and composition laws of Definition 1.194, with the specified derived-limit and effectivity hypotheses wherever an actual limit is claimed.

(L8) The compatible degeneration, quotient orientation, Weyl and wall-atlas data $(D0)$, $(O1)$, $(O1)^{+}$, $(O2)$ and the finite Pfaffian datum P_{fin} of Theorem 7.19.

Then the stages define the Dirac–Igusa pro-object. Their supplied primitive maps identify $P_X^{\Pi,\mathrm{red}}$ with $\mathfrak{g}_{\Delta_S}$ at the stated algebraic or completed scope. Their associative source and Pfaffian comparisons have the respective conclusions of their comparison theorems. An operator trace requires the independent cyclic state datum of Appendix 7.

Proof. The component maps in (L7) define the cofiltered diagram of structured finite stages. It is the asserted pro-object by Proposition 1.196. The geometric representatives of (L6) define the presentation map; its stated kernel equality and generation give the primitive isomorphism. The maps and inverses in (L5) give the associative comparison. The skew sections, line comparisons and normalizations in (L4) and (L8) give the Pfaffian theorem. Each passage to a completion uses that component's exactness and effectivity hypotheses. None of these constructions selects the cyclic state needed for the final trace assertion. □

The parameter datum (S, H, σ, E, o_E) also requires its actual base-change maps. Definitions 8.27 and 8.29 specify those maps and the polarization and origin conventions. The origin change of Proposition 8.30 compares determinant Picard classes; an equivariant root character requires its further lift. A scalar denominator determines none of these base-change or pro-object morphisms.

Part III

Orientations and Pfaffians

Chapter 7

Pfaffians, orientation characters and primitive recognition

The automorphic form and the BKM denominator are already related by an explicit theorem:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5})(Z) = 64^{-1}\Delta_5(2Z).$$

The compact $K3 \times E$ problem asks for more. It asks for a geometric Pfaffian section, a coherent action on its orientation, and a realization of the BKM bracket on the geometric primitive spaces. These constructions have different sources and require different comparison maps. This chapter states the consequences of supplying them, while keeping the missing compact constructions visible.

The real simple generators have the usual Kac–Moody relations [66]. The generalized construction of [8] admits imaginary simple roots; the parity and multiplicity data for this denominator are those of Chapter 4. The normalization and doubled modular variable in the displayed formula are those of the type-II lattice in Chapter 4. The Pfaffian comparison uses Δ_5 itself, after a specified parameter map and an isomorphism of lines. Its reflection character is compared with the homogeneous orthogonal character $\widetilde{\nu}_{\Delta_5}$, not directly with the symplectic Maass character ν_{Δ_5} of a lift [84].

1 The geometric data and their comparisons

1.1 *A system of finite geometric objects*

Fix a cofiltered system of finite geometric moduli problems indexed by R, with their actual transition maps. When constructed, a Dirac–Igusa system records the following objects and maps:

$$\begin{pmatrix} \mathcal{A}_{X,R},\ F^{\mathrm{hyb}}_{X,R},\ \Gamma_{X,R},\ \Pi_X,\ \Phi_R,\ o_R,\ H_R,\ P^{\Pi}_R, \\ C_{X,R},\ \Theta_{\mathrm{Kos},R},\ \mathscr{L}_{\mathrm{Pf},R},\ \mathrm{pf}_{X,R},\ \epsilon_{o,R},\ \mathrm{Rec}_R \end{pmatrix}_R .$$

Here F^{hyb} is the specified formal or geometric coefficient object, Γ its charge group, Π_X the specified normal-ordered charge homomorphism, and Φ its critical coefficient realization. The symbol o denotes its orientation, H the algebra carrying the supplied Hall operations, and P^{Π} its descended primitive Lie algebra. The coalgebra C and map Θ_{Kos} belong to a separately specified bar–cobar comparison. The remaining entries are the Pfaffian line and section, an orientation character when extractable from its action, and the primitive recognition map. An enhanced factorization algebra structure on $\mathcal{A}$, including any claimed E_3 structure, is further data with its operations and coefficient category; the list does not construct it.

If all these data and their transitions are given, they define a pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ in their common category of structured objects. A pro-object is a diagram, rather than an assertion that a geometric inverse limit exists. For ordinary pro-objects the formula is

$$\mathrm{Hom}_{\mathrm{Pro}(C)}((X_i),(Y_j)) = \varprojlim_j \varinjlim_i \mathrm{Hom}_C(X_i, Y_j);$$

in a derived setting the corresponding mapping spaces and homotopy limits are used. Cofinal reindexing preserves the pro-object. Passing from it to an effective geometric line, a completed algebra or an analytic operator requires the relevant comparison in each case. A finite root set is not a finite moduli space, and a Harder–Narasimhan bound is not automatically a compatible root-height quotient.

Remark 1.1 *(Three objects attached to Δ_5).* The automorphic section $\mathcal{D}_X = \Delta_5$, the denominator of $\mathfrak{g}_{\Delta_5}$, and a putative compact Pfaffian section have different definitions. The first two are identified by the classical automorphic and BKM theorems [9, 52–54]; the third needs the geometric line and section comparison below. Writing the same scalar expansion does not identify their bases, coefficient systems or operations.

1.2 Reduction, orientation, transport and walls

We retain the names $(D0)$, $(O1)$, $(O1)^+$, $(O2)$ for four geometric requirements. They refer to data and equations, rather than four universal scalar obstruction classes.

The $(D0)$ datum is a specified degeneration from the compact problem to the chosen endpoint, together with finite object, extension and complete-flag moduli, their derived enhancements and transitions. The reduced complexes, coefficient specialization, orientation and Pfaffian comparisons must be supplied on that family. A semiregularity functional on Ext^2 can help construct a reduced obstruction theory; it does not alone give a nondegenerate reduced shifted symplectic form or a vanishing-cycle specialization functor. Similarly, a reduced virtual-dimension shift does not by itself identify orientation lines by tensoring with a parity symbol.

The $(O1)$ datum is the strong reduced orientation of Definition 1.75. Its extension formula contains the cross determinant $L(F, G)$, the computed reduction excess and the complete-flag coherence. It descends through the common elliptic quotient as a root pair. Stabilizer characters are allowed on the quotient stack. If the chosen theory also asks for a coarse or inertia-trivial orientation, their vanishing is an additional condition. Its global Kummer class cannot in general be recovered solely from pointwise stabilizer restrictions or a chosen representative of a Borel filtration term.

The $(O1)^+$ datum consists of the geometric Weyl transports and coherent root lifts of Definition 1.81. It includes a trivialization of the sign cocycle on the actual action groupoid, the involutive relations and transport of all extension and quotient identifications. An orbit groupoid has the cohomology of its stabilizer; it is not automatically the classifying space of the whole Weyl group. Extracting a scalar character further needs the reference fibre identifications of Definition 1.85.

The $(O2)$ datum is a complete type-II wall atlas with compatible normal skew blocks

$$D_{u_i} = \begin{pmatrix} 0 & u_i \\ -u_i & 0 \end{pmatrix}, \qquad \mathrm{Pf}(D_{u_i}) = u_i.$$

Each stipulated base reflection sends u_i to $-u_i$. Its reference comparison preserves the tangential section and unit; the actual root lift carries the Pfaffian section to its pullback. These charts must cover the relevant components and boundary strata, and their orientation, coefficient and integration comparisons must agree on overlaps. A generic rank-one calculation alone supplies neither coverage nor those equations. In particular the middle wall $\delta_2 \leftrightarrow (0, 1, 1)$ needs a geometric object in that degree. If the chosen charge comparison assigns it positive geometric elliptic degree, the support obstruction of Section 1.1 requires a wrapped source object. The Gram label alone does not establish that geometric degree.

The finite geometric Pfaffian datum (P_{fin}) of Definition 1.146 is additional. It supplies the skew realization, actual section, oriented block factorization, every residual unit, the parameter map and the line comparison. Neither a list of ranks nor an orientation alone supplies that section.

Proposition 1.2 *(What remains when a comparison is absent).* A supplied orientation without Weyl lifts remains a root pair; there is then no specified W-action on it. Coherent lifts without wall charts give an action but no computed simple-wall signs. These orientation data without a Pfaffian realization do not specify a section. A Pfaffian section without a parameter and line comparison is not yet a section of the automorphic line. Finally, a section identity without primitive representatives and their relations does not identify the source Lie algebra.

Proof. Each conclusion names the object or map supplied by the corresponding datum. The missing constructions are not defined by the remaining ones. For example the two signs of a square root have the same square, while the abelian and Heisenberg algebras on the same three-dimensional graded space have the same scalar skew blocks. Thus squaring forgets a root lift and graded block dimensions forget a bracket. These examples establish the stated limitations of those particular data; they do not assert independence of every geometric hypothesis on a compact family. □

Proposition 1.3 *(Limits of an independence argument).* The preceding requirements have different mathematical roles, but some depend on the existence of others. In particular a root transport includes root objects, and a geometric Pfaffian datum includes its line and square map. Treating the requirement spaces as independent Cartesian factors does not prove that their vanishing conditions are geometrically independent.

Proof. A transport $o_U \to g^* o_V$ is not an object of a category independent of o_U, o_V; its source and target are those root objects. Likewise the square equation for a Pfaffian section already involves its line. A product of formal predicates in which these objects can be varied independently therefore need not parametrize any geometric data. A genuine nonimplication would need a counterexample satisfying the other geometric hypotheses in the same category. The concrete examples above prove only the specified scalar and algebraic failures. □

1.3 The scalar branch convention

Set $D_5 = 64^{-1}\Delta_5$ and $\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$, with the complete chamber factors of Chapter 3. The OP scalar branch is

$$Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\Delta_5^{-2}.$$

The sign belongs to the signed chamber convention for the reduced count [101]. The number $4096 = 64^2$ is the squared theta normalization. Both statements concern the scalar invariant. They do not calculate the action on a geometric orientation line. An identification with a protected trace requires the state and sewing comparison of the theory in question.

2 The conditional Pfaffian section identity

Theorem 2.1 *(Pfaffian–Dirac comparison).* Suppose the compact system supplies $(D0)$, $(O1)$, $(O1)^+$, $(O2)$ and (P_{fin}) on the specified cofinal geometric system. Assume its oriented block factorizations cover all active target degrees and preserve both parities, with scalar

normalization $64x^{\rho}$. Suppose also that its compatible Pfaffian lines and sections are effective on a base B, with parameter map $\kappa : B \to \mathcal{A}_2$ and line comparison

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \kappa^*(\mathcal{L}^5 \otimes \nu_{\Delta_5}).$$

Assume the formal block product converges on a nonempty cusp domain and the compared sections extend holomorphically to a connected reduced analytic base, with that domain meeting every irreducible component. Then

$$\iota_{\mathrm{aut}}(\mathrm{pf}_X) = \kappa^*\Delta_5.$$

The formal equality already holds in the specified height completion; the analytic conclusion uses the additional convergence and extension hypotheses.

Proof. At each finite height the oriented skew-block comparison gives

$$64q^{1/2}r^{1/2}s^{1/2}\prod_{\gamma\in A_R}(1-x^{\gamma})^{\mathrm{sdim}\,V_{\gamma}}$$

by Proposition 1.147. In a fixed height jet only finitely many active factors contribute. Therefore the products have a unique limit in the positive-height completion. No assertion about general geometric inverse limits is used in this coefficient stabilization.

The classical Borcherds–Gritsenko–Nikulin product identifies this formal series, with its boundary factors and normalization, with the cusp expansion of Δ_5. The actual line comparison puts both sides in the same line over B. Effectivity supplies its geometric section; convergence makes the two expansions equal on the stated open domain. Their difference is a holomorphic section vanishing there, so the identity theorem on each reduced irreducible analytic component gives the result. Alternatively, when both sections descend to the Siegel modular stack, the usual expansion principle gives the same equality [58, 59]. A real-root character alone would not supply that modular descent. □

Remark 2.2 *(The role of the geometric hypotheses).* The proof uses the actual block comparison to obtain the product, positive height to take its formal limit, and the parameter and line comparison to interpret its analytic continuation. The orientation and wall conditions control the geometric choices and reflection action entering those comparisons. Some of these conditions may already be included in a stronger formulation of (P_{fin}); no assertion of logical minimality is needed. The theorem is conditional because those compact geometric constructions are not consequences of the target product.

If a protected state theory independently satisfies

$$Z_X = -4096\,\iota_{\mathrm{aut}}(\mathrm{pf}_X)^{-2},$$

then the theorem yields $Z_X = -4096\,\kappa^*\Delta_5^{-2}$. Under the rescaling $Z_{\mathrm{BPS}} = -4096^{-1}Z_X$, this is $Z_{\mathrm{BPS}} = \kappa^*\Delta_5^{-2}$. These are consequences of the additional sewing identity, rather than constructions of a compact microscopic state space.

3 The orientation character

THEOREM 3.1 *(Conditional orientation character identity).* Suppose $(O1)^+$ gives coherent orientation transport and the reference identifications needed to extract ϵ_o. Suppose $(O2)$ gives a compatible rank-one Pfaffian crossing on each of the three simple type-II walls. Then

$$\epsilon_o = \det_W = \widetilde{\nu}_{\Delta_5}|_W.$$

Proof. In the reference frames, pullback sends the section $v_i u_i$ to $-v_i u_i$, since the unit is invariant. Compatibility of the actual root lift with that section therefore forces its residual sign to be -1. This gives all three generator values. Coherent transport and the reference identifications make these values a character. They equal the determinant character on the generators, and the homogeneous automorphic character has the same values by Lemma 10.2. □

Remark 3.2 *(The OP sign and the reflection sign).* The normal factor has Pfaffian u and determinant u^2. Squaring erases its reflection sign. The OP branch sign cannot recover it. Likewise a real path joining $u > 0$ to $u < 0$ would cross the deleted wall; the comparison here is the specified reflection lift, not parallel transport along such a path.

Remark 3.3 *(Imaginary root divisors).* Imaginary simple roots have no associated simple Weyl reflections. Any geometric monodromy around a divisor bearing such a label needs the punctured analytic family and locally constant coefficient system of Definition 1.90. The oriented coefficient is twisted by its orientation sign torsor, not by tensoring a constructible sheaf with a coherent line. Neither the target multiplicity nor the Maass multiplier establishes this local constancy or defines the meridian action.

Remark 3.4 *(Chamber symmetry).* The homogeneous automorphic character is trivial on the type-I chamber group S_3. The abstract extension of $\det_W$ to $W \rtimes S_3$ compatible with it is therefore $(w, \sigma) \mapsto \det_W(w)$. The other abstract extension is its product with $\operatorname{sgn}(\sigma)$. Realizing either on the compact geometry still requires chamber transports and the semidirect relations of Proposition 1.89.

4 Primitive recognition

Let $P_X^{\Pi,\mathrm{desc}}$ be an actually constructed descended primitive Lie superalgebra. When a coefficient realization gives

$$P_X^{\Pi,\mathrm{desc}} = H^\circ \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X),$$

the equality includes that realization, its differential, the primitive operation and charge descent. Let B_X be a specified even invariant Lie form, and put

$$R_X = \operatorname{Rad} B_X, \qquad P_X^{\Pi,\mathrm{red}} = P_X^{\Pi,\mathrm{desc}}/R_X.$$

Invariance makes R_X a Lie ideal. Its generated ideal in the ordinary enveloping algebra is a Hopf ideal. The Lie form itself is not required to be adjoint to the primitive Hopf coproduct.

Theorem 4.1 *(Recognition from the compact representatives).* Suppose $P_X^{\Pi,\mathrm{red}}$ satisfies the ten hypotheses of Theorem 1.160: a constructed source and descent; the Cartan and root grading; representatives of every real and imaginary simple generator with both parities; the Borcherds–Kac relations; the invariant-form comparison; the exact presentation kernel; generation; the filtered enveloping comparison; a specified continuous completion when used; and the full parity dimensions with zero spaces in non-root degrees. Then there is an isomorphism of graded Lie superalgebras

$$\mathfrak{g}_{\Delta_5} \xrightarrow{\sim} P_X^{\Pi,\mathrm{red}}$$

preserving the invariant forms. It induces a filtered Hopf isomorphism of their ordinary enveloping algebras. If the maps and inverses extend continuously to the specified completions, the completed objects are also identified. The superdenominator is $64^{-1}\Delta_5(2Z)$.

Proof. Let $\mathfrak{F}$ be the free Lie superalgebra on the chosen Cartan basis and simple symbols, and J_{BK} the ideal of the presentation in Definition 3.3. The geometric representatives define $\pi : \mathfrak{F} \to P_X^{\Pi,\mathrm{red}}$. The relations give $J_{\mathrm{BK}} \subset \ker \pi$, the no-extra-relations hypothesis gives equality, and generation gives surjectivity. Therefore

$$\mathfrak{F}/J_{\mathrm{BK}} = \mathfrak{g}_{\Delta_{\mathrm{s}}} \simeq P_X^{\Pi,\mathrm{red}}.$$

The form comparison and functoriality of the ordinary enveloping algebra give the remaining algebraic assertions. The target invariant form is already nondegenerate; no further target radical is added to the presentation ideal.

Compatible finite partial-operation comparisons determine this map only when they cover all the relevant inputs, outputs and relations. For genuine inverse systems of isomorphisms, the inverses are also compatible and give the inverse limit isomorphism. Passing a radical quotient through a limit requires exactness of its kernel tower; finite-dimensional towers give that exactness under the hypotheses of Proposition 1.184. None of these linear statements equips a two-sided root product with a continuous bracket. The completion hypothesis specifies that bracket and its domain. Finally the denominator follows from the known target presentation and its full graded dimensions. □

The associative Ran comparison is another assertion. In the coefficient category of Section 1.5, let

$$V_E = \mathsf{Den}_{\Delta_{\mathrm{s}},E}, \qquad \mathcal{A}_{\Delta,E} = U_\star(\iota_* V_E).$$

The first object is the BD current envelope; the second is its associative Ran envelope. A source datum supplies an augmented algebra $A_{X,R}$, a coalgebra map $b_{X,R} : C_{X,R} \to BA_{X,R}$, and an associative comparison $\vartheta_R : A_{X,R} \to T_R$, with their acyclic cones and operation compatibilities. Its map is

$$\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon_{A_{X,R}} \circ \Omega b_{X,R} : \Omega C_{X,R} \longrightarrow T_R.$$

Here B, Ω use the specified Ran tensor $\star$ and empty unit. If these maps are equivalences on the admitted bar–cobar domain, preserve the transitions, and pass to the specified limit with target $\mathcal{A}_{\Delta,E}$, they give

$$\Omega C_X \simeq \mathcal{A}_{\Delta,E}.$$

A direct statement about the BD chiral algebra V_E requires the corresponding chiral comparison functor. Primitive recognition alone supplies neither comparison. A word-cut coalgebra also needs a map preserving the full collision operations before it can be identified with this Ran coalgebra.

Remark 4.2 *(Identical dimensions with a different bracket).* Give the target graded super-vector space the zero bracket. Its two parity dimensions in every degree, and hence its scalar superproduct, remain unchanged. But $[e_i, f_i] = h_i$ fails for each nonzero real Cartan vector. The assignments of generators do not factor through the BKM presentation. A zero form would moreover have the entire space as radical. This example proves that parity dimensions and a scalar denominator do not recognize a Lie algebra. Adding an arbitrary nondegenerate form still does not supply the required invariant form and bracket comparison.

Remark 4.3 *(Algebraic recognition and geometric orientation).* The algebraic recognition theorem needs the stated source algebra, representatives and relations. It can be applied to an algebra constructed by any method satisfying those hypotheses. In the compact geometric construction considered here, orientation and coefficient descent are among the requirements for producing that source and its operations. A Frobenius identity does not by itself construct the coproduct, prove a radical coideal condition, or identify the compact primitive algebra.

5 Compatibility with chiral Koszul duality and Calabi–Yau quantum groups

The three theorems sit inside chiral Koszul duality and Calabi–Yau quantum groups as the View ② / View ③ face of the Calabi–Yau-to-chiral comparison. Three compatibilities are required.

5.1 Primitive-to-shadow comparison

The primitive-to-shadow comparison is the cross-level identity ②↔③ applied to the primitive-to-shadow part of the chain. The Vol II primitive datum on (X, D, τ) is the bicoloured nine-tuple

$$\big(C\big|_{(X,D,\tau)},\ \mathcal{D}^{\mathrm{cl}}\big|_X;\ b,\ A_b,\ F^{\mathrm{cl}};\ \sigma_{\mathrm{half}},\ \mathrm{tr}^{\mathrm{cl}}_X;\ \mathrm{Tr}_C\big).$$

The chart of vacuum b is the α-datum; the chart algebra $A_b = \mathrm{End}_C(b)$ is the open primitive algebra; the closed-colour vacuum factorization algebra F^{cl} gives the closed-side trace $\mathrm{tr}^{\mathrm{cl}}_X$ which evaluates at the type-II cusp $\mathfrak{c}_\infty$ to the weight-5 Borcherds–Gritsenko–Nikulin section $\mathcal{D}_X = \Delta_5$ (View ①). The comparison $A_b = \Phi_d^{(\Sigma_{d-1},C)}(C_X)$ is the chain-fusion datum of chiral Koszul duality and Calabi–Yau quantum groups at level 2: under that datum the Calabi–Yau-side chiral algebra A_X is compared with the open-side chart algebra A_b on the same curve C with boundary vacuum induced by (C_X, Σ_{d-1}, C). In the present setting, the comparison concerns the chiral algebra on E attached to the Hall source of $K3 \times E$; conditional on the chiral Koszul source datum, the finite Koszul comparison datum, and strict Koszul Mittag–Leffler exactness, the chiral cobar $\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$ of Theorem 4.1 is the source-side chiral expression at level 2, and its constant-current Lie superalgebra is $\mathfrak{g}_{\Delta_5}$ (View ②). The protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ is the pre-trace Pfaffian section. After squaring and applying the retained level-Z datum, Vol I Theorem H (chiral-Hochschild concentration on the Koszul locus) and Vol II's cyclic-Hochschild Beilinson stratification (i)+(ii) identify the derived-centre trace pairing of degree $-\dim_{\mathbb{C}}(K3 \times E)$ whose scalar is Δ_5^{-2} (Appendix 7, Theorem 7.23). The level-A gravity-line operator algebra acting on the boundary is the open Hall–Borcherds residual of Vol II and is not constructed by the level-Z trace criterion.

5.2 Vol III chain fusion at level 2

The Vol III κ-stratification supplies the κ-tuple $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ at $K3 \times E$; the row $\kappa_{\mathrm{BKM}} = 5$ is the cusp-form weight realized by Δ_5, and the row $\kappa_{\mathrm{fiber}} = 24$ is $\chi_{\mathrm{top}}(K3) = 24$. The additive expression $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ has no invariant meaning: the Hodge reading gives $0 + 2 = 2$ at $N = 1$, and the Heisenberg reading gives the accidental equality $3 + 2 = 5$ only on that row. The CHL frame has Borcherds weights $(5, 3, 2, \frac{3}{2}, 1)$, fixed by $c_N(0)/2$ with $c_N(0) = (10, 6, 4, 3, 2)$ per Jatkar–Sen and Govindarajan–Krishna [47, 48, 60], not by a fixed additive formula. Thus κ_{fiber} is the Euler characteristic of the topological fibre and not of $\mathcal{O}_{\mathrm{fiber}}$; for K3, $\chi_{\mathrm{top}}(K3) = 24$ but $\chi(\mathcal{O}_{K3}) = 2$. Under the Vol III chain-fusion datum at level 2, the chiral algebra A_X on the CY face is compared with the chart algebra A_b on the Vol II open face. In the $K3 \times E$ specialization this comparison becomes $A_X = A_b = \Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$ only under the hypotheses of Theorem 4.1.

Remark 5.1 *(Coefficient projection is not a Hall comparison).* A Hall–Borcherds comparison map cannot be defined by coefficient projection alone, and a coefficient projection respecting charge addition because both products use the same cone is not a recognition theorem. A coefficient projection is not a Hall product/bracket matrix comparison, and does not prove Borcherds–Serre rows, radical descent, no-extra-relations, or PBW. The recognition criterion (S4)–(S10) of Theorem 4.1 is the mathematical statement:

the source-side bracket matrices, Hopf radical, kernel equality, and PBW associated-graded comparison are independent data, none of which is inferable from a coefficient projection.

5.3 BCOV / mixed holomorphic-topological strings

The mixed holomorphic-topological strings companion provides the local model $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ with brane completion as one realization of the factorization-algebra datum Φ_d^{FA}; the global obstruction is the holomorphic de Rham obstruction class. The level-n scalar shadow $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ on the closed $K3 \times E \mapsto$ point cobordism is the BPS partition function (Chapter 8). A three-dimensional gravity-line interpretation with internal compact factor $K3 \times E$ requires the Hall–Borcherds residual of Vol II and is not claimed by Theorem 2.1. The identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$ is a protected Pfaffian identity before trace; its inverse square is the level-n protected scalar trace. The Pfaffian and orientation parts of the operator-level datum $\mathfrak{D}_X^{\mathrm{DI}}$ are defined on $K3 \times E$ relative to the retained data of Appendix 7; the primitive Lie-superalgebra part is the compact-source recognition problem.

6 FINITE PROGRESS AND THE REMAINING COMPACT CONSTRUCTIONS

The target degree calculations give concrete comparison problems. At height three, the target has real simple dimensions 1|0, isotropic dimensions 10|0, and interior dimensions 29|93. The isotropic space consists of the real bracket and nine even imaginary simple directions. The interior space admits a rank-28 differential in the positive height-three quotient, leaving cohomology 1|65. That differential extends through height five and fails at height six for the unchanged Borcherds relations, by Theorem 5.3. Thus a proposed geometric differential must be compared with the relations on its actual domain, rather than inferred from the signed dimension -64.

The first enlarged degree set has 35 degrees, including zero codomains of relations. With $a_{ij} = \delta_i + \delta_j$, $C_{k,s} = a_{ij} + s\delta_k$, and $D_k = \delta_k + 2a_{ij}$, it is

$$W_{\mathrm{rel}}^+ = W_3 \cup \{3\delta_i + \delta_j : i \neq j\} \cup \{2a_{ij} : i < j\} \\ \cup \{C_{k,s} : 2 \le s \le 5\} \cup \{2\delta_{123}\}.$$

The three missing nonzero proper subdegrees are D_k, giving 38 degrees after adjoining them. Their full target dimensions are known:

$$2a_{ij} : 10|0, \quad C_{k,2} : 201|93, \quad C_{k,3} : 29|93, \quad C_{k,4} : 10|0, \quad C_{k,5} : 0|0, \\ D_k : 510|1023, \qquad 2\delta_{123} : 10136|6120.$$

These refine the signed coefficients 10, 108, -64, 10, 0, -513, 4016. Their derivation from the full presentation is given in Chapter 4; the compact-source representatives and operation comparisons remain separate.

For the degree-enlargement rule

$$T_i(\beta) = \beta + (1 - (\delta_i, \beta))\delta_i \quad \text{when } (\delta_i, \beta) < 0,$$

the first enlargement of the specified ten degrees in that 38-degree set adds 21 new output degrees. Eighteen have nonzero signed coefficient. Their nonzero proper subdegrees contribute 126 incidences across 30 new degrees, giving $38 + 21 + 30 = 89$. These are the finite counts proved in Section 1.6, using elliptic reduction of every Jacobi coefficient. Assigning zero to a coefficient beyond a truncated Fourier expansion would change those counts.

This rule is the real Serre output degree on an imaginary simple generator. On a composite root it merely specifies a degree; it need not annihilate the whole root space. Under this rule and nonzero proper-subdegree closure, the affine families $A_N \Rightarrow B_N \Rightarrow A_{N+2}$ of Proposition 1.186 force an unbounded set. Finite partial-operation diagrams remain useful: they specify exactly which inputs, codomains and relations they contain. The infinite enlargement argument is not an obstruction to all finite comparison calculations, and a finite collection of target dimensions is not a compact recognition theorem.

The orientation calculations likewise produce definite local results. Formal coordinate changes admit the all-arity functors and full Artin determinant roots of Section 1.2. Complete flags retain their cross determinants and common trace excess. Finite stabilizer root problems reduce to actual character square roots, while general gerbes can have additional mixed cohomology classes. The first missing compact orientation construction is a single reduced critical coefficient system whose root comparisons agree on all these flags, quotient overlaps and transitions.

For the wall count, the only unconditional combinatorial statement is $|\{\pm 1\}^{12}| = 4096$. To identify this set with the geometric configurations of a half-Hilbert wall atlas requires a bijection, coverage and the correct stabilizer weights. If those hypotheses are supplied, the count agrees numerically with 64^2. That equality of integers does not identify the wall integration map with the OP scalar normalization.

The original compact comparison consequently still requires the reduced coefficient realization and Hall operations, the geometric Pfaffian family, the state and sewing map, and the modular parameter and line comparison. The ordinary connected cocommutative Hopf recognition theorem applies only when that recipient and its hypotheses are constructed. Localized, spectral or braided Hall operations use other tensors and coefficient domains; their positive results do not automatically furnish this ordinary compact structure.

The three-dimensional gravitational interpretation and the Hall–Borcherds operator comparison need their actual state spaces and sewing operations. Numerical levels do not construct them. For example the cusp-form weight is five, while the K3 elliptic genus has weight zero, index one and value 24 at the trivial elliptic variable [51]. A notation that groups categorical, Hodge, Heisenberg, BKM and fibre values into a tuple such as (0, 0, 3, 5, 24) must define the five entries separately. The numerical equality $5 = 3 + 2$ supplies no comparison among their categories or traces. The finite constructions above provide specific local maps and falsifiers; the global compact section and state statements remain conditional on the stated geometric constructions.

Part IV

Elliptic states and critical geometry

Chapter 8

Elliptic states, joining and the inverse Igusa form

The inverse Igusa form occurs in two geometric series. Hilbert schemes of a K3 surface give an elliptic-index series after its boundary factors have been restored. Reduced curve counts on $X = S \times E$, with S a K3 surface and E an elliptic curve, give the reciprocal cusp form itself, with a specified normalization. To compare operations behind these formulas, one needs the complexes on which they act.

We first construct elliptic states from the chiral de Rham sheaf. Its actual Enriques coefficient has parity dimensions (1|65), although the corresponding BKM root space has dimensions (29|93). Both have superdimension -64. Hilbert correspondences then give a concrete joining operator, whose double commutator with creators measures multiplication of surface classes. These constructions make the comparison problem explicit. They do not identify a reduced threefold Hall action with a field action.

The second half of the chapter relates the primitive curve counts to Δ_5^{-2}, determines the spin L-function of the weight-ten eigenline, and specifies the additional trace and sewing data needed for a compact realization. A Pfaffian comparison retains a line, a section and its symmetry action. Squaring that section and taking a numerical trace each forget information, by different operations.

1 Geometric fields before their index

Let Y be a smooth projective complex variety of dimension d. On a coordinate chart its chiral de Rham sheaf Ω_Y^{ch} has even fields γ^i, β_i and odd fields c^i, b_i, with singular products

$$\beta_i(z)\gamma^j(w) \sim \frac{\delta_i^j}{z-w}, \qquad b_i(z)c^j(w) \sim \frac{\delta_i^j}{z-w}.$$

Their conformal weights are 0, 1, 0, 1. The fields c^i, b_i have fermionic charges 1, -1; the even fields have charge zero. We write $a_{(m)}b$ for the coefficient of $(z-w)^{-m-1}$ in the field product, and ∂ for vertex translation.

For a coordinate change $\widetilde{\gamma} = f(\gamma)$ with inverse g, put $f_i^a = \partial_i f^a$ and $g_a^i = \partial_a g^i$. The fields transform by

$$\widetilde{c}^a = f_i^a(\gamma)c^i, \qquad \widetilde{b}_a =: g_a^i(f(\gamma))b_i :,$$
$$\widetilde{\beta}_a =: \beta_i g_a^i(f(\gamma)) : + : \partial_k(g_a^i \circ f)(\gamma)c^k b_i : .$$

The colons denote the normal product in the indicated order. The last summand is essential. For example, when $\widetilde{\gamma} = \gamma^2$ on a punctured chart,

$$\widetilde{\beta} =: \beta(2\gamma)^{-1} : -(2\gamma^2)^{-1} : cb : .$$

The coordinate theorem proves both preservation of the singular products and exact composition on overlaps.[1]

[1] F. Malikov, V. Schechtman and A. Vaintrob, *Chiral de Rham complex*, (3.17), Theorems 3.7–3.8.

Choose a finite affine cover with affine intersections. The Thom–Whitney totalization of its cosimplicial sections is denoted

$$\mathbb{V}_Y = \mathrm{TW}\big(\check{C}^\bullet(Y, \Omega^{\mathrm{ch}}_Y)\big).$$

Concretely, a component is a polynomial differential form on a simplex with values in sections on the corresponding intersection; face and degeneracy maps impose the compatibility conditions. For a form ω and field v, total parity is $\deg \omega + \epsilon(v)$, where ϵ is intrinsic fermionic parity. Thus

$$(\omega \otimes v)_{(m)}(\eta \otimes w) = (-1)^{\epsilon(v)\deg\eta}\omega\eta \otimes v_{(m)}w.$$

The differential is the simplex de Rham differential. No additional differential on the fields is included.

PROPOSITION 1.1. The complex $\mathbb{V}_Y$ is a dg vertex superalgebra computing $R\Gamma(Y, \Omega^{\mathrm{ch}}_Y)$ in each conformal weight. Its vacuum is the compatible constant unit.

Proof. Tensoring a vertex superalgebra with a commutative dg algebra gives the displayed product, and the Koszul sign makes the differential a derivation. Compatible cosimplicial components preserve the vertex identities. Integration on simplices compares this complex with the Cech complex; the polynomial Poincare contraction proves that comparison is a quasi-isomorphism. At each weight, the coordinate construction has a finite filtration by coherent tensor bundles. They are acyclic on the affine intersections. Induction through the filtration identifies the Cech complex with derived sections. □

Let L_0, J_0 be the conformal-weight and charge operators, with formal variables q, y. The associated-graded supercharacter of the field sheaf is

$$y^{-d/2}\bigotimes_{a\geq 1}\Big(\Lambda_{-yq^{a-1}}T_Y^* \otimes \Lambda_{-y^{-1}q^a}T_Y \otimes \mathrm{Sym}_{q^a}\,T_Y^* \otimes \mathrm{Sym}_{q^a}\,T_Y\Big). \tag{8.1}$$

The exterior signs encode intrinsic parity. The coordinate filtration need not split as a field sheaf. Its Euler characteristic nevertheless gives

$$\sum_k (-1)^k \,\mathrm{str}\Big(q^{L_0}y^{J_0-d/2}; H^k(Y, \Omega^{\mathrm{ch}}_Y)\Big) = \mathrm{Ell}(Y; q, y).$$

This equality retains the product of the sheaf-degree sign and the intrinsic-parity sign. It does not recover either grading from the resulting Laurent series.

1.1 *The first Enriques coefficient*

Let T be an Enriques surface and $K = K_T$. A trivialization of $K^{\otimes 2}$ permits local volume coordinates whose transition determinants are constant ± 1. The coordinate change of $J = \sum_i : c^i b_i :$ adds the vertex derivative of the logarithm of that determinant, so J is global. Its zero mode is fermionic charge. Set

$$\mathscr{V} = (\Omega^{\mathrm{ch}}_T)_{L_0=1, J_0=2}.$$

Since normalized charge is $J_0 - 1$, this is the coefficient of qy. The weight-zero, charge-two sheaf is K.

PROPOSITION 1.2. There is a splitting of sheaves of complex vector spaces

$$\mathscr{V} = \partial K \oplus \ker J_{(1)}, \qquad P = \tfrac{1}{2}\partial J_{(1)}, \qquad P^2 = P.$$

The maps need not be $\mathcal{O}_T$-linear.

Proof. The translation commutator is $[J_{(1)}, \partial] = J_{(0)}$. For $a \in K$, weight gives $J_{(1)}a = 0$, while charge gives $J_{(0)}a = 2a$. Thus $J_{(1)}\partial a = 2a$. Translation is injective on K, with left inverse $J_{(1)}/2$, proving the splitting. Locally the two summands of $\partial(c^1c^2) = (\partial c^1)c^2 + c^1\partial c^2$ each map to c^1c^2 under $J_{(1)}$. □

Theorem 1.3. The dimensions of $H^k(T, \mathscr{V})$, for $k = 0, 1, 2$, are $(0, 65, 1)$. The summand $\ker J_{(1)}$ has only H^1, of dimension 65. On a K3 surface the corresponding field coefficient has dimensions $(1, 130, 1)$.

Proof. Writing $\Omega = \Omega^1_T$, the coefficient of (8.1) has internally even associated graded

$$\Omega \otimes \Omega \oplus K \otimes \Omega \oplus K \otimes T_T \cong \operatorname{Sym}^2 \Omega \oplus K \oplus \Omega \oplus T_T.$$

Use $\Omega \cong T_T \otimes K$ and $K^2 \cong \mathcal{O}_T$. Pullback along the etale K3 cover is injective on global sections. The vanishing of positive symmetric holomorphic tangent and cotangent tensors on K3, together with Serre duality, leaves only H^1 in the three noncanonical summands.[2] Their dimensions are 45, 10, 10. For instance,

$$\operatorname{ch}(\operatorname{Sym}^2 \Omega) = 3 - 4c_2(T), \qquad \operatorname{td}(T) = 1 + c_2(T)/12, \qquad \int_T c_2(T) = 12,$$

so Riemann–Roch gives $\chi(\operatorname{Sym}^2 \Omega) = -45$. The canonical summand has $H^2(K) = \mathbb{C}$ and no other cohomology.

The finite filtration gives $h^0 = 0$, $h^1 \leq 65$, $h^2 \leq 1$, and Euler characteristic -64. The projector preserves $H^2(K)$, so $h^2 = 1$ and $h^1 = 65$. On K3 the three degree-one dimensions are 90, 20, 20, and the canonical summand contributes one dimension in degrees zero and two. The same splitting and filtration bounds give $(1, 130, 1)$. □

Total parity therefore gives $(1|65)$ on Enriques. The BKM coefficient $(29|93)$ in Chapter 4 has the same superdimension, but a comparison would have to account for the additional 28 dimensions of each parity. The charge-current projector acts on the geometric field sheaf; it supplies no cancellation differential on the BKM algebra.

There is a natural extra differential on chiral de Rham fields. Locally put $Q = \sum_i : \beta_i c^i :$ and $G = \sum_i : b_i \partial\gamma^i :$. Their modes satisfy

$$[Q_{(0)}, G_{(1)}]_+ = L_0.$$

The operators in this formula are global, even where Q itself changes by a total vertex derivative.[3] At weight $m > 0$, $G_{(1)}/m$ contracts $Q_{(0)}$. With the total cochain signs it also contracts the sum of $Q_{(0)}$ and the sheaf-resolution differential. Moreover, $Q_{(0)}$ raises charge by one. This extra differential thus changes both the positive-weight cohomology and its charge grading; it cannot be inserted while preserving the state theory just constructed.

[2] S. Kobayashi, *The first Chern class and holomorphic symmetric tensor fields*, Corollary 8; I. Dolgachev, *A brief introduction to Enriques surfaces*, §3.

[3] Malikov, Schechtman and Vaintrob, loc. cit., Theorems 4.2–4.4.

2 CYCLE SECTORS AND THE CHIRAL COMPARISON

The twisted chiral de Rham construction associates a sheaf $\Omega^{\mathrm{ch},g}_{Y^N}$ to the fixed locus of each $g \in \mathfrak{S}_N$, with a compatible centralizer action.[4] Writing $C(g)$ for the centralizer of g, define the unshifted sector complex

$$\mathcal{H}_N(Y) = \bigoplus_{[g]\subset\mathfrak{S}_N} R\Gamma\big((Y^N)^g, \Omega^{\mathrm{ch},g}_{Y^N}\big)^{C(g)}.$$

Only the sheaf-resolution differential is used. Coefficients are in $\mathbb{C}$; finite-group invariants are exact. For an n-cycle σ_n, set

$$\mathcal{K}_n(Y) = R\Gamma\big(Y, \Omega^{\mathrm{ch},\sigma_n}_{Y^n}\big)^{\mu_n}.$$

Its local oscillators have indices in $\frac{1}{n}\mathbb{Z}$, and the cyclic invariants retain the centralizer projection.

Let s record cycle length.

THEOREM 2.1. There is a natural equivalence of supercomplexes

$$\prod_{N\ge 0} s^N \mathcal{H}_N(Y) \simeq \widehat{\mathrm{Sym}}_{\mathrm{super}}\left(\bigoplus_{n\ge 1} s^n \mathcal{K}_n(Y)\right).$$

The product completion is coefficientwise in length and conformal weight, and the symmetry uses total cochain and intrinsic parity.

Proof. A permutation with a_n cycles of length n has fixed locus $Y^{\sum a_n}$ and centralizer $\prod_n(\mu_n^{a_n} \rtimes \mathfrak{S}_{a_n})$. Disjoint coordinate sets give external products of the twisted sheaves. The finite coherent filtrations justify coefficientwise Kunneth. Taking cyclic invariants first gives $\bigotimes_n \mathcal{K}_n(Y)^{\otimes a_n}$; the remaining symmetric groups give the super symmetric powers. Each length has finitely many partitions, proving the stated completion. □

In the symmetric-power model, use coinvariants and identify them with invariants by averaging. Concatenation and assignment of cycles to two groups define a commutative, cocommutative dg super Hopf algebra. Explicitly,

$$\Delta(v_1 \cdots v_a) = \sum_{I\subset\{1,\dots,a\}} (-1)^{\epsilon(I)} v_I \otimes v_{I^c}.$$

Subwords keep their order; $\epsilon(I)$ is the total Koszul permutation sign. The empty cycle configuration is the unit and the counit component. Single-cycle vectors are primitive and have antipode -1. In invariant coordinates multiplication is the normalized shuffle. These conventions fix the factorials. For multiplication M_v and the derivation D_λ associated to a homogeneous continuous single-cycle functional,

$$[D_\lambda, M_v]_{\mathrm{super}} = \lambda(v).$$

The identity is the graded Leibniz rule. The Hopf identities follow by assigning cycles to three groups in either order.

The seed fields give another action. Let $V_Y = H^0(Y, \Omega^{\mathrm{ch}}_Y)$. Its mode Lie superalgebra is

$$\mathfrak{m}(V_Y) = \frac{V_Y \otimes \mathbb{C}[t,t^{-1}]}{\mathrm{im}(\partial\otimes 1 + 1\otimes\partial_t)}, \qquad [a[m], b[n]] = \sum_{j\ge 0}\binom{m}{j}(a_{(j)}b)[m+n-j].$$

[4] E. Frenkel and M. Szczesny, *Chiral de Rham complex and orbifolds*, Theorems 4.2–4.3.

Here $a[m]$ is the class of $a \otimes t^m$. The diagonal field $a^\Delta = \sum_{i=1}^N a^{(i)}$ is invariant under permutations and acts on every twisted sector. Distinct factors have no singular products, so this is a mode Lie action ρ, with central vacuum mode $1[-1]$ acting by N. Under the cycle decomposition it acts on each cycle in turn; hence

$$[\rho(A), M_v]_{\mathrm{super}} = M_{\rho(A)v}.$$

The diagonal map is not a unital vertex map: its vacuum has multiplicity N, and regular products have cross terms. Neither this length-preserving mode action nor cycle concatenation is an extension correspondence for sheaves on $S \times E$.

2.1 *An obstruction at weight one*

Take $Y = S$ to be K3 and write

$$C_{H,N} = R\Gamma(S^{[N]}, \Omega^{\mathrm{ch}}_{S^{[N]}}), \qquad C_{O,N} = \mathcal{H}_N(S).$$

Retain sheaf degree k, conformal weight w, charge J and intrinsic parity ϵ. Total parity is $k + \epsilon$, and normalized charge at length N is $J - N$. An untwisted field monomial satisfies $\epsilon \equiv J \pmod 2$: the odd generators have charges ± 1, and the even generators have charge zero. This holds sheafwise and therefore in cohomology.

The age of a permutation is the sum of the numbers $\theta \in (0, 1)$ in its normal eigenvalues $e^{2\pi i\theta}$, with multiplicity. On a surface product it is $N - c(g)$, where $c(g)$ counts cycles. Age contributes to the twisted charge; the constant ground section has intrinsic parity zero.

In the transposition sector, let τ be the constant ground section. It has $(k, w, J, \epsilon) = (0, 0, 1, 0)$; the charge one is its age. The global invariant field $G^\Delta = G^{(1)} + G^{(2)}$ gives $v = (G^\Delta)_{(0)}\tau$. The difference coordinates have half-integral modes. Directly,

$$v = \frac{1}{2}\sum_{a=1}^{2} \gamma^{a,-}_{-1/2} b^-_{a,-1/2}\tau, \qquad 2\beta^-_{1,1/2} c^{1,-}_{1/2} v = \tau.$$

The rightmost detector acts first. Its two contractions use $[\beta_r, \gamma_t] = \delta_{r+t,0}$ and $\{b_r, c_t\} = \delta_{r+t,0}$. The coefficient $1/2$ is the derivative of the half-integral coordinate mode. Globality follows from the invariant field action before this local calculation is used. The detector proves nonvanishing.

Theorem 2.2 *(Unshifted chiral comparison).* There is no cochain-degree-zero quasi-isomorphism, in either direction or by a zigzag, between $C_{H,2}$ and $C_{O,2}$ that preserves weight, charge and total parity. The same obstruction occurs at every even cycle length.

Proof. The class v lies in $H^0(C_{O,2})$ with $(w, J, \epsilon) = (1, 0, 1)$. It cannot be a boundary in a sheaf-resolution model concentrated in nonnegative degrees. The corresponding Hilbert component is zero by the untwisted charge-parity relation. A map preserving k and total parity also preserves ϵ. Its cone therefore retains this class, in degree zero for a map from Hilbert to orbifold and in degree -1 for the reverse direction.

For an N-cycle, let τ_N be the constant ground section. The global descendant $v_N = (\sum_{i=1}^N G^{(i)})_{(0)}\tau_N$ has the Fourier-coordinate expression

$$v_N = \sum_{a=1}^{2}\sum_{r=1}^{N-1} \frac{r}{N}\gamma^{a,r}_{-r/N} b_{a,r;-(1-r/N)}\tau_N.$$

The positive detector $(N/r)\beta_{a,r;r/N}c_{1-r/N}^{a,r}$ sends it to τ_N. Each term has weight one and centralizer eigenvalue one. The ground age is $N-1$, so the class has $(k,w,J,\epsilon)=(0,1,N-2,1)$. For even N, its charge is even and its intrinsic parity odd, again excluding a Hilbert counterpart. Homogeneous resolution changes and contractible additions preserve this cohomology mismatch. A zigzag would identify the two homogeneous cohomology groups, and coefficientwise completion leaves them unchanged. □

This argument does not obstruct the corresponding odd-length component. It also specifies a different possible comparison. Let $C[a]^k=C^{k+a}$, and let Π reverse intrinsic parity. For a permutation g, its surface age is $a(g)=N-\ell(g)$, where $\ell(g)$ counts cycles. Define

$$C_{O,N}^{\mathrm{age}}=\bigoplus_{[g]}\Pi^{a(g)}R\Gamma\big((S^N)^g,\Omega_{S^N}^{\mathrm{ch},g}\big)^{C(g)}[-a(g)].$$

Both sheaf degree and intrinsic parity increase by the age, so total parity and the elliptic supertrace are unchanged. The transposition class moves to $(k,w,J,\epsilon)=(1,1,0,0)$. The mismatch used in the theorem disappears. Constructing quasi-isomorphisms with $C_{H,N}$, together with supported field, creation and joining maps and their coherent compatibilities, remains a separate question.

3 Hilbert correspondences and joining

Let Y be K3 or Enriques, put $X_N=Y^{[N]}$, with $X_0=\mathrm{pt}$, and use rational cohomology with its complex orientation. Write $A=H^*(Y;\mathbb{Q})$. Its odd part is zero, and $c_1(T_Y)=0$ rationally. The punctual incidence variety for $N\ge 0$ and $r>0$ is

$$\mathfrak{Q}_{N,r}=\{(Z',x,Z):Z\subset Z',\ \ell(Z)=N,\ \ell(Z')=N+r,\ \mathrm{Supp}(I_Z/I_{Z'})=\{x\}\}$$

lies in $X_{N+r}\times Y\times X_N$. Its fundamental cycle and projections π_+,s_Y,π_- define

$$C_r(a)v=(\pi_+)_*\big([\mathfrak{Q}_{N,r}]\cup s_Y^*a\cup\pi_-^*v\big).$$

Cap product and Poincare duality are understood. The pushforward is proper. If $\dagger$ is the ordinary Poincare adjoint, set $A_r(a)=(-1)^{r-1}C_r(a)^\dagger$. Then

$$[A_r(a),C_t(b)]=r\delta_{rt}\int_Y ab,\qquad [C_r(a),C_t(b)]=[A_r(a),A_t(b)]=0.$$

The Nakajima–Grojnowski theorem identifies the polynomial space

$$\mathrm{Sym}\left(\bigoplus_{r\ge1}s^rA\right)\cong\bigoplus_{N\ge0}H^*(X_N;\mathbb{Q})$$

by sending the generator $p_r(a)$ to $C_r(a)|0\rangle$, where $|0\rangle=1_{X_0}$. Creation has degree $\deg a+2(r-1)$ and Hodge shift $(r-1,r-1)$.[5] Over distinct points the iterated length-one correspondence orders the points in $N!$ ways. Hence

$$\frac{1}{N!}C_1(1)^N|0\rangle=1_{X_N}.$$

[5] See [94] and M. Lehn–C. Sorger, *The cup product of the Hilbert scheme for K3 surfaces*, Theorems 3.5–3.6. Their trace on the shifted surface algebra is $-\int_Y$; the annihilator above uses $+r\int_Y ab$.

These correspondences have a simple Hall antecedent. A point of X_N is a cyclic quotient $\mathcal{O}_Y \twoheadrightarrow Q$. Constructible functions on zero-dimensional sheaves act on those quotients by

$$(h \cdot f)(\mathcal{O}_Y \twoheadrightarrow Q) = \int_{U \subset Q} h(U) f(\mathcal{O}_Y \twoheadrightarrow Q/U)\, d\chi.$$

Associativity integrates over the same flags $0 \subset U \subset W \subset Q$ in either order. The zero subobject gives the unit. For $Z \subset Z'$, the extension is

$$0 \longrightarrow I_Z/I_{Z'} \longrightarrow \mathcal{O}_{Z'} \longrightarrow \mathcal{O}_Z \longrightarrow 0.$$

The framing generates the middle object. Euler pushforward proves this constructible-function action; critical cohomology requires its own pullback and pushforward maps. Passing to $Y \times E$ also admits nilpotent normal operators, so the surface category is not extension-closed in the threefold category.

3.1 The surface diagonal

Choose a basis $u_1, \dots, u_b$ of $H^2(Y;\mathbb{Q})$, where $b = b_2(Y)$, and let ω be the point class with $\int_Y \omega = 1$. Writing $Q_{ij} = \int_Y u_i u_j$,

$$A = \mathbb{Q}1 \oplus \bigoplus_i \mathbb{Q}u_i \oplus \mathbb{Q}\omega, \qquad u_i u_j = Q_{ij}\omega, \qquad u_i\omega = \omega^2 = 0.$$

Multiplication m and its adjoint diagonal Δ satisfy

$$\Delta 1 = 1 \otimes \omega + \omega \otimes 1 + \sum_{i,j} Q^{ij} u_i \otimes u_j,$$

$$\Delta u_i = u_i \otimes \omega + \omega \otimes u_i, \qquad \Delta\omega = \omega \otimes \omega,$$

where $(Q^{ij}) = Q^{-1}$. Testing against basis pairs proves these formulas. Consequently

$$e_A = m\Delta 1 = (b+2)\omega = c_2(T_Y), \qquad m\Delta(a) = e_A a.$$

The coefficient is 12 for Enriques and 24 for K3. A split followed by a join inserts this Euler class.

The full permutation product explains this insertion. Label each cycle of $\sigma \in \mathfrak{S}_N$ by an element of A. For a joint orbit O of $\langle \sigma, \tau \rangle$, let $c_\sigma(O)$ count its σ-cycles, and set

$$g_O = \frac{|O| + 2 - c_\sigma(O) - c_\tau(O) - c_{\sigma\tau}(O)}{2}.$$

Riemann–Hurwitz identifies it with the nonnegative genus of the transitive three-branch covering. If a_O, b_O are the products of the input labels on that orbit, its output labels are

$$\Delta_{c_{\sigma\tau}(O)}(a_O b_O e_A^{g_O}).$$

Here Δ_k is the iterated diagonal, with $\Delta_1 = \mathrm{id}$. Tensoring over orbits gives the Frobenius permutation product. Its associativity follows from the projection formula and $m\Delta(a) = e_A a$, which identify the decompositions of each connected covering surface. For $s_2 = (12)$ and $\rho = (123)$,

$$(as_2)(bs_2) = \Delta(ab)\,\mathrm{id}, \qquad (a\rho)(b\rho) = abe_A\rho^2, \qquad (a\rho)(b\rho^{-1}) = \Delta_3(ab)\,\mathrm{id}.$$

Genus at least two gives zero because $e_A^2 = 0$.

For the Hilbert product, multiply this law by

$$(-1)^{(|\sigma|+|\tau|-|\sigma\tau|)/2}, \qquad |\sigma| = N - c(\sigma).$$

Permutation parity makes the exponent integral; telescoping proves the two-cocycle identity. The invariant algebra with this sign is $H^*(X_N;\mathbb{Q})$, since K_Y is numerically trivial.[6] A cycle-labelled permutation summed over all conjugates with repetitions corresponds to the product of its creators on the vacuum. The identity sector then has coefficient $N!$. Over $\mathbb{C}$, the factor $i^{|\sigma|}$ removes the sign twist; over $\mathbb{Q}$, the sign remains explicit.

3.2 The second-order operator

For a homogeneous basis a_α of A, write $a_\alpha a_\beta = m_{\alpha\beta}^{\gamma} a_\gamma$ and $\Delta a_\gamma = \Delta_\gamma^{\alpha\beta} a_\alpha \otimes a_\beta$. Let $\partial_{n,\alpha}$ differentiate $p_n(a_\alpha)$. Define

$$J = \frac{1}{2} \sum_{i,j\geq 1} ij\, m_{\alpha\beta}^{\gamma} p_{i+j}(a_\gamma)\, \partial_{i,\alpha}\partial_{j,\beta},$$

$$S = \frac{1}{2} \sum_{i,j\geq 1} (i+j) \Delta_\gamma^{\alpha\beta} p_i(a_\alpha)\, p_j(a_\beta)\, \partial_{i+j,\gamma}, \qquad D_H = J - S.$$

All sums are finite at fixed length. Joining two cycles gives ij endpoint choices. Splitting a cycle gives $i+j$ choices, counted twice by the ordered sum. Thus the transposition sum acts by $J+S$ in the ordinary orbifold product and by D_H with the Hilbert sign. Geometrically D_H is cup multiplication by $-c_1(\mathcal{O}_Y^{[N]})$; the canonical-class term vanishes.[7] For $N \geq 2$, the transposition class is $C_2(1)C_1(1)^{N-2}|0\rangle/(2(N-2)!)$, with the denominator given by its centralizer order.

PROPOSITION 3.1. If $M_i(a)$ multiplies by $p_i(a)$, then

$$[[D_H, M_i(a)], M_j(b)] = ij M_{i+j}(ab), \qquad [[[D_H, M_i(a)], M_j(b)], M_k(c)] = 0.$$

The operator preserves length and raises cohomological degree by two and normalized Hodge charge by one. Moreover,

$$D_H^2 p_2(a) = -p_2(e_A a), \qquad D_H^{2N+1}|_{H^*(X_N)} = 0.$$

Proof. The first-order splitting term has zero double commutator with multiplication. Differentiating both derivative slots of J gives $ij M_{i+j}(ab)$, which commutes with every creator. Joining increases the age degree by two; splitting inserts a degree-four diagonal and decreases the age degree by two. Both have Hodge increase $(1,1)$. A label of Hodge degree (r,s) on a length-i cycle has normalized charge $r+i-1-i = r-1$, so the increase is one. At length two, splitting and joining gives $-m\Delta(a)$, proving the square formula. The degree range $0,\dots,4N$ proves the nilpotence bound. □

In particular $D_H^2 p_2(1) \neq 0$. This joining operator is even and second order; it is not a square-zero differential. Its double commutator gives an operation that the scalar elliptic genus alone does not specify.

[6] Lehn–Sorger, loc. cit., §2 and Theorem 3.2.

[7] Lehn–Sorger, Proposition 3.9.

3.3 *Preparing the elliptic matrix coefficient*

Let $\mathcal{E}_{q,y}(X_N)$ be the tensor class in (8.1), and put

$$\varepsilon_N = \operatorname{td}(T_{X_N})\operatorname{ch}(\mathcal{E}_{q,y}(X_N)), \qquad R = \mathbb{Q}(y)[[q]].$$

Riemann–Roch gives $\int_{X_N}\varepsilon_N = \operatorname{Ell}(X_N; q, y)$. These are even, generally inhomogeneous cohomology classes. Their degree-zero parts r_N have constant q coefficient $y^{-N}(1-y)^{2N}$, a unit over $\mathbb{Q}(y)$. Since $\nu_N = \varepsilon_N - r_N$ has positive degree,

$$\varepsilon_N^{-1} = r_N^{-1}\sum_{j=0}^{2N}(-r_N^{-1}\nu_N)^j.$$

Write U_N for multiplication by ε_N.

A cohomological kernel from X_M to X_N is a class on $X_N \times X_M$. It acts by pullback, cup product and pushforward; convolution uses the same operations on the triple product. Transport it by

$$\widetilde{K} = \pi_N^*\varepsilon_N \cup K \cup \pi_M^*\varepsilon_M^{-1}.$$

The middle classes cancel in every convolution, so

$$\widetilde{L\circ K} = \widetilde{L}\circ\widetilde{K}, \qquad T_{\widetilde{K}} = U_N T_K U_M^{-1}, \qquad \widetilde{[\Delta_{X_N}]} = [\Delta_{X_N}].$$

The transported creators and annihilators obey the same Heisenberg relations. Cup multiplication commutes with U_N, so D_H is unchanged. Telescoping and the vacuum normalization give

$$\frac{1}{N!}\widetilde{C}_1(1)^N|0\rangle = \varepsilon_N.$$

With $\langle I|$ denoting integration in each length,

$$\left\langle I\middle|\exp(s\widetilde{C}_1(1))\middle|0\right\rangle = \sum_{N\ge 0} s^N \operatorname{Ell}(Y^{[N]}; q, y). \tag{8.2}$$

This is a specified matrix coefficient. Conjugation leaves ordinary traces unchanged, and q here is a coefficient parameter. The inverse used above is an inverse in cohomology, not a tensor inverse of a coefficient complex.

The twisted-sector index theorem and the elliptic McKay theorem identify this scalar with the index of the cycle sectors.[8] If $\operatorname{Ell}(Y; q, y) = \sum c(m,\ell)q^m y^\ell$, then

$$\sum_{N\ge 0} s^N \operatorname{Ell}(Y^{[N]}; q, y) = \prod_{n>0,\, m\ge 0,\, \ell}(1 - s^n q^m y^\ell)^{-c(nm,\ell)}.$$

For Enriques and K3 the seeds are $\phi_{0,1}$ and $2\phi_{0,1}$. With the boundary factor $b(q, y)$ of Section 8, the two series are

$$F_T = \frac{s^{1/2}b(q,y)}{D_5(q,y,s)}, \qquad F_S = \frac{s\,b(q,y)^2}{D_5(q,y,s)^2}, \qquad F_T^2 = F_S.$$

[8] Frenkel–Szczesny, Theorem 6.1, with the intrinsic-parity signs in (8.1); L. Borisov–A. Libgober, *McKay correspondence for elliptic genera*, Corollary 6.7.

The boundary factors cancel before specialization at $y = 1$. The unshifted obstruction of Theorem 2.2 shows why this scalar equality cannot be read as a comparison of all the underlying gradings and operations.

There is also a domain issue at $y = 1$. At $q = 0$, set $t = y - 1$, $z = y + y^{-1} - 2 = t^2/(1+t)$, and $a = 1 + z/12$. On the surface,

$$\varepsilon_Y(0, y) = z\,1 + a\,c_2(T_Y).$$

Writing $e = \chi_{\mathrm{top}}(Y)$, its multiplication matrix on $(1, \omega)$ is

$$\begin{pmatrix} z & 0 \\ ae & z \end{pmatrix}.$$

The transported length-one annihilator therefore has value

$$\widetilde{A}_1(1)(1) = \int_Y \varepsilon_Y^{-1} = -ae\,z^{-2},$$

a fourth-order pole at $t = 0$, although the index $\int_Y \varepsilon_Y = ae$ is regular. The image lattice $\varepsilon_Y H^*(Y;\mathbb{Q})[[t]]$ is abstractly free, while its embedded special image has rank one. Its cokernel is

$$\mathbb{Q}[[t]]/(t^4) \oplus \big(\mathbb{Q}[[t]]/(t^2)\big)^{b_2(Y)}.$$

Indeed the outer matrix has Smith form $\mathrm{diag}(1, t^4)$, and multiplication on $H^2(Y)$ is a unit times t^2. The derived special fibre of the cokernel consequently has $b_2(Y) + 1$ dimensions in each of degrees $-1, 0$. The later degeneration chapters develop this distinction between the lattice, its embedded fibre and the derived cokernel.

Finally, any invertible even classes α_N, with $\alpha_0 = 1$, give the same conjugation construction with matrix coefficient $\sum_N s^N \int_{X_N} \alpha_N$. For $N > 0$, taking $\alpha_N = 1 + c_N\,[\mathrm{pt}]$ realizes an arbitrary coefficient $c_N \in R$. A prescribed scalar series therefore does not determine a critical Hall action on independently defined field complexes.

4 Primitive curve counts on the threefold

Return to $X = S \times E$. Let β_h be a nonzero primitive K3 class with $\beta_h^2 = 2h - 2$, and let d be the degree along E. Translation by E and the K3 symplectic reduction enter the reduced theory. Its stable-pair and ideal-sheaf moduli are $P_n(X, (\beta_h, d))$ and $I_n(X, (\beta_h, d))$. For pairs, $n = \chi(F)$; for an ideal sheaf of a curve C, $n = \chi(\mathcal{O}_C)$. The obstruction theories and virtual classes live on these moduli. The target remains a complex threefold.

Convention 4.1 *(Rational representatives and variables).* Let $R_{h,d}(v)$ be the rational function whose Laurent expansion at $v = 0$ is the reduced DT/PT series in class (β_h, d). Use

$$(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}}) = (s, q, r), \qquad v = -r,$$

and define the rational representative of the full primitive series by

$$Z^X_{\mathrm{OP}}(r, q, s) = \sum_{h,d \geq 0} R_{h,d}(-r)\, q^{h-1} s^{d-1}.$$

The DT/PT expansion uses $r = 0$; the Borcherds product chamber uses the expansion at $r = \infty$, coefficientwise in q, s. The GW substitution is $r = e^{iu}$, hence $v = -e^{iu}$, followed by Laurent expansion at $u = 0$.

Oberdieck proves equality of reduced DT and PT series for nonzero K3 class and rationality of the PT series. The degree-zero DT correction is trivial because $\chi_{\mathrm{top}}(S\times E)=0$ [99, Theorem 3]. The primitive PT/GW comparison is [100, Proposition 5]. The K3 reduction and primitive-class convention also agree with the reduced K3 theory of [88].

Theorem 4.2 *(The primitive inverse Igusa series).* With Convention 4.1, the reduced numerical invariants satisfy

$$Z^X_{\mathrm{DT}}=Z^X_{\mathrm{PT}}=Z^X_{\mathrm{OP}}=-(\chi^{\mathrm{OP}}_{10})^{-1}=-4096\,\Delta_5(Z)^{-2}. \tag{8.3}$$

Here the equality of DT/PT series means equality of their rational representatives. Define the unscaled scalar by

$$Z^{K3\times E}_{\mathrm{BPS}}:=-4096^{-1}Z^X_{\mathrm{OP}}=\bigl(\Phi^{\mathrm{un}}_{10}\bigr)^{-1}=\Delta_5^{-2}. \tag{8.4}$$

This definition does not supply an operator trace.

Proof. The reduced GW series is

$$Z(u,q,s)=\sum_{g,h,d\geq 0}N_{g,h,d}u^{2g-2}q^{h-1}s^{d-1}.$$

Oberdieck–Pixton identify it with the Laurent expansion of $-1/\chi^{\mathrm{OP}}_{10}$ under $r=e^{iu}$ [101, Theorem 1]. Their product is

$$\chi^{\mathrm{OP}}_{10}(r,q,s)=rqs\prod_{(h,k,d)>0}(1-r^kq^hs^d)^{c(4hd-k^2)},\qquad c(4hd-k^2)=2f(hd,k).$$

Positivity means $h,d\geq 0$ and $h+d>0$, or $h=d=0$ and $k<0$. The primitive PT/GW comparison, including $v=-e^{iu}$, and the reduced DT/PT theorem identify the rational representatives. In the theta convention of this book,

$$\chi^{\mathrm{OP}}_{10}=D_5^2=2^{-12}\Delta^{\theta}_{10}=4096^{-1}\Delta_5^2,\qquad \Delta^{\theta}_{10}=\Phi^{\mathrm{un}}_{10}=\Delta_5^2. \tag{8.5}$$

This is the product normalization of Chapter 3, with the classical Igusa identification [58, 59]. Inverting proves (8.3) and the rescaling gives (8.4). □

The theorem concerns the primitive K3-class branch. It makes no assertion for an arbitrary imprimitive class. Changing stability chamber in a proposed compact Hall realization requires the correspondence comparison of Definition 8.55; the rational variable substitution alone does not transport a Hall bracket.

4.1 *Coefficients, roots and normalization*

The seed has expansion

$$\phi_{0,1}=y^{-1}+10+y+q\bigl(10y^{-2}-64y^{-1}+108-64y+10y^2\bigr)+O(q^2).$$

Its coefficients have several uses with different objects behind them. They are Euler superdimensions of the elliptic coefficient bundles. At the root $\alpha(n,\ell,m)=2nf_2-\ell f_3+2mf_{-2}$, the value $f(nm,\ell)$ is the signed root multiplicity in the BKM denominator. The K3 genus has coefficients $2f$, which produce the squared automorphic product. These equalities are equalities of indices; they do not identify the ordinary parity dimensions or construct bracket maps.

In the product coordinates of Chapter 3,

$$\Delta_5(Z) = 64(qrs)^{1/2} \prod_{(n,\ell,m)>0} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

The chamber includes the factor $1 - r^{-1}$ at $n = m = 0$. For $P = D_5/(qrs)^{1/2}$, one has

$$[qs]P = (1 - r^{-1})\big(-9(r^2 + r^{-2}) + 84(r + r^{-1}) - 6\big), \qquad [qrs]P = 93.$$

Indeed the pure q and pure s factors contribute $(1 - qA)(1 - sA)$, where $A = r + 10 + r^{-1}$, and the mixed factor contributes $-qs(10(r^2 + r^{-2}) - 64(r + r^{-1}) + 108)$. Their product gives the displayed expression. The isolated factor $(1 - qrs)^{-64}$ supplies 64; the remaining factors supply 29. The coefficient 93 is therefore different from the root superdimension -64.

Squaring gives

$$\Delta_5^2 = 4096qrs \prod_{(n,\ell,m)>0} (1 - q^n r^\ell s^m)^{2f(nm,\ell)}.$$

Thus, with $(p, q, \widetilde{q}) = (r, q, s)$ in the OP notation,

$$\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2, \qquad [pq\widetilde{q}]\chi_{10}^{\mathrm{OP}} = 1, \qquad [(qrs)^{1/2}]D_5 = 1.$$

The leading theta coefficient is 64, and its square is 4096. The minus sign in (8.3) is the OP convention. None is an orientation character.

Remark 4.3 *(The sign data).* The homogeneous orthogonal character $\widetilde{\nu}_{\Delta_5}$ of Definition 6.2 belongs to the weight-five automorphic line. A geometric character ϵ_o requires an actual equivariant root line and coherent wall transport. Its local sign must be computed from the action on that root, as in Chapter 7. The formal set $\Omega_{12} = \{\pm 1\}^{12}$ also has cardinality 4096, but a geometric sign orbit identified with that set is additional data. Its cardinality cannot be inferred from the theta coefficient.

At a generic smooth point of the diagonal divisor on Siegel space, write $\Delta_5 = ua$ in a local frame, with a a unit. Then $\Delta_5^{-2} = u^{-2}a^{-2}$ has a double pole. This divisor calculation leaves the geometric mirror-discriminant comparison of Chapter 15 open. The coefficient of $q^{-1}s^{-1}$, expanded at $r = 0$, is

$$[q^{-1}s^{-1}]\Delta_5^{-2} = \frac{r}{4096(1 - r)^2} = \sum_{n\geq 1} \frac{n}{4096} r^n.$$

A fixed Euler degree extracts $[r^n]$; it is not the complete series. Evaluation at a Siegel point outside the polar divisor is yet another operation. The target Euler characteristic is $\chi(O_X) = \chi(O_S)\chi(O_E) = 0$, which is distinct from the Euler degree of a curve or stable pair.

5 The weight-ten Hecke eigenline

The arithmetic of Δ_5^2 is independent of its scalar normalization. Put

$$g(\tau) = \Delta(\tau)E_6(\tau), \qquad \Delta = q\prod_{n\geq 1}(1 - q^n)^{24}, \qquad E_6 = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n.$$

Here $\sigma_5(n) = \sum_{d|n} d^5$. The space $S_{18}(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional, so g is its normalized Hecke eigenform. The Saito–Kurokawa correspondence identifies its lift with the one-dimensional weight-ten Siegel cusp space. Thus its eigenline is $\mathbb{C}\Delta_5^2 = \mathbb{C}\chi_{10}^{\mathrm{OP}}$ [118, Theorem 2 and its corollary].

Proposition 5.1 *(The full Hecke operators).* Let a_p be the coefficient of q^p in g, and use Andrianov's full operators $T(p)$ and $T(p^2)$, normalized by [2, Theorem 3.1.1]. On the weight-ten eigenline,

$$\lambda(p) = a_p + p^8 + p^9, \qquad \lambda(p^2) = a_p^2 + a_p(p^8 + p^9) + p^{18}.$$

In particular,

p	a_p	$\lambda(p)$	$\lambda(p^2)$
2	-528	240	135424
3	-4284	21960	293343849

Proof. For a positive even Siegel weight K, the lift of a weight-$2K-2$ eigenform has local polynomial

$$(1 - p^{K-2}t)(1 - p^{K-1}t)(1 - a_p t + p^{2K-3}t^2).$$

Andrianov's polynomial for the full operators is

$$1 - \lambda(p)t + \bigl(\lambda(p)^2 - \lambda(p^2) - p^{2K-4}\bigr)t^2 - \lambda(p)p^{2K-3}t^3 + p^{4K-6}t^4.$$

Set $x = p^{K-2}$, $y = p^{K-1}$, so $xy = p^{2K-3}$. The linear and quadratic coefficients of the factored polynomial are $-(a_p + x + y)$ and $a_p(x + y) + 2xy$. Therefore

$$\lambda(p^2) = (a_p + x + y)^2 - x^2 - a_p(x + y) - 2xy = a_p^2 + a_p(x + y) + y^2.$$

Taking $K = 10$ gives the formulas. Expanding the defining product for Δ and the divisor series for E_6 gives $g = q - 528q^2 - 4284q^3 + \cdots$, from which the table follows. □

The notation $T(p^2)$ here denotes the full Hecke operator; it must not be replaced by a single double-coset generator with the same informal degree. The elliptic bound $|a_p| \le 2p^{17/2}$ shows $\lambda(p)/p^9 \to 1$. The Satake factor p^9, normalized by $p^{17/2}$, has absolute value $p^{1/2}$, expressing the non-tempered nature of this lift.

Remark 5.2 *(The Maass subspace).* For positive even K, the full-level lift takes $S_{2K-2}(\mathrm{SL}_2(\mathbb{Z}))$ to the Maass subspace in weight K, with Hecke eigenvalues prescribed by the local polynomial above. At $K = 10$, both the source and the entire Siegel cusp space have dimension one [118, pp. 375, 379–381]. The product normalization (8.5) changes its generator but leaves all Hecke eigenvalues unchanged.

Theorem 5.3 *(The spin factorization and its completion).* For $F = \Delta_{10}^{\theta} = \Delta_5^2$,

$$L_{\mathrm{spin}}(s, F) = \zeta(s - 8)\zeta(s - 9)L(s, g), \qquad g = \Delta E_6. \tag{8.6}$$

The local polynomial and completed function are

$$Q_{p,F}(t) = 1 - \lambda(p)t + (\lambda(p)^2 - \lambda(p^2) - p^{16})t^2 - \lambda(p)p^{17}t^3 + p^{34}t^4,$$
$$\Psi_F(s) = (2\pi)^{-2s}\Gamma(s)\Gamma(s - 8)L_{\mathrm{spin}}(s, F).$$

The completion is meromorphic and satisfies $\Psi_F(18 - s) = \Psi_F(s)$.

Proof. The Saito–Kurokawa local factor just computed gives (8.6) in a right half-plane and hence by meromorphic continuation [118, (10)–(11), Theorem 2]. Andrianov's Theorem 3.1.1 supplies the completion and functional equation [2]. These are separate uses of the two results: the general genus-two functional equation does not by itself imply the Saito–Kurokawa factorization. □

PROPOSITION 5.4 *(The pole at $s = 10$).* The possible pole at $s = 9$ in (8.6) is cancelled by $L(9, g) = 0$. At $s = 10$ there is a simple pole with positive residue

$$\operatorname*{Res}_{s=10} L_{\mathrm{spin}}(s, F) = \zeta(2) L(10, g) = \frac{\pi^2}{6} L(10, g) > 0. \tag{8.7}$$

Proof. The Mellin transform

$$\Lambda(g, s) = (2\pi)^{-s} \Gamma(s) L(s, g) = \int_0^\infty g(iy) y^{s-1} \, dy$$

continues to an entire function. Modularity of weight eighteen gives $g(i/y) = -y^{18} g(iy)$. Splitting the integral at one and substituting $y \mapsto 1/y$ gives

$$\Lambda(g, s) = \int_1^\infty g(iy) \left(y^{s-1} - y^{17-s} \right) dy, \qquad \Lambda(g, s) = -\Lambda(g, 18 - s).$$

Thus $L(9, g) = 0$. At $s = 10$ the factor in parentheses is $y^9 - y^7 > 0$ for $y > 1$.

The product for $\Delta(iy)$ is positive. Modularity of E_6 gives $E_6(i) = 0$, and its Fourier series gives

$$\frac{d}{dy} E_6(iy) = 1008\pi \sum_{n \geq 1} n \sigma_5(n) e^{-2\pi n y} > 0.$$

Hence $g(iy) > 0$ for $y > 1$, proving $\Lambda(g, 10) > 0$ and $L(10, g) > 0$. The pole of $\zeta(s - 9)$ has residue one; multiplying the remaining factors gives (8.7). □

The rational-period theorem supplies a nonzero even-critical period $\Omega^+(g)$, defined up to $\mathbb{Q}^\times$, such that

$$\frac{L(10, g)}{(2\pi i)^{10} \Omega^+(g)} \in \mathbb{Q}.$$

Here the superscript specifies the even critical integers, rather than the parity of the index in $r_n(g) = \int_0^\infty g(iy) y^n \, dy$. The rational structures come from Eichler–Shimura–Manin theory [71, 85], with the period convention of [32]. The positivity proof is additional to that rationality statement. Neither argument uses a geometric Pfaffian or a compact Hall realization.

6 TRACE DOMAINS AND SEWING

A scalar series becomes a trace only after a state space, an endomorphism and a trace functional have been specified. At a finite geometric truncation R, suppose one has charge-graded perfect complexes $V_{R,\gamma}$, where $\gamma = (h, d, n)$, and even degree-zero chain endomorphisms $U_{R,\gamma}$. For a bounded finite-dimensional model their normalized trace is

$$t_{R,\gamma} = \lambda_{R,\gamma} \sum_{k,\epsilon} (-1)^{k+\epsilon} \operatorname{tr}(U_{R,\gamma} \mid V_{R,\gamma}^{k,\epsilon}).$$

The intrinsic parity ϵ is included only when that grading is part of the state theory. Cyclicity of the finite supertrace makes chain-homotopic endomorphisms have the same value. The normalization $\lambda_{R,\gamma}$ is essential: the coefficient $n/4096$ above need not be an integer superdimension. An inverse limit requires coefficientwise stabilization or a trace-convergence theorem; a formal limit alone does not supply it.

If the construction starts from a boundary dg-category C_R and a chosen object b_R, then $A_{b,R} = \mathrm{End}_{C_R}(b_R)$ is an associative algebra by composition. For a chosen coefficient line ℓ_R, a cyclic functional $A_{b,R} \to \ell_R[-3]$, a cyclic pairing on its Hochschild chains, and a pairing on a chiral centre are differently typed data. Passing between them requires the relevant comparison. An object b_R also need not generate the boundary category. A proposed compact realization must specify its trace domain and prove compatibility with the state and all sewing maps.

6.1 *A finite-trace obstruction on a continuous carrier*

The domain matters even for a positive heat operator. Write M_f for multiplication by f, and let $\mathcal{H} = L^2((0,\infty), dP)$ and $C_t = M_{e^{-tP^2}}$, with $t > 0$.

Proposition 6.1. There is no two-sided ideal $\mathcal{J} \subset B(\mathcal{H})$ containing C_t and carrying a complex-linear functional $\tau : \mathcal{J} \to \mathbb{C}$ with $\tau(AB) = \tau(BA)$ for $A \in \mathcal{J}$, $B \in B(\mathcal{H})$, which agrees with the ordinary trace on finite-rank operators. No continuity or positivity assumption on the functional is needed.

Proof. Let $q = M_{\mathbf{1}_{[1,2]}}$. Multiplication by $\mathbf{1}_{[1,2]}e^{tP^2}$ is bounded, so $q \in \mathcal{J}$. The range $q\mathcal{H}$ is infinite-dimensional. Choose a unilateral shift V on it, extended by zero on its orthogonal complement. Then

$$V^*V = q, \qquad VV^* = q - p_1,$$

where p_1 is a rank-one projection. As $V = qV \in \mathcal{J}$, cyclicity and linearity give $\tau(q) = \tau(q - p_1) = \tau(q) - 1$, a contradiction. □

The commutative multiplier algebra has the positive semifinite weight $M_f \mapsto \int_0^\infty f(P)\, dP$ for $f \geq 0$, finite on C_t. The proposition rules out a finite cyclic extension to the specified full operator ideal. It is not an obstruction to every trace domain, and it says nothing about the Igusa coefficients without an actual identification of their operator carrier with this one.

6.2 *Marked kernels retain finite amplitudes*

Choose a nonnegative measurable ρ such that e^{-tP^2} and $\rho(P)e^{-tP^2}$ lie in $\mathcal{H}$ for every $t > 0$. Put

$$v_0(t) = e^{-tP^2}, \qquad v_1(t) = \rho(P)e^{-tP^2}, \qquad K_{ij}(t,s) = |v_i(t)\rangle\langle v_j(s)|.$$

These rank-one kernels have the sewing law

$$K_{ij}(t,s)K_{k\ell}(u,v) = \langle v_j(s), v_k(u)\rangle K_{i\ell}(t,v).$$

It follows directly by applying both sides to a vector. The middle overlap is part of the product. In particular,

$$\langle v_0(t), v_1(s)\rangle = \int_0^\infty \rho(P)e^{-(t+s)P^2}\, dP, \qquad \langle v_1(t), v_1(s)\rangle = \int_0^\infty \rho(P)^2 e^{-(t+s)P^2}\, dP.$$

One and two marked boundaries insert ρ and ρ^2, respectively. Each $K_{ii}(t,t)$ is positive; when $v_i(t) \neq 0$, division by $\|v_i(t)\|^2$ gives a positive contraction. For $a > 0$, $C_a v_i(t) = v_i(a+t)$, so the heat action is compatible with these marked states. This construction preserves finite matrix coefficients without assigning a finite trace to C_t in the forbidden ideal.

6.3 *A closed scalar does not fix a cap*

A finite example separates normalization from sewing. On $A = \mathbb{C}e_1 \oplus \mathbb{C}e_2$, take

$$e_i e_j = \delta_{ij} e_i, \qquad 1 = e_1 + e_2, \qquad \varepsilon(e_1) = w, \quad \varepsilon(e_2) = 1 - w, \qquad 0 < w < 1.$$

This is a positive Frobenius algebra for its coordinatewise real structure. If $p_1 = w$, $p_2 = 1 - w$, its coproduct and cup are

$$\Delta(e_i) = p_i^{-1} e_i \otimes e_i, \qquad \mathrm{coev}(1) = \sum_i p_i^{-1} e_i \otimes e_i.$$

The inverse weights are forced by the snake identity. A connected surface of genus g with q outgoing circles sends e_i through multiplication, handles and coproducts with the corresponding factor p_i^{1-g-q}; in particular its closed value is

$$Z_g = \sum_i p_i^{1-g}.$$

Both $w = 1/2$ and $w = 1/3$ give $Z_0 = 1$ and $Z_1 = 2$, but their genus-two values are 4 and $9/2$. Thus even a unit, a normalized sphere and the same torus value do not determine all sewing data. This example does not compare two theories with the same complete family of closed operations.

6.4 *When a supertrace is positive*

Let a graded Hilbert space carry an odd self-adjoint operator D and an even nonnegative self-adjoint Hamiltonian H strongly commuting with D. Fix a domain of $\beta > 0$ on which $e^{-\beta H}$ is trace class. Each finite-dimensional H-eigenspace decomposes into $\ker D$ and paired even/odd states on its complement. Their contributions cancel, giving

$$\mathrm{STr}(e^{-\beta H}) = \mathrm{STr}(e^{-\beta H}|_{\ker D}).$$

If the physical kernel is entirely even, this becomes an ordinary positive trace and hence a Laplace transform of a positive spectral measure. Without that parity hypothesis the remaining measure is signed. A protected index therefore supplies neither positivity nor an ordinary thermal interpretation merely by being written as a supertrace.

7 THE COMPACT REALIZATION QUESTION

The geometric states, Hilbert correspondences and reduced curve counts constructed above have different coefficient theories. A compact realization must compare those theories by maps preserving their operations. In particular, the prepared state in (8.2), the empty cut unit, the local field vacuum and a scalar augmentation are distinct data. Equality of their eventual scalar series cannot identify them.

Scope 7.1 (The physical interpretation). The primitive inverse Igusa identity is a theorem about reduced numerical invariants. A protected D-brane interpretation asks for

$$\Omega_X(h, d, n) \overset{\mathrm{phys}}{=} \mathbf{DT}^X_{n,(\beta_h, d)}$$

in a specified physical chamber. This is additional to the DT/PT/GW theorem. Dyon counting provides a physical setting for inverse automorphic products [35]; the automorphic construction itself is the singular theta lift [10]. Neither result constructs the state and sewing maps of a three-dimensional gravitational sector obtained from $\mathrm{AdS}_3 \times S^3 \times K3$. A metric path integral or a boundary operator algebra requires those additional data.

7.1 The Hall and state comparison

Fix a reduced oriented Hall coefficient theory on the desired threefold moduli, including its extension flags, supported pullbacks and pushforwards, relative unit and a geometric charge map. Denote its completed associative algebra by $\widehat{\mathcal{H}}_X^{\mathrm{red}}$. This notation presupposes the construction of those operations; it does not assert that every reduction admits them.

Let $\mathfrak{n}_+$ be the positive part of $\mathfrak{g}_{\Delta_5}$. At the ordinary coefficient level the target is $\widehat{U(\mathfrak{n}_+)}$, with a specified positive-height completion. For a chiral comparison on E, one must instead supply a geometric associative Ran algebra F_X and the corresponding current target $\mathcal{A}_{+,E}$, with their Ran convolution tensor and empty unit. A current envelope and an ordinary enveloping algebra have different domains.

Conjecture 7.2 (Compact Hall and state comparison). For a suitable reduced coefficient theory and compatible charge dictionary, construct algebra and supported current comparisons

$$\Theta_{\mathrm{Hall}} : \widehat{\mathcal{H}}_X^{\mathrm{red}} \longrightarrow \widehat{U(\mathfrak{n}_+)}, \qquad \Theta_E : F_X \longrightarrow \mathcal{A}_{+,E}.$$

Their primitive maps must preserve the defining relations, invariant pairing and its radical. Their state maps must intertwine the extension operations with independently defined geometric fields and joining correspondences, including the supported collision operations and vacuum insertions. For the Hilbert/orbifold field comparison, use the age and parity data specified after Theorem 2.2.

The charge dictionary is part of the problem. The Enriques seed has half the K3 elliptic genus, but dividing an index by two does not construct a functor from K3 threefold states to Enriques states. Nor does matching the coefficient -64 construct the extra differentials or primitive brackets. The ordinary cycle Hopf algebra has zero primitive commutator; the nonzero cohomological joining commutator is an even second-order operation. A Hall comparison must identify the actual operation complexes before comparing these expressions.

The gravity-line extension of this problem, as formulated in [82], further asks for a boundary action and bulk sewing. A Drinfeld double requires a compatible Hopf pairing, integral form and completion. Current enveloping along E requires its own coefficient and support functor. An open–closed comparison must preserve the operations of the two-coloured Swiss-cheese chiral-topological operad, including its boundary and bulk compositions. None is a formal consequence of an isomorphism of graded vector spaces. In particular, an ambient fusion or reconstruction object cannot be identified with a unitary spectral presentation without a comparison of those structures.

Once source and target operation complexes are fixed, their higher comparison equations are concrete. If b_A, b_B are their bar coderivations and the components below arity n have been constructed, put

$$R_n = (b_B F_{<n} - F_{<n} b_A)_n.$$

The next equation is $\partial_{\mathrm{Hom}} F_n = -R_n$. For a contraction $\partial_{\mathrm{Hom}} h + h\,\partial_{\mathrm{Hom}} = 1 - P$, the choice $F_n = -h R_n$ solves it when $P R_n = 0$. The normal-coordinate construction of Chapter 5 solves its own coordinate equations; it does not compute these Hall-supported residuals. This locates the next comparison without assuming its answer.

7.2 A Pfaffian and its inverse square

Suppose an independently constructed family over a base B has a Pfaffian line $\mathscr{L}_{\mathrm{Pf}}$, section p_X, and squaring map

$$\sigma : \mathscr{L}_{\mathrm{Pf}}^{\otimes 2} \xrightarrow{\sim} \mathscr{L}_{\det}.$$

For a rank-two skew block $\left(\begin{smallmatrix} 0 & u \\ -u & 0 \end{smallmatrix}\right)$, the Pfaffian is u and the determinant is u^2. The square therefore forgets the sign of its root. A comparison with a modular form also needs a parameter map $\kappa : B \to \mathcal{A}_2$ and an isomorphism

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf}} \xrightarrow{\sim} \kappa^*(\mathcal{L}^5 \otimes \nu_{\Delta_5}).$$

Here $\mathcal{L}$ is the Hodge line on the Siegel modular stack and ν_{Δ_5} is its multiplier character.

THEOREM 7.3 *(Conditional square-root consequence).* Assume the complete geometric hypotheses of Theorem 2.1: the reduced orientation and wall data, the finite oriented block comparisons in every active degree with both parities, compatible effective lines and sections, and the stated parameter, convergence and extension conditions. Then, as meromorphic sections over B,

$$\iota_{\mathrm{aut}}(p_X)^{-2} = \kappa^*\Delta_5^{-2} = \kappa^*(\Phi_{10}^{\mathrm{un}})^{-1} = \kappa^* Z_{\mathrm{BPS}}^{K3\times E}.$$

If a protected state theory independently satisfies the sewing identity $Z_X = -4096\, \iota_{\mathrm{aut}}(p_X)^{-2}$, then its scalar equals $-4096\, \kappa^*\Delta_5^{-2}$.

Proof. The Pfaffian comparison gives $\iota_{\mathrm{aut}}(p_X) = \kappa^*\Delta_5$. Squaring and inverting gives the first two equalities; Theorem 4.2 gives the last. The additional sewing identity yields the state-theoretic conclusion. It is not implied by taking the reciprocal of a section. □

The expression $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ abbreviates p_X only when this geometric system has been constructed. A proposed system may be organized as

$$\mathfrak{D}_X^{\mathrm{DI}} = \big\{\big(\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, p_{X,R}, \epsilon_{o,R}, \mathrm{Rec}_R\big)\big\}_R.$$

The entries name, respectively, the chosen factorization refinement and hybrid Ran carrier, charge indices and map, coefficient complexes, orientation, extension operations, primitive pieces, source coalgebra and Koszul comparison, Pfaffian line and section, symmetry action and recognition map. This denotes the pro-system, with its specified transition maps. Each entry has its own construction and compatibility conditions. In particular, a finite root window is not automatically a proper geometric truncation, and a formal height limit need not be an effective family.

The conditions $(D0)$, $(O1)$, $(O1)^+$, $(O2)$ express distinct geometric requirements. A degeneration system must compare reduced coefficients, orientations, their specialization and integration. The reduced orientation must descend through stabilizers, with coherent symmetry lifts. A wall model must compare the actual root action and its local Pfaffian divisor. The finite Pfaffian condition (P_{fin}) further requires oriented block factorizations, units, all active parity dimensions and the leading coefficient $64x^{\rho}$. These conditions control the conditional theorem; they are not existence results for the compact family.

If ϵ_o denotes the resulting root-line character, the reflection comparison asks for

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \widetilde{\nu}_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The rank-one wall calculation determines a sign only after the actual root lift, the section transformation and the local frame comparison have been specified. A ratio of scalar formulas in unrelated frames does not determine that character. Global modular descent also requires the full multiplier and boundary normalization, beyond the reflection subgroup.

An inverse-square scalar is invariant under $p_X \mapsto -p_X$. Likewise, the identity has supertrace one on spaces of dimensions (1|0) and (2|1). These examples exhibit information lost by the two operations. A virtual determinant constructed from Euler superdimensions sees the product exponents, but does not supply brackets, Hopf-pairing radicals, a Pfaffian line or a root of its symmetry action. The automorphic product, its BKM denominator and a geometric Pfaffian meet only after the stated comparisons. In the Gritsenko–Nikulin coordinates the algebraic denominator is $64^{-1}\Delta_5(2Z)$; the change of argument is part of that lattice convention.

Remark 7.4 *(Five numerical invariants).* Let r_{Gram} be the rank of the three-coordinate Gram lattice. Define $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_X)$, let $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}}$ denote the same alternating Hodge sum in the indicated convention, and put $\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Delta_5)$ and $\kappa_{\mathrm{fiber}} = \chi_{\mathrm{top}}(S)$. Then

$$(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, r_{\mathrm{Gram}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24).$$

The two zeroes come from $1 - 1 + 1 - 1$. The modular weight is $f(0,0)/2 = 5$ [9, 51], while $\chi(\mathcal{O}_S) = 2$. Thus an additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_S)$ would assert $5 = 2$. A genus-one Heisenberg bar coefficient instead requires a chosen complex, trace and normalized class; rescaling its trace changes it without changing r_{Gram}. An identification with the Gram rank requires its own map.

8 Further structures

An associative source comparison must keep the carrier and its tensor product fixed. If C_X is a chiral coalgebra, a centre comparison goes through a specified cobar algebra or comodule category. A filtered quasi-isomorphism of underlying complexes $A_b \simeq C_X$ does not compare their multiplications, cooperations or cyclic functionals. Applying chiral Hochschild or boundary-realization constructions [81–83] therefore requires an associative comparison preserving those operations, completion, support and transitions. Lie–Com chiral Koszul duality cannot provide it by a change of notation.

A local mixed holomorphic–topological theory raises another comparison problem. A model on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ specifies both topological and holomorphic directions. Its coefficient and operator complexes must descend, and the quantum master equation and anomaly class must be defined in the actual global deformation complex. Kunneth gives

$$H^1(S \times E, \mathcal{O}_{S\times E}) \cong \mathbb{C}.$$

This nonzero group neither identifies the full obstruction complex nor proves that an obstruction class is nonzero. A quartic invariant such as $\mathrm{Tr}_{\mathrm{ad}}(x^4)$ likewise specifies an anomaly question only after its BV descent and form degrees are fixed. Formal local Hamiltonian data alone do not produce the compact sewing identity.

For a gravitational boundary algebra, the Brown–Henneaux relation $c = 3\ell/(2G_N)$ fixes a Virasoro parameter. With the usual normalization its bracket is

$$[T_\lambda T] = \partial T + 2\lambda T + \frac{c}{12}\lambda^3.$$

A comparison with a K3 Heisenberg construction must identify the actual bracket, state and trace normalization. The Borcherds weight 5 does not determine that central charge. Constructing an acting algebra with determinant section Δ_5^2, proving that its partition function is the reciprocal section, and relating it to p_X remain separate requirements.

8.1 Mathieu and umbral structures

The K3 elliptic genus $2\phi_{0,1}$ has an $N = 4$ superconformal-character decomposition [38]. After the massless terms are separated, its mock-modular series has form

$$\Sigma(\tau) = -2q^{-1/8} + \sum_{n\geq 1} A_n q^{n-1/8}.$$

The Mathieu twining functions of [42] determine class functions, and the module theorem proves that the massive coefficients are actual M_{24}-characters [45]. This character theorem does not supply a simultaneous M_{24}-action on every geometric K3 state model.

Umbral moonshine begins with the six-group construction of [19] and extends to the twenty-three Niemeier lattices with nonempty root systems.[9] The A_1^{24} case gives Mathieu moonshine. A twining construction changes the coefficient data and uses the powers of the group element. For the Mathieu inputs, the full lifts are meromorphic Borcherds products [116]. Chapter 16 gives their lattices, weights and the odd-divisor obstruction to particular square roots. These results use the complete twining and cusp data; the untwined coefficients $f(n, \ell)$ alone do not define the lift. An equivariant denominator algebra or a geometric Hall action requires further constructions.

A compact comparison must identify the coefficient and state theories on their actual supports, then compare the oriented modular lines and sewing functionals. The inverse Igusa series fixes the scalar that these maps must recover.

BIBLIOGRAPHIC NOTES

The primitive numerical identity uses Oberdieck–Pixton [101], Oberdieck–Pandharipande [100], and the reduced DT/PT comparison [99]; direct vertex calculations in DT theory are developed in [15]. The Saito–Kurokawa factorization is described in [118]; Andrianov [2] fixes the full Hecke polynomial and its analytic completion. Their arithmetic statements concern the weight-ten eigenline, independently of the Pfaffian realization. The periods in [32, 71, 85] express rationality of critical values, while the Mellin argument above proves the nonvanishing needed for the pole.

[9] M. Cheng, J. Duncan and J. Harvey, *Umbral moonshine and the Niemeier lattices*, §1.

Chapter 9

Elliptic classes and Euler degeneration

The elliptic characteristic class acts on cohomology by multiplication. Where this multiplication is invertible, it transports geometric correspondence operators by conjugation. At the specialization $y = 1$, however, the class becomes the top Chern class. On a connected positive-dimensional variety with nonzero Euler characteristic, its multiplication has rank one. To understand the specialization of the representation, we must therefore retain the image of multiplication and its position inside cohomology.

We shall identify this image with an equivariant Euler ideal. Its quotient is the cohomology of the projectivized tangent bundle, with the parameter acting by the tautological class. This description turns the orders of vanishing into lengths of nilpotent Jordan blocks.

Surfaces provide the first example. The Hilbert square of a K3 surface exhibits several distinct lengths. We then determine which operators preserve the image. The usual Heisenberg vacuum relation obstructs an extension to the unchanged ambient cohomology at $y = 1$.

1 The characteristic class and its Euler factor

Let X be a smooth projective complex variety of complex dimension $d > 0$. We use rational singular cohomology, with its cup product and complex orientation, and put

$$H_X = H^*(X;\mathbb{Q}), \qquad R_0 = \mathbb{Q}[[\zeta]], \qquad R = \mathbb{Q}[[q,\zeta]], \qquad H_{X,R} = H_X \otimes_{\mathbb{Q}} R.$$

The coefficient variables are even. We retain cohomological parity. Multiplication by an inhomogeneous even class need not preserve the integer cohomological grading. All power series use the indicated formal topology. The variable q records elliptic energy, and $y = e^{\zeta}$ is a formal coordinate near $y = 1$. We write $\chi(X) = \sum_i (-1)^i \dim_{\mathbb{Q}} H^i(X;\mathbb{Q})$ for the topological Euler characteristic.

For a vector bundle V, write $\Lambda_t V = \sum_{j\ge 0} t^j \Lambda^j V$ and $\operatorname{Sym}_t V = \sum_{j\ge 0} t^j \operatorname{Sym}^j V$. The normalized elliptic tensor class is

$$\mathcal{E}_{q,y}(X) = y^{-d/2} \bigotimes_{r\ge 1} \Big(\Lambda_{-yq^{r-1}} T_X^* \otimes \Lambda_{-y^{-1}q^r} T_X \otimes \operatorname{Sym}_{q^r} T_X^* \otimes \operatorname{Sym}_{q^r} T_X \Big).$$

The expression belongs to $K^0(X)_{\mathbb{Q}}[[q,\zeta]]$, where $K^0(X)$ is the Grothendieck group of algebraic vector bundles. The factor $y^{-d/2}$ means $e^{-d\zeta/2}$, also when d is odd. Its cohomological characteristic class and multiplication operator are

$$\epsilon_X(q,e^{\zeta}) = \operatorname{td}(T_X)\operatorname{ch}(\mathcal{E}_{q,e^{\zeta}}(X)), \qquad E_X(a) = \epsilon_X(q,e^{\zeta}) \smile a.$$

The Todd class supplies the Riemann–Roch normalization: integration of ϵ_X gives the holomorphic Euler characteristic of the elliptic tensor class. The operator E_X retains more information than this integral.

Let $x_1, \dots, x_d$ denote formal Chern roots of T_X. Define

$$e_\zeta(T_X) = \prod_{i=1}^{d}(x_i - \zeta) = \sum_{k=0}^{d}(-\zeta)^{d-k}c_k(T_X).$$

This is the equivariant Euler class when fiber scaling has weight $-\zeta$. We shall use the even power series $s(w) = \sinh(w/2)/(w/2)$, whose constant term is one. The factor at energy zero belonging to a single root is

$$e^{-\zeta/2}\frac{x(1-e^{\zeta-x})}{1-e^{-x}} = \frac{x\sinh((x-\zeta)/2)}{\sinh(x/2)} = (x-\zeta)\frac{s(x-\zeta)}{s(x)}. \tag{9.1}$$

The remaining factors for that root form the series

$$\psi(q,\zeta,x) = \prod_{r\geq 1}\frac{(1-e^{\zeta}q^re^{-x})(1-e^{-\zeta}q^re^{x})}{(1-q^re^{-x})(1-q^re^{x})}.$$

THEOREM 1.1. The elliptic class factors as

$$\epsilon_X(q,e^\zeta) = e_\zeta(T_X)U_X(q,\zeta), \qquad U_X \in H^{\times}_{X,R}, \qquad U_X(q,0) = 1. \tag{9.2}$$

The unit is given by

$$U_X(q,\zeta) = \prod_{i=1}^{d}\frac{s(x_i-\zeta)}{s(x_i)}\psi(q,\zeta,x_i). \tag{9.3}$$

In particular, the embedded ideals satisfy

$$\epsilon_X H_{X,R} = e_\zeta(T_X)H_{X,R}. \tag{9.4}$$

The two multiplication complexes $[H_{X,R} \xrightarrow{\epsilon_X} H_{X,R}]$ and $[H_{X,R} \xrightarrow{e_\zeta} H_{X,R}]$ lie in resolution degrees $-1, 0$. Multiplication by U_X on the source and the identity on the target define an isomorphism from the first complex to the second.

Proof. For each power of q, only finitely many factors contribute. Their denominators are q-adic units. Positive cohomological degrees are nilpotent, so substitution of the Chern roots into the series is defined. The factors involving s are units as well. At $\zeta = 0$, every ratio in (9.3) equals one. The splitting principle and (9.1) prove (9.2).

Multiplication by U_X is an automorphism, which proves the equality of ideals. The identity $e_\zeta U_X = \epsilon_X$ gives the chain map. □

Thus the ideal, its quotient, and the multiplication complex are independent of elliptic energy after the displayed identification. The conjugated correspondence operators will still depend on q.

Example 1.2 *(A surface with rationally trivial canonical class).* Let Y be a K3 or Enriques surface. Write $\omega \in H^4(Y;\mathbb{Q})$ for the point class normalized by $\int_Y \omega = 1$. Since $c_1(T_Y) = 0$ rationally and $c_2(T_Y) = \chi(Y)\omega$, we have

$$e_\zeta(T_Y) = \zeta^2 + \chi(Y)\omega.$$

Both surfaces have vanishing odd rational cohomology. On the span of $1, \omega$, multiplication has matrix

$$\begin{pmatrix} \zeta^2 & 0 \\ \chi(Y) & \zeta^2 \end{pmatrix}, \tag{9.5}$$

and on $H^2(Y;\mathbb{Q})$ it is ζ^2 times the identity.

The nonzero rational number $\chi(Y)$ is a unit of R_0. Elementary row and column operations therefore give diagonal entries $1, \zeta^4$ for (9.5), up to units. Consequently

$$\operatorname{coker} E_Y \cong R/(\zeta^4) \oplus \big(R/(\zeta^2)\big)^{b_2(Y)}, \qquad b_2(Y) = \begin{cases} 22, & Y \text{ a K3 surface,} \\ 10, & Y \text{ an Enriques surface.} \end{cases}$$

The length-four summand records the coupling of degree zero to top degree. The rank-one specialization alone does not detect its length.

2 The quotient as a projective bundle

Let $\pi : \mathbb{P}(T_X) \to X$ parametrize one-dimensional subspaces of tangent fibers. Its tautological line is $L = \mathcal{O}(-1) \subset \pi^* T_X$. Put $\lambda = c_1(L)$. This convention determines the sign in the comparison with ζ.

Theorem 2.1. There is a canonical isomorphism of R-superalgebras

$$H_{X,R}/(\epsilon_X(q, e^{\zeta})) \;\cong\; H^*(\mathbb{P}(T_X);\mathbb{Q})[[q]], \qquad a \longmapsto \pi^* a, \quad \zeta \longmapsto \lambda. \tag{9.6}$$

On the right, ζ acts by cup product with the nilpotent class λ.

Proof. The inclusion $L \hookrightarrow \pi^* T_X$ is a nowhere-zero section of $\pi^* T_X \otimes L^*$. Hence its top Chern class vanishes:

$$\sum_{k=0}^{d} (-\lambda)^{d-k} \pi^* c_k(T_X) = 0.$$

The classes $1, \lambda, \dots, \lambda^{d-1}$ restrict to a basis of the cohomology of every fiber $\mathbb{P}^{d-1}$. Leray–Hirsch shows that these classes form a basis over H_X. Since the relation is monic up to sign, it gives the entire presentation

$$H^*(\mathbb{P}(T_X);\mathbb{Q}) \cong H_X[\zeta]/(e_\zeta(T_X)).$$

In this graded presentation, ζ has degree two. The quotient has finite top degree, so ζ is nilpotent. Completing in ζ therefore leaves the quotient unchanged. Extend in q and apply (9.4). □

The same Euler polynomial occurs in zero-section self-intersection for the tangent bundle with fiber weight $-\zeta$. Removing the zero section and quotienting by scalar multiplication gives $\mathbb{P}(T_X)$. Formula (9.6) identifies the elliptic quotient with the cohomology of this geometry. It does not require a choice of Chern roots or a basis of cohomology.

Example 2.2 *(The sign for $\mathbb{P}^2$).* Put $X = \mathbb{P}^2$ and $h = c_1(\mathcal{O}(1))$, so $H_X = \mathbb{Q}[h]/(h^3)$ and $c(T_X) = 1 + 3h + 3h^2$. The relation is

$$\lambda^2 - 3h\lambda + 3h^2 = 0. \tag{9.7}$$

It can also be read directly on the complete flag variety. A tangent line at a line $A \subset \mathbb{C}^3$ determines a plane B containing A, and is $\operatorname{Hom}(A, B/A)$. Thus, if $a = c_1(A)$ and $b = c_1(B/A)$, then $h = -a$ and $\lambda = b - a$.

The flag relations include $a^2 + ab + b^2 = 0$ and $a^2 b + ab^2 = 0$. Substitution gives (9.7). Changing the middle sign leaves $6h\lambda$, which is nonzero since $1, \lambda$ form a basis over H_X. This example detects the convention that is invisible on surfaces with $c_1 = 0$.

3 Elementary divisors and nilpotent lengths

Write $D = \dim_{\mathbb{Q}} H_X$. A matrix over the discrete valuation ring R_0 with nonzero determinant admits invertible row and column operations reducing it to $\operatorname{diag}(\zeta^{\ell_1}, \dots, \zeta^{\ell_D})$, where $0 \le \ell_1 \le \cdots \le \ell_D$. The integers ℓ_i are its Smith exponents. A zero exponent contributes no cokernel. A positive exponent contributes $R_0/(\zeta^{\ell_i})$.

We use this theorem over R_0, then extend the resulting matrices to R. We do not require a Smith theorem over a ring with two parameters.

PROPOSITION 3.1. Multiplication by $e_\zeta(T_X)$ on $H_X \otimes R_0$ is injective, with determinant $(-\zeta)^{dD}$. Its positive Smith exponents are the Jordan-block lengths of cup product by λ on $H^*(\mathbb{P}(T_X); \mathbb{Q})$, and their sum is dD. If X is connected and $\chi(X) \neq 0$, there is one zero Smith exponent and $D - 1$ positive exponents.

Proof. Order a homogeneous basis of H_X by increasing degree. Every positive Chern class raises degree, so the multiplication matrix is triangular with diagonal $(-\zeta)^d$. Its determinant is nonzero in R_0. The adjugate identity proves injectivity.

Each summand $R_0/(\zeta^\ell)$ is a single nilpotent Jordan block for multiplication by ζ. Theorem 2.1 identifies this endomorphism with cup product by λ. Leray–Hirsch gives total dimension dD, which is also the valuation of the determinant.

At $\zeta = 0$ the matrix is multiplication by $c_d(T_X)$. For connected X, let $\omega_X \in H^{2d}(X; \mathbb{Q})$ satisfy $\int_X \omega_X = 1$. The operator vanishes on positive degrees and sends 1 to $\chi(X)\omega_X$. Its rank is one when $\chi(X) \neq 0$, so precisely one Smith exponent is zero. □

Let $C_X = \operatorname{coker} E_X$. Its derived fiber at $\zeta = 0$ means $C_X \otimes^{\mathbf{L}}_R R/(\zeta)$, computed from a free resolution. Theorem 1.1 and the preceding proposition give the resolution $[H_{X,R} \xrightarrow{\epsilon_X} H_{X,R}]$. Thus a block of length $\ell > 0$ has derived fiber

$$[\mathbb{Q}[[q]] \xrightarrow{0} \mathbb{Q}[[q]]]$$

in resolution degrees $-1, 0$. In the connected case with $\chi(X) \neq 0$, each of these degrees has rank $D - 1$. The ordinary fiber and the derived fiber have different information, but neither records the positive lengths without retaining the parameter action.

For $X = \mathbb{P}^2$, the multiplication matrix in the basis $1, h, h^2$ is

$$\begin{pmatrix} \zeta^2 & 0 & 0 \\ -3\zeta & \zeta^2 & 0 \\ 3 & -3\zeta & \zeta^2 \end{pmatrix}.$$

The entries generate the unit ideal, the two-by-two minors generate (ζ^2), and the determinant is ζ^6. The exponents are $0, 2, 4$. Equivalently, cup product by the tautological class on the six-dimensional cohomology of the flag variety has blocks of lengths two and four.

4 The Hilbert square of a K3 surface

Let S be a projective K3 surface and $X = S^{[2]}$ its Hilbert scheme of length-two subschemes. The degree shifts in the Hilbert-scheme cohomology decomposition[94] give, at length two, $\operatorname{Sym}^2 H^*(S)$ together with a second copy of $H^*(S)$ shifted up by two. Therefore the even Betti numbers of X are

$$(b_0, b_2, b_4, b_6, b_8) = (1, 23, 276, 23, 1),$$

and its odd cohomology vanishes. The Hodge shift on the second summand raises both indices by one, so $h^{0,0} = h^{0,2} = h^{0,4} = 1$ and $\chi(\mathcal{O}_X) = 3$.

We use the Beauville–Bogomolov bilinear form q_X on $H^2(X;\mathbb{Q})$, normalized by $\int_X \alpha^4 = 3q_X(\alpha,\alpha)^2$ and $q_X(\alpha,\alpha) > 0$ for an ample class α. The Chern–Fujiki identities give

$$\int_X c_4 = 324, \qquad \int_X c_2\alpha\beta = 30 q_X(\alpha,\beta), \qquad \int_X c_2^2 = 828.$$

The first number also follows from the Betti numbers. The middle identity is the length-two coefficient of the Chern–Fujiki generating formula.[1]

For the last number, Hirzebruch–Riemann–Roch and $c_1 = 0$ give $720\chi(\mathcal{O}_X) = 3\int_X c_2^2 - \int_X c_4$. Nondegeneracy of q_X implies that multiplication by c_2 is an isomorphism $H^2(X) \to H^6(X)$.

Theorem 4.1. For $X = S^{[2]}$, the Smith exponents, with multiplicities, are

$$0^{1}, \qquad 2^{23}, \qquad 4^{276}, \qquad 6^{23}, \qquad 8^{1}. \tag{9.8}$$

At every elliptic energy,

$$C_X \cong (R/(\zeta^2))^{23} \oplus (R/(\zeta^4))^{276} \oplus (R/(\zeta^6))^{23} \oplus R/(\zeta^8).$$

Its derived fiber has rank 323 over $\mathbb{Q}[[q]]$ in each of resolution degrees -1, 0.

Proof. The symplectic form identifies T_X with T_X^*, so $c_1 = c_3 = 0$ over $\mathbb{Q}$. Hence $e_\zeta(T_X) = \zeta^4 + \zeta^2 c_2 + c_4$.

Choose $\omega \in H^8(X;\mathbb{Q})$ with $\int_X \omega = 1$. On the invariant span of 1, c_2, ω, the matrix is

$$\begin{pmatrix} \zeta^4 & 0 & 0 \\ \zeta^2 & \zeta^4 & 0 \\ 324 & 828\zeta^2 & \zeta^4 \end{pmatrix}. \tag{9.9}$$

Its entries generate the unit ideal. All two-by-two minors are divisible by ζ^4, and the last two rows and first two columns have minor $504\zeta^4$. Its determinant is ζ^{12}. The exponents of this block are consequently 0, 4, 8.

Choose a basis $a_1, \dots, a_{23}$ of $H^2(X)$. The classes $c_2 a_i$ form a basis of $H^6(X)$. Each pair $a_i, c_2 a_i$ gives

$$\begin{pmatrix} \zeta^4 & 0 \\ \zeta^2 & \zeta^4 \end{pmatrix},$$

whose exponents are 2, 6.

Finally, the kernel of $\int_X c_2(-)$ on $H^4(X)$ has dimension 275 and is complementary to $\mathbb{Q}c_2$, because $\int_X c_2^2 \neq 0$. Multiplication on each of these directions is ζ^4. These invariant subspaces exhaust cohomology and yield (9.8). The 323 positive exponents sum to $1296 = 4 \cdot 324$. The free resolutions of the positive blocks give the derived fiber, and (9.2) extends the decomposition from R_0 to R. □

[1] Y. Cao, G. Oberdieck and Y. Toda, *Gopakumar–Vafa type invariants of holomorphic symplectic 4-folds*, Theorems 4.1–4.2, https://doi.org/10.1007/s00220-023-04882-8.

There is a geometric description of the same lengths. Write C_i for cup product by $c_i(T_X)$ on H_X. In the projective-bundle basis $1, \lambda, \lambda^2, \lambda^3$, cup product by λ has matrix

$$\begin{pmatrix} 0 & 0 & 0 & -C_4 \\ I & 0 & 0 & 0 \\ 0 & I & 0 & -C_2 \\ 0 & 0 & I & 0 \end{pmatrix}.$$

The last column is the relation $\lambda^4 + c_2\lambda^2 + c_4 = 0$.

The three-dimensional block (9.9) contributes Jordan lengths $4, 8$. Each pair from H^2 and H^6 contributes $2, 6$. Each remaining middle-degree direction contributes 4.

For higher Hilbert schemes, the same projective-bundle theorem applies, but the lengths require the corresponding Chern multiplication operators. The length-zero Hilbert scheme is a point: its elliptic class is one, its cokernel is zero, and its rank-zero tangent projectivization is empty.

5 TRANSPORT OF CORRESPONDENCES

The determinant calculation shows that E_X becomes invertible over $R^{\mathrm{loc}} = R[\zeta^{-1}]$. Indeed, $e_\zeta = (-\zeta)^d(1 + N)$ with N in positive cohomological degree. N is nilpotent, so $(1 + N)^{-1}$ is a finite sum. The factor U_X is already invertible over R.

For smooth projective complex varieties X, Y, a cohomology kernel $K_{YX} \in H^*(Y \times X; \mathbb{Q}) \otimes R$ defines

$$T_K(a) = (p_Y)_*(K_{YX} \smile p_X^* a).$$

Pushforward uses the complex orientation. We compose kernels in the same order as operators: on $Z \times Y \times X$,

$$K_{ZY} * K_{YX} = (p_{ZX})_*(p_{ZY}^* K_{ZY} \smile p_{YX}^* K_{YX}).$$

All projections used for pushforward are proper. These formulas extend R^{loc}-linearly, and the diagonal class is the identity kernel.

PROPOSITION 5.1. The transformed kernel

$$\widetilde{K}_{YX} = p_Y^* \epsilon_Y \smile K_{YX} \smile p_X^* \epsilon_X^{-1}$$

acts by $E_Y T_K E_X^{-1}$. This transformation preserves convolution, identity kernels, and every operator relation over R^{loc}. If the original operators are defined over R, their transforms act over R on the image modules

$$M_X = E_X(H_{X,R}) \subset H_{X,R}.$$

Proof. The projection formula gives the operator identity. In a convolution, the two factors pulled back from the middle variety are $\epsilon_Y^{-1}\epsilon_Y = 1$. They are even, so no sign is introduced. This proves preservation of convolution. Restriction to the diagonal similarly cancels the two factors and preserves its class.

Finally, injectivity of E_X identifies the free module $H_{X,R}$ with its image M_X. Through this isomorphism the transformed action is simply $E_Y T_K E_X^{-1} : M_X \to M_Y$, which is defined over R. □

The image module is free, although its quotient in the ambient module has ζ-torsion. Regularity on the image therefore does not imply regularity on the ambient module. Moreover, multiplication by ϵ_X is a module map. It is generally not a unital algebra map for the original cup product. The proposition is an operator transport statement, with no multiplicative identification of coefficient algebras.

6 Simultaneous preservation and the vacuum

Let $L_0 = R_0^D$ and let $M \subset L_0$ be a submodule of full rank. Choose a Smith basis so that $M = D_\ell L_0$, where $D_\ell = \operatorname{diag}(\zeta^{\ell_1}, \dots, \zeta^{\ell_D})$. The algebra of endomorphisms preserving both modules is

$$\mathscr{O}_\ell = \{T \in \operatorname{End}_{R_0}(L_0) : T(M) \subset M\}.$$

Proposition 6.1. An ambient endomorphism belongs to $\mathscr{O}_\ell$ precisely when

$$T_{ij} \in \zeta^{\max(0,\ell_i-\ell_j)} R_0. \tag{9.10}$$

For $\ell = (0, m)$ with $m > 0$, the algebra $\mathscr{O}_\ell/\zeta\mathscr{O}_\ell$ has dimension four. It has two orthogonal idempotents and two opposite arrows, with both arrow composites zero. Its image acting on $L_0/\zeta L_0$ has dimension three.

Proof. Preservation of M is equivalent to regularity of $D_\ell^{-1} T D_\ell$. Its (i, j) entry is $\zeta^{\ell_j-\ell_i} T_{ij}$. Combining this with ambient regularity gives (9.10).

For two exponents the algebra is

$$\begin{pmatrix} R_0 & R_0 \\ \zeta^m R_0 & R_0 \end{pmatrix}$$

with free basis $e_{11}, e_{22}, e_{12}, \zeta^m e_{21}$. The off-diagonal products are $\zeta^m e_{11}$ and $\zeta^m e_{22}$. Both vanish modulo $\zeta\mathscr{O}_\ell$. The lower arrow has nonzero class in this abstract fiber, but acts as zero on $L_0/\zeta L_0$. The other three basis elements act independently. □

Applied to the Euler image, this condition determines which regular ambient operators descend to the projective-bundle quotient. The unit factorization makes the same calculation valid after extension to R. The distinction between the abstract fiber and its action is essential: a generator may survive in the algebra while its specialized operator vanishes.

For a K3 or Enriques surface Y, put $X_n = Y^{[n]}$ and $H_n = H^*(X_n; \mathbb{Q}) \otimes R$, with empty vacuum $|0\rangle = 1 \in H_0 = R$. Normalize the length-one Nakajima creator and annihilator so that

$$C_1(\omega)|0\rangle = \omega, \qquad [A_1(1), C_1(\omega)]|0\rangle = |0\rangle.$$

The annihilator lowers length and kills the empty vacuum. Here both insertions are even, so the bracket is the ordinary commutator. Let $\widetilde{C}_1$, $\widetilde{A}_1$ denote their transports from Proposition 5.1.

Proposition 6.2. The transported creator and the specified empty vacuum cannot be part of a regular Heisenberg action on the unchanged modules H_n at $\zeta = 0$. The transported original correspondence action is regular on the image modules $M_n = E_{X_n}(H_n)$.

Proof. Since $E_{X_0} = 1$, the transported creator sends the vacuum to

$$\widetilde{C}_1(\omega)|0\rangle = \epsilon_Y \omega = \zeta^2 u(q, \zeta)\omega, \qquad u(q, \zeta) \in R^\times.$$

Only the degree-zero component of the unit U_Y contributes to this product, and $c_2(T_Y)\omega = 0$. Any R-linear annihilator $H_1 \to H_0$ sends this vector into $\zeta^2 R$. The reversed composition on the vacuum is zero, because there is no negative length. The normalized Heisenberg relation would make their difference equal to $1 \in R$, a contradiction. Regularity on M_n follows from Proposition 5.1. □

These results determine the degeneration of the cohomological correspondence representation. A compact critical Hall realization requires an additional comparison with its coefficient complexes, preserving the supported products, vacuum, orientations and reduction maps. Likewise, the formal parameter used here does not construct the modular line over the genus-two moduli stack. The Euler quotient provides a concrete recipient for these comparisons and a constraint on any comparison that retains the ambient cohomology and the vacuum.

Chapter 10

Derived Euler fibers and higher operations

The Euler quotient of Chapter 9 is the cohomology of a projective bundle. Its parameter acts by the first Chern class of the tautological line. Taking the derived fiber at parameter zero recovers the cochain algebra of the unit sphere bundle. The resulting cohomology need not determine its multiplication up to homotopy. For the tangent bundle of a K3 surface, a nonzero triple Massey product records information lost by passing to the cohomology algebra.

We shall construct the comparison with sphere-bundle cochains and compute all the transferred products. We then retain the parameter action on the Euler quotient. Its Jordan blocks give a second family of higher operations, now on a module over an exterior algebra. Finally, split bundle embeddings provide pullback and Gysin maps between the Euler models. These maps exhibit the compatibility a geometric correspondence must possess before it acts on a derived Euler fiber.

Two gradings enter these constructions. In the geometric cochain model, the parameter z has cohomological degree two. In the module calculation, it has resolution degree zero and may retain degree two as a separate internal grading. The unit in Theorem 1.1 of Chapter 9 is generally inhomogeneous in cohomological degree. Its equality of completed Euler and elliptic ideals therefore does not itself give a comparison of graded cochain algebras.

1 The projective-bundle cochain algebra

Let X be a compact smooth manifold and let $E \to X$ be a complex vector bundle of rank $r \geq 1$. Write $\pi : P = \mathbf{P}(E) \to X$ for its bundle of lines and $L = \mathcal{O}_P(-1) \subset \pi^* E$ for the tautological line. Over $\mathbb{R}$ use the de Rham algebra $A_X = \Omega^*(X)$. Over $\mathbb{Q}$ use $A_X = A_{\mathrm{PL}}(X)$, the commutative algebra of rational polynomial forms. Choose closed representatives $c_j(E) \in A_X^{2j}$ of the Chern classes, with $c_0(E) = 1$.

The Euler relation is the homogeneous polynomial

$$e_z(E) = \sum_{j=0}^{r} (-z)^{r-j} c_j(E), \qquad |z| = 2.$$

To impose this relation at the cochain level, adjoin one odd generator:

$$\mathcal{P}_E = A_X[z] \otimes \Lambda(\eta_E), \qquad |\eta_E| = 2r - 1, \qquad D\eta_E = e_z(E), \quad Dz = 0. \tag{10.1}$$

Here $\Lambda(\eta_E)$ is the exterior algebra, and D extends the differential of A_X by the graded Leibniz rule. All differentials in this section raise cohomological degree by one.

An explicit comparison over $\mathbb{R}$ determines the geometric meaning of the generator. Choose Hermitian connections on E and L, normalized by

$$c(\nabla) = \det\left(1 + \frac{i}{2\pi} F_\nabla\right), \qquad \lambda = c_1(\nabla L).$$

The tautological inclusion is a nowhere-zero section of $W = \pi^*E \otimes L^*$. Let ∇_0 preserve the trivial line spanned by this section and be trivial on it. Let ∇_1 be the tensor-product connection. Their top Chern forms satisfy

$$c_r(\nabla_0) = 0, \qquad c_r(\nabla_1) = e_\lambda(\pi^*E).$$

Put $a = \nabla_1 - \nabla_0$ and $\nabla_t = \nabla_0 + ta$. If F_t is the curvature and P_r is the polarized invariant polynomial normalized by $P_r(F, \dots, F) = c_r(\nabla)$, define

$$\Theta_E = r \int_0^1 P_r(a, F_t, \dots, F_t)\, dt.$$

The Bianchi identity gives $\frac{d}{dt} c_r(\nabla_t) = r\, dP_r(a, F_t, \dots, F_t)$. Consequently $d\Theta_E = e_\lambda(\pi^*E)$, and there is a cochain-algebra map

$$\Phi_E : \mathcal{P}_E \longrightarrow \Omega^*(P), \qquad \alpha \longmapsto \pi^*\alpha, \quad z \longmapsto \lambda, \quad \eta_E \longmapsto \Theta_E.$$

This is the Chern–Simons transgression for the tautological section.[1]

THEOREM 1.1. The map Φ_E is a quasi-isomorphism. Over $\mathbb{Q}$, $\mathcal{P}_E$ is a commutative rational cochain model of $\mathbf{P}(E)$, with z representing $c_1(L)$.

Proof. As a complex, $\mathcal{P}_E$ is the extension defined by multiplication by $e_z(E)$ on $A_X[z]$, with the shift in (10.1). This multiplication is injective on $H^*(X)[z]$. Indeed, its leading coefficient is $(-1)^r$, so the highest nonzero coefficient of a polynomial cannot disappear. Zero divisors in $H^*(X)$ do not affect this argument. The long exact sequence of the extension gives

$$H^*(\mathcal{P}_E) = H^*(X)[z]/(e_z(E)).$$

The classes $1, c_1(L), \dots, c_1(L)^{r-1}$ restrict to a basis on each projective fiber. The projective-bundle theorem therefore identifies this quotient with $H^*(P)$ by base pullback and $z \mapsto c_1(L)$. These are exactly the maps induced by Φ_E.

For rational polynomial forms, choose a closed representative of $c_1(L)$. The top Chern relation is exact on P, so choose a primitive as the image of η_E. The same cohomology calculation proves that the resulting rational map is a quasi-isomorphism. If a representative e changes to $e + D\sigma$, the generator change $\eta'_E \mapsto \eta_E + \sigma$ compares the two models. □

We use the polynomial model for the derived construction below. Completion in z means degreewise completion with $|z| = 2$. Its cohomology quotient agrees with the polynomial quotient because $c_1(L)$ is nilpotent on $H^*(P)$. A degree-zero ring $\mathbb{Q}[[q]]$ may be adjoined by completed scalar extension.

2 The unit sphere bundle

Choose a Hermitian metric on E. A unit vector determines its line, so the unit circle bundle of L is the unit sphere bundle $S(E)$. Thus $S(E)$ is the homotopy fiber of the classifying map $P \to BU(1)$ of L. We write $C^*(-;\mathbb{Q})$ for a commutative rational cochain model, and give its parameter map $\mathbb{Q}[z] \to C^*(P;\mathbb{Q})$ by this classifying map.

[1] D. L. Johnson, *Chern–Simons forms on associated bundles, and boundary terms*, math/0601182, Theorem 1.2 and §2.2.

Theorem 2.1. There is a multiplicative derived comparison

$$C^*(\mathbf{P}(E);\mathbb{Q}) \otimes^{\mathbb{L}}_{\mathbb{Q}[z]} \mathbb{Q} \simeq C^*(S(E);\mathbb{Q}), \qquad |z| = 2. \tag{10.2}$$

A cochain model for this fiber is

$$\mathcal{S}_E = A_X \otimes \Lambda(\eta'_E), \qquad |\eta'_E| = 2r-1, \qquad D\eta'_E = c_r(E). \tag{10.3}$$

Proof. Resolve the augmentation of $\mathbb{Q}[z]$ by adjoining ν of degree one with $D\nu = z$. The polynomial

$$Q_E(z) = \frac{e_z(E) - c_r(E)}{z}$$

is closed and has degree $2r-2$. In the resulting relative model set $\eta'_E = \eta_E - \nu Q_E(z)$. Its differential is $c_r(E)$. The remaining factor $\mathbb{Q}[z] \otimes \Lambda(\nu)$ contracts onto its constants. This computes the derived tensor product without inverting the parameter.

The same change of generator has a geometric interpretation. On the circle bundle put $\alpha = \frac{i}{2\pi} A_L$, where A_L is the unitary connection form. With the complex orientation of the circle, $\int_{S^1} \alpha = -1$ and $d\alpha$ is the pullback of λ. If $\rho : S(E) \to P$ is the projection, the form

$$\rho^*\Theta_E - \alpha Q_E(\rho^*\lambda)$$

has differential equal to the pullback of $c_r(E)$ from X.

On a sphere fiber the first term vanishes by dimension. The second is $(-1)^{r+1}\alpha\lambda^{r-1}$, with pullbacks suppressed. Since $\int_{\mathbf{P}^{r-1}} \lambda^{r-1} = (-1)^{r-1}$, its integral on the complex-oriented sphere is -1. It represents a fiber generator with the sign fixed by $D\eta'_E = c_r(E)$. The sphere-bundle filtration, whose fiber cohomology has two rows, now proves the de Rham comparison.

Over $\mathbb{Q}$ the relative model of an oriented odd-sphere bundle adjoins a generator with differential its Euler class. The complex structure makes its orientation local system trivial, and its Euler class is $c_r(E)$. This gives (10.3). Equivalently, the cochain comparison for the homotopy pullback over the simply connected space $BU(1)$ gives (10.2). □

3 The multiplication retained by the fiber

Let X now be a connected smooth projective complex variety of dimension $d > 0$, with $H^{\mathrm{odd}}(X;\mathbb{Q}) = 0$. Put $A = H^*(X;\mathbb{Q})$ and assume that the topological Euler characteristic $\chi = \chi(X)$ is nonzero. Choose $\omega \in A^{2d}$ with $\int_X \omega = 1$, and let $\tau : A \to \mathbb{Q}$ be integration, extended by zero outside degree $2d$. The top Chern class of TX is $\chi\omega$.

A cochain algebra is *formal* if it is connected to its cohomology algebra with zero differential by a chain of quasi-isomorphisms. Compact Kähler formality, including its rational form, permits the replacement of A_X by A.[2]

An exact change in the representative of the Euler class is absorbed by the generator change used above. The sphere bundle $S(TX)$ therefore has the finite cochain model

$$B = A \otimes \Lambda(\eta), \qquad |\eta| = 2d-1, \qquad d\eta = \chi\omega.$$

[2] P. Deligne, P. Griffiths, J. Morgan and D. Sullivan, *Real homotopy theory of Kähler manifolds*, Inventiones Mathematicae **29** (1975), 245–274. See §6, Main Theorem. Rational formality is proved by D. Sullivan, *Infinitesimal computations in topology*, Publications Mathématiques de l'IHÉS **47** (1977), 269–331. See Theorem 12.1.

There is only one contractible pair, namely $\eta \mapsto \chi\omega$. Set $A' = \bigoplus_{0<j<2d} A^j$. Representatives for cohomology form the subspace

$$H = \mathbb{Q}1 \oplus A' \oplus \eta A^{>0} \subset B. \tag{10.4}$$

Let $i : H \to B$ be inclusion and let $p : B \to H$ kill η and ω. Define the degree-minus-one homotopy h by

$$h(\omega) = \eta/\chi, \qquad h = 0 \quad \text{on the other summands.}$$

Then $pi = 1$, $dh + hd = 1 - ip$, and $h^2 = hi = ph = 0$.

The contraction transfers multiplication to higher maps on H. To fix their signs, let sV lower the degree of a graded space V by one. Give the tensor coalgebra $T^c(sV) = \bigoplus_{n\geq 0}(sV)^{\otimes n}$ its deconcatenation coproduct. An uncurved A_∞ structure on V is a degree-one coderivation b with $b(1) = 0$ and $b^2 = 0$. Its Taylor components are the suspended multiplications.

We use

$$b_1(sa) = -s(da), \qquad b_2(sa, sb) = (-1)^{|a|}s(ab), \qquad b_3(sa, sb, sc) = (-1)^{2|a|+|b|}s\,m_3(a, b, c).$$

Extending a Taylor component across preceding suspended inputs carries their Koszul sign. A minimal structure has $b_1 = 0$, and a strict unit acts as the ordinary unit in arity two and gives zero in higher arity.

THEOREM 3.1. The space H has a strictly unital minimal A_∞ structure quasi-isomorphic to B. Its binary product is $p(i(-)i(-))$. The only possibly nonzero higher multiplication is

$$m_3(a, b, c) = \frac{\tau(bc)a - \tau(ab)c}{\chi}\,\eta, \qquad a, b, c \in A'. \tag{10.5}$$

It is zero on all other homogeneous input patterns, and $m_n = 0$ for $n \geq 4$. The suspended components of a quasi-isomorphism $I : H \to B$ are

$$I_1 = sis^{-1}, \qquad I_2(sa, sb) = s\big(\tau(ab)\eta/\chi\big) \quad (a, b \in A'), \qquad I_n = 0 \quad (n \geq 3),$$

with I_2 zero on the remaining input patterns.

Proof. The projected binary product removes exactly the top-class component $\tau(ab)\omega$. Its primitive is $h(ab) = \tau(ab)\eta/\chi$. For $a, b \in A'$ the binary bar equation is therefore

$$-s\,d\big(\tau(ab)\eta/\chi\big) + s(ab) = s\big(ab - \tau(ab)\omega\big).$$

The two placements of this primitive in arity three give $p\big(a\,h(bc) - h(ab)\,c\big)$, which is (10.5). If an input contains η, a prospective contribution either has zero homotopy or contains η^2. Normalization of the contraction gives strict unitality.

In a transferred multiplication tree, every nonzero homotopy edge has output proportional to η. A second homotopy above it vanishes, and the product of two such outputs vanishes. Every tree with at least four leaves has one of these forms. The trees for I_n, $n \geq 3$, vanish for the same reason.

Write b_B, b_H for the two bar coderivations. The remaining terms in $b_B I = I b_H$ are products of two I_2 outputs or applications of I_2 to an m_3 output. Both are zero. The linear term of I is injective, hence so is I by tensor length. The equality $I b_H^2 = b_B^2 I = 0$ gives all the A_∞ identities.

Since I_1 induces the identity on cohomology, I is a quasi-isomorphism.[3] □

[3] B. Keller, *Introduction to A-infinity algebras and modules*, math/9910179, §§3.3–3.6 and 4.2. The displayed contraction fixes the signs here.

For example, the binary product of $a, b \in A'$ is $ab - \tau(ab)\omega$, whereas $a(\eta b) = \eta ab$. The product of two classes containing η is zero. The ternary product measures the difference between using the two possible primitives when a triple product passes through the omitted top class.

4 Nonformality and examples

If even cocycles a, b, c have $ab = dU$ and $bc = dV$, the cocycle $aV - Uc$ defines a triple Massey product. Its indeterminacy consists of the products obtained by changing U and V by closed classes. A nonzero Massey product with zero indeterminacy obstructs formality. The contraction above supplies both primitives and their signs.

Corollary 4.1. Under the hypotheses of Section 3, if $\dim H^2(X;\mathbb{Q}) \geq 2$, the rational cochain algebra of $S(TX)$ is not formal. This applies to every $X = Y^{[n]}$ with $n > 0$ and Y a projective K3 or Enriques surface.

Proof. Choose independent $a, c \in A^2$. Poincaré duality gives $b \in A^{2d-2}$ with $\tau(ab) = 1$ and $\tau(bc) = 0$. Then $ab = \omega = d(\eta/\chi)$ and $bc = 0$, so

$$\langle [a], [b], [c] \rangle = -[c\eta]/\chi \neq 0.$$

The indeterminacy is $[a]H^{2d-1}(B) + H^{2d-1}(B)[c]$, which is zero. Indeed, the base has no odd cohomology and η is not closed. The nonzero class $c\eta$ occurs in (10.4).

The Hilbert schemes in the assertion are smooth projective. In the Hilbert-scheme cohomology formula, a surface class of degree j contributes a generator of degree $j + 2m - 2$ in length m. For these surfaces j is even, and the degree-two surface classes persist in every positive length.

The Euler generating series is

$$\sum_{n \geq 0} \chi(Y^{[n]})\, p^n = \prod_{m \geq 1} (1 - p^m)^{-\chi(Y)}.$$

Its coefficients are positive since $\chi(Y)$ is 24 or 12. Thus all hypotheses hold. The cohomology formula used here is the geometric Heisenberg decomposition[94]. □

For a surface Y satisfying the same hypotheses, write $u_i u_j = Q_{ij}\omega$ in a basis of $H^2(Y;\mathbb{Q})$. The cohomology of $S(TY)$ has dimensions $1, b_2(Y), b_2(Y), 1$ in degrees $0, 2, 5, 7$. Its products include

$$u_i(u_j\eta) = Q_{ij}\omega\eta, \qquad m_3(u_i, u_j, u_k) = \frac{Q_{jk}u_i - Q_{ij}u_k}{\chi(Y)}\,\eta.$$

In a hyperbolic plane take $u^2 = v^2 = 0$, $uv = \omega$, $a = u + v$, and $c = u - v$. Then $a^2 = 2\omega$ and $ac = 0$, giving

$$\langle a, a, c \rangle = -\frac{2}{\chi(Y)}c\eta = \begin{cases} -c\eta/6, & Y \text{ an Enriques surface}, \\ -c\eta/12, & Y \text{ a K3 surface}. \end{cases}$$

Both Massey products have zero indeterminacy.

Example 4.2. The condition $b_2(X) \geq 2$ is sufficient but not necessary. Let $X = \mathrm{Gr}(2, 4)$, the Grassmannian of two-dimensional subspaces of $\mathbb{C}^4$, and let S be its tautological rank-two bundle. Put $h = c_1(S^*)$ and $p = c_2(S^*)$. Its rational cohomology is

$$A = \mathbb{Q}[h, p]/(h^3 - 2hp,\ h^4 - 3h^2p + p^2), \qquad |h| = 2, \quad |p| = 4.$$

The Schubert classes are $1, h, p, h^2 - p, hp, p^2$. The relations are the degree-six and degree-eight terms of $c(S)^{-1}$, because the quotient bundle has rank two. They give $p^2 = h^2 p = \omega$, with $\int_X \omega = 1$, and $h^4 = 2\omega$. The six Schubert cells give $\chi(X) = 6$.

Although $b_2(X) = 1$, take $a = b = p$ and $c = h^2 - p \neq 0$. Then $ab = \omega$ and $bc = 0$. In the sphere model $|\eta| = 7$ and $d\eta = 6\omega$, so

$$\langle [p], [p], [h^2 - p] \rangle = -[(h^2 - p)\eta]/6 \neq 0.$$

Its indeterminacy vanishes because $H^7(B) = 0$. Thus $S(T\,\mathrm{Gr}(2,4))$ is also nonformal.

The general sphere-bundle formality criterion imposes a condition on the full cohomology algebra of the base. A single Betti number cannot replace it.[4]

5 The higher action of the parameter

The preceding products belong to the cochain algebra of a sphere bundle. To retain the action of z on the projective-bundle quotient, we instead take a derived Hom over the parameter ring. Set $R_0 = \mathbb{Q}[[z]]$, with z in resolution degree zero, and let C be a finite-length R_0-module. Multiplication by z is a nilpotent operator N on C. The Koszul dual algebra is $\mathcal{K} = \mathrm{Ext}^*_{R_0}(\mathbb{Q}, \mathbb{Q}) = \Lambda(\xi)$, where ξ has resolution degree one.

A dg $\mathcal{K}$-module representing $\mathrm{RHom}_{R_0}(\mathbb{Q}, C)$ is

$$M_C = C \otimes \Lambda(\xi), \qquad d(a) = \xi N a, \quad d(\xi a) = 0. \tag{10.6}$$

The action of ξ is exterior multiplication. Using the free resolution of $\mathbb{Q}$ with differential $-z$ gives the displayed sign after applying Hom. If z has internal degree two as well, give ξ internal degree -2. Then d raises resolution degree by one and preserves internal degree.

For the cyclic block $C_m = R_0/(z^m)$ choose a basis $e_0, \dots, e_{m-1}$ with $N e_j = e_{j+1}$ and $N e_{m-1} = 0$. The cohomology of M_{C_m} has one generator u in resolution degree zero and one generator v in degree one. A contraction is

$$\begin{aligned} i(u) &= e_{m-1}, & i(v) &= (-1)^{m-1} \xi e_0, \\ h(\xi e_j) &= e_{j-1} \quad (j > 0), & h &= 0 \quad \text{otherwise.} \end{aligned}$$

The projection is $p(e_{m-1}) = u$, $p(\xi e_0) = (-1)^{m-1} v$, and zero on the other basis vectors. It satisfies $dh + hd = 1 - ip$.

For modules use the reduced bar comodule $T^c(s\overline{\mathcal{K}}) \otimes M$, where $\overline{\mathcal{K}} = \mathbb{Q}\xi$ and s now lowers resolution degree by one. An A_∞ module structure on M is a square-zero degree-one differential on this comodule compatible with the algebra bar differential. Here the reduced algebra bar differential is zero.

Fix the left-module sign so that the action term on $(s\xi) \otimes a$ is ξa. The operation m_{j+1} has j algebra inputs, one module input, and resolution degree $1 - j$. The adjective reduced excludes inputs equal to the algebra unit.

THEOREM 5.1. The minimal strictly unital A_∞ module of M_{C_m} has exactly one nonzero reduced operation:

$$m_{m+1}(\underbrace{\xi, \dots, \xi}_{m}, u) = v. \tag{10.7}$$

[4] J. Zhou, *Formality of sphere bundles*, Algebraic & Geometric Topology **25** (2025), 2297–2315, 2304.09594, Theorem 1.2. For a nonzero top Euler class, formality of the total space forces the formal base cohomology to be generated by one element.

All reduced operations on v vanish. A bar comparison with M_{C_m} has components

$$I_j((s\xi)^{\otimes j}, u) = (-1)^j e_{m-1-j} \quad (0 \le j < m), \qquad I_0(v) = (-1)^{m-1}\xi e_0,$$

and all other components zero. In particular, the higher module structure determines m.

Proof. The transferred component with j algebra inputs is $p\xi(-h\xi)^{j-1}i$. Each application of $h\xi$ lowers the basis index by one. Projection first survives at $j = m$, when the sign chosen for v gives (10.7). It vanishes for larger j. Starting from v, the first exterior multiplication already vanishes.

Since $s\xi$ has degree zero, the comparison equation on these words is

$$dI_j + \xi I_{j-1} = \sum_{k=1}^{j} I_{j-k} m_{k+1}.$$

For $j < m$, the two terms on the left cancel. At $j = m$, their sum is $I_0(v)$. Above m, both sides vanish. The minimal bar differential squares to zero since its only nonzero output is v, on which every reduced operation is zero. Strict units give the remaining identities.

The coefficient dual to the Taylor map in (10.7) is z^m. The corresponding two-term free presentation is $[R_0 \xrightarrow{z^m} R_0]$, up to an overall shift. Its only nonzero cohomology is $R_0/(z^m)$. Deleting the higher map would change this module while leaving the two cohomology dimensions of M_{C_m} equal to one. □

For $m = 1$ the operation is the ordinary binary action. For $m > 1$ the binary action on cohomology is zero, and the first nonzero reduced operation has arity $m + 1$. If e_0 has internal degree k, then u has internal degree $k + 2m - 2$ and v has degree $k - 2$. Equation (10.7) preserves that grading.

Apply the theorem to $C = H^*(\mathbf{P}(TX); \mathbb{Q})$, with $N = c_1(L) \cup -$. The Jordan lengths computed in Chapter 9 (Proposition 3.1 and Theorem 4.1) give the following multiplicities:

X	m_3	m_5	m_7	m_9
Enriques surface	10	1	0	0
K3 surface	22	1	0	0
K3 Hilbert square	23	276	23	1

.

For the Hilbert square there are 323 blocks, but their lengths sum to 1296. The two resolution cohomology dimensions each see only 323. The higher module operations distinguish all four lengths. Although their displayed decomposition uses a Jordan basis, the dg module (10.6) is defined directly by the tautological operator.

This construction also retains maps. If $T : C \to C'$ satisfies $TN = N'T$, then $T \otimes 1$ is a strict dg $\mathcal{K}$-module map. Choose contractions and transfer them to the bar modules, writing I, P, H for the resulting maps with $IP - 1 = -dH - Hd$. The transferred map is $F(T) = P'TI$.

Let $T_1 : C_1 \to C_2$ and $T_2 : C_2 \to C_3$ commute with the respective parameter operators. Subscripts on I, P, H label these modules. Then

$$F(T_2)F(T_1) - F(T_2T_1) = dK + Kd, \qquad K = -P_3T_2H_2T_1I_1.$$

This follows by substituting $I_2P_2 - 1 = -dH_2 - H_2d$. The perturbation formulas are finite at each bar length, since each action lowers that length. Thus changes of contraction retain the module maps and their composition homotopies.

For comparison, a free resolution of a cyclic block gives

$$\begin{aligned} \operatorname{Hom}_{R_0}(C_m, C_n) &= z^{\max(n-m,0)} R_0/(z^n), \\ \operatorname{Ext}^1_{R_0}(C_m, C_n) &= R_0/(z^{\min(m,n)}), \qquad \operatorname{Ext}^j_{R_0}(C_m, C_n) = 0 \quad (j > 1). \end{aligned} \tag{10.8}$$

Indeed, applying Hom to $[R_0 \xrightarrow{z^m} R_0] \to C_m$ gives the kernel and cokernel of z^m on C_n.

Finite-length R_0-modules form the extension closure of $\mathbb{Q}$. Derived Hom from $\mathbb{Q}$ sends this generator to the free $\mathcal{K}$-module and identifies their derived endomorphisms. Exact triangles extend the identification to all finite-length modules. The minimal modules therefore retain the groups in (10.8).

6 Split embeddings and Gysin maps

Return to cohomological grading. Let E and F be complex bundles over X, of ranks $r \geq 1$ and $s \geq 0$. Choose Chern representatives from direct-sum connections, so that $e_z(E \oplus F) = e_z(E)e_z(F)$ holds as a cochain identity. For the inclusion $j : \mathbf{P}(E) \hookrightarrow \mathbf{P}(E \oplus F)$, the normal bundle is $L^* \otimes \pi^* F$. Its Euler class is $e_z(F)$ after identifying z with $c_1(L)$.

THEOREM 6.1. The pullback and the degree-$2s$ Gysin map admit the following $A_X[z]$-linear representatives on Euler cochain models:

$$\begin{aligned} j^*(a) &= a, & j^*(\eta_{E\oplus F} b) &= \eta_E e_z(F) b, \\ j_!(a) &= e_z(F) a, & j_!(\eta_E b) &= \eta_{E\oplus F} b. \end{aligned} \tag{10.9}$$

They satisfy the projection formula, the self-intersection formula $j^* j_! = e_z(F)$, and composition for flags of split bundles.

Proof. Write odd generators on the left. For homogeneous b, $D(\eta_E b) = e_z(E)b - \eta_E Db$. Substitution into (10.9) proves the chain-map identities, including coefficients of odd degree or nonzero differential. The same formulas give $j_!(x\, j^* y) = j_!(x)\, y$ and $j^* j_!(x) = e_z(F)x$. For $E \subset E \oplus F \subset E \oplus F \oplus G$, the even factors multiply as $e_z(F)e_z(G)$, while the odd generator is sent to the final odd generator. This proves flag compatibility.

For pullback, retain the fixed connection on $L^* \otimes \pi^* F$. On $\pi^* E \otimes L^*$, choose the connection ∇_0 trivial on the tautological section. Their direct sum preserves the tautological section of the restricted bundle. The top Chern transgression is then $\Theta_E e_\lambda(F)$. This realizes the first line of (10.9) geometrically.

For the Gysin map, first work over the universal base $BU(r) \times BU(s)$. Its rational cohomology is the polynomial algebra in the Chern classes of E and F, all in even degrees. Together with z, call this algebra R. The two projective-bundle models resolve $R/(e_z(E))$ and $R/(e_z(E)e_z(F))$. The projection $L \to \pi^* F$ cuts out $\mathbf{P}(E)$ transversely. Its Euler class gives $j_!(1) = e_z(F)$.

A degree-$2s$ R-linear map from the free Euler resolution of $R/(e_z(E))$ to $R/(e_z(E)e_z(F))$ is determined by its value on 1. Its value on η_E would have odd degree $2(r+s) - 1$, and the target has no odd part. The relation is respected precisely because $e_z(E)e_z(F) = 0$ there. Thus the Gysin class has the representative whose value on 1 is $e_z(F)$. Lifting this representative to the target Euler resolution gives exactly the second line of (10.9).

The Thom construction commutes with pullback of the pair of bundles. Derived base change from the universal, simply connected base therefore gives the same Gysin map over X. This identifies its class in the derived category of $A_X[z]$-modules. □

These formulas apply to a split normal bundle and its Thom class. A Hall correspondence has a deformation complex involving cross derived Hom terms. To use the same construction there, one must compare that complex, its orientation, and its coefficient pullback and pushforward with the Euler models. The split calculation specifies the required compatibilities but does not supply this comparison.

7 Maps of Euler cones and the vacuum

The odd generators above also express a general condition on derived quotient maps. Let $f : A \to B$ and $g : C \to D$ be degree-zero cochain maps. Their cones have underlying spaces $B \oplus A[1]$ and $D \oplus C[1]$, with differentials

$$d_f = \begin{pmatrix} d_B & f \\ 0 & -d_A \end{pmatrix}, \qquad d_g = \begin{pmatrix} d_D & g \\ 0 & -d_C \end{pmatrix}.$$

For multiplication by a homogeneous Euler class of degree $2r$, the source shift $A[-2r]$ makes this convention applicable.

A degree-k map preserving the target subcomplexes B and D has the form

$$T = \begin{pmatrix} b & h \\ 0 & (-1)^k a \end{pmatrix}, \qquad |a| = |b| = k, \quad |h| = k - 1.$$

These maps form a Hom complex with differential $\partial T = d_g T - (-1)^k T d_f$. Direct multiplication gives $\partial^2 = 0$ and the Leibniz rule for composition. In degree zero, closedness says that a and b are chain maps and

$$d_D h + h d_A = bf - ga. \tag{10.10}$$

The off-diagonal entry is therefore the specified homotopy making the coefficient square commute. Composing two closed degree-zero maps gives $h_{21} = b_2 h_1 + h_2 a_1$.

There is a simple obstruction at the empty object. Use only resolution grading for this example, with all coefficient modules in degree zero. The empty Euler map is $f_0 = 1 : R_0 \to R_0$. For a connected positive-dimensional smooth projective Y, the nonempty map is $f_1 = e_z(TY) : H^*(Y)[[z]] \to H^*(Y)[[z]]$. An ordinary creator with $b(1) = 1$ would require $1 = e_z(TY)a(1)$ in (10.10). This is impossible: reduction at $z = 0$ and then to cohomological degree zero sends the right side to zero and the left side to one.

The Euler-transported creator with $b(1) = e_z(TY)$ instead has the strict lift $a(1) = 1$, $h = 0$. Its source cone is nevertheless Cone(1), which is acyclic, so the induced derived map is zero. This does not alter the image-module action constructed in Section 5 of Chapter 9. It shows why that action and an action on the Euler quotient require different vacuum data.

To identify either higher structure with chiral collision operations, one must construct a map from the relevant supported operation complex. It must preserve the intrinsic sphere-fiber products or the parameter module operations on their respective carriers, together with the vacuum comparisons.

Chapter 11

Chiral diagonals and curvature

Cohomological joining uses pullback and Gysin maps of a target diagonal. An operation on chiral states must also account for the momentum fields conjugate to its normal directions. The simplest operator product prevents restriction by a quotient: killing a coordinate would kill the vacuum. A differential can instead remove the coordinate and its conjugates together.

We construct this differential first in an affine chart and then on a normal bundle. Its cohomology retains the tangential chiral de Rham fields. Passing from the normal bundle to the actual formal diagonal introduces curvature. A relative vertex complex retains that formal geometry and its fermionic charge, while a second complex, obtained by finite conjugation, resolves the central Dolbeault algebra. The final calculation records the parameter torsion that appears when the normal contraction is made compatible with Laurent charge. These constructions specify what a chiral version of the cochain maps in Chapter 10 would have to preserve.

1 The normal quartet

Work over $\mathbb{C}$. In one normal coordinate the chiral de Rham system has even fields v, β and odd fields c, b, with singular operator products

$$\beta(z)v(w) \sim \frac{1}{z-w}, \qquad b(z)c(w) \sim \frac{1}{z-w}. \tag{11.1}$$

The variables z, w are insertion variables, distinct from target coordinates such as v.

All other generating singular products vanish, and vertex skew symmetry determines the reversed products. In particular $v(z)\beta(w) \sim -(z-w)^{-1}$.

We use the following weights and gradings:

	v	β	c	b
conformal weight	0	1	0	1
fermionic charge J	0	0	1	-1
cochain degree	0	0	-1	1.

For these fields, cochain degree is $-J$ and its reduction modulo two is their parity. These are not two independent contributions to a tensor-product sign. Later, Dolbeault form degree will be added to cochain degree.

The vertex products $a_{(n)}b$ give the singular part

$$a(z)b(w) \sim \sum_{n\geq 0} \frac{(a_{(n)}b)(w)}{(z-w)^{n+1}}.$$

Colons denote normal ordering, and T denotes vertex translation.

The local systems (II.1), with the chiral coordinate-change laws, define the sheaf Ω^{ch}_Y of vertex superalgebras on a smooth complex manifold Y.[1] Here it has zero differential. In particular, its usual chiral de Rham differential is not included.

Suppose that a unital vertex homomorphism ϕ kills the normal coordinate. The relation $\beta_{(0)}v = 1$ would give

$$1 = \phi(\beta_{(0)}v) = \phi(\beta)_{(0)}\phi(v) = 0.$$

Thus a chiral restriction cannot be this quotient.

Define instead

$$\delta = (:vb:)_{(0)}, \qquad \delta c = v, \quad \delta\beta = -b, \quad \delta v = \delta b = 0. \tag{II.2}$$

Every zero mode is a derivation of every vertex product and commutes with T. The current $:vb:$ has regular self-product, hence $\delta^2 = 0$. The sign on the momentum field is forced by

$$\delta(\beta_{(0)}c) = (-b)_{(0)}c + \beta_{(0)}v = -1 + 1 = 0.$$

The operator δ raises cochain degree by one, preserves conformal weight, and lowers J by one.

We use two carriers. The polynomial carrier consists of finite normally ordered states with polynomial normal coordinates.

For the formal carrier, complete the coefficient functions in the undifferentiated normal coordinates at each fixed conformal weight. Then take the direct sum over conformal weights. A state consequently has bounded conformal weight. Positive-weight creation variables remain polynomial, with bounded multiplicities in each weight. Singular products differentiate coefficient functions only finitely many times for a fixed pair of states, and are continuous in this topology.

THEOREM 1.1. Tensor a tangential chiral de Rham algebra with finitely many normal quartets (II.1). Give the tensor product the sum of the normal differentials (II.2), and give the tangential factor zero differential and cochain grading $-J$. The tangential inclusion is a quasi-isomorphism of differential graded vertex superalgebras on both the polynomial and the fixed-weight formal carriers.

Proof. Use the Poincaré–Birkhoff–Witt (PBW) state presentation, whose underlying vector space has polynomial and exterior creation variables. Write v_j, β_j, c_j, b_j for the creation variables corresponding to the normalized derivatives $T^j/j!$ of the four fields, $j \geq 0$.

On one normal factor,

$$\delta = \sum_{j\geq 0}(v_j\partial_{c_j} - b_j\partial_{\beta_j}), \qquad h_0 = \sum_{j\geq 0}(c_j\partial_{v_j} - \beta_j\partial_{b_j}).$$

Fermionic derivatives are left derivatives. The anticommutator $\delta h_0 + h_0\delta$ is a derivation of the PBW polynomial and exterior algebra and is the identity on every generator. It is therefore the total normal-number operator N. For several normal factors, sum the two operators over the factors.

On positive normal number put $h = h_0/N$, and put $h = 0$ on the normal vacuum. Let i include that vacuum and let π extract its coefficient. Then

$$\delta h + h\delta = 1 - i\pi. \tag{II.3}$$

[1] F. Malikov, V. Schechtman and A. Vaintrob, *Chiral de Rham complex*, Sections 2–3. The coordinate transformations and their composition are in Sections 3.6–3.9.

Each polynomial involves finitely many modes.

On formal coefficient functions the same identity holds coefficientwise. Differentiation lowers adic order by at most one. Division by a positive integer preserves the topology. Thus the normal cohomology is its scalar vacuum.

Tensoring this contraction with the tangential factor proves the cohomology assertion. The inclusion preserves the vacuum, translation and all vertex products, so it is the asserted vertex quasi-isomorphism. □

The projection π is a map of complexes. The equation $\beta_{(0)}v = 1$ prevents it from being a vertex homomorphism. The homotopy is likewise a linear operator, not a vertex derivation. These distinctions allow the contraction to exist without contradicting the obstruction to a quotient.

The weight bound is essential for locality. An unrestricted oscillator completion would admit both $\sum_{m\geq 0} : \beta^m :$ and $\sum_{m\geq 0} : v^m :$. Their complete contractions include $m!(z-w)^{-m}$ for arbitrarily large m, so no finite locality order would remain. The first series has unbounded conformal weight and is excluded.

Finally, ∂ does not preserve an individual Laurent coefficient of the elliptic state, because it lowers fermionic charge. It removes only normal fields. Tangential states of positive conformal weight survive. A differential contracting all positive conformal weights has a different cohomology.

2 Affine diagonals and regrouping

Consider $r \geq 1$ copies of one affine target coordinate, with field columns γ, c, β, b. Let $\mathbf{1}$ be the column of r ones and set

$$P_r = I_r - \frac{1}{r}\mathbf{1}\mathbf{1}^{\mathrm{t}}, \qquad \delta_r = \left(: \sum_{i,j} (P_r)_{ij}\gamma_i b_j : \right)_{(0)}. \tag{11.4}$$

The fields tangent to the full diagonal are

$$\bar{\gamma} = \frac{1}{r}\sum_i \gamma_i, \qquad \bar{c} = \frac{1}{r}\sum_i c_i, \qquad \bar{\beta} = \sum_i \beta_i, \qquad \bar{b} = \sum_i b_i.$$

Their two nonzero generating products have coefficient one, as in (11.1). The complementary fields form $r-1$ normal quartets. On the original columns, $\delta_r c = P_r\gamma$ and $\delta_r\beta = -P_r b$. Theorem 1.1 contracts their total normal number.

Proposition 2.1. Successive affine diagonal operations associated to any partition and refinement of the r factors give the differential δ_r. Their changes of normal basis compose and preserve the tangential fields and all vertex products.

Proof. For a set partition $\mathcal{P}$, let $Q_{ij} = 1/|B|$ when i, j belong to the same block $B \in \mathcal{P}$, and let $Q_{ij} = 0$ otherwise. It is the orthogonal projector onto block-constant vectors. There is an orthogonal decomposition

$$P_r = (I_r - Q) + \left(Q - \frac{1}{r}\mathbf{1}\mathbf{1}^{\mathrm{t}}\right).$$

The first term records the normal directions within each block. The second records the differences of block means, with the block-size weights in their induced pairing. Their currents sum to the current in (11.4). Repeated refinement decomposes the same projector.

A normal-basis change sends v, c by an invertible matrix and β, b by its inverse transpose. It preserves the two pairings in (11.1) and the current $\sum_a : v^a b_a :$. Composition of these transformations is matrix multiplication, proving the compatibility of all successive regroupings. □

This calculation concerns the diagonal in a product of affine target charts. To apply it to a curved target, one must first understand its behavior under changes of normal frame.

3 The absolute normal-bundle complex

Let $E \to Y$ be a holomorphic vector bundle, and let $M = \mathrm{Tot}(E)$. Locally write x^i for base coordinates and v^a for fiber coordinates. There is a chiral de Rham quartet for every base and fiber coordinate. Let $\mathcal{V}_E$ be the sheaf over Y of states on M polynomial in the fiber directions, or its fixed-weight completion along the zero section.

Give every c, including each c^i_x, degree -1, and every b degree 1. This total fermionic grading respects a varying frame: under $v' = g(x)v$ one has

$$c'_v = g(x)c_v + (dg)_x(v)c_x. \tag{II.5}$$

A degree assigned only to normal fermions would not respect (II.5).

The fiber Euler vector field $\mathsf{E} = v^a \partial_{v^a}$ is intrinsic. Chiral contraction with a vector field is compatible with the chiral de Rham coordinate laws. Consequently

$$\delta = (\iota_{\mathsf{E}})_{(0)}, \qquad \iota_{\mathsf{E}} = \sum_a : v^a b_a : \tag{II.6}$$

is a global differential on $\mathcal{V}_E$. In a local frame it has the normal action (II.2) and kills tangential fields.

The following homotopy controls the nonconstant part of a frame change. For a matrix $A(x)$, write $W = A^b_a(x) v^a \partial_{v^b}$ and put

$$H_A =: A^b_a c^a_v \beta_b : - : \partial_i A^b_a \, c^i_x c^a_v b_b : . \tag{II.7}$$

The chiral Lie derivative along W, denoted by $\mathcal{L}_W$, is the infinitesimal action of its coordinate change.

LEMMA 3.1. The zero mode of (II.7) satisfies $[\delta, (H_A)_{(0)}] = \mathcal{L}_W$.

Proof. Applying the odd derivation δ gives the current

$$: A^b_a v^a \beta_b : + : A^b_a c^a_v b_b : + : \partial_i A^b_a \, v^a c^i_x b_b : .$$

The last sign follows from $\delta(c^i_x c^a_v b_b) = -c^i_x v^a b_b$. The three terms are the vector-field term and the two fermionic terms in the chiral Lie derivative of W. Taking zero modes proves the identity. □

THEOREM 3.2. The cohomology vertex sheaf of $(\mathcal{V}_E, \delta)$ is canonically Ω^{ch}_Y, with zero differential and cochain grading equal to minus its fermionic charge. This holds for polynomial normal states and for the fixed-weight formal completion.

Proof. In each frame, Theorem 1.1 identifies the cohomology with the tangential fields. It remains to compare these identifications. After shrinking around a base point, a varying frame matrix is a constant matrix times $\exp A(x)$, using the holomorphic matrix logarithm near the identity. A constant frame matrix fixes the tangential fields and the normal vacuum.

Let F_s be the chiral frame action of $\exp(sA)$. It commutes with δ because $[\mathsf{E}, W] = 0$, and $\frac{d}{ds} F_s = \mathcal{L}_W F_s$. Lemma 3.1 gives

$$F_1 - 1 = \delta K + K\delta, \qquad K = \int_0^1 (H_A)_{(0)} F_s \, ds.$$

Thus every vertical frame change induces the identity on the local tangential cohomology. A change of base coordinates induces the base chiral de Rham transformation. These identifications satisfy the cocycle identity and preserve vertex products, giving the canonical cohomology sheaf. □

The theorem constructs a canonical cohomology sheaf without requiring a global strict inclusion of tangential vertex fields. For the full diagonal in Y^r, its linear normal bundle is $T_Y \otimes (\mathbb{C}^r/\mathbb{C}_{\mathrm{diag}})$. The theorem applies to this bundle, and the decompositions used in Proposition 2.1 give its compatible linear regroupings. The next issue is comparison with the formal diagonal itself.

4 Curvature of the formal diagonal

Write two nearby target points as $(x, x+v)$. Under $x' = f(x)$, their difference becomes

$$v' = f(x+v) - f(x) = df_x(v) + \tfrac{1}{2} d^2 f_x(v,v) + O(v^3). \tag{11.8}$$

The quadratic term is the first discrepancy from the linear normal transition.

Proposition 4.1. Let Y be a K3 or Enriques surface. Its formal diagonal has no holomorphic linearization over the first projection inducing the identity on the normal bundle.

Proof. The quadratic term of such a linearization, written as an exponential, would have the form

$$\mathrm{Exp}_x(v) = x + v - \tfrac{1}{2}\Gamma_x(v,v) + O(v^3)$$

with Γ_x symmetric. Comparing $f(\mathrm{Exp}_x(v))$ and $\mathrm{Exp}_{f(x)}(df_x(v))$ gives

$$\Gamma_{f(x)}(df_x(v), df_x(v)) = df_x \Gamma_x(v,v) - d^2 f_x(v,v).$$

This is the transformation law of a torsion-free holomorphic connection on T_Y.

Extend that connection by the bundle's $\bar\partial$ operator. Its curvature has type $(2,0)$, so the Chern–Weil form representing c_2 has type $(4,0)$ and vanishes on a complex surface. This would give $c_2(T_Y) = 0$, contrary to $\int_Y c_2(T_Y) = 24$ for a K3 surface and 12 for an Enriques surface. □

Equivalently, the quadratic discrepancies in (11.8) represent the symmetric Atiyah class of T_Y, which obstructs a holomorphic connection. This obstruction leaves the ordinary diagonal kernel intact. The sheaf $\mathcal{O}_\Delta$ is perfect because the local coordinate differences are a regular sequence. For a unital affine coordinate algebra B, the bar complex has terms $B^{\otimes(n+2)}$, outer bimodule action and alternating adjacent multiplication. It resolves B as a B-bimodule functorially. A chiral comparison must retain the additional coordinate-change data of its normal fields.

5 A relative lift retaining the formal neighborhood

Assume that Y is Kähler. Choose a Kähler connection and its smooth formal exponential. The transported Dolbeault algebra of the formal diagonal over the first projection is

$$\Big(\mathcal{A}_Y^{0,*}(\widehat{\mathrm{Sym}} T_Y^*), D_B\Big), \qquad D_B = \bar\partial + R, \qquad R = \sum_{n\geq 2} R_n. \tag{11.9}$$

Here $\mathcal{A}_Y^{0,*}$ is the Dolbeault algebra of smooth $(0,*)$-forms. Each R_n is a $(0,1)$-form with values in homogeneous vertical polynomial vector fields of degree n.

We define R_n as the homogeneous Taylor coefficient of this transported differential. This fixes the factorial convention.

The formal-exponential theorem gives

$$\bar\partial R + \tfrac{1}{2}[R, R] = 0, \tag{II.10}$$

and R_2 represents the Atiyah class.[2]

Define the relative formal bc–$\beta\gamma$ system $\mathcal{V}_{\mathrm{rel}}$ of T_Y as follows. Its normal fields have products (II.1), its coefficients $\mathcal{A}_Y^{0,*}$ are central vertex elements with $T = 0$, and it has no base momentum fields. Linear frames act on v, c and contragrediently on β, b. Complete in the normal coordinates at each conformal weight, as in Section I.

Total cochain degree is Dolbeault degree minus J. Forms are placed to the left of the fiber fields. The tensor product uses total degree for its signs.

For a vertical vector field $V = V^i(v)\,\partial_{v^i}$, its chiral Lie action is

$$\mathcal{L}_V = \Big(: V^i(v)\beta_i : + : \partial_j V^i(v) c^j b_i :\Big)_{(0)}. \tag{II.11}$$

It acts by V on functions and by $c^i \mapsto \partial_j V^i c^j$ on the coordinate fermions. This action is the derivative of the formal-coordinate action in the chiral de Rham construction. Its bosonic and fermionic anomalies cancel, giving the exact identity

$$[\mathcal{L}_V, \mathcal{L}_W] = \mathcal{L}_{[V,W]}. \tag{II.12}$$

We extend it to form-valued vector fields with the tensor-product signs.

THEOREM 5.1. The operator

$$D_{\mathrm{form}} = \bar\partial + \mathcal{L}_R \tag{II.13}$$

is a global square-zero derivation of $\mathcal{V}_{\mathrm{rel}}$. It preserves conformal weight and fermionic charge, and restricts to D_B on fiber functions. Changes of formal exponential induce chiral coordinate isomorphisms intertwining these differentials, with the coordinate composition law.

Proof. Equations (II.10) and (II.12) give

$$D_{\mathrm{form}}^2 = \mathcal{L}_{\bar\partial R + [R,R]/2} = 0.$$

The current in (II.11) has weight one and charge zero. Its zero mode is a derivation, preserves these gradings, and acts by V on functions.

Under a formal coordinate change, the chiral de Rham transformation transports the Lie action. For a base-dependent change, its Dolbeault derivative supplies the extra term that also appears when conjugating $\bar\partial + R$ on functions. Hence the chiral transformation intertwines (II.13). Composition of the coordinate transformations gives the cocycle identity.

At fixed conformal weight, a state has boundedly many momentum fields. Only boundedly many contractions can therefore lower the normal order of a coefficient. The sum of homogeneous terms is continuous in the specified formal completion. Thus these local formulas define the global formal operator. □

[2] S. Yu, *The Dolbeault dga of the formal neighborhood of the diagonal*, Theorem 3.2 and Section 3. The homogeneous polynomial convention in (II.9) is obtained from its symmetric-tensor Taylor convention.

The relative Euler contraction δ of (11.6) is still defined on this carrier. Its commutator with the new differential measures the curvature error. For a form-valued vertical field V, write $\iota_V =: V^i b_i :$.

Proposition 5.2. With the total graded commutator,

$$[D_{\mathrm{form}}, \delta] = -\sum_{n\geq 2} (n-1)(\iota_{R_n})_{(0)}. \tag{11.14}$$

In particular, $D_{\mathrm{form}} + \delta$ need not square to zero.

Proof. The chiral Cartan identity gives $[\mathcal{L}_V, \iota_{\mathsf{E},(0)}] = \iota_{[V,\mathsf{E}],(0)}$. For a homogeneous degree-n vector field, $[V, \mathsf{E}] = -(n-1)V$. The holomorphic Euler contraction commutes with $\bar\partial$ with the total sign. Extending the Cartan identity to forms yields (11.14). □

6 A conjugated Koszul resolution

The Euler contraction can also be used to construct a global resolution of the central base. For a degree-n vertical field put

$$H_{V_n} = \frac{1}{n} : \partial_j V_n^i(v) c^j \beta_i : . \tag{11.15}$$

The odd derivation rule and the homogeneous Euler identity give

$$\delta H_{V_n} = K_{V_n} =: V_n^i \beta_i : + \frac{1}{n} : \partial_j V_n^i c^j b_i : .$$

The fermionic coefficient here is $1/n$, whereas the Lie action (11.11) has coefficient one.

Apply (11.15) to the tensors R_n and let $\mathscr{S} = \sum_n H_{R_n}$, with forms placed first. Its form degree is one and its fiber cochain degree is -1. Thus $S_0 = \mathscr{S}_{(0)}$ is an even derivation of total degree zero. Define

$$d_0 = \bar\partial + \delta, \qquad D_K = e^{S_0} d_0 e^{-S_0}. \tag{11.16}$$

On a complex d-fold, $S_0^{d+1} = 0$ by form degree, so both exponentials are finite.

Theorem 6.1. The operator D_K is a global square-zero vertex derivation. The central inclusion

$$i : (\mathcal{A}_Y^{0,*}, \bar\partial) \longrightarrow (\mathcal{V}_{\mathrm{rel}}, D_K)$$

is a quasi-isomorphism. For choices S_0, S_0' on this carrier, the automorphism $e^{S_0'} e^{-S_0}$ intertwines their resolutions. These intertwiners compose exactly.

Proof. The currents (11.15) are tensorial under relative linear frame changes. Their zero modes are global derivations, and finite exponentiation gives vertex automorphisms fixing the central base. Conjugation therefore proves the square-zero and derivation assertions.

The normal homotopy (11.3) is equivariant under linear frame changes with central coefficients. It defines a global relative homotopy h and vacuum projection π. Extend h to forms with the tensor-product sign. Then $\bar\partial h + h\bar\partial = 0$ and $d_0 h + h d_0 = 1 - i\pi$.

Transporting the contraction gives

$$h_K = e^{S_0} h e^{-S_0}, \qquad \pi_K = \pi e^{-S_0}, \qquad D_K h_K + h_K D_K = 1 - i\pi_K.$$

This proves the quasi-isomorphism of complexes, and the central inclusion is a vertex morphism. The projection and the homotopy are only required to be linear maps of the indicated degrees. Finally, cancellation of adjacent exponentials proves both the intertwining identity and composition of the changes of choice. □

The square-zero argument uses homogeneous form-valued tensors to define $\mathscr{S}$, but does not require their Maurer–Cartan equation. That equation is needed for the geometric interpretation of D_{form}, which restricts to the formal-neighborhood Dolbeault differential on functions. Theorem 6.1 compares $(\mathcal{V}_{\mathrm{rel}}, D_K)$ with the central Dolbeault algebra by a quasi-isomorphism. It does not identify the two differentials.

7 The second correction on a surface

On a surface the conjugation has only two nontrivial form degrees. Write $K_1 = \sum_n K_{R_n}$ as a form-valued current. The total sign is $d_0\mathscr{S} = \bar\partial\mathscr{S} - K_1$. Expanding (11.16) gives

$$D_K = d_0 + (K_1)_{(0)} - (\bar\partial\mathscr{S})_{(0)} + \tfrac{1}{2}[S_0, (K_1)_{(0)}].$$

All further terms have form degree at least three.

The first-form action on functions is R. The second-form term can contain fermionic fields of negative cochain degree.

Example 7.1. Take fiber coordinates v_1, v_2 and exterior form variables η_1, η_2, placed to the left of fiber fermions. Set

$$V = v_1^2\partial_{v_2}, \qquad W = v_2^2\partial_{v_1}, \qquad H_V =: v_1c_1\beta_2 :, \qquad H_W =: v_2c_2\beta_1 :,$$

and suppress $\bar\partial$. For $S_0 = \eta_1(H_V)_{(0)} + \eta_2(H_W)_{(0)}$, put $A_1 = [S_0, \partial]$ and $A_2 = \frac{1}{2}[S_0, A_1]$. Then

$$A_1(v_1) = \eta_2 v_2^2, \qquad A_1(v_2) = \eta_1 v_1^2,$$

while

$$A_2(v_1) = -\eta_1\eta_2\left(\tfrac{3}{2}v_1v_2c_1 + \tfrac{1}{2}v_1^2c_2\right),$$
$$A_2(v_2) = \eta_1\eta_2\left(\tfrac{1}{2}v_2^2c_1 + \tfrac{3}{2}v_1v_2c_2\right).$$

Indeed $S_0(v_1) = \eta_2v_2c_2$, and the two compositions are

$$S_0A_1(v_1) = -2\eta_1\eta_2v_1v_2c_1, \qquad A_1S_0(v_1) = \eta_1\eta_2(v_1v_2c_1 + v_1^2c_2).$$

Their half-difference gives the first formula. Interchanging the two indices gives the second. Each action on these coordinate polynomials has only a single contraction, so these are identities of zero modes.

Thus the algebra of fiber functions need not be a subcomplex of D_K. The example uses specified polynomial tensors. They are not assumed to satisfy (11.10) or to arise from a K3 metric. They show that the second correction can be nonzero for arbitrary tensors.

8 Charge restoration and parameter torsion

Introduce an even parameter u of fermionic charge $+1$ and cochain degree zero. The combined charge includes its exponent. Cochain degree still uses the original field grading and Dolbeault degree. In particular, charge is no longer parity after adjoining u. The differential $u\partial$ preserves combined charge.

Let K be a finite normal PBW subcomplex stable under ∂ and the contraction h, with cohomology consisting of its scalar vacuum in degree zero. Write $B^k = \mathrm{im}(\partial : K^{k-1} \to K^k)$.

Proposition 8.1. There are vector-space splittings giving

$$H^k(K[u], u\partial) \cong B^k \otimes \mathbb{C}[u]/(u) \oplus \begin{cases} \mathbb{C}[u]|0\rangle, & k = 0, \\ 0, & k \neq 0. \end{cases} \tag{11.17}$$

The derived fiber at $u = 0$ is $(K, 0)$. The same statements hold coefficientwise in the fixed-weight formal completion, with coefficientwise completed base change.

Proof. Multiplication by u is injective on $K[u]$, so the cycles of $u\partial$ are $(\ker \partial)[u]$ and its boundaries are $uB[u]$. Split $\ker \partial$ into B and the scalar vacuum to obtain (11.17). The finite complex is termwise free over $\mathbb{C}[u]$, so derived base change sets its differential to zero. In the fixed-weight completion the normal-number contraction gives the same continuous splitting coefficientwise. Completed base change has the asserted zero differential on each coefficient. □

For two normal directions, keep no derivative jets and truncate total normal number at three. There are four bosonic and four fermionic variables. Their normal-number series and its first four coefficients are

$$\frac{(1+t)^4}{(1-t)^4}, \qquad 1 + 8 + 32 + 88 = 129.$$

The contraction leaves one vacuum and 64 acyclic pairs. After scaling the differential there are therefore 64 torsion cohomology generators, besides the free vacuum. The terms of the parameter complex are free even though its cohomology is not flat.

For the curved resolution, charge restoration begins over $\mathbb{C}[u, u^{-1}]$. The finite conjugation is

$$e^{S_0/u}(\bar\partial + u\partial)e^{-S_0/u} = \bar\partial + u\partial + (K_1)_{(0)} + u^{-1}K_2 + \cdots .$$

On a surface the last term is

$$K_2 = -(\bar\partial \mathscr{S})_{(0)} + \tfrac{1}{2}[S_0, (K_1)_{(0)}].$$

A correction of form degree r has charge $r - 1$, so its coefficient is u^{1-r}. A regular extension on the unchanged carrier requires these negative-power coefficients to vanish, or to acquire the required divisibility in a parameter-dependent family. Example 7.1 exhibits a nonzero K_2 for polynomial tensors. Its value for a specified K3 curvature is a separate geometric calculation.

9 The diagonal transfer still to be constructed

For a compact complex surface Y and $N \geq 0$, the geometric state object at length N is

$$\mathcal{H}_N(Y) = \bigoplus_{[g] \subset \mathfrak{S}_N} R\Gamma\big(Y^N, \Omega^{\mathrm{ch},g}_{Y^N}\big)^{C(g)}. \tag{11.18}$$

Here $C(g)$ is the centralizer of g. The sheaf $\Omega^{\mathrm{ch},g}_{Y^N}$ consists of twisted chiral de Rham modules on the fixed locus $(Y^N)^g$. We view it as a sheaf on Y^N.

Its coordinate and centralizer actions are those of the twisted-module construction.[3]

[3] E. Frenkel and M. Szczesny, *Chiral de Rham complex and orbifolds*, Sections 4.8–4.10, Theorems 4.2–4.3. The index is treated in Section 6.

The derived global sections in (11.18) use only the sheaf-resolution differential. Finite-group invariants are exact over $\mathbb{C}$. We use individual length and conformal-weight components. Their generating series are taken coefficientwise.

The absolute Euler complex has cohomology Ω^{ch}_Y. The relative formal-diagonal complex retains the transported Dolbeault algebra on functions and the normal fields. The conjugated Koszul complex resolves the central base. Each construction therefore has a specified use in a comparison. The supported diagonal transfer on (11.18) remains to be constructed.

Let S be a K3 surface and E an elliptic curve. For the threefold $S \times E$, the zero object in the relative translation quotient lies over BE, the classifying stack of E. The comparison must retain that relative unit and compatible orientation square roots along extension flags.

It must also identify the reduced state with a specified geometric module coefficient. Let H, R be unital algebras and $I_0 : H \to R$ a unital homomorphism. A reduced functional $D : H \to R$ satisfying

$$D(ab) = I_0(a)D(b) + D(a)I_0(b), \qquad D(1) = 0$$

defines the unital action

$$a \longmapsto \begin{pmatrix} I_0(a) & D(a) \\ 0 & I_0(a) \end{pmatrix}$$

on the right R-module $R \oplus R$. Multiplicativity is exactly the displayed derivation identity. Thus D is its off-diagonal coefficient. Its unit data differ from an empty-vacuum matrix coefficient of value one.

A chiral McKay comparison with the Hilbert-scheme state complex $R\Gamma(Y^{[N]}, \Omega^{\mathrm{ch}}_{Y^{[N]}})$ must likewise identify the complexes and their operations beyond their equal elliptic indices.

Question 9.1. Construct a diagonal transfer on the actual chiral state sheaves, through defects or bimodules preserving the prescribed Laurent charge. Construct the compact reduced critical Hall coefficient functor, with its oriented extension pullbacks and supported pushforwards, and compare its diagonal operation with this transfer. Can the comparison preserve the curvature corrections, derived specialization, supported products, vacuum, and the common elliptic torsor of the reduction?

The next coherence can be stated once its supports and coefficients are fixed. Let (B_A, b_A) and (B_B, b_B) be specified supported bar coalgebras with degree-one coderivations squaring to zero. Suppose that the degree-zero components $F_1, \dots, F_{n-1}$ satisfy the comparison equations through arity $n-1$. Let ∂ be the differential on the arity-n Hom complex induced by the unary coderivation components. The residual is

$$\mathfrak{r}_n = (b_B F_{<n} - F_{<n} b_A)_n, \qquad \partial F_n = -\mathfrak{r}_n.$$

The preceding equations imply $\partial \mathfrak{r}_n = 0$.

Suppose that this Hom complex contracts onto chosen cohomology representatives, with projection P and homotopy h. Then $\partial h + h\partial = 1 - P$, so $F_n = -h\mathfrak{r}_n$ solves the equation when $P\mathfrak{r}_n = 0$. The normal contraction in (11.3) computes its own vertex complex. Applying this criterion to compact Hall operations requires their supported Hom complex and the value of its residual.

Chapter 12

Elliptic Thom lines and derived incidence

The elliptic class of Chapter 9 transports cohomological correspondences. A sheaf-valued coefficient requires the line carrying that class and the map defined by its section. Already for two interacting colors, the line has nonzero degree. Its section vanishes twice at collision. A change of line objects can remove a tensor twist while preserving this doubled zero.

We follow the pair coefficient into its simultaneous derived intersections. A spanning tree reduces each connected graph to a finite algebra. The first cycle gives a nonzero ternary operation, and a repeated color changes the character of its perfect trace.

We then study two parameter families of these algebras. Independent quadratic parameters produce discriminant modules. The actual elliptic displacement gives a different family, with exceptional fibers over the elliptic curve and a square root on a halving cover. These calculations specify coefficient data for the compact comparison posed at the end of the chapter.

1 The pair coefficient and its image

Fix a smooth complex elliptic curve E with origin, and put $\Theta = \mathcal{O}_E(0)$ with divisor section ϑ. This E parametrizes spectral variables. Its role differs from the elliptic factor of a threefold and from a curve of field insertions.

Take one point of each of two distinct colors i, j, with two arrows from i to j. Choose weights $(2, 0)$ and $(0, 2)$. Their compatibility imposes $2t_1 = 2t_2$ on the elliptic equivariant parameters. In the formal neighborhood of the origin this forces $t_1 = t_2$.

We work on the component $t_1 = t_2 = t$ and set $h = 2t$. With $u = z_j - z_i$, the forward arrow factor is

$$s_h(u) = \vartheta(u)\vartheta(u+h), \qquad \mathcal{S} = u^*\Theta \otimes (u+h)^*\Theta, \qquad \mathcal{S}^{-1} \xrightarrow{s_h} \mathcal{O}.$$

The maps u and $u + h$ have domain $E_u \times E_h$. There is no same-color factor for this dimension vector.

These are the compatible weights and arrow factor of Yang–Zhao.[1]

At $h = 0$, write $d : E^2 \to E$ for the difference map and $\Delta = d^{-1}(0)$. The coefficient map becomes

$$\mathcal{L} = d^*\Theta^{-2} = \mathcal{O}_{E^2}(-2\Delta) \xrightarrow{\vartheta(d)^2} \mathcal{O}_{E^2}.$$

Its image is the ideal of 2Δ, and its cokernel is $\mathcal{O}_{2\Delta}$.

Proposition 1.1. The line $\mathcal{L}$ is nontrivial. On a normal formal disk to Δ, its coefficient cokernel is $\mathbb{C}[[u]]/(u^2)$. The sequence

$$0 \longrightarrow \mathcal{O}_\Delta(-\Delta) \longrightarrow \mathcal{O}_{2\Delta} \longrightarrow \mathcal{O}_\Delta \longrightarrow 0 \tag{12.1}$$

does not split over the ambient local ring.

[1] Y. Yang and G. Zhao, *Quiver varieties and elliptic quantum groups*, Assumption 2.1, Remark 2.2(2) and equations (2)–(3). Their Theorem 8.4 compares algebras of rational sections. The isomorphism in Remark 8.5 requires localization.

Proof. Restrict $\mathcal{L}$ to $\{z_1\} \times E$. Its degree is -2, whereas a trivial line has degree zero. Choose an invariant differential on E and its normalized formal logarithm u. In a theta frame, $\vartheta(u) = u\,v(u)$ with $v(0) \neq 0$. Rescaling the source frame removes $v(u)^2$, leaving the map u^2.

The ideal $(u)/(u^2)$ is the first term of (12.1). A module section would lift 1 to an element with nonzero constant term annihilated by u. Such an element does not exist in $\mathbb{C}[[u]]/(u^2)$. □

Thus restriction to reduced support loses a normal extension class. Even a regular scalar change of frame preserves its doubled vanishing. At nonzero displacement, the two orders also have different images. In an odd theta frame the reversed factor is $\vartheta(u)\vartheta(u-h)$. Writing $D_a = \{u = a\}$ on E_u, the two image ideals are

$$\mathcal{I}_+ = \mathcal{O}(-D_0 - D_{-h}), \qquad \mathcal{I}_- = \mathcal{O}(-D_0 - D_h).$$

Their sum is the image of the two maps together.

PROPOSITION 1.2. On a fixed fiber, $\mathcal{J}_h = \mathcal{I}_+ + \mathcal{I}_-$ is

$$\mathcal{J}_h = \begin{cases} \mathcal{O}(-D_0), & 2h \neq 0, \\ \mathcal{O}(-D_0 - D_h), & 2h = 0,\ h \neq 0, \\ \mathcal{O}(-2D_0), & h = 0. \end{cases}$$

Near $(u, h) = (0, 0)$ the image ideal is

$$J = (u(u+h), u(u-h)) = (u^2, uh) \subset \mathbb{C}[[u, h]].$$

Its quotient contains a nonzero class $[u]$ killed by h. Formation of the image fails to commute with specialization at $h = 0$.

Proof. The sum of two divisor ideals on a smooth curve has the smaller vanishing order at every point. The common zeros are precisely those listed in the three cases, with their displayed multiplicities.

Formal logarithms and invertible theta frames give $u(u+h)$ and $u(u-h)$. Their sum and difference generate u^2, uh because 2 is invertible. Hence $u \notin J$ and $hu \in J$.

Moreover uh has nonzero class in J/hJ. An equality $uh = hj$ with $j \in J$ would imply $u = j$, contrary to $u \notin J$. This class maps to zero in $\mathbb{C}[[u, h]]/(h)$. The specialized maps themselves have image (u^2). □

When $2h \neq 0$, the quotient $\mathcal{J}_h/\mathcal{I}_+$ is supported at $-h$, and $\mathcal{J}_h/\mathcal{I}_-$ is supported at h. At $2h = 0$ the two image divisors agree. This identifies divisor lines. An identification of ordered Hall operations must also retain their scalar and equivariant data.

2 FLAGS, LINE TWISTS AND SIMULTANEOUS EQUATIONS

Set $h = 0$. Let G be a finite simple graph with ordered vertex set V. An edge records an interacting pair of multiplicity two. For disjoint subsets $A, B \subset V$, define

$$D_A = \sum_{\substack{i<j,\ i,j\in A \\ \{i,j\}\in E(G)}} \Delta_{ij}, \qquad D_{A,B} = \sum_{\substack{i\in A,\ j\in B \\ \{i,j\}\in E(G)}} \Delta_{ij}.$$

Here Δ_{ij} is the equality divisor on the product of spectral curves. Pull all lines to the common product and put

$$\mathcal{I}_A = \mathcal{O}(-2D_A), \qquad \mathcal{L}_{A,B} = \mathcal{O}(-2D_{A,B}).$$

The divisor sections define their embeddings into $\mathcal{O}$. Write $\mathrm{Perf}(X)$ for the category of complexes locally quasi-isomorphic to bounded complexes of finite-rank vector bundles. For $F \in \mathrm{Perf}(E^A)$ and $G' \in \mathrm{Perf}(E^B)$, define $F \star G' = (F \boxtimes G') \otimes \mathcal{L}_{A,B}$. Here $\boxtimes$ is the derived external tensor product. No shuffle symmetrization is included in this definition.

Proposition 2.1. Multiplication of the pair sections factors through

$$\mathcal{I}_A \otimes \mathcal{I}_B \otimes \mathcal{L}_{A,B} \xrightarrow{\sim} \mathcal{I}_{A\cup B}.$$

These isomorphisms are compatible with every ordered flag. The line identifications

$$\mathcal{L}_{A,B} \otimes \mathcal{L}_{A\cup B,C} \cong \mathcal{L}_{B,C} \otimes \mathcal{L}_{A,B\cup C} \tag{12.2}$$

preserve the sections and all regroupings. In particular,

$$\mathcal{L}_{A,B} \cong \mathcal{I}_{A\cup B} \otimes \mathcal{I}_A^{-1} \otimes \mathcal{I}_B^{-1}.$$

The functors $\mathsf{T}_A(F) = F \otimes \mathcal{I}_A^{-1}$ untwist $\star$. Halving the divisors gives compatible square roots of the line identifications.

Proof. The identity $D_{A\cup B} = D_A + D_B + D_{A,B}$ separates internal and crossing edges. It gives the first isomorphism, its section and the coboundary formula. For three blocks, each side of (12.2) contains every edge between distinct blocks once. The same count applies to any finite ordered flag. Thus every composite is the identification of the same divisor line.

Cancelling the internal lines gives $\mathsf{T}_{A\cup B}(F \star G') \cong \mathsf{T}_A(F) \boxtimes \mathsf{T}_B(G')$. The empty sector has $\mathcal{I}_\varnothing = \mathcal{O}$. Replacing each coefficient -2 by -1 preserves every divisor addition identity and gives the square roots. □

The doubled-divisor cokernel belongs to the final embedding $\mathcal{I}_{A\cup B} \hookrightarrow \mathcal{O}$. Under T, the two-color coefficient map becomes $\mathcal{O} \to \mathcal{O}(2\Delta)$, with its doubled zero unchanged. The square roots above concern the specified labeled divisor lines at $h = 0$. Compact orientation lines, same-color factors and shuffle maps require their own comparisons.

To retain simultaneous equations, fix a commutative ring R and elements $f_1, \dots, f_r$. Their Koszul algebra is

$$K_R(f_1, \dots, f_r) = \big(R \otimes \Lambda(\epsilon_1, \dots, \epsilon_r), d\big), \qquad |\epsilon_i| = -1, \quad d\epsilon_i = f_i.$$

The differential is a graded derivation of degree one. All tensor signs below use this cochain grading.

Over $R = \mathbb{C}[[u]]$, $K_R(u^2)$ has only $H^0 = R/(u^2)$. In $K_R(u, u)$, subtract the first odd generator from the second. Resolving the remaining regular equation gives

$$K_R(u, u) \simeq R/(u) \otimes \Lambda(\epsilon), \qquad d\epsilon = 0.$$

The first construction retains a nilpotent coordinate. The second retains a negative excess class.

For a triangle put $x = z_2 - z_1$, $y = z_3 - z_2$ in logarithmic coordinates. The product coefficient has cone $K_{\mathbb{C}[[x,y]]}(x^2y^2(x+y)^2)$, supported on three doubled lines and concentrated in cohomological degree zero. The simultaneous pair intersection is

$$K_{\mathbb{C}[[x,y]]}(x^2, y^2, (x+y)^2). \tag{12.3}$$

It is supported at the origin and has negative cohomology. This additional incidence object does not measure a failure of associativity of the original product.

For nonzero f, g in a domain there is instead an exact sequence

$$0 \longrightarrow R/(f) \xrightarrow{g} R/(fg) \longrightarrow R/(g) \longrightarrow 0.$$

Its global version carries the corresponding line twists. This filtration of a product cone is different from the derived tensor product of the two cones.

3 A global spectral coefficient and its finite model

Suppose that G is connected, with $n \geq 1$ vertices and m edges. Common translation acts freely on E^n, and differences from the first point identify

$$B_n = E^n / E_{\mathrm{diag}} \cong E^{n-1}.$$

Each equality divisor descends to a divisor D_e on B_n. Write $L_e = \mathcal{O}_{B_n}(2D_e)$ and let s_e be its divisor section. Define the commutative differential graded sheaf

$$\mathscr{K}_G = \bigotimes_{e \in E(G)} [L_e^{-1} \xrightarrow{s_e} \mathcal{O}_{B_n}], \tag{12.4}$$

with L_e^{-1} in degree -1. Its coefficient algebra is $R\Gamma(B_n, \mathscr{K}_G)$.

LEMMA 3.1. The complex $\mathscr{K}_G$ is perfect and its cohomology sheaves have finite length at the origin of B_n. Restriction to the formal completion there computes its derived global sections as a differential graded algebra.

Proof. Every factor is a two-term complex of line bundles. Away from their common zero locus, an invertible section makes one factor contractible. Connectedness identifies the common zero locus with the origin in B_n. The coherent cohomology sheaves therefore have finite length.

They have no higher cohomology and are unchanged by completion. The hypercohomology spectral sequence proves that restriction induces a cohomology isomorphism. Restriction also preserves multiplication and the differential. □

For $n = 1$, this algebra is $\mathbb{C}$ on a point. The empty configuration has translation quotient $[\mathrm{pt}/E] = BE$ and is outside this $n \geq 1$ construction. The spectral quotient also differs from the common translation torsor on compact sheaf moduli.

Formula (12.4) also defines complexes for disconnected subgraphs on the same vertex set. Adding an edge e gives the exact triangle

$$\mathscr{K}_G \otimes L_e^{-1} \xrightarrow{s_e} \mathscr{K}_G \longrightarrow \mathscr{K}_{G+e} \longrightarrow \mathscr{K}_G \otimes L_e^{-1}[1].$$

The middle map is inclusion of the even summand and preserves the differential graded algebra structure. Locally place the new generator first and write $a + \epsilon_e b$. The last map sends this element to b in the shifted term. Its sign follows from $d(\epsilon_e b) = s_e b - \epsilon_e db$.

Permuting edges permutes odd generators with exterior signs. Thus the edge additions and all subset cubes commute on the full Koszul object, including disconnected intermediate subgraphs.

Choose an oriented spanning tree T and let $u_1, \dots, u_{n-1}$ be its logarithmic edge differences. These are coordinates in the completed local ring of B_n. For a chord $c \notin T$, its difference is

$$\ell_c = \sum_a \sigma_{ca} u_a, \qquad \sigma_{ca} \in \{0, 1, -1\}.$$

The signs record traversal of the tree path. Put $\beta = m - n + 1$ and $A_T = \mathbb{C}[u_1, \dots, u_{n-1}]/(u_1^2, \dots, u_{n-1}^2)$.

Theorem 3.2. After these coordinate and frame choices, $R\Gamma(B_n, \mathscr{K}_G)$ is represented by

$$C_{G,T} = A_T \otimes \Lambda(\epsilon_c : c \notin T), \qquad d\epsilon_c = 2 \sum_{a<b} \sigma_{ca}\sigma_{cb} u_a u_b. \tag{12.5}$$

Its dimension is 2^m and its degrees lie in $[-\beta, 0]$. Different trees give quasi-isomorphic algebras through the full edge Koszul complex.

Proof. In a theta frame, each section is the square of its logarithmic difference times a unit. Divide its Koszul generator by that unit. The tree equations form the regular sequence $u_1^2, \dots, u_{n-1}^2$. Their Koszul resolution maps multiplicatively to A_T. Tensoring with the finite free chord factors preserves the quasi-isomorphism.

In A_T the squares vanish, leaving the displayed differential. Squarefree coordinate monomials and exterior chord monomials give $2^{n-1}2^\beta = 2^m$ basis elements. Every tree reduction starts from the same completed full edge complex, giving the asserted zigzag. □

A change of tree need not strictly identify the two finite models. Symmetries act first on the full edge complex. A symmetry preserving the chosen tree acts directly on its quotient, with the induced exterior signs.

Proposition 3.3. Fix orders of the tree coordinates and chord generators. The top-coefficient functional

$$\tau(a) = [u_1 \cdots u_{n-1}\epsilon_1 \cdots \epsilon_\beta]a$$

gives a perfect cochain pairing $C_{G,T} \otimes C_{G,T} \to \mathbb{C}[\beta]$ by multiplication. Here $C[r]^i = C^{i+r}$. Consequently

$$\dim H^{-j}(C_{G,T}) = \dim H^{-(\beta-j)}(C_{G,T}).$$

Proof. Each basis monomial pairs only with its complementary monomial. The exterior ordering supplies its sign, so the matrix is a signed permutation matrix. The differential lowers exterior length, which gives $\tau(da) = 0$. Applying this identity to $d(ab)$ proves compatibility with the differential. A perfect pairing of finite complexes over a field induces a perfect pairing on cohomology. □

The normalization of this trace is part of the construction. A graph symmetry may change it by a character. We first compute the simplest cycle, then determine this character in a repeated-color example.

4 The triangle and its minimal algebra

For the triangle, (12.3) reduces along the tree equations x^2, y^2 to

$$A = \mathbb{C}[x, y]/(x^2, y^2) \otimes \Lambda(\eta), \qquad |x| = |y| = 0, \quad |\eta| = -1, \quad d\eta = xy. \tag{12.6}$$

Indeed the remaining equation is $2xy$. Dividing its odd generator by two gives η. The algebra has eight basis elements. Write $H = H^*(A)$, represented by

$$H^0 = \langle 1, x, y \rangle, \qquad H^{-1} = \langle x\eta, y\eta, xy\eta \rangle. \tag{12.7}$$

The radical $V = \langle x, y \rangle \subset H^0$ has square zero. Its only nonzero actions on the displayed negative classes are

$$x \cdot y\eta = y \cdot x\eta = xy\eta.$$

Products of two negative classes vanish.

Let $i : H \to A$ include these representatives, and let $p : A \to H$ kill xy and η and retain the other six basis elements. Set

$$\mathsf{h}(xy) = \eta, \qquad \mathsf{h} = 0 \quad \text{on the other basis elements.}$$

Then $pi = 1$, $d\mathsf{h} + \mathsf{h}d = 1 - ip$, and $\mathsf{h}^2 = \mathsf{h}i = p\mathsf{h} = 0$. Define a symmetric form q on V by $q(x, y) = q(y, x) = 1$ and $q(x, x) = q(y, y) = 0$.

Augment A by $x, y, \eta \mapsto 0$, and give H the induced augmentation. The augmentation ideal is the kernel of this map.

We fix the bar signs before giving the higher product. Write sa for suspension, with $|sa| = |a| - 1$. On the tensor coalgebra of suspended augmentation ideals use the degree-one components

$$\begin{aligned} b_1(sa) &= -s(da), \\ b_2(sa, sb) &= (-1)^{|a|} s(ab), \\ b_3(sa, sb, sc) &= (-1)^{2|a|+|b|} s\, m_3(a, b, c). \end{aligned} \tag{12.8}$$

Their extension as coderivations uses the Koszul sign for passing a degree-one map across the preceding suspended inputs. The source algebra A has $b_3 = 0$. The minimal algebra below has $b_1 = 0$.

THEOREM 4.1. The binary product in (12.7) extends to a strictly unital minimal A_∞ algebra by

$$m_3(a, b, c) = \big(q(b, c)a - q(a, b)c\big)\eta, \qquad a, b, c \in V, \tag{12.9}$$

and by setting $m_3 = 0$ on every other pattern of inputs. All m_n with $n \geq 4$ vanish. A strictly unital quasi-isomorphism from this algebra to A has suspended components

$$I_1(sa) = s\, i(a), \qquad I_2(sa, sb) = s\, \mathsf{h}(i(a)i(b)), \qquad I_{n\geq 3} = 0.$$

The second component is nonzero only for inputs in V.

Proof. The binary product is $p(i(a)i(b))$. The only nonzero homotopy replaces xy by η. In arity two the bar comparison equation is exactly

$$d\mathsf{h}(i(a)i(b)) = i(a)i(b) - i\, m_2(a, b).$$

For three inputs in V, the two remaining terms are

$$p\big(i(a)\mathsf{h}(i(b)i(c)) - \mathsf{h}(i(a)i(b))i(c)\big),$$

which gives (12.9) with the signs of (12.8). An input of degree -1 already contains η, so both terms vanish.

In every higher transfer tree, a further homotopy above an η-marked expression is zero, or two homotopy outputs are multiplied and give $\eta^2 = 0$. Thus all higher products vanish. The same two facts kill all comparison equations above arity three and all composites of two ternary operations.

In arity four, for $a, b, c, d \in V$ the remaining equation is

$$m_2(m_3(a,b,c),d) + m_2(a, m_3(b,c,d)) = 0.$$

The two coefficients of $xy\eta$ cancel by (12.9). If an input has degree -1, every term vanishes. These formulas give the transfer along the stated contraction and verify its bar-square equations. Every higher component vanishes on a unit, while $I_1(1) = 1$. □

The ternary product is detected without choosing a minimal-model basis. Since $xy = d\eta$ and $y^2 = 0$, the triple Massey set is

$$\langle [x], [y], [y] \rangle = -[y\eta] + \langle [xy\eta] \rangle. \tag{12.10}$$

Its indeterminacy is $xH^{-1} + H^{-1}y = \langle xy\eta \rangle$. It does not contain zero, so A is nonformal. This operation belongs to the simultaneous derived intersection of the doubled pair divisors. It does not assert that the original Hall multiplication fails associativity.

Remark 4.2. The multiplicity of the equations matters. With linear pair equations the corresponding formal complex is $K(x, y, x + y)$. Subtracting the first two odd generators from the third makes the latter closed. Resolving x, y then gives $\mathbb{C} \otimes \Lambda(\epsilon)$ with zero differential. The closed generator records the linear dependence among the three equations. The ternary operation in (12.9) records the doubled equations.

There is also an exact algebraic comparison with an Euler extension. Put $X = \mathbb{P}^1 \times \mathbb{P}^1$ and choose the two degree-two generators x, y with $\int_X xy = 1$. Then $H^*(X;\mathbb{C}) = \mathbb{C}[x, y]/(x^2, y^2)$ and $e(TX) = 4xy$. In the algebra $H^*(X;\mathbb{C}) \otimes \Lambda(e)$ with $|e| = 3$ and $de = 4xy$, set $\eta = e/4$.

Its multiplication and differential are exactly those of (12.6), after changing the integer degrees of x, y, η to 0, 0, -1. Parity is unchanged. This comparison of algebras does not identify their geometric sources.

5 Two equal colors

Take four labeled vertices 0, 1, 2, 3 with colors (1, 1, 2, 3), and join precisely the pairs of unequal colors. The tree with edges 02, 12, 23 is preserved by the transposition $\sigma = (01)$. In logarithmic coordinates put

$$x = z_0 - z_2, \qquad y = z_1 - z_2, \qquad z = z_3 - z_2.$$

The two chords give $(x - z)^2$ and $(y - z)^2$. After the tree quotient and rescaling their generators, the finite model is

$$C = \mathbb{C}[x, y, z]/(x^2, y^2, z^2) \otimes \Lambda(\eta_1, \eta_2), \qquad d\eta_1 = xz, \quad d\eta_2 = yz.$$

All three coordinates have degree zero and both odd generators degree -1. The actual graph transposition acts by

$$\sigma(x) = y, \quad \sigma(y) = x, \quad \sigma(z) = z, \qquad \sigma(\eta_1) = \eta_2, \sigma(\eta_2) = \eta_1.$$

In particular $\sigma(\eta_1\eta_2) = -\eta_1\eta_2$. This action commutes with d and is the action induced from the full edge complex. Write C^+ and C^- for its invariant and anti-invariant subcomplexes.

THEOREM 5.1. The cohomology dimensions, in degrees 0, −1, −2, are

$$\dim H^*(C) = (5, 10, 5), \qquad \dim H^*(C^+) = (4, 5, 1), \qquad \dim H^*(C^-) = (1, 5, 4).$$

The invariant cohomology algebra has bases

$$\begin{aligned} H^0(C^+) &= \langle 1, \alpha, z, \chi \rangle, & \alpha &= x + y, & \chi &= xy, \\ H^{-1}(C^+) &= \langle p, q, r, s, t \rangle, \\ p &= x\eta_1 + y\eta_2, & q &= z(\eta_1 + \eta_2), \\ r &= xy(\eta_1 + \eta_2), & s &= xz\eta_1 + yz\eta_2, \\ t &= yz\eta_1 + xz\eta_2, \\ H^{-2}(C^+) &= \langle u \rangle, & u &= z(x - y)\eta_1\eta_2. \end{aligned}$$

Apart from the unit and graded commutativity, its nonzero products of these basis elements are

$$\alpha^2 = 2\chi, \qquad \alpha p = r, \qquad zp = s, \qquad \alpha q = s + t, \qquad pq = u. \tag{12.11}$$

Proof. The dimensions of C in degrees $0, -1, -2$ are $8, 16, 8$. The image of $d : C^{-1} \to C^0$ is $\langle xz, yz, xyz \rangle$. The other differential sends $a\eta_1\eta_2$ to $a(xz\eta_2 - yz\eta_1)$. Its image has basis

$$xz\eta_2 - yz\eta_1, \quad xyz\eta_1, \quad xyz\eta_2.$$

Both differentials have rank three, proving the total dimensions.

The invariant degree-zero space has basis $1, \alpha, z, \chi, \alpha z, xyz$, and its boundaries span $\alpha z, xyz$. The invariant cycles of degree -1 have basis p, q, r, s, t, w, where $w = xyz(\eta_1 + \eta_2)$. The invariant degree--2 space has basis $(x - y)\eta_1\eta_2$ and u. Their differentials are $-w$ and zero. This proves the displayed invariant bases.

Since taking invariants is exact over $\mathbb{C}$, subtraction gives the anti-invariant dimensions. Direct multiplication gives (12.11). Every omitted product contains a squared coordinate or is a multiple of the boundary w. □

COROLLARY 5.2. The invariant algebra C^+ is nonformal. More precisely,

$$\langle [\alpha], [z], [z] \rangle = -[q] + \langle [r], [s], [t] \rangle$$

does not contain zero.

Proof. The invariant primitive $\eta_1 + \eta_2$ has differential αz, and $z^2 = 0$. It gives the representative $-z(\eta_1 + \eta_2) = -q$. By (12.11), the indeterminacy $\alpha H^{-1}(C^+) + H^{-1}(C^+)z$ is $\langle r, s, t \rangle$. The cohomology basis shows that it does not contain q. □

PROPOSITION 5.3. The perfect trace of this labeled model, $\tau(a) = [xyz\eta_1\eta_2]a$, satisfies $\tau(\sigma a) = -\tau(a)$. Its pairing restricts to perfect pairings between opposite characters, and vanishes between equal characters. The invariant cohomology has no perfect scalar Frobenius pairing supported in total degree -2.

Proof. The coordinate product xyz is fixed and the exterior product changes sign. Thus $\tau(\sigma a\, \sigma b) = -\tau(ab)$. If a, b have characters ε, δ, a nonzero pairing requires $\varepsilon\delta = -1$. The full pairing is perfect by Proposition 3.3. Decomposition into the two characters proves perfectness of the opposite blocks. Such a scalar Frobenius trace on the invariant cohomology would pair H^0 perfectly with H^{-2}, but these have dimensions four and one. □

The character is part of this explicit descent problem. It does not rule out a comparison retaining a nontrivial orientation line or both character sectors. It rules out discarding them while keeping the same perfect scalar trace.

6 Every graph with one cycle

The triangle dimensions extend to all connected graphs with one cycle. Suppose that the cycle has length c and that the graph has n vertices. Removing a cycle edge gives a spanning tree. Orient its $c-1$ edges along the remaining cycle path and put

$$A_s = \mathbb{C}[u_1, \dots, u_s]/(u_1^2, \dots, u_s^2), \qquad s = c-1, \qquad L = u_1 + \cdots + u_s.$$

The one chord has equation L^2. The $n-c$ tree coordinates outside the cycle occur in no chord equation.

For polynomial degree, give every u_a degree one. This grading differs from cochain degree.

Theorem 6.1. For this graph, cohomology occurs only in degrees 0, -1, and

$$\dim H^0(C_{G,T}) = \dim H^{-1}(C_{G,T}) = 2^{n-c}\binom{c}{\lfloor c/2 \rfloor}.$$

For the cycle alone, the polynomial-degree-k part of H^0 has dimension

$$\max\left\{\binom{s}{k} - \binom{s}{k-2}, 0\right\},$$

where a binomial coefficient outside $0, \dots, s$ is zero.

Proof. On the squarefree monomials u_I define $F(u_I) = \sum_{i \in I} u_{I\setminus\{i\}}$. Multiplication by L and this operator satisfy

$$[F, L](u_I) = (s - 2|I|)u_I.$$

With $H(u_I) = (s-2|I|)u_I$, one also has $[H, L] = -2L$ and $[H, F] = 2F$. These identities can be checked one variable at a time. The algebra A_s is the tensor product of the resulting two-dimensional representations.

For completeness, give its monomial basis the orthonormal positive Hermitian form. Then $F = L^*$. A primitive vector v of degree j, meaning $Fv = 0$, generates a string satisfying

$$FL^r v = r(s - 2j - r + 1)L^{r-1}v.$$

The formula follows inductively from the commutators.

Its norm identity shows that the nonzero string runs from degree j to degree $s - j$. Its span is stable under F, L, H, so its orthogonal complement is stable too. Repetition decomposes A_s into such symmetric strings.

Consequently $L^2 : (A_s)^{k-2} \to (A_s)^k$ has maximal rank. On a string it is nonzero exactly when both positions occur. The sets of strings containing these two positions are nested. This proves the cokernel formula.

Summing its positive differences telescopes to

$$\binom{s}{\lfloor s/2 \rfloor} + \binom{s}{\lfloor (s-1)/2 \rfloor} = \binom{s+1}{\lfloor (s+1)/2 \rfloor}.$$

The kernel and cokernel of the endomorphism L^2 have the same dimension. The remaining $n - c$ squarefree variables contribute the factor 2^{n-c}. □

Thus a tree on four vertices has only H^0 of dimension eight, a triangle has dimensions $(3, 3)$, and a four-cycle has $(6, 6)$. The repeated-color graph above has $(5, 10, 5)$.

For comparison take the complete graph on four distinct colors. Its star-tree model is

$$A_3 \otimes \Lambda(\alpha, \beta, \gamma), \qquad d\alpha = xy, \quad d\beta = xz, \quad d\gamma = yz,$$

where $A_3 = \mathbb{C}[x, y, z]/(x^2, y^2, z^2)$. The three differential ranks, from degrees $-1, -2, -3$, are $4, 6, 4$. The first image is $\langle xy, xz, yz, xyz \rangle$.

For the middle image use

$$xy\beta - xz\alpha, \quad xy\gamma - yz\alpha, \quad xz\gamma - yz\beta, \quad xyz\alpha, \quad xyz\beta, \quad xyz\gamma.$$

These six independent vectors span all products of the three basic images. The last kernel is the common annihilator of xy, xz, yz, namely $\langle xy, xz, yz, xyz \rangle$. Since the term dimensions are $8, 24, 24, 8$, the cohomology dimensions are $(4, 14, 14, 4)$. This is a four-color example. All these numbers measure the specified incidence coefficients, rather than root multiplicities of an Igusa algebra.

7 Independent quadratic parameters

The finite graph complex can occur as a derived fiber of a family whose total cohomology is concentrated in degree zero. We first give each edge its own quadratic parameter. The actual elliptic displacement will then provide a different comparison.

Retain the connected graph G, spanning tree T, tree coordinates u_a and chord forms ℓ_c of Theorem 3.2. For a tree edge a also write $\ell_a = u_a$. Give every edge an independent parameter t_e, and put

$$R_G = \mathbb{C}[t_e : e \in E(G)], \qquad B_T = R_G[u_1, \dots, u_{n-1}]/(u_a^2 - t_a : a \in T).$$

The squarefree monomials u_I form an R_G-basis of B_T. Define

$$C_{G,T} = B_T \otimes \Lambda(\epsilon_c : c \notin T), \qquad d\epsilon_c = \ell_c^2 - t_c, \qquad |\epsilon_c| = -1. \tag{12.12}$$

This is a polynomial deformation of the finite formal model. The parameters are independently assigned to the edges. No identification with a universal compact Hall deformation is part of its definition.

THEOREM 7.1. The complex (12.12) is finite free over R_G, with term ranks $2^{n-1}\binom{\beta}{k}$ in degree $-k$. Its only cohomology is

$$H^0(C_{G,T}) \cong \mathbb{C}[u_1, \dots, u_{n-1}], \qquad t_e \longmapsto \ell_e^2. \tag{12.13}$$

Its derived fiber at $t_e = 0$ is the graph algebra $C_{G,T}$. The coefficient functional

$$a \longmapsto [u_1 \cdots u_{n-1}\epsilon_1 \cdots \epsilon_\beta]a$$

gives a perfect pairing $C_{G,T} \otimes_{R_G} C_{G,T} \to R_G[\beta]$ of complexes.

Proof. Eliminate the tree parameters to identify B_T with the polynomial ring $\mathbb{C}[u_1, \dots, u_{n-1}, t_c : c \notin T]$. Each remaining equation $\ell_c^2 - t_c$ eliminates its own parameter. In any order they form a regular sequence, with quotient the ring in (12.13). Its Koszul resolution is precisely (12.12). The displayed free basis gives the term ranks and computes derived base change by ordinary tensoring. At the origin its differential reduces to (12.5).

For the pairing, a product $u_I u_J$ contributes to the top squarefree coefficient exactly when $J = I^c$. An overlap replaces a square by a parameter and loses that coordinate from the squarefree part. The exterior factors likewise pair complementary subsets. The pairing matrix is therefore a signed permutation matrix over R_G. Its trace vanishes on the differential because exterior length decreases. The derivation rule proves compatibility with d. □

The theorem also holds after completion at the parameter origin. The image ideal contains every u_a^2, so the completed degree-zero cohomology is $\mathbb{C}[[u_1, \dots, u_{n-1}]]$. The family need not be flat in cohomology: its central derived fiber retains the negative classes computed in Section 4. On a cover adjoining square roots of the parameters, the equations factor as $(\ell_e - \sqrt{t_e})(\ell_e + \sqrt{t_e})$. No such cover is needed to define the family itself.

7.1 *The triangle discriminant*

For three edges set $t_{12} = a$, $t_{23} = b$, and $t_{13} = a + b + 2c$. After rescaling the chord generator, the family is

$$R = \mathbb{C}[a, b, c], \qquad B = R[x, y]/(x^2 - a, y^2 - b), \qquad \mathcal{A}_R = B \otimes \Lambda(\eta), \quad d\eta = xy - c. \tag{12.14}$$

Thus $\mathcal{A}_R$ is free of rank four in each of degrees $-1, 0$. Its total cohomology is $\mathbb{C}[x, y]$, with $a = x^2$, $b = y^2$, $c = xy$. In the basis $1, x, y, xy$, multiplication by $xy - c$ is

$$F = \begin{pmatrix} -c & 0 & 0 & ab \\ 0 & -c & b & 0 \\ 0 & a & -c & 0 \\ 1 & 0 & 0 & -c \end{pmatrix}, \qquad \det F = (c^2 - ab)^2. \tag{12.15}$$

Put $\delta = c^2 - ab$ and $S = R/(\delta)$.

Theorem 7.2. As an S-module,

$$H^0(\mathcal{A}_R) \cong S \oplus I, \qquad I = (a, c) \subset S.$$

The second summand has rank one and is nonfree at the cone vertex. It has the two-periodic free resolution over S determined by

$$M = \begin{pmatrix} -c & b \\ a & -c \end{pmatrix}, \qquad N = \begin{pmatrix} -c & -b \\ -a & -c \end{pmatrix}, \qquad MN = NM = \delta I_2. \tag{12.16}$$

The map $\operatorname{Spec} \mathbb{C}[x, y] \to \operatorname{Spec} S$ is the quotient double cover by $(x, y) \mapsto (-x, -y)$. The ring S is already normal.

Proof. Split the four basis elements by this involution. On $\langle 1, xy\rangle$, the matrix is $\left(\begin{smallmatrix} -c & ab \\ 1 & -c \end{smallmatrix}\right)$. Its unit entry permits invertible row and column operations reducing it to $\operatorname{diag}(1, -\delta)$. On $\langle x, y\rangle$, the matrix is M. The invariant subring of $\mathbb{C}[x, y]$ is generated by x^2, y^2, xy, with relation $c^2 = ab$. It is S.

Multiplication by x identifies the anti-invariant summand with (a, c), sending x to a and y to c. This multiplication is injective in $\mathbb{C}[x, y]$. The odd block of (12.15) supplies its presentation by M.

The ideal I is nonzero in the domain S, so it has rank one. At the maximal ideal (a, b, c) its classes a, c are independent modulo $(a, b, c)I$. It therefore requires two generators there and cannot be a free rank-one module.

The products in (12.16) follow by multiplication. To prove exactness, suppose $M\bar{v} = 0$ over S. Lift to R, so $Mv = \delta w = MNw$. The determinant of M is the nonzero polynomial δ, hence M is injective over R and $v = Nw$. Thus $\ker \bar{M} = \operatorname{im} \bar{N}$. Interchanging M, N proves the other equality.

Finally, let an element of the invariant fraction field be integral over the invariant ring. It is also integral over the normal ring $\mathbb{C}[x, y]$, so it belongs to that ring. It is still invariant, proving that the invariant ring is normal. The invariant field has index two in $\mathbb{C}(x, y)$. The cover is consequently not a normalization map. □

In this decomposition, invariant and anti-invariant refer to the simultaneous sign action, not to cochain degree or super parity. A generic rank-two module has here a nonfree rank-one summand. Replacing it by two free copies would lose the module at the vertex.

PROPOSITION 7.3. The derived fibers of (12.14) have the following dimensions:

parameter	rank F	$\dim H^0$	$\dim H^{-1}$
$\delta \neq 0$	4	0	0
$\delta = 0,\ (a, b, c) \neq 0$	2	2	2
$(a, b, c) = 0$	1	3	3.

Every smooth-discriminant fiber is formal, with cohomology algebra $\mathbb{C}^2 \otimes \Lambda(\xi)$, $|\xi| = -1$. The vertex fiber is the nonformal triangle algebra (12.6).

Proof. Off the discriminant, F is invertible. At a nonzero point of $\delta = 0$, at least one of a, b is nonzero. If $a \neq 0$, split $x^2 = a$ into the two roots $r, -r$. Each component is

$$\mathbb{C}[y]/(y^2 - b) \otimes \Lambda(\eta), \qquad d\eta = r_j y - c, \qquad r_j \in \{r, -r\}.$$

Multiplication by $r_j y - c$ has rank one.

Its kernel is the line generated by $r_j y + c$, since $r_j^2 b = c^2$ and the coefficient r_j is nonzero. Sending ξ to $(r_j y + c)\eta$ and 1 to 1 gives a differential graded algebra quasi-isomorphism from $\Lambda(\xi)$ to this component. The case $b \neq 0$ is symmetric. At the vertex, F has only its bottom-left entry nonzero. Its fiber is exactly (12.6), whose Massey product was computed in (12.10). □

Ordinary specialization of $H^0(\mathcal{A}_R)$ at the vertex gives only $\mathbb{C}[x, y]/(x^2, y^2, xy)$. It loses both the negative classes and their ternary operation. The relative pairing can also be seen directly. Extraction of the coefficient of xy in B has Gram matrix

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad F^t P = PF.$$

Extracting additionally η gives the perfect pairing $\mathcal{A}_R \otimes_R \mathcal{A}_R \to R[1]$. The jump in fiber cohomology therefore occurs inside a fixed free complex with a perfect chain pairing.

7.2 A nonnormal four-cycle image

For a four-cycle choose consecutive tree coordinates u, v, w. Its final edge has difference $u + v + w$. Write its parameter as $a + b + c + 2z$, where the three tree parameters are a, b, c. Its finite free family is

$$\begin{aligned} R_4 &= \mathbb{C}[a,b,c,z], \qquad B_4 = R_4[u,v,w]/(u^2-a, v^2-b, w^2-c), \\ \mathcal{A}_4 &= B_4 \otimes \Lambda(\eta), \qquad |\eta| = -1, \qquad d\eta = uv + uw + vw - z. \end{aligned} \tag{12.17}$$

It resolves $\mathbb{C}[u,v,w]$ with parameter map

$$(a,b,c,z) = (u^2, v^2, w^2, uv + uw + vw).$$

Define the quartic polynomial

$$P_4 = (z^2 + ab - c(a+b))^2 - 4ab(z+c)^2.$$

Theorem 7.4. The parameter image in (12.17) has coordinate ring

$$A_4 = \mathbb{C}[a,b,c,z]/(P_4),$$

and P_4 is irreducible. If F_4 is multiplication by $uv + uw + vw - z$ on the rank-eight R_4-module B_4, then $\det F_4 = P_4^2$. The ring A_4 is not normal. Its normalization is

$$D_4 = \mathbb{C}[u,v,w]^{\{\pm 1\}},$$

where the sign acts simultaneously on all three variables. The map from $\mathbb{A}^3$ to $\operatorname{Spec} D_4$ still has generic degree two.

Proof. Put $z = uv + uw + vw$ and observe that $(z - uv)^2 - c(u+v)^2 = 0$. The two polynomials $(z-uv)^2 - c(u+v)^2$ and $(z+uv)^2 - c(u-v)^2$ are conjugate under the sign change of uv. Their product, after using $u^2 = a, v^2 = b, w^2 = c$, is

$$\big(z^2 + ab - c(a+b) - 2uv(z+c)\big)\big(z^2 + ab - c(a+b) + 2uv(z+c)\big) = P_4.$$

Thus the four possible values of $uv + uw + vw$ under independent signs, modulo simultaneous sign, are the roots of P_4.

Over $\mathbb{C}(a,b,c)$ the extension obtained by adjoining u, v, w is multiquadratic of degree eight. Its sign group fixes $uv + uw + vw$ exactly when all three signs are equal: the three monomials uv, uw, vw are independent over $\mathbb{C}(a,b,c)$. Its orbit therefore has four elements, proving that P_4 is the minimal polynomial. Since P_4 is monic in z, polynomial division shows that the kernel of the parameter map is exactly (P_4).

The determinant of multiplication by $uv + uw + vw - z$ is its norm. Each of the four orbit values occurs twice among the eight sign choices, so this norm is P_4^2. The identity over the fraction field is an identity over R_4.

The invariant ring D_4 is generated by $u^2, v^2, w^2, uv, uw, vw$. Each cross product is integral over A_4, for example $(uv)^2 = ab$. They also lie in its fraction field, since

$$uv = \frac{z^2 + ab - c(a+b)}{2(z+c)}, \tag{12.18}$$

and cyclically for uw, vw. The denominators are nonzero elements of that field. Hence D_4 is finite and birational over A_4. It is normal by the same finite-invariant argument used for the triangle, and is therefore the integral closure of A_4.

Finally, grade u, v, w by degree one. The degree-two part of the image ring A_4 is spanned by

$$u^2, \quad v^2, \quad w^2, \quad uv + uw + vw.$$

It does not contain uv. This integral element is missing from A_4, proving nonnormality. Simultaneous sign has order two on the fraction field of $\mathbb{C}[u, v, w]$, so the last assertion follows. □

For the triangle the squares and the single cross product already generate the sign-invariant ring. For the four-cycle the parameter records only the sum of three cross products. Formula (12.18) recovers them generically, while normalization restores them across the divisors where those fractions are not regular expressions. The normalization and the double cover thus record different geometric operations.

8 The actual elliptic displacement

Take three ordered distinct colors with arrows pointing from lower to higher color. Remove the common spectral translation and use differences $x = z_2 - z_1$, $y = z_3 - z_2$. In a formal logarithm the third difference is $x + y$. Write h also for the formal displacement parameter and set $R = \mathbb{C}[[h]]$. Consider the Koszul dg algebra over $R[x, y]$ of

$$x(x + h), \qquad y(y + h), \qquad (x + y)(x + y + h). \tag{12.19}$$

Its extension to $\mathbb{C}[[h, x, y]]$ is the formal elliptic incidence complex after removing invertible theta factors. Each Koszul generator has cohomological degree -1 and differential equal to its equation. The differential is a graded derivation.

THEOREM 8.1. The Koszul algebra of (12.19) is multiplicatively quasi-isomorphic to

$$B_h = R[x, y]/(x^2 + hx, y^2 + hy), \qquad A_h = B_h \otimes \Lambda(\eta), \qquad |\eta| = -1, \quad d\eta = xy. \tag{12.20}$$

It is free of rank four in degrees 0 and -1. Its cohomology $H = H^*(A_h)$ is free over R, with bases

$$H^0 = R\{1, x, y\}, \qquad H^{-1} = R\{k_1, k_2, k_3\}, \qquad \begin{aligned} k_1 &= [(x + h)\eta], \\ k_2 &= [(y + h)\eta], \\ k_3 &= [(xy - h^2)\eta]. \end{aligned} \tag{12.21}$$

The degree-zero relations are $x^2 = -hx$, $y^2 = -hy$, $xy = 0$. The action on negative cohomology is

	k_1	k_2	k_3
x	0	$k_3 + hk_1$	$-hk_3 - h^2k_1$
y	$k_3 + hk_2$	0	$-hk_3 - h^2k_2$.

(12.22)

Every product of two negative classes vanishes.

Proof. The first two equations form a regular sequence: each is monic in its own variable, also after quotienting by the other. Their Koszul algebra resolves B_h. Tensoring with the remaining free Koszul factor preserves this quasi-isomorphism. In B_h the third equation is $2xy$. Dividing its odd generator

by two gives η. The same reduction applies to the completed coordinate ring, since the resulting finite R-module is complete.

In the ordered basis $1, x, y, xy$ of B_h, multiplication by xy has only one nonzero row, namely $(1, -h, -h, h^2)$, in the xy row. Its cokernel is free on $1, x, y$. The elements $x + h$, $y + h$, $xy - h^2$ are killed by xy. Subtracting their multiples from any kernel vector leaves a scalar killed by the unit entry of the displayed row. Thus they form a free basis of the kernel, proving (12.21).

Multiplying these representatives in $B_h \otimes \Lambda(\eta)$ gives (12.22). The relation $\eta^2 = 0$ gives the last claim. □

The free cohomology has a contraction defined over R. Let $i : H \to A_h$ include the displayed representatives. On the basis $1, x, y, xy, \eta, x\eta, y\eta, xy\eta$, let $p : A_h \to H$ kill xy, η, send the last three elements to k_1, k_2, k_3, and fix $1, x, y$. Put

$$\mathsf{H}(xy) = \eta, \qquad \mathsf{H} = 0 \text{ on the other basis elements.}$$

Then $pi = 1$, $d\mathsf{H} + \mathsf{H}d = 1 - ip$, and $\mathsf{H}^2 = \mathsf{H}i = p\mathsf{H} = 0$. For example, $\mathsf{H}d(x\eta) = -h\eta = x\eta - i(k_1)$, so the parameter term is part of the contraction.

Proposition 8.2. The functional $\tau : A_h \to R[1]$ extracting the coefficient of $xy\eta$ defines a perfect cyclic pairing of complexes by $(a, b) \mapsto \tau(ab)$. The induced pairing between H^0 and H^{-1} is perfect over R.

Proof. The multiplication-trace matrix on B_h, with trace equal to the coefficient of xy, is

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -h \\ 0 & 1 & 0 & -h \\ 1 & -h & -h & h^2 \end{pmatrix}.$$

Its determinant is 1. It pairs the degree-zero and degree-minus-one copies of B_h in A_h. The trace kills dA_h, and the derivation rule gives $\tau((da)b) + (-1)^{|a|}\tau(a\, db) = 0$. Graded commutativity gives cyclicity. On the cohomology bases of (12.21), the pairing matrix is

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -h \\ 1 & 0 & -h \end{pmatrix},$$

of determinant -1. □

8.1 *The ternary class at coalescence*

The contraction determines the multiplication beyond cohomology. Put $V = Rx \oplus Ry \subset H^0$, and define $q : V \otimes_R V \to R$ by $q(x, y) = q(y, x) = 1$, $q(x, x) = q(y, y) = 0$. Let $\psi : V \to H^{-1}$ send x to k_1 and y to k_2.

Augment A_h by $x, y, \eta \mapsto 0$ and H by $x, y, k_1, k_2, k_3 \mapsto 0$. For $A = A_h$ or H, put $\overline{A} = \ker(A \to R)$. On $T^c_R(s\overline{A}) = \bigoplus_{n \geq 1} (s\overline{A})^{\otimes_R n}$ use the suspension $|sa| = |a| - 1$ and bar components

$$b_1(sa) = -s(da), \quad b_2(sa, sb) = (-1)^{|a|} s(ab), \quad b_3(sa, sb, sc) = (-1)^{2|a|+|b|} s\, m_3(a, b, c).$$

Their coderivation extensions use the Koszul sign for passing a degree-one map across the preceding suspended inputs. For A_h set $b_3 = 0$, and for H set $b_1 = 0$.

THEOREM 8.3. The multiplication (12.22) extends to a strictly unital minimal A_∞ algebra on H by

$$m_3(a,b,c) = q(b,c)\psi(a) - q(a,b)\psi(c), \qquad a,b,c \in V. \tag{12.23}$$

The ternary product is zero on all other basis input patterns, and $m_n = 0$ for $n \geq 4$. A strictly unital quasi-isomorphism to A_h has suspended components

$$I_1(sa) = s\,i(a), \qquad I_2(sa,sb) = s\,\mathsf{H}(i(a)i(b)), \qquad I_{n\geq 3} = 0.$$

Proof. The binary transfer is $p(i(a)i(b))$. In arity two the comparison equation is $d\mathsf{H}(i(a)i(b)) = i(a)i(b) - i\,m_2(a,b)$. For three degree-zero inputs, the two transfer trees give

$$p\big(i(a)\mathsf{H}(i(b)i(c)) - \mathsf{H}(i(a)i(b))i(c)\big).$$

Since H extracts the xy coefficient and $p(x\eta) = k_1$, $p(y\eta) = k_2$, this is (12.23). A negative input makes both terms zero. Unit inputs vanish because $p\mathsf{H} = \mathsf{H}i = 0$.

Two internal homotopies in a transfer tree either lie on the same branch or on different branches. On the same branch the second acts on an expression containing η and vanishes. On different branches their outputs multiply to zero because $\eta^2 = 0$. All higher transfer products therefore vanish.

For the comparison, the final homotopy also kills each expression containing η, so $I_{n\geq 3} = 0$. The bar comparison in arity three is the displayed two-tree formula together with $d\mathsf{H} + \mathsf{H}d = 1 - ip$. In every higher arity its terms contain either two I_2 outputs or a homotopy applied to a negative input and vanish. Thus the bar comparison holds in every arity.

Its leading tensor-length term is $i^{\otimes n}$, so the induced coalgebra map is injective. Compatibility with the square-zero bar differential on A_h implies $b_H^2 = 0$. Finally I_1 induces the identity on cohomology, and every higher component vanishes on a unit. □

The ternary operation has a more precise relative invariant than its nonvanishing formula. Use the normalized Hochschild complex of the graded R-algebra (H, m_2). An arity-two cochain of internal degree -1 can be nonzero only on $V \otimes V$, with values in H^{-1}. Define such a cochain by

$$G_2(x,x) = 0, \quad G_2(x,y) = G_2(y,x) = k_1, \quad G_2(y,y) = k_2. \tag{12.24}$$

For degree-zero inputs its coboundary is

$$(\delta_H G_2)(a,b,c) = aG_2(b,c) - G_2(ab,c) + G_2(a,bc) - G_2(a,b)c.$$

PROPOSITION 8.4. The arity-three, internal-degree-minus-one Hochschild class of m_3 is nonzero and has annihilator exactly (h). More precisely,

$$\delta_H G_2 = h\,m_3. \tag{12.25}$$

Proof. The arity-four A_∞ identity says that m_3 is a Hochschild cocycle. On the ordered triples $xxx, xxy, xyx, xyy, yxx, yxy, yyx, yyy$, the coboundary of (12.24) is

$$h(0, k_1, 0, -k_2, -k_1, 0, k_2, 0),$$

by (12.22). These are exactly the values of hm_3. Other normalized input patterns vanish by degree, proving (12.25).

Reduce modulo h, and let G be a normalized two-cochain of internal degree -1. Since $xy = y^2 = 0$,

$$(\delta_H G)(x, y, y) = xG(y, y) - G(x, y)\, y \in \mathbb{C}k_3 :$$

the action of x, y on negative classes has image in that line. But $m_3(x, y, y) = -k_2$. Hence m_3 is not a boundary after reduction, and therefore is not a boundary over R. Every scalar outside (h) is a unit in R. Equation (12.25) therefore gives the exact annihilator. □

Equivalently, at $h = 0$ the triple Massey set is

$$\langle [x], [y], [y] \rangle = -k_2 + \mathbb{C}k_3.$$

Indeed $xy = d\eta$, $y^2 = 0$, and the indeterminacy is $xH^{-1} + H^{-1}y = \mathbb{C}k_3$. It excludes zero. Thus the central algebra is nonformal even though the relative cohomology is free and commutes with base change.

Remark 8.5. The torsion assertion concerns this particular ternary class. It does not extend to all positive Hochschild arities. Indeed let D act by zero on H^0 and by the identity on H^{-1}. It is an R-linear derivation of internal degree zero: multiplication by H^0 preserves H^{-1}, and two negative classes multiply to zero.

In the Hochschild complex of (H, m_2), D is therefore a unary cocycle. Unary boundaries of internal degree zero are inner commutators, which vanish because H is graded commutative. Since H^{-1} is free and nonzero over R, no nonzero scalar annihilates this class. It remains nonzero after inverting h.

8.2 Localization and the independent quadratic family

Let $K = \mathbb{C}((h))$. The tree algebra $B_h \otimes_R K$ has four orthogonal idempotents

$$\begin{aligned} e_{00} &= (x + h)(y + h)/h^2, \quad & e_{10} &= -x(y + h)/h^2, \\ e_{01} &= -(x + h)\, y/h^2, & e_{11} &= xy/h^2. \end{aligned} \tag{12.26}$$

They sum to 1 and evaluate at the four points with $x, y \in \{0, -h\}$. On the first three, $xy = 0$, and on the last, $xy = h^2$.

Proposition 8.6. Projection to the first three idempotent factors gives a multiplicative quasi-isomorphism

$$\mathcal{A}_h \otimes_R K \longrightarrow \prod_{\substack{(i,j)\in\{0,1\}^2 \\ (i,j)\neq(1,1)}} K \otimes \Lambda(\eta_{ij}), \qquad |\eta_{ij}| = -1, \quad d = 0.$$

The kernel contracts by multiplication by $h^{-2}e_{11}\eta$.

Proof. On the e_{11} factor, $d(h^{-2}e_{11}\eta) = e_{11}$, so the derivation rule gives a contracting homotopy. The other factors have zero differential. The product decomposition is by central idempotents, hence its projection is multiplicative. □

The poles in (12.26) prevent this decomposition from extending across $h = 0$. There is also an explicit logarithmic connection on the punctured model:

$$\nabla_{\partial_h} = \partial_h + h^{-1}(x\partial_x + y\partial_y + 2\eta\partial_\eta).$$

It preserves $x^2 + hx$, $y^2 + hy$ and commutes with $d\eta = xy$. The elements x/h, y/h, η/h^2 are horizontal. The residue on the cohomology basis has weights $0, 1, 1, 3, 3, 4$. In the analytic realization on a punctured disk its monodromy is trivial.

This specifies an Euler connection. An identification with a Gauss–Manin connection would require an additional comparison.

The equations $X^2 = a, Y^2 = b$ with chord differential $XY - c$, for independent complex parameters a, b, c, give a different family. The discrepancy already appears under recentering: with $X = x + h/2$, $Y = y + h/2$ one has

$$x^2 + hx = X^2 - h^2/4, \quad y^2 + hy = Y^2 - h^2/4, \quad xy = XY - \frac{h}{2}(X + Y) + \frac{h^2}{4}.$$

The linear term in the last equation is indispensable. There is in fact an obstruction to every base map, not only to this attempted change of coordinates.

PROPOSITION 8.7. No map $\mathbb{C}[a, b, c] \to R$ makes the derived pullback of

$$\mathbb{C}[a, b, c][X, Y]/(X^2 - a, Y^2 - b) \otimes \Lambda(\zeta), \qquad |\zeta| = -1, \quad d\zeta = XY - c,$$

quasi-isomorphic to A_h as a dg R-algebra.

Proof. Over K, the differential of the pulled-back complex, in the basis $1, X, Y, XY$, is

$$F = \begin{pmatrix} -c & 0 & 0 & ab \\ 0 & -c & b & 0 \\ 0 & a & -c & 0 \\ 1 & 0 & 0 & -c \end{pmatrix}.$$

Both cohomology dimensions can be 3 only if F has rank 1. Its two-by-two minors include a, b, c up to sign, so this forces $a = b = c = 0$ in K and therefore in R. The pullback would then be the constant central algebra. Its Massey set $-k_2 + Kk_3$ still excludes zero, so it remains nonformal over K. Proposition 8.6 makes the generic fiber of A_h formal, a contradiction. □

9 DISPLACEMENT ALONG SIGNED GRAPH PATHS

The triangle reduction extends to every connected finite simple graph G on ordered vertices $0, \dots, n - 1$, with $R = \mathbb{C}[[h]]$ as in the formal triangle. Orient each edge from its lower to its higher vertex, and assign it the pair equation $\ell_e(\ell_e + h)$, where $\ell_{ij} = z_j - z_i$. Set $z_0 = 0$ to remove common spectral translation, choose an oriented spanning tree T, and use its $k = n - 1$ edge differences $u_1, \dots, u_k$ as coordinates. For a chord c write

$$\ell_c = \sum_{a=1}^{k} \sigma_{ca} u_a, \qquad \sigma_{ca} \in \{-1, 0, 1\}, \qquad \beta = |E(G)| - n + 1.$$

The signs record the traversal of its tree path.

THEOREM 9.1. The Koszul algebra of the shifted edge equations has a multiplicative quasi-isomorphism to

$$C_{G,h} = R[u_1, \dots, u_k]/(u_a(u_a + h) : 1 \le a \le k) \otimes \Lambda(\epsilon_c : c \notin T), \qquad |\epsilon_c| = -1,$$

with differential

$$d\epsilon_c = 2 \sum_{a<b} \sigma_{ca}\sigma_{cb} u_a u_b + h \sum_a (\sigma_{ca} - \sigma_{ca}^2) u_a. \tag{12.27}$$

Its term in degree $-j$ is free of rank $2^k\binom{\beta}{j}$ over R. The total rank is $2^{|E(G)|}$. Define the finite set

$$\mathcal{A}_G = \left\{ s \in \{0,1\}^k : \sum_a \sigma_{ca} s_a \in \{0,1\} \text{ for every chord } c \right\}.$$

Over $K = \mathbb{C}((h))$ there is a multiplicative quasi-isomorphism

$$C_{G,h} \otimes_R K \longrightarrow \prod_{s \in \mathcal{A}_G} \Lambda_K(\epsilon_1, \dots, \epsilon_\beta), \qquad d = 0. \tag{12.28}$$

Proof. Tree differences are integral unimodular coordinates on the translation quotient. Their monic quadratics are a regular sequence. Quotient by them and send the tree Koszul generators to zero, retaining the chord generators. This is a multiplicative quasi-isomorphism, as in Theorem 8.1.

Expanding $\ell_c^2 + h\ell_c$ and substituting $u_a^2 = -hu_a$ gives (12.27). In particular a negatively traversed tree edge contributes $-2hu_a$, whose linear term cannot be discarded. The squarefree monomials in the u_a, tensored with exterior monomials, give the asserted ranks.

After inverting h, the tree algebra splits into the points $u_a = -hs_a$, $s_a \in \{0,1\}$. On that factor the chord differential is $h^2 L_c(s)(L_c(s) - 1)$, where $L_c(s) = \sum_a \sigma_{ca} s_a$. If one such scalar is nonzero it is invertible in K. Multiplication by $\epsilon_c / (h^2 L_c(s)(L_c(s) - 1))$ contracts that factor. On the remaining factors every chord differential is zero. Projection onto their product is (12.28). □

Cohomology over R distinguishes graphs with the same generic description. Write K_n for the complete graph on n vertices, ij for the edge $\{i, j\}$, and $C_{G,0} = C_{G,h} \otimes_R R/(h)$. In the following table the dimensions are listed in degrees $0, -1, \dots$. Torsion refers to cohomology modules over R.

Proposition 9.2. For the graphs specified below, the central dimensions, generic dimensions and torsion submodules are

G	$H^*(C_{G,0})$	$H^*(C_{G,h} \otimes_R K)$	R-torsion
K_3	$(3, 3)$	$(3, 3)$	0
$01, 12, 23, 03$	$(6, 6)$	$(4, 4)$	$(R/(h))^2$ in H^0
$K_4 \setminus \{01\}$	$(5, 10, 5)$	$(5, 10, 5)$	0
K_4	$(4, 14, 14, 4)$	$(4, 12, 12, 4)$	$(R/(h))^2$ in H^{-1}.

In the third row the colors of vertices 0, 1, 2, 3 are $(1, 1, 2, 3)$, and only intercolor pair factors are included. The last row is an auxiliary four-color graph.

Proof. Use the following trees and chord path vectors, in the orders shown:

G	T	$(\sigma_c)_c$
K_3	$(01, 12)$	$(1, 1)$
$01, 12, 23, 03$	$(01, 03, 23)$	$(-1, 1, -1)$
$K_4 \setminus \{01\}$	$(02, 12, 13)$	$(1, -1, 1),\ (0, -1, 1)$
K_4	$(01, 12, 23)$	$(1, 1, 0),\ (1, 1, 1),\ (0, 1, 1)$.

There are respectively 3, 4, 5, 4 admissible assignments. Theorem 9.1 gives all generic dimensions.

For the calculation over R, order squarefree monomials by the bit mask of their tree coordinates, with u_1 the least significant bit. Order exterior subsets lexicographically. Multiplication by u_a is the tensor

product of identity matrices with $\left(\begin{smallmatrix} 0 & 0 \\ 1 & -h \end{smallmatrix}\right)$ in the ath position. Equation (12.27) therefore specifies every differential matrix. Removing the rth exterior factor has sign $(-1)^{r-1}$.

Cancel a constant nonzero entry w at row r, column c, replacing the other entries by

$$D'_{ij} = D_{ij} - D_{ic}w^{-1}D_{rj} \quad (i \neq r,\ j \neq c).$$

Apply the inverse column change to the preceding differential and the inverse row change to the following one. Their components in the cancelled summand vanish because consecutive differentials compose to zero. Thus each cancellation splits off the acyclic complex $[R \xrightarrow{1} R]$ over R.

Choose the first available constant entry, reading differentials from lowest degree upwards and each matrix by rows. There are 1, 2, 6, 14 such cancellations in the four cases. For the triangle and the third graph the remaining differentials are zero, with respective term ranks $(3, 3)$ and $(5, 10, 5)$.

For the four-cycle the residual map has size 6 by 6. Its only nonzero rows, after factoring out h, are

$$\begin{pmatrix} -2h & 2h^2 & 4h & -4h^2 & 2h^2 & -2h^3 \\ -2 & 2h & -2 & 2h & 2h & -2h^2 \\ -2 & 2h & 4 & -4h & 2h & -2h^2 \end{pmatrix}.$$

The first row is h times the third. The other two rows have a constant invertible minor in columns 1, 3. Every remaining column is an R-linear combination of these. Hence the two nonzero Smith factors of the original residual map are h, h.

For K_4 the remaining term ranks are $(4, 14, 14, 4)$. The sole nonzero differential is from degree -2 to degree -1. After factoring out h, deleting its zero rows and columns gives

$$\begin{pmatrix} -2 & 2h & 2 & 0 & -2h & -2h & 0 \\ 2 & -2h & 0 & -2 & 2h & 0 & 2h \\ 0 & 0 & -2 & 2 & 0 & 2h & -2h \end{pmatrix}.$$

The rows sum to zero, and the first two rows have a constant invertible minor in columns 1, 3. All other columns lie in their span. Again the nonzero Smith factors are h, h. The cokernel therefore has two $R/(h)$ summands, now in degree -1. The residual differentials vanish modulo h, giving the central dimensions in the table. This also shows that no higher power of h occurs in the torsion. □

The absence of cohomological torsion for the triangle is consistent with Proposition 8.4: that proposition concerns a Hochschild class of its multiplication. In the repeated-color row a full Hall coefficient must also include the same-color factor. The number of admissible graph branches by itself defines neither a compact BPS multiplicity nor an Igusa coefficient.

10 The family over the spectral elliptic curve

The formal calculation has a global source. On $E_x \times E_y \times E_h$ put $u_1 = x, u_2 = y, u_3 = x + y$, and let

$$\mathcal{S}_i = u_i^*\Theta \otimes (u_i + h)^*\Theta, \qquad s_i = \vartheta(u_i)\vartheta(u_i + h).$$

Define

$$\mathcal{K} = \bigotimes_{i=1}^{3} [\mathcal{S}_i^{-1} \xrightarrow{s_i} \mathcal{O}], \qquad \mathcal{C} = R\pi_*\mathcal{K}, \quad \pi : E_x \times E_y \times E_h \to E_h. \tag{12.29}$$

The displayed factors have degrees -1, 0. Write $E[n] = \ker([n] : E \to E)$ for the n-torsion subgroup.

Theorem 10.1. The complex $\mathscr{C}$ is represented by a two-term complex of locally free sheaves of rank four on E_h. Its derived fiber dimensions are

h	$\dim H^0$	$\dim H^{-1}$
0	3	3
$h \neq 0,\ 2h \neq 0$	3	3
$h \neq 0,\ 2h = 0$	4	4.

Near a nonzero point $h_0 \in E[2]$, three summands have zero differential. The fourth has local form $[O \xrightarrow{v} O]$, with v a parameter at h_0 times a unit. It contributes a skyscraper to $H^0(\mathscr{C})$. Its extra negative fiber class is a Tor_1 class.

Proof. The divisors of s_1, s_2 cut out

$$Z = \{x = 0 \text{ or } -h\} \times_{E_h} \{y = 0 \text{ or } -h\},$$

with their Cartier multiplicities. Each factor is finite flat of degree two over E_h: the two sections have no vertical component, and their sum has constant degree two. Equivalently, near their meeting point its equation is a monic quadratic in a relative local coordinate. Their product is finite flat of degree four. The first two sections are a regular sequence and their Koszul complex resolves O_Z.

Consequently (12.29) reduces to the finite pushforward of $[\mathcal{S}_3^{-1}|_Z \xrightarrow{s_3} O_Z]$. An invertible sheaf on a finite flat degree-four scheme has a locally free rank-four pushforward. Both terms are therefore locally free, and specialization computes every derived fiber.

For $h \neq 0$, the four points of Z_h are distinct. The third section is zero at $(0,0)$, $(0,-h)$, $(-h,0)$. At $(-h,-h)$ its value, in the corresponding line, is

$$\vartheta(-2h)\vartheta(-h). \tag{12.30}$$

It is nonzero unless $2h = 0$.

At a nonzero two-torsion point the first factor has a simple zero, since $[2]$ is etale, and the second is nonzero. Local line frames give the fourth complex in the statement. Its cokernel is $O/(v)$ and its kernel is zero. Derived specialization creates the extra class in degree -1.

At $h = 0$, formal logarithms and invertible theta frames give (12.20), with dimensions $(3,3)$. □

For $h \neq 0$ the fiber is formal: each surviving reduced tree point carries one exterior factor with zero differential, and a nonsurviving point is contractible. At the three nonzero two-torsion points there are four such exterior factors. No trivialization of the three persistent line summands over the parameter curve is needed for this local description.

On the original cover $h = 2t$, the exceptional extra fibers lie at $E[4] \setminus E[2]$, a set of twelve points. The nonformal doubled fiber lies at the four points of $E[2]$. A formal neighborhood of $t = 0$ does not contain the twelve exceptional points.

10.1 A root line on the halving cover

Choose a rigidification of the symmetric line Θ at the origin. Regard $\mathcal{S} = u^*\Theta \otimes (u+h)^*\Theta$ as a line on $E_u \times E_h$. For each fixed h it has degree two on E_u, so it has four degree-one square roots. Their variation is a separate question.

THEOREM 10.2. There is no line $\mathcal{R}$ on $E_u \times E_h$ with $\mathcal{R}^{\otimes 2} \cong \mathcal{S}$. There is also no continuous relative choice of a degree-one root, even after tensoring $\mathcal{S}$ by a line pulled back from E_h. On the cover $h = 2t$ there is an isomorphism

$$u^*\Theta \otimes (u + 2t)^*\Theta \cong (u + t)^*\Theta^{\otimes 2} \otimes t^*\Theta^{\otimes 2}, \tag{12.31}$$

and hence the root

$$\mathcal{R}_t = (u + t)^*\Theta \otimes t^*\Theta. \tag{12.32}$$

For every nonzero $\xi \in E[2]$, the deck transformation $t \mapsto t + \xi$ changes its restriction to a u-fiber by the nontrivial two-torsion line associated to ξ. For $2t \neq 0$, the section $\vartheta(u)\vartheta(u + 2t)$ is not the square of a regular section of $\mathcal{R}_t$.

Proof. Restrict $\mathcal{S}$ to $\{u_0\} \times E_h$. Its first factor is constant and its second has degree one. The square of a line has even degree, which proves the first assertion. Under $\mathrm{Pic}^1(E) \cong E$, a root on a fixed fiber has a point m with $2m = -h$. A continuous choice over E_h would induce $2m_* = -\mathrm{id}$ on $\pi_1(E) = \mathbb{Z}^2$, which is impossible. A line pulled back from the base does not change these relative Picard classes.

For a symmetric rigidified line the cube theorem gives the parallelogram identity

$$(f + g)^*\Theta \otimes (f - g)^*\Theta \cong (f^*\Theta)^{\otimes 2} \otimes (g^*\Theta)^{\otimes 2}$$

for maps f, g to E. The rigidification removes the constant fiber factor. Apply the identity with $f = u + t$, $g = t$ to obtain (12.31).[2] This proves (12.32) with its square isomorphism.

On a u-fiber the ratio under a deck shift is the translate of $T_\xi^*\Theta \otimes \Theta^{-1}$, where T_ξ is translation by ξ. The degree-one polarization $E \to \mathrm{Pic}^0(E)$ is an isomorphism, so this line is nontrivial for $\xi \neq 0$ and has order two.

Finally the two zeros $0, -2t$ of the displayed section are distinct and simple when $2t \neq 0$. A square of a regular section has even vanishing order at every point. □

The isomorphism (12.31) is a line identity. The scalar equation $\vartheta(u)\vartheta(u + 2t) = (\vartheta(u + t)\vartheta(t))^2$ is false: its right-hand side vanishes identically at $t = 0$.

For an ordered flag, tensor (12.32) over its selected interacting cross-pairs, using the same fixed square isomorphism for every labeled pair. Every parenthesization contains each pair exactly once. Associativity and symmetry of tensor products of lines therefore give all flag coherences, and their squares are the corresponding coefficient-line identifications. For the negative kernel lines use the inverse roots. These constructions concern the specified arrow factors. Same-color and dynamical factors require their own lines.

11 THE COMPACT COEFFICIENT COMPARISON

The pair map, its simultaneous derived intersection, and its parameter module retain different data. A multiplicative coefficient comparison must specify which of these objects is its target. For the graph intersection it must also preserve the higher products, the trace character and derived specialization. The independent quadratic family cannot replace the actual elliptic displacement family by a change of base.

Let S be a smooth projective K3 surface and let E_{geom} be an elliptic curve. A reduced critical Hall theory on $S \times E_{\mathrm{geom}}$ requires its own compact moduli, orientations and extension maps. Extension flags

[2] J. S. Milne, *Abelian Varieties*, Section 5, the cube theorem and its symmetric-line consequences. Stacks Project, Theorem of the cube, tag 0BF4.

retain their common translation torsor. Their empty object lies over BE_{geom}, the classifying stack of these torsors. The quotient of nonempty spectral configurations constructed above does not identify this geometric base.

An operator representation has a different preservation requirement. Multiplication by t on the cochain complex $\mathbb{C}[t]$, with zero differential, is a cochain endomorphism. It is neither a unital algebra homomorphism nor a derivation. Thus an action on a state complex is governed by its endomorphism algebra. It need not act by automorphisms of an auxiliary multiplication on that state space.

Question 11.1. Construct the reduced oriented critical coefficient functor for the common-torsor extension geometry of $S \times E_{\mathrm{geom}}$. Specify its extension pullbacks and supported pushforwards. Give a map to a spectral base and determine its recipient: the product coefficient cone, the simultaneous pair intersection, or an independently defined state complex.

Can a multiplicative comparison retain the ordered coefficient lines, their ambient images, the class $[m_3]$ and $\delta_H G_2 = h m_3$? Can it extend over the whole spectral elliptic curve while preserving the exceptional fibers and the halving cover with its action on root choices? For a state representation, construct the corresponding supported operators and identify its marked reduced state.

A regular multiplicative comparison cannot replace G_2 by G_2/h without changing the coefficient ring. The cyclic trace of the spectral model also requires a comparison with compact reduced integration. The latter has its own dual-number-valued state, as expressed by the off-diagonal module coefficient in Section 9. The definition of reduced integration retains this distinction already at the numerical level.[3]

Higher comparison terms must be constructed in the actual supported Hom complex. The bar residual criterion at the end of Chapter 11 applies once that complex and its cohomology projection are supplied. The finite incidence contractions compute the operations of their own coefficient algebras.

Finally, the root line on a fixed elliptic curve is not yet a weight-five automorphic line. The Igusa comparison requires a genus-two modular family and an identification of its geometric line and section with the denominator line and its automorphic section. The local theta factors and the chosen graph trace specify inputs for this construction, rather than the required modular comparison.

[3] G. Oberdieck and J. Shen, *Reduced Donaldson–Thomas invariants and the ring of dual numbers*, Introduction. See the definition of the reduced integration map.

Chapter 13

Critical completion and constructible coefficients

The incidence equations of Chapter 12 define a coherent algebra. A vanishing-cycle coefficient requires a potential. For the triangle, three equations constrain two coordinates, so the given derived zero locus cannot itself carry a nondegenerate (-1)-shifted form. One multiplier for each equation produces a critical space. Its additional directions determine the coefficient to be compared with compact Hall geometry.

We first construct this completion, including its coordinate changes, orientation and constraint correspondences. The function λz^m then separates three kinds of data: coherent multiplicity, transverse vanishing monodromy and compact integration. For the triangle, we determine the entire rational central perverse sheaf and its two monodromy summands. One nonzero summand has zero compact integral. The displacement family also retains a nontrivial specialization map when nearby monodromy is the identity.

All schemes are over $\mathbb{C}$. Differentials have degree 1, and $C[r]^i = C^{i+r}$. Fix a smooth separated scheme B of finite type and pure dimension m, a rank-r vector bundle $V \to B$, and a section s of V. Its derived zero locus $Z = Z^{\mathrm{der}}(s)$ has the Koszul algebra $\mathrm{Sym}_{O_B}(V^\vee[1])$, with differential given by contraction with s. Write $Z_{\mathrm{cl}} = t_0 Z$ for its classical truncation.

The cotangent complex $\mathbb{L}_Z$ represents derived first-order differentials. Its dual is

$$\mathbb{T}_Z = R\mathcal{H}om(\mathbb{L}_Z, O_Z).$$

A nondegenerate (-1)-shifted two-form induces a quasi-isomorphism $\mathbb{T}_Z \simeq \mathbb{L}_Z[-1]$.

1 The tangent complex and the critical space

Proposition 1.1. The cotangent complex of Z, restricted to Z_{cl}, has the two-term presentation

$$\mathbb{L}_Z|_{Z_{\mathrm{cl}}} = \left[V^\vee|_{Z_{\mathrm{cl}}} \xrightarrow{ds} \Omega_B|_{Z_{\mathrm{cl}}}\right] \quad \text{in degrees } -1, 0. \tag{13.1}$$

If Z_{cl} is nonempty and Z has a nondegenerate (-1)-shifted two-form, then $m = r$. In particular, the fixed-parameter derived incidence scheme of a connected graph with a cycle, using the pair sections $\vartheta(u_e)\vartheta(u_e + h)$, has no such form.

Proof. The free Koszul presentation gives (13.1). At any classical point its Euler rank is $m - r$. Nondegeneracy would identify $\mathbb{T}_Z$ with $\mathbb{L}_Z[-1]$. Duality preserves Euler rank and this shift reverses it, so $m - r = -(m - r)$.

For a connected graph on n vertices and e edges, quotienting simultaneous spectral translation leaves $m = n - 1$ coordinates and $r = e$ equations. The incidence locus contains the configuration with every difference zero. Thus $m - r = -\beta$, where $\beta = e - n + 1$ is the number of independent cycles. □

Nonemptiness is necessary: the section 1 of the trivial line over $\mathbb{A}^2$ has empty derived zero locus, since its Koszul algebra has $d\epsilon = 1$. Nondegeneracy on the empty scheme imposes no rank equality.

This calculation concerns the specified derived enhancement. It permits another critical enhancement of the same classical support. It also permits an absolute parameter family with a different Euler rank. For a displacement parameter h, the assertion applies fiberwise, or to the relative cotangent complex over the h-curve. Including h among the coordinates adds one to the rank calculation.

The shifted cotangent space is $T^*[-1]Z = \operatorname{Spec}_Z \operatorname{Sym}_{O_Z}(\mathbb{T}_Z[1])$.

Let $\mathcal{M} = \operatorname{Tot}_B(V^\vee)$, with projection π. The evaluation pairing defines the regular function

$$F(x, \lambda) = \langle \lambda, s(x) \rangle.$$

Its derived critical locus, denoted $\widehat{Z}$, is the derived zero locus of dF on $\mathcal{M}$.

THEOREM 1.2. There is an equivalence of (-1)-shifted symplectic derived schemes

$$\widehat{Z} = \operatorname{Crit}^{\mathrm{der}}(F) \simeq T^*[-1]Z.$$

In local coordinates $x_1, \dots, x_m$ and a frame $s_1, \dots, s_r$, its algebra is

$$\widehat{A} = O_B[\lambda_1, \dots, \lambda_r] \otimes \Lambda(\xi_1, \dots, \xi_m, \epsilon_1, \dots, \epsilon_r),$$

where $|\lambda_a| = 0$, $|\xi_i| = |\epsilon_a| = -1$, and

$$d\epsilon_a = s_a, \qquad d\xi_i = \sum_a \lambda_a \partial_i s_a, \qquad dx_i = d\lambda_a = 0. \tag{13.2}$$

The inclusion of the original Koszul algebra represents $T^*[-1]Z \to Z$. Setting $\lambda_a = \xi_i = 0$ represents its zero section. Their composite on Z is the identity.

Proof. The equations in (13.2) are the partial derivatives of F with respect to the coordinates of $\mathcal{M}$. They therefore give its critical Koszul algebra. They also present $\operatorname{Sym}_{O_Z}(\mathbb{T}_Z[1])$: the shifted duals of the odd Koszul generators give the degree-zero multipliers, and those of the base coordinates give the ξ_i.

The common presentation identifies the standard shifted cotangent form with the critical form. This is the local comparison of Kinjo, Lemma 2.7.[1]

The two displayed algebra maps commute with the differential and have the stated composite. The coordinate changes below give their descent. □

2 DESCENT AND ORIENTATION

The multipliers transform in the dual bundle. The odd base coordinates must transform as well. Let $s' = G(x)s$ for an invertible matrix of regular functions. Use column vectors for s, λ, ϵ and put

$$\lambda' = G^{-T}\lambda, \qquad \epsilon' = G\epsilon, \qquad \xi_i' = \xi_i + \lambda'^T(\partial_i G)\epsilon. \tag{13.3}$$

PROPOSITION 2.1. Equations (13.3), together with ordinary cotangent coordinate changes on B, glue the critical algebra and its shifted symplectic form. Adding one coordinate z and one equation $s_{r+1} = z$ leaves the derived zero locus unchanged and adds the hyperbolic term $\lambda_{r+1}z$ to its critical potential.

[1] T. Kinjo, *Dimensional reduction in cohomological Donaldson–Thomas theory*, arXiv:2102.01568v2, Lemma 2.7. For the shifted cotangent form, see D. Calaque, *Shifted cotangent stacks are shifted symplectic*. arXiv:1612.08101v2, Theorem 2.2.

Proof. The identity $\lambda^T s = \lambda'^T s'$ preserves F. Moreover,

$$d\epsilon' = Gs = s', \qquad d\xi'_i = \lambda'^T G\partial_i s + \lambda'^T(\partial_i G)s = \lambda'^T \partial_i s'.$$

Thus the correction in (13.3) is exactly the product-rule term required for a cochain map.

In the critical-chart convention the odd Liouville form is

$$\theta = \sum_i \xi_i \, d_{\mathrm{dR}} x_i + \sum_a \epsilon_a \, d_{\mathrm{dR}} \lambda_a.$$

Since $d_{\mathrm{dR}}\lambda' = G^{-T} d_{\mathrm{dR}}\lambda - G^{-T}(d_{\mathrm{dR}} G)^T \lambda'$, the extra term in $\epsilon'^T d_{\mathrm{dR}}\lambda'$ cancels the correction in $\sum_i \xi'_i d_{\mathrm{dR}} x_i$. Hence θ, and therefore its de Rham differential, is preserved. Under two frame changes the correction satisfies the product rule for their product matrix, proving the cocycle.

For the additional presentation pair, the four new generators have differential

$$d\epsilon_{r+1} = z, \qquad d\xi_z = \lambda_{r+1}.$$

Their algebra is the Koszul resolution of the origin in $\mathbb{A}^2_{z,\lambda_{r+1}}$. It contracts to $\mathbb{C}$. This argument does not justify deleting an original equation. □

An orientation of a critical locus is a square root of its critical canonical line on the reduced support. A global critical chart $\mathrm{Crit}(F) \subset M$ identifies that line with $K_M^{\otimes 2}$ on this support, so it supplies the root K_M. If $B \to H$ is smooth and the construction is relative to H, the cotangent sequence of $M \to B$ gives

$$K_{M/H} \cong \pi^*(K_{B/H} \otimes \det V). \tag{13.4}$$

Restricted to the reduced critical locus, its square is the critical canonical line.

The corresponding absolute formula uses K_B. These are orientations for the respective critical charts. Identifying them with the intrinsic shifted-cotangent orientation uses the determinant conventions of the comparison.

For the global spectral family, let E be an elliptic curve, $\Theta = \mathcal{O}_E(0)$ with its divisor section ϑ, and $B = E^{n-1} \times E_h$. For an oriented graph edge e, let $u_e : B \to E$ be its vertex difference after quotienting simultaneous translation. Define

$$\mathcal{S}_e = u_e^*\Theta \otimes (u_e + h)^*\Theta, \qquad s_e = \vartheta(u_e)\vartheta(u_e + h), \qquad V = \bigoplus_e \mathcal{S}_e.$$

The pairing with $s = (s_e)$ defines a global potential on $\mathrm{Tot}_B(V^\vee)$.

Its relative critical locus uses derivatives along B/E_h and the multipliers. The same local construction and descent give the relative shifted cotangent completion. The absolute critical locus also has an odd coordinate ξ_h with $d\xi_h = \partial_h F$ in a parameter chart. These are two different enhancements of the family.

Formula (13.4) uses $\det V = \bigotimes_e \mathcal{S}_e$. It needs no square root of an individual $\mathcal{S}_e$. This critical orientation does not decide whether an individual pair line has a root over the displacement base. It does not decide whether the distinguished section is a square.

3 Adding constraints

Fix finitely many indexed line bundles L_a over the same smooth base B, with sections s_a. For a subset I put $V_I = \bigoplus_{a \in I} L_a$, and let Z_I be the derived zero locus of $(s_a)_{a \in I}$. Let $\widehat{A}_I$ denote its critical algebra. If $I \subset J$, forgetting equations defines $f_{JI} : Z_J \to Z_I$. Cotangent pullback gives the correspondence

$$T^*[-1]Z_I \longleftarrow Z_J \times_{Z_I} T^*[-1]Z_I \longrightarrow T^*[-1]Z_J, \tag{13.5}$$

where all fiber products of derived schemes are derived.

THEOREM 3.1. The correspondence (13.5) is Lagrangian for the difference of the two shifted symplectic forms. In a common affine chart its middle algebra is

$$W_{I,J} = \widehat{A}_I \otimes \Lambda(\epsilon_a : a \in J \setminus I), \qquad d\epsilon_a = s_a.$$

For $I \subset J \subset K$, composition has the equivalence

$$W_{I,J} \otimes^{\mathbb{L}}_{\widehat{A}_J} W_{J,K} \simeq W_{I,K}. \tag{13.6}$$

These equivalences agree for every nested flag of subsets.

Proof. The left algebra map is inclusion from $\widehat{A}_I$. The map from $\widehat{A}_J$ sets $\lambda_a = 0$ for $a \in J \setminus I$ and retains their ϵ_a. It fixes all other generators. Equation (13.2) verifies that this is a cochain map, including the differentials of the ξ_i.

The pulled-back critical forms agree because each extra term contains $d_{\mathrm{dR}}\lambda_a = 0$. The correspondence is the shifted conormal of the graph of f_{JI}. Its Lagrangian nondegeneracy is Calaque's shifted conormal theorem, Theorem 2.8 of the cited paper.

As a module over $\widehat{A}_J$, the algebra $W_{J,K}$ has a finite filtration by the exterior length of its new generators, with free graded pieces. It is therefore K-flat, so ordinary tensor product computes the derived tensor product in (13.6). It sets to zero precisely the multipliers in $K \setminus I$ and retains precisely their odd generators and equations. This is $W_{I,K}$.

Every longer flag gives the same algebra. Reordering the odd generators uses the exterior Koszul signs. Frame changes respecting the indexed line summands preserve these subset maps. Proposition 2.1 therefore glues the equivalences. An arbitrary change mixing an included and an excluded equation changes the specified flag. □

This constructs geometric correspondences and their composition. A map on their unintegrated vanishing-cycle coefficients still requires a coefficient operation with the appropriate support and orientation. The geometric composition alone does not supply that operation.

4 The completed triangle

For a fixed complex parameter h, take the local shifted pair equations

$$s_1 = x(x+h), \qquad s_2 = y(y+h), \qquad s_3 = (x+y)(x+y+h).$$

They are the formal equations of the selected forward pair sections $\vartheta(u)\vartheta(u+h)$.

For the two-arrow weights of Yang–Zhao, the original equal equivariant parameter is t and $h = 2t$.[2]

Integrating an invariant differential gives a holomorphic local logarithm. The theta sections are their linear factors times invertible holomorphic functions. Rescaling the dual multipliers by those functions gives these equations and an analytic isomorphism of their potential germs near zero. The same changes may also be completed formally.

The invertible row change replacing s_3 by $(s_3 - s_1 - s_2)/2 = xy$ changes the full potential to

$$W_h = Px(x+h) + Qy(y+h) + Txy, \qquad P = p + r, \quad Q = q + r, \quad T = 2r. \tag{13.7}$$

[2] Y. Yang and G. Zhao, *Quiver varieties and elliptic quantum groups*. arXiv:1708.01418v1, Remark 2.2(2) and equation (3). Only these ordered intercolor pair factors are used here.

Thus its critical algebra has five odd generators and

$$d\epsilon_P = x(x+h), \qquad d\epsilon_Q = y(y+h), \qquad d\epsilon_T = xy,$$
$$d\xi_x = (2x+h)P + Ty, \qquad d\xi_y = (2y+h)Q + Tx. \tag{13.8}$$

Proposition 4.1. For $h \neq 0$, the reduced critical locus of W_h is the disjoint union of the three affine lines

$$(x, y; P, Q, T) = (0, 0; 0, 0, T), \quad (-h, 0; 0, T, T), \quad (0, -h; T, 0, T).$$

The Hessian has rank four at every point of these lines. For $h = 0$, the reduced critical support is $\mathbb{A}^3_{P,Q,T}$ at $x = y = 0$. Put $\Delta = T^2 - 4PQ$. Its tangent and obstruction dimensions are

multiplier stratum	rank Hess W_0	$(\dim H^0\mathbb{T}, \dim H^1\mathbb{T})$
$\Delta \neq 0$	2	$(3, 3)$
$\Delta = 0,\ (P, Q, T) \neq 0$	1	$(4, 4)$
$(P, Q, T) = 0$	0	$(5, 5)$.

Proof. The first three equations in (13.8) give the three listed pairs (x, y) when $h \neq 0$. The last two then give the listed multiplier lines. The principal Hessian minor on x, y, P, Q has determinant h^4 on every line. Its tangent vector is in the Hessian kernel, so the rank is exactly four.

At $h = 0$, the equations $x^2 = y^2 = xy = 0$ force $x = y = 0$ on the reduced support. The only nonzero Hessian block is

$$H_{\text{normal}} = \begin{pmatrix} 2P & T \\ T & 2Q \end{pmatrix}, \qquad \det H_{\text{normal}} = -\Delta.$$

The critical tangent complex is the Hessian map between two five-dimensional spaces. Its kernel and cokernel have dimension five minus this rank. □

The completion has introduced noncompact multiplier directions. Its critical tangent dimensions therefore differ from the cohomology dimensions of the coherent incidence algebra. Vanishing cycles on those multiplier directions provide a third, explicitly computable coefficient object.

5 Vanishing cycles and a single multiplier

For a complex algebraic variety Y, write $\mathrm{D}^b_c(Y; \mathbb{Q})$ for the bounded derived category of rational sheaves on its analytic space, with finite-dimensional algebraically constructible cohomology. All constructible coefficients below lie in this category.

Write $\varphi_f = \phi_f[-1]$ for shifted vanishing cycles, with $\varphi_0 = \mathrm{id}$. On a smooth complex N-fold M, put

$$\mathcal{P}_f = \varphi_f \mathbb{Q}_M[N].$$

This is the perverse normalization. Its stalk at a critical point p is the reduced cohomology of a local Milnor fiber, shifted so that

$$\mathcal{H}^k(\mathcal{P}_f)_p = \widetilde{H}^{k+N-1}(\mathrm{MF}_p(f); \mathbb{Q}). \tag{13.9}$$

When Tate twists are displayed, coefficients are understood in the corresponding mixed-Hodge or monodromic refinement.

The middle perverse convention places a local system on a smooth complex d-fold in degree $-d$. The functor φ_f is exact for this perverse structure. For an open inclusion j, the middle extension $j_{!*}K$ is the perverse image of ${}^pH^0(j_!K) \to {}^pH^0(Rj_*K)$. Here ${}^pH^0$ is degree-zero perverse cohomology and K is perverse on the open stratum. This extension has no nonzero perverse subobject or quotient supported on the boundary.

Hyperbolic stabilization of a critical chart, with the induced orientation, preserves this normalized coefficient. The comparison includes the determinant double cover of the added quadratic form.[3] In the coordinate-equation stabilization above, the specified hyperbolic pair gives that orientation.

We first justify the use of global fibers in the homogeneous examples below. Let g be a nonconstant homogeneous polynomial of positive degree on $\mathbb{C}^N$ and suppose that $g^{-1}(1)$ is smooth. Dilation identifies

$$g^{-1}(\delta) \cap B_\varepsilon \cong g^{-1}(1) \cap B_{\varepsilon|\delta|^{-1/\deg g}}.$$

The squared norm on the smooth affine fiber is proper and semialgebraic. Its critical-value set is semialgebraic and, by Sard's theorem, has measure zero, hence is finite. Above its last critical value, increasing the sublevel radius does not change homotopy type. Thus the local Milnor fiber at the origin has the homotopy type of $g^{-1}(1)$. This argument allows a nonisolated critical locus.

Let $f_m(\lambda, z) = \lambda z^m$ on $\mathbb{A}^2$, for $m \geq 1$. For $m \geq 2$ its reduced critical support is the λ-axis. Let $j : \mathbb{C}^* \hookrightarrow \mathbb{C}$ and $i : \{0\} \hookrightarrow \mathbb{C}$ be its two strata. On the punctured axis define the local system L_m by the reduced cyclic permutation representation

$$(L_m)_\lambda = \{(a_1, \dots, a_m) \in \mathbb{Q}^m : \textstyle\sum_j a_j = 0\}, \qquad T(a_1, \dots, a_m) = (a_m, a_1, \dots, a_{m-1}). \quad (13.10)$$

Reversing the loop orientation replaces T by its inverse. For $m = 1$ put $L_1 = 0$.

THEOREM 5.1. On its reduced support, the underlying rational perverse sheaf of f_m is

$$\mathcal{P}_{f_m} \cong j_{!*}L_m[1] \oplus i_*\mathbb{Q}. \quad (13.11)$$

The extension is clean: $j_!L_m[1] \simeq Rj_*L_m[1]$. At the origin the only nonzero rational stalk group is $\mathbb{Q}$ in degree zero.

Proof. For $\lambda \neq 0$, the transverse fiber has m points. Their reduced degree-zero cohomology gives (13.10). The smooth axis supplies the shift by one. The characteristic polynomial of T on this space is

$$1 + t + \cdots + t^{m-1}, \qquad \det(1 - T) = m.$$

Consequently the punctured-disc cochain complex $[L_m \xrightarrow{T-1} L_m]$ is acyclic. This proves $j_!L_m \simeq Rj_*L_m$. Both its stalk and costalk at the origin vanish, so adjunction gives no morphisms, in any degree, between this extension and an origin-supported complex. The localization triangle therefore splits off the origin part.

The homogeneous fiber $\lambda z^m = 1$ is $\mathbb{C}^*$, parametrized by $z \neq 0$. The preceding homogeneous argument identifies the local Milnor fiber with it up to homotopy. Its only reduced cohomology is $H^1(\mathbb{C}^*; \mathbb{Q}) = \mathbb{Q}$. Equation (13.9) puts this in degree zero. For $m = 1$ the critical point is isolated and the same calculation gives the entire sheaf. This proves (13.11) in rational coefficients. □

[3] C. Brav, V. Bussi, D. Dupont, D. Joyce and B. Szendrői, *Symmetries and stabilization for sheaves of vanishing cycles*. arXiv:1211.3259, Theorem 5.4.

The circle complex also gives $R\Gamma_c(\mathbb{C}^*, L_m) = 0$, by duality or by the same invertibility of $T - 1$. Thus compact integration over the multiplier retains the origin term alone. The unintegrated local system still has rank $m - 1$ and cyclic monodromy. The coherent algebra $\mathbb{C}[z]/(z^m)$, these vanishing cycles and their compact integral are three different coefficient objects.

6 The triangle stalks

For the polynomial $W_h = Px(x + h) + Qy(y + h) + Txy$ of (13.7), write $\mathcal{P}_h = \varphi_{W_h}\mathbb{Q}_{\mathbb{A}^5}[5]$. At $h = 0$ it is supported on $\mathbb{A}^3_{P,Q,T}$. The discriminant $\Delta = T^2 - 4PQ$ defines the quadratic rank strata.

Theorem 6.1. At a point $(0, 0; P, Q, T)$ the nonzero rational stalk groups of $\mathcal{P}_0$ are as follows:

stratum	cohomological degree	rank
$\Delta \neq 0$	-3	1
$\Delta = 0,\ (P, Q, T) \neq 0$	-2	1
$(P, Q, T) = 0$	-1	1.

For $h \neq 0$, its restriction to each of the three critical lines is isomorphic to $\mathbb{Q}_{\mathbb{A}^1}[1]$.

Proof. At quadratic rank two, the holomorphic Morse lemma with parameters gives a nondegenerate quadratic in two normal variables and three smooth spectator coordinates. Its Milnor fiber has reduced cohomology in degree one. The shift in (13.9) therefore gives $1 - (5 - 1) = -3$.

At rank one, assume $P \neq 0$. The patch with $Q \neq 0$ is identical after interchanging x, y. Completing the square gives

$$W_0 = P\left(x + \frac{T}{2P}y\right)^2 + \left(Q - \frac{T^2}{4P}\right)y^2.$$

Choose a local square root of the unit P. The new coefficient $s = Q - T^2/(4P)$ is a transverse parameter to the rank-one stratum. The germ becomes $u^2 + sv^2$, with two spectator coordinates.

Theorem 5.1 for sv^2 has reduced Milnor cohomology in degree one at its origin. Thom–Sebastiani for the additional nondegenerate square suspends it to degree two. Formula (13.9) then gives -2.

At the zero quadratic form, project the global fiber $W_0 = 1$ to (x, y). The point $(0, 0)$ is excluded. At every other point, $Px^2 + Txy + Qy^2 = 1$ is a nonzero affine linear equation in three multipliers, so its solutions form an affine plane. On $x \neq 0$ one solves for P, and on $y \neq 0$ one solves for Q. This is a locally trivial affine-bundle torsor over $\mathbb{C}^2 \setminus \{0\}$, with contractible fiber. It has the homotopy type of S^3.

The homogeneous argument applies to W_0 of degree three, so this is also the local Milnor homotopy type. Its unique reduced class gives degree $3 - (5 - 1) = -1$.

For $h \neq 0$, Proposition 4.1 gives a smooth critical line and a nondegenerate Hessian in four normal directions. The Morse lemma with parameters gives a rank-one local system shifted by one on that line. The line is simply connected, so this local system is constant. □

Stalk ranks alone do not describe how a perverse sheaf glues across these strata. Even on the open quadratic stratum the rank-one coefficient has nontrivial monodromy.

Proposition 6.2. On $\Delta \neq 0$, the normal vanishing cohomology of W_0 is the sign local system associated with the double cover

$$w^2 = \Delta. \tag{13.12}$$

It is nontrivial.

Proof. A change of orthonormal frame for a nondegenerate complex quadratic form acts on its one-dimensional vanishing cohomology by the determinant character of $O(2,\mathbb{C})$. The two choices of determinant frame give the double cover (13.12). The determinant of the normal Hessian is $-\Delta$ and multiplication of a square root by i identifies the two sign conventions. For an explicit loop, set $Q = 1$, $T = 0$ and move P once around zero. A local normal coordinate $\sqrt{P}\,x$ returns with the opposite sign. Reflection of that coordinate acts by -1 on the vanishing class. □

This double cover lies over multiplier space. It persists at zero spectral displacement and is independent of the cover $h = 2t$ used to take roots of spectral pair lines. It also coexists with the global critical orientation (13.4). An orientation and a constant trivialization of every normal vanishing-cycle local system are different data.

7 The central perverse sheaf

Put $B_{\mathrm{mult}} = \mathbb{A}^3_{P,Q,T}$ and $f = W_0 = Px^2 + Txy + Qy^2$. Regard $\Phi = \mathcal{P}_0$ as a perverse sheaf on its reduced support B_{mult}. Write $\mathcal{T}_f$ for raw geometric monodromy in the value of f, with no dimension-dependent sign. This is distinct from continuation in multiplier space and from monodromy in the displacement h. Let $D = \{\Delta = 0\}$, $D^\circ = D \setminus \{0\}$ and $U = B_{\mathrm{mult}} \setminus D$.

LEMMA 7.1. The automorphism $\mathcal{T}_f$ satisfies $\mathcal{T}_f^2 = 1$. The only nonzero stalk cohomology on each stratum is

stratum	stalk of Φ	$\mathcal{T}_f$
U	$H^{-3} = \mathbb{Q}$	$+1$
D°	$H^{-2} = \mathbb{Q}$	-1
$\{0\}$	$H^{-1} = \mathbb{Q}$	$+1$.

Proof. The path $(x, y) \mapsto (e^{\pi i t}x, e^{\pi i t}y)$, with P, Q, T fixed, rotates the value of f once. It preserves small balls centered on B_{mult}. Its endpoint is the involution $(x, y) \mapsto (-x, -y)$, and two circuits give the identity. Thus it defines the stated involution on vanishing cycles.

The local fibers are those of Theorem 6.1. At rank two, a nondegenerate binary quadratic form has Milnor fiber of homotopy type S^1. At rank one, completing a square gives $u^2 + sv^2$, with two additional smooth parameters. The fiber of sv^2 has homotopy type $\mathbb{C}^*$. Adding u^2 suspends its reduced cohomology. Its function monodromy on that cohomology is the product of $+1$ for sv^2 and -1 for u^2.

These are the Thom–Sebastiani monodromy conventions.[4]

At the origin, projection of $f^{-1}(1)$ to $\mathbb{C}^2 \setminus \{0\}$ is an affine-plane torsor: for each nonzero (x, y), the equation is a nonzero linear equation in P, Q, T. It is locally trivial and has contractible fibers. Hence $f^{-1}(1) \simeq S^3$. The homogeneous comparison proved in Section 5 applies to this cubic, whose fiber over one is smooth.

The involution covers the antipodal map of $\mathbb{C}^2 \setminus \{0\}$ and acts by $+1$ on H^3. At rank two its action on $H^1(S^1)$ is also $+1$. The shift $\phi_f[-1]\mathbb{Q}[5]$ places reduced Milnor cohomology of degree k in stalk degree $k - 4$. □

[4] C. Brav, V. Bussi, D. Dupont, D. Joyce and B. Szendrői, *Symmetries and stabilization for sheaves of vanishing cycles*. arXiv:1211.3259, Definition 2.12 and Theorem 2.13. Their twisted monodromy includes the sign $(-1)^{\dim U}$. The symbol $\mathcal{T}_f$ here is their untwisted function monodromy.

7.1 The two local systems

Consider the double covers

$$\nu : \widetilde{B}_{\mathrm{mult}} = \{w^2 = \Delta\} \longrightarrow B_{\mathrm{mult}}, \qquad \rho : \mathbb{A}^2 \longrightarrow D, \quad (a,b) \longmapsto (a^2, b^2, 2ab). \tag{13.13}$$

Let L_{det} be the sign local system of ν on U, and let L_{sq} be the sign local system of ρ on D°. The rank-two vanishing line is L_{det} by Proposition 6.2. On D°, the form is $(ax + by)^2$. The loop $P = e^{2\pi i t}$, $Q = T = 0$ reverses its chosen square coordinate and hence its suspended vanishing class. Since $D^\circ = (\mathbb{C}^2 \setminus \{0\})/\{\pm 1\}$ has fundamental group $\mathbb{Z}/2$, this identifies the rank-one vanishing line with L_{sq}.

Write $j : U \hookrightarrow B_{\mathrm{mult}}$, $k : D^\circ \hookrightarrow D$ and $i : D \hookrightarrow B_{\mathrm{mult}}$. For a pure-dimensional variety Y, the intersection complex IC_Y is the middle extension of the constant complex on its smooth locus, normalized by $\mathrm{IC}_Y|_{Y_{\mathrm{sm}}} = \mathbb{Q}[\dim Y]$. For a finite double-cover pushforward, the superscript $-$ denotes its deck anti-invariant direct summand.

7.2 The vertex and its involution

Lemma 7.2. The stalk of $\mathrm{IC}_{\widetilde{B}_{\mathrm{mult}}}$ at the origin has a copy of $\mathbb{Q}$ in each of degrees -3 and -1. Deck monodromy acts by $+1$ and -1 on them, respectively.

Proof. The matrix

$$\begin{pmatrix} 2P & T + w \\ T - w & 2Q \end{pmatrix}$$

has determinant $w^2 - \Delta$. It identifies $\widetilde{B}_{\mathrm{mult}}$ with the rank-at-most-one locus of two-by-two matrices. Choosing the line containing the image gives a proper resolution

$$r : \mathrm{Tot}_{\mathbb{P}^1}(\mathcal{O}(-1) \oplus \mathcal{O}(-1)) \longrightarrow \widetilde{B}_{\mathrm{mult}}.$$

It is an isomorphism away from the origin, with exceptional fiber $\mathbb{P}^1$.

Since $2 \dim \mathbb{P}^1 = 2 < 3$, the map is small, and $Rr_*\mathbb{Q}[3] = \mathrm{IC}_{\widetilde{B}_{\mathrm{mult}}}$. Proper base change gives the two stalk groups.[5]

The deck involution is matrix transpose. It exchanges the two small resolutions, so its action must be computed intrinsically. The link of the node is the circle bundle of $\mathcal{O}(-1,-1)$ over $\mathbb{P}^1 \times \mathbb{P}^1$. If a, b are the two degree-two generators, its Euler class is $a + b$ up to sign. The Gysin sequence gives

$$H^2(\mathrm{link}; \mathbb{Q}) = \frac{\mathbb{Q}a \oplus \mathbb{Q}b}{\mathbb{Q}(a+b)}.$$

Transpose interchanges the two factors and acts by -1 on this quotient. Intersection extension across an isolated threefold singularity retains this link group as stalk degree -1, and the constant link group as degree -3. The latter is invariant. □

Theorem 7.3. There is a decomposition of rational perverse sheaves

$$\Phi = \mathcal{P}_+ \oplus \mathcal{P}_-, \qquad \mathcal{P}_\pm = \mathrm{im}\, \frac{1 \pm \mathcal{T}_f}{2},$$

[5] M. A. de Cataldo and L. Migliorini, *The decomposition theorem, perverse sheaves and the topology of algebraic maps*. arXiv:0712.0349, Remark 4.2.4.

with

$$\mathcal{P}_+ \cong j_{!*}(L_{\det}[3]) \cong (R\nu_* \operatorname{IC}_{\widetilde{B}_{\mathrm{mult}}})^-, \tag{13.14}$$

$$\mathcal{P}_- \cong i_* k_{!*}(L_{\mathrm{sq}}[2]) \cong i_*(R\rho_* \mathbb{Q}_{\mathbb{A}^2}[2])^-. \tag{13.15}$$

Raw function monodromy acts by $+1$ and -1 on these summands, respectively.

Proof. Lemma 7.1 gives the two projectors in the perverse category. Vanishing cycles commute with Verdier duality, which replaces monodromy by its inverse. Each summand is therefore self-dual.

The plus summand restricts to $L_{\det}[3]$ on U and has zero stalks and costalks on D°. At the origin, its stalk has degree -1 and its costalk degree $+1$. It has no perverse quotient or subobject supported on the boundary. These are precisely the middle-extension conditions.

The minus summand has support D, restricts to $L_{\mathrm{sq}}[2]$, and has zero stalk and costalk at the origin. The same criterion identifies it.[6]

Finite pushforward is perverse exact. For ρ, the origin is a fixed point of the deck involution, which acts trivially on the constant stalk. Its anti-invariant stalk and, by duality, costalk vanish. This proves (13.15).

For ν, the anti-invariant stalk vanishes at every smooth branch point. Its origin stalk is $\mathbb{Q}$ in degree -1, with dual costalk in degree $+1$, by Lemma 7.2. Thus its anti-invariant pushforward satisfies the same middle-extension conditions as $\mathcal{P}_+$. □

8 Dimensional reduction and constraint maps

Return to a smooth complex m-fold B, a rank-r bundle V, and a section s. Put $M = \operatorname{Tot}_B(V^\vee)$, let $\pi : M \to B$ be its projection, and let $F = \langle \lambda, s \rangle$. Write $i : Z_{\mathrm{cl}} \hookrightarrow B$ for the ordinary zero locus with its underlying complex topology, and $\widetilde{i} : M|_{Z_{\mathrm{cl}}} \hookrightarrow M$. The symbol $R\pi_!$ means derived direct image with proper support. No properness of the multiplier bundle is assumed.

There is a natural coefficient map

$$\begin{aligned} \alpha_{\mathcal{G}} : R\pi_! \varphi_F \pi^* \mathcal{G} &\longrightarrow R\pi_! \widetilde{i}_* \widetilde{i}^* \pi^* \mathcal{G} \\ &\cong i_* i^* \mathcal{G}[-2r](-r), \qquad \mathcal{G} \in \mathrm{D}^b_c(B; \mathbb{Q}). \end{aligned} \tag{13.16}$$

To specify the first arrow, let $i_0 : F^{-1}(0) \hookrightarrow M$. The support description

$$\varphi_F \mathcal{H} = i_{0*} i_0^* R\Gamma_{\{\operatorname{Re} F \le 0\}} \mathcal{H}$$

gives its natural map to coefficients on the zero fiber. Restrict this map to $M|_{Z_{\mathrm{cl}}}$, then apply $R\pi_!$. The second arrow in (13.16) uses the complex orientation of the rank-r fibers.

THEOREM 8.1. The map $\alpha_{\mathcal{G}}$ is an isomorphism, natural in $\mathcal{G}$. Thus

$$R\pi_! \varphi_F \pi^* \mathcal{G} \simeq i_* i^* \mathcal{G}[-2r](-r). \tag{13.17}$$

For rational sheaves the Tate twist is omitted. The comparison commutes with raw function monodromy, which is the identity on the right-hand side. For two linear potentials it carries the Thom–Sebastiani map to the compactly supported Künneth map.

[6] L. G. Maxim and J. Schürmann, *Constructible sheaf complexes in complex geometry and applications*. arXiv:2105.13069, Sections 2.2, 2.4 and 4.2, for intermediate extension, finite pushforward and Verdier duality.

Proof. Where a component of s is nonzero, replace its multiplier coordinate by F. A coefficient pulled back from B is constant in this coordinate, so its vanishing cycles are zero. Thus the source is supported over Z_{cl}. On a trivializing chart, the linear-potential reduction theorem identifies the map just defined with an isomorphism.[7]

The local cancellation can be seen in one multiplier. Homogeneity makes the nearby term a derived direct image from its punctured multiplier line. For $j : \mathbb{C}^* \hookrightarrow \mathbb{C}$ and a local system L with fiber A and monodromy T, both $R\Gamma(\mathbb{C}^*, L)$ and $(Rj_*L)_0$ are the circle complex $[A \xrightarrow{T-1} A]$. Their restriction map is an isomorphism. The localization triangle gives $R\Gamma(\mathbb{C}, j_!L) = 0$, and Verdier duality gives $R\Gamma_c(\mathbb{C}, Rj_*L) = 0$. This also holds for bounded complexes with locally constant cohomology and requires no finite-order hypothesis on T.

For several multipliers, the blow-up of the zero section is the tautological line over the projective multiplier bundle. On its ath chart the potential is $\lambda(s_a + \sum_{b\neq a} s_b v_b)$. The same punctured-line cancellation, with proper base change for the blow-up, is the local reduction argument. The remaining coefficient is $i_*i^*\mathcal{G} \otimes R\Gamma_c(\mathbb{A}^r, \mathbb{Q})$, and the fiber complex is $\mathbb{Q}[-2r](-r)$.

The support map is invariant under bundle frames, so these local comparisons glue. Its monodromic refinement has constant value-parameter monodromy on the reduced side. The support inclusion defining Thom–Sebastiani commutes with restriction, giving the last assertion. For $r = 0$, the formula is $\varphi_0 = \mathrm{id}$. □

The pullback condition is essential. For $F = t$ on $\mathbb{A}^1$ and the origin-supported coefficient $\mathcal{H} = i_{0*}\mathbb{Q}$, nearby cycles vanish and $\varphi_t\mathcal{H} = \mathcal{H}$. Its compact integral is $\mathbb{Q}$, whereas the coefficient section 1 over a point has empty zero locus. Replacing $\pi^*\mathcal{G}$ by this arbitrary coefficient would therefore make (13.17) false.

Corollary 8.2. For the subset I of a fixed collection of sections, let $r_I = |I|$, let π_I be its multiplier bundle and let $i_I : (Z_I)_{\mathrm{cl}} \hookrightarrow B$. The normalized integrated coefficient

$$C_I = R\pi_{I!}\varphi_{F_I}\mathbb{Q}[2r_I](r_I) \simeq i_{I*}\mathbb{Q}_{(Z_I)_{\mathrm{cl}}} \tag{13.18}$$

has restriction maps $C_I \to C_J$ for $I \subset J$, compatible with every nested flag.

Proof. The inclusion $a : (Z_J)_{\mathrm{cl}} \hookrightarrow (Z_I)_{\mathrm{cl}}$ has the ordinary coefficient unit

$$\mathbb{Q}_{(Z_I)_{\mathrm{cl}}} \longrightarrow Ra_*\mathbb{Q}_{(Z_J)_{\mathrm{cl}}}.$$

Push it to B and transport it through (13.18). The units for successive inclusions compose to the unit for their composite. This proves the flag identities. The normalization removes the separate shift and twist for every added multiplier. □

These are maps after multiplier integration. They are not unintegrated vanishing-cycle Gysin maps on a sheaf-extension stack. In particular, they do not equip the geometric correspondences of Theorem 3.1 with a Hall product before integration.

The right side of (13.17) sees the underlying classical space. It retains neither nilpotent thickening nor the negative coherent Koszul cohomology as such. For $s = z^m$, this agrees with Theorem 5.1: the integral is independent of m, although the transverse local system and coherent length both depend on m.

[7] B. Davison, *The critical CoHA of a quiver with potential*. arXiv:1311.7172, Theorem A.1, Corollary A.7 and Proposition A.8.

For $f_m = \lambda z^m$, the perverse normalization is $[2]$. Dimensional reduction therefore makes its compact integral $\mathbb{Q}(-1)$ in degree zero in the monodromic mixed-Hodge refinement. The clean punctured local system has zero compact integral. Consequently the origin term of Theorem 5.1 has that same Tate class. This specifies the Hodge class without identifying local and global mixed-Hodge structures merely from a homotopy equivalence of their fibers.

For a fixed triangle parameter, $m = 2$, $r = 3$, and the perverse normalization is $[5]$. Hence

$$R\Gamma_c(\mathrm{Crit}(W_h), \mathcal{P}_h) \simeq R\Gamma_c((Z_h)_{\mathrm{cl}}, \mathbb{Q})[-1](-3). \tag{13.19}$$

At $h = 0$ the right side has one class, and at a generic nonzero displacement it has three classes, all in degree one.

For the global elliptic pair sections, take $B = E_x \times E_y$ and the three differences x, y, $x + y$. The first two pair sections give $x, y \in \{0, -h\}$ when $h \neq 0$. Three of the four pairs satisfy the last equation for every h. At $(-h, -h)$ that section has value $\vartheta(-2h)\vartheta(-h)$. It vanishes precisely at nonzero two-torsion displacements. Thus the global version of (13.19) has four classes there.

The compactness of E^2 does not make its multiplier bundle proper. The symbol ! in these formulas remains essential. Verdier duality converts it to an ordinary direct image only while changing the coefficients to their duals.

9 Compact integration and monodromy

PROPOSITION 9.1. Compact integration over multiplier space satisfies

$$R\Gamma_c(B_{\mathrm{mult}}, \mathcal{P}_-) = 0, \qquad R\Gamma_c(B_{\mathrm{mult}}, \mathcal{P}_+) \cong \mathbb{Q}[-1].$$

In particular, it annihilates a nonzero perverse sheaf.

Proof. By (13.15), the first integral is the anti-invariant part of $R\Gamma_c(\mathbb{C}^2, \mathbb{Q})[2]$. The only compact cohomology of $\mathbb{C}^2$ is $H^4_c = \mathbb{Q}$, and $(a, b) \mapsto (-a, -b)$ preserves its complex orientation. That anti-invariant part is zero.

For the plus summand, the resolution in Lemma 7.2 retracts to its exceptional $\mathbb{P}^1$. Ordinary global cohomology of its pushforward therefore maps isomorphically to the origin stalk. This restriction is intrinsic and commutes with the deck involution on the intersection complex. No action on a chosen resolution is required. Consequently $R\Gamma(B_{\mathrm{mult}}, \mathcal{P}_+) = \mathbb{Q}[1]$. Verdier duality and self-duality give the second formula. □

These statements are in rational constructible sheaves. For the total complex Φ, dimensional reduction gives $\mathbb{Q}[-1](-3)$ with its monodromic mixed Hodge normalization, as in (13.19) and Theorem 8.1. This does not specify separate Hodge lifts of (13.14) and (13.15).

The same reduction has a useful monodromy consequence. Let X be smooth, $s_1, \dots, s_r$ regular functions, $\pi : X \times \mathbb{A}^r \to X$, and $F = \sum_{a=1}^r \lambda_a s_a$.

PROPOSITION 9.2. Let $\mathcal{K}$ be a monodromy-invariant direct summand of $\phi_F[-1]\mathbb{Q}$ on which $\mathcal{T}_F - 1$ is invertible. Then $\pi_!\mathcal{K} = 0$.

Proof. The natural reduction map identifies $\pi_!\phi_F[-1]\mathbb{Q}$ with the constant coefficient on $Z = \{s_a = 0\}$, shifted by $-2r$. It commutes with raw function monodromy, and the target monodromy is the identity. Hence on the direct summand $\pi_!\mathcal{K}$ the endomorphism $\mathcal{T}_F - 1$ is both zero and invertible.

The required equivariance is part of the natural monodromic reduction map.[8] □

The pullback hypothesis and its counterexample are those of Theorem 8.1.

9.1 The integral cyclic boundary

Over $\mathbb{Z}$, division by two is unavailable. The cyclic local system of the monomial example gives a direct test of the resulting boundary terms. For $m \geq 2$, let

$$I_m = \{(a_1, \dots, a_m) \in \mathbb{Z}^m : \sum a_i = 0\},$$

with cyclic permutation T_m.

Proposition 9.3. The map $T_m - 1 : I_m \to I_m$ is injective and has cokernel $\mathbb{Z}/m$.

Proof. Let C_m be the cyclic group of order m. Multiplication by $t - 1$ identifies $\mathbb{Z}[C_m]/\mathbb{Z}(1 + \cdots + t^{m-1})$ with its augmentation ideal I_m. The invariant vectors in I_m are constant vectors with sum zero, hence vanish. Taking coinvariants in the quotient presentation gives $\mathbb{Z}/(1 + \cdots + 1) = \mathbb{Z}/m$. Equivalently, the Smith factors are $1, \dots, 1, m$. □

For the associated local system on a punctured disk, the boundary stalk of its derived direct image is the circle complex $[I_m \xrightarrow{T_m - 1} I_m]$ in degrees 0, 1. It is nonzero over $\mathbb{Z}$ and vanishes after tensoring with $\mathbb{Q}$. In particular, a sign local system has an integral $\mathbb{Z}/2$ boundary term. This computation does not give an integral direct-sum version of Theorem 7.3.

10 Three branches specializing to one point

Work over a small complex disk $\mathbb{D}$ around $h = 0$. The ordinary total zero locus of the polynomial triangle has ring

$$\mathbb{C}[h, x, y]/(x(x + h), y(y + h), xy).$$

Define

$$z_0 = h + x + y, \qquad z_1 = -x, \qquad z_2 = -y.$$

Then $h = z_0 + z_1 + z_2$, and its ideal is (z_0z_1, z_0z_2, z_1z_2). Thus the total classical space is reduced: it is the union of three coordinate axes. Each axis maps isomorphically to the parameter disc and all three meet at its origin.

The nonreduced central scheme fiber has just one underlying topological point.

Let $f : Z_{\mathrm{cl}} \to \mathbb{D}$ be this finite map and put $\mathcal{V} = f_*\mathbb{Q}_{Z_{\mathrm{cl}}}$. Write $i : \{0\} \hookrightarrow \mathbb{D}$ for the origin inclusion.

Theorem 10.1. There is an exact sequence of constructible sheaves

$$0 \longrightarrow \mathcal{V} \longrightarrow \mathbb{Q}^3_{\mathbb{D}} \xrightarrow{(a,b,c)\mapsto(b-a,c-a)} i_*\mathbb{Q}^2 \longrightarrow 0. \tag{13.20}$$

The nearby cycles at zero are $\mathbb{Q}^3$ with identity monodromy. The specialization map is

$$\mathbb{Q} \longrightarrow \mathbb{Q}^3, \qquad a \longmapsto (a, a, a), \tag{13.21}$$

[8] B. Davison, *The critical CoHA of a quiver with potential*. arXiv:1311.7172, Theorem A.1 and Corollary A.7. In the proof of the latter, the comparison over the punctured value parameter is constant on the reduced side.

and consequently

$$\varphi_h(\mathcal{V}[1])_0 \cong \mathbb{Q}^2. \tag{13.22}$$

Branch permutations induce the standard two-dimensional representation of S_3 on this last vector space. There is no nonzero subsheaf of $\mathcal{V}$ supported only at the origin.

Proof. Normalization separates the three axes into three discs. A locally constant function descends exactly when its three values at the common point agree. On the punctured disc normalization is an isomorphism on each branch. At the origin its coefficient map is the diagonal inclusion. These stalk calculations prove (13.20) and (13.21).

Each branch is a section over the whole disc, so continuation around zero fixes it. The vanishing-cycle triangle therefore identifies (13.22) with $\mathbb{Q}^3/\mathbb{Q}(1,1,1)$. Permutation of the three branch values gives the stated S_3 representation. Finally, a section supported only at the common point must vanish along every incident branch. Its common value is then zero. This proves the last assertion. □

In particular, $\mathcal{V}$ cannot be replaced by a punctured constant sheaf together with an independent central skyscraper while preserving its specialization map. Identity nearby monodromy has not made specialization an isomorphism.

Let $p : (x, y, h) \mapsto h$ be the base projection, so that $f = p|_{Z_{\mathrm{cl}}}$. Let π project the multiplier bundle to this three-dimensional base. For the absolute total-family potential F on that six-dimensional bundle, dimensional reduction before parameter restriction gives

$$R(p\pi)_!\varphi_F\mathbb{Q} \simeq \mathcal{V}[-6](-3). \tag{13.23}$$

Indeed f is finite, and proper base change to its finite fibers shows that its higher constructible direct images vanish. Equation (13.17) then gives (13.23). This uses the absolute potential with its h derivative. It asserts no arbitrary base-change identity for unintegrated vanishing cycles. The fixed-h perverse coefficients have the separate normalization in (13.19).

The coherent family and the constructible family therefore have distinct defects. Monic reduction of the first two Koszul equations gives

$$A_h = \mathbb{C}[[h]][x, y]/(x(x+h), y(y+h)) \otimes \Lambda(\eta), \qquad d\eta = xy.$$

Multiplication by xy on the basis $1, x, y, xy$ has sole nonzero row $(1, -h, -h, h^2)$. Its unit entry gives free cohomology of rank three in each of degrees $0, -1$.

At zero it is the nonformal algebra of Theorem 4.1. Meanwhile the constructible coefficient has the rank-two specialization defect (13.22). These statements coexist because their coefficient functors retain different data.

There is a concrete evaluation map between the coherent degree-zero ring and its branch normalization. For $R = \mathbb{C}[[h]]$, evaluate the basis $1, x, y$ at $(0,0), (-h, 0), (0, -h)$. The resulting matrix is

$$E_h = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -h & 0 \\ 1 & 0 & -h \end{pmatrix}, \qquad \det E_h = h^2.$$

It is injective over R, and its image consists exactly of triples whose pairwise differences are divisible by h. Indeed subtracting the first row from the other two makes the matrix $\mathrm{diag}(1, -h, -h)$.

Thus its cokernel is $(R/(h))^2$, and its specialization has rank one. After complexifying the rational stalks, the surviving diagonal is the one in (13.21). This compares the displayed evaluation maps. It does not identify coherent and constructible coefficient functors or their higher products.

Near a nonzero two-torsion displacement, the fourth elliptic tree branch meets the last divisor in a simple isolated point, disjoint from the three persistent branches. Its constructible contribution is an actual parameter-supported skyscraper. The corresponding coherent local complex is $[O \xrightarrow{v} O]$, where v is a parameter times a unit. Its extra negative derived-fiber class comes from Tor. This isolated contribution differs from the collision of the three persistent branches at zero.

11 The compact coefficient comparison

The surface Hall theory supplies an independent comparison source. For a projective K3 surface S, let $\mathfrak{M}_S(\gamma)$ be the stack of coherent sheaves of Mukai vector γ. Write v_γ for its virtual dimension. Borel–Moore homology is defined by the dualizing complex, with proper direct image as its covariant operation.

The shifted rational groups

$$H_{S,n}(\gamma) = H^{\mathrm{BM}}_{-n+v_\gamma}(\mathfrak{M}_S(\gamma);\mathbb{Q})$$

carry the Hall multiplication and the cocommutative coproduct constructed by Davison–Hennecart–Kinjo–Schiffmann–Vasserot.

The coproduct takes values in the product completion over charge decompositions, while multiplication uses the locally finite part. Let $\chi(\gamma_1,\gamma_2)$ be the Euler pairing, the alternating dimension of the surface Ext groups. The charge braiding includes $(-1)^{\chi(\gamma_1,\gamma_2)}$, together with the cochain Koszul sign. The primitives for this coproduct are their affinized BPS Lie algebra.[9]

Relative dimensional reduction also exists for perverse pullbacks along locally finitely presented geometric morphisms of derived Artin stacks, with classical base locally of finite type.[10] For a derived zero locus of rank r on a smooth m-dimensional base, this dimension is $m - r$. It agrees with the shift in (13.19) for $m = 2$, $r = 3$.

An application to a compact translation quotient must identify its coefficient and extension maps with those in the theorem. An Ext-quiver comparison must also intertwine the surface coproduct before any primitive identification follows.

Question 11.1. For the common-torsor extension geometry of sheaves on $S \times E_{\mathrm{geom}}$, construct the reduced oriented critical coefficient functor and its maps on extension correspondences. Give the map to a spectral base. Specify its recipient: a coherent incidence algebra, an unintegrated vanishing-cycle coefficient, its constructible reduction, or an independently defined state complex.

Can this comparison preserve the common translation torsor, relative empty object, orientations, supports and every flag composition? Determine which higher products, monodromy summands and specialization maps survive it. Construct its marked state and compare compact reduced integration with the selected spectral trace. Finally, identify the geometric modular line and section required by the Igusa denominator.

[9] B. Davison, L. Hennecart, T. Kinjo, O. Schiffmann and E. Vasserot, *Hecke operators on symplectic surfaces and χ-independence*. arXiv:2607.01355v1, equation (4.1), Section 6.1, Proposition 6.4, Corollary 6.5 and Definition 6.6.

[10] A. A. Khan, T. Kinjo, H. Park and P. Safronov, *Period sheaves via perverse pullbacks*. arXiv:2510.18610v1, Theorem 4.1. The dimension is the relative virtual dimension of the derived source.

Chapter 14

Critical categories and cyclic realization

The potential of Chapter 13 has three incidence branches when the displacement is nonzero. Each branch yields a matrix factorization. To compare this category with a geometric coefficient, we must retain the morphisms between the objects and prove that the chosen objects generate the category. The return products remain nonzero before the displacement is inverted. After inversion, every complex of morphisms between distinct branches contracts.

We construct all nine morphism complexes and their products in a finite Clifford model. Multiplier derivatives then prove generation for the polynomial family. The same morphism comparison survives arbitrary base change, but generation does not. A monomial example records its multiplicity in a higher operation. Absolute Hochschild and cyclic calculations show which of these data survive passage to global periodic invariants.

Let S be a commutative ring in even parity and $W \in S$. A matrix factorization consists of a finite projective $\mathbb{Z}/2$-graded S-module and an odd endomorphism D satisfying $D^2 = W\mathrm{I}$. For two such objects,

$$\partial_{ab}(c) = D_b c - (-1)^{|c|} c D_a$$

is the differential on their morphisms. It squares to zero.

Composition of endomorphisms satisfies the graded Leibniz rule. All explicit factorizations below are finite free. All categorical gradings in this chapter are parity gradings unless an integer Koszul degree is specified.

Write $\mathrm{MF}(S, W)^{\pi}$ for the idempotent completion of this dg category. A collection *split-generates* if finite sums, shifts, cones and direct summands produce every object of its homotopy category. The closure under these operations is its *thick closure*. For a dg algebra A, $\mathrm{Perf}(A)$ denotes perfect right dg modules: the thick closure of the free module A. A Morita equivalence is an equivalence on perfect modules with their dg enhancements.

1 A relative Clifford comparison

Let R be a commutative $\mathbb{Q}$-algebra and $A = (A_{ij})$ a symmetric n-by-n matrix over R. Put $S = R[x_1, \dots, x_n]$.

On $S \otimes \Lambda^* R^n$, let e_i be exterior multiplication and ι_i contraction. Thus

$$[e_i, e_j]_+ = [\iota_i, \iota_j]_+ = 0, \qquad [e_i, \iota_j]_+ = \delta_{ij},$$

where $[u, v]_+ = uv + vu$ for odd operators. Define

$$D = \sum_i x_i e_i + \sum_i (Ax)_i \iota_i, \qquad \gamma_i = e_i - \sum_j A_{ij} \iota_j. \tag{14.1}$$

Then

$$D^2 = x^t A x, \quad [D, \gamma_i]_+ = 0, \quad [D, \iota_i]_+ = x_i, \quad [\gamma_i, \gamma_j]_+ = -2A_{ij}.$$

Let $\mathrm{Cl}_R(-A)$ be the parity-graded R-algebra on odd generators γ_i with the last relations and zero differential.

THEOREM 1.1. The indicated operators define a quasi-isomorphism of R-linear dg algebras

$$\mathrm{Cl}_R(-A) \longrightarrow \bigl(\mathrm{End}_S(S \otimes \Lambda^* R^n), [D, -]\bigr).$$

It commutes with arbitrary commutative base change of R. Neither nondegeneracy of A nor reducedness of R is required.

Proof. The exterior-contraction operators $e_I \iota_J$, with each index set ordered increasingly, form an S-basis of the endomorphisms. Replacing each e_i by $\gamma_i = e_i - \sum_j A_{ij}\iota_j$ is triangular in exterior length, with diagonal entries one. Consequently the $\gamma_I \iota_J$ are another basis. In particular, the ordered γ_I are linearly independent and realize the stated Clifford algebra.

In this basis the differential kills γ_I and replaces each ι_i by x_i, with the Koszul sign past γ_I. The underlying complex is therefore $\mathrm{Cl}_R(-A)$ tensored with the polynomial Koszul complex of $x_1, \dots, x_n$. That sequence is regular over every coefficient ring. Its augmentation evaluates at $x = 0$ and removes positive contraction length.

For polynomial degree d and contraction length q, the Koszul homotopy away from $d + q = 0$ is

$$\frac{\sum_i \iota_i \partial_{x_i}}{d + q}.$$

On the Clifford factor it carries the sign $(-1)^{|\gamma_I|}$. Its anticommutator with the differential is the identity away from degree $d + q = 0$. This proves the quasi-isomorphism and its compatibility with base change. The inclusion is multiplicative. The projection and homotopy are only linear maps. □

Over a field with a nondegenerate form, this is the usual Clifford description of quadratic matrix factorizations.[1] The theorem computes the chosen object. Generation of a larger nonisolated category is a separate question.

The coefficient hypothesis matters for this presentation and contraction. Over $\mathbb{Z}$, the anticommutator relation with $n = 1$ and $A = 0$ gives $\mathbb{Z}[g]/(2g^2)$, whereas the exterior operator satisfies $g^2 = 0$. The element g^2 in that presented algebra is nonzero, as its image in $\mathbb{F}_2[g]$ shows. Characteristic zero alone does not justify either division by two or the displayed Euler homotopy.

For the central triangle take

$$R = \mathbb{C}[P, Q, T], \qquad A = \begin{pmatrix} P & T/2 \\ T/2 & Q \end{pmatrix}.$$

Then

$$\gamma_1^2 = -P, \qquad \gamma_2^2 = -Q, \qquad \gamma_1\gamma_2 + \gamma_2\gamma_1 = -T. \tag{14.2}$$

The even element $w = 2\gamma_1\gamma_2 + T$ satisfies

$$w^2 = T^2 - 4PQ = \Delta, \qquad w\gamma_i = -\gamma_i w.$$

[1] T. Dyckerhoff, *Compact generators in categories of matrix factorizations*, arXiv:0904.4713, Section 5.5, following the quadratic theory of Buchweitz–Eisenbud–Herzog. The isolated-singularity generation theorem is not needed for the endomorphism comparison proved here.

Since the even basis is 1, $\gamma_1\gamma_2$, its subalgebra is exactly $R[w]/(w^2-\Delta)$, the coordinate ring of the cover in (13.13).

It is not the center of the full universal Clifford algebra. On the locus $\Delta \neq 0$, adjoining $\sqrt{\Delta}$ gives idempotents $\frac{1}{2}(1 \pm w/\sqrt{\Delta})$, exchanged by the deck involution. This recovers its sign local system.

It does not supply a functor from matrix factorizations to perverse sheaves.

2 The three objects and every morphism

Now put $R = \mathbb{C}[h, P, Q, T]$ and retain the actual displaced potential

$$W_h = Px(x+h) + Qy(y+h) + Txy.$$

Let C_R be the free rank-four Clifford algebra (14.2) over this ring. Give it the odd derivation $\partial_0\gamma_1 = hP$, $\partial_0\gamma_2 = hQ$. Set

$$t_0 = 0, \qquad t_1 = h\gamma_1, \qquad t_2 = h\gamma_2.$$

Each satisfies $\partial_0 t_a + t_a^2 = 0$.

Definition 2.1. The marked dg category C_h has objects 0, 1, 2. For every ordered pair, put

$$\mathrm{Hom}_{C_h}(a,b) = C_R, \qquad \partial_{ab}(c) = \partial_0 c + t_b c - (-1)^{|c|} c t_a. \tag{14.3}$$

Composition from $\mathrm{Hom}(b,c) \otimes_R \mathrm{Hom}(a,b)$ to $\mathrm{Hom}(a,c)$ is Clifford multiplication in that order. The units are the copies of 1.

The displayed identities for t_a give $\partial_{ab}^2 = 0$. Expanding (14.3) gives the Leibniz rule for every composable pair. Thus the definition gives a strict dg category, finite free in every morphism complex.

On $R[x,y] \otimes \Lambda^* R^2$, use the Clifford operators from (14.1) and put

$$\begin{aligned} D_0 &= xe_1 + ye_2 + (Px + Ty/2 + hP)\iota_1 + (Tx/2 + Qy + hQ)\iota_2, \\ D_1 &= D_0 + h\gamma_1, \qquad D_2 = D_0 + h\gamma_2. \end{aligned} \tag{14.4}$$

All three squares are $W_h 1$. They are Koszul factorizations at the respective sections $(0,0)$, $(-h,0)$ and $(0,-h)$ of the incidence locus.

A dg functor is a *quasi-equivalence* if it induces quasi-isomorphisms on morphism complexes and is essentially surjective on the homotopy category.

Theorem 2.2. The constant-in-x, y Clifford operators give a quasi-equivalence from C_h to the full dg subcategory of matrix factorizations of W_h on D_0, D_1, D_2. It preserves composition and units strictly and is compatible with arbitrary commutative base change of R.

Proof. Write $a_0 = (0,0)$, $a_1 = (1,0)$ and $a_2 = (0,1)$. In every matrix morphism complex use the basis $c\iota_J$, with c a Clifford word. The differential on c is (14.3), and for a single contraction generator it is

$$\partial(c\iota_i) = \partial_{ab}(c)\iota_i + (-1)^{|c|} c\big(x_i + h(a_a)_i\big), \qquad (x_1,x_2) = (x,y).$$

Indeed $[D_a, \iota_i]_+ = x_i + h(a_a)_i$.

The regular sequence belongs to the source object a. Iteration gives the tensor product with its polynomial Koszul complex. Translating coordinates reduces its contraction to the one in Theorem 1.1.

Thus inclusion is a quasi-isomorphism on every morphism complex. It preserves multiplication, and the three objects already exhaust the stated target subcategory. The contractions are defined over R, so they survive base change, including $h = 0$ and nilpotent parameters. □

This also permits the base change $R \to \mathbb{C}[P, Q, T][[h]]$, with x, y still polynomial variables. No analytic convergence or generation of the entire matrix-factorization category follows from the morphism comparison alone. The generation theorem proved below supplies the latter assertion while the multiplier variables remain in the ring. Arbitrary specialization can destroy it.

3 Cohomology before and after localization

PROPOSITION 3.1. For $a \neq b$ there is an odd R-linear operator H_{ab} on the morphism complex such that

$$\partial_{ab} H_{ab} + H_{ab} \partial_{ab} = h\mathrm{I}. \tag{14.5}$$

Moreover

$$H^{\bar{0}} \operatorname{Hom}(a, b) \cong R/(h), \qquad H^{\bar{1}} \operatorname{Hom}(a, b) \cong R/(h). \tag{14.6}$$

Proof. Write $\delta_{ab} = \partial_{ab}/h$. The odd Clifford derivatives ∂_{γ_i} are defined by $\partial_{\gamma_i}\gamma_j = \delta_{ij}$ and the super Leibniz rule. They preserve the Clifford relations, and

$$\delta_{ab}\partial_{\gamma_i} + \partial_{\gamma_i}\delta_{ab} = ((a_b)_i - (a_a)_i)\mathrm{I}.$$

For distinct objects one coordinate difference is ± 1. Divide by that sign to define H_{ab}. Thus δ_{ab} is split exact.

In the bases $1, \gamma_1\gamma_2$ and γ_1, γ_2, each of its two parity matrices has determinant zero and an entry ± 1. Its kernel and image are therefore the same free rank-one summand. Multiplying the differential by the non-zero-divisor h keeps that kernel and multiplies its image by h. This proves (14.6). □

For the three diagonal complexes define, respectively,

$$(r_1, r_2) = (P, Q), \quad (-P, Q - T), \quad (P - T, -Q), \qquad \kappa_a = r_1\gamma_2 - r_2\gamma_1. \tag{14.7}$$

PROPOSITION 3.2. For each object a,

$$H^{\bar{0}} \operatorname{End}(a) = R/(hr_1, hr_2), \qquad H^{\bar{1}} \operatorname{End}(a) = (R/(h))\kappa_a.$$

Their odd generators satisfy

$$\kappa_a^2 = PQ(T - P - Q) \tag{14.8}$$

in the even cohomology ring.

Proof. The differentials of γ_1, γ_2 and $v = \gamma_1\gamma_2$ are hr_1, hr_2 and $h\kappa_a$. Each pair (r_1, r_2) is regular. Hence every odd cycle is a multiple of its Koszul syzygy κ_a, and the odd boundaries are its h-multiples. An even cycle is scalar. Its boundaries form (hr_1, hr_2). Finally

$$\kappa_a^2 = -Qr_1^2 - Pr_2^2 + Tr_1r_2,$$

which equals (14.8) for all three pairs in (14.7). □

After inverting h, all off-diagonal morphism complexes contract by H_{ab}/h. The diagonal cohomology rings become $R[h^{-1}]/(r_1, r_2)$, with zero odd cohomology. These are the three multiplier lines of Proposition 4.1. The localized category is quasi-equivalent to the disjoint union of its diagonal dg algebras. This does not infer formality of those algebras from their cohomology rings.

At $h = 0$, by contrast, every differential in (14.3) vanishes. The endomorphism algebra of the direct sum of the three identical objects is $\mathrm{Mat}_3(C_R/(h))$. Each off-diagonal derived fiber has two directions in each parity. One comes from cohomology modulo h, the other from h-torsion in the opposite parity.

For object zero, the odd specialization retains the syzygy sequence

$$0 \longrightarrow R_0\kappa_0 \longrightarrow R_0\gamma_1 \oplus R_0\gamma_2 \xrightarrow{(a,b)\mapsto Pa+Qb} (P,Q) \longrightarrow 0, \qquad R_0 = \mathbb{C}[P,Q,T].$$

The additional even vector v has connecting class $\partial v/h = \kappa_0$. Cohomology followed by ordinary specialization would lose these connecting maps.

4 Return and transit products

We now fix generators, so every composition has a specified sign. With $v = \gamma_1\gamma_2$, take the following closed representatives:

ab	α_{ab} (even)	β_{ab} (odd)
01	$Q + v$	γ_1
10	$T - Q + v$	γ_1
02	$T - P + v$	γ_2
20	$P + v$	γ_2
12	$T + v$	$\gamma_2 - \gamma_1$
21	v	$\gamma_2 - \gamma_1$.

(14.9)

They generate the two groups in (14.6). Put $K = P + Q - T$ and $\ell_{01} = P$, $\ell_{02} = Q$, $\ell_{12} = K$. The notation ℓ_{ab} is symmetric in its indices.

Proposition 4.1. For $a \neq b$, the return products in $H^*\,\mathrm{End}(a)$ are

$$\begin{aligned} [\beta_{ba}\beta_{ab}] &= -\ell_{ab}, \\ [\alpha_{ba}\alpha_{ab}] &= -\prod_{\{i,j\}\neq\{a,b\}} \ell_{ij}, \\ [\alpha_{ba}\beta_{ab}] &= \varepsilon_{ab}\kappa_a, \qquad [\beta_{ba}\alpha_{ab}] = -\varepsilon_{ab}\kappa_a. \end{aligned} \tag{14.10}$$

The product ranges over the other two edges. Set $\varepsilon_{ab} = 1$ for $a < b$ and -1 for $a > b$. All these classes are nonzero and killed by h.

For distinct a, b, c, write the four transit products as

$$\begin{aligned} \alpha_{bc}\alpha_{ab} &= u_1\alpha_{ac}, \quad \beta_{bc}\beta_{ab} = u_2\alpha_{ac}, \\ \alpha_{bc}\beta_{ab} &= u_3\beta_{ac}, \quad \beta_{bc}\alpha_{ab} = u_4\beta_{ac}. \end{aligned}$$

Their coefficients are

abc	u_1	u_2	u_3	u_4
012	Q	-1	P	K
021	$-P$	-1	$-Q$	K
102	$-K$	-1	P	$-Q$
120	P	1	$-K$	Q
201	K	1	Q	$-P$
210	$-Q$	1	$-K$	$-P$.

(14.11)

These identities hold already for the chosen chain representatives.

Proof. The relations (14.2) give

$$v^2 = -Tv - PQ, \qquad v\gamma_1 = P\gamma_2 - T\gamma_1, \qquad \gamma_1 v = -P\gamma_2, \qquad v\gamma_2 = -Q\gamma_1, \qquad \gamma_2 v = Q\gamma_1 - T\gamma_2.$$

Substituting (14.9) proves (14.10) and every entry of (14.11). For example, the path $0 \to 1 \to 0$ gives

$$-P, \quad -QK, \quad \kappa_0, \quad -\kappa_0$$

in the order odd–odd, even–even, even–odd, odd–even. The transit product $(\gamma_2 - \gamma_1)\gamma_1 = P - T - v = -\alpha_{02}$ gives the second entry in the first transit row.

Every boundary has a factor h, whereas each displayed return is a nonzero polynomial Clifford vector modulo h. Thus none is a boundary. Equation (14.5) supplies a primitive for h times either off-diagonal factor. Multiplying that primitive by the other closed factor, with its Leibniz sign, proves that each return is killed by h. □

The three plane forms have a symmetric expression in the original constraint multipliers:

$$P = p + r, \qquad Q = q + r, \qquad T = 2r, \qquad \begin{aligned} \Delta &= -4(pq + qr + rp), \\ -\kappa_a^2 &= (p + q)(q + r)(r + p). \end{aligned}$$

Indeed the potential becomes

$$p\,x(x + h) + q\,y(y + h) + r(x + y)(x + y + h).$$

The transformations $(x, y) \mapsto (y, x)$ with $p \leftrightarrow q$, and $(x, y) \mapsto (-(x + y + h), y)$ with $p \leftrightarrow r$, preserve it. They permute the three branch points and the three plane forms. This is the S_3 symmetry of the potential and its marked support. An equivariant categorical comparison must additionally specify the maps intertwining its chosen factorizations.

5 Generation from smooth branches

Let S be a regular Noetherian ring of finite Krull dimension, let W be a non-zero-divisor, and set $A = S/(W)$. The singularity category is $D_{\mathrm{sg}}(A) = D^b_{\mathrm{coh}}(A)/\mathrm{Perf}(A)$. The regular-ambient equivalence identifies it with matrix factorizations. The image of an A-module under this equivalence is its *stabilization*.[2]

Proposition 5.1. Let $f_1, \dots, f_r \in S$ act null-homotopically on every finite matrix factorization, and put $I = (f_1, \dots, f_r)$. Suppose ideals $J_1, \dots, J_k$ contain W, satisfy $J_1 \cdots J_k \subset I$, and have regular quotients S/J_a. Then the stabilizations of the S/J_a split-generate $\mathrm{MF}(S, W)^\pi$.

Proof. If $DH + HD = f\,1$, the cone of multiplication by f has differential and change of basis

$$D_f = \begin{pmatrix} D & f\,1 \\ 0 & -D \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & -H \\ 0 & 1 \end{pmatrix}, \qquad D_f U = U \begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}.$$

[2] D. Orlov, *Matrix factorizations for nonaffine LG-models*, arXiv:1101.4051, Theorem 3.5. An affine regular Noetherian scheme of finite Krull dimension has the required locally free resolutions.

For the local and nonlocal ring formulations, see T. Dyckerhoff, *Compact generators in categories of matrix factorizations*. arXiv:0904.4713, Sections 2.1–2.4.

The second summand is shifted, so U is even and invertible. Thus the original object is a direct summand of its cone.

Repeat for every f_i. On a split sum with differential $\operatorname{diag}(D, -D)$, a subsequent homotopy is $\operatorname{diag}(H_i, -H_i)$. Conjugate it back through the preceding splitting. No anticommutation of the initially chosen homotopies is needed.

Choose a bounded coherent representative N in $D_{\mathrm{sg}}(A)$. Its ordinary Koszul tensor

$$N \otimes K(f_1, \dots, f_r)$$

represents these iterated cones. Every cohomology module V is killed by I.

Filter V by successive products of the J_a. The product containment makes the last term zero, and each quotient is a finite module over one regular ring S/J_a. It therefore has a finite projective resolution over that ring. Each projective term is a summand of a finite sum of S/J_a as an A-module.

The filtration and the bounded truncation triangles put the Koszul tensor in the thick closure of the branch modules. Its original summand belongs to that closure after idempotent completion. □

The derivatives of W supply the homotopies. For a free factorization,

$$D(\partial_z D) + (\partial_z D)D = (\partial_z W)1. \tag{14.12}$$

For a projective module, choose a connection along the derivation ∂_z and differentiate the endomorphism D covariantly. Such a connection exists by writing the module as an idempotent summand of a free module and projecting its coefficientwise connection. The same Leibniz identity proves (14.12). Flatness of the connection is unnecessary.

5.1 The polynomial triangle

Put $R = \mathbb{C}[h, P, Q, T]$, $S = R[x, y]$, and retain

$$W_h = Px(x + h) + Qy(y + h) + Txy.$$

Let D_0, D_1, D_2 be the factorizations (14.4), and let $\mathcal{C}_h$ be the finite dg category of Definition 2.1. Write $\mathcal{E}_h = \bigoplus_{a,b=0}^{2} \operatorname{Hom}_{\mathcal{C}_h}(a, b)$, with matrix composition as multiplication. Its underlying R-module has rank 36.

Theorem 5.2. The object $G = D_0 \oplus D_1 \oplus D_2$ split-generates $\mathrm{MF}(S, W_h)^{\pi}$. There is a Morita equivalence

$$\operatorname{Perf}(\mathcal{E}_h) \simeq \mathrm{MF}(S, W_h)^{\pi}. \tag{14.13}$$

At $h = 0$, with P, Q, T still variables, $D_0|_{h=0}$ alone split-generates. Its Morita algebra is the rank-four Clifford algebra (14.2) over $R_0 = \mathbb{C}[P, Q, T]$, with zero differential.

Proof. The multiplier derivatives generate

$$I = (x(x + h), y(y + h), xy).$$

Set $J_0 = (x, y)$, $J_1 = (x + h, y)$ and $J_2 = (x, y + h)$. The invertible change

$$z_0 = h + x + y, \qquad z_1 = -x, \qquad z_2 = -y$$

identifies I with (z_0z_1, z_0z_2, z_1z_2). It follows that $I = J_0 \cap J_1 \cap J_2$ and $J_0J_1J_2 \subset I$. All three quotients are R.

For a module killed by I, the filtration

$$0 \subset J_1J_2V \subset J_1V \subset V$$

has quotients killed by J_0, J_2, J_1, respectively. Their projective resolutions over R have length at most four. Equations (14.12) and Proposition 5.1 apply.

Each D_a is the Koszul stabilization of its branch module. Indeed its exterior coefficients generate J_a and its contraction coefficients express W_h in that regular sequence. The Koszul stabilization is obtained by adding the corresponding multiplication-by-W_h homotopy to the Koszul resolution.[3]

The full morphism comparison of Theorem 2.2 is multiplicative. It identifies $\mathcal{E}_h$ with the derived endomorphisms of G. The two Morita functors are $\mathrm{RHom}(G, -)$ and $- \otimes^{\mathbb{L}}_{\mathcal{E}_h} G$.

At $h = 0$ the derivative ideal becomes $(x, y)^2$. The two-step (x, y)-adic filtration reduces every module killed by this ideal to modules over the regular ring R_0. Proposition 5.1, with the branch ideal repeated twice, proves the central assertion. Theorem 1.1 supplies its endomorphism algebra. □

The proof also applies to finite projective factorizations over $\mathbb{C}[P, Q, T][[h]][x, y]$. The branch quotients remain regular, the filtrations are finite, and the morphism contractions are coefficient-linear. The cyclic comparisons below concern the finite-type algebraic models.

6 Failure under multiplier specialization

PROPOSITION 6.1. For $R \to \mathbb{C}$ given by $h = P = Q = T = 0$, the natural functor

$$\mathrm{Perf}(\mathcal{E}_h \otimes^{\mathbb{L}}_R \mathbb{C}) \longrightarrow \mathrm{MF}(\mathbb{A}^2, 0)^{\pi}$$

is not essentially surjective.

Proof. All three images are $K_0 = (\mathbb{C}[x, y] \otimes \Lambda^*\mathbb{C}^2, xe_1 + ye_2)$. On $x \neq 0$, contraction by ι_1/x contracts K_0. On $y \neq 0$, use ι_2/y. Its thick closure vanishes away from the origin. The free even factorization $(\mathcal{O}_{\mathbb{A}^2}, 0)$ does not vanish there. It therefore lies outside the image. □

The coefficient-linear contractions in Theorem 2.2 do survive this base change, so the functor still computes the full subcategory generated by K_0. What fails is generation of the new ambient category: an arbitrary zero-potential object has no multiplier derivatives from which to obtain the homotopies (14.12).

7 A monomial and its higher products

One object already distinguishes a higher operation from its periodic invariant. The potential λz^m has multiplicity m, while its binary cohomology algebra is the same for every $m \geq 3$. We keep λ as a parameter and compute the complete minimal algebra of its chosen factorization.

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[3] T. Dyckerhoff, *Compact generators in categories of matrix factorizations*. arXiv:0904.4713, Section 2.3, Lemma 2.6 and Corollary 2.7. Section 2.4 extends the construction to regular nonlocal rings.

Exterior–contraction duality changes the stabilization by a parity shift, which leaves its thick closure unchanged.

Put $R = \mathbb{C}[\lambda]$, $S = R[z]$ and $m \geq 2$. All complexes in this section are $\mathbb{Z}/2$-graded, with even scalar coefficients. On the free module $S^{1|1}$ define

$$D_m = \begin{pmatrix} 0 & z \\ \lambda z^{m-1} & 0 \end{pmatrix}, \qquad D_m^2 = \lambda z^m I.$$

Let $A_m = \mathrm{End}_S(S^{1|1})$, with differential $\delta a = D_m a - (-1)^{|a|} a D_m$. The square of this differential is zero because D_m^2 is scalar.

7.1 The parameter-linear contraction

Set

$$J = \begin{pmatrix} 0 & -1 \\ \lambda z^{m-2} & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The first two matrices are odd and the third is even. Direct multiplication gives

$$\delta J = 0, \qquad \delta H = zI, \qquad \delta E = zJ, \qquad J^2 = -\lambda z^{m-2} I. \tag{14.14}$$

Every even matrix has a unique expression $aI + bE$, and every odd matrix a unique expression $aJ + bH$, with $a, b \in R[z]$.

Proposition 7.1. The cohomology of A_m is the free R-module $R1 \oplus R\chi$, with 1 even and $\chi = [J]$ odd. For $a_0 = a|_{z=0}$, define

$$\begin{aligned} p(aI + bE) &= a_0 1, & p(aJ + bH) &= a_0 \chi, \\ i(1) &= I, & i(\chi) &= J, \\ \mathsf{h}(aI + bE) &= \frac{a - a_0}{z} H, & \mathsf{h}(aJ + bH) &= \frac{a - a_0}{z} E. \end{aligned}$$

Then $pi = 1$, $\delta\mathsf{h} + \mathsf{h}\delta = 1 - ip$ and $\mathsf{h}^2 = \mathsf{h}i = p\mathsf{h} = 0$. These identities survive every commutative base change of R.

Proof. Equations (14.14) split the underlying complex into two copies of multiplication by z: one sends H to zI, and the other sends E to zJ. This computes the cohomology. The polynomial $a - a_0$ is divisible by z, so the displayed homotopy uses no localization. Applying $\delta\mathsf{h} + \mathsf{h}\delta$ to $aI + bE$ gives $(a - a_0)I + bE$, and applying it to $aJ + bH$ gives $(a - a_0)J + bH$. The other identities follow from the same decomposition. All formulas are polynomial in λ and require no flatness of its specialization. □

7.2 Every arity from one polynomial identity

Let s denote parity suspension, so $|sa| = |a| - 1$. Our suspended differential and multiplication on a dg algebra are

$$b_1(sa) = -s(\delta a), \qquad b_2(sa, sb) = (-1)^{|a|} s(ab).$$

The maps b_n extend by the tensor signs to a degree-one coderivation on the tensor coalgebra of the suspension. An A_∞ structure means that this coderivation squares to zero. A morphism means a coalgebra map commuting with these coderivations.

Introduce an even auxiliary variable τ and put

$$D_m(\tau) = \begin{pmatrix} 0 & z-\tau \\ \lambda \sum_{j=0}^{m-1} z^{m-1-j}\tau^j & 0 \end{pmatrix}.$$

The geometric sum gives

$$D_m(\tau)^2 = \lambda(z^m - \tau^m)I.$$

Writing $a(\tau) = D_m(\tau) - D_m = \sum_{j=1}^{m-1} \tau^j A_j$, we obtain

$$A_1 = J, \qquad A_j = \lambda z^{m-1-j} H \quad (2 \le j < m), \qquad \delta a(\tau) + a(\tau)^2 = -\lambda \tau^m I. \tag{14.15}$$

THEOREM 7.2. On $R1 \oplus R\chi$, put the strict unit products and the single additional reduced coefficient

$$b_m((s\chi)^{\otimes m}) = \lambda s1. \tag{14.16}$$

A reduced coefficient has no unit among its inputs. All other reduced coefficients vanish. These operations give a strictly unital minimal A_∞ algebra.

There is an A_∞ quasi-isomorphism to A_m with

$$F_1(s1) = sI, \qquad F_j((s\chi)^{\otimes j}) = sA_j \quad (1 \le j < m).$$

All remaining components are zero, including every F_j with $j > 1$ and a unit input. The formulas commute with arbitrary commutative base change of R.

Proof. Every word without units is a power of $s\chi$, which has even parity. On such words the equation $b_{A_m} F = Fb$ has generating series

$$-s\big(\delta a(\tau) + a(\tau)^2\big) = \lambda \tau^m sI.$$

This is (14.15). On the right, a source b_m creates $s1$. A higher component of F with that unit input is zero. Only $F_1(s1)$ at arity m remains.

For words containing a unit, strict unitality fixes $b_2(s1, sa) = sa$ and $b_2(sa, s1) = (-1)^{|a|} sa$. Deleting a unit on its left or on its right gives opposite terms after the tensor sign. Internal terms therefore cancel in pairs. At either endpoint the remaining term equals multiplication by the target unit I. Every higher coefficient with a unit input vanishes. This proves the morphism equation on all words.

The same cancellation proves $b^2 = 0$. For $m \ge 3$, an inner b_m creates a unit, so it cannot be an input to another normalized higher coefficient. The only remaining terms are the two adjacent binary unit terms, which cancel. For $m = 2$, the assertion is associativity of the algebra with unit and relation $\chi^2 = -\lambda$. Finally $F_1 = i$ induces the cohomology isomorphism of Proposition 7.1. The polynomial formulas and that contraction persist after base change. □

For $m = 2$, (14.16) is the Clifford product $\chi^2 = -\lambda$. For $m \ge 3$, the binary product has $\chi^2 = 0$, while the first nonzero reduced higher operation has arity m when $\lambda \ne 0$. At $\lambda = 0$, every reduced coefficient vanishes. The model becomes the exterior algebra on χ.

The field version relating higher multiplications to potential coefficients is Dyckerhoff's theorem.[4]

[4] T. Dyckerhoff, *Compact generators in categories of matrix factorizations*. arXiv:0904.4713, Section 5.6, Theorem 5.8. The present formulas specify the coefficient parameter, the suspension signs and the complete comparison for the chosen object.

Corollary 7.3. Fix a nonzero complex value of λ and $m \geq 3$. Among these strictly unital minimal algebras, the arity of the first nonzero reduced higher operation is preserved by A_∞ isomorphisms. In particular, it determines m.

Proof. A parity-preserving unital isomorphism has linear term $\chi \mapsto c\chi$ with $c \neq 0$. On a pure $s\chi$ word, every Taylor component has even parity in the suspension. Its value therefore lies in the line spanned by $s\chi$.

Before the first higher operation, the morphism equations involve only products of two odd elements, which are zero. At that first arity, the nonzero $s1$ coefficient can only be matched by a higher operation of the same arity in the other algebra. Its coefficient changes by the nonzero factor c^m. The first arity is therefore unchanged. □

Proposition 7.4. With λ retained as a variable, D_m split-generates $\mathrm{MF}(\mathbb{C}[\lambda, z], \lambda z^m)^\pi$.

Proof. The multiplier derivative of any factorization differential contracts z^m by (14.12). Use m copies of the branch ideal (z) in Proposition 5.1. Their product is (z^m) and the quotient is the regular ring $\mathbb{C}[\lambda]$. Its Koszul stabilization is D_m. □

Thus the minimal algebra above gives a Morita model of the total polynomial family. After specializing λ, its explicit endomorphism comparison remains valid. That comparison alone does not prove generation of the specialized ambient category.

8 Absolute Hochschild coefficients

All invariants in this section are over $\mathbb{C}$. For a complex dg category $\mathcal{A}$, its Hochschild cochain complex is the derived endomorphism complex of the diagonal bimodule:

$$C^\bullet(\mathcal{A}) = R\,\mathrm{Hom}_{\mathcal{A}^{\mathrm{op}} \otimes_{\mathbb{C}} \mathcal{A}}(\mathcal{A}, \mathcal{A}).$$

The diagonal has component $\mathrm{Hom}_{\mathcal{A}}(a, b)$ at the object pair (a, b). Its cohomology is $\mathrm{HH}^*(\mathcal{A})$.

Suppose a smooth Landau–Ginzburg model has all critical points in the zero fiber. Its absolute Hochschild cohomology is computed by polyvectors with differential given by contraction with dW.[5]

Return to $R_0 = \mathbb{C}[P, Q, T]$ and $S_0 = R_0[x, y]$. For $W_0 = Px^2 + Txy + Qy^2 \in S_0$, the identity $W_0 = P\partial_P W_0 + Q\partial_Q W_0 + T\partial_T W_0$ verifies the critical-value hypothesis.

Proposition 8.1. The absolute Hochschild cohomology of $\mathrm{MF}(S_0, W_0)^\pi$ is the parity folding of the cohomology of

$$\mathcal{J} = \frac{R_0[x, y]}{(x^2, y^2)} \otimes \Lambda(\alpha, \beta, \gamma), \qquad \begin{cases} d\alpha = xy, \\ d\beta = 2Px + Ty, \\ d\gamma = Tx + 2Qy, \end{cases} \tag{14.17}$$

where $|\alpha| = |\beta| = |\gamma| = -1$. As an R_0-complex, its free ranks in degrees $0, -1, -2, -3$ are $4, 12, 12, 4$. Moreover,

$$H^0(\mathcal{J}) = R_0 \oplus M, \qquad M^2 = 0, \qquad M = \mathrm{coker}\begin{pmatrix} 2P & T \\ T & 2Q \end{pmatrix}. \tag{14.18}$$

[5] K. H. Lin and D. Pomerleano, *Global matrix factorizations*. arXiv:1101.5847, Theorem 3.1 and Lemma 3.3. The complex is absolute, so it includes multiplier derivatives.

The dimensions of its derived fibers are

rank of the quadratic form	H^0	H^{-1}	H^{-2}	H^{-3}
2	1	3	3	1
1	2	6	6	2
0	3	9	9	3.

Proof. The five gradient entries are x^2, y^2, xy, $2Px + Ty$, $Tx + 2Qy$. The first two are a regular sequence. Tensoring their Koszul resolution with the other free Koszul factors and then augmenting gives (14.17). In degree zero all products of x, y vanish, while their linear relations are the columns in (14.18).

At a rank-two form the two linear gradients are independent coordinates. Eliminating them leaves three closed exterior generators over $\mathbb{C}$.

At a rank-one form, make a linear change to a nonzero square. One linear gradient and the square of the remaining coordinate form a regular sequence, with a length-two quotient. The three excess Koszul generators are closed after elementary generator changes.

At the zero form, multiplication by xy on $\mathbb{C}[x, y]/(x^2, y^2)$ has rank one. Its kernel and cokernel both have dimension three. The other two exterior generators are closed. These three computations give the table. □

Here H^0 means integer Koszul degree zero, rather than the entire even group after folding. The module M is supported on $4PQ - T^2 = 0$, has rank one at its generic point, and needs two generators at the vertex.

The strict supercenter consists of elements that supercommute with every Clifford generator. For the universal Clifford algebra it is R_0. An even central coefficient of $\gamma_1\gamma_2$ must vanish. An odd central vector lies in the kernel of the displayed Hessian, which is zero over the polynomial domain. At $P = Q = T = 0$ the Clifford algebra becomes exterior and is entirely supercentral. Thus strict supercenters also need not commute with base change.

The table describes fibers of the fixed absolute complex. It makes no identification with invariants of every newly specialized matrix-factorization category.

8.1 The monomial Hochschild modules

For $F_m = \lambda z^m$, the absolute gradient has two entries. They retain the multiplier derivative even when one later takes a fiber over the multiplier base.

PROPOSITION 8.2. For $m \geq 2$, the absolute gradient Koszul complex has

$$\begin{aligned} H^0 &= \mathbb{C}[\lambda, z]/(z^m, \lambda z^{m-1}), \\ H^{-1} &= \big(\mathbb{C}[\lambda, z]/(z^{m-1})\big)\theta, \qquad\qquad \theta = m\lambda\epsilon_\lambda - z\epsilon_z, \\ H^{-2} &= 0. \end{aligned} \tag{14.19}$$

Its parity folding gives absolute Hochschild cohomology.

Proof. The differential is $d\epsilon_\lambda = z^m$, $d\epsilon_z = m\lambda z^{m-1}$. Cancel z^{m-1} in the syzygy equation in the polynomial domain. The regular pair $(z, m\lambda)$ has syzygies generated by $(m\lambda, -z)$. The degree-minus-two boundary is $-z^{m-1}\theta$, with zero kernel. The degree-zero quotient gives the remaining assertion. □

9 The mixed cyclic comparison

For a complex dg category $\mathcal{A}$, let $CC_\bullet(\mathcal{A})$ be its absolute Hochschild bar complex, with internal and Hochschild differential b and Connes differential B. These odd operators satisfy $b^2 = B^2 = bB + Bb = 0$. Its periodic cyclic homology is the cohomology of $(CC_\bullet(\mathcal{A})((u)), b + uB)$, where u is an even formal variable. Laurent series have finitely many negative powers and arbitrary nonnegative powers of u. For $\mathcal{A} = C_h$, the bar words range over all composable cycles of its three objects.

A u-connection is an operator $\nabla_u = \partial_u + A(u)$, where $A(u)$ is $\mathbb{C}((u))$-linear and $[\nabla_u, b + uB] = (b + uB)/(2u)$. A mixed-complex map f is weakly compatible with connections if there is an odd $\mathbb{C}((u))$-linear map H such that

$$\nabla'_u f - f\nabla_u = (b' + uB')H + H(b + uB).$$

Here f also denotes its extension to Laurent series.

Return to $R = \mathbb{C}[h, P, Q, T]$ and $S = R[x, y]$. Efimov's comparison, applied to the Morita equivalence (14.13), gives a zigzag of quasi-isomorphisms of mixed complexes

$$(CC_\bullet(\mathcal{E}_h), b, B) \simeq (\Omega^\bullet_S, -dW_h\wedge, d).$$

The comparison is weakly compatible with the connections.

It passes from curved modules through the completed curved bar complexes for $(S, -W_h)$ and (S, W_h). Their completion is the product over bar-word lengths. It allows the curvature term to insert an additional bar entry while retaining an infinite sequence of lengths.

The final curved Hochschild–Kostant–Rosenberg map is

$$a_0[a_1|\cdots|a_k] \longmapsto \frac{a_0}{k!}\, da_1 \wedge \cdots \wedge da_k.$$

Its inputs are scalar curved-bar words after those comparisons, not Clifford words.[6]

On the de Rham side the differential and connection are

$$Q_u = -dW_h \wedge + ud, \qquad \nabla_u = \partial_u + \frac{\Gamma}{u} + \frac{W_h}{u^2}, \qquad \Gamma|_{\Omega^p} = -p/2.$$

They satisfy $Q_u^2 = 0$ and $[\nabla_u, Q_u] = Q_u/(2u)$. Indeed the four nonzero commutator terms are d, $(dW_h\wedge)/(2u)$, $-d/2$ and $-(dW_h\wedge)/u$. Their sum is the stated expression. All six affine coordinates, including h, P, Q, T, are differentiated. For h fixed, the corresponding five-variable absolute model is a separate application.

Even a length-one bar boundary uses both return paths. In the convention where it is the supercommutator, closed odd maps $f : 0 \to 1$ and $g : 1 \to 0$, both represented by γ_1, give

$$b(g[f]) = gf - (-1)^{|f||g|} fg = -P_{1_0} - P_{1_1}.$$

This relates the two returns. It does not kill each term separately or permit the intermediate object to be deleted.

[6] A. I. Efimov, *Cyclic homology of categories of matrix factorizations*. arXiv:1212.2859v2, Theorems 1.1–1.4, Proposition 3.14 and Section 4, equations (4.2)–(4.6). The hypotheses are smoothness, separation, finite type and characteristic zero, with the critical locus contained in the zero fiber. The linear multiplier identity verifies the last condition.

10 THE GRADING OF THE PERIODIC REALIZATION

Let M be a smooth quasi-projective complex d-fold with potential F. Form the quotient of its dg category of coherent matrix factorizations by the thick subcategory generated by totalizations of short exact sequences. Write $\mathrm{MF}(M, F)$ for its idempotent completion. Assume that every critical point lies in $F^{-1}(0)$.

Write ϕ_F for classical vanishing cycles and normalize $\Phi_F = \phi_F[-1]\mathbb{Q}_M[d]$. We regard this complex on M by extension by zero. Efimov's periodic comparison identifies HP_k with the parity folding of $H^{k-1}(F^{-1}(0), \phi_F\mathbb{C}_M)((u))$, with connection monodromy $T_F(-1)^k$.

Consequently

$$H^j(M, \Phi_F) \longleftrightarrow H^{j+d-1}(F^{-1}(0), \phi_F\mathbb{Q}_M), \qquad k = j + d. \tag{14.20}$$

Here T_F is monodromy in the value of F. The support convention is ordinary cohomology. The multiplier projection need not be proper.

For the Morse function xy on $\mathbb{A}^2$, the Milnor fiber is homotopy equivalent to a circle. Its reduced cohomology is in degree one, giving even periodic degree $k = 2$. For x^2 on $\mathbb{A}^1$, the two-point Milnor fiber has reduced cohomology in degree zero, giving odd periodic degree $k = 1$. These examples fix the shift.

For the central triangle, Proposition 9.1 gives $R\Gamma(\mathbb{A}^3, \Phi_{W_0}) = \mathbb{Q}[1]$. The single normalized degree is $j = -1$ and $d = 5$. Thus the raw vanishing degree is 3 and the periodic degree is 4. We obtain

$$\mathrm{HP}_{\bar{0}}\,\mathrm{MF}(\mathbb{A}^5, W_0) = \mathbb{C}((u)), \qquad \mathrm{HP}_{\bar{1}} = 0.$$

The nonzero summand $\mathcal{P}_-$ of Theorem 7.3 is self-dual. Its compact cohomology vanishes by Proposition 9.1, so Verdier duality also makes its ordinary cohomology zero. It contributes nothing to these global periodic groups.

Raw function monodromy is $+1$ on this class. The even periodic degree leaves that sign unchanged.

At a fixed nonzero numerical h, the three multiplier lines of Proposition 4.1 are Morse–Bott with four hyperbolic normal coordinates. Their orientation local systems are constant on the affine lines. Each contributes one even class, giving periodic ranks $(3, 0)$. These are separately specified absolute categories.

COROLLARY 10.1. For every $m \geq 2$, the absolute periodic cyclic ranks of $\mathrm{MF}(\mathbb{A}^2, \lambda z^m)^\pi$ are $(1, 0)$.

Proof. For $\Phi_m = \varphi_{F_m}\mathbb{Q}[2]$, dimensional reduction over the multiplier has $R\pi_!\Phi_m = \mathbb{Q}_{\{z=0\}}$ in rational sheaves. Verdier duality gives the same ordinary pushforward. Its normalized ordinary cohomology therefore has one class in degree zero. The ambient dimension is two, so (14.20) puts it in even periodic degree. □

The absolute Hochschild modules in (14.19) depend on m, as does the first reduced operation in Theorem 7.2. The constant periodic ranks determine neither datum. No assertion about conservativity of a categorical functor follows from this equality of ranks alone.

11 GLOBAL ELLIPTIC SECTORS AND PERIODIC RANKS

Let E be a complex elliptic curve, $\Theta = \mathcal{O}_E(0)$ its theta line, and ϑ its divisor section. Fix $n \geq 1$, set $h = [n]t$, and put $B_n = E_x \times E_y \times E_t$. For $u_1 = x, u_2 = y, u_3 = x + y$, define

$$\mathcal{S}_i = u_i^*\Theta \otimes (u_i + [n]t)^*\Theta, \qquad s_i = \vartheta(u_i)\vartheta(u_i + [n]t).$$

Let $V_n = \bigoplus_i \mathcal{S}_i$, $\mathcal{M}_n = \mathrm{Tot}_{B_n}(V_n^\vee)$, and $F_n = \langle \lambda, s \rangle$. This is a smooth quasi-projective sixfold with a global regular function. Fiberwise multiplier dilation shows that every critical point lies in its zero fiber.

Proposition 11.1. Above each point with $[n]t \in E[2] \setminus \{0\}$ and $x = y = -[n]t$, the potential at $\lambda = 0$ has analytic local form $pu + qv + rw$. The factorization

$$D_{\mathrm{extra}} = ue_1 + ve_2 + we_3 + p\iota_1 + q\iota_2 + r\iota_3 \tag{14.21}$$

on $\mathcal{O} \otimes \Lambda^* \mathbb{C}^3$ is a generator over the convergent local ring of this chart, and over its formal completion. Its endomorphism cohomology is $\mathbb{C}$ in even parity and zero in odd parity. The three moving branch objects vanish near these points. There are $3n^2$ such local sectors.

Proof. The fourth branch and its torsion condition are those of Theorem 10.1 and (12.30). The first two sections have simple roots at the indicated fourth branch point. Use them as coordinates u, v. Subtract their multiples from the third section. On their common branch its restriction is $\vartheta(-2h)\vartheta(-h)$.

At nonzero two-torsion this has a simple zero in h, since $[2]$ is étale and $\vartheta(-h) \neq 0$. It is therefore a third coordinate w. The dual change of multiplier frame gives $F_n = pu + qv + rw$. The point of the t-curve is also unramified under $[n]$.

Exterior–contraction relations prove $D_{\mathrm{extra}}^2 = F_n \mathrm{I}$. The endomorphism differential sends ι_i to $(u, v, w)_i$ and e_i to $(p, q, r)_i$. The resulting six-coordinate Koszul complex resolves $\mathbb{C}$. Generation follows by the smooth-branch argument with (u, v, w), or by iterated hyperbolic stabilization. The three other branch supports are disjoint from this point, so their objects restrict to zero there. Finally each of the three nonzero points of $E[2]$ has n^2 preimages under $[n]$. □

At fixed $w = 0$ the relative chart has two hyperbolic pairs and a free multiplier direction r. The proposition uses the absolute chart, in which w is a coordinate. It constructs local generators. A global finite generator with all descent maps is a further construction.

Write Z_n for the reduced common zero locus of the s_i. Its curve part consists of three copies of E,

$$C_0 = (0, 0, t), \qquad C_1 = (0, -[n]t, t), \qquad C_2 = (-[n]t, 0, t).$$

They meet at the n^2 points $t \in E[n]$. The remaining points are exactly the $3n^2$ points of Proposition 11.1.

Theorem 11.2. The Betti numbers and absolute periodic cyclic ranks are

$$(b_0, b_1, b_2)(Z_n) = (1 + 3n^2,\ 2n^2 + 4,\ 3), \tag{14.22}$$

$$\dim_{\mathbb{C}((u))} \mathrm{HP}_{\bar{0}}\, \mathrm{MF}(\mathcal{M}_n, F_n) = 3n^2 + 4,$$

$$\dim_{\mathbb{C}((u))} \mathrm{HP}_{\bar{1}}\, \mathrm{MF}(\mathcal{M}_n, F_n) = 2n^2 + 4. \tag{14.23}$$

Proof. The first two constraints force $x, y \in \{0, -[n]t\}$. Evaluating the third gives exactly the stated locus. For $C = C_0 \cup C_1 \cup C_2$, normalization has the exact sheaf sequence

$$0 \longrightarrow \mathbb{Q}_C \longrightarrow \bigoplus_{i=0}^{2} (\nu_i)_* \mathbb{Q}_E \longrightarrow \bigoplus_{t \in E[n]} \mathbb{Q}_t^2 \longrightarrow 0.$$

The last map is $(a, b, c) \mapsto (b - a, c - a)$ at every intersection. Its map on global constants has rank two. The long exact sequence therefore gives $b_0(C) = 1$, $b_1(C) = 6 + (2n^2 - 2)$ and $b_2(C) = 3$.

The isolated points give (14.22). In particular $\chi(Z_n) = n^2$.

Let $\pi : M_n \to B_n$ and $i : Z_n \hookrightarrow B_n$. For $\Phi_n = \phi_{F_n}[-1]\mathbb{Q}_{M_n}[6]$, linear-potential dimensional reduction gives

$$R\pi_!\Phi_n \simeq i_*\mathbb{Q}_{Z_n}.$$

The base dimension and vector-bundle rank are both three.

This formula is for rational sheaves. Its mixed-Hodge version includes the Tate twist (-3).[7]

Verdier self-duality of Φ_n turns the compact pushforward into $R\pi_*\Phi_n \simeq i_*\omega_{Z_n}$, where ω_{Z_n} is the rational dualizing complex. Since Z_n is proper,

$$H^j(M_n, \Phi_n) = H^{-j}(Z_n, \mathbb{Q})^\vee.$$

The normalized degrees are 0, -1, -2. Equation (14.20) gives raw vanishing degrees 5, 4, 3 and periodic degrees 6, 5, 4. Thus the even periodic rank is $b_0 + b_2$ and the odd rank is b_1, proving (14.23). □

The ranks are $(7, 6)$ on the displacement base $n = 1$ and $(16, 12)$ on the half-displacement cover $n = 2$. They are invariants of the absolute global spectral category. A relative cyclic complex over the t-curve, its base-change maps and its connection in t require their own construction. The periodic u-connection has the comparison above. The rank formula does not compute its full monodromy matrix.

12 The coherent comparison and its grading

The same equations have the coherent Koszul algebra

$$K_h = \mathbb{C}[h, x, y] \otimes \Lambda(\epsilon_1, \epsilon_2, \epsilon_3), \qquad \begin{cases} d\epsilon_1 = x(x + h), \\ d\epsilon_2 = y(y + h), \\ d\epsilon_3 = xy, \end{cases} \qquad |\epsilon_i| = -1. \tag{14.24}$$

Its coherent derived category is related to an equivariant singularity category of the potential, with the multiplier grading retained. Precisely, let $X = \mathbb{A}^3_{h,x,y}$ and let $Z = \{W_h = 0\} \subset X \times \mathbb{A}^3_{P,Q,T}$. The multiplicative group dilates P, Q, T with weight one. The equivariant singularity category is the quotient of bounded equivariant coherent complexes on Z by equivariant perfect complexes.

Isik's Koszul duality identifies it with the bounded coherent derived category of K_h. The nonregular section requires the Koszul dg algebra in place of its degree-zero cohomology.[8]

This graded equivalence has a different source of grading from the parity category in Definition 2.1. To use it for the marked objects, one must specify their graded lifts, the parameter-linear functor and its effect on the objects and their compositions.

Question 12.1. Construct the graded lifts and the $\mathbb{C}[h]$-linear Koszul-duality comparison. Require it to carry the point modules at $(0, 0)$, $(-h, 0)$ and $(0, -h)$ of (14.24) to the objects (14.4). Specify the shifts and periodization. How does it identify their Yoneda products with (14.10) and (14.11), and the coherent higher operations with the relative operations of the marked category?

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[7] B. Davison, *The critical CoHA of a quiver with potential*, arXiv:1311.7172, Theorem A.1. T. Kinjo, *Dimensional reduction in cohomological Donaldson–Thomas theory*, arXiv:2102.01568, equation (1.1) and Theorem 4.14. Use the critical-chart orientation and its comparison with the canonical shifted-cotangent orientation.

[8] M. U. Isik, *Equivalence of the derived category of a variety with a singularity category*. arXiv:1011.1484, Theorem 3.6 and Remark 3.7. Here the smooth ambient scheme is the absolute space $\mathbb{A}^3_{h,x,y}$. The section is $(x(x + h), y(y + h), xy)$.

13 The compact categorical comparison

The Morita equivalence (14.13) identifies two descriptions of the specified polynomial potential. An identification with a compact sheaf deformation potential requires an additional geometric map. The half-displacement cover, the quadratic determinant cover and the Enriques canonical cover have different bases. Comparing their lines requires maps between those bases. The genus-two parameter family and weight-five automorphic line also remain separate constructions.

Question 13.1. Construct the comparison from compact translation-reduced critical Hall coefficients to the spectral category or to an independently specified state endomorphism category. Which maps preserve extension pullback and pushforward, orientation, the common translation torsor, the relative empty object and the reduced state? How do they carry the return products of (14.10), the unintegrated summand $\mathcal{P}_{-}$ of Theorem 7.3, and the extra local objects (14.21)?

Construct the coefficient comparison before global sections and before inverting h. Can it retain the arity in Theorem 7.2 and the parameter specialization map of Theorem 10.1? Give a finite global generator of the elliptic spectral category with its descent maps, including the extra local sectors. Identify the reduced marked state and the modular line and section required by the Igusa denominator.

Part V

Periods and the compact realization

Chapter 15

K3 periods and the mirror discriminant

1 A geometric source for the Siegel period domain

The Siegel threefold already parametrizes a classical family of lattice-polarized K3 surfaces. Its decomposable-abelian-surface divisor is also a locus where the K3 polarization enlarges. This gives a geometric interpretation of the zero divisor of Δ_5. The further question is whether an independently defined compact $K3 \times E$ construction produces the same period map, line and meromorphic section. That comparison must retain the elliptic variation and the geometric source of the section.

We first construct the K3 lattice and its period coordinates. The two orbits of roots then distinguish the Humbert surfaces H_1 and H_4. Finally we compare the local divisor on the period cover with its coarse quotient. These calculations specify the target of the compact mirror problem without assuming that problem solved.

1.1 Polarization and the mirror lattice

Write

$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \Lambda_{K3} = 3U \oplus 2E_8(-1), \qquad \widetilde{\Lambda}_{K3} = U_{\mathrm{Muk}} \oplus \Lambda_{K3}.$$

The signatures are $(3, 19)$ and $(4, 20)$, respectively; positive entries are counted first. The second lattice is the Mukai lattice, with pairing

$$((r, c, s), (r', c', s')) = c \cdot c' - rs' - r's$$

[91, 92].

Let M be an even lattice of signature $(1, t)$. An M-polarization of a K3 surface S is a primitive embedding $M \hookrightarrow \mathrm{Pic}(S)$; we require its image to contain a nef and big class, and call the polarization *pseudoample*. Requiring an ample class is stronger. A choice of a primitive hyperbolic plane in $M^\perp \subset \Lambda_{K3}$ is additional data. Since U is unimodular, such a choice splits

$$M^\perp = U_{\mathrm{mir}} \oplus N.$$

The lattice N, of signature $(1, 18 - t)$, is the mirror lattice for this choice [36, §§1–3, 5]. Thus lattice polarization does not itself include a mirror cusp.

In the same K3 lattice, the complementary relation is

$$N^\perp = U_{\mathrm{mir}} \oplus M, \qquad \mathrm{sign}(N^\perp) = (2, t + 1).$$

Indeed, unimodularity of the chosen U_{mir} splits it from Λ_{K3}. In the remaining summand, M is primitive and $N = M^\perp$; taking the orthogonal complement again recovers M.

For a fixed primitive embedding, the marked period domain is

$$\Omega_M = \{[\omega] \in \mathbb{P}(M^\perp \otimes \mathbb{C}) : (\omega,\omega) = 0, (\omega,\overline{\omega}) > 0\}^+.$$

The superscript selects one connected component. Its complex dimension is $20 - \operatorname{rk} M = 19 - t$. The arithmetic group for polarizations fixed pointwise is the image of the isometries of Λ_{K3} fixing M. Equivalently it is the component-preserving discriminant kernel in $O(M^\perp)$, with the relevant ample-chamber convention. It need not be the full orthogonal group.

The lattice used here is

$$M = U \oplus \langle -2 \rangle.$$

Embed its rank-one summand as a root in one $E_8(-1)$ factor of Λ_{K3}, and its hyperbolic plane in one of the three U's. The root complement is $E_7(-1)$, so

$$M^\perp = 2U \oplus E_8(-1) \oplus E_7(-1), \qquad N = U \oplus E_8(-1) \oplus E_7(-1).$$

In particular, N has rank 17 and signature $(1, 16)$. Its orthogonal complement in the same K3 lattice is

$$T = N^\perp = 2U \oplus \langle -2 \rangle, \qquad \operatorname{sign}(T) = (2, 3).$$

The domain Ω_M has dimension 17, whereas $\Omega_N \subset \mathbb{P}(T \otimes \mathbb{C})$ has dimension 3. They parametrize different lattice-polarized K3 surfaces.

1.2 The integral gluing

The root complement above is integral, but the root and its orthogonal complement do not split E_8 integrally. An explicit basis exhibits the difference. In Euclidean $\mathbb{R}^8$, set

$$\begin{aligned} a_1 &= \tfrac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 + e_8), \\ a_2 &= e_1 + e_2, \qquad a_3 = -e_1 + e_2, \\ a_j &= -e_{j-2} + e_{j-1} \qquad (4 \le j \le 8). \end{aligned}$$

Their Gram matrix is the positive E_8 matrix, of determinant 1. The first seven roots have the E_7 Gram matrix, of determinant 2. They are orthogonal to $r = e_7 + e_8$, and

$$r = 2a_1 + 3a_2 + 4a_3 + 6a_4 + 5a_5 + 4a_6 + 3a_7 + 2a_8. \tag{15.1}$$

Consequently

$$[E_8 : E_7 \oplus \mathbb{Z}r] = 2, \qquad \det(E_7 \oplus \mathbb{Z}r) = 4.$$

Indeed, replacing the last basis vector a_8 by r has determinant 2. Equation (15.1) also displays the missing glue vector. Reversing the form gives the asserted embedding into $E_8(-1)$.

This distinction matters when a root becomes algebraic. A primitive closure can enlarge $E_7(-1) \oplus \langle -2 \rangle$ to $E_8(-1)$. Adding the two orthogonal lattices without their glue would give the wrong discriminant group.

2 The period map and an explicit K3 family

Choose a basis (e_1, f_1, e_2, f_2, g) of T with $(e_i, f_i) = 1$, $g^2 = -2$, and all other pairings zero. For

$$Z = \begin{pmatrix} \tau & z \\ z & \sigma \end{pmatrix} \in \mathbb{H}_2, \qquad w(Z) = e_1 + (z^2 - \tau\sigma) f_1 + \tau e_2 + \sigma f_2 + z g,$$

one has

$$(w(Z), w(Z)) = 0, \qquad (w(Z), \overline{w(Z)}) = 4 \det \operatorname{Im} Z > 0.$$

The first identity follows by canceling $2(z^2 - \tau\sigma) + 2\tau\sigma - 2z^2$. For the second, write $\operatorname{Im}\tau = b$, $\operatorname{Im}\sigma = d$, $\operatorname{Im} z = f$; the pairing is $4(bd - f^2)$. The tube-domain parametrization $Z \mapsto [w(Z)]$ identifies $\mathbb{H}_2$ with the selected component Ω_N.

The discriminant group of T is $\mathbb{Z}/2$, whose quadratic-form automorphism group is trivial. Thus the component-preserving discriminant kernel here is $O^+(T)$. Its effective projective action is the genus-two Siegel modular action, with real exceptional isomorphism $\mathrm{PSp}_4(\mathbb{R}) \simeq \mathrm{SO}_0(T \otimes \mathbb{R})$. The K3 Torelli theorem gives the coarse period isomorphism

$$\mathcal{M}_N \simeq O^+(T)\backslash\Omega_N \simeq \mathrm{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2, \tag{15.2}$$

where $\mathcal{M}_N$ denotes the coarse moduli space of pseudoample N-polarized K3 surfaces [22, Proposition 3.1]. The quotient on the right also parametrizes principally polarized abelian surfaces. The associated Hodge correspondence is the classical K3–abelian-surface correspondence of [22, Proposition 3.2].

Here is a family realizing the K3 side. For complex parameters $(\alpha, \beta, \gamma, \delta)$ with $(\gamma, \delta) \neq (0, 0)$, consider the quartic in $\mathbb{P}^3_{[x:y:z:w]}$

$$y^2 z w - 4x^3 z + 3\alpha x z w^2 + \beta z w^3 + \gamma x z^2 w - \tfrac{1}{2}(\delta z^2 w^2 + w^4) = 0. \tag{15.3}$$

In this equation z is a projective coordinate, distinct from the off-diagonal period coordinate used above. The minimal resolution is a K3 surface with a canonical pseudoample N-polarization. Every such polarized surface occurs, and its isomorphism class is determined by

$$[\alpha : \beta : \gamma : \delta] \in \mathbb{P}(2, 3, 5, 6), \qquad (\gamma, \delta) \neq (0, 0)$$

[22, Theorems 1.2 and 1.4]. The construction identifies the two exceptional root configurations of types E_8 and E_7 together with the hyperbolic plane of an elliptic fibration. The minimal resolution, rather than the singular quartic itself, is the smooth K3 surface.

The weights can be checked directly. Replacing the parameters by $(t^4\alpha, t^6\beta, t^{10}\gamma, t^{12}\delta)$ and the coordinates by $(t^2 x, t^3 y, t^{-6} z, w)$ leaves every term in (15.3) unchanged. Over $\mathbb{C}$, this realizes the weighted scaling $(2, 3, 5, 6)$.

In the modular normalizations of Clingher–Doran, the inverse period map is

$$Z \longmapsto [\mathcal{E}_4(Z) : \mathcal{E}_6(Z) : 2^{12}3^5 C_{10}(Z) : 2^{12}3^6 C_{12}(Z)] \tag{15.4}$$

[22, Theorem 3.5]. Their proof compares two elliptic fibrations on the associated Kummer surface: one obtained by a Nikulin involution of (15.3), the other from genus-two theta functions. Matching the Weierstrass coefficients gives (15.4). The normalization of the weight-ten coordinate is

$$C_{10} = -2^{-14}\Delta_5^2 = -\tfrac{1}{4}\Xi, \qquad \Xi = 64^{-2}\Delta_5^2.$$

This follows from their squared theta-product formula [22, (34)]; the phase change in Section 4 has square one. Thus their C_{10} must not be substituted for the monic counting form Ξ without its scalar.

A family of smooth resolutions is taken on a suitable marked or level cover. Its product with a fixed elliptic curve is a compact family of threefolds with trivial canonical bundle. This is already a positive geometric construction. Comparing it with a prescribed compact Hall theory, or allowing a specified varying elliptic factor, requires further maps.

2.1 Forms, sign conventions and Hodge lines

The orthogonal lattice of the preceding calculation has signature $(2, 3)$. The exterior-square model of Section 2.1 uses signature $(3, 2)$. To pass between them, negate the entire bilinear form and then identify $-U \simeq U$ in each hyperbolic factor. This sends T to a lattice isometric to $2U \oplus \langle 2\rangle$. It is an identification with $T(-1)$, not an isometry from T to a lattice of different signature. The domain inequality changes from positive to negative, and a root of square -2 becomes a vector of square 2. The orthogonal hyperplane is unchanged.

On the marked domain, the K3 Hodge line is $\lambda_N = \mathbb{C}w(Z)$. The primitive exterior-square construction identifies this line with the determinant of the abelian surface's holomorphic one-forms. An equivariant line comparison requires the corresponding arithmetic actions. In particular, the coarse isomorphism (15.2) does not by itself identify the two moduli stacks or all of their stabilizer actions on Hodge lines. We use a common level cover whenever an actual line is required.

Let λ_{aut} be the weight-one automorphic line on that cover, and let ε_5 be the character line of Δ_5. Then

$$\Delta_5 \in H^0(\lambda_{\mathrm{aut}}^{\otimes 5} \otimes \varepsilon_5), \qquad \Delta_5^2 \in H^0(\lambda_{\mathrm{aut}}^{\otimes 10}), \qquad \varepsilon_5^{\otimes 2} \simeq \mathcal{O}.$$

The character belongs to the full automorphy law. Its slash-action value need not equal the sign of a function in a chosen linear tube frame; Section 4 and the character calculation of Chapter 3 fix that convention.

3 The two root divisors

For a primitive root $v \in T$, put

$$H_v = \{[\omega] \in \Omega_N : (\omega, v) = 0\}, \qquad \operatorname{div}(v) = \gcd\{(v, x) : x \in T\}.$$

These are rational quadratic, or Heegner, divisors [14, §2]. The roots of T form two orthogonal-group orbits, distinguished by divisibility [22, §3.3]. Their simplest representatives are

$$v_2 = g, \qquad v_1 = e_2 - f_2.$$

Both have square -2, while

$$\operatorname{div}(v_2) = 2, \quad (w(Z), v_2) = -2z; \qquad \operatorname{div}(v_1) = 1, \quad (w(Z), v_1) = \sigma - \tau.$$

Their orthogonal complements in T are $2U$ and $U \oplus \langle 2\rangle \oplus \langle -2\rangle$, respectively. On the Siegel quotient their images are

$$H_1 : \ z = 0, \qquad H_4 : \ \tau = \sigma,$$

with the full arithmetic orbits understood. The subscript is the Humbert discriminant. The decomposable locus is H_1, not H_2.

To retain the root-type notation, write

$$\mathfrak{H}_{\mathrm{II}} = \Big(\bigcup_{v^2=-2,\ \mathrm{div}(v)=2} H_v \Big) / O^+(T) = H_1.$$

A point of H_1 corresponds to a product of two elliptic curves. On the K3 side, its N-polarization extends to $U \oplus E_8(-1) \oplus E_8(-1)$. The index-two glue above explains the second E_8. The other root orbit gives the $N \oplus \langle -2 \rangle$ enhancement along H_4. Thus the full root arrangement has two types, whereas Δ_5 detects the divisibility-two type. Neither type should be identified with singular fibres of every chosen K3 family: an extra algebraic root and a singular uncontracted K3 surface are different conditions.

3.1 Orders on the period cover and the coarse quotient

Let $\widetilde{H}_1$ be the reduced union of the divisibility-two hyperplanes in $\mathbb{H}_2$. The Borcherds product and theta normalization give

$$\mathrm{div}_{\mathbb{H}_2}(\Delta_5) = \widetilde{H}_1, \qquad \mathrm{div}_{\mathbb{H}_2}(\Delta_5^{-2}) = -2\widetilde{H}_1.$$

These are divisors in the open domain, before adding a toroidal boundary [10, 52, 54]. If $\mathrm{Pole}(s)$ denotes the effective polar divisor of a meromorphic section s, then

$$\mathrm{Pole}_{\mathbb{H}_2}(\Delta_5^{-2}) = 2\widetilde{H}_1.$$

The sign differs from the signed divisor of the section.

Near a generic point of $z = 0$, take the normal coordinate $u = z$. The root reflection sends u to $-u$, and the local theta product has the form

$$\Delta_5 = u\, a(u), \qquad a(0) \neq 0.$$

Its odd transformation in this frame is part of the line action. On the effective coarse reflection quotient the normal coordinate is $t = u^2$, so

$$\Delta_5^2 = t\, b(t), \qquad b(0) \neq 0.$$

Consequently the square has zero order one, and its inverse has pole order one, along the generic coarse Humbert divisor. On the period cover, or as a stack divisor pulled back to it, the corresponding orders are two. A root line with nontrivial reflection action does not descend as an ordinary line on the same coarse chart.

The distinction is visible without a global product formula: $\mathrm{ord}_u(t) = 2$. The reflection records the ramification. It does not determine a distinguished meromorphic section or its order. In particular, Picard–Lefschetz monodromy alone cannot prove an order-two pole for a geometric discriminant.

The two integers appearing here have different sources. The Borcherds weight $f(0,0)/2 = 5$ comes from the constant Jacobi coefficient. The Weyl vector specifies the cusp exponents. Neither the weight nor the reflection alone replaces the full boundary factor of Chapter 3.

4 What remains for a compact mirror comparison

For $X = S \times E$, Künneth gives

$$H^3(X, \mathbb{Z}) = H^2(S, \mathbb{Z}) \otimes H^1(E, \mathbb{Z}), \qquad \mathrm{rk}\, H^3(X) = 44.$$

This is a weight-three Hodge structure with alternating polarization. In contrast,

$$H^{\text{even}}(X,\mathbb{Z}) = \widetilde{H}(S,\mathbb{Z}) \otimes \big(H^0(E,\mathbb{Z}) \oplus H^2(E,\mathbb{Z})\big)$$

is an integral group of rank 48. The two even elliptic summands have constant Hodge type. They do not contain the varying elliptic modulus, which belongs to $H^1(E)$. Identifying their underlying rank-two lattice with a hyperbolic plane supplies no elliptic period map.

PROPOSITION 4.1 *(A restriction on a Hodge-quotient construction).* A polarized weight-three variation cannot have a rank-five polarized quotient with nondegenerate induced polarization. No integral Tate twist of a weight-three variation has weight two.

Proof. The polarization in odd weight is alternating. If A is an alternating matrix of odd size over a field of characteristic zero, then $\det A = \det A^t = \det(-A) = -\det A$, so $\det A = 0$. A rank-five alternating polarization is therefore degenerate. An integral Tate twist changes the weight by an even integer, which proves the second assertion. □

This excludes the specified polarized-quotient route from $H^3(X)$ to T. It does not exclude a different geometric construction of a period map. The classical N-polarized K3 family is already such a period construction, with a different input. Nor does the dimension difference between Ω_M and Ω_N prohibit every correspondence. It shows why mirror lattices alone cannot specify one.

We use the notation

$$\mathcal{P}^{(3)}_{K3\times E} \longrightarrow \mathcal{M}_N \simeq \mathcal{A}_2$$

only for a proposed three-dimensional compact period construction with this actual comparison map supplied. The superscript denotes a dimension, not the rank of a weight-two period lattice; that lattice has rank five. It is not a definition of a quotient of $H^3(X)$ or of the rank-48 even cohomology group.

4.1 The line and section problem

A compact mirror comparison would start with a geometric family, a period morphism $p : B \to \mathcal{A}_2$, and a geometric Hodge or determinant line λ_{per} on an explicitly chosen base or stack. It would then supply a meromorphic section s_{mir} and a line isomorphism relating it to the inverse Igusa section. This order prevents the desired modular form from serving as the definition of its own geometric source.

Conjecture 4.2 *(Compact mirror discriminant comparison).* For a suitable mirror construction associated with the compact $K3 \times E$ theory, there exist the preceding family, period map and geometric section, together with

$$\iota : p^*\lambda_{\text{aut}}^{-10} \xrightarrow{\sim} \lambda_{\text{per}}^{-10}, \qquad \iota(p^*\Delta_5^{-2}) = s_{\text{mir}}.$$

The comparison is required to extend across the chosen discriminant and boundary, with the same cusp normalization. The elliptic parameter and the comparison with the original compact geometric source are part of the construction.

On a common period cover on which the reduced discriminant $\widetilde{\mathfrak{D}}_{\text{mir}}$ pulls back to $\widetilde{H}_1$, and where the period map is unramified normally, the predicted polar-divisor equality is

$$\text{Pole}(s_{\text{mir}}) = 2\widetilde{\mathfrak{D}}_{\text{mir}}.$$

For a ramified period map the orders acquire its ramification indices. On the generic effective coarse reflection quotient the corresponding order is one. These statements fix the base before assigning a multiplicity.

Matching divisors is weaker than matching sections. After the comparison of lines, their quotient can be a nonconstant invertible function on an open base. A boundary extension and a normalization must eliminate that ambiguity. Likewise a square-line comparison does not construct a square root with the required character.

4.2 The four independent period conditions

The geometric comparison separates into the following conditions.

(M1) Construct the period morphism and the comparison of lines, with their coefficient, marking and stabilizer conventions. On the open complement of the discriminant this trivializes $p^*\lambda_{\mathrm{aut}}^{-10} \otimes \lambda_{\mathrm{per}}^{10}$.

(M2) Extend the section comparison over the discriminant and a specified boundary. Compute its local orders on that base and its full cusp expansion; the weight five alone is insufficient.

(M3) Relate the chosen mirror construction to the original compact source, including the varying elliptic factor. Dolgachev's lattice construction and the complexified Kähler interpretation of [3] do not furnish a canonical map between moduli spaces of dimensions seventeen and three, or an involution of either space.

(M4) For another polarization, specify the new orthogonal lattice, arithmetic group, automorphic input and geometric line. Its higher-rank extension is a new comparison problem. The classical rank-three denominator examples of [54], the reflective-product constructions of [50], and the proposed lattice-polarized families of [78] do not impose a universal Picard-rank criterion for this comparison.

The broader Borcherds construction supplies many possible orthogonal automorphic targets [8, 9]. For example, the fake-monster root lattice has signature $(25, 1)$, whereas its weight-twelve orthogonal form lives on a lattice of signature $(26, 2)$, or $(2, 26)$ after reversing the form. It is a different target from T. Any restriction or quasi-pullback must specify its embedding and the extra root factors; it is not the period comparison of Conjecture 4.2 by itself.

4.3 Curve counts, Pfaffians and states

The primitive reduced curve-counting theorem gives, in the conventions fixed in Chapter 8,

$$Z_{\mathrm{OP}}^{X} = -\Xi^{-1} = -4096\Delta_5^{-2}$$

[15, 101]. The underlying generating series and its rational expansion are the enumerative assertion. The rescaled scalar $-4096^{-1}Z_{\mathrm{OP}}^{X} = \Delta_5^{-2}$ does not on its own specify an operator trace or a geometric discriminant section. Conjecture 4.2 would give a period-side interpretation of that scalar if its geometric construction and normalization were supplied.

The automorphic product, the denominator superalgebra and the regularized theta lift are established at their stated recipients. The compact Pfaffian identity additionally needs a geometric skew family, an orientation and a comparison of sections. Its four conditions (D0), (O1), (O1$^+$), (O2) govern those objects, whereas (M1)– (M4) govern the period family and its section. Neither collection of conditions supplies the other. A gravitational or boundary interpretation further requires its state, supported operations, trace domain and sewing maps [79, 80, 82].

The numerical tuple $(0, 0, 3, 5, 24)$ retains these distinctions. The first two entries are the holomorphic Euler characteristics of the compact product, the third is the Cartan rank of the reference root algebra, the fourth is its Borcherds weight, and the last is the Mukai rank of a K3 surface. A proposed

Hall specialization must compare its own Cartan object before using the third entry as a geometric rank. The tuple is not a relation among these invariants and does not construct a mirror period map.

The classical K3 period model and its root divisors therefore remain available as positive geometric input. The compact problem asks for the additional map from independently defined geometry, with its line, section and elliptic variation, to that explicit target.

Chapter 16

The remaining geometric comparisons

1 From the denominator to a compact realization

The automorphic product and the Borcherds superalgebra give two constructions of the same normalized denominator. The compact realization problem asks for geometric objects and maps behind that identity. Its source is a system of moduli and extension correspondences on $X = S \times E$; its target includes the root spaces with both parities, their brackets, a modular line and a state carrying the required operators. Equality of scalar products cannot supply these structures.

The established reference identities are

$$D_5 = 64^{-1}\Delta_5, \qquad \Xi = D_5^2 = 4096^{-1}\Delta_5^2, \qquad \operatorname{den}(\mathfrak{g}_{\Delta_5})(Z) = D_5(2Z).$$

Their product, theta and singular-lift descriptions are developed in Chapters 3, 4 and 5 from [9, 52]. The geometric Pfaffian would be a section in an independently constructed line. Identifying it with Δ_5 requires the line map, its character and the normalized section comparison.

Several genuine geometric constructions already lie between the compact problem and these targets. The spherical triple has a cyclic formal deformation model and compatible formal flags. Ordinary critical Hall multiplication has its orientation and support hypotheses. Hilbert schemes carry elliptic coefficient bundles, geometric incidence operators and scalar indices. The spectral section has explicit critical coefficients and categorical realizations. Each supplies a particular object or map; none alone identifies the complete reduced compact Hall system with the denominator algebra.

The finite geometric datum of Section 1.7 is therefore kept as a system

$$\{\mathfrak{D}^{\mathrm{DI}}_{X,R}\}_R.$$

Its fourteen components include the fields, hybrid supports, charges and projection, critical coefficients and roots, Hall operations and primitives, coalgebra and associative comparison, Pfaffian line and section, Weyl action and recognition map. The morphism conditions (M1)–(M8) specify how these components commute. This notation denotes a pro-object once its maps exist; it does not assert an effective infinite moduli object or an ordinary componentwise inverse limit.

1.1 The four orientation and degeneration conditions

The names $(D0)$, $(O1)$, $(O1)^+$, $(O2)$ refer to four construction conditions. They are not four classes in one cohomology group. The finite Pfaffian comparison P_{fin} is additional. Appendix 7 gives their precise recipients and proves the consequences when the data exist.

1.1.1 D0 specialization

The D0 problem asks for specialization of reduced deformation complexes, oriented critical coefficients, flag operations and Pfaffian sections to zero-dimensional objects on the threefold. The first requirement is an actual family and its charge map. The stability and moduli constructions of [75, 105] motivate the

problem; a compact retained Hall slice still needs its own boundedness proof. For sheaves on a fixed polarized X, flatness preserves the Hilbert polynomial. A nonzero curve class cannot become a pure Do sheaf in such a family. A comparison involving those two charges must change the geometric or categorical recipient and construct its specialization functor.

The ambient shifted symplectic structure [104] does not remove scalar automorphisms or their dual obstruction directions. A reduced two-term complex needs the actual reduction and its amplitude proof. Cosection localization [67] applies to its specified obstruction theory; it does not automatically produce a sheaf of reduced vanishing cycles. Surjectivity in nonzero curve class cannot simply be continued to the Do class.

A Do specialization system must retain object, extension, mixed, wrapped and higher flag loci, the coefficient specialization maps, root pairs and their cross determinants, support conditions and stage transitions. The critical-sheaf and orientation theorems [12, 64] enter after those hypotheses are met. Theorem 7.2 constructs the resulting pro-system from the supplied maps; effectivity remains an additional limit statement. A K3 Hilbert-scheme index tests its scalar consequence but does not construct this threefold specialization.

1.1.2 Orientation on the translation quotient

For a retained prequotient locus B, root the actual reduced determinant line $\mathscr{D}$ on $[B/E]$:

$$(o, \rho : o^{\otimes 2} \simeq \mathscr{D}).$$

The obstruction is its Kummer class in $H^2([B/E], \mu_2)$. When it vanishes, the isomorphism classes of root pairs form an $H^1([B/E], \mu_2)$-torsor. Neither rational Grothendieck–Riemann–Roch nor translation invariance supplies the required root with its square map.

There are two separate descent issues. A universal object on a scalar gerbe has weight one and need not descend; its endomorphism complex has weight zero and can descend with its natural maps. A root on the quotient stack can carry a nontrivial stabilizer character, while coarse descent requires trivial stabilizer action. Removing common scalars also need not make all remaining inertia finite.

The finite-stabilizer calculations in Lemmas 7.6 and 7.7 identify exact equations after the obstruction has been reduced to their classifying-space edge. For $E[2]$, all three cyclic restrictions detect a quadratic class. For the two-primary group $(\mathbb{Z}/2^a)^2$, $a \geq 2$, the mixed class x_1x_2 is invisible on cyclic subgroups. In either case, the character of the chosen root is additional degree-one data. Coherent primitives and flag comparisons are needed to glue these local choices.

1.1.3 Weyl transport of the root

The type-II Coxeter subgroup

$$W = \langle s_1, s_2, s_3 \mid s_i^2 = 1 \rangle$$

acts on the abstract root datum. Its geometric action requires correspondences or functors on the actual compact recipient and lifts to the root pairs. The lifts have an action-groupoid sign defect c_o. A chosen primitive for that defect, compatible with quotient, flag and stage maps, makes the action honest. Vanishing of an abstract class without these choices does not supply the transported orientation.

If the residual generator actions are constant signs and the section-compatible skew walls give -1, then Theorem 7.10 identifies the character with $\det_W$. This is the restriction of the homogeneous orthogonal character of Δ_5. The chamber-symmetry extension and the symplectic automorphy factor are separate. The abelianization $W_{\mathrm{ab}} = (\mathbb{Z}/2)^3$ classifies constant sign characters; it does not classify the geometric root gerbe.

1.1.4 Skew wall geometry

A simple Pfaffian wall has the rank-two skew normal block

$$D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad \operatorname{pf}(D_u) = u, \qquad \det D_u = u^2.$$

The complex is placed in degrees $[-1, 0]$, with Pfaffian line $\det V^\vee$. A reflection lift and an invariant tangential unit then give the local sign -1. The scalar complex with differential u has determinant u and is a different object. Appendix 5 proves the skew calculation and its precise character consequence.

The global condition requires actual wall objects, full charge matches, reduced charts, overlaps and their cocycles, Weyl and stabilizer maps, boundary coverage and compatible transitions. Any assertion that only type-II walls contribute must also exclude other geometric components. One normal chart does not provide this atlas.

The abstract set of twelve signs has $2^{12} = 4096$ elements. The squared theta-leading coefficient is also $64^2 = 4096$. A geometric wall count needs a bijection or a weighted integration calculation on the actual atlas. Neither of those numerical equalities supplies it. In particular, $L_{\mathrm{II}}/2L_{\mathrm{II}}$ has eight elements.

With all four conditions and P_{fin} supplied, Theorem 7.19 yields the geometric Pfaffian and its W-character. Primitive recognition still requires (S1)–(S10). The trace still requires a state and cyclic comparison. The hypothesis-deletion statements of Proposition 1.2 retain these distinctions.

2 Mathieu products and the equivariant question

The K3 elliptic genus is $2\phi_{0,1}$. Its expansion in small $N = 4$ superconformal characters led Eguchi, Ooguri and Tachikawa to Mathieu moonshine [38]. Gannon proves existence of the graded M_{24}-module with the prescribed character multiplicities [45]. The mock modular form and its nonholomorphic completion are functions associated with that module; the completion is not itself a graded module. Its massive level grading also differs from a full Jacobi or root grading.

Umbral moonshine has twenty-three cases, indexed by Niemeier lattices with nonempty root systems X, with groups

$$G^X = \operatorname{Aut}(N^X)/W^X.$$

The A_1^{24} case is Mathieu moonshine. Duncan, Griffin and Ono prove module existence for the other twenty-two cases [37], following the Niemeier formulation of [21], which extends the six-group construction of [19]. The Leech lattice has no roots and lies outside this list. These module-existence theorems do not construct a Lie bracket, a vertex algebra or a compact Hall action. The K3 interpretations of [20, 42] address additional geometric and conformal-field-theoretic structure.

2.1 The full multiplicative lifts

Use the full twined genus normalization

$$\varphi_g(\tau, z) = \sum_{n \geq 0,\, l \in \mathbb{Z}} c_g(4n - l^2) q^n r^l, \qquad \varphi_{1A} = 2\phi_{0,1}.$$

The scalar functions are known for all twenty-six conjugacy classes; in particular the order-seven and order-fifteen genera are covered by [44]. Their multiplicative product uses all powers of g:

$$\Phi_g(Z) = qrs \prod_{(n,l,m) > 0} \exp\left(- \sum_{a \geq 1} \frac{c_{g^a}(4nm - l^2)}{a} (q^n r^l s^m)^a \right).$$

Here positivity is the cusp convention of Chapter 3. At the identity,

$$\Phi_{1A} = \Xi = 4096^{-1}\Delta_5^2.$$

The half-genus convention divides the seed by two; it must not be mixed with this full-genus product.

Wang–Williams prove that every Φ_g is a meromorphic Borcherds product on

$$L_g = U(N_g) \oplus U \oplus A_1, \qquad \operatorname{sign}(L_g) = (3,2),$$

where N_g is the level of the twining datum [116, Theorem 1.3]. The input is the complete tuple $(\varphi_{g^d})_{d\mid N_g}$, with its cusp data. For cycle shape $\prod_t t^{b_t}$, the weight is

$$k_g = \frac{1}{2}\sum_t b_t - 2.$$

Thus a single constant coefficient at the infinity cusp of φ_g does not determine the answer by the untwined level-one rule. The lattice, the power tuple and all relevant cusps are part of the theorem.

THEOREM 2.1 *(The order-seven and order-fifteen products).* The four classes $7A, 7B, 15A, 15B$ give two scalar products:

$$\Phi_{7A} = \Phi_{7B}, \qquad \Phi_{15A} = \Phi_{15B}.$$

Their levels and weights are respectively $(7,1)$ and $(15,0)$. Neither scalar product has a meromorphic square root on $\mathbb{H}_2$.

Proof. The paired classes have identical twining functions and identical power tuples for this construction. Their cycle shapes are $1^3 7^3$ and $1\cdot 3\cdot 5\cdot 15$; the weight formula gives $6/2-2=1$ and $4/2-2=0$.

The obstruction to a root can be read from the actual divisor. In $U(N)\oplus A_1$, choose $(e,f)=N$, $h^2=2$, with the other pairings zero. Then

$$Q(ue+wh+vf) = Nuv + w^2.$$

The primitive dual-lattice vectors

$$v_7 = \frac{e}{7} + \frac{h}{2} - \frac{f}{7}, \qquad v_{15} = \frac{e}{15} + \frac{h}{2} - \frac{2f}{15}$$

have norms

$$Q(v_7) = \frac{3}{28}, \qquad Q(v_{15}) = \frac{7}{60}.$$

The corresponding principal-part coefficients in the vector-valued inputs are -1 [116, Appendix A]. The largest negative index magnitude there is $1/4$, while $4Q(v_7) > 1/4$ and $4Q(v_{15}) > 1/4$. Consequently no positive multiple av_g, $a \geq 2$, contributes to the divisor multiplicity. The Borcherds multiplicity formula therefore gives a simple pole along each indicated generic hyperplane. A meromorphic square has even divisor order, so a meromorphic square root cannot exist. This is also the square-root classification in [116, Theorem 1.3(2)]. □

Remark 2.2 *(What the obstruction excludes).* A multiplier or an unramified level cover does not change the odd local divisor order. A branched cover could change it, but would change the recipient. Thus a construction requiring these full products to be squares of meromorphic Pfaffian denominators on $\mathbb{H}_2$ is obstructed. The full products themselves exist. Their weights 1 and 0 are not a failure of modularity, and the cyclotomic degrees $\varphi(7)=6$ and $\varphi(15)=8$ are not such an obstruction either.

The four conjugacy-class labels still matter for representation theory even though this scalar construction identifies each pair. They have six and four orbits, respectively, in the twenty-four point permutation representation. The geometric K3 symmetry criterion requires at least five orbits: the order-seven classes are realized by geometric symplectic automorphisms, while the order-fifteen classes fail this criterion [43, §2.1]. This has a different scope from existence of their scalar twining functions. Neither equality of paired functions nor a finite Fourier comparison constructs a geometric action.

2.2 The remaining equivariant algebra

The open comparison is an equivariant lift with actual M_{24}-representations in the required Jacobi and root degrees, compatible brackets or Hall operations, and the full power-trace formula above. An action on an independently defined state must produce the twining operations rather than be assigned after the scalar functions are known. If a denominator algebra is claimed, its simple generators, both parities, relations and invariant form must be constructed in the relevant lattice and completion.

The untwined half-genus algebra $\mathfrak{g}_{\Delta_5}$, the full products Φ_g, and possible umbral algebraic realizations are therefore separate targets. The theorem rules out the stated meromorphic square-root requirement for orders seven and fifteen; it does not rule out every equivariant algebraic construction. Nor does module existence for the umbral groups construct twenty-three Borcherds superalgebras or a map from the Mathieu classes to Niemeier root systems.

The selected low-level frame of Theorem 10.4 retains its exact hypotheses. Its list is not a classification of all twined Borcherds products. Likewise the Monster and Fake Monster denominator constructions provide comparison examples, not compact realizations on $K3 \times E$ or $K3 \times K3 \times E$ merely from their lattice ranks. The root lattice $\mathrm{II}_{25,1}$ and the orthogonal automorphic lattice $\mathrm{II}_{26,2}$ have different roles in the Fake Monster example.

3 States, fusion and acting algebras

3.1 The Hall–Borcherds state comparison

The trace target is Δ_5^{-2} in the cusp normalization used here. It is already a well-defined scalar meromorphic function. To derive it as a compact protected trace one needs an independently constructed state, operators, cyclic structure and normalization. Appendix 7 gives a sufficient state comparison and derives the product from its oscillators. A degree -3 pairing on a centre alone is not that trace construction.

The algebraic and categorical recipients must be typed. If C_X is a counital conilpotent coalgebra, its cobar ΩC_X is an associative algebra. A boundary endomorphism algebra can be compared with that cobar, and its bar with C_X. Ordinary Hochschild cochains, an internal factorization centre, cyclic chains and endomorphisms of the state are different objects. Any proposed comparison must preserve the coefficient tensor, unit, support and completion as well as the indicated operations [81, 82].

Scalar nonuniqueness has a simple exact witness. The underlying graded super vector space of $\mathfrak{g}_{\Delta_5}$ can be given the zero bracket without changing its root superdeterminant, as in Lemma 1.187. An even-odd pair with zero differential also has zero superdimension but nonzero cohomology. Thus even the full scalar product does not determine the bracket or the state complex. An acyclic summand, by contrast, would not by itself distinguish quasi-isomorphism types.

A Hall–Borcherds operator comparison additionally requires the positive Hall map, its actual tensor and cooperation, an opposite half and appropriate pairing when a double is claimed, and the current or factorization maps on E. The invariant Lie form on primitives is not a full Hopf pairing on their

ordinary nonabelian enveloping algebra. The geometric realization and normalization of a cyclic trace are further equations, not consequences of those names.

3.2 A compact chain-fusion comparison

Fix a Calabi–Yau category C_X, an admissible chart (Σ_{d-1}, C) and a specified open factorization category. A proposed chain-fusion comparison asks for a boundary condition $b(X, \Sigma_{d-1}, C)$ and a map

$$\Phi_d^{(\Sigma_{d-1},C)}(C_X) \longrightarrow R\,\mathrm{End}_C\big(b(X, \Sigma_{d-1}, C)\big),$$

with a definition of the source functor Φ_d, its tensor, and its clutching, cyclic and Calabi–Yau compatibilities. An equivalence requires an inverse on the claimed domain. A boundary object does not automatically generate that domain; both supported actions and the relevant generation or Morita hypotheses must be proved.

For $X = K3 \times E$, the proposed chiral specialization has reference target

$$U^{\mathrm{ch}}(\mathfrak{heis}_{\mathrm{Muk}}) \otimes U^{\mathrm{ch}}(\mathfrak{g}_{K3}^{\mathrm{BPS}}).$$

The recipient must specify whether these are Beilinson–Drinfeld current envelopes or corresponding objects on $\mathrm{Ran}(E)$, and which tensor is used. This target notation does not define $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$. The map requires the compact field complex, local and wrapped supports, coefficient comparisons and factorization operations. If an E_3- or BV model is used, its quantum master equation, anomaly, framing and compact-support descent must hold on those same complexes. The K3 surface, the geometric elliptic curve, the equivariant elliptic parameter and the world-sheet curve retain their distinct roles until compared by actual maps.

The outstanding boundary object is $b(K3 \times E, \Sigma_2, E)$ in this specified category. Local affine or toric models do not construct it on the compact threefold. Nor is a divisor on the geometric curve E automatically an Igusa cusp in Siegel space; that identification would require its own parameter map. Other compact non-toric threefolds, including Borcea–Voisin and Schoen examples, require their own boundary constructions.

The finite first-root comparisons do not settle this problem. The height bound of a root diagram, a geometric HN bound and a formal precision are different indices. A comparison at one finite stage does not give an equivalence at every stage, and Francis–Gaitsgory Lie–commutative duality does not supply the required associative-to-associative comparison.

3.3 A gravity-line action

A gravity-line realization would consist of an acting algebra, a module and cyclic sewing operations, with a determinant section identified with Δ_5^2 and a protected state trace identified with its reciprocal. The determinant has weight ten; its primitive square root has weight five. These are automorphic weights, not by themselves a central charge or a count of metric degrees of freedom. If a pentagon sewing diagram is used, its five composites, homotopies and determinant comparison must be constructed before assigning that scalar section to it.

Several numerical invariants coexist in this problem:

$$\big(\chi(\mathcal{O}_{S\times E}), \sum_q (-1)^q h^{0,q}(S \times E), \mathrm{rk}\,\mathfrak{h}_{\Delta_5}, \mathrm{wt}\,\Delta_5, \mathrm{rk}\,\widetilde{H}^*(S, \mathbb{Z})\big) = (0, 0, 3, 5, 24).$$

The first two entries are zero because the elliptic curve has $\chi(\mathcal{O}_E) = 0$. The third is the rank of the reference Cartan; calling it a geometric Heisenberg rank requires the Hall comparison. The fourth is

$f(0,0)/2 = 10/2$, and the last is the Mukai rank, equal to $\chi_{\mathrm{top}}(S)$. This tuple consists of different invariants. The numerical equality $3 + 2 = 5$ is not a construction relating their categories or lines, and $\chi(\mathcal{O}_S) = 2$ differs from $\chi_{\mathrm{top}}(S) = 24$.

A gravitational path-integral interpretation further needs a specified physical theory and the applicable asymptotic symmetry, modularity, vacuum and saddle hypotheses. The examples and constraints of [55] do not derive such a theory from the Igusa product. Likewise a prepared state is additional structure; normalizing a trace or adjoining a scalar vacuum does not solve its cyclic and operator compatibility equations.

4 The first geometric maps still required

The compact construction begins with the global reduced oriented coefficient pull–push on extension flags. It must preserve the common translation torsor, cross determinants, support conditions and full higher coherences. A compatible coproduct then needs its own coefficient maps and tensor. The charge-two obstruction for ordinary-Koszul cooperation on the specified polynomial shuffle carrier does not rule out every Hall recipient, but it prevents using that carrier without changing the obstructed requirement. A meromorphic tensor, localization or different coefficient category is a new construction to be compared.

Primitive recognition requires actual representatives and their product, coproduct, unit, counit, bracket and invariant-form matrices. The relation kernel, radical quotient, generation and PBW maps must be proved on finite word spaces containing every intermediate operation. In the first interior degree the target is $29 \mid 93$, not merely superdimension -64. The rank-28 differential works through height five and fails at height six in its unchanged presentation. A different derived presentation must retain its own comparison and cannot silently inherit the obstructed one.

Compatible finite comparisons must exhaust the operations and respect their genuine transition maps. Levelwise isomorphisms are sufficient for a pro-isomorphism, but not necessary in general. Cohomology commutes with a countable derived inverse limit only with control of the relevant $\varprojlim^1$ term. Root-line effectivity, primitive kernels and PBW completions each have their own comparison requirements.

The period construction of Chapter 15 gives a classical family of lattice-polarized K3 surfaces on a suitable marked or level cover of the Siegel threefold. The remaining compact mirror map must still connect the chosen threefold source, its Hodge or determinant line and its geometric section to that period family. The even cohomology of $S \times E$ cannot recover a varying elliptic period, and a rank-five polarized weight-three quotient is obstructed by its alternating polarization. These exclude the specified routes, not every possible geometric comparison.

The resulting problem remains to construct the compact source, its operations, modular line and cyclic state together, with maps to the established formal, algebraic and automorphic objects. The classical period model, full Mathieu products, local critical models and finite operator calculations sharpen that problem. They leave the original geometric realization to be supplied by those operation-preserving maps.

1 Lattice computations

The seven integral lattices are recorded here in one place: the hyperbolic plane $U = \mathrm{II}_{1,1}$, the Lorentzian root lattice $\Lambda_{II}^{2,1}$ of the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$, the K3 cohomology lattice $\Lambda_{K3} = \mathrm{II}_{3,19}$, the lattice-polarized K3 sublattice Λ_{pol}, the Mukai lattice $\widetilde{H}(S,\mathbb{Z}) \simeq \mathrm{II}_{4,20}$, the formal dyonic charge lattice $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$, and the signature-$(3,2)$ lattice $\Lambda^{3,2}$ that bridges $\mathbf{Sp}_4(\mathbb{Z})$ to the type-IV orthogonal group via the projective exterior-square image. Each is given by its Gram matrix or root data, signature, discriminant, and automorphism group, with primary citations (Conway–Sloane 1999 §2.5, §4, §16; Nikulin 1980 Theorem 1.1.2 and Theorem 1.14.4; Mukai 1984; Gritsenko–Nikulin 1998 Part II §2, §5). Ranks, signatures, and the Cartan matrix of the type-II simple roots are checked by the rational computations displayed below.

The identities about Δ_5, $\mathfrak{g}_{\Delta_5}$, the homogeneous orthogonal character $\widetilde{\nu}_{\Delta_5}$, and the Pfaffian wall normal form on $W^{(2)}(\Lambda_{II}^{2,1})$ use the lattice data fixed here. The geometric orientation character ϵ_o is compared with this extended character as in Theorem 3.1.

Convention 1.1 *(Sign of the bilinear form; orientation of the time-like cone).* The standard mathematical convention for the hyperbolic plane U is $U = (\mathbb{Z}^2, ((0,1),(1,0)))$, even unimodular of signature $(1,1)$ and discriminant -1 (Conway–Sloane 1999 §2.5). This convention is primary.

The exterior-square model of $\mathbf{Sp}_4(\mathbb{Z})$ in §2.1 employs the negated form

$$\Lambda^{(1,1)} = \left(\mathbb{Z}^2, \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\right),$$

The time-like cone $C(\Lambda^{2,1})_+ \subset \Lambda^{2,1} \otimes \mathbb{R}$ has negative square $(x,x) < 0$. The relation is $\Lambda^{(1,1)} = U(-1)$, and $f \mapsto -f$ identifies this hyperbolic plane with U. The cohomological lattices use U. The root and exterior-square lattices use the displayed BKM bases. Negating an entire lattice form is a separate operation: it exchanges the two signature entries and multiplies the determinant by $(-1)^{\mathrm{rk}\,L}$.

1.1 *The hyperbolic plane $U = \mathrm{II}_{1,1}$*

The hyperbolic plane is the unique even unimodular indefinite lattice of signature $(1,1)$ (Conway–Sloane 1999 §2.5; Nikulin 1980 Theorem 1.1.2).

Definition 1.2 *($U = \mathrm{II}_{1,1}$).* [definitional] Let $U := \mathbb{Z}e \oplus \mathbb{Z}f$ with Gram matrix

$$G_U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad e^2 = f^2 = 0, \quad e \cdot f = 1.$$

Verification 1.3 *(Rank, signature, discriminant of U).* [computed] $\mathrm{rk}\,U = 2$. The eigenvalues of G_U are ± 1, so the signature is $(1,1)$. The discriminant is

$$\mathrm{disc}\,U = \det G_U = -1.$$

U is even (every G_U-diagonal entry is $0 \equiv 0 \pmod 2$) and unimodular ($|\det G_U| = 1$). These are the defining invariants of $\mathrm{II}_{1,1}$ in the Conway–Sloane classification (Conway–Sloane 1999 §2.5, Theorem 2.5.1).

Proposition 1.4 *($\mathrm{Aut}(U)$).* [proved] The orthogonal group of U is

$$\mathrm{O}(U) = \{\pm \mathbf{I}\} \times \langle \sigma \rangle \simeq (\mathbb{Z}/2)^2, \qquad \sigma : e \leftrightarrow f.$$

The subgroup preserving the positive-cone component containing $e + f$ is $\mathrm{O}^+(U) = \{\mathbf{I}, \sigma\} \cong \mathbb{Z}/2$.

Proof. An isometry $\phi \in \mathrm{O}(U)$ preserves the isotropic vectors of U. In rank 2 over $\mathbb{Z}$, the isotropic primitive vectors are exactly $\pm e, \pm f$, so ϕ permutes these four vectors. The permutations preserving the bilinear form give the four-element group above (Conway–Sloane 1999 §4.1). □

Remark 1.5 *(Standard sign convention versus BKM convention).* The BKM convention uses $\Lambda^{(1,1)} = U(-1)$ in the cusp polarization of §2.1 and in the exterior-square model of the automorphic group, where the time-like cone of $\Lambda^{2,1}$ has negative square. The map $U \to U(-1)$, $(e, f) \mapsto (e, f)$ is a bijection of underlying $\mathbb{Z}$-modules; only the bilinear form flips sign. The lattices $\Lambda^{2,1}$ and $\Lambda^{3,2}$ are presented in BKM coordinates; even-unimodular ambient lattices $(\Lambda_{K3}, \widetilde{H}(S, \mathbb{Z}))$ are presented in the standard Conway–Sloane convention.

1.2 *The Lorentzian root lattice* $\Lambda^{2,1}_{II}$

The Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$ of [54, Theorem 2.1, Section 5.1.1, case $(t = 1, II, \overline{6})$] is graded by an integral lattice of signature $(2, 1)$. This is the lattice on which the type-II simple-root Cartan matrix is $2(2\mathbf{I}_3 - \mathbf{J}_3)$ where $\mathbf{J}_3$ is the all-ones matrix.

Definition 1.6 *(Ambient* $\Lambda^{2,1} = U(-1) \oplus \langle 2 \rangle$*).* [definitional] Set

$$\Lambda^{2,1} := U(-1) \oplus \langle 2 \rangle = \mathbb{Z}e \oplus \mathbb{Z}f \oplus \mathbb{Z}h, \qquad e \cdot f = -1,\ e^2 = f^2 = 0,\ h^2 = 2.$$

The Gram matrix in the basis (e, f, h) is

$$G_{2,1} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad \det G_{2,1} = -2, \qquad \operatorname{disc} \Lambda^{2,1} = -2.$$

The BKM lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ in (f_2, f_3, f_{-2})-coordinates uses exactly this Gram, with $\Lambda^{(1,1)} = U(-1)$ on the (f_2, f_{-2})-block and $[2] = \langle 2 \rangle$ on the f_3-axis.

Definition 1.7 *(*$\Lambda^{2,1}_{II}$ *as the index-four type-II sublattice of* $\Lambda^{2,1}$*).* [definitional] The type-II sublattice

$$\Lambda^{2,1}_{II} \;:=\; \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv 0\,(2),\ n \equiv 0\,(2)\} \;\subset\; \Lambda^{2,1}$$

is the index-4 sublattice on which the type-II Cartan matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$ is the Gram of the simple roots $(\delta_1, \delta_2, \delta_3)$ of Proposition 1.10. Abstractly it is the lattice

$$\Lambda^{2,1}_{II} \;\simeq\; U(4) \oplus \langle 2 \rangle, \qquad \operatorname{disc} \Lambda^{2,1}_{II} = -32,$$

matching the terminology of Chapter 1 and the type-II sublattice convention of Chapter 4.

Verification 1.8 *(Rank, signature, discriminant of* $\Lambda^{2,1}_{II}$*).* [computed] $\operatorname{rk} \Lambda^{2,1}_{II} = 3$. The Gram of $\Lambda^{2,1}_{II}$ in $(\delta_1, \delta_2, \delta_3)$ of Proposition 1.10 is the Cartan matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$ of equation (4.2); its determinant is

$$\det\big(2(2\mathbf{I}_3 - \mathbf{J}_3)\big) = -32.$$

The hyperbolic-plus-rank-one block of the ambient $\Lambda^{2,1}$ has signature $(2, 1)$; the index-4 sublattice $\Lambda^{2,1}_{II}$ inherits this signature, so

$$\sigma(\Lambda^{2,1}_{II}) = (2, 1).$$

The discriminant group $(\Lambda^{2,1}_{II})^* / \Lambda^{2,1}_{II}$ has order 32 as required. The lattice $\Lambda^{2,1}_{II}$ is even because every $\delta_i^2 = 2$ and the off-diagonal couplings $(\delta_i, \delta_j) = -2$ are integral. Changing the sign of one hyperbolic basis

vector gives the standard hyperbolic-plane convention $U = ((0,1),(1,0))$ and presents $\Lambda^{2,\mathrm{I}} \simeq U \oplus \langle 2\rangle$ and $\Lambda^{2,\mathrm{I}}_{II} \simeq U(4) \oplus \langle 2\rangle$; rank, signature, discriminant group order, automorphism group, and Cartan datum are unchanged.

Remark 1.9 *(Identification with $\Lambda^{2,\mathrm{I}}$).* The BKM convention uses $\Lambda^{2,\mathrm{I}} = \Lambda^{(1,1)} \oplus [2] = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$ with Gram in (f_2, f_3, f_{-2})

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

matching $G_{2,1}$ of Definition 1.6 up to the basis swap $e \mapsto f_2, f \mapsto f_{-2}, h \mapsto f_3$. The type-II sublattice $\Lambda^{2,\mathrm{I}}_{II} \subset \Lambda^{2,\mathrm{I}}$ of Definition 1.7 is the index-4 sublattice on which the simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ of Proposition 1.10 live; it is the even Lorentzian root lattice of $\mathfrak{g}_{\Delta_5}$.

PROPOSITION 1.10 *(Type-II simple-root Cartan).* [proved] In the coordinates (f_2, f_3, f_{-2}) fixed by the Weyl chamber, set

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Then $\delta_i^2 = 2$ for $i = 1, 2, 3$ and

$$\big((\delta_i, \delta_j)\big)_{i,j=1,2,3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix} = 2(2\mathbf{I}_3 - \mathbf{J}_3),$$

where $\mathbf{J}_3$ is the 3×3 all-ones matrix. Each δ_i is type-II in the sense that $(\delta_i, \Lambda^{2,\mathrm{I}}) = 2\mathbb{Z}$.

Proof. Direct computation in the BKM Gram gives the matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$, equal to the displayed Cartan matrix. The real-root presentation uses the Cartan formalism of [66]. The type-II divisibility statement follows because each δ_i pairs against $\Lambda^{2,\mathrm{I}}$ with even values: (δ_i, f_2), (δ_i, f_3), $(\delta_i, f_{-2}) \in 2\mathbb{Z}$ for each i. This is the GN root datum of [54, Section 5.1.1, case $(1, II, \overline{0})$]. □

PROPOSITION 1.11 *(Weyl vector ρ).* [proved] The Weyl vector

$$\rho := \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \tfrac{1}{2}f_3 + f_{-2} \in \Lambda^{2,\mathrm{I}} \otimes \mathbb{Q}$$

satisfies $(\rho, \delta_i) = -1$ for $i = 1, 2, 3$ and

$$(\rho, \rho) = -\tfrac{3}{2}, \qquad \rho \in C(\Lambda^{2,\mathrm{I}})_+.$$

Choose the component of the negative cone containing ρ. On its unit hyperboloid, the chamber $\mathcal{P}_{II} = \{x : (x, \delta_i) \le 0,\ i = 1, 2, 3\}$ is an ideal hyperbolic triangle and has finite area.

Proof. In the basis (f_2, f_3, f_{-2}) the vector is $\rho = (1, -1/2, 1)$. Pairing it with $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$ and $\delta_3 = f_3$ gives -1 in each case. This is the Weyl-vector convention of [54, (2.7)]. Direct computation in the BKM Gram of Definition 1.7 gives

$$(\rho, \rho) = 2(f_2, f_{-2}) + \tfrac{1}{4}(f_3, f_3) = 2(-1) + \tfrac{1}{4}(2) = -\tfrac{3}{2}.$$

The three isotropic rays generated by $\delta_1 + \delta_2$, $\delta_2 + \delta_3$ and $\delta_1 + \delta_3$ lie in the boundary of the chosen cone: their pairing with ρ is -2. Each is perpendicular to its two indicated roots and has pairing -4 with the third. They are therefore the three ideal vertices of the chamber. Its interior contains the normalized ρ. An ideal triangle has area π for curvature -1. □

The basis $\delta_1+\delta_3,\ \delta_2+\delta_3,\ \delta_3$ has Gram matrix $U(-4)\oplus\langle 2\rangle$, with determinant -32. Since $(\rho,\delta_i)=-1$ for each generator, $\rho\in(\Lambda^{2,1}_{II})^\vee$. These are arithmetic properties of the root lattice.

Proposition 1.12 *(Reflection group; Nikulin's theorem).* [proved; Nikulin 1980 Theorem 1.4.1] Let $W^{(2)}(\Lambda^{2,1})$ be the subgroup of $\mathbf{O}(\Lambda^{2,1})$ generated by reflections in primitive roots of square 2. Then

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}),$$

where the subscript $+$ denotes the index-two subgroup preserving the time-like cone $C(\Lambda^{2,1})_+$. Equivalently, every isometry of $\Lambda^{2,1}$ preserving the chosen positive cone is a product of square-2 reflections.

Proof. Nikulin [97, Theorem 1.4.1] establishes for the rank-three hyperbolic lattice $\Lambda^{(1,1)}\oplus[2]$ the equality $\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1})$; see also Gritsenko–Nikulin [52, Proposition 1.5]. □

Verification 1.13 *(Type-II reflection orbit).* [computed] For each $\delta\in\{\delta_1,\delta_2,\delta_3\}$, the reflection $s_\delta: x\mapsto x-(x,\delta)\delta$ satisfies $s_\delta(\delta)=-\delta$ and preserves the Cartan matrix on the δ-orbit. Concretely, $s_{\delta_1}(\delta_1)=-\delta_1$, $s_{\delta_1}(\delta_2)=\delta_2+2\delta_1$, etc.; the bilinear form on $\{\delta_1,\delta_2,\delta_3\}$ is reflection-invariant by direct matrix computation.

Remark 1.14 *(The two type-2 divisibility classes).* Each $\delta\in\Lambda^{2,1}$ with $\delta^2=2$ has divisibility $(\delta,\Lambda^{2,1})\in\{\mathbb{Z},2\mathbb{Z}\}$. The two cases are recorded as *type I* ($(\delta,\Lambda^{2,1})=\mathbb{Z}$) and *type II* ($(\delta,\Lambda^{2,1})=2\mathbb{Z}$); both classes generate index-finite normal subgroups of $W^{(2)}(\Lambda^{2,1})$:

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_I)] = 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{II})] = 6,$$

following [52, Sections 4–5]. The denominator algebra $\mathfrak{g}_{\Delta_5}$ uses the type-II reflection chamber.

1.3 *The K3 lattice* $\Lambda_{K3}=\mathrm{II}_{3,19}$

The integral cohomology of a K3 surface carries the ambient lattice for its divisors and periods. The distinction between an orthogonal sum over $\mathbb{Q}$ and one over $\mathbb{Z}$ is essential: discriminant gluing usually joins the algebraic and transcendental summands.

Definition 1.15 *(K3 lattice).* For a complex K3 surface S, its K3 lattice is

$$\Lambda_{K3}=H^2(S,\mathbb{Z})\cong U^{\oplus 3}\oplus E_8(-1)^{\oplus 2},$$

with the cup-product pairing. Here E_8 is the positive definite root lattice and $E_8(-1)$ has the negated form.

Verification 1.16 *(K3 invariants).* Each summand is even and unimodular. Their ranks and signatures give

$$\operatorname{rk}\Lambda_{K3}=22, \qquad \sigma(\Lambda_{K3})=(3,19), \qquad \det\Lambda_{K3}=-1.$$

The classification of indefinite even unimodular lattices therefore identifies it with $\mathrm{II}_{3,19}$ [57, Chapter 14, §1].

For a nondegenerate integral lattice L, put

$$L^\vee=\{x\in L\otimes\mathbb{Q} : (x,L)\subset\mathbb{Z}\}, \qquad A_L=L^\vee/L.$$

If L is even, its discriminant quadratic form is $q_L(\bar x)=(x,x)\bmod 2\mathbb{Z}$. An anti-isometry of discriminant forms is an isomorphism $\phi: A_L\to A_T$ with $q_T(\phi(\bar x))=-q_L(\bar x)$.

PROPOSITION 1.17 *(Primitive vectors and algebraic classes).* The group $\mathbf{O}(\Lambda_{K_3})$ is transitive on primitive vectors of each fixed square. For a projective K3 surface of Picard rank ρ, put $N = \mathrm{NS}(S)$ and $T = N^{\perp}$. These are primitive sublattices, with

$$\sigma(N) = (1, \rho - 1), \qquad \sigma(T) = (2, 20 - \rho), \qquad \Lambda_{K_3} \otimes \mathbb{Q} = (N \otimes \mathbb{Q}) \oplus (T \otimes \mathbb{Q}).$$

There is an anti-isometry $\phi : A_N \to A_T$ for which

$$\Lambda_{K_3}/(N \oplus T) = \{(\bar{x}, \phi(\bar{x})) : \bar{x} \in A_N\} \subset A_N \oplus A_T.$$

In particular, $[\Lambda_{K_3} : N \oplus T] = |\det N|$.

Proof. In a lattice containing $U \oplus U$, Eichler's criterion determines a primitive vector's orbit by its square and discriminant class. The latter is trivial here [57, Chapter 14, Corollary 1.10].

The equality $N = H^{1,1}(S) \cap H^2(S, \mathbb{Z})$ proves primitivity. An orthogonal complement in an integral lattice is also primitive. The Hodge index theorem gives the signature of N, hence that of T and the rational decomposition.

Let $K = \Lambda_{K_3}/(N \oplus T) \subset A_N \oplus A_T$. Primitivity makes each projection of K injective. Unimodularity gives $|K|^2 = |A_N||A_T|$, so both projections are surjective. Thus K is the graph of an isomorphism. Evenness of Λ_{K_3} says that this graph is isotropic for $q_N + q_T$, which is the anti-isometry assertion. □

For example, if $N = \mathbb{Z}h$ with $h = e + df$ in a hyperbolic plane and $d > 0$, then $h^2 = 2d$ and the complementary line is $\mathbb{Z}(e - df)$. These two lines have index $2d$ in U. This gives the same index for $N \oplus T \subset \Lambda_{K_3}$ and rules out an integral splitting in this example. Lattice isometries also need not be automorphisms of a fixed surface: such an automorphism preserves its Hodge structure and ample cone.

1.4 *Lattice polarizations and discriminant gluing*

Definition 1.18 *(Lattice polarization).* Let $N = \Lambda_{\mathrm{pol}}$ be an even lattice of signature $(1, \rho - 1)$, where $1 \le \rho \le 20$. An ample N-polarization of a K3 surface is a primitive embedding $N \hookrightarrow \mathrm{NS}(S)$ whose image contains an ample class. A marking identifies this with an embedding $N \hookrightarrow \Lambda_{K_3}$ and places the period line in

$$\mathcal{D}_{N^{\perp}} = \{[\omega] \in \mathbb{P}(N^{\perp} \otimes \mathbb{C}) : (\omega, \omega) = 0,\ (\omega, \bar{\omega}) > 0\}.$$

The period map to this domain is for marked surfaces. Forgetting the marking requires the corresponding arithmetic quotient.

VERIFICATION 1.19 *(Complement and gluing index).* For a primitive embedding of N into the K3 lattice, its complement $T = N^{\perp}$ has rank $22 - \rho$ and signature $(2, 20 - \rho)$. The argument of Proposition 1.17 gives

$$(A_N, q_N) \cong (A_T, -q_T), \qquad |\det N| = |\det T| = [\Lambda_{K_3} : N \oplus T].$$

The finite-index extension is specified by the graph of this anti-isometry in $A_N \oplus A_T$.

PROPOSITION 1.20 *(Embedding by discriminant gluing).* Let N be even of signature $(1, \rho - 1)$. It embeds primitively in Λ_{K_3} precisely when an even lattice T of signature $(2, 20 - \rho)$ admits an anti-isometry $A_N \cong A_T$.

A sufficient condition for existence and uniqueness of the embedding up to $\mathbf{O}(\Lambda_{K_3})$ is

$$\rho \le 19, \qquad \ell(A_N) + 2 \le 22 - \rho,$$

where $\ell(A_N)$ is the minimal number of generators of A_N.

Proof. Necessity follows from the preceding gluing calculation. Conversely, given T and an anti-isometry ϕ, form

$$L_\phi = \{(x, y) \in N^\vee \oplus T^\vee : \bar{y} = \phi(\bar{x})\}.$$

The graph is isotropic, so L_ϕ is even and integral. Its index $|A_N|$ over $N \oplus T$ makes it unimodular. Its signature is $(3, 19)$, hence $L_\phi \cong \Lambda_{K3}$. The graph condition also shows $L_\phi \cap (N \otimes \mathbb{Q}) = N$, so the embedding is primitive.

The sufficient condition is Nikulin's indefinite embedding theorem [96, Theorem 1.14.4]. Its stated form is [57, Chapter 14, Theorem 1.12]. The inequalities $1 < 3$ and $\rho - 1 < 19$ ensure that the complement has both signs. The displayed bound is the required discriminant-length bound. □

Thus a primitive embedding requires an actual complement lattice, not merely a proposed discriminant group. Once an embedding exists, the K3 period theorem realizes N as the full Néron–Severi lattice of a projective K3 surface. Choosing a positive chamber determines which classes are ample. The explicit embeddings and ample classes used here were constructed in §2.

Remark 1.21 *(Mirror lattice convention).* Suppose $N^\perp$ contains a hyperbolic plane U_{mir}. Its unimodularity gives an integral splitting

$$N^\perp = U_{\mathrm{mir}} \oplus \check{N}, \qquad \check{N} = U_{\mathrm{mir}}^\perp \cap N^\perp.$$

Then $\check{N}$ has rank $20 - \rho$ and signature $(1, 19 - \rho)$. This is the lattice used for the mirror polarization. The choice of U_{mir} is part of the construction.

For $N = U \oplus \langle -2 \rangle$, embed U in one hyperbolic summand of Λ_{K3} and the vector of square -2 as a root in one $E_8(-1)$ summand. The root complement there is $E_7(-1)$. Removing one of the two remaining hyperbolic planes gives

$$\check{N} \cong U \oplus E_7(-1) \oplus E_8(-1), \qquad \sigma(\check{N}) = (1, 16).$$

The interpretation of Δ_5^{-2} as a mirror section of the inverse Hodge line with reduced type-II Heegner support remains a conjectural comparison. The lattice splitting alone does not construct that section or the compact object of Chapter 6.

1.5 *The Mukai lattice* $\widetilde{H}(S, \mathbb{Z}) = \mathrm{II}_{4,20}$

Definition 1.22 *(Cohomological and algebraic Mukai lattices).* The cohomological Mukai lattice and its pairing are

$$\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}), \qquad \langle (r, D, s), (r', D', s') \rangle = D \cdot D' - rs' - r's.$$

For a projective K3 surface, the algebraic sublattice is $\widetilde{H}_{\mathrm{alg}}(S, \mathbb{Z}) = \mathbb{Z} \oplus \mathrm{NS}(S) \oplus \mathbb{Z}$.

Verification 1.23 *(Mukai invariants).* The two outer cohomology groups form a hyperbolic plane. Consequently

$$\widetilde{H}(S, \mathbb{Z}) \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2}, \qquad \mathrm{rk} = 24, \quad \sigma = (4, 20), \quad \det = 1.$$

The algebraic sublattice instead has rank $\rho + 2$ and signature $(2, \rho)$. Its determinant has absolute value $|\det \mathrm{NS}(S)|$, which need not be one.

PROPOSITION 1.24 *(Mukai vector and Euler form).* For $F \in D^b \operatorname{Coh}(S)$,

$$v(F) = \operatorname{ch}(F)\sqrt{\operatorname{td}(S)} = (\operatorname{rk} F, c_1(F), \operatorname{rk} F + \operatorname{ch}_2(F)) \in \widetilde{H}_{\mathrm{alg}}(S, \mathbb{Z}),$$

where the last coordinate is integrated over S. For any F, G,

$$\chi_S(F, G) = -\langle v(F), v(G)\rangle.$$

This identifies the numerical Grothendieck group with the algebraic Mukai lattice.

Proof. The identities $\sqrt{\operatorname{td}(S)} = 1 + [\mathrm{pt}]$ and Hirzebruch–Riemann–Roch give the formulas. The vectors of line bundles and point sheaves generate the algebraic lattice, whose pairing is nondegenerate. Thus the kernel of the Mukai vector is exactly the Euler radical, as proved in Proposition 1.2. □

Remark 1.25 *(The hyperbolic sign).* On H^2 the Mukai pairing is the cup product. On $H^0 \oplus H^4$ its matrix in the rank and point bases is

$$J_- = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

The integral matrix $D = \operatorname{diag}(1, -1)$ satisfies $D^t J_- D = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Changing the off-diagonal sign therefore leaves the lattice isometry class and signature unchanged. Riemann–Roch fixes the minus sign for the specified coordinates of $v(F)$. For instance, $\langle v(\mathcal{O}_S), v(\mathcal{O}_p)\rangle = -1$, whereas $\chi_S(\mathcal{O}_S, \mathcal{O}_p) = 1$. For a nonisotropic vector $v \in \widetilde{H}(S, \mathbb{R})$, the orthogonal reflection is $r_v(x) = x - 2\langle x, v\rangle v/\langle v, v\rangle$.

PROPOSITION 1.26 *(Isometries of the Mukai lattice).* The group $\mathbf{O}(\widetilde{H}(S, \mathbb{Z}))$ is transitive on primitive vectors of each fixed square. A $\mathbb{C}$-linear exact derived autoequivalence of S induces an integral Hodge isometry of this lattice, with H^0 and H^4 assigned Hodge type $(1, 1)$.

Proof. The first assertion follows from the same primitive-vector criterion as Proposition 1.17. A Fourier–Mukai kernel $\mathcal{P}$ gives the action $\alpha \mapsto p_{2*}(p_1^* \alpha\, v(\mathcal{P}))$. Integrality of $v(\mathcal{P})$, the projection formula and the inverse kernel make this an integral isometry. Algebraicity of its Chern character preserves the Mukai Hodge structure [57, Chapter 16, Lemma 3.1 and Proposition 3.2]. □

The vectors $v(\mathcal{O}_S) = (1, 0, 1)$ and $v(\mathcal{O}_p) = (0, 0, 1)$ have squares -2 and 0, respectively. Each represents its own primitive lattice orbit. Transitivity of the full orthogonal group does not assert transitivity of derived autoequivalences on objects or on a fixed Hodge structure.

1.6 The formal paired charge lattice

Definition 1.27 *(Paired charges and Gram map).* For $X = S \times E$, define

$$\Gamma_X^{\mathrm{form}} = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z}), \qquad c = (Q, P).$$

Its componentwise form is

$$\langle (Q, P), (Q', P')\rangle_\Gamma = \langle Q, Q'\rangle + \langle P, P'\rangle,$$

and its quadratic Gram map is

$$\Pi_X(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2}\right) \in \Gamma_{\mathrm{gram}} = \mathbb{Z}^3.$$

Evenness makes the half-squares integral. The polarization of Π_X is the bilinear map of Lemma 6.5.

Verification 1.28 *(Paired-lattice invariants).* With the componentwise form, Γ_X^{form} is even unimodular of rank 48, signature $(8, 40)$ and determinant 1. Thus it is isometric to $\mathrm{II}_{8,40}$.

Remark 1.29 *(Three pairings on paired charges).* Write ω_E for the point class of integral one. The Künneth isomorphism identifies $c = (Q, P)$ with

$$v_X = P \otimes 1_E + Q \otimes \omega_E.$$

Besides the componentwise form, this group carries the symmetric cross-pairing

$$b_{\mathrm{cross}}(c, c') = Q \cdot P' + P \cdot Q'.$$

It is the tensor of the Mukai form with the hyperbolic form on $H^0(E, \mathbb{Z}) \oplus H^2(E, \mathbb{Z})$, and has signature $(24, 24)$. For actual coherent complexes, the Euler form is instead

$$\chi_X(c, c') = Q \cdot P' - P \cdot Q'.$$

This alternating form is derived in §6.

The two symmetric forms are auxiliary lattice structures. Coherent charges lie in the paired algebraic sublattice, and the Gram map is an additional quadratic operation on those charges.

Lemma 1.30 *(Two hyperbolic planes and primitive lifts).* Choose orthogonal hyperbolic summands U_1, U_2 of $\widetilde{H}(S, \mathbb{Z})$, with bases e_i, f_i and $e_i \cdot f_i = 1$. For every $(n, l, m) \in \mathbb{Z}^3$, the vectors

$$Q = e_1 + nf_1, \qquad P = lf_1 + e_2 + mf_2$$

span a primitive rank-two sublattice and satisfy $\Pi_X(Q, P) = (n, l, m)$.

Proof. The pairings are $Q^2 = 2n, Q \cdot P = l$, and $P^2 = 2m$. The e_1, e_2 rows of the two-column coordinate matrix form the identity matrix. Its maximal minors therefore have greatest common divisor one, which proves primitivity. □

Verification 1.31 *(Root coordinates of a lift).* In the cusp coordinates fixed for the root lattice,

$$\alpha(\Pi_X(Q, P)) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The coefficient of $e_1 \wedge e_2$ in $Q \wedge P$ is one. These are integral lattice identities for every triple. The chosen hyperbolic planes need not be algebraic for a fixed surface. A coherent representative, and then a semistable one, require additional geometric conditions.

1.7 *The signature-*$(3, 2)$ *lattice* $\Lambda^{3,2}$ *and the* $\mathbf{PSp}_4 \simeq \mathbf{SO}(3, 2)_+$ *bridge*

The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}_+(3, 2)$ identifies the Siegel domain with the type-IV orthogonal domain of an integral lattice $\Lambda^{3,2}$ of signature $(3, 2)$, and it identifies $\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$ with the exterior-square arithmetic group $\Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+$ ([52, Section 3]; Borcherds 1995 §5). The Igusa cusp form Δ_5 lives on the Siegel side; the Borcherds product expansion lives on the orthogonal side.

Definition 1.32 *($\Lambda^{3,2}$ in the exterior-square model).* [definitional] Let $\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4$ and consider the exterior square $\wedge^2\Lambda^4$ with the symmetric pairing

$$u \wedge v \cdot u' \wedge v' = [\det(u, v, u', v')]_{e_1\wedge e_2\wedge e_3\wedge e_4} \in \mathbb{Z}.$$

Then $(\wedge^2\Lambda^4, (\ ,\ )) \simeq \mathrm{II}_{3,3}$ as an even unimodular lattice of signature $(3, 3)$. Fix the primitive vector $q_0 := e_1 \wedge e_3 + e_2 \wedge e_4 \in \wedge^2\Lambda^4$, for which $q_0^2 = -2$, and define

$$\Lambda^{3,2} := q_0^{\perp} \subset \wedge^2\Lambda^4.$$

VERIFICATION 1.33 *(Rank, signature, discriminant of $\Lambda^{3,2}$).* [computed] $\operatorname{rk}\wedge^2\Lambda^4 = \binom{4}{2} = 6$. Taking the orthogonal complement of the primitive vector q_0 gives $\operatorname{rk}\Lambda^{3,2} = 5$. The explicit Gram matrix of $\Lambda^{3,2}$ in the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ is

$$G_{3,2} = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where $f_1 = e_1 \wedge e_2$, $f_2 = e_2 \wedge e_3$, $f_3 = e_1 \wedge e_3 - e_2 \wedge e_4$, $f_{-2} = e_4 \wedge e_1$, $f_{-1} = e_4 \wedge e_3$. Direct computation: the matrix is antidiagonal except for the central $h^2 = 2$ entry, with off-diagonal entries $-1, -1, -1, -1$. Expansion gives

$$\det G_{3,2} = 2 \cdot (-1)^4 \cdot \operatorname{sgn}(1, 2, 3, 4 \mapsto 4, 3, 2, 1) = 2 \cdot 1 \cdot (+1) = +2,$$

since the permutation $(1, 2, 3, 4) \mapsto (4, 3, 2, 1)$ has six inversions and even parity. Therefore

$$\sigma(\Lambda^{3,2}) = (3, 2), \qquad \operatorname{disc}\Lambda^{3,2} = 2.$$

This is a rational computation in the exterior-square lattice.

PROPOSITION 1.34 *($\Lambda^{3,2} \simeq U \oplus U \oplus \langle 2\rangle$ in the fixed convention; $\Lambda^{3,2} \simeq U(-1) \oplus U(-1) \oplus \langle 2\rangle$ in the standard convention).* [proved] In the BKM convention of the exterior-square model,

$$\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2],$$

with $\Lambda^{(1,1)} = U(-1)$ and $[2]$ the rank-1 lattice with Gram (2). The Gram presentation displayed in Verification 1.33 arises from the f-basis: (f_1, f_{-1}) pairs as a copy of $\Lambda^{(1,1)}$ ($(f_1, f_{-1}) = -1$), (f_2, f_{-2}) pairs as a second copy of $\Lambda^{(1,1)}$ ($(f_2, f_{-2}) = -1$), and f_3 contributes $(f_3, f_3) = 2$.

Proof. The pairings $(f_1, f_{-1}) = -1$, $(f_2, f_{-2}) = -1$, $(f_3, f_3) = 2$ are read off from $G_{3,2}$. The orthogonal decomposition is direct: $\mathbb{Z}f_1 + \mathbb{Z}f_{-1} \perp \mathbb{Z}f_2 + \mathbb{Z}f_{-2} \perp \mathbb{Z}f_3$ inside $\Lambda^{3,2}$ because every cross-pair vanishes in $G_{3,2}$. The identification with $\Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ follows because each rank-2 block has Gram $((0, -1), (-1, 0)) = \Lambda^{(1,1)}$ and the rank-1 block has Gram $(2) = [2]$. □

PROPOSITION 1.35 *(The projective exterior-square arithmetic group).* [proved; exterior-square model of $\mathbf{Sp}_4(\mathbb{Z})$] The exterior-square representation $\wedge^2$ induces an isomorphism

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \xrightarrow{\sim} \Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+, \qquad \Gamma_{3,2} := \wedge^2\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\}.$$

Concretely, the integral generators

$$g_0 = \begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix}, \quad g_M = \begin{pmatrix} \mathbf{I}_2 & M \\ 0 & \mathbf{I}_2 \end{pmatrix}, \quad g_A = \begin{pmatrix} {}^tA^{-1} & 0 \\ 0 & A \end{pmatrix} \quad (A \in \mathbf{GL}_2(\mathbb{Z}))$$

of $\mathbf{Sp}_4(\mathbb{Z})$ map under $\wedge^2$ to integral isometries of $\Lambda^{3,2}$, displayed as the three explicit 5×5 matrices $\wedge^2(g_0)$, $\wedge^2(g_M)$, $\wedge^2(g_A)$ in the exterior-square model. The image is the arithmetic group through

which the Siegel modular group acts on the type-IV model. The Siegel space $\mathbb{H}_2$ embeds into the type-IV upper half-space $\mathbb{H}_+^{\mathbf{IV}} \subset \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C})$ via

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \mapsto {}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1) z_0,$$

and the action of $\mathbf{Sp}_4(\mathbb{Z})$ on $\mathbb{H}_2$ corresponds to the action of $\Gamma_{3,2}$ on $\mathbb{H}_+^{\mathbf{IV}}$.

Proof. The Pfaffian pairing on $\wedge^2\Lambda^4$ is preserved by $\wedge^2\mathbf{SL}_4$. The subgroup fixing q_0 is $\mathbf{Sp}_4(\mathbb{Z})$ by definition of the symplectic form $B_{q_0}(x, y)e_1 \wedge e_2 \wedge e_3 \wedge e_4 = -x \wedge y \wedge q_0$. The kernel on $q_0^\perp = \Lambda^{3,2}$ is $\{\pm\mathbf{I}_4\}$. The exceptional Lie-algebra isomorphism $\mathfrak{sp}_4(\mathbb{R}) \simeq \mathfrak{so}(3, 2)$ extends to the integral form on the displayed generators. The discriminant group of $\Lambda^{3,2}$ has one non-zero element; hence the stable and full projective components coincide in the Gritsenko–Nikulin normalization. This is the content of [52, Section 3] and the exterior-square model of the automorphic group. □

Remark 1.36 *(Embedding $\Lambda^{2,1}_{II} \subset \Lambda^{3,2}$).* The primitive sublattice

$$\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}$$

of the Weyl-chamber section is the BKM root lattice: discarding the (f_1, f_{-1})-block of $\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ leaves $\Lambda^{(1,1)} \oplus [2] \simeq \Lambda^{2,1}$. The type-II sublattice $\Lambda^{2,1}_{II} = \{mf_2 + lf_3 + nf_{-2} \mid m, n \equiv 0 \pmod 2\}$ is the lattice of even integral type-II classes, on which the simple-root Cartan

$$\big((\delta_i, \delta_j)\big) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

of Proposition 1.10 is realized via $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$. Every isometry of $\Lambda^{2,1}$ extends to an isometry of $\Lambda^{3,2}$ by the identity on $(\Lambda^{2,1})^\perp \subset \Lambda^{3,2}$.

1.8 *Cross-references and consistency check*

The lattices recorded above enter as follows.

- $U = \mathrm{II}_{1,1}$: every U-summand of $\Lambda^{3,2}$, Λ_{K3}, and $\widetilde{H}(S, \mathbb{Z})$. The relation $\Lambda^{(1,1)} = U(-1)$ records the BKM negated convention.
- $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2 \rangle$ (with $U(4)$ written as $U(-4)$ in the BKM sign convention): the BKM root lattice on which the type-II simple-root Cartan $2(2\mathbf{I}_3 - \mathbf{J}_3)$ and the Weyl vector $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$ are realized. The denominator algebra $\mathfrak{g}_{\Delta_5} = \mathfrak{g}_{\Delta_5}$ is graded by $\Lambda^{2,1}_{II}$ ([54, Theorem 2.1]).
- $\Lambda_{K3} = \mathrm{II}_{3,19}$: K3 cohomology with intersection pairing. Polarized K3 sublattices Λ_{pol} embed primitively into Λ_{K3}.
- Λ_{pol}: signature $(1, \rho - 1)$ for $\rho \le 20$. It is the lattice-polarization sublattice of Λ_{K3}, not the full Mukai lattice.
- $\widetilde{H}(S, \mathbb{Z}) = \mathrm{II}_{4,20}$: the cohomological Mukai lattice containing each component Q_Y, P_Y of $v_X(\mathcal{I}_Y)$. Their three pairings give the Gram map (Theorem 3.8).
- $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z})$: the formal additive dyonic charge lattice of $X = S \times E$, on which the quadratic Gram-shadow projection Π_X and the symmetric bilinear 2-cocycle B are defined (Definition 1.1).

- $\Lambda^{3,2} \simeq U(-1) \oplus U(-1) \oplus \langle 2 \rangle$: the signature-$(3,2)$ orthogonal lattice that bridges $\mathbf{PSp}_4(\mathbb{Z})$ to the type-IV orthogonal group via the exterior-square representation; on this lattice the Borcherds product expansion of Δ_5 is written.

VERIFICATION 1.37 *(End-to-end computation check).* [computed] The lattice data satisfy, in one self-contained $\mathbb{Q}$-arithmetic computation: (i) the rank-5 Gram matrix $G_{3,2}$ in the $(f_1, f_2, f_3, f_{-2}, f_{-1})$-basis; (ii) the orthogonality of the symplectic invariant $q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$ to the basis; (iii) the type-II simple-root Cartan $2(2\mathbf{I}_3 - \mathbf{J}_3)$; (iv) the Weyl vector $\rho = (1, -1/2, 1)$ in (f_2, f_3, f_{-2}) coordinates and the $(\rho, \delta_i) = -1$ relation; (v) the type-II reflection action and Cartan invariance under each s_{δ_i}. Thus the Gram, Cartan, and Weyl-vector data of §1.2 and §1.7 agree in the common rational lattice.

Remark 1.38 *(Negating a lattice form).* The lattice $\Lambda^{3,2} = (\Lambda^{(1,1)})^{\oplus 2} \oplus \langle 2 \rangle$ has signature $(3,2)$ and determinant 2. Negating its whole form gives

$$\Lambda^{3,2}(-1) \cong U^{\oplus 2} \oplus \langle -2 \rangle,$$

with signature $(2,3)$ and determinant -2. Their underlying discriminant groups are both $\mathbb{Z}/2$, but their discriminant quadratic forms take values $1/2$ and $-1/2$ on the nonzero class, respectively. Thus changing the hyperbolic basis and negating the whole form must be distinguished.

Similarly, $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus \langle 2 \rangle$ has determinant -2, while its negative has determinant 2. The type-II sublattice has index four, hence determinant -32 in the chosen BKM convention. Its discriminant group has order 32, whereas the ambient lattice has discriminant group of order two. Divisibility of a primitive ambient vector is controlled by the latter group. The sign of the determinant does not change this underlying group.

Remark 1.39 *(Lattice references).* The cohomological lattices, discriminant gluing and primitive-vector orbits are treated in [57, Chapter 14]. The cohomological action of derived equivalences is in its Chapter 16. The embedding theorem used above is [96, Theorem 1.14.4]. For the even unimodular classification and root lattices see [23]. The type-II root datum, Weyl vector and denominator normalization are those of [54, Theorem 2.1].

2 JACOBI COEFFICIENTS AND AUTOMORPHIC PRODUCTS

The product is determined by its Jacobi input, coefficient cone and cusp normalization. This appendix gives the explicit calculations used in Chapters 3 and 5. It then places the Igusa form in the eight-form classification of diagonal-divisor forms, keeping the Jacobi index, congruence level and character of each form.

2.1 *The Fourier coefficients of $\phi_{0,1}$*

Use $q = e^{2\pi i \tau}$, $r = e^{2\pi i z}$ and

$$\phi_{0,1} = 4 \sum_{a=2}^{4} \frac{\theta_a(\tau, z)^2}{\theta_a(\tau, 0)^2} = \sum_{n \geq 0, \ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell.$$

The genus-one theta series are those in Chapter 3. Their initial terms are

$$\begin{aligned} q^{-1/8}\theta_2(\tau, z) &= r^{1/2} + r^{-1/2} + q(r^{3/2} + r^{-3/2}) + O(q^3), \\ \theta_3(\tau, z) &= 1 + q^{1/2}(r + r^{-1}) + q^2(r^2 + r^{-2}) + O(q^{9/2}), \\ \theta_4(\tau, z) &= 1 - q^{1/2}(r + r^{-1}) + q^2(r^2 + r^{-2}) + O(q^{9/2}). \end{aligned}$$

The same expansions at $r = 1$ give the denominators.

For the first quotient, cancellation of $q^{1/4}$ leaves a denominator $4(1 + 2q + q^2 + O(q^3))$. Its inverse through q^2 is $\frac{1}{4}(1 - 2q + 3q^2)$. For the other two quotients the terms with odd powers of $q^{1/2}$ cancel in their sum.

Put $R_j = r^j + r^{-j}$. Expansion gives

	$4\theta_2(\tau,z)^2/\theta_2(\tau,0)^2$	$4\theta_3(\tau,z)^2/\theta_3(\tau,0)^2 + 4\theta_4(\tau,z)^2/\theta_4(\tau,0)^2$
q^0	$R_1 + 2$	8
q^1	$2R_2 - 4$	$8R_2 - 64R_1 + 112$
q^2	$R_3 - 4R_2 - R_1 + 8$	$112R_2 - 512R_1 + 800$

and hence

$$\phi_{0,1} = R_1 + 10 + q(10R_2 - 64R_1 + 108) + q^2(R_3 + 108R_2 - 513R_1 + 808) + O(q^3).$$

Equivalently,

	$\ell = -3$	-2	-1	0	1	2	3
$n = 0$	0	0	1	10	1	0	0
$n = 1$	0	10	-64	108	-64	10	0
$n = 2$	1	108	-513	808	-513	108	1

The first row fixes the Borcherds weight and divisor. The next rows determine the first imaginary-root exponents.

The elliptic transformation law gives

$$f(n,\ell) = f(n + a\ell + a^2, \ell + 2a), \qquad f(n,\ell) = f(n,-\ell).$$

Reducing ℓ to 0 or 1 shows that $f(n,\ell)$ depends only on $4n - \ell^2$, and that it vanishes for $4n - \ell^2 < -1$. In particular,

$$f(0,0) = f(1,\pm 2) = 10, \quad f(1,0) = f(2,\pm 2) = 108,$$
$$f(0,\pm 1) = f(2,\pm 3) = 1, \quad f(1,\pm 3) = 0.$$

In the theta decomposition $\phi_{0,1} = h_0\theta_{1,0} + h_1\theta_{1,1}$, the functions h_0, h_1 depend on τ alone:

$$h_0 = 10 + 108q + 808q^2 + \cdots, \qquad h_1 = q^{-1/4}(1 - 64q - 513q^2 + \cdots).$$

The negative numbers are Euler superdimensions in the product interpretation. They do not determine the separate parity dimensions of an arbitrary coefficient complex.

2.2 *The active cone and the simple walls*

For $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right)$, write $q = e^{2\pi i z_1}$, $r = e^{2\pi i z_2}$, $s = e^{2\pi i z_3}$. The ordering convention is

$$\Gamma_{\text{eff}} = \{(n,\ell,m) : m > 0,\ n \geq 0\}$$
$$\cup \{(n,\ell,0) : n > 0\} \cup \{(0,\ell,0) : \ell < 0\}.$$

The unrestricted coordinate ℓ is integral in each sector. These sectors form a semigroup under addition. Only indices in

$$\Gamma_{\text{act}} = \{(n,\ell,m) \in \Gamma_{\text{eff}} : f(nm,\ell) \neq 0\}$$

contribute nontrivial factors.

In the basis (f_2, f_3, f_{-2}), fix $(f_2, f_{-2}) = -1$, $f_3^2 = 2$, with all other pairings zero. The product-root map and its norm are

$$\alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2}, \qquad \alpha(n, \ell, m)^2 = 2\ell^2 - 8nm.$$

For $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$, this becomes

$$\alpha(n, \ell, m) = n\delta_1 + m\delta_2 + (n + m - \ell)\delta_3.$$

The support inequality $\ell^2 \le 4nm + 1$ implies $n + m - \ell \ge 0$ at every active index with $n + m > 0$. The remaining active index is $(0, -1, 0)$. Thus the total root height

$$h = 2(n + m) - \ell$$

is positive on every factor, and only finitely many factors occur below a fixed height. The associated completed ring is $\mathbb{Z}[[qr, sr, r^{-1}]]$. The larger ordering semigroup itself does not have finite decomposition sets.

PROPOSITION 2.1 *(Images of the simple type-II walls).* The inverse images of the simple roots are

$$\gamma_{\text{wall}}(\delta_1) = (1, 1, 0), \qquad \gamma_{\text{wall}}(\delta_2) = (0, 1, 1), \qquad \gamma_{\text{wall}}(\delta_3) = (0, -1, 0).$$

Their product exponents are all 1.

Proof. The coefficients in the simple-root basis are $(n, m, n + m - \ell)$. Setting these successively equal to $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ gives the three triples. Their exponents are $f(0, 1)$, $f(0, 1)$ and $f(0, -1)$, respectively. □

Some further product degrees illustrate the distinction between a root norm and its signed multiplicity:

(n, ℓ, m)	$\alpha(n, \ell, m)^2$	$f(nm, \ell)$
$(1, 1, 0)$	2	1
$(0, 1, 1)$	2	1
$(0, -1, 0)$	2	1
$(0, 0, m)$, $m \ge 1$	0	10
$(1, 0, 1)$	−8	108
$(1, 1, 1)$	−6	−64
$(1, 2, 1)$	0	10
$(1, 3, 1)$	10	0
$(2, 1, 1)$	−14	−513
$(1, 2, 2)$	−8	108

The inactive row $(1, 3, 1)$ is allowed by the ordering convention but contributes no factor. The active set need not itself be an additive monoid. Its containment in the positive simple-root monoid is what makes the product and its logarithm well-defined.

2.3 *The product and its theta phase*

Theorem 2.2 *(Borcherds–Gritsenko–Nikulin product).* In the theta convention

$$\nu_{a,b}(Z)=\sum_{u\in\mathbb{Z}^2} e^{\pi i((u+a/2)^t Z(u+a/2)+b^t u)},\qquad \Delta_5=\prod_{a^t b\ \text{even}}\nu_{a,b},$$

one has

$$\Delta_5(Z)=64(qrs)^{1/2}\prod_{\Gamma_{\mathrm{eff}}}(1-q^n r^\ell s^m)^{f(nm,\ell)}.$$

The product is a formal total-height expansion and an absolutely convergent analytic expansion sufficiently near the standard cusp in the type-II chamber. Its continuation is the holomorphic weight-five cusp form with Maass character.

Proof. The Jacobi product theorem applies to $\phi_{0,1}$ [54, Theorem 2.1 and Example 2.4]. The classical product identification, with the theta normalization in the statement, is [52, Theorem 4.1]. The active-cone argument above gives the formal completion. The six nonconstant theta factors have leading coefficients

$$2q^{1/8},\ 2q^{1/8},\ 2s^{1/8},\ 2s^{1/8},\ 2q^{1/8}s^{1/8}(r^{1/4}+r^{-1/4}),\ 2q^{1/8}s^{1/8}(r^{1/4}-r^{-1/4}).$$

Together with the other four leading constants, they give $64q^{1/2}s^{1/2}(r^{1/2}-r^{-1/2})$. This fixes the coefficient 64 and its sign. □

The monic form is $D_5=\Delta_5/64$. If theta characteristics are instead defined with phase $b^t(u+a/2)$, the product of the ten constants is $-\Delta_5$. The square and the divisor are unchanged. The sign of the square-root section is fixed only after this phase convention is specified.

For analytic convergence, the chamber inequalities are

$$y_2<0,\qquad y_1+y_2>0,\qquad y_3+y_2>0,\qquad y_j=\mathrm{Im}z_j.$$

The product theorem applies sufficiently far toward the cusp within this chamber. In particular it applies on a neighborhood of $(y_1,y_2,y_3)=(T,-T/2,T)$ for large T. The frequently used ordering $0<|s|\ll|q|\ll|r|^{-1}\ll 1$ selects a further subregion. Its interpretation does not replace the support or convergence hypotheses.

2.4 *The coefficient 93*

The leading coefficient of Δ_5 and the first interior coefficient of its normalized product are different quantities. Put

$$P(q,r,s)=\frac{D_5}{(qrs)^{1/2}}.$$

Its constant coefficient in q,s is $1-r^{-1}$.

To compute the coefficient of qs, only the factors of q-degree and s-degree at most one are needed. Let $A=r+10+r^{-1}$. The boundary factors give $(1-qA)(1-sA)$ to these degrees. The mixed factors give

$$1-qs\big(10(r^2+r^{-2})-64(r+r^{-1})+108\big).$$

Therefore

$$[qs]P=(1-r^{-1})\big(-9(r^2+r^{-2})+84(r+r^{-1})-6\big),\qquad [qrs]P=93.$$

The factor $(1-qrs)^{-64}$ contributes 64 to this coefficient. The other factors contribute 29.

The same calculation is particularly transparent in root variables $x_1 = qr$, $x_2 = sr$, $x_3 = r^{-1}$:

$$P = \prod_{i=1}^{3}(1-x_i)\prod_{i<j}(1-10x_ix_j)(1+64x_1x_2x_3) \pmod{(x_1^2, x_2^2, x_3^2)}.$$

Thus

$$[x_ix_j]P = -9, \qquad [x_1x_2x_3]P = -1+30+64 = 93.$$

The standard minimal Gritsenko–Nikulin presentation assigns nine even imaginary simple generators to each primitive null pair. At the first interior root $\beta = \delta_1+\delta_2+\delta_3$, the coefficient 93 gives 93 odd imaginary simple generators.

The earlier even generators contribute 29 even vectors. Two are multilinear brackets of the real generators. The other $3\cdot 9$ bracket a null simple generator with the omitted real generator. The first real Serre relations have height four. Consequently the whole root space has parity dimensions

$$\dim \mathfrak{g}_{\Delta_5\beta}^{\bar{0}} = 29, \qquad \dim \mathfrak{g}_{\Delta_5\beta}^{\bar{1}} = 93, \qquad \operatorname{sdim} \mathfrak{g}_{\Delta_5\beta} = -64.$$

This uses the defining Lie presentation, not just the signed product exponent. Its construction is given in Chapter 4.

In Gritsenko–Nikulin's Fourier notation $e^{\pi i(nz_1+\ell z_2+mz_3)}$, the coefficient with $(n,\ell,m) = (1,1,1)$ is 64 [52, (1.6)–(1.7)]. In the present q, r, s notation that is $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5$. It is consistent with the coefficient 93 of P.

2.5 *Fourier–Jacobi and separating factors*

Factoring all $m = 0$ terms gives

$$D_5 = s^{1/2}b(q,r)\prod_{m>0,\ n\geq 0,\ \ell\in\mathbb{Z}}(1-q^nr^\ell s^m)^{f(nm,\ell)},$$

where

$$b(q,r) = q^{1/2}(r^{1/2}-r^{-1/2})\prod_{a\geq 1}(1-q^a)^{10}(1-rq^a)(1-r^{-1}q^a) = \eta(\tau)^9\vartheta(\tau,z).$$

The odd Jacobi theta function ϑ is normalized by this product and has first term $q^{1/8}(r^{1/2}-r^{-1/2})$. The squared boundary factor is

$$\mathcal{B}(q,r) = b(q,r)^2 = q(r+r^{-1}-2)\prod_{a\geq 1}(1-q^a)^{20}(1-rq^a)^2(1-r^{-1}q^a)^2.$$

These are the factors needed in the normalized half and full elliptic-index series.

The separating limit is

$$D_5 = 2\pi i z_2\,\eta(z_1)^{12}\eta(z_3)^{12} + O(z_2^3).$$

It follows by dividing out $r^{1/2}-r^{-1/2}$, pairing the ℓ and $-\ell$ factors, and using $\phi_{0,1}(\tau,0) = 12$. The quotient is even in z_2, which gives the stated cubic remainder.

Squaring yields

$$\Xi = D_5^2 = (2\pi i z_2)^2\eta(z_1)^{24}\eta(z_3)^{24} + O(z_2^4).$$

The product with leading coefficient one is Ξ, the χ_{10} of [100, 101]. The theta square is $\Delta_5^2 = 4096\,\Xi$. Accordingly the primitive reduced scalar series is

$$Z_{\mathrm{OP}}^X = -\Xi^{-1} = -4096\,\Delta_5^{-2}.$$

The expansion prescriptions and primitive-class hypotheses are those of Chapter 1.

2.6 *The two denominator exponentials*

Lemma 2.3 *(Two exponential normalizations).* For $\zeta = z_3 f_2 + z_2 f_3 + z_1 f_{-2}$,

$$q^n r^\ell s^m = e^{-\pi i(\alpha(n,\ell,m),\zeta)_L}.$$

The denominator written with $e^{-2\pi i(\alpha,\zeta)_L}$ is

$$\operatorname{den}(\mathfrak{g}_{\Delta_5})(Z) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

Proof. The pairing is $(\alpha(n,\ell,m),\zeta)_L = -2(nz_1 + \ell z_2 + mz_3)$, proving the first formula. The Gritsenko–Nikulin denominator theorem uses the doubled exponential for each root and for $\rho = f_2 - \frac{1}{2}f_3 + f_{-2}$ [52, §3]. Substitution $Z \mapsto 2Z$ makes precisely this change. □

In these coordinates the additive denominator is

$$\sum_{w \in W^{(2)}(M_{II})} \det(w)\left(e^{-2\pi i(w\rho,\zeta)_L} - \sum_a m(a) e^{-2\pi i(w(\rho+a),\zeta)_L}\right).$$

For a primitive isotropic ray a, its coefficients are related by

$$1 - \sum_{t\geq 1} m(ta)u^t = \prod_{t\geq 1}(1-u^t)^{\tau(ta)}.$$

The numbers $m(a)$ on negative-norm rays and $\tau(ta)$ on null rays specify the imaginary simple-root data. They enter the Lie presentation in addition to the real-root Cartan matrix.

The determinant signs in this alternating sum belong to the linear tube action. On a chosen symplectic reflection lift, the weight-five factor $j(g,Z)^5$ must also be included. Lemma 6.3 computes the resulting relation between the tube character ε_5 and the Maass character ν_5.

2.7 *The eight diagonal-divisor forms*

For positive integers t, N, let $\Gamma_t(N)$ be the Hecke congruence subgroup of the paramodular group:

$$\Gamma_t(N) = \left\{ \begin{pmatrix} * & t* & * & * \\ * & * & * & t^{-1}* \\ N* & Nt* & * & * \\ Nt* & Nt* & t* & * \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Q}) : \text{each } * \in \mathbb{Z} \right\}.$$

In particular $\Gamma_1(1) = \mathbf{Sp}_4(\mathbb{Z})$, and $\Gamma_1(N) = \Gamma_0^{(2)}(N)$. The integer t is the polarization type and the index of the weight-zero Jacobi input. The integer N is the congruence level. A diagonal-divisor form has precisely the $\Gamma_t(N)$-translates of $z_2 = 0$ as its zero divisor, each with multiplicity one.

Cléry–Gritsenko classify eight such forms, up to scalar [50, Theorem 1.4]. The table uses their names and the order of the character or multiplier on $\Gamma_t(N)$:

j	F_j	(t,N)	k_j	$\mathrm{ord}(\nu_j)$	c_j^{tot}
1	Δ_5	$(1,1)$	5	2	10
2	Δ_2	$(2,1)$	2	4	4
3	Δ_1	$(3,1)$	1	6	2
4	$\Delta_{1/2}$	$(4,1)$	$\frac{1}{2}$	8	1
5	∇_3	$(1,2)$	3	2	6
6	∇_2	$(1,3)$	2	2	4
7	$\nabla_{3/2}$	$(1,4)$	$\frac{3}{2}$	4	3
8	Q_1	$(2,2)$	1	4	2

The final column denotes the total constant coefficient entering the weight formula, so $k_j = c_j^{\mathrm{tot}}/2$. Its meaning at congruence level is specified below. The monic convention for the first form is D_5. Our symbol Δ_5 restores the factor 64.

In this order the weights and total constants are

$$(5, 2, 1, \tfrac{1}{2}, 3, 2, \tfrac{3}{2}, 1), \qquad (10, 4, 2, 1, 6, 4, 3, 2).$$

If the pairs are instead ordered

$$(1,1), (2,1), (1,2), (3,1), (1,3), (4,1), (1,4), (2,2),$$

the same columns read

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1), \qquad (10, 4, 6, 2, 4, 1, 3, 2).$$

Keeping the parameter pair prevents an interchange of the paramodular and congruence families.

2.7.1 The first Fourier–Jacobi coefficients

With monic leading terms, the eight Jacobi coefficients are

F	$b_F(\tau,z)$	Jacobi-parabolic multiplier
D_5	$\eta(\tau)^9\vartheta(\tau,z)$	$v_\eta^{12}\times v_H$
Δ_2	$\eta(\tau)^3\vartheta(\tau,z)$	$v_\eta^{6}\times v_H$
Δ_1	$\eta(\tau)\vartheta(\tau,z)$	$v_\eta^{4}\times v_H$
$\Delta_{1/2}$	$\vartheta(\tau,z)$	$v_\eta^{3}\times v_H$
∇_3	$\eta(\tau)\eta(2\tau)^4\vartheta(\tau,z)$	$\chi_2^{(2)}\times v_H$
∇_2	$\eta(3\tau)^3\vartheta(\tau,z)$	$\chi_2^{(3)}\times v_H$
$\nabla_{3/2}$	$\dfrac{\eta(2\tau)\eta(4\tau)^2}{\eta(\tau)}\vartheta(\tau,z)$	$\chi_4^{(4)}\times v_H$
Q_1	$\dfrac{\eta(2\tau)^2}{\eta(\tau)}\vartheta(\tau,z)$	$\chi_4^{(2)}\times v_H$

These coefficients and their lifts are given in [50, (13)–(19), §3]. The symbol $\chi_d^{(N)}$ denotes the order-d character or multiplier determined by the displayed seed on $\Gamma_0(N)$. The first coefficient of the theta-normalized Δ_5 is $64b_{D_5}$.

For clarity, use Heisenberg elements

$$[\lambda,\mu;\kappa]=\begin{pmatrix}1&0&0&\mu\\ \lambda&1&\mu&\kappa\\ 0&0&1&-\lambda\\ 0&0&0&1\end{pmatrix},\qquad \lambda,\mu,\kappa\in\mathbb{Z}.$$

Their character on the extended index-$\frac{1}{2}$ theta function $\vartheta(\tau,z)e^{\pi i\sigma}$ is

$$v_H([\lambda,\mu;\kappa])=(-1)^{\lambda+\mu+\lambda\mu+\kappa}.$$

Indeed z changes by $\lambda\tau+\mu$, and σ changes by $\lambda^2\tau+2\lambda z+\lambda\mu+\kappa$. The odd theta transformation cancels the first two terms in the latter exponential and supplies $(-1)^{\lambda+\mu}$. This also specifies the convention for every v_H in the table.

2.7.2 Products at paramodular level

For $t=1,2,3,4$, the index-t weak Jacobi inputs have constant rows

$$r+10+r^{-1},\quad r+4+r^{-1},\quad r+2+r^{-1},\quad r+1+r^{-1},$$

respectively.

Besides $\phi_{0,1}$, they can be written as

$$\begin{aligned}\phi_{0,4}(\tau,z)&=\frac{\vartheta(\tau,3z)}{\vartheta(\tau,z)},\\ \phi_{0,3}(\tau,z)&=\left(\frac{\vartheta(\tau,2z)}{\vartheta(\tau,z)}\right)^2,\\ \phi_{0,2}(\tau,z)&=\frac{\vartheta(\tau,2z)^4}{\vartheta(\tau,z)^3\vartheta(\tau,3z)}+\frac{\vartheta(\tau,4z)\vartheta(\tau,z)}{\vartheta(\tau,2z)\vartheta(\tau,3z)}.\end{aligned}$$

The last expression follows by substituting the first two into

$$\phi_{0,2}\phi_{0,4}-\phi_{0,3}^2=\frac{\vartheta(\tau,4z)}{\vartheta(\tau,2z)}.$$

The apparent poles cancel in these weak Jacobi forms [54, Lemma 2.5 and (2.10), (2.24)]. Their initial coefficients satisfy

$$\phi_{0,2}=r+4+r^{-1}+q\big(r^3+r^{-3}-8(r^2+r^{-2})-(r+r^{-1})+16\big)+O(q^2).$$

Write $\phi_{0,t}=\sum f_t(n,\ell)q^nr^\ell$. Their monic products are

$$\begin{aligned}\Delta_2&=q^{1/4}r^{1/2}s^{1/2}\prod_{(n,\ell,m)>0}(1-q^nr^\ell s^{2m})^{f_2(nm,\ell)},\\ \Delta_1&=q^{1/6}r^{1/2}s^{1/2}\prod_{(n,\ell,m)>0}(1-q^nr^\ell s^{3m})^{f_3(nm,\ell)},\\ \Delta_{1/2}&=q^{1/8}r^{1/2}s^{1/2}\prod_{(n,\ell,m)>0}(1-q^nr^\ell s^{4m})^{f_4(nm,\ell)}.\end{aligned}$$

Here $(n,\ell,m)>0$ is the cusp ordering, with zero exponents omitted. These identities are [54, (2.11), (2.20)–(2.21)].

They extend to Γ_t^+, generated by Γ_t and the Fricke involution, with the specified multipliers. The projective Fricke action is

$$\begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \longmapsto \begin{pmatrix} t z_3 & z_2 \\ z_2 & z_1/t \end{pmatrix}.$$

2.7.3 The constant terms at all cusps

Let $\phi \in J^!_{0,t}(\Gamma_0(N))$ be holomorphic on $\mathbb{H}_1\times\mathbb{C}$ and meromorphic at the cusps. Choose a scaling matrix $M_{f/e}$ for each cusp f/e, with $e \mid N$. Put

$$h_e = \frac{N}{\gcd(e^2,N)}, \qquad N_e = \frac{N}{e}, \qquad \phi|_{0,t}M_{f/e} = \sum_{n\in h_e^{-1}\mathbb{Z},\ \ell\in\mathbb{Z}} c_{f/e}(n,\ell)q^n r^\ell.$$

The width is h_e.

Assume at each cusp that

$$\frac{h_e}{N_e}c_{f/e}(u,\ell)\in\mathbb{Z} \qquad \text{whenever } 4tu-\ell^2\le 0.$$

Cléry–Gritsenko's product theorem [50, Theorem 3.1] then gives a meromorphic modular form on $\Gamma_t(N)^+ = \langle \Gamma_t(N), V_t\rangle$, where V_t is the Fricke involution above. Its weight is

$$k = \frac{1}{2}c^{\mathrm{tot}}, \qquad c^{\mathrm{tot}} = \sum_{f/e}\frac{h_e}{N_e}\,c_{f/e}(0,0).$$

Its factors are

$$B_\phi = q^A r^B s^C \prod_{f/e}\prod_{(n,\ell,m)>0}\left(1-(q^n r^\ell s^{tm})^{N_e}\right)^{(h_e/N_e)c_{f/e}(nm,\ell)}.$$

For this possibly polar input, $(n,\ell,m)>0$ means $m>0$ with $n,\ell\in\mathbb{Z}$, or $m=0, n>0$, or $m=n=0, \ell<0$. For a weak input the nonzero factors have $n\ge 0$.

The cusp prefactors are

$$A=\frac{1}{24}\sum_{f/e,\ell}h_e c_{f/e}(0,\ell), \quad B=\frac{1}{2}\sum_{f/e,\ell>0}\ell h_e c_{f/e}(0,\ell), \quad C=\frac{1}{4}\sum_{f/e,\ell}\ell^2 h_e c_{f/e}(0,\ell).$$

The singular coefficients at all cusps specify the divisor. The complete cusp expansions also determine the multiplier and weight.

For the four congruence forms, the constant contributions are

F	cusps	$c_{f/e}(0,0)$	c^{tot}
∇_3	∞, 0	2, 4	$2+4=6$
∇_2	∞, 0	1, 3	$1+3=4$
$\nabla_{3/2}$	∞, 1/2, 0	0, 2, 2	$0+\frac{1}{2}(2)+2=3$
Q_1	∞, 0	0, 2	$0+2=2$

These are the inputs of [50, §3]. In particular ∇_3 has weight 3 although its constant coefficient at infinity is 2. The single infinity coefficient suffices when $N=1$, but does not give the congruence weight.

2.7.4 Half-integral automorphy

The orders 8 and 4 in the table are orders of multiplier systems. They do not specify the degree of a covering space. For a half-integral weight, one can choose holomorphic square roots of $j(g,Z) = \det(CZ+D)$ and retain their multiplication cocycle in the multiplier. Equivalently, use the group

$$\widetilde{\Gamma} = \{(g,\varphi) : g \in \Gamma,\ \varphi(Z)^2 = j(g,Z)\},$$

with product $(g,\varphi)(h,\psi) = (gh, (\varphi\circ h)\psi)$. On this cover the half-integral automorphy factor is single-valued.

This is an output group acting on Siegel space. The metaplectic input group $\mathbf{Mp}_2(\mathbb{Z})$ acts on the Jacobi theta components and must be specified separately.

2.8 Denominator algebras and CHL parameters

The first four paramodular forms have the Gritsenko–Nikulin denominator presentations [54, §5.1.1]. The other four forms also occur with specified root data in Govindarajan's twisted-CHL constructions [46, §4 and Table 3]. For CHL orbifold orders one, two and three, the dyon interpretation is developed in [18].

To relate the two parameter systems, write N_{orb} for the CHL orbifold order and M_{tw} for the twisting order. For the entries with $N_{\mathrm{orb}} \le M_{\mathrm{tw}}$, the catalog pair is

$$t = \gcd(N_{\mathrm{orb}}, M_{\mathrm{tw}}), \qquad N = \frac{M_{\mathrm{tw}}}{t}.$$

The dictionary is

F	$(N_{\mathrm{orb}}, M_{\mathrm{tw}})$	number of real simple roots
Δ_5	(1, 1)	3
Δ_2	(2, 2)	4
Δ_1	(3, 3)	6
$\Delta_{1/2}$	(4, 4)	countably infinite
∇_3	(1, 2)	3
∇_2	(1, 3)	3
$\nabla_{3/2}$	(1, 4)	3
Q_1	(2, 4)	4

The root lattice has rank three even when the set of real simple roots is larger. At fixed N_{orb} the Weyl vector and real-root system agree, while the imaginary multiplicities vary with M_{tw}.

The forms at exchanged CHL parameters are related by a change of cusp and can have different denominator chambers [46, (3.12), §4]. The Weyl chamber, parity prescription and additive denominator must therefore accompany the product.

The half-weight 3/2 associated with the cyclic order-four CHL eta product is consistent with the preceding table. For cyclic orders 1, 2, 3, 4, 6, let g_N be the eta products

N	$g_N(\tau)$	$\mathrm{wt}(g_N)$
1	$\eta(\tau)^{24}$	12
2	$\eta(\tau)^8\eta(2\tau)^8$	8
3	$\eta(\tau)^6\eta(3\tau)^6$	6
4	$\eta(\tau)^4\eta(2\tau)^2\eta(4\tau)^4$	5
6	$\eta(\tau)^2\eta(2\tau)^2\eta(3\tau)^2\eta(6\tau)^2$	4

Multiplication by the Jacobi factor ϑ^2/η^6 subtracts two from the weight, as in the CHL lifting prescription [48, Table 1 and §3.2]. Half of the resulting seed weights is

$$5, \quad 3, \quad 2, \quad \frac{3}{2}, \quad 1.$$

This eta-weight calculation does not identify all these parameters with the pairs (t, N), nor does it construct a compact Hall source for any row.

2.9 The weight and the geometric source

For each form in the catalog, define $\kappa_{\mathrm{BKM}}(F_j) = \mathrm{wt}(F_j)$. Then the precise common weight identity is

$$\kappa_{\mathrm{BKM}}(F_j) = \frac{c_j^{\mathrm{tot}}}{2}.$$

For the Igusa input, $c_1^{\mathrm{tot}} = f(0,0) = 10$.

The number 5 is a coordinate determined by the automorphic construction. The categorical and Hodge quantities attached to $S \times E$ come from other constructions. So do the rank 3 of the root-degree lattice, $24 = e(S)$, and $2 = \chi(\mathcal{O}_S)$. A normalized Heisenberg trace may introduce another comparison invariant. Its identification with a lattice rank requires the comparison data stated in Chapter 1. The numerical equality $3 + 2 = 5$ does not replace the product weight formula.

The catalog supplies automorphic lines, sections and, with the cited root presentations, denominator algebras. The primitive reduced $S \times E$ counting theorem supplies the scalar $-\Xi^{-1}$ for the first form. Constructing a compact source with the appropriate coefficient complexes, Hall operations, reduction, orientation and state is an additional requirement. The conditional Pfaffian criteria in Appendix 7 retain those data. The comparison must preserve the complete boundary factor, the character and the divisor multiplicities. It must also preserve the exponential convention and the normalized section.

3 MUKAI VECTORS AND GRAM DEGREES

Let S be a smooth complex projective K3 surface and E a smooth complex elliptic curve. Consider a one-dimensional subscheme $Y \subset S \times E$ with curve class (β, d). Here β is its projected class on S, identified with its Poincaré dual, and d is its degree over E. The two Mukai components of its ideal give

$$\Pi_X(Q_Y, P_Y) = \left(\frac{\beta^2}{2}, \chi(\mathcal{O}_Y), d - 1\right).$$

We derive the formula, give sheaf representatives of the three simple Gram degrees, and describe the quadratic correction required when charges are added.

3.1 Surface vectors and pairings

Definition 3.1 *(Mukai vector).* Write ω_S for the point class with $\int_S \omega_S = 1$. The cohomological Mukai lattice is

$$\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}),$$

with coordinates (r, D, s) denoting $r + D + s\omega_S$. For $F \in D^b\,\mathrm{Coh}(S)$, set

$$v_S(F) = \mathrm{ch}(F)\sqrt{\mathrm{td}(S)} = (\mathrm{rk}\,F, c_1(F), \mathrm{rk}\,F + \mathrm{ch}_2(F)).$$

Here $\mathrm{ch}_2(F)$ is integrated over S, and $\sqrt{\mathrm{td}(S)} = 1 + \omega_S$. Actual coherent vectors lie in $\widetilde{H}_{\mathrm{alg}}(S, \mathbb{Z}) = \mathbb{Z} \oplus \mathrm{NS}(S) \oplus \mathbb{Z}$.

Proposition 3.2 *(Mukai form).* The symmetric form

$$\langle (r, D, s), (r', D', s') \rangle = D \cdot D' - rs' - r's = -\int_S (r, -D, s)(r', D', s')$$

is even unimodular of signature $(4, 20)$ on the full lattice. On the algebraic sublattice it has signature $(2, \rho(S))$, and

$$\chi_S(F, G) = -\langle v_S(F), v_S(G) \rangle.$$

Proof. The $H^0 \oplus H^4$ part is a hyperbolic plane. Adjoining it to $H^2(S, \mathbb{Z}) \cong U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$ proves the assertion about the full lattice. Adjoining it to $\mathrm{NS}(S)$, whose signature is $(1, \rho - 1)$, gives the algebraic signature. Hirzebruch–Riemann–Roch gives the last identity, as in Proposition 1.2. □

Remark 3.3 *(Symmetry and additional algebraic structures).* The Mukai form is symmetric on both the algebraic and transcendental parts. Its real infinitesimal isometry algebra is $\mathfrak{so}(4, 20)$. Its complexification is $\mathfrak{so}_{24}(\mathbb{C})$. This statement concerns a quadratic vector space. A separate supergrading does not change that bilinear form, and the form alone defines neither a Yangian nor a Hall algebra.

A comparison with a quantum group requires its multiplication, coproduct and deformation parameter, together with a map respecting those structures. For example, the Hilbert-scheme construction of [107, §2] uses the equivariant cohomology of Hilbert schemes of $\mathbb{A}^2$, while the construction in [73, §2] uses quiver representation stacks. Neither construction by itself supplies such a comparison for the full K3 Mukai lattice.

The surface examples in Example 1.3 include point ideals and divisor ideals, which have different vectors and self-pairings. They will supply the threefold representatives below.

3.2 Paired charges on the threefold

Proposition 3.4 *(Künneth decomposition and Euler form).* Let $X = S \times E$ and normalize ω_E by $\int_E \omega_E = 1$. Every even integral cohomology class has a unique expression

$$v_X = P \otimes 1_E + Q \otimes \omega_E, \qquad Q, P \in \widetilde{H}(S, \mathbb{Z}).$$

For a perfect complex $\mathcal{F}$ on X, the components of $v_X(\mathcal{F}) = \mathrm{ch}(\mathcal{F})\sqrt{\mathrm{td}(X)}$ satisfy

$$P = v_S(Li_t^*\mathcal{F}), \qquad Q = v_S(Rp_{S*}\mathcal{F}).$$

Here $i_t : S \hookrightarrow X$ is a fiber inclusion and $p_S : X \to S$ is projection. If $c = (Q, P)$ and $c' = (Q', P')$ are charges of $\mathcal{F}, \mathcal{G}$, then

$$\chi_X(\mathcal{F}, \mathcal{G}) = Q \cdot P' - P \cdot Q'.$$

Proof. The cohomology of both factors is torsion free, and the odd cohomology of S vanishes. The integral Künneth theorem therefore gives the decomposition. Derived restriction extracts the 1_E coefficient. Grothendieck–Riemann–Roch for p_S, with $\mathrm{td}(E) = 1$, extracts the ω_E coefficient. In particular, both components of a coherent charge are integral and algebraic.

For the Euler form, the duality involution sends $P \otimes 1_E + Q \otimes \omega_E$ to $P^\vee \otimes 1_E - Q^\vee \otimes \omega_E$. Thus Riemann–Roch and $\int_S v^\vee w = -\langle v, w \rangle$ give

$$\chi_X(\mathcal{F}, \mathcal{G}) = \int_S (P^\vee Q' - Q^\vee P') = Q \cdot P' - P \cdot Q'.$$

□

There are also two useful symmetric forms on this rank-48 group:

$$b_{\oplus}(c,c') = Q\cdot Q' + P\cdot P',$$
$$b_{\mathrm{cross}}(c,c') = Q\cdot P' + P\cdot Q'.$$

Their signatures are $(8,40)$ and $(24,24)$, respectively. The second is the tensor of the Mukai form with the hyperbolic form on $H^0(E,\mathbb{Z})\oplus H^2(E,\mathbb{Z})$. Both are auxiliary to the alternating threefold Euler form.

Definition 3.5 *(Gram degrees).* Order the formal charge lattice as $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z})\oplus\widetilde{H}(S,\mathbb{Z})$, with $c=(Q,P)$. Its Gram map is

$$\Pi_X : \Gamma_X^{\mathrm{form}} \longrightarrow \Gamma_{\mathrm{gram}} = \mathbb{Z}^3, \qquad \Pi_X(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

We call $\Pi_X(c)$ the Gram degree of c. It determines the exponent of $q^n r^l s^m$ when $(n,l,m)=\Pi_X(c)$ in the chosen Igusa cusp coordinates.

3.3 The ideal-sheaf dictionary

THEOREM 3.6 *(Threefold ideal-sheaf charge).* Let $Y\subset X$ be a one-dimensional closed subscheme. Write its fundamental curve class as $[Y]=(\beta,d)\in H_2(S,\mathbb{Z})\oplus H_2(E,\mathbb{Z})$ and put $n=\chi(\mathcal{O}_Y)$. Identify β with its Poincaré dual in $H^2(S,\mathbb{Z})$. Then

$$v_X(I_Y) = (1,0,1-d)\otimes 1_E + (0,-\beta,-n)\otimes \omega_E.$$

Proof. The first nonzero Chern character of $\mathcal{O}_Y$ is

$$\mathrm{ch}_2(\mathcal{O}_Y) = \beta\otimes\omega_E + d\omega_S\otimes 1_E.$$

Since $c_1(X)=0$, Riemann–Roch gives $\int_X \mathrm{ch}_3(\mathcal{O}_Y)=n$. The ideal sequence and $\sqrt{\mathrm{td}(X)} = (1+\omega_S)\otimes 1_E$ now give the displayed formula. Zero-dimensional embedded components contribute to n and require no change in the calculation. □

Remark 3.7 *(The Euler coordinate).* The equality $\int_X \mathrm{ch}_3(\mathcal{O}_Y)=n$ follows because the only possible extra term in degree six is $\mathrm{ch}_2(\mathcal{O}_Y)c_1(X)/2$, which vanishes. The ideal sequence fixes the sign $-n$ in Q_Y.

THEOREM 3.8 *(D6–D2–D0 Gram dictionary).* Under the preceding hypotheses, put $P_Y=(1,0,1-d)$ and $Q_Y=(0,-\beta,-n)$. Then

$$\Pi_X(Q_Y,P_Y) = \left(\frac{\beta^2}{2}, n, d-1\right).$$

If S is elliptically fibered with section s_1 and fiber F, the class $\beta_h = s_1 + hF$, $h\geq 0$, gives $\Pi_X(Q_Y,P_Y) = (h-1,n,d-1)$.

Proof. The three pairings are

$$Q_Y^2 = \beta^2, \qquad Q_Y\cdot P_Y = n, \qquad P_Y^2 = 2(d-1).$$

For the elliptic class, adjunction gives $s_1^2=-2$. Also, $s_1\cdot F=1$ and $F^2=0$. Hence $\beta_h^2 = 2h-2$. □

Remark 3.9 *(Algebraicity and primitivity).* The projection of an effective curve cycle on X to S is an effective algebraic cycle, possibly zero. Thus $\beta \in \mathrm{NS}(S)$ and $d \geq 0$. The formula holds whether or not β is primitive. Here primitive means nonzero and indivisible in the integral lattice, rather than merely indivisible among classes represented by a chosen family of curves. The class β_h is primitive because $\beta_h \cdot F = 1$.

These numerical conditions do not determine the existence of a subscheme for every prescribed triple (β, d, n). Enumerative comparisons impose their own hypotheses, including the primitive K3-class regime in [100]. The charge calculation does not extend that regime to imprimitive classes.

Remark 3.10 *(Additive charges).* An exact triangle adds Mukai vectors, so a Hall operation is graded by (Q, P) whenever its source category and coefficient theory have been specified. Gram degrees add with an additional cross term. Consequently a sum over equal Gram degrees is not by itself a regrading by a group homomorphism.

3.4 Polarization of the Gram map

Lemma 3.11 *(The cross term).* For $c = (Q, P)$ and $c' = (Q', P')$, define

$$B(c, c') = (Q \cdot Q',\ Q \cdot P' + Q' \cdot P,\ P \cdot P').$$

Then B is symmetric, integral and bilinear, and

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c'), \qquad B(c, c) = 2\Pi_X(c).$$

With $(\delta f)(c, c') = f(c) + f(c') - f(c + c')$, one has $B = -\delta\Pi_X$ and $\delta B = 0$.

Proof. Expand the two squares and the middle pairing in $\Pi_X(c + c')$. Bilinearity gives the cocycle equation

$$B(c, c') + B(c + c', c'') = B(c', c'') + B(c, c' + c'').$$

The remaining identities follow by subtraction and by setting $c' = c$. □

Remark 3.12 *(Quadratic and linear corrections).* The equality $B(c, c) = 2\Pi_X(c)$ also follows from $\Pi_X(2c) = 4\Pi_X(c)$. Thus the group-cohomology class of B is zero, with a quadratic trivialization. An additive homomorphism f has $\delta f = 0$, so it cannot trivialize a nonzero B. This is the obstruction to an additive linear correction.

3.5 An additive extension of Gram degrees

Definition 3.13 *(Normal-ordered charge group).* Let $\widehat{\Gamma}_X$ be the set $\Gamma_X^{\mathrm{form}} \times \Gamma_{\mathrm{gram}}$ with law

$$(c, T) \star (c', T') = (c + c', T + T' + B(c, c')).$$

Its identity is $(0, 0)$ and its inverse operation is $(c, T)^{-1} = (-c, -T + 2\Pi_X(c))$. The split section and the raw placement are, respectively,

$$s(c) = (c, \Pi_X(c)), \qquad i_0(c) = (c, 0).$$

Proposition 3.14 *(Additive Gram map).* The cocycle law makes $\widehat{\Gamma}_X$ an abelian group. The map

$$(c, T) \longmapsto (c, T - \Pi_X(c))$$

is an isomorphism to the ordinary direct sum $\Gamma_X^{\mathrm{form}} \oplus \Gamma_{\mathrm{gram}}$. In particular,

$$\overline{\Pi}_X(c, T) = \Pi_X(c) - T$$

is a homomorphism, s is a homomorphism, and $\overline{\Pi}_X \circ s = 0$, whereas $\overline{\Pi}_X \circ i_0 = \Pi_X$.

Proof. Associativity is the cocycle equation, and symmetry of B gives commutativity. For the displayed splitting, the identity $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$ cancels the cross term. The other assertions follow immediately. The raw placement fails to be additive precisely where B is nonzero. □

The threefold sheaves of Proposition 7.9 realize the three simple Gram degrees $(1, 1, 0)$, $(0, 1, 1)$ and $(0, -1, 0)$. They are external sums of point ideals or point sheaves on S. Their construction requires no special Picard lattice. Their support dimensions differ, and a common stability condition with the required Hall operations remains a separate comparison problem.

Remark 3.15 *(Choice of lift).* Products of raw lifts accumulate the cross term in the central coordinate. For example,

$$i_0(c) \star i_0(c') = (c + c', B(c, c')).$$

The additive map $\overline{\Pi}_X$ sends this product to $\Pi_X(c) + \Pi_X(c')$. The split lift $s(c + c')$ instead has additive Gram degree zero. These are different lifts of the same underlying charge. The lattice extension is associative, but lifting an actual Hall product requires a rule compatible with its objects and extension correspondences.

Remark 3.16 *(Effective support).* The formal lattice contains nonalgebraic vectors. Even in its algebraic sublattice, a class need not be represented by an object in a prescribed heart or stability chamber. Such support must be specified before forming its Hall algebra, as in Definition 8.25. The ideal-sheaf dictionary and the three explicit sheaves above supply particular representatives, not a realization of every root class with all the required operations.

3.6 Automorphic and protected interpretations

The automorphic form Δ_5 is a section of its weight-five line with character ν_{Δ_5}. In the root-lattice normalization, the Borcherds–Kac–Moody denominator is

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

The Gram dictionary explains the associated Fourier exponents. It does not construct a compact protected Pfaffian section. Such a section requires the oriented geometric data of Chapter 6, including the Weyl-transport condition 1.81, and the finite-stage hypotheses of Theorem 1.200.

Likewise, symmetry of the Mukai lattice does not identify a Hall algebra with its orthogonal isometry algebra. Both comparisons require maps respecting the relevant operations. The lattice and charge identities proved here remain valid independently of those constructions.

4 HALL CORRESPONDENCES, ORIENTATIONS AND REDUCED CLASSES

An extension correspondence can support several different operations. Ordinary cohomology, compactly supported cohomology and Borel–Moore homology require different coefficient maps. A virtual cycle supplies none of these maps by itself. We state the critical Hall theorem with its coefficient comparison,

then explain the reduced stable-pair construction on $X = S \times E$, where S is a smooth projective K3 surface and E is an elliptic curve.

We work over $\mathbb{C}$ with rational constructible coefficients. A chosen nonzero form Ω_S on S and a nonzero one-form on E trivialize K_X. Thus X has complex dimension three. The shifted symplectic structure on its derived moduli of perfect complexes has degree -1: the Calabi–Yau formula is $2 - \dim_{\mathbb{C}} X$ [104]. For a fourfold it would instead be -2. The virtual-class theories of [11, 102, 103] and the toric fourfold DT/PT correspondence of [17] concern that different dimension. The algebraic charges of coherent sheaves lie in the support described in Definition 8.25, with the Mukai pairing on the K3 factor [91]. Neither a virtual-cycle construction nor a formal quiver chart enlarges that support automatically.

4.1 Oriented critical coefficients

A d-critical locus (M, s) is a classical space locally presented as the critical locus of a function on a smooth space, with compatible critical structures on overlaps. It has a canonical line $K_{M,s}$ on M^{red}. An orientation is a line L and an isomorphism $L^{\otimes 2} \simeq K_{M,s}$. For a critical chart $R \hookrightarrow U$ with function f, the restriction of $K_{M,s}$ is $K_U^{\otimes 2}|_{R^{\mathrm{red}}}$.

Joyce–Song construct analytic critical charts for sheaf moduli under their Calabi–Yau hypotheses, including $H^1(\mathcal{O}_X) = 0$ [62, Theorems 5.4–5.5]. That local construction does not supply the global oriented coefficient object in the following theorem. In particular, $S \times E$ does not satisfy the displayed vanishing hypothesis.

Theorem 4.1 *(Oriented vanishing cycles).* An oriented complex algebraic d-critical locus (M, s, L) carries a perverse sheaf $\Phi_{M,s,L}$ whose restriction to each critical chart is the perverse vanishing-cycle sheaf of f, twisted by the sign local system of compatible identifications $L|_{R^{\mathrm{red}}} \simeq K_U|_{R^{\mathrm{red}}}$. These local sheaves glue canonically with their duality and monodromy structures [12, Theorem 6.9].

The twist in this statement is a rank-one rational local system associated to a μ_2-torsor. It is not tensor product with the underlying algebraic line L. The construction for oriented d-critical Artin stacks uses smooth descent and the stack version of these coefficient comparisons; we use the conventions of [68, §4]. In particular, the phrase "vanishing-cycle coefficients" includes the orientation and its chosen descent isomorphisms.

For a continuous family of real elliptic operators on a compact manifold, the determinant line instead has an orientation double cover [63, Definition 1.2]. A comparison with critical orientation data must identify these two orientation problems; a trivialization in one does not specify that comparison.

The Thom–Sebastiani theorem compares the critical coefficients of $f \boxplus g$ on a product with the external tensor product of the coefficients of f and g [12, Theorem 2.13]. For constructible derived coefficients this is the natural monodromy-compatible comparison of [86, §0]. An extension stack is generally not that product. Its Hall operation uses an additional comparison on the extension correspondence.

4.2 The critical Hall product

Let $\mathcal{C}$ be a finite-type stable dg category with a 3-Calabi–Yau structure, and let $\mathcal{M}_{\mathcal{C}}$ denote its derived moduli stack of pseudo-perfect modules. Choose an open substack $\mathcal{M}$ closed under direct sums and containing zero. Write $\Gamma = \pi_0(\mathcal{M})$ and $\mathcal{M} = \coprod_{\alpha \in \Gamma} \mathcal{M}_\alpha$. Direct sum makes Γ a commutative monoid. The two-step filtration stack gives the correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{F}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Here p records the associated graded objects in the fixed order, and q records the filtered object's underlying object.

An orientation of $\mathcal{M}$ induces an orientation on a graded pair through this Lagrangian correspondence. Orientation data identifies it with the external product of the two orientations and identifies the orientation at zero with the trivial one. Such data are *strong* when the two induced identifications on three factors agree, and both unit diagrams commute [68, Definition 8.4].

We use the cohomological convention $K[d]^i = K^{i+d}$. Let $\kappa(\alpha, \beta)$ denote the integer shift in the hyperbolic coefficient comparison

$$(\Phi_\alpha \boxtimes \Phi_\beta)[\kappa(\alpha,\beta)] \xrightarrow{\sim} R p_* q^! \Phi_{\alpha+\beta}. \tag{1}$$

This definition fixes the sign of every shift below. The index satisfies

$$\kappa(\alpha,\beta) + \kappa(\alpha+\beta,\gamma) = \kappa(\beta,\gamma) + \kappa(\alpha,\beta+\gamma), \qquad \kappa(0,\alpha) = \kappa(\alpha,0) = 0.$$

For Γ-graded complexes define the corresponding tensor product by

$$(V \otimes_\kappa W)_\gamma = \bigoplus_{\alpha+\beta=\gamma} (V_\alpha \otimes W_\beta)[\kappa(\alpha,\beta)].$$

The cocycle identity supplies its associativity constraint.

THEOREM 4.2 *(Critical cohomological Hall multiplication).* Assume that $\mathcal{M}^{\mathrm{cl}}$ is a quasi-separated Artin stack with affine stabilizers, that it is Θ-reductive, and that its zero object is a closed point substack. Equip $\mathcal{M}$ with strong orientation data. Then the coefficients in (1) exist and define a unital associative multiplication

$$H \otimes_\kappa H \longrightarrow H, \qquad H = \bigoplus_{\alpha\in\Gamma} H^\bullet(\mathcal{M}_\alpha, \Phi_\alpha).$$

The unit comes from the zero object. The multiplication is the critical Hall construction of [68, Theorem 8.6, Corollary 8.8 and Proposition 8.29], with shifts specified by (1).

Proof. The cited theorem constructs the hyperbolic comparison and its compatibility with three-step filtrations. Its adjoint is

$$p^*(\Phi_\alpha \boxtimes \Phi_\beta)[\kappa(\alpha,\beta)] \longrightarrow q^! \Phi_{\alpha+\beta}.$$

The Θ-reductivity hypothesis makes q proper on each component used here. Apply ordinary cohomology, then the counit $Rq_! q^! \Phi_{\alpha+\beta} \to \Phi_{\alpha+\beta}$, using $Rq_! = Rq_*$ on that component. The resulting map has source $(H_\alpha \otimes H_\beta)[\kappa(\alpha,\beta)]$. The two compositions for a triple are induced by the same three-step filtration correspondence. The associativity assertion for the coefficient comparison and the strong orientation identification make these maps equal. The two zero-endpoint filtrations give the unit identities. □

For a smooth projective threefold with trivial canonical bundle, Joyce–Upmeier construct orientation data on the moduli of sheaves and perfect complexes [64, Theorem 4.4]. The existence of strong orientation data is a further condition; its obstruction is formulated in [64, Theorem 3.10]. Without this condition the critical multiplication has the sign ambiguity described in [68, Remark 8.7 and Example 8.9]. For Ginzburg categories of quivers with potential, strong orientation data and comparison with the quiver product of [73] are proved in [68, Proposition 8.34, Corollary 8.36 and Proposition 8.37].

The theorem uses ordinary critical cohomology. A product on compactly supported cohomology, on dual compactly supported cohomology, or on a reduced translation quotient needs its own coefficient comparison and support functoriality. In particular, a properness assertion about a Quot scheme must refer to coherent sheaves, a fixed Hilbert polynomial, and closed conditions on the quotients. It does not prove properness for an arbitrary open stability condition or an arbitrary tilted heart.

4.3 Collision coproducts and primitives

A multiplication does not specify a coproduct. A candidate collision correspondence has the reverse form

$$\mathcal{M}_\gamma \xleftarrow{r} C_{\alpha,\beta} \xrightarrow{s} \mathcal{M}_\alpha \times \mathcal{M}_\beta, \qquad \alpha+\beta=\gamma.$$

To use it one must give a coefficient morphism with the variance and shift required by the chosen homology theory, and identify its iterates on triple collisions. The extension correspondence alone provides neither of these data.

Theorem 4.3 *(A supplied collision coproduct).* Suppose the collision correspondences induce maps $\Delta : H \to H \otimes H$ in a specified symmetric monoidal category, and that their coefficient comparisons satisfy coassociativity, both counit identities, and compatibility with the Hall product and its unit. Suppose also that the counit is an algebra map. Then the resulting unital algebra and counital coalgebra form a bialgebra. This conclusion applies to the compact reduced Hall object when the geometric and coefficient hypotheses of Definition 1.113 are satisfied.

Proof. Compose the coefficient morphisms along the two triple-collision correspondences. Their identification gives coassociativity. The zero-output correspondences give the counit equations. The compatibility correspondence for multiplication and collision gives

$$\Delta m = (m \otimes m)(1 \otimes \tau \otimes 1)(\Delta \otimes \Delta),$$

where τ is the symmetry of the chosen category. The stated unit and counit compatibilities are the remaining bialgebra equations. □

The symmetry and tensor product in this statement are part of its hypotheses. An algebra in the index-twisted category of Theorem 4.2 is not, without a compatible regrading or comparison, an ordinary Hopf superalgebra.

4.4 Primitive Lie superalgebras

Now let H be a bialgebra in super vector spaces over a field of characteristic zero. Put $\overline{H} = \ker \varepsilon$ and let $\pi = 1 - \eta\varepsilon : H \to \overline{H}$ be the augmentation projection. The reduced coproduct on $\overline{H}$ is $\overline{\Delta} = (\pi \otimes \pi)\Delta$.

Theorem 4.4 *(Primitive closure).* The space

$$P(H) = \ker(\overline{\Delta} : \overline{H} \longrightarrow \overline{H} \otimes \overline{H})$$

is a Lie superalgebra under $[x, y] = xy - (-1)^{|x||y|} yx$. A compatible differential makes it a dg Lie superalgebra. No antipode is needed for this conclusion.

Proof. For $x \in \overline{H}$, the counit and coaugmentation equations give $\Delta x = x \otimes 1 + 1 \otimes x + \overline{\Delta} x$. For homogeneous primitives x, y, multiplicativity of Δ gives

$$\Delta(xy) = xy \otimes 1 + x \otimes y + (-1)^{|x||y|} y \otimes x + 1 \otimes xy.$$

Subtract the same formula with x, y interchanged and multiplied by $(-1)^{|x||y|}$. The mixed terms cancel, proving primitive closure. The super Jacobi identity follows by expanding the commutator in the associative algebra. A differential compatible with m and Δ preserves both $\overline{H}$ and its primitive kernel. □

This result concerns the primitive kernel of a specified coproduct. It does not identify that kernel with BPS cohomology, the primitive part of a word-cut bar construction, or the denominator Lie superalgebra. Those comparisons retain the hypotheses of Theorem 1.160.

4.5 Cosection localization and reduction

Definition 4.5 *(Cosection and degeneracy locus).* Let M be a Deligne–Mumford stack with a perfect obstruction theory $\mathbb{E} \to \mathbb{L}_M$ of amplitude $[-1, 0]$. Its obstruction sheaf is $\mathrm{Ob}_M = h^1(\mathbb{E}^\vee)$. A cosection is an $\mathcal{O}_M$-linear map $\sigma : \mathrm{Ob}_M \to \mathcal{O}_M$. Let $U \subset M$ be the open locus where it is surjective, and let $Z = M \setminus U$ be its degeneracy locus.

Write $\mathfrak{E} = h^1/h^0(\mathbb{E}^\vee)$ for the obstruction bundle stack. The relevant cone stack is

$$\mathfrak{E}(\sigma) = \mathfrak{E}|_Z \cup \ker(\sigma : \mathfrak{E}|_U \longrightarrow \mathbb{A}^1_U).$$

It contains the full obstruction bundle stack over Z. It is not the kernel of the obstruction sheaf regarded as a perfect complex.

THEOREM 4.6 *(Kiem–Li localization).* The intrinsic normal cone lies in $\mathfrak{E}(\sigma)$, and the localized Gysin map gives

$$[M]^{\mathrm{vir}}_{\mathrm{loc}} \in A_{\mathrm{vd}}(Z), \qquad i_*[M]^{\mathrm{vir}}_{\mathrm{loc}} = [M]^{\mathrm{vir}},$$

where $i : Z \hookrightarrow M$ and $\mathrm{vd} = \mathrm{rk}\,\mathbb{E}$ [67, Theorem 5.1]. If σ is everywhere surjective, then $[M]^{\mathrm{vir}} = 0$.

Proof. The cone containment and localized Gysin construction are the cited theorem. Compatibility of the localized Gysin map with the ordinary zero-section Gysin map gives the pushforward identity. If $U = M$, then Z is empty, so its Chow group and the localized class vanish. □

Removing a trivial obstruction direction is a different construction. It requires a map of obstruction complexes and a compatible reduced perfect obstruction theory on the specified moduli space. When this removes a rank-one obstruction summand, the virtual dimension rises by one. The localized class above retains the original virtual dimension and is supported on Z; it is not this reduced class.

For a perfect complex F on X, the familiar candidate semiregularity functional is

$$\mathrm{Ext}^2_X(F, F) \longrightarrow H^3(X, \Omega^1_X) \xrightarrow{\Omega_S \wedge -} H^3(X, K_X) \xrightarrow{\int_X} \mathbb{C}.$$

The first arrow is composition with the Atiyah class followed by trace. Its restriction to the relevant obstruction complex, its surjectivity in families and the resulting reduced theory must be specified. The nonvanishing of Ω_S alone proves none of these assertions. The formal Borcherds variable s does not define the degeneracy locus of this cosection.

THEOREM 4.7 *(Reduced stable pairs on $S \times E$).* Let $\beta \in \mathrm{Pic}(S)$ be a nonzero curve class, $d \geq 0$, and $n \in \mathbb{Z}$. Set $P = P_n(S \times E, (\beta, d))$. There is a reduced class $[P]^{\mathrm{red}} \in A_1(P)$. The translation action has finite stabilizers, and its Deligne–Mumford quotient P/E carries a zero-dimensional virtual class such that

$$\pi^*[P/E]^{\mathrm{vir}} = [P]^{\mathrm{red}}, \qquad \pi : P \longrightarrow P/E.$$

Its degree is the orbifold Euler characteristic weighted by the Behrend function of P/E. It equals the reduced invariant obtained by integrating the normalized incidence insertion on $[P]^{\mathrm{red}}$ [99, Theorems 1–2].

Proof. Oberdieck constructs an isotrivial map $P \to E$, equivariant for a nonzero multiple of translation. Its fibre K over zero has a symmetric perfect obstruction theory, and $P/E \simeq K/G$ for a finite group G. The class $|G|^{-1}(K \to P/E)_*[K]^{\mathrm{vir}}$ is independent of the chosen isotrivial map. The virtual-class formula in terms of the Fulton Chern class gives its pullback as $[P]^{\mathrm{red}}$. Behrend's formula on K, together with finite descent, identifies its degree with the weighted Euler characteristic of the quotient. The normalized incidence insertion intersects an elliptic orbit with degree one, giving the second description of that degree. These are the constructions and comparisons of [99, §3]. □

The theorem holds for nonzero β without requiring it to be primitive. The primitive-class hypothesis enters the particular Igusa generating-series identity discussed in Chapter 3. Its normalization is $Z^X_{\mathrm{OP}} = -4096\Delta_5^{-2}$, after the specified DT/PT and GW/PT comparisons and expansion conventions. A numerical identity for these reduced invariants does not supply critical Hall coefficient morphisms.

Proposition 4.8 *(Reduced critical coefficients require a critical structure).* Suppose a reduced translation quotient is equipped with a d-critical structure and an orientation compatible with its local critical charts. Then Theorem 4.1, or its Artin stack version on the stated domain, gives its critical coefficient sheaf. A Hall multiplication on these coefficients further requires the oriented coefficient comparison on extension flags and its associativity and unit coherences.

Proof. Apply the oriented critical-sheaf theorem to the supplied structure. The Hall maps are then constructed only when the coefficient comparison and support conditions in Theorem 4.2, or the specified reduced analogue, are available. A symmetric perfect obstruction theory or a virtual class alone is not among the hypotheses that construct these comparison morphisms. □

The reduced stable-pair theorem supplies a virtual class on its particular moduli spaces. It does not state that every reduced semistable Hall stack is an oriented d-critical stack, or that removing one obstruction direction preserves a shifted symplectic structure. The desired compact coefficient theory must construct these structures and their maps on the actual extension and collision diagrams.

4.6 Stable envelopes and quantum groups

Let M be a smooth holomorphic symplectic variety with a proper map to an affine variety. Let $A \subset T$ be tori acting on M, with A preserving the symplectic form, and assume the equivariance and T-formality hypotheses of [87, §3.1]. The nonzero A-weights in the normal bundle of M^A determine hyperplanes in $\mathrm{Cochar}(A) \otimes \mathbb{R}$. Their complement has chambers. Choose a chamber C and a polarization of the fixed locus. The fixed components need not be points.

Definition 4.9 *(Stable envelope).* The stable envelope is the $H_T^\bullet(\mathrm{pt})$-linear map

$$\mathrm{Stab}_C : H_T^\bullet(M^A) \longrightarrow H_T^\bullet(M)$$

characterized by attracting support, the signed Euler class of the negative normal bundle on each fixed component, and a strict bound on the degree in the A-parameters on lower fixed components.

THEOREM 4.10 *(Maulik–Okounkov stable envelopes).* Under these hypotheses and choices, the stable envelope exists and is unique [87, Theorem 3.3.4]. After equivariant localization it is invertible, and chamber comparisons

$$R_{C',C} = \mathrm{Stab}_{C'}^{-1}\,\mathrm{Stab}_{C}$$

satisfy the chamber composition identities. For the framing factorizations of Nakajima quiver varieties [93, 95], these identities give the spectral-parameter Yang–Baxter equation [87, §4.1.8–9]. The resulting geometric Yangian Y_Q acts on the direct sums of quiver-variety cohomologies used in that construction.

Proof. Existence and uniqueness are the cited stable-envelope theorem. Localization makes the triangular restriction matrix invertible. Successive chamber comparisons cancel their intermediate stable envelopes. On a three-factor framing decomposition, the two routes between opposite chambers factor into the three pairwise R-matrices, giving the Yang–Baxter equation. The Yangian is constructed from these operators in [87, Chapters 5–6]. □

The Lie algebra $\mathfrak{g}_Q$ obtained from the classical R-matrix contains the quiver Kac–Moody algebra and can be larger; see [87, §1.2.8]. It must not be identified with that Kac–Moody algebra for an arbitrary quiver. The Schiffmann–Vasserot construction uses framed rank-r torsion-free sheaves on $\mathbb{P}^2$, framed along its line at infinity. Let $M_{r,n}$ be their instanton moduli, and localize equivariant cohomology over $K_r = \mathbb{C}(x, y, e_1, \ldots, e_r)$. Their specialized algebra $\mathbf{SH}^{(r)}_{K_r}$ acts faithfully on $L^{(r)}_{K_r} = \bigoplus_n H^{\widetilde{T}}_*(M_{r,n}) \otimes K_r$. It embeds into the specified degreewise completed current algebra of $W_{\kappa-r}(\mathfrak{gl}_r)$, with $\kappa = -y/x$, and induces its Verma-module action on $L^{(r)}_{K_r}$ [107, Introduction, Theorems A–C]. This preserves the distinction between a convolution algebra, its completion, and a representation of a vertex algebra. These theorems provide concrete quantum-group models; a comparison with a compact reduced $S \times E$ Hall algebra requires further geometric and coefficient maps.

A compact comparison would have to specify an integral form, a coproduct and pairing, the completed Drinfeld double, its Cartan subalgebra and the geometric operators defining the comparison. Equality of an index or of a denominator does not determine these structures. Definition 1.142 keeps this problem separate from the construction of the reference Yangian.

4.7 The remaining compact comparisons

Three established constructions enter the compact problem: the critical Hall product under strong orientation hypotheses, the reduced stable-pair virtual class, and the quantum groups acting on quiver-variety cohomology. Their comparison requires maps on coefficients. The formal cyclic charts provide a local domain on which to construct and test those maps; they do not identify the three constructions globally.

For the reduced $S \times E$ theory, the first equations are the coefficient comparison on extensions and its equality on the two three-step flag diagrams, including the orientation identifications. The collision coproduct has a separate three-output equation and a compatibility equation with multiplication. Elliptic descent is relative to a common E-torsor and has unit BE. A final comparison with the Igusa denominator also needs the primitive Lie form, operator realization, protected state and modular line described in the main text. These requirements concern different mathematical objects; satisfying one leaves the others to be constructed.

5 Pfaffian crossings and reflection characters

A simple zero of a Pfaffian changes sign under reflection of its normal coordinate. To interpret that sign as the action on an orientation line, one must specify a lift of the reflection and require it to preserve the section. This appendix gives the local skew model, compares its sign with the automorphic character, and states the gluing needed for a global compact comparison.

5.1 The three type-II walls

Use the root lattice

$$L = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}, \qquad (f_2, f_{-2}) = -1, \quad f_3^2 = 2,$$

with other pairings zero. Its type-II sublattice is

$$L_{\mathrm{II}} = \{mf_2 + lf_3 + nf_{-2} : m, n \in 2\mathbb{Z}\} \simeq U(4) \oplus \langle 2 \rangle.$$

It has index four in L and determinant -32. The three simple roots and their Gram matrix are

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3, \qquad A = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The eigenvalues $4, 4, -2$ give signature $(2, 1)$. For a root δ, reflection is $s_\delta(x) = x - (x, \delta)\delta$. The Weyl vector is

$$\rho = \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \tfrac{1}{2}f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

These formulas use the same lattice and active cone as Chapter 4.

The walls bound an ideal hyperbolic triangle. Its reflection group $W = W^{(2)}(L_{\mathrm{II}})$ has presentation

$$W = \langle s_1, s_2, s_3 \mid s_i^2 = 1 \rangle, \qquad s_i = s_{\delta_i} \tag{2}$$

[52, Proposition 2.2]. A product of distinct simple reflections is parabolic, not hyperbolic. On their two-dimensional root span its matrix is

$$s_i s_j = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix} = I + \begin{pmatrix} 2 & -2 \\ 2 & -2 \end{pmatrix}, \qquad (s_i s_j - I)^2 = 0.$$

It has infinite order because its n-th power is $I + n(s_i s_j - I)$.

The type-I normals $f_{-2} - f_2, f_2 - f_3, f_{-2} - f_3$ have divisibility one in L; their reflections give the chamber symmetry group S_3. The δ_i have divisibility two. The index six of the reflection subgroup in the full norm-two reflection group is independent of the index four of the lattices. The denominator on L_{II} is $64^{-1}\Delta_5(2Z)$. Doubling changes the exponential coordinates as in Proposition 10.1; it does not change these integral indices.

5.2 One normal Pfaffian factor

Take a complex normal chart $U = T_0 \times \mathbb{A}^1_u$, or its analytic analogue, with base reflection $s(t, u) = (t, -u)$. Let $V = \mathcal{O}_U^2$ and put

$$D_u : V \longrightarrow V^\vee, \qquad D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}.$$

The complex $K_D = [V \xrightarrow{D_u} V^\vee]$ lies in degrees $[-1, 0]$. Its Pfaffian line and section are

$$\mathscr{L}_{\mathrm{Pf}} = \det V^\vee, \qquad \mathrm{pf}(D_u) = u, \qquad \mathrm{pf}(D_u)^2 = \det D_u = u^2$$

in the chosen frame. There is one hyperbolic Pfaffian factor; the skew matrix has rank two away from the wall. The scalar complex $[O \xrightarrow{u} O]$ has determinant section u, so it is a different determinant problem.

A compatible reflection lift is explicit. With $R = \mathrm{diag}(-1, 1)$,

$$R^t D_u R = D_{-u}, \qquad \det R = -1.$$

Hence the induced comparison of Pfaffian frames has sign -1. An identity lift on the rank-two bundle would fail the first identity. For a tangential skew factor, assume its Pfaffian has a nonvanishing reference section v fixed by the chosen comparison. The total local coefficient is then vu.

PROPOSITION 5.1 *(Sign of a section-compatible crossing).* In this model, pullback by the base reflection changes the Pfaffian expression by -1 and preserves its square:

$$s^*(vu) = -vu, \qquad s^*((vu)^2) = (vu)^2.$$

Relative to the specified reference frames, a lift carrying the section to its pullback has residual sign $\epsilon_o^{\mathrm{loc}}(s) = (-1)^{N^{\mathrm{Pf}}} = -1$, where $N^{\mathrm{Pf}} = 1$.

Proof. The Pfaffian of the displayed matrix is u. Pullback replaces u by $-u$ and leaves v fixed. The transformation law $\mathrm{Pf}(R^t D R) = \det(R)\,\mathrm{Pf}(D)$ identifies this change with the stated lift. Squaring removes the sign. □

The functional identity is not, by itself, a construction of that lift on a geometric orientation line. Nor does a real path through $u = 0$ compute monodromy in the wall complement. The skew model, tangential unit and section comparison are the hypotheses of the local assertion.

5.3 The automorphic character

Let $\widetilde{\nu}_{\Delta_5}$ be the homogeneous orthogonal character of Definition 6.2. It retains both the weight-five line and its modular character.

LEMMA 5.2 *(Values on the six reflections).* The values of $\widetilde{\nu}_{\Delta_5}$ are -1 on the three simple type-II reflections and $+1$ on the three type-I chamber reflections. Multiplying Δ_5 by a nonzero constant changes none of these values.

Proof. The symplectic lifts in Lemma 6.3 satisfy $\sigma(g_\beta) = -s_\beta$ and $j(g_\beta, Z) = -1$. Maass's formula [52, (1.5)] gives modular character $+1$ on the type-II lifts and -1 on the type-I lifts. Homogeneity of degree -5 gives $\widetilde{\nu}_{\Delta_5}(-I) = -1$, reversing the signs for s_β. These are the linear reflection signs of [52, Proposition 2.1]. A scalar cancels from every transformation law. □

THEOREM 5.3 *(Local orientation comparison).* Suppose each simple type-II wall has the section-compatible skew crossing above. Then, in those reference comparisons,

$$\epsilon_o^{\mathrm{loc}}(s_i) = -1 = \widetilde{\nu}_{\Delta_5}(s_i), \qquad 1 \le i \le 3.$$

Proof. Apply Proposition 5.1 and Lemma 5.2 to each generator. □

5.4 *From three local signs to an action*

The group W has a determinant character $\det_W$, taking value -1 on every s_i. Equation (2) shows directly that these assignments respect its relations. The preceding lemma proves

$$\widetilde{\nu}_{\Delta_5}|_W = \det_W .$$

To compare a geometric orientation with this character, local lifts must first form an action on the same root pair.

For chosen lifts ϕ_w of $w \in W$, their composition defect is

$$c(w,v) = \phi_w \, w^*(\phi_v) \, \phi_{wv}^{-1}.$$

When all lifts preserve the specified square map, this defect is μ_2-valued. On a connected base with constant coefficients it is a group two-cocycle. More generally its coefficients are locally constant signs on the relevant components, with the induced group action. A one-cochain η with $\delta\eta = c$ replaces the lifts by coherent ones. This cochain equation does not determine their character: different coherent lifts can differ by a character.

Assume further that the coherent action has constant scalar values in the reference comparisons and that it preserves the local Pfaffian sections. Those values give a homomorphism $\epsilon_o : W \to \{\pm 1\}$. Since its generator values are -1,

$$\epsilon_o = \det_W = \widetilde{\nu}_{\Delta_5}|_W .$$

This is the conditional global conclusion. It uses the action, reference comparisons and sections, rather than just an equality of determinant squares.

5.5 *The global wall-atlas condition*

Definition 5.4 *(Compatible wall atlas).* For each finite geometric stage R, a compatible type-II wall atlas consists of charts $U_{w,i,R}$, for $w \in W$ and $1 \le i \le 3$, with skew-complex comparisons to the normal model above; Pfaffian-line and section isomorphisms on overlaps; and coherent lifts of the Weyl action. The charts cover the specified geometric locus and its required boundary pieces. Their comparisons commute with the transition maps in R.

Equivalently, when lifts are initially projective, include their actual sign two-cocycle and a one-cochain trivializing it. That trivialization must agree on components, boundaries and overlaps, and preserve the local section comparisons.

The condition is denoted $(O2_{\mathrm{atlas}})$. Its local hypotheses can be expressed by the chart, wall, splitting, tangential-unit and normal-rank residuals

$$\mathfrak{o}^{\mathrm{chart}}_{\partial,R}, \quad \mathfrak{o}^{\mathrm{wall}}_{\partial,R}, \quad \mathfrak{o}^{\mathrm{split}}_{\partial,R}, \quad \mathfrak{o}^{\mathrm{unit}}_{\partial,R}, \quad \mathfrak{o}^{\mathrm{rank}}_{\partial,R}.$$

Their vanishing means that the corresponding geometric maps exist with the stated comparisons. The transition residual $\mathfrak{o}^{\mathrm{tr}}_{\partial,R'R}$ additionally requires commutativity with the stage maps. A local zero in one chart gives neither a global covering nor a trivialization of the global cocycle.

Together with quotient orientation and coherent Weyl transport, this atlas yields the character comparison above. The conditional transport statement of Theorem 7.16 uses these hypotheses. Constructing them on the compact reduced Hall locus remains a geometric problem.

5.6 The two numerical appearances of 4096

The abstract set $\Omega_{12} = \{\pm 1\}^{12}$ has $2^{12} = 4096$ elements. A geometric interpretation as a count of wall configurations requires an actual bijection with this set, including the identifications, coverage and stabilizer weights. Definition 5.4 alone gives no such bijection.

Independently, the theta-leading coefficient is 64, so

$$\Xi = (64^{-1}\Delta_5)^2, \qquad Z^X_{\mathrm{OP}} = -\Xi^{-1} = -4096\Delta_5^{-2}$$

in the primitive curve-counting convention [100, 101]. Here $4096 = 64^2$ is a normalization constant. Equality with $|\Omega_{12}|$ supplies no map from a geometric orbit to Ω_{12}, and it does not determine an orientation character. The separate hypotheses in Theorem 7.14 must be retained when either geometric interpretation is claimed.

5.7 Parity and the first timelike root

A reflection character acts on a line. Primitive recognition compares a Lie superalgebra and therefore needs the even and odd dimensions as well as its brackets. Put $\beta = \delta_1 + \delta_2 + \delta_3$.

LEMMA 5.5 *(The first timelike parity split).* For the fixed Gritsenko–Nikulin presentation,

$$\operatorname{sdim} \mathfrak{g}_{\Delta_5,\beta} = -64, \qquad (\dim \mathfrak{g}_{\beta,\bar{0}}, \dim \mathfrak{g}_{\beta,\bar{1}}) = (29, 93).$$

Proof. The Jacobi coefficient is $f(1,1) = -64$ [39]. In the presentation, the even part at β contains two independent Lie words in the three real generators and nine brackets for each of the three real–null decompositions. Thus its dimension is $2 + 3 \cdot 9 = 29$. The odd part consists of the 93 imaginary simple generators read from the chamber correction. No real Serre relation occurs in this multidegree. This is the word calculation of Proposition 4.6, and $29 - 93 = -64$. □

The integer 93 comes from the simple-generator correction, not from assigning one parity to a product exponent. More explicitly, with the complete boundary factors retained,

$$m(\beta) = -[qrs]\big((qrs)^{-1/2}64^{-1}\Delta_5\big) = -93.$$

Witt's multigraded formula gives the two real words as $3!/3 = 2$; the null-ray multiplicity nine is the first coefficient of the corresponding eta-nine correction [52, 54].

COROLLARY 5.6 *(Parity dimensions do not determine brackets).* Even the full list of root parity dimensions does not recognize the positive Borcherds algebra without an operation comparison.

Proof. Give the underlying positive graded super vector space of $\mathfrak{g}_{\Delta_5}$ the zero bracket. Every even and odd root dimension is unchanged, but the real-root bracket $[e_i, e_j] \neq 0$ for $i \neq j$ has been lost. The two graded Lie superalgebras cannot be identified while preserving these specified generators and their brackets. A single root component is not itself the recipient for all those brackets. □

Higher degrees sharpen the comparison. Write $a_{ij} = \delta_i + \delta_j$. At $2a_{ij}$, the signed root multiplicity ten consists of the null simple multiplicity nine and the one-dimensional affine contribution. At $a_{ij} + 2\delta_k$, with distinct i, j, k, the simple correction is 90 and the signed multiplicity is 108. The full parity pair is (201, 93), by Proposition 4.7. Thus the residual $108 - 90 = 18$ is a signed bracket contribution; it is not a parity block. Also

$$s_{\delta_k}(\beta) = a_{ij} + 3\delta_k,$$

so the height-four degree is not that reflected timelike degree. The real Serre exponents are three for a real–real pair, five for a complementary real–null pair, and three for a real generator acting on a timelike simple generator of degree β. The all-height parity recursion of Proposition 4.3 determines both target parities in the fixed presentation. It does not construct a compact source with those dimensions or operations.

5.8 The remaining geometric comparisons

The local skew calculation supplies a Pfaffian section with a specified reflection lift. The automorphic calculation independently supplies the homogeneous character. A compact identification needs a covering geometric wall atlas, compatible square-root and section comparisons, quotient descent, and coherent Weyl transport at every stage and under every transition. These are the hypotheses in Theorems 7.9 and 7.10 and the wall-atlas condition above.

A geometric primitive comparison must further preserve both parities and the actual Lie and Hall operations. The scalar normalization and the abstract twelve-sign count solve none of those existence problems. They remain independent even after the local reflection signs have been matched.

6 Finite comparison maps for the compact realization

The finite data of Section 1.7 become useful when their maps can be calculated. This appendix fixes the geometric supports, coefficient morphisms and matrix equations needed for that calculation. It also separates the formal target from the compact source: a target degree can be listed before a sheaf of that degree or a Hall operation has been constructed.

Throughout, $X = S \times E$, with S a smooth projective K3 surface and E a smooth elliptic curve. The stage parameter R bounds the chosen geometric data. It is distinct from root height, word length, cohomological degree and any formal deformation parameter. When several bounds are used, their comparison is part of the construction.

The proposed compact datum includes a field complex $\mathcal{A}^{E_3}_{X,R}$, its local/wrapped specialization, critical coefficients and orientation, Hall operations, primitives, a coalgebra and its associative comparison, Pfaffian sections and recognition maps. If an E_3- or BV description is required, its formal target charts, field differential and operations must be constructed. The quantum master equation, anomaly and framing, compact-support maps and factorization descent are additional conditions on those same complexes. A map denoted $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$ must preserve the specified operations and stage transitions; it is not defined by naming a current-envelope target.

6.1 Charges, finite supports and target degrees

Fix the Abramovich–Polishchuk heart and its Liu tilt at the scope of Appendix 8 [1, 75]. The polynomial charge has the form

$$L_M(n) = a(M)n + b(M) + i\big(c(M)n + d(M)\big).$$

The weak charge, torsion pair and tilted heart must be kept with the central charge; retaining only the polynomial does not retain the stability condition. A finite subobject- and quotient-closed collection with a strict integer length admits the finite HN argument of Proposition 8.6: choose maximal phase and then maximal length at each quotient. Its factor count is bounded by the initial length. This is an argument in the chosen heart, not a theorem that the moduli of all such factors is finite or proper.

Let Γ_{act} be the nonzero Jacobi support in the cusp convention of Chapter 3, for the index-one form of [39]. A finite set can be obtained by taking

$$A_R^{\circ} = \{(n, l, m) \in \Gamma_{\mathrm{act}} : n + |l| + m \le R\}$$

and its downward saturation in the positive root order. This gives an exhaustion of the active target degrees. The actual positive cone and its finite height fibres establish finiteness; they do not supply geometric representatives.

For geometric charge data, the normal-ordered lift is

$$\widehat{\Gamma}_R = \{(c, T) : c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\}, \qquad \Gamma_R^{\Pi} = \{\Pi_X(c) + T : (c, T) \in \widehat{\Gamma}_R\}.$$

The translate set $\mathcal{T}_R(c)$, its bounds and the cocycle B are those of Definition 1.189. The formal lift of every Gram triple in Lemma 6.7 is a lattice construction. It does not prove effectivity, stability or existence of a compact object at that lift.

At a retained positive root β, the target has separately specified even and odd dimensions e_β, o_β. The minimal Gritsenko–Nikulin presentation and its ordinary and super denominators [52–54] determine them by Proposition 4.3. Signed exponents alone do not: adding an even and an odd generator can leave their difference unchanged. The first interior target is $29 \mid 93$, and the next height-four timelike target is $201 \mid 93$. Geometric recognition requires maps to these spaces, not just the numbers -64 and 108.

A finite relation diagram enlarges its degree set to retain the output of every intermediate bracket. Let W_3 be the thirteen positive root degrees of height at most three, and put

$$R_4 = \{3\delta_i + \delta_j : i \ne j\}, \qquad I_4 = \{2a_{ij} : i < j\},$$

$$C_{\le 7} = \{a_{ij} + s\delta_k : \{i, j, k\} = \{1, 2, 3\},\ 2 \le s \le 5\}.$$

Here $a_{ij} = \delta_i + \delta_j$, as in the target presentation. The degree set

$$W_{\mathrm{rel}}^{+} = W_3 \cup R_4 \cup I_4 \cup C_{\le 7} \cup \{2\delta_{123}\}, \qquad D_k = \delta_k + 2a_{ij},$$

has 35 elements, including the nine zero root spaces at the terminal real and complementary Serre degrees. Adding the D_k gives $W_{\mathrm{sat}}^{+} = W_{\mathrm{rel}}^{+} \cup \{D_1, D_2, D_3\}$, but the effect of a further enlargement must be computed from its actual set and operations. The first further enlargement described after Proposition 1.185 has 89 degrees. Later finite totals cannot be inferred by retaining a count calculated from a different initial set.

The six-degree families A_N, B_N of Proposition 1.186 satisfy $A_N \Rightarrow B_N \Rightarrow A_{N+2}$ under its specified degree-enlargement rule. To realize such degrees geometrically, one must give R_N, c_N, T_N satisfying $\Pi_X(c_N) + T_N = \gamma_N$ for every relevant member of the family. Definition 1.190 keeps this existence assertion separate from the target calculation.

6.2 Finite moduli and universal complexes

A bounded geometric class can be described by

$$c = (\widehat{c}, [a_c, b_c], (P_{c,i}), N_c),$$

where $\widehat{c}$ is its charge, $[a_c, b_c]$ its prescribed cohomological range, $P_{c,i}$ its Hilbert polynomials and N_c a regularity bound. A supplied finite collection C_R of such data determines the moduli problem

$$\mathfrak{M}_{R,c}^{ss} = \{A \in \mathrm{Perf}(X) : A \text{ has the specified class and bounds and is semistable}\}.$$

One must prove that the chosen locus is algebraic and of finite type. Semistability is not generally a closed condition under every specialization. If a proper correspondence argument uses a closed locus inside a bounded Quot space, that closedness is an extra hypothesis on the locus being used.

The tangent complex at $A \in \mathrm{Perf}(X)$ is $R\,\mathrm{Hom}(A, A)[1]$. On a CY3 it can have cohomology in degrees -1, 0, 1, 2. Scalar automorphisms and their Serre-dual trace direction are present before reduction. A two-term reduced complex therefore needs an actual reduction map and proof of its amplitude. Restriction of the ambient shifted symplectic form to an arbitrary substack is not automatically nondegenerate [104].

On the scalar gerbe, the universal complex $\mathcal{U}$ has weight one. The complex $R\mathcal{H}om(\mathcal{U}, \mathcal{U})$ has weight zero and can descend along the gerbe with its natural maps. Descent of $\mathcal{U}$ itself needs a trivialization or compensating weight, or else a twisted recipient. The distinction is already visible on $B\mathbb{G}_m$: the standard representation does not descend to a line on a point, although its endomorphism line does.

Removing the common scalar subgroup does not make all stabilizers finite. For a direct sum of two distinct stable objects, a relative $\mathbb{G}_m$ remains; repeated summands retain larger linear groups. A finite-inertia locus and its closed strata must be specified if the coefficient or descent theorem requires them. Quotient by translation in E is a further geometric operation, with its own stabilizers and support conditions.

The retained object set, HN factors and chosen extension diagrams must be compatible. This is not closure under all extensions inside a bounded nonzero positive charge set: repeated sums leave any fixed positive bound. Likewise, reversal of a finite HN word and a formal charge involution does not construct derived duality on the chosen heart. A geometric duality must land in the specified recipient and preserve the retained support and orientation data. The hypotheses and their valid finite consequences are separated in Appendix 8.

6.3 Hybrid support and correspondence maps

The elliptic support degree is defined from the actual one-cycle:

$$(p_E)_* \,\mathrm{cyc}_1(F) = b_R^{\mathrm{geom}}(F)[E].$$

For effective one-dimensional support, a positive degree projects onto all of E. At degree zero, every curve component has a point as its image in E. These support statements are independent of the normal-ordered Borcherds coordinate $m_R^{\mathrm{Bch}} = \mathrm{pr}_3\, \overline{\Pi}_X$. An equality between those two degree functions is a comparison to be proved, not a convention for the support.

The local closed-configuration stack for a finite index set I has pairs $(A, (e_i)_{i \in I})$ with

$$\mathrm{Supp}(A) \subset \bigcup_{i \in I} S \times \{e_i\}.$$

Repeated points are allowed in this closed configuration recipient. The open disjoint locus and the collision locus have different restriction maps. A local section on one does not itself define the operation across the other.

The wrapped sector begins with an E-equivariant prequotient $\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}$ and its anchor $\lambda_{\eta,R}$. A determinant anchor of Euler weight χ transforms under translation with that weight. It can be constant along orbits when $\chi = 0$. Jordan–Hölder spectral data may distinguish, for example, $\mathcal{O}_E^2$ from $L \oplus L^{-1}$, but translation acts trivially on the isomorphism classes of degree-zero line bundles. Thus that datum alone does not separate every orbit.

Changing the origin from o to a changes the relevant Picard class by

$$\lambda_a \otimes \lambda_{\mathrm{o}}^{-1} \simeq \mathcal{O}_E\big(\chi(\mathrm{o}-a)\big)$$

under the stated determinant convention. It does not canonically choose a stabilizer character. Even a trivial line can carry different linearizations. An equivariant comparison must retain that choice.

The four binary correspondences LL, LW, WL, WW are defined before quotient, keeping both input anchors and the output anchor. For two wrapped inputs, the common-translation quotient satisfies

$$[M \times N/E] \simeq [M/E] \times_{BE} [N/E]$$

with diagonal translation. The ordinary product of quotient stacks would permit two unrelated torsors and changes the operation. Quotienting an extension diagram therefore requires a pseudofunctor $Q_{E,R}$, its operation comparisons $\theta^{Q}_{\mu,R}$, and their base-change coherences. If $\mathcal{B}_R^{E,\mathrm{hyb}}$ denotes the specified pre-quotient correspondence construction, the descended one is $Q_{E,R}\mathcal{B}_R^{E,\mathrm{hyb}}$ only after those comparisons are supplied.

The two-step flag atlas has eight words of length three in $\{L, W\}^3$. Their associativity squares are finite tests of the actual correspondences. Full factorization needs all higher flags and collision coherences as well. A finite table of binary maps does not provide those higher maps.

On a sheaf locus, properness of the target extension map can follow from its description as a closed relative Quot locus. The argument requires the bounded Hilbert data and the closed source condition. It applies separately to LL, the two ordered mixed cases, and WW when their respective hypotheses hold. It supplies no properness of the other map in the correspondence and does not prove Thom–Sebastiani, quotient descent or a protected coefficient identity. Section 1.1 gives the actual local models and the remaining global comparisons.

6.4 Reduced orientation and equivariant descent

Let K_{o} be a specified perfect deformation complex with a cosection morphism $\sigma : K_{\mathrm{o}} \to \mathcal{O}[-2]$. Its fibre is perfect:

$$K^{\mathrm{fib}} \longrightarrow K_{\mathrm{o}} \xrightarrow{\sigma} \mathcal{O}[-2].$$

The notation K^{red} further specifies which scalar, orbit and dual obstruction directions are removed. A morphism to $H^2(S, \mathcal{O}_S)[-1]$ would remove a different degree. The precise shift is therefore part of the cosection construction.

For a nonzero elliptic translation direction u_F, Serre duality pairs $\mathrm{Ext}^1(F,F)/\mathbb{C}u_F$ with $\ker \sigma_F \subset \mathrm{Ext}^2(F,F)$. This gives a linear reduced pairing. A family of reduced critical charts, its orientations and coefficient gluing must still be constructed. The positive reduced stable-pair comparison and the ordinary critical CoHA construction in the preceding chapters retain their own geometric domains; neither automatically provides this complete reduced compact coefficient system.

The determinant convention is

$$\det K = \bigotimes_i (\det K^i)^{(-1)^i}, \qquad \det(K[1]) = (\det K)^{-1}, \qquad \det(K^\vee) = (\det K)^{-1}.$$

A root is the pair $(o, \rho : o^{\otimes 2} \simeq \det K)$. If a further perfect cosection factor C_{cs} is used, root the actual tensor product $\mathscr{D} = \det K^{\mathrm{red}} \otimes \det C_{\mathrm{cs}}$. Its obstruction is $\delta(\mathscr{D}) \in H^2(B, \mu_2)$. An underlying cokernel sheaf need not have a determinant until a perfect model is supplied.

For a complex line,

$$w_2(\mathscr{D}_{\mathbb{R}}) = c_1(\mathscr{D}) \bmod 2.$$

This uses the integral Chern class. A rational Chern-character calculation cannot detect all its torsion. If a cocycle a^{red} represents the obstruction, an actual trivialization is a cochain h^{red} with $\delta h^{\mathrm{red}} = a^{\mathrm{red}}$. Vanishing of its cohomology class does not select that cochain. The isomorphism classes of root pairs form an $H^1(B, \mu_2)$-torsor, rather than merely the two-torsion subgroup of the Picard group.

For an E-action, solve the root problem on $[B/E]$. The group $H^2(BE; \mathbb{F}_2)$ has basis u_1, u_2, but an expression $a_1u_1 + a_2u_2$ contributes only through the actual quotient or spectral-sequence map. That map need not be injective. For the translation action of E on itself, $[E/E]$ is a point. Thus requiring both coefficients to vanish in $H^2(BE)$ is not a universal free-action descent criterion.

Finite stabilizers

At a finite stabilizer H, distinguish the equivariant Kummer class from the character of the chosen root. For $H = E[2] \simeq (\mathbb{Z}/2)^2$, write

$$\beta = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad |x_i| = 1.$$

Restriction to the three order-two subgroups gives coefficients

$$r_1 = b_{20}, \qquad r_2 = b_{02}, \qquad r_3 = b_{20} + b_{11} + b_{02}.$$

Hence

$$b_{20} = r_1, \quad b_{02} = r_2, \quad b_{11} = r_1 + r_2 + r_3, \qquad \beta = 0 \iff r_1 = r_2 = r_3 = 0.$$

Indeed, the two coordinate restrictions send (x_1, x_2) to $(t, 0)$ and $(0, t)$, while the diagonal restriction sends it to (t, t).

For a two-primary subgroup $(\mathbb{Z}/2^a)^2$ with $a \geq 2$,

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |y_i| = 2.$$

A degree-two class is $A_1y_1 + A_{12}x_1x_2 + A_2y_2$. The mixed class restricts to zero on every cyclic subgroup: on a cyclic group of order divisible by four, both degree-one factors are multiples of an exterior generator, while on an order-two subgroup their pullbacks from $\mathbb{Z}/2^a$ vanish. The functional

$$\pi_{12}(y_1) = \pi_{12}(y_2) = 0, \qquad \pi_{12}(x_1x_2) = 1$$

detects the missing coefficient A_{12}. For $E[N]$, the exterior-generator case requires $4 \mid N$; when its two-primary exponent is two, the polynomial $E[2]$ calculation applies instead. Odd-order groups have no positive-degree mod-two group cohomology, by averaging.

Even when the degree-two class vanishes, a chosen equivariant root can have a nontrivial character $\lambda^H \in H^1(BH; \mathbb{F}_2)$. For $E[2]$, write $\lambda^H = \lambda_1x_1 + \lambda_2x_2$; its three restriction values are $\lambda_1, \lambda_2, \lambda_1 + \lambda_2$. The pair $(1, 0)$ gives a nontrivial character despite a zero Kummer class. Coarse descent requires that every stabilizer act trivially on the root fibre. Stack descent and coarse descent are therefore different requirements.

For an inclusion $K \hookrightarrow H$, chosen cocycle primitives require comparisons of the form

$$\operatorname{res} h^H - h^K = \delta q^{K,H}.$$

The comparisons must satisfy their own compatibility for chains of subgroups and for overlap and stage maps. Cohomological vanishing at each stabilizer separately supplies no such coherent choice.

Cross determinants and critical coefficients

Write $\mathscr{D}_A^{\mathrm{full}} = \det R\,\mathrm{Hom}(A, A)$. The block decomposition of this full, unshifted complex gives

$$\mathscr{D}_{A\oplus B}^{\mathrm{full}} \simeq \mathscr{D}_A^{\mathrm{full}} \otimes \mathscr{D}_B^{\mathrm{full}} \otimes \det R\,\mathrm{Hom}(A, B) \otimes \det R\,\mathrm{Hom}(B, A).$$

For a reduced complex, one must first identify its corresponding determinant triangle and any additional reduction factors. When the displayed cross factor is the one in that triangle, the orientation comparison requires

$$o_{A\oplus B} \simeq o_A \otimes o_B \otimes q_{A,B}, \qquad q_{A,B}^{\otimes 2} \simeq \det R\,\mathrm{Hom}(A, B) \otimes \det R\,\mathrm{Hom}(B, A).$$

For a filtered extension, the corresponding determinant triangle and its coherent higher-flag identifications replace the split formula. Omitting $q_{A,B}$ generally changes the square map.

The critical coefficient Φ_o is a constructible object formed using the orientation and its sign torsor [12, 63, 64]. It is not the ordinary tensor $\Phi \otimes o$ of a constructible complex with a coherent line. A coefficient map on an extension correspondence must be an actual morphism in its constructible or mixed-Hodge recipient, with the required shifts and supports. The line-level determinant identity alone supplies no such morphism. Thom–Sebastiani comparisons and their full flag associativity must be checked for these coefficient maps.

Weyl cocycles and their compatible primitives

At a finite action groupoid $\mathcal{G}_R$, let the lifted arrows have sign defect $c_{o,R}$. With a fixed finite cochain model, its trivialization is the linear equation

$$d_R^{1} b_R = v(c_{o,R})$$

over $\mathbb{F}_2$. It has a solution exactly when

$$\operatorname{rank} d_R^{1} = \operatorname{rank}[d_R^{1} \mid v(c_{o,R})],$$

provided the represented defect is actually a cocycle in that cochain complex. The matrix ranks test solvability; the vector b_R is the chosen trivialization.

Transition maps must act on the cochain models, not only their cohomology. If $\rho_{RR'}^{i} : C^i(\mathcal{G}_{R'}) \to C^i(\mathcal{G}_R)$, require

$$d_R^{1}\rho_{RR'}^{1} = \rho_{RR'}^{2} d_{R'}^{1}, \quad \rho_{RR'}^{2} v(c_{o,R'}) = v(c_{o,R}), \quad \rho_{RR'}^{1} b_{R'} = b_R.$$

A tower of solvable equations need not admit the last compatible choice without an inverse-system argument.

Suppose $\eta_{R,S}$ trivializes the quotient orientation on a stratum S, and $\Theta_{i,R,S}$ is its chosen transport. The remaining torsor defect is represented by

$$\Delta_{i,R,S}^{\omega} = \Theta_{i,R,S}^{1}\eta_{R,S} - \eta_{R,s_iS}.$$

It must be closed, have zero class in the actual degree-one cohomology, and have a compatible primitive when a chain-level identification is required. The reduced degree-two defects require the corresponding cochain comparisons as well. For identified finite stabilizers, the matrices on their cohomology must intertwine both classes and characters:

$$M_i^H v(\beta_S^H) = v(\beta_{s_iS}^{H'}), \qquad N_i^H v(\lambda_S^H) = v(\lambda_{s_iS}^{H'}).$$

The same assertion for any BE-component uses its actual induced map M_t^E.

Once coherent lifts have constant residual generator values σ_i^o, their character on a word is

$$\epsilon_o(s_{i_1}\cdots s_{i_m}) = \prod_{a=1}^{m} \sigma_{i_a}^o.$$

Section-compatible skew crossings give $\sigma_i^o = -1$, hence $\epsilon_o = \det_W$, as in Appendix 5. The automorphic homogeneous character is trivial on the chamber symmetry S_3. Extending a geometric W-action to the full semidirect product requires actual chamber-symmetry correspondences and their lifts. The two scalar extensions

$$\widetilde{\epsilon}_\chi(w,\sigma) = \epsilon_o(w)\chi(\sigma), \qquad \chi \in \{1, \mathrm{sgn}\},$$

show that the W-restriction alone cannot choose that extension.

At another divisor, a coefficient local system $\mathcal{K}_R$ on the complement has monodromy

$$T_{R,S,\eta} = \rho_{\mathcal{K}_R}(\gamma_{R,S,\eta})$$

around a specified meridian $\gamma_{R,S,\eta}$. This requires local constancy and the stated representation of the fundamental group. A coherent root line has no such representation until a flat structure is supplied, and an arbitrary constructible critical complex need not be locally constant on the proposed complement. Monodromy at an imaginary-root divisor need not be a Weyl reflection.

Parity also does not choose a root-line action. The Berezinian of a super vector space and a chosen square root with its equivariance are different data. Similarly, a duality $K_\gamma \simeq K_{-\gamma}^\vee[-3]$, when actually supplied, gives

$$\det K_\gamma \simeq \det K_{-\gamma}.$$

The dual and the odd shift each invert the determinant, so their effects cancel. A claim about opposite root lines needs the chosen square-map and duality comparison; it cannot be inferred by inserting an inverse determinant or an unexplained sign line.

Finally, a compatible family of root pairs is first an object of the limit of the relevant Picard groupoids. Its effectivity as a line on an infinite geometric recipient needs an actual comparison

$$\mathrm{Pic}^{or}(B_\infty) \longrightarrow \varprojlim_R \mathrm{Pic}^{or}(B_R).$$

An equivalence here is stronger than a statement about individual classes in π_0. Taking isomorphism classes can lose the coherent isomorphisms and their automorphisms.

6.5 Hall maps, primitives and pairings

Let $\mathcal{H}_{R,c}$ be the specified coefficient realization of the oriented critical object $\Phi_{o,R,c}$ on the retained moduli stack. This can be compactly supported critical cohomology or a declared Borel–Moore model; its grading, duals and shifts are part of the definition. Do not identify these realizations by notation alone.

An extension correspondence has maps

$$\mathfrak{M}_a \times \mathfrak{M}_b \xleftarrow{p} \mathfrak{E}_{a,b} \xrightarrow{q} \mathfrak{M}_{a+b}.$$

For a product, specify the pullback or Gysin map appropriate to p, the critical coefficient morphism $\theta_{a,b}$, and the proper or compact-support pushforward appropriate to q. The notation

$$m_{a,b} = q_! \theta_{a,b} p^\dagger$$

records these chosen maps; $p^{\dagger}$ includes its degree shift. Ordinary pullback on sheaves does not automatically define a map on compactly supported cohomology. The coefficient and support comparisons must supply that map in the chosen realization.

For a separated finite-type map with a compactification $q = \overline{q}\, j$, the sheaf-theoretic expression is

$$q_! K = \overline{q}_* j_! K.$$

A common proper refinement of two compactifications identifies these objects through the standard proper base-change comparison. This proves independence of the compactification at the functorial level; it does not manufacture a missing morphism from the resulting coefficient object to the desired target coefficient.

On the three-input flag correspondence, associativity compares

$$m_{a+b,c}(m_{a,b} \otimes \mathrm{I}) \quad \text{and} \quad m_{a,b+c}(\mathrm{I} \otimes m_{b,c}).$$

The geometric proof must identify the two composed diagrams, their coefficients and shifts. Higher flags give coherence among these identifications. Likewise a coproduct requires a separate coefficient map and admissible pushforward in the reverse diagram. It is not obtained by formally inverting $\theta_{a,b}$ unless that map is an equivalence on the stated recipient.

For a bialgebra comparison, the four-block flag decomposes by

$$\mathsf{S}(a, b; u, v) = \{(a_1, a_2, b_1, b_2) : a = a_1 + a_2,\ b = b_1 + b_2, u = a_1 + b_1,\ v = a_2 + b_2\}.$$

With M and D denoting the product and coproduct matrices, its required identity is

$$D^{a+b;u,v} M^{a,b} = \sum_{\mathsf{S}(a,b;u,v)} (M^{a_1,b_1} \otimes M^{a_2,b_2})(\mathrm{I} \otimes \tau \otimes \mathrm{I}) \cdot (D^{a;a_1,a_2} \otimes D^{b;b_1,b_2}).$$

The braiding τ must be the one belonging to the actual tensor. An ordinary Koszul equation cannot be substituted for a meromorphic braided equation. Unit and counit diagrams are additional parts of a bialgebra structure.

For a connected positively graded ordinary bialgebra, the antipode recursion is

$$S(\mathrm{I}) = \mathrm{I}, \qquad S(x) = -x - \sum S(x')x'', \qquad \overline{\Delta} x = \sum x' \otimes x''.$$

Positive degree makes it finite on each homogeneous element. The convolution inverse identities are $m(S \otimes \mathrm{I})\Delta = m(\mathrm{I} \otimes S)\Delta = \eta\epsilon$. This antipode preserves total charge; it is not a geometric charge-reversing duality.

Let $I = \ker \epsilon$. On this augmentation ideal,

$$\overline{\Delta}(x) = \Delta x - \mathrm{I} \otimes x - x \otimes \mathrm{I}, \qquad P = \ker(\overline{\Delta} : I \to I \otimes I).$$

For the ordinary Koszul tensor, the supercommutator $[x, y] = xy - (-\mathrm{I})^{|x||y|} yx$ of primitives is primitive. Expanding the bialgebra identity cancels the two cross terms. Associativity also gives the super Jacobi identity

$$(-\mathrm{I})^{|x||z|}[x, [y, z]] + (-\mathrm{I})^{|y||x|}[y, [z, x]] + (-\mathrm{I})^{|z||y|}[z, [x, y]] = \mathrm{O}.$$

Thus P is a Lie superalgebra. After a proved charge descent,

$$P^{\Pi}_{\gamma} = \bigoplus_{\overline{\Pi}_X(\widehat{c})=\gamma} P_{\widehat{c}}, \qquad [P^{\Pi}_{\gamma,\tilde{p}}, P^{\Pi}_{\eta,\tilde{q}}] \subset P^{\Pi}_{\gamma+\eta,\tilde{p}+\tilde{q}}.$$

The additivity of the descended degree is a hypothesis on the operations; the central cocycle alone does not establish it.

A canonical primitive projector requires further assumptions. For the connected cocommutative ordinary Hopf realization over characteristic zero, the Eulerian logarithm gives the projector proved in Chapter 4. A chosen linear splitting of $P \hookrightarrow I$ is not that projector, and its naturality under transition maps must be checked independently.

Coherent charge descent

An additive charge map regrades supplied operations. A chain-level repair of a nonadditive comparison asks for more: a specified deformation complex, the cochain representing the defect, and its nullhomotopy. If that recipient is a Hochschild cochain complex, the proposed equation has the form

$$d_{\mathrm{Hoch}}\Theta_{\Pi} = B_{\mathrm{ch}}, \qquad |\Theta_{\Pi}| = 1, \quad |B_{\mathrm{ch}}| = 2.$$

Here B_{ch} must first be constructed as a closed cochain from the geometric charge defect. The equation is soluble exactly when its class vanishes in that specified complex. Triple homotopies, cyclic refinements, Jacobi compatibility and radical preservation are further equations on the chosen solution. They may be denoted $\mathfrak{o}^{\mathrm{top}}_{\Theta}$, $\mathfrak{o}^{\mathrm{Hoch}}_{\Theta}$, $\mathfrak{o}^{\mathrm{tri}}_{\Theta}$, $\mathfrak{o}^{\mathrm{Jac}}_{\Theta}$, $\mathfrak{o}_{F}$, $\mathfrak{o}^{\mathrm{cyc}}_{\Theta}$, $\mathfrak{o}^{\mathrm{rad}}_{\Theta}$: these names record the carrier, Hochschild, triple, Jacobi, invariant-form, cyclic and radical requirements, not seven classes in an unspecified group.

The Loday–Quillen–Tsygan comparison concerns stable matrix Lie homology and cyclic homology [76, 77, 112]. Applying it to this problem requires a map from the actual compact coefficient system to its recipient. Neither that stable theorem nor the critical Hall construction of [73] supplies Θ_{Π}, a reduced orientation or all the listed operation homotopies.

Invariant Lie forms and a separate Hopf pairing

An even opposite-degree invariant Lie form has matrices

$$B(e_{\gamma,\tilde{p},a}, e_{-\gamma,\tilde{p},b}) = G_{\gamma,\tilde{p};ab}, \qquad G_{\gamma,\tilde{p}} = (-1)^{\tilde{p}} G^{t}_{-\gamma,\tilde{p}}.$$

Other degree or parity pairings vanish. Invariance is

$$B([x, y], z) = B(x, [y, z]),$$

or equivalently $B([x, y], z) + (-1)^{|x||y|} B(y, [x, z]) = 0$. If this form is defined through a geometric product and a charge-zero functional, both that functional and the coefficient pairing must be constructed. A formal integral symbol does not supply them.

For supplied homogeneous bases and dual output functionals, the product and pairing entries are then

$$M^{c}_{ab} = \langle e^{c}, q_! \theta\, p^{\dagger}(e_a \otimes e_b)\rangle, \qquad G_{\gamma;ab} = \mathrm{tr}_{\mathrm{o}}\, m(e_{\gamma,a}, e_{-\gamma,b}).$$

The first expression is the coefficient realization of the correspondence; the second additionally requires its charge-zero trace and the claimed invariant-form identity. Integration and duality must be defined on those same coefficient objects.

The radical R_B is a Lie ideal by invariance. Its generated ideal in $U(P)$ is a Hopf ideal because every radical generator is primitive. This is the positive quotient construction of Proposition 8.2. If matrices Q_γ have columns lifting bases of the radical quotients, then the descended form is

$$\overline{G}_\gamma = Q_\gamma^t G_\gamma Q_{-\gamma}.$$

The left and right radicals must be killed before this expression is independent of the lifts. Nondegeneracy requires

$$\operatorname{rank} \overline{G}_\gamma = \dim(P_\gamma / R_{B,\gamma}) = \dim(P_{-\gamma} / R_{B,-\gamma}).$$

The Q's here are sections represented by columns, not arbitrary quotient matrices denoted by the same symbol.

A full Hopf pairing, if supplied on a different appropriate recipient, has the additional adjointness equation

$$\langle xy, z\rangle = \langle x \otimes y, \Delta z\rangle.$$

For homogeneous basis elements it reads

$$\sum_c M_{ab}^c G_{cd} = \sum_{u,v} D_d^{uv} (-1)^{|e_b||e_u|} G_{au} G_{bv},$$

together with its unit and counit equations. This is not a further identity of the invariant Lie form on an ordinary nonabelian primitive algebra. Indeed, Hopf adjointness and $\langle 1, x\rangle = \epsilon(x)$ make $\langle [x, y], z\rangle = 0$ for primitives, whereas the normalized invariant form on $\mathfrak{sl}_2$ gives $B([h, e], f) = 2B(e, f) = 2$. Those two requirements cannot be imposed on the same form in that recipient.

Matrix transport

For a transition $T_\gamma : H_{R,\gamma} \to H_{R',\gamma}$, product and coproduct preservation are the equations

$$T_{\alpha+\beta} M_R^{\alpha,\beta} = M_{R'}^{\alpha,\beta} (T_\alpha \otimes T_\beta),$$
$$(T_\mu \otimes T_\nu) D_R^{\rho;\mu,\nu} = D_{R'}^{\rho;\mu,\nu} T_\rho, \qquad \rho = \mu + \nu.$$

Preserving unit and counit then intertwines the reduced coproduct on augmentation ideals and hence maps primitives to primitives. Radical preservation also needs the actual pairing comparison. For example, if an opposite-degree map L satisfies

$$B_{R'}(T_\gamma x, y') = B_R(x, L_{-\gamma} y'),$$

then a radical vector maps to a radical vector provided the displayed pairings test all target vectors. This is a statement about the specified maps; equal matrix ranks alone do not imply it.

The super bracket matrix is obtained from the two ordered products and their Koszul signs. Kernel bases, quotient sections and the maps $A_{\beta,\tilde{p}}$ to the target must satisfy these transport equations. The relation complex and its spectral sequence must use the same representatives and maps. PBW comparisons concern the actual filtered enveloping algebras and their finite weighted pieces, not a formal symmetric algebra on a root fragment.

A Hall–Drinfeld double additionally needs a compatible opposite half, a full pairing in its declared tensor, cross relations and completion. Serre duality of the geometric complexes and the weight-five denominator do not supply those constructions [79].

6.6 *Finite skew blocks and the modular section*

For a bundle V of rank $2n$ and a skew map $D : V \to V^\vee$, the complex $[V \xrightarrow{D} V^\vee]$ in degrees $[-1, 0]$ has Pfaffian line $\det V^\vee$ and section $\mathrm{pf}(D)$, with square $\det D$. A geometric comparison must include this shift and skew structure; a fibrewise Serre-dual pairing alone does not supply a family of skew operators or a section.

For one positive degree γ, the block

$$J_\gamma = \begin{pmatrix} 0 & 1 - x^\gamma \\ -(1 - x^\gamma) & 0 \end{pmatrix}$$

has Pfaffian $1 - x^\gamma$. With e_γ even and o_γ odd copies, the formal super ratio is

$$(1 - x^\gamma)^{e_\gamma - o_\gamma}.$$

At finite height, the product of these blocks gives a finite jet. Local finiteness in positive height makes the jets compatible. It constructs a formal reference product, not an independently defined compact geometric skew complex.

For geometric Pfaffians, require a parameter map to the relevant modular domain and line isomorphisms

$$\iota_{\mathrm{aut},R} : \mathscr{L}_{\mathrm{Pf},R} \xrightarrow{\sim} (\lambda_{\mathrm{aut}}^5 \otimes \varepsilon_5)|_R,$$

commuting with square maps, stage transitions and cusp frames. In those frames the desired finite expansion is

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma \in A_R} (1 - x^\gamma)^{f_\gamma} \pmod{\text{higher positive height}},$$

with the entire boundary factor included. If a protected integration formula is used to obtain this section, it must act on the actual coefficient objects and line-valued sections. Defining every input block to be the desired scalar factor does not construct that geometric integration map.

The normal wall block is $D_u = \left(\begin{smallmatrix} 0 & u \\ -u & 0 \end{smallmatrix}\right)$. An invariant tangential unit and a section-compatible reflection lift give the residual sign -1. The global atlas and root transport are the separate requirements of Appendix 5. The quotient $L_{\mathrm{II}}/2L_{\mathrm{II}}$ has eight elements, while the abstract twelve-sign set has 4096. Neither supplies a geometric wall-atlas count without its comparison.

The reference normalization is

$$[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64, \qquad [qrs]\Delta_5^2 = 4096, \qquad D_5 = 64^{-1}\Delta_5, \quad \Xi = D_5^2.$$

The homogeneous character on the type-II Weyl group is $\det_W$ and is trivial on chamber symmetries. The symplectic multiplier is kept with the weight-five automorphy factor. On the period cover, $\mathrm{div}(\Delta_5) = \widetilde{H}_1$, so its inverse square has pole order two; the generic coarse quotient has order one, as in Chapter 15. These target data do not construct the geometric section or its mirror interpretation.

Analytic convergence in a cusp region, continuation and boundary extension are further requirements beyond the formal product. Under the independent geometric line and section comparisons, the finite Pfaffian formula yields Δ_5 at that normalization. An identity for a protected trace additionally requires its own state and sewing map; the scalar normalization alone is insufficient.

6.7 The associative source coalgebra

Fix the coefficient category, its unit and convolution tensor. The source $C_{X,R}$ is a counital coaugmented coalgebra with differential and a conilpotent or specified positive filtration. Its cobar is the associative algebra $\Omega C_{X,R}$. Conversely an augmented associative source $A_{X,R}$ has a bar coalgebra $BA_{X,R}$. A twisting cochain and the bar–cobar comparison preserve these types. A coalgebra is not identified with an endomorphism algebra merely because their underlying complexes have the same graded ranks.

For an ordinary augmented dg algebra A over a field k, the underlying bar coalgebra is the tensor coalgebra on the suspension of its augmentation ideal. Its length filtration is

$$F_pBA = k \oplus \bigoplus_{1 \le j \le p} (s\overline{A})^{\otimes j},$$

and deconcatenation is

$$\Delta[a_1|\cdots|a_p] = 1 \otimes [a_1|\cdots|a_p] + [a_1|\cdots|a_p] \otimes 1 + \sum_{i=1}^{p-1} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_p].$$

Internal differentials and adjacent products give its bar differential with the suspended Koszul signs. If every factor has positive height, a bounded total-height piece has bounded word length. This does not identify word cuts with every geometric collision operation.

Given coalgebra, source-counit and algebra maps

$$b_R : C_{X,R} \longrightarrow BA_{X,R}, \qquad \varepsilon_R : \Omega BA_{X,R} \longrightarrow A_{X,R}, \qquad \vartheta_R : A_{X,R} \longrightarrow \mathcal{A}_{\Delta,E,R},$$

their composite is $\Theta_{\mathrm{Kos},R} = \vartheta_R \varepsilon_R \Omega b_R$. Its equivalence requires the hypotheses and inverses for these actual maps. A source-internal counit cannot replace the independent map ϑ_R to the target. The same notation in a Ran recipient uses its specified convolution and completed bar.

On the positive Ran coefficient recipient of Section 1.5, a proposed comparison has the form

$$C_{X,R} \longrightarrow C_{\Delta,R}, \qquad \Omega C_{X,R} \longrightarrow \mathcal{A}_{\Delta,E,R},$$

with the actual differentials, supports, completions and inverse maps. The ordinary bar of the reference enveloping algebra is a separate object. An envelope of a Beilinson–Drinfeld current Lie algebra and its main-diagonal Ran object also require their specified functors and units. Francis–Gaitsgory Lie–commutative duality [40] does not identify these associative constructions; the chiral operations of [7] retain their specified tensor and support as well.

The finite comparison must intertwine the geometric primitive map, Weyl transport, root-pair and Pfaffian comparisons and the invariant Lie form wherever these structures occur. On ordinary enveloping algebras it must preserve the PBW filtration. Additional full Hopf pairings, collision maps and factorization operations have their own comparison equations. Commuting finite maps give a comparison of pro-objects; passing to an effective geometric or analytic limit uses separate hypotheses.

6.8 Recognition from geometric representatives

Let $\mathfrak{F}$ be the free Lie superalgebra on the target simple and Cartan symbols, and let J_{BK} be the Borcherds relation ideal. Geometric representatives define a map

$$\pi : \mathfrak{F} \longrightarrow P_X^{\Pi,\mathrm{red}}.$$

Its factorization through the target requires every Cartan, Chevalley, real Serre, orthogonality and super sign relation. The desired identification follows when

$$\ker \pi = J_{\mathrm{BK}}, \qquad \operatorname{im} \pi = P_X^{\Pi,\mathrm{red}},$$

with the stated degree, parity and invariant-form compatibilities. This is Theorem 1.160.

A finite proof retains bounded word spaces and their actual intermediate degrees. Its matrices include the product M, coproduct D, unit and counit, bracket B, invariant form G, radical inclusion K, quotient section Q, and comparison maps $A_{\beta,\tilde{p}}$. The same symbols must not hide different domains: B here is a bracket matrix, whereas the charge normal-ordering $B(c, c')$ is a cocycle. The Hall boundary complex, its spectral sequence and the relation kernel calculations are performed on the chosen geometric representatives, not on an independently manufactured target vector space.

For the target positive roots, the height windows

$$W_\nu = \{\beta \in \mathcal{R}_+ : \operatorname{ht} \beta \le \nu\}, \qquad \bigcup_\nu W_\nu = \mathcal{R}_+,$$

give the Gram degree $\gamma_\beta = (b_1, b_1 + b_2 - b_3, b_2)$ for $\beta = b_1\partial_1 + b_2\partial_2 + b_3\partial_3$. Thus $\Gamma(W_\nu) = \{\gamma_\beta : \beta \in W_\nu\}$ is a target degree window. Its exhaustion is proved by $\beta \in W_{\operatorname{ht}\beta}$; source support exhaustion requires the separate geometric representatives.

The finite conditions (R0)–(R8) in Definition 1.161 retain source supports, representatives, relation identities, parity bijections, the invariant form and radical, kernel equality, generation, PBW and transition compatibility. Every relation word must be evaluated in a stage containing its intermediate outputs. The all-level assertion follows only when these bounded word spaces exhaust the presentation with compatible comparisons. A finite prefix or a list of signed multiplicities does not prove it.

At δ_{123}, recognition compares 29 even and 93 odd vectors together with their brackets. The rank-28 differential of Chapter 4 extends through height five in the stated presentation and fails at height six. A derived replacement changes its presentation or its recipient and must provide the corresponding map. It cannot silently keep the original relations while retaining the obstructed differential.

For a topological conclusion, specify the completion of both source and target. Positive-height completion uses genuine positive Lie quotients. A product over both positive and negative root degrees does not automatically have continuous brackets: a fixed output degree can receive infinitely many cancelling input pairs. A different completion requires its own finite-support or convergence argument.

An ordinary primitive isomorphism induces the filtered enveloping Hopf isomorphism under its stated hypotheses. It does not by itself identify a chiral field algebra, internal centre, boundary category or physical state. Those are further operation-preserving maps [79, 81, 82].

6.9 Dependencies, transitions and effectivity

The geometric dependencies form a partial order. The Hall operations need their correspondence and coefficient maps; recognition needs the actual source primitives and their presentation map. A geometric Pfaffian comparison needs a skew family, root data and a section. These are dependencies of this proposed compact construction, not a universal forced order among all the objects. A determinant root can be studied before any Hall product, and the bar of an augmented algebra is defined without an orientation line.

The finite morphisms (M1)–(M8) of Definition 1.194 express the required compatibility. They must preserve charge, coefficients, operations, root pairs, Pfaffian sections, coalgebras and recognition maps.

Root-line transitions are isomorphisms after the appropriate pullback. Kernel and pairing transitions use their actual source and target maps. Cohomological ranks alone do not define them.

The collection $\{\mathfrak{D}^{\mathrm{DI}}_{X,R}\}_R$ is a pro-object once the maps and coherence are supplied. Its Hom formula is the limit-colimit formula of Proposition 1.196. Compatible levelwise inverses are sufficient for a pro-isomorphism, but not necessary in a general pro-category. The zero-bonding nonzero-stage tower in that subsection gives a counterexample to the converse.

For each countable coefficient tower, the exact obstruction to commuting cohomology with derived inverse limit is $\varprojlim{}^1 H^{q-1}$ [117, §3.5]. Finite-dimensional field-valued towers are Mittag–Leffler; finite-rank integral towers need not be. Primitive-space, pairing-kernel and PBW towers require this check on their own coefficients. A nonuniform collection of finite cutoffs gives no all-level conclusion without a cofinality argument.

The resulting conditional comparison is the one in Theorem 1.201. It retains the geometric degeneration and coefficient, quotient orientation, coherent Weyl transport and wall-atlas conditions, together with independent Pfaffian, primitive and associative comparison data. Its remaining compact construction is to supply these objects and maps on the same geometric system. Their formal reference models, finite matrix identities and automorphic scalar are necessary comparison data, but do not replace that construction.

7 Orientation, wall and trace comparisons

A square root of a determinant, a Pfaffian section and a protected trace require different constructions. The square root must descend through the geometric quotient. The section must arise from a skew complex and agree with the automorphic section in a specified line. The trace requires a state and a cyclic operation on that state. This appendix gives the comparison criteria and their consequences. It does not assume that the complete compact datum exists.

7.1 The geometric recipients

Fix a smooth projective K3 surface S, an elliptic curve E and $X = S \times E$. Choose a holomorphic two-form σ_S on S and a nonzero one-form on E, hence a Calabi–Yau three-form on X. The stability heart and bounded geometric loci are those of Appendix 8. The symmetric Mukai form on $H^*(S, \mathbb{Z})$ is distinct from the skew Euler form of the threefold category.

At each stage R, specify the object, extension, mixed, wrapped and higher flag stacks, their coefficient categories and transition maps. A notation such as $\mathfrak{M}^{\mathrm{red}}_R$ includes an actual reduction construction. The ambient shifted symplectic structure on $\mathrm{Perf}(X)$ does not make every retained substack quasi-smooth, reduced or oriented. In particular, the scalar and dual scalar directions must be accounted for before using a two-term obstruction theory.

7.2 Specialization to zero-dimensional objects

The threefold Do sector consists of zero-dimensional objects on X; for sheaves its charge is their length. It is different from $\mathrm{Hilb}^n(S)$, which supplies a surface comparison. A degeneration from curve classes to this sector must first specify its geometric recipient.

There is an elementary obstruction in a fixed recipient. In a flat family of coherent sheaves on a fixed polarized projective threefold, the Hilbert polynomial is constant. A nonzero effective one-cycle has a positive linear Hilbert coefficient, while a zero-dimensional sheaf has a constant Hilbert polynomial. Such a flat family cannot specialize the former to the latter. Varying only stability does not change

this conserved charge. A proposed Do comparison must therefore either stay within the Do charge or construct a different geometric or categorical degeneration and its charge map.

Definition 7.1 (A Do specialization system). A datum Do_{HN} consists of the following compatible objects and maps.

(Do.1) For every stage R, a specified family of geometric or categorical recipients over a pointed curve, with a chosen Do subcategory in the special fibre. If the family is geometric and flatness is required, flatness is part of the datum. Its charge specialization is specified and respects the preceding Hilbert-polynomial constraint in a fixed sheaf recipient.

(Do.2) Object and full flag loci over that curve, with transition maps $\rho_{RR'}$ in their actual variance, composition isomorphisms, and the properness or support conditions needed by their coefficient functors.

(Do.3) Perfect reduced deformation complexes and oriented critical charts on the loci where critical coefficients are used, together with comparison maps from the ambient theory. Their amplitude and nondegeneracy are hypotheses on these loci.

(Do.4) The actual semiregularity morphisms, their fibres and their compatibility with flags and specialization. A surjectivity theorem for nonzero curve class is used only there; it does not supply surjectivity in the Do sector.

(Do.5) Reduced constructible coefficients Φ_R^{red} and specialization morphisms, including an equivalence wherever one is required by the comparison. Their extension and higher flag maps commute with specialization.

(Do.6) Root pairs $(o_R, o_R^{\otimes 2} \simeq \mathscr{D}_R^{\mathrm{red}})$, with their cross-determinant factors, square-compatible transition maps and higher flag coherences.

(Do.7) Compatible skew complexes, Pfaffian lines and sections, quotient-after-correspondence maps, compact-support operations and protected integration maps wherever the proposed comparison uses them.

(Do.8) The exactness and effectivity hypotheses needed for each asserted limit: derived inverse-limit comparisons for coefficient complexes, and effective descent of root pairs and sections if an actual infinite geometric recipient is claimed.

These clauses define a system in the specified recipient. They do not assert a flat degeneration of nonzero curve charge to Do sheaves on the unchanged threefold.

Theorem 7.2 *(Compatible Do specialization).* A datum Do_{HN} defines a pro-system of reduced coefficients, orientation pairs, Pfaffian sections and the supplied operations, together with its specialization comparison. On any recipient where (Do.8) gives effectivity, the corresponding limit carries these structures and maps.

Proof. The composition and higher flag comparisons make the stagewise objects a diagram in their coefficient and Picard categories. The specialization maps give a morphism of these diagrams. The same is true for every operation square by (Do.5)–(Do.7). This constructs the pro-system. Effectivity and passage through cohomology then use precisely the comparisons in (Do.8). For countable coefficient towers, the possible extra term is $\varprojlim{}^1 H^{q-1}$, as in Section 1.7. There is no general Mittag–Leffler assertion about a gerbe or a functor without first specifying the corresponding diagram. □

The ambient shifted symplectic theorem [104], orientation theory [63, 64], the oriented critical-sheaf construction [12] and cosection localization [67] enter at different steps. Cosection localization of a virtual cycle does not itself construct reduced vanishing-cycle coefficients or their specialization.

Remark 7.3 *(The surface scalar comparison).* The Hilbert–Chow map $\mathrm{Hilb}^n(S) \to \mathrm{Sym}^n(S)$ and the Hilbert-scheme elliptic classes of Chapter 8 give an actual K3 state and scalar calculation. The χ_y-genus is a specialization of that elliptic genus, not the whole genus. Reduced K3 curve invariants [88] and threefold Do specialization are further, distinct constructions. None of these scalar formulas provides the maps in $D\mathrm{o}_{\mathrm{HN}}$.

7.3 Roots on the translation quotient

Let B_R be a retained locus before translation quotient and $\mathscr{D}_R$ its actual reduced determinant line. Its root problem belongs to $[B_R/E]$. Denote the Kummer obstruction by $\delta(\mathscr{D}_R) \in H^2([B_R/E], \mu_2)$. If this class vanishes, the isomorphism classes of root pairs form an $H^1([B_R/E], \mu_2)$-torsor. A choice of pair is still needed.

Definition 7.4 *(Quotient orientation).* The datum $O\mathrm{I}_{\mathrm{quot}}$ consists of roots of the actual reduced determinant lines on the quotient object and flag stacks, with their square maps, cross-determinant comparisons and coherent transition isomorphisms. Equivalently, on a presenting groupoid it consists of root cocycles and chosen primitives for their defects, compatible with every overlap, flag and stage map. Coarse descent, when required, additionally asks that stabilizers act trivially on the chosen root fibres.

Definition 7.5 *(Strong reduced orientation).* A strong reduced orientation is such a quotient root system whose extension comparisons are associative on all retained flags and compatible with the reduced critical coefficient maps. No Spin^c or Pin^c structure is included unless its quadratic bundle and lift are separately constructed.

A free E-action does not justify setting an independently written class in $H^2(BE, \mu_2)$ to zero. It is its image in the actual quotient problem that matters. For example, $[E/E] = \mathrm{pt}$. Likewise, an edge computation at a finite stabilizer applies only after ordinary, mixed and differential terms in the equivariant obstruction have been accounted for.

LEMMA 7.6 *(The Klein-four detection equations).* For $E[2] = (\mathbb{Z}/2)^2$, write

$$\beta = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2 \in H^2(BE[2], \mathbb{F}_2), \qquad |x_i| = 1.$$

If r_1, r_2, r_3 are its coefficients on the two coordinate subgroups and the diagonal subgroup, then

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3.$$

Thus $\beta = 0$ exactly when all three restrictions vanish. For a chosen root character $\lambda = \lambda_1 x_1 + \lambda_2 x_2$, the three restriction values are instead $\lambda_1, \lambda_2, \lambda_1 + \lambda_2$. The two vanishing problems are independent.

Proof. The ring is $H^*(BE[2], \mathbb{F}_2) = \mathbb{F}_2[x_1, x_2]$. The restrictions substitute $(t, 0), (0, t), (t, t)$. These substitutions give both formulas. A zero degree-two class admits, for example, the nonzero character $(1, 0)$. □

LEMMA 7.7 *(The full two-primary stabilizer).* For $E[N] = (\mathbb{Z}/N)^2$, odd N gives no positive-degree mod-two group cohomology. If $2^a \parallel N$, the calculation reduces to $(\mathbb{Z}/2^a)^2$. For $a = 1$ the preceding lemma applies. For $a \geq 2$,

$$H^*(B(\mathbb{Z}/2^a)^2, \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2.$$

Consequently an edge obstruction and a root character have the forms

$$\beta = A_1 y_1 + A_{12} x_1 x_2 + A_2 y_2, \qquad \lambda = \lambda_1 x_1 + \lambda_2 x_2.$$

Their simultaneous vanishing requires all five coefficients to vanish. The coefficient A_{12} is invisible on every cyclic subgroup.

Proof. Averaging removes the odd-primary part. The cyclic mod-two cohomology ring for order 2^a, $a \geq 2$, is $\Lambda(x) \otimes \mathbb{F}_2[y]$; Künneth gives the ring above. On a cyclic subgroup of order at least four, both restricted degree-one classes are multiples of one exterior class, whose square vanishes. On an order-two subgroup their restrictions are zero. Hence $x_1 x_2$ is invisible in either case. □

Remark 7.8 *(Detection and gluing).* These are equations for the indicated classifying-space edge. They do not provide its geometric reduction, a primitive for a cocycle, or compatibility of chosen primitives on intersecting strata. The exterior-generator calculation requires $4 \mid N$; for $N \equiv 2 \pmod 4$, the Klein-four calculation applies. The degree-one condition is needed for the requested trivial stabilizer action, not for every root on a quotient stack.

Theorem 7.9 *(Quotient orientation criterion).* The coherent roots and flag comparisons of $O_{1\mathrm{quot}}$ descend to a strong reduced orientation on the quotient stack. If their stabilizer characters vanish and the stated coarse descent theorem applies to the retained locus, they descend further to its coarse recipient.

Proof. Descent of line bundles with square maps is descent in the Picard groupoid. The supplied overlap cocycles give its descent datum; the flag coherences descend the cross-determinant comparisons. This proves stack descent. The extra trivial stabilizer action and the coarse descent hypotheses give the second assertion. There is no multiplication formula omitting the cross determinant. □

7.4 *Weyl transport*

Use the type-II Coxeter subgroup $W = \langle s_1, s_2, s_3 \mid s_i^2 = 1 \rangle$. Its full lattice isometry group also contains chamber symmetries; these two groups have different character comparisons. A geometric W-action requires maps of the actual root pairs on the quotient action groupoid, with their square maps and coefficient and flag compatibilities.

Theorem 7.10 *(Equivariant root transport).* Suppose the chosen lifts of Definition 1.81 have a specified trivialization of their action-groupoid defect and zero remaining quotient-torsor defects. They define an honest W-equivariant root system. If its residual scalar generator values are constant and each section-compatible simple skew wall has value -1, its character is

$$\epsilon_o(w) = \det_W(w).$$

This is the restriction of the homogeneous orthogonal character $\widetilde{\nu}_{\Delta_5}$ to W.

Proof. Correcting the lifts by the chosen primitive makes their compositions satisfy the action law. The quotient-torsor equations keep the root and coefficient comparisons in that same action. After its residual action has been identified with a scalar character, its value on a reduced word is the product of the values on the generators. Cancellation of s_i^2 preserves that product. Every value -1 therefore gives $(-1)^{\ell(w)}$, which is the determinant character of the reflection representation. The character of Δ_5 has these generator values by Appendix 5. □

Remark 7.11 *(Which information a character retains).* The abelianization $W_{\mathrm{ab}} = (\mathbb{Z}/2)^3$ classifies the constant sign characters. It does not classify root obstructions on the geometric action groupoid. The products of two distinct simple reflections are parabolic of infinite order; there is no order-three braid relation. Extending the root action to $W \rtimes S_3$ requires chamber-symmetry lifts. The two extensions $\epsilon_o(w)$ and $\epsilon_o(w)\,\mathrm{sgn}(\sigma)$ have the same restriction to W, while the automorphic homogeneous character chooses the trivial S_3-factor. Its symplectic multiplier must still be kept with the weight-five automorphy factor.

7.5 *Skew wall charts and their global comparison*

Definition 7.12 *(A type-II wall atlas).* A datum $O2_{\mathrm{atlas}}$ consists of actual wall objects with their full charges, wall semistability and reduced deformation maps. At each simple wall it gives a chart with normal coordinate u, a skew normal block

$$D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad [V \xrightarrow{D_u} V^\vee] \text{ in degrees } [-1, 0],$$

an invertible tangential Pfaffian v, and a specified reflection lift carrying the total section vu to the reflected section. The tangential unit is invariant in this comparison. The charts have root-pair and section isomorphisms on overlaps, a trivialized sign defect on triple overlaps, and compatible flag, Weyl, boundary and stage maps. A proposed twelve-sign geometric parametrization is additional data: it is a bijection from its actual geometric index set to $\{\pm 1\}^{12}$, with the asserted multiplicities and coverage.

THEOREM 7.13 *(Transport of a simple skew wall).* A wall atlas with the preceding Weyl transport has a simple Pfaffian zero and residual reflection sign -1 at every transported type-II wall.

Proof. In the normal frame, $\mathrm{pf}(D_u) = u$. The rank-two lift $R = \mathrm{diag}(-1, 1)$ satisfies $R^t D_u R = D_{-u}$ and has determinant -1 on the Pfaffian frame. Multiplication by the invariant tangential unit preserves the simple zero and the sign. The supplied chart isomorphisms preserve the divisor order, and character conjugacy gives the same sign at $ws_i w^{-1}$. A scalar complex $[\mathbb{C} \xrightarrow{u} \mathbb{C}]$ has determinant u and is not this skew normal block. □

THEOREM 7.14 *(The two scalar counts).* The abstract sign set $\Omega_{12} = \{\pm 1\}^{12}$ has 4096 elements, and the normalized theta product satisfies

$$[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64, \qquad [qrs]\Delta_5^2 = 4096.$$

A geometric wall index set has cardinality 4096 if it is supplied with the bijection in $O2_{\mathrm{atlas}}$. The equality of the two integers does not construct that bijection.

Proof. There are two independent choices in each of twelve coordinates. The theta coefficient is Lemma 2.3; squaring the lowest monomial gives 64^2. The geometric assertion follows only from the stated bijection. □

Remark 7.15 *(A count, a character and a scalar).* The root-lattice quotient $L_{\mathrm{II}}/2L_{\mathrm{II}}$ has eight elements. It is neither Ω_{12} nor a constructed wall atlas. A sign character, a geometric multiplicity and the normalization $D_5 = \Delta_5/64$ retain different data, even when their resulting scalar formulas agree.

Theorem 7.16 *(Gluing the transported wall sections).* The overlap, stabilizer and action coherences of $O2_{\mathrm{atlas}}$ glue the transported wall sections on the declared geometric recipient. A twelve-sign parametrization, when supplied, retains its cardinality under this transport.

Proof. The overlap maps glue the lines and sections because their triple composites are the identity after the chosen correction. Two Weyl paths with the same endpoint differ by the specified stabilizer comparison; its coherence identifies the transported sections. This is the required path independence. An orbit-stabilizer count without those geometric identifications would not prove it. □

7.6 The finite Pfaffian and the assembled criterion

Definition 7.17 *(Finite geometric Pfaffian data).* The datum P_{fin} is that of Definition 1.146: actual reduced skew perfect complexes, root lines, Pfaffian sections and their stage comparisons, together with an independently defined map to the modular domain and line isomorphisms to $\lambda_{\mathrm{aut}}^{5} \otimes \varepsilon_{5}$. The finite expansions include the full boundary factors and the leading coefficient 64. Every inverse-limit or analytic conclusion uses its stated exactness, convergence and continuation hypotheses. A manufactured product of hyperbolic blocks gives the formal reference section but not this geometric comparison.

Definition 7.18 *(The complete comparison datum).* Write

$$\mathfrak{R}_{X}^{\mathrm{DI}} = (D\mathrm{o}_{\mathrm{HN}}, O1_{\mathrm{quot}}, \mathcal{T}_{W}, O2_{\mathrm{atlas}}, P_{\mathrm{fin}}, Z_{\mathrm{HB}}),$$

where $\mathcal{T}_{W}$ is the coherent root transport just specified and Z_{HB} is the state and trace comparison defined below. All components live on compatible geometric systems. The first five concern the Pfaffian and its orientation; the last is additional trace data.

Theorem 7.19 *(Relative Pfaffian criterion).* Suppose the first five components of $\mathfrak{R}_{X}^{\mathrm{DI}}$ satisfy their comparison conditions. Then the geometric Pfaffian has the root transport and normalized automorphic section of Theorem 2.1. Its character on W is $\det_{W}$. Primitive recognition remains conditional on (S1)–(S10), and the trace requires Z_{HB}.

Proof. The Do system supplies its specialization maps, quotient orientation descends the roots, and Weyl transport identifies their wall actions. The wall atlas glues the local skew sections. The independent Pfaffian-line comparison and compatible normalized finite expansions in P_{fin} then identify the section with Δ_{5} on their declared formal or analytic domain. The character assertion is Theorem 7.10. None of these steps constructs primitive representatives or a cyclic state. □

7.7 Algebras, centres and cyclic states

The coalgebra C_{X} and its associative cobar $A_{X} = \Omega C_{X}$ have different types. A boundary algebra A_{b} can be compared with A_{X}; its bar can be compared with C_{X}. A filtered quasi-isomorphism from an algebra to that coalgebra is not an algebra comparison.

For an ordinary dg algebra, the derived centre is

$$Z^{\mathrm{der}}(A_{b}) = R\,\mathrm{Hom}_{A_{b} \otimes A_{b}^{\mathrm{op}}}(A_{b}, A_{b}).$$

A chiral or factorization centre needs the corresponding enriched bimodule category and internal operations. The ordinary formula does not identify it with a physical bulk. The interface comparison of [81] uses an actual coefficient-linear strong monoidal equivalence with the required enrichment. Those

hypotheses, and both supported actions when a junction is used, remain conditions on the compact source.

Even a nondegenerate degree -3 pairing

$$Z \otimes Z \longrightarrow \mathbb{C}[-3]$$

is a bilinear map, not a prepared state, a sewing operation or a partition trace. A cyclic refinement and a state on which the charged operators act must be supplied. The finite modes of Chapter 8 illustrate the distinction: the elliptic coefficient bundles and their geometric joining maps exist before a compact boundary intertwiner is known.

Definition 7.20 (A Hall–Borcherds trace comparison). A trace comparison is an operation-preserving map from an independently defined compact Hall state and its cyclic operations to a specified boundary state, preserving charges, parity, support, completion and normalization. If a centre acts on the boundary state, its action and the compatibility of this map with that action are part of the comparison. The resulting protected partition function is the trace on the state, with its stated vacuum and regularization conventions.

Remark 7.21 (A trace and an acting algebra). A trace comparison does not construct a gravity-line operator algebra. Conversely, an acting algebra does not select a cyclic state or a trace functional. Centre cochains, cyclic chains and endomorphisms of the state have separate roles; no equality of their scalar characters identifies them.

7.8 A conditional trace calculation

Definition 7.22 (A compatible cyclic state datum). The datum Z_{HB} specifies:

(Z1) The compact coalgebra C_X, an associative source $A_X = \Omega C_X$, and a boundary algebra A_b, with actual filtered comparisons in the declared ordinary or factorization recipient. Any bar comparison has coalgebra type.

(Z2) The coefficient tensor, unit, support and completion, the stage transition maps, and the convergence or exactness needed for the asserted trace and limit.

(Z3) The enriched centre comparison and its action on an independently defined compact state $\mathcal{H}_X$, whenever a centre interpretation is required. Its hypotheses and maps are supplied on that same recipient.

(Z4) A cyclic chain functional or equivalent specified cyclic structure, the chosen state or boundary condition, and its sewing maps. A degree -3 Calabi–Yau pairing, if used, has a comparison to these maps; it does not replace them.

(Z5) A charge- and parity-preserving comparison of the completed state, or of a convergent associated graded, with

$$\widehat{\mathrm{Sym}}_{\mathrm{sup}}\left(\bigoplus_{\gamma>0} V_\gamma^{\oplus 2}\right), \qquad \mathrm{sdim}\, V_\gamma = f_\gamma,$$

where each V_γ is a finite-dimensional super vector space and f_γ is the exponent in the product for D_5 in Chapter 3. It intertwines the cyclic trace and fixes the normal-ordering scalar and monomial to $64^{-2}x^{-2\rho}$, with $x^\rho = q^{1/2}r^{1/2}s^{1/2}$. The comparison retains the axis factors and every relevant stage map.

Clause (Z5) is a substantive state comparison. It is not implied by matching a cyclic pairing or by prescribing a desired scalar partition function.

Theorem 7.23 *(Trace from the state comparison).* Given Z_{HB}, its normalized protected trace is

$$\mathrm{Tr}_{\mathrm{prot},\mathcal{H}_X} = 64^{-2}x^{-2\rho}\prod_{\gamma>0}(1-x^{\gamma})^{-2f_{\gamma}} = \Delta_5^{-2}$$

in the stated cusp completion, and on any analytic domain to which the supplied comparison continues it.

Proof. An even oscillator of degree γ contributes $(1-x^{\gamma})^{-1}$ to the supertrace of its symmetric algebra. An odd oscillator contributes $1-x^{\gamma}$. The doubled spaces in (Z5) therefore contribute $(1-x^{\gamma})^{-2f_{\gamma}}$. Positive-height local finiteness makes every fixed-height coefficient a finite calculation. The state and cyclic comparisons identify this trace with the geometric one; their convergence hypotheses justify any associated-graded or analytic passage. The prescribed vacuum monomial and scalar then give the inverse square of the complete Borcherds product. □

Theorem 7.24 *(Joint Pfaffian and trace consequence).* If all components of $\mathfrak{R}_X^{\mathrm{DI}}$ exist with the stated compatibilities, the geometric Pfaffian section is Δ_5, its W-character is $\det_W$, and the protected state trace is Δ_5^{-2}. In the separate convention of Convention 4.1, the corresponding OP scalar representative is $-4096\Delta_5^{-2}$. These conclusions do not assert primitive recognition or construct a gravity-line operator algebra.

Proof. Apply Theorems 7.19 and 7.23 to their compatible data. The OP factor is the declared change of scalar convention. Recognition needs its presentation map, and an acting gravity algebra needs its operator and module construction. □

Remark 7.25 *(The additional input in a trace theorem).* A centre theorem identifies a specified enriched endomorphism object. An orientation theorem provides a root with coherent transports. Neither theorem supplies $\mathcal{H}_X$, the cyclic functional, the sewing maps or the state comparison in (Z5). These remain compact geometric constructions even when the scalar Borcherds product is already known.

7.9 Compatibility and the remaining construction

The compatibility problem is to construct the preceding objects on one system of compact moduli and correspondence stacks. Orientation and critical-sheaf theorems apply on their stated recipients. The fourfold orientation calculation of [17] does not replace the threefold reduced orientation problem. The reduced stable-pair comparison of [98] and the curve-counting formula of [101] retain their numerical scope and normalization; they do not construct a chiral state intertwiner.

The first missing coefficient map is the global reduced oriented pull–push on compact extension flags, with the common translation torsor and higher coherences. A coproduct needs its own support and coefficient comparison. The ordinary-Koszul cooperation obstruction on the specified polynomial shuffle carrier does not disappear by naming a bialgebra; a meromorphic or localized cooperation changes the tensor or recipient and requires a new comparison. This keeps the positive formal and ordinary critical Hall constructions separate from the unresolved reduced compact structure.

After that construction, primitive recognition still asks for actual representatives, relations, both parity dimensions, invariant forms, radicals, generation and PBW compatibility. The modular comparison still asks for the geometric line and section. The trace comparison still asks for the cyclic state and its sewing

map. These are the maps by which the established local, algebraic and automorphic constructions could become one compact realization.

The standard current and conformal examples have their own domains: affine Kac–Moody algebras at noncritical level, principal W-algebras, Virasoro algebras, rational $c = 24$ models including the Monster theory, and cosets with the required finiteness hypotheses. A theorem on those objects does not place the compact hybrid Hall recipient in its domain. The Saito–Kurokawa arithmetic of Proposition 5.1 and the selected input pairs of Theorem 10.4 remain positive scalar and automorphic comparisons at their separately stated scopes.

8 Stability, finite charge spaces, and elliptic support

A Hall operation uses families of exact sequences, together with a coefficient theory on their moduli. The lattice calculations determine neither of these. A product stability condition gives the filtration problem on $S \times E$. Moduli and descent conditions then determine which geometric operations can be defined.

8.1 A product stability condition

Definition 8.1 *(Product heart and central charge).* Fix a numerical Bridgeland stability condition $\sigma_S = (\mathcal{A}_S, Z_S)$ on $D^b\operatorname{Coh}(S)$ with $Z_S(\Lambda_S^{\mathrm{alg}}) \subset \mathbb{Q} + i\mathbb{Q}$. Choose a degree-one line bundle L_E on E. The Abramovich–Polishchuk heart [1] is

$$\mathcal{A}_E^{\mathrm{AP}} = \{M \in D^b\operatorname{Coh}(X) : Rp_{S*}(M \otimes p_E^* L_E^n) \in \mathcal{A}_S \text{ for } n \gg 0\}.$$

Write

$$Z_S(Rp_{S*}(M \otimes p_E^* L_E^n)) = a(M)n + b(M) + i(c(M)n + d(M)).$$

For rational $t > 0$, the weak charge and slope are

$$Z_t = at - d + ict, \qquad \nu_t(M) = \begin{cases} (-at + d)/(ct) & c(M) \neq 0, \\ +\infty & c(M) = 0. \end{cases}$$

For nonzero objects, the minimum and maximum slopes of the Harder–Narasimhan factors define the torsion pair

$$\mathcal{T}_t = \{M : \nu_{t,\min}(M) > 0\}, \qquad \mathcal{F}_t = \{M : \nu_{t,\max}(M) \leq 0\}.$$

Both subcategories include zero. Set

$$\mathcal{A}_X^{\mathrm{AP/Liu}} = \mathcal{A}_E^t = \langle \mathcal{T}_t, \mathcal{F}_t[1] \rangle, \qquad Z_E^{s,t} = cs + b + i(-at + d).$$

Proposition 8.2 *(The product stability condition).* For rational $s, t > 0$, the pair

$$\sigma_{s,t} = (\mathcal{A}_E^t, Z_E^{s,t})$$

is a stability condition whose central charge factors through Γ_X^{alg}. Its support property holds on

$$\Lambda_S^{\mathrm{alg}} \oplus \left(\Lambda_S^{\mathrm{alg}}/\ker Z_S\right), \qquad (Q, P) \longmapsto (P, \overline{Q}).$$

These pairs extend continuously to positive real parameters. No support property on the larger full paired lattice is asserted.

Proof. The rational central charge has discrete image because it factors through the finite-rank lattice Λ_S^{alg}. The assertion is Liu's product theorem, including its torsion-pair construction and support property [75, Theorems 4.7 and 5.9]. □

In the coordinates of Chapter 2, these coefficients have a direct interpretation. Grothendieck–Riemann–Roch gives

$$v(Rp_{S*}(M \otimes p_E^* L_E^n)) = Q(M) + nP(M).$$

Indeed, $\mathrm{ch}(L_E^n) = 1 + n\omega_E$ and $\mathrm{td}(E) = 1$, so integration over E selects $Q + nP$. Consequently

$$a(M) + ic(M) = Z_S(P(M)), \qquad b(M) + id(M) = Z_S(Q(M)).$$

This identifies the polynomial coefficients with the actual numerical charges. The product stability theorem does not assert finite type or properness for every moduli locus considered below.

8.2 Finite windows and Harder–Narasimhan filtrations

At finite height, termination of subobject chains can be imposed by a nonnegative integer length that strictly increases under strict inclusions. This gives an elementary noetherianity criterion for the chosen subcategory, without a global noetherianity assertion for $\mathcal{A}_X^{\mathrm{AP/Liu}}$.

A finite height bound is generally incompatible with closure under direct sums. Indeed, for a nonzero object A, the split inclusions

$$A \subset A^{\oplus 2} \subset A^{\oplus 3} \subset \cdots$$

are strict. An isomorphism-invariant length strictly increasing on these inclusions takes infinitely many values. Such an additive subcategory cannot have finitely many isomorphism classes. A finite window therefore retains exact sequences from the ambient heart, without being an exact category in its own right.

Definition 8.3 *(Retained noetherian window).* Fix the heart $\mathcal{A}_X^{\mathrm{AP/Liu}}$. A retained finite window at height R is a full subcategory

$$\mathcal{A}_{X,R}^{\mathrm{ret}} \subset \mathcal{A}_X^{\mathrm{AP/Liu}}$$

containing zero and closed under isomorphisms, with finitely many isomorphism classes. Require every ambient subobject and quotient of an object in the window to lie in the window. A retained exact sequence is a short exact sequence in the ambient heart whose three terms lie in the window. No closure under direct sums or arbitrary extensions is assumed.

The window carries an isomorphism-invariant length function

$$\ell_R : \mathrm{Obj}(\mathcal{A}_{X,R}^{\mathrm{ret}}) \longrightarrow \mathbb{Z}_{\geq 0}$$

with $\ell_R(A) = 0$ exactly for zero. Require $\ell_R(A) < \ell_R(B)$ for each non-isomorphism ambient monomorphism $A \hookrightarrow B$ between retained objects.

Proposition 8.4 *(Retained finite windows are noetherian).* [proved] Every retained finite window $\mathcal{A}_{X,R}^{\mathrm{ret}}$ in the sense of Definition 8.3 satisfies the ascending chain condition on ambient subobjects of each retained object.

Proof. Let

$$A_0 \subset A_1 \subset A_2 \subset \cdots$$

be a chain of retained subobjects of an object A in $\mathcal{A}^{\mathrm{ret}}_{X,R}$. After deleting repetitions, each inclusion is strict. The length function then gives a strictly increasing sequence

$$\ell_R(A_0) < \ell_R(A_1) < \ell_R(A_2) < \cdots .$$

All A_i lie in the finite retained window and all are subobjects of A, so $\ell_R(A_i) \le \ell_R(A)$. A strictly increasing sequence of non-negative integers bounded above by $\ell_R(A)$ is finite. Thus the original ascending chain stabilizes. □

A chain of strict subobjects of A has at most $\ell_R(A)$ inclusions. Each inclusion increases an integer length by at least one. Stability supplies a second ordering, by phase, which selects the successive quotients of a Harder–Narasimhan filtration.

Definition 8.5 *(Retained finite Harder–Narasimhan filtration datum).* Let $\mathcal{A}^{\mathrm{ret}}_{X,R}$ be a retained finite window. For a nonzero object M of the heart, its stability phase is

$$\phi_\sigma(M) = \pi^{-1} \arg Z^{s,t}_E(M) \in (0, 1].$$

An object M is semistable if every nonzero ambient subobject $B \subset M$ has $\phi_\sigma(B) \le \phi_\sigma(M)$.

Let $\mathcal{S}_R$ be the nonzero semistable objects in the window. Their phases form a finite ordered set. Choose an order-preserving identification of that set with a finite subset $\Phi_R \subset \mathbb{Q}$, and denote the resulting phase label by $\phi_R : \mathcal{S}_R \to \Phi_R$. These rational labels encode the order. They need not equal the stability phases.

A retained finite Harder–Narasimhan filtration datum assigns to every $A \in \mathcal{A}^{\mathrm{ret}}_{X,R}$ a filtration

$$0 = A_0 \subset A_1 \subset \cdots \subset A_m = A$$

in the ambient heart. Every A_i and every quotient $E_i = A_i/A_{i-1}$ must belong to the window, with $E_i \in \mathcal{S}_R$ and

$$\phi_R(E_1) > \phi_R(E_2) > \cdots > \phi_R(E_m).$$

The zero object has the empty filtration.

PROPOSITION 8.6 *(Construction of finite HN filtrations).* [proved] Every retained finite window in the heart of $\sigma_{s,t}$ admits a datum of Definition 8.5. For each A, the filtration has at most $\ell_R(A)$ nonzero quotients.

Proof. For any nonzero quotient C of A, choose a nonzero subobject $B \subset C$ of maximal phase. Among subobjects of that phase, choose one with maximal $\ell_R(B)$. The choices exist because all subobjects of C belong to the finite window, and phase and length are constant on isomorphism classes. Every subobject of B is also a subobject of C, so B is semistable.

We claim that every nonzero subobject $D \subset C/B$ has phase strictly smaller than $\phi_\sigma(B)$. Let $B' \subset C$ be its inverse image. The ambient exact sequence

$$0 \longrightarrow B \longrightarrow B' \longrightarrow D \longrightarrow 0$$

has all terms in the window. Additivity of the central charge gives $Z^{s,t}_E(B') = Z^{s,t}_E(B) + Z^{s,t}_E(D)$.

For nonzero numbers with distinct arguments in $(0, \pi]$, the sum has argument strictly between them. Equal arguments remain equal. Thus $\phi_\sigma(D) > \phi_\sigma(B)$ contradicts maximality of the phase of B. Equality would give $\phi_\sigma(B') = \phi_\sigma(B)$, whereas the strict inclusion $B \subset B'$ gives $\ell_R(B') > \ell_R(B)$.

This contradicts the second maximality condition and proves the claim.

Start with $C = A$, retain the chosen B, and repeat the construction on A/B. Taking inverse images in A produces a chain of ambient subobjects. Subobject and quotient closure keeps all its terms and semistable factors inside the window. The claim makes their phases strictly decrease. Each nonzero factor strictly increases the length of the inverse-image subobject, so after at most $\ell_R(A)$ steps the quotient is zero. The resulting chain is the required filtration. □

Example 8.7 *(Bounded semisimple windows).* Let $S_1, \dots, S_k$ be pairwise nonisomorphic simple objects of the ambient heart, and let N be a nonnegative integer. The objects

$$\bigoplus_{i=1}^{k} S_i^{\oplus n_i}, \qquad n_i \geq 0, \qquad \sum_i n_i \leq N,$$

and their isomorphic copies form a retained finite window. Subobjects and quotients of a semisimple object are semisimple, and their simple multiplicities are bounded by those of the object.

The composition length is $\sum_i n_i$, and strictly increases under strict monomorphisms. Each S_i is stable because it has no proper nonzero subobject. Grouping the summands of equal phase and ordering the groups by decreasing phase gives the HN filtration explicitly. For $k = 1$, this is the one-factor filtration of every nonzero object.

For $N > 0$, the window need not contain the direct sum of two objects of length N. It therefore illustrates the distinction between ambient exact sequences and an additive exact category. In the AP/Liu heart this construction requires a chosen finite collection of simple objects. It does not supply compact Hall moduli for more general branes.

The construction uses only the stability function, finite object classes, subobject and quotient closure, and the strict length bound. If the length is additive on retained short exact sequences, it also gives $\ell_R(A) = \sum_i \ell_R(E_i)$. Finite-type moduli and extension correspondences remain geometric constructions beyond this filtration theorem.

Definition 8.8 *(Retained bounded HN type datum).* Let $\mathcal{A}^{\mathrm{ret}}_{X,R}$ be a retained finite window with a retained finite Harder–Narasimhan filtration datum. Its *retained HN type* is

$$\mathrm{HNType}_R(A) = \big(([E_1], \phi_R(E_1)), \dots, ([E_m], \phi_R(E_m))\big)$$

where $E_i = A_i/A_{i-1}$ are the retained semistable quotients. For zero, use the empty word.

A retained bounded HN type datum consists of a finite set $\mathfrak{T}_R$ containing these words and a decoration of semistable factor classes:

$$[E] \longmapsto \big(\widehat{c}_E, [a_E, b_E], P_E, N_E, \mathcal{F}_E\big).$$

Here $\widehat{c}_E$ is the retained numerical class and $[a_E, b_E]$ its cohomological-amplitude interval. The symbols $P_E, N_E, \mathcal{F}_E$ specify a Hilbert polynomial, a regularity bound, and a chosen family closed under the required operations. This datum is a finite bound on active-window HN types. It is not a finite-type semistable substack, not a quasi-smooth derived enhancement, and not a compact Hall stage.

Proposition 8.9 *(HN type is bounded in a retained active window).* [proved] Let $\mathcal{A}^{\mathrm{ret}}_{X,R}$ be a retained finite window with a retained finite Harder–Narasimhan filtration datum. Then the set of retained HN types occurring in $\mathcal{A}^{\mathrm{ret}}_{X,R}$ is finite. More precisely, if

$$N_R = \max\{\ell_R(A) \mid A \in \mathcal{A}^{\mathrm{ret}}_{X,R}\},$$

then every retained HN type has length at most N_R.

Proof. The retained window has finitely many isomorphism classes, hence the set of retained semistable HN factors is finite. If

$$0 = A_0 \subset A_1 \subset \cdots \subset A_m = A$$

is the retained HN filtration of A, then each inclusion $A_{i-1} \subset A_i$ is strict unless $m = 0$. The retained length rank gives

$$0 = \ell_R(A_0) < \ell_R(A_1) < \cdots < \ell_R(A_m) \leq N_R,$$

so $m \leq N_R$. Thus every retained HN type is a word of length at most N_R in a finite alphabet of semistable factor classes. The set of such words is finite, and the strict phase condition selects a finite subset. A finite Liu–Hilbert decoration of the factor classes therefore gives a finite decorated HN type set. □

The bound $m \leq N_R$ controls the number of factors in each HN word. The finite factor alphabet and its Liu–Hilbert decorations then give finitely many decorated words. Representing their semistable factors by algebraic stacks requires additional geometric data. Finiteness of the word set alone does not supply those families.

8.3 Moduli hypotheses and derived amplitudes

The finite-window argument concerns a finite set of isomorphism classes. A finite-type moduli stack can instead contain infinitely many such classes. To pass from that argument to geometry, one must specify the relevant families and prove their boundedness.

Definition 8.10 *(Bounded moduli data).* For a finite set C_R of numerical types, choose algebraic substacks $\mathfrak{M}^{ss}_{R,c}$ of the stack of objects of $\mathcal{A}^t_E$. Require each to parametrize exactly the selected $\sigma_{s,t}$-semistable families of type c. Bounded moduli data include finite-type Quot or Postnikov parameter spaces covering these stacks and finite sets of Hilbert polynomials. They also include amplitude bounds $[a_c, b_c]$ and regularity bounds N_c for cohomology sheaves, relative to a fixed ample bundle on X. When a closed locus in a chosen ambient stack is needed, closedness is a further hypothesis on the inclusion.

PROPOSITION 8.11 *(Consequences of the bounded moduli hypotheses).* Suppose the covering maps in the bounded moduli data are smooth and surjective. Then every $\mathfrak{M}^{ss}_{R,c}$ is of finite type. Its objects satisfy the specified class, polynomial, amplitude and regularity bounds. If the chosen inclusions into ambient stacks are closed immersions, their images are specialization closed.

Proof. Finite type is smooth-local on an algebraic stack. The bounds hold on the covering parameter spaces and descend to their represented families. Specialization-closedness is a condition on the chosen locus, independent of the finiteness of its set of numerical types. □

Derived moduli introduce a different issue. At an object F of the heart, the tangent complex of the derived stack of perfect complexes is

$$\mathbb{T}_{\mathrm{Perf}(X),F} = R\,\mathrm{Hom}_X(F, F)[1].$$

Negative self-extensions vanish in a heart. Serre duality on the threefold then bounds this tangent complex in degrees $[-1, 2]$. Its cotangent complex has amplitude $[-2, 1]$ in general. In particular, scalar automorphisms give a nonzero degree--1 tangent space and a nonzero degree-one cotangent space.

Definition 8.12 *(Derived enhancement and reduced chart).* A derived enhancement of $\mathfrak{M}^{ss}_{R,c}$ is a derived stack $\mathfrak{M}^{\mathrm{der}}_{R,c}$ with that classical truncation and a comparison with the relevant moduli of perfect complexes.

A quasi-smooth reduced chart means a separately constructed derived Deligne–Mumford stack $\mathfrak{N}_{R,c}$ with perfect cotangent complex in degrees $[-1, 0]$. Its use as a Calabi–Yau chart requires a specified (-1)-shifted symplectic form and a compatible comparison with the reduced deformation complexes of the objects.

Proposition 8.13 *(The scalar obstruction to a two-term chart).* The unrigidified stack of nonzero objects cannot have cotangent amplitude $[-1, 0]$ at an object with scalar automorphisms. A chosen reduced model is quasi-smooth when its cotangent complex is perfect with that amplitude. It is (-1)-shifted symplectic only if its closed two-form induces an equivalence $\mathbb{T} \simeq \mathbb{L}[-1]$.

Proof. The infinitesimal scalar automorphism gives a copy of $\mathbb{C}$ in $H^{-1}(\mathbb{T})$, hence in the dual of $H^1(\mathbb{L})$. This excludes amplitude $[-1, 0]$ on the unrigidified stack. The assertions about the reduced chart are the quasi-smoothness and nondegeneracy criteria for its specified structures. □

The Calabi–Yau trace gives the ambient shifted symplectic form [104]. Its restriction along an arbitrary substack need not be nondegenerate. Removing scalar automorphisms changes one end of the tangent complex. A two-term self-dual model must also account for the Serre-dual trace direction. Neither operation is supplied by boundedness of the classical moduli stack.

8.4 Scalar and translation quotients

Definition 8.14 *(Scalar rigidification).* Suppose the scalar subgroup $\mathbb{G}_m$ is central in the inertia of a moduli stack $\mathfrak{M}$. A scalar rigidification is a $\mathbb{G}_m$-gerbe

$$\pi : \mathfrak{M} \longrightarrow \mathfrak{M}^{\mathrm{rig}}$$

that replaces every automorphism group by its quotient by the scalar subgroup. Require the residual automorphism groups to be finite on the locus under consideration.

Proposition 8.15 *(Residual inertia).* At an object F, scalar rigidification has residual stabilizer $\mathrm{Aut}(F)/\mathbb{G}_m$. It is finite precisely when this quotient is finite. In particular, removing scalars from $\bigoplus_i L_i \otimes W_i$ generally leaves a positive-dimensional group.

Proof. The first assertion is the inertia property of rigidification. For the displayed polystable sheaf, distinct stable summands have no Hom groups between them. Thus $\mathrm{Aut}(F) = \prod_i \mathbf{GL}(W_i)$, whose quotient by the diagonal scalars has dimension $\sum_i (\dim W_i)^2 - 1$ when $F \neq 0$. □

Rigidification removes an existing central subgroup of automorphisms. It is not an external quotient $[\mathfrak{M}/\mathbb{G}_m]$. For example, $B\mathbb{G}_m \to \operatorname{Spec} \mathbb{C}$ is the scalar rigidification of $B\mathbb{G}_m$. The quotient for a trivial external $\mathbb{G}_m$ action instead has another $B\mathbb{G}_m$ factor.

Definition 8.16 *(Normalized universal complex).* Let $\mathcal{U} \in \mathrm{Perf}(X \times \mathfrak{M})$ be the tautological complex, on which scalar inertia has weight one. A normalization is a line bundle $\mathcal{L}$ on $\mathfrak{M}$ of scalar weight -1. The normalized complex is $\mathcal{U} \otimes \mathrm{pr}_{\mathfrak{M}}^{*} \mathcal{L}$. If no such line exists, one may instead retain the corresponding twisted family on the gerbe, with its twist specified.

Proposition 8.17 *(The scalar weight of a universal family).* A nonzero tautological complex of scalar weight one does not descend unchanged across π. If a normalizing line exists, the weight-zero tensor product descends to a perfect complex $\mathcal{U}^{\mathrm{rig}}$ on $X \times \mathfrak{M}^{\mathrm{rig}}$. This descent is unique up to equivalence and is compatible with base change.

Proof. The pullback of any complex on $\mathfrak{M}^{\mathrm{rig}}$ has trivial action by the kernel $\mathbb{G}_m$ of the inertia map. A nonzero weight-one complex has a different action, so it cannot be such a pullback.

Tensoring by a weight--1 line produces weight zero. Locally for the faithfully flat topology, the gerbe is $B\mathbb{G}_m \times U \to U$. A weight-zero complex on this product is precisely the pullback of a complex on U: the weight-zero representation has trivial $\mathbb{G}_m$ action. These local pullback equivalences glue, and are compatible with base change. Perfectness descends in the same topology. □

A fixed Tor-amplitude bound $[a_c, b_c]$ is further data on a class-c locus. This means that $\mathcal{U} \otimes^L M$ has cohomology sheaves only in these degrees for every quasicoherent sheaf M. Tensoring by the normalizing line preserves this bound. The descended complex $\mathcal{U}^{\mathrm{rig}}$ and all its derived pullbacks then have Tor amplitude in $[a_c, b_c]$. Indeed, the bound is local for faithfully flat descent and is preserved by derived pullback.

On $B\mathbb{G}_m$, the standard one-dimensional representation is the simplest example of the obstruction. Its underlying vector space is a line, but its inertia action prevents descent to a point. A normalization is additional data and need not exist globally. Determinants of cohomology give an explicit sufficient condition.

PROPOSITION 8.18 *(Normalization from Euler weights).* Let $\mathcal{U}$ be a perfect family of scalar weight one on $X \times \mathfrak{M}$, where X is smooth and projective. Suppose there are perfect complexes $A_1, \dots, A_k$ on X for which the integers

$$d_j = \chi(X, \mathcal{U}_m \otimes^L A_j)$$

are independent of $m \in \mathfrak{M}$ and have greatest common divisor one. Choose integers b_j with $\sum_j b_j d_j = -1$. For the projection $p : X \times \mathfrak{M} \to \mathfrak{M}$, the line

$$\mathcal{L} = \bigotimes_j \Big(\det R p_*(\mathcal{U} \otimes^L \mathrm{pr}_X^* A_j)\Big)^{b_j}$$

has scalar weight -1 and normalizes the universal family.

Proof. Proper smooth pushforward makes each displayed complex perfect. A scalar z acts by z on every cohomology group of its fiber. It therefore acts by z^{d_j} on its determinant, since the exponent is the alternating sum of the cohomology dimensions. The tensor product has weight $\sum_j b_j d_j = -1$. Proposition 8.17 then gives the descended normalized family. □

For a family of virtual rank one, take $A_1 = \mathcal{O}_x$ at any fixed smooth point $x \in X$. Derived restriction has Euler characteristic one, so the inverse of its determinant is a normalizing line. This provides a normalization on a rank-one moduli locus without a choice of a frame of the universal family.

Definition 8.19 *(Elliptic quotient and finite closed cover).* Suppose translation in E acts on $\mathfrak{M}^{\mathrm{rig}}$ and preserves the selected locus. Let $\mathfrak{Q} = [\mathfrak{M}^{\mathrm{rig}}/E]$. When a free quotient is used, require the action to be free in the stack-theoretic sense. When finite stabilizers are allowed, specify them on the quotient. A finite closed cover consists of closed immersions $\mathfrak{Z}_j \hookrightarrow \mathfrak{Q}$ whose images cover that locus, with finite inertia on each $\mathfrak{Z}_j$.

PROPOSITION 8.20 *(Quotient and covering consequences).* The map $\mathfrak{M}^{\mathrm{rig}} \to \mathfrak{Q}$ is an E-torsor in the stack sense. The $\mathfrak{Z}_j$ form the stated finite closed cover with the prescribed inertia. Equivariant coefficients descend only with a cocycle compatible with this quotient and its stabilizers.

Proof. A map $T \to [\mathfrak{M}^{\mathrm{rig}}/E]$ specifies an E-torsor on T and an equivariant map from that torsor to $\mathfrak{M}^{\mathrm{rig}}$. This is the defining pullback property of the quotient. The covering and inertia conclusions are precisely the hypotheses on the closed immersions. Descent of coefficients uses their equivariant cocycle, which is separate from the existence of the quotient stack. □

8.5 A finite word model for closure

The following calculation records three closure conditions without asserting that a geometric correspondence has been constructed. Let s_3, s_2, s_1 be symbols with strictly decreasing phases and set

$$\mathcal{W} = \{\mathrm{o}, s_3, s_2, s_1, e_{32}, e_{21}, f_{321}\}, \quad e_{32} = (s_3, s_2), \quad e_{21} = (s_2, s_1), \quad f_{321} = (s_3, s_2, s_1).$$

Definition 8.21 *(Selected extensions and factor closure).* Retain the four concatenation rules

$$s_3 * s_2 = e_{32}, \quad s_2 * s_1 = e_{21}, \quad e_{32} * s_1 = f_{321}, \quad s_3 * e_{21} = f_{321}.$$

For a geometric interpretation, each rule must be represented by an actual exact sequence with the indicated types. Factor closure requires a chosen semistable moduli locus for every factor occurring in a retained word. Each factor retains its numerical class and, when a finite closed cover is used, its chosen closed locus. Require every factor occurrence to belong to these loci.

Proposition 8.22 *(Closure of the word set).* The four selected concatenations lie in $\mathcal{W}$. Every nonzero factor is one of s_3, s_2, s_1, and the six nonempty words have ten factor occurrences in total. Under the stated factor-closure requirements, taking these factors stays in the chosen substack system.

Proof. The two words of length two contribute four occurrences, the three words of length one contribute three, and f_{321} contributes three. The displayed rules list their retained concatenations. □

Definition 8.23 *(Duality on words).* Set $D(s_3) = s_1$, $D(s_2) = s_2$, $D(s_1) = s_3$ and $D(t_1, \dots, t_k) = (D(t_k), \dots, D(t_1))$. Set $D(\mathrm{o}) = \mathrm{o}$.

Proposition 8.24 *(Dual closure).* The map D is an involution of $\mathcal{W}$. It exchanges e_{32} and e_{21} and fixes f_{321}.

Proof. Reversing the word twice restores its order, and dualizing each symbol twice restores the symbol. Applying the rule to the three longer words gives the asserted values. □

These statements concern selected words and selected extensions. They assert neither closure under every extension nor existence of an extension stack. A sheaf duality must preserve the selected heart, or be compared with a second heart. It must also act on families and extension flags.

8.6 Effective charges and parameter transport

Definition 8.25 *(Geometric charge support).* Choose an extension-closed full subcategory $\mathcal{B}$ of the heart $\mathcal{A}^t_E$, containing zero. Its effective charge semigroup is

$$\Gamma^{\mathrm{eff}}_{X,\mathcal{B}} = \{v_X(F) : F \in \mathcal{B}\} \subset \Gamma^{\mathrm{alg}}_X.$$

For a fixed phase ϕ, the semistable subcategory $\mathcal{P}(\phi)$ is one such choice. The notation $\Gamma^{\mathrm{eff}}_{X,\sigma}$ presupposes a specified heart or single-phase subcategory of this kind. A bounded window instead has a charge set, without an assertion of additive closure.

The semigroup property follows from direct sums, which are split extensions. Semistable objects of different phases need not have a semistable direct sum. Their union is a support set and is not assumed to define a semigroup.

Remark 8.26 *(Formal, algebraic and effective charges).* The inclusions

$$\Gamma^{\mathrm{eff}}_{X,\mathcal{B}} \subset \Gamma^{\mathrm{alg}}_{X} \subset \Gamma^{\mathrm{form}}_{X}$$

separate existence of objects from algebraicity of numerical classes and from cohomological lattice arithmetic. Only the last two groups are fixed by their lattice definitions. The first depends on the chosen category and stability condition.

Definition 8.27 *(Admissible parameter transport).* A parameter datum is

$$\mathfrak{p} = (S, H, \sigma, E, \mathrm{o}_E, \tau_X, \mathcal{R}_{\mathrm{HN}}),$$

where τ_X trivializes K_X and $\mathcal{R}_{\mathrm{HN}}$ is a specified cofiltered system of finite stages. Every stage includes those moduli, correspondences, coefficients and orientation data that its operations use. An admissible isomorphism includes maps $f : S \to S'$, $u : E \to E'$ preserving these structures. It also specifies compatible isomorphisms of the finite-stage data after a common cofinal refinement. Write $\mathsf{Par}^{\mathrm{DI}}$ for this groupoid.

PROPOSITION 8.28 *(Transport of a specified system).* An admissible isomorphism transports the finite-stage constructions and induces their pro-isomorphism. It preserves the Mukai pairing, Π_X, B, and $\overline{\Pi}_X$. It preserves Hall, Koszul, primitive and Pfaffian structures when their comparison maps and coherence identities are part of the given stage isomorphisms.

Proof. The map f induces a Mukai isometry, while u preserves the two integral generators of $H^{\mathrm{even}}(E)$. The formulas for the charge maps are therefore invariant. Derived transport along $f \times u$ carries the given families and correspondences to their primed versions. The supplied coefficient and orientation isomorphisms identify the operations. Compatibility with transitions then gives an isomorphism of the cofiltered systems after their common cofinal refinement. □

Definition 8.29 *(Polarization and origin).* Changing (H, σ) changes the selected moduli and their Harder–Narasimhan types. A comparison across stability chambers requires specified wall-crossing maps with their coefficient and orientation compatibilities. For the origin change from o_E to $a \in E$, define the determinant anchor by

$$\lambda^{\det}_{a}(\mathcal{F}) = \det R p_{E*}\mathcal{F} \otimes \mathcal{O}_E(-\chi(\mathcal{F})a),$$

for a perfect complex on X. Here $\chi(\mathcal{F}) = \chi(X, \mathcal{F})$.

PROPOSITION 8.30 *(Origin change of the determinant anchor).* The anchors satisfy the Picard-group identity

$$[\lambda^{\det}_{a}(\mathcal{F})] - [\lambda^{\det}_{\mathrm{o}}(\mathcal{F})] = [\mathcal{O}_E(\chi(\mathcal{F})(\mathrm{o}_E - a))].$$

This identity does not assign a character to a finite stabilizer. If an equivariant line and compatible transport of all operations are specified, their chosen character twist is part of that transport.

Proof. Canceling the two copies of $\det R p_{E*}\mathcal{F}$ gives the line-bundle ratio. A stabilizer character requires an action on a fiber of a line bundle, hence a linearization. The underlying line does not determine it. On a point with trivial action by a finite group H, every character of H linearizes the trivial line. Thus a Picard class alone cannot specify the character needed for equivariant transport. □

Even after the anchor ratio is fixed, orientation lines can carry independent characters and higher compatibility conditions. A coherent equivariant comparison must supply these explicitly. The determinant translation law below describes the numerical part of this problem.

8.7 Finite regrading and radicals

Remark 8.31 *(Finite charge spaces).* Choose a finite set of source charges together with the maps whose images are needed for a given calculation. Closure under those maps is part of the choice. A finite charge set, an HN type bound and a finite-type moduli stack are different objects. A denominator identity supplies none of their transition maps.

Definition 8.32 *(Gram fibers and parity).* Write $\Gamma_X(\gamma) = \Pi_X^{-1}(\gamma)$ and

$$K_{\Pi,0} = \Gamma_X(0) = \{(Q, P) : Q^2 = P^2 = Q \cdot P = 0\}.$$

For a finite-dimensional super vector space $V = \bigoplus_c V_c$, set

$$(\Pi^{\mathrm{fib}}_{X,*}V)_{\gamma,\bar{p}} = \bigoplus_{c\in\Gamma_X(\gamma)} V_{c,\bar{p}}, \qquad \bar{p} \in \mathbb{Z}/2.$$

Proposition 8.33 *(The zero Gram fiber).* The set $K_{\Pi,0}$ is not a subgroup of Γ_X^{form}.

Proof. In two orthogonal hyperbolic planes, set $k_1 = (e_1, e_2)$ and $k_2 = (f_1, -f_2)$. Both have Gram degree zero, whereas

$$\Pi_X(k_1 + k_2) = \Pi_X(e_1 + f_1, e_2 - f_2) = (1, 0, -1).$$

□

Definition 8.34 *(A radical in the zero-degree space).* For $V_{\Pi,0} = \bigoplus_{c\in K_{\Pi,0}} V_c$ and a bilinear form h on this finite-dimensional space, define

$$\mathrm{Rad}_{\Pi,0}(V, h) = \ker\big(V_{\Pi,0} \longrightarrow V^*_{\Pi,0}\big), \qquad v \longmapsto h(v, -).$$

Remark 8.35 *(Radical and cancellation).* This radical is a vector subspace. It gives no quotient of the charge group by $K_{\Pi,0}$. Nor does it automatically define an ideal or a coideal in an algebra carried by V. For example, the form with matrix $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ on $\mathbb{Q}z_0 \oplus \mathbb{Q}z_1$ has radical $\mathbb{Q}z_0$.

A different loss of information occurs when one takes superdimension. An even line and an odd line have parity dimensions $(1, 1)$ and superdimension zero. Such lines can occur in one Gram degree with distinct source charges. For instance,

$$\Pi_X(e_1 + f_1, f_1 + e_2) = \Pi_X(e_1 + f_1, e_1) = (1, 1, 0).$$

Assigning opposite parities to these two summands gives this example.

Exchanging Q and P in $(e_1 + f_1, f_1 + e_2 + f_2)$ gives another pair, both of degree $(1, 1, 1)$. The two zero-degree charges in the previous proof give the same parity phenomenon inside $V_{\Pi,0}$.

PROPOSITION 8.36 *(Exactness and parity of finite regrading).* Regrading finite-support super vector spaces along any set map is exact and preserves the even and odd dimensions separately, by summing them in each fiber. This applies to Π_X and to the homomorphism $\overline{\Pi}_X$ on its different source group.

Proof. Every target component is a finite direct sum of source components. Direct sums preserve exact sequences, with no change of parity. Additivity of the set map is unnecessary for this vector-space statement. □

PROPOSITION 8.37 *(Finite bar and bracket operations).* Let $f = \overline{\Pi}_X$. Let A be a $\widehat{\Gamma}_X$-graded augmented differential graded algebra with finite support. Regrading commutes with the reduced tensor coalgebra on $\bar{A}[1]$, truncated to lengths 1 through N. It preserves deconcatenation and the bar differential. Regrading a homogeneous Lie superalgebra preserves its bracket. A degree-zero pairing remains of degree zero.

Proof. Proposition 7.7 gives the tensor and differential identifications, including the cohomological shift $[1]$. On a word $[a_1|\cdots|a_j]$, additivity gives target degree $\sum_i f(\deg a_i)$. Cutting the word or multiplying adjacent factors preserves this degree. The same equality proves bracket homogeneity. If $h(V_a, V_{a'})$ is nonzero only for $a + a' = 0$, then it is nonzero only for $f(a) + f(a') = 0$ after regrading. □

The unshifted tensor coalgebra has the same degree calculation. The shift is necessary when it is used as the cohomological bar complex. Regrading does not construct a chiral Koszul comparison or establish a pairing identity that was absent in the source.

PROPOSITION 8.38 *(Filtrations and height decompositions).* Let V be a finite-support $\widehat{\Gamma}_X$-graded vector space with a filtration F_dV by homogeneous subspaces. Then

$$\operatorname{gr}_d(f_*V)_\gamma = \bigoplus_{f(a)=\gamma} \operatorname{gr}_d V_a.$$

A homogeneous filtered multiplication with $F_iV\ F_jV \subset F_{i+j}V$ retains this property after regrading. For any homomorphism $h : \Gamma_{\text{gram}} \to \mathbb{Z}$, the direct sum decomposition by negative, zero and positive values of $h \circ f$ is also preserved. A bracket has height equal to the sum of its input heights.

Proof. Exactness commutes the quotient F_d/F_{d-1} with the sum in each fiber. Multiplication keeps its filtration bound because the underlying map is unchanged. The height sign is constant on a fiber of f, and $h(f(a + a')) = h(f(a)) + h(f(a'))$. □

This includes a supplied Poincaré–Birkhoff–Witt filtration. When the filtered pieces have finite dimension, put $b_d(\gamma) = \dim F_d(f_*V)_\gamma$. Their graded dimension is $b_d(\gamma) - b_{d-1}(\gamma)$.

Positive-positive brackets stay positive, negative-negative brackets stay negative, and zero-height inputs preserve the other input's height. For opposite signs, the actual sum decides the output. The zero-height space need not be a Cartan subalgebra. A completed triangular decomposition requires its own algebraic and continuity hypotheses.

8.8 Local and wrapped moduli

Definition 8.39 *(Hybrid parameter space).* At a finite stage R, choose local color classes with projection-finite moduli $\mathfrak{M}^{L}_{R,c}$ and wrapped color classes with prequotient moduli $\mathfrak{M}^{W}_{R,c}$. A test family contains a finite configuration of maps to E with local colors. It also contains finitely many wrapped families with their specified anchors.

Maps identify repeated local points and transport all labels and families coherently. Denote this prestack by $\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$. When its descent data for a topology τ are specified, write $\mathrm{Ran}^{\mathrm{hyb}}_{R,\tau}(E)$ for the resulting stack. Empty configurations are included when a unit is required.

The local and wrapped subprestacks are denoted $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ and $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$. For effective curves, a positive projection degree gives a wrapped object by Lemma 8.1. For more general complexes the support condition is part of the chosen moduli problem. The degree-labeled set $\bigsqcup_{b\geq 0}\mathrm{Ran}(E)\times\{b\}$ does not retain these wrapped families.

Remark 8.40 *(Factorization is additional structure).* A projection-finite moduli problem can be organized over the Ran prestack. A factorization object on that prestack still requires restriction, collision and disjoint-support maps with their compatibilities. Wrapped families have their own parameter stacks. The hybrid parameter space records both types and does not itself construct their operations.

8.9 Mixed extension correspondences

Definition 8.41 *(Selected extension spans).* For $A, B \in \{L, W\}$ and classes c_1, c_2, c_3 with $c_3 = c_1 + c_2$, specify a stack $\mathfrak{E}^{AB}_{R,c_1,c_2;c_3}$ of exact sequences $0 \to F_1 \to F_3 \to F_2 \to 0$ in the chosen heart. It has a span

$$\mathfrak{M}^{A}_{R,c_1} \times \mathfrak{M}^{B}_{R,c_2} \xleftarrow{p} \mathfrak{E}^{AB}_{R,c_1,c_2;c_3} \xrightarrow{q} \mathfrak{M}^{C}_{R,c_3},$$

where p takes the subobject and quotient, q takes the middle object, and C is its specified support type. Each operation requires the pullback or Gysin map along p, the pushforward along q, and compatible coefficient maps with their degree shifts. In a Borel–Moore theory, properness of q is one sufficient pushforward hypothesis.

There are four ordered binary types LL, LW, WL, WW. The two mixed types have different input orders even when their charge sums agree. For local configurations indexed by finite sets I, J, the target configuration is the union of the two configurations, with repeated points identified. In a mixed span the wrapped source anchor and the wrapped target anchor are both retained. In a WW span there are two wrapped source anchors. These functions belong to the prequotient correspondence and require their own compatibility laws.

A local–wrapped–local flag has three source factors and a wrapped target. Its two binary decompositions involve the ordered types LW then WL, or WL then LW, with the appropriate intermediate class in each case. The corresponding flag stacks must identify the two composites together with their coefficient and orientation maps. This is the content required of the two-sided mixed correspondence.

If translation acts equivariantly on such a span, quotient all its terms by the same copy of E. For E-stacks M_1, M_2, there is a canonical equivalence

$$[(M_1 \times M_2)/E] \simeq [M_1/E] \times_{BE} [M_2/E].$$

An object on either side is one E-torsor with two equivariant maps. Thus the source of the quotient extension span is this fiber product over BE. Two unrelated torsors, as in the ordinary product of the two

quotients, do not give the same source. The empty-object quotient is BE, which is the corresponding relative unit.

Remark 8.42 *(Flags and higher coherence).* The eight words

$$LLL,\ LLW,\ LWL,\ LWW,\ WLL,\ WLW,\ WWL,\ WWW$$

list the ternary support types. An exact category has canonical comparisons between the two ways of forming a three-step extension flag. Restricted moduli and their coefficients must inherit these comparisons. Units, permutations, longer flags, collisions and descent impose further identities. They are indexed by colored trees and their contractions, not by the eight ternary words alone.

8.10 Determinant anchors and translation

Definition 8.43 *(Determinant anchor).* Let $\mathcal{F}$ be a perfect family on $S \times E \times T$, and let $\pi_{E,T} : S \times E \times T \to E \times T$ be the projection. Suppose $\chi(\mathcal{F}_t) = \chi$ is constant on T. Set

$$\mathscr{D}(\mathcal{F}) = \det R\pi_{E,T*}\mathcal{F}, \qquad \lambda^{\det}(\mathcal{F}) = \mathscr{D}(\mathcal{F}) \otimes \mathrm{pr}_E^{*}\, \mathcal{O}_E(-\chi\, \mathrm{o}_E).$$

We use $E \simeq \mathrm{Pic}^{\mathrm{o}}(E)$ given by $a \mapsto \mathcal{O}_E(a - \mathrm{o}_E)$ to regard the relative Picard class of $\lambda^{\det}$ as a map $T \to E$.

Proper smooth pushforward makes $R\pi_{E,T*}\mathcal{F}$ perfect, so its determinant is defined and compatible with derived base change [33, 70]. For a perfect complex C on a genus-one curve, $\deg \det C = \chi(C)$. Hence the normalized line has relative degree zero. Its class in the relative Picard functor disregards lines pulled back from T. It does not choose a trivialization of such lines.

PROPOSITION 8.44 *(Translation weight).* Use support translation $a \cdot \mathcal{F} = (\mathrm{id}_S \times \tau_a)_*\mathcal{F}$. Then

$$\lambda^{\det}(a \cdot \mathcal{F}) = \lambda^{\det}(\mathcal{F}) + \chi a \quad \text{in } \mathrm{Pic}^{\mathrm{o}}(E) \simeq E.$$

For a section $a : T \to E$, the ratio in the relative Picard functor is represented by $\mathcal{O}_{E\times T}(\chi(\Gamma_a - \Gamma_{\mathrm{o}}))$, where Γ_a is its graph. The kernel of the translation weight is $E[|\chi|]$ for $\chi \neq 0$, and is all of E for $\chi = 0$.

Proof. Base change identifies the determinant after translation with $\tau_{-a}^{*}\mathscr{D}(\mathcal{F})$. For a degree-$\chi$ line L on E, the relative Picard class of $\tau_{-a}^{*}L \otimes L^{-1}$ is $\chi(a - \mathrm{o}_E)$. One sees this first for a divisor by translating each point, then for any line by subtracting divisors. The same identity holds in families in the relative Picard functor. Normalizing both sides by $-\chi\, \mathrm{o}_E$ gives the result. The sign is positive for support pushforward and negative for translation pullback. □

PROPOSITION 8.45 *(Unit-weight cover and finite slice).* Let $\mathfrak{M}$ be an E-stack with a morphism $\lambda : \mathfrak{M} \to E$ satisfying $\lambda(a \cdot F) = \lambda(F) + \chi a$ for a fixed nonzero integer χ. Put

$$\widetilde{\mathfrak{M}} = \mathfrak{M} \times_{E,[\chi]} E, \qquad \mathfrak{M}_{\mathrm{o}} = \lambda^{-1}(\mathrm{o}_E).$$

Then $\widetilde{\mathfrak{M}} \to \mathfrak{M}$ is finite *étale* of degree χ^2. The anchor $(F, b) \mapsto b$ has translation weight one under $a \cdot (F, b) = (a \cdot F, b + a)$. Moreover,

$$\widetilde{\mathfrak{M}} \simeq E \times \mathfrak{M}_{\mathrm{o}}, \qquad [\mathfrak{M}/E] \simeq [\mathfrak{M}_{\mathrm{o}}/E[|\chi|]].$$

Proof. Multiplication by χ on a complex elliptic curve is an *étale* isogeny of degree χ^2. Its base change gives the first assertion. The equation $\chi b = \lambda(F)$ is preserved by the displayed E action. The mutually inverse maps

$$(F, b) \longmapsto (b, (-b) \cdot F), \qquad (b, F_0) \longmapsto (b \cdot F_0, b)$$

identify the cover with $E \times \mathfrak{M}_0$.

Put $K = E[|\chi|]$. The isogeny $[\chi]$ is a K-torsor, so $\mathfrak{M}$ is the quotient of this cover by its deck action. Under the displayed identification that action is

$$k \cdot (b, F_0) = (b - k, k \cdot F_0).$$

It commutes with the E action on the first factor. Quotienting $E \times \mathfrak{M}_0$ first by E leaves $\mathfrak{M}_0$ with its K action. Quotienting in the other order gives $[\mathfrak{M}/E]$. This proves the equivalence of stacks, including their automorphism groups. □

This constructs the normalization on the nonzero-χ locus from the actual anchor map. Orientation and coefficient descent still have to be compatible with the finite residual group. The cover does not choose its stabilizer characters.

Proposition 8.46 *(The zero-Euler branch).* When $\chi = 0$, the determinant anchor is constant along every translation orbit. On a locus where translation acts freely, it therefore fails to distinguish all points of that one-dimensional orbit. An equivariant anchor of weight one cannot be a function of this determinant anchor alone.

Proof. The translation formula becomes $\lambda^{\det}(a \cdot F) = \lambda^{\det}(F)$. A function of this invariant map is invariant, whereas a weight-one anchor changes by a. These requirements disagree for $a \neq 0$ on a free orbit. □

Proposition 8.47 *(Jordan–Hölder data beyond the determinant).* For a semistable rank-r, degree-zero vector bundle V on E, its Jordan–Hölder factors are degree-zero lines $L_1, \dots, L_r$. The divisor

$$\mathrm{sp}^{\mathrm{JH}}(V) = \sum_i [L_i] \in \mathrm{Sym}^r \mathrm{Pic}^0(E)$$

is independent of the filtration and has Abel sum $\det V$. It distinguishes $\mathcal{O}_E^{\oplus 2}$ from $L \oplus L^{-1}$ for every nontrivial $L \in \mathrm{Pic}^0(E)$, although their determinants are equal.

Proof. The classification of semistable bundles on an elliptic curve gives the stated stable factors and identifies their unordered collection with the S-equivalence class [4, 113]. The determinant is the tensor product of these factors, giving the Abel sum. If $[L] + [L^{-1}] = 2[\mathcal{O}_E]$, equality of the two unordered multisets forces $L = \mathcal{O}_E$. □

This divisor distinguishes the two S-equivalence classes, but not points of a translation orbit. Translation acts trivially on the isomorphism class of each degree-zero line, so it also fixes the divisor. The zero-χ orbit-separation problem remains after adding this particular invariant.

8.11 Topological obstruction groups

We compute the cohomology of classifying spaces with coefficients $\mathbb{F}_2$. Here BE means the topological classifying space of the complex Lie group E. To use these groups for an algebraic orientation problem, one must specify the comparison carrying its actual equivariant obstruction classes into them.

LEMMA 8.48 *(Connected translations).* There is an isomorphism

$$H^*(BE;\mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \qquad |u_i| = 2.$$

Thus $H^1(BE;\mathbb{F}_2) = 0$, whereas a degree-two class has the form $\alpha = a_1u_1 + a_2u_2$ and vanishes exactly when $a_1 = a_2 = 0$.

Proof. Topologically $E \simeq (S^1)^2$, so $BE \simeq (\mathbb{CP}^\infty)^2$. The two universal first Chern classes and the Künneth formula give the polynomial ring. □

Vanishing of H^1 excludes a continuous sign character of the connected group. It does not force a degree-two gerbe class to vanish.

LEMMA 8.49 *(Finite translation groups).* For $E[2] \cong (\mathbb{Z}/2)^2$,

$$H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \qquad |x_i| = 1.$$

Write a degree-two class as $\beta = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$. Its restrictions to the cyclic subgroups generated by $(1,0)$, $(0,1)$ and $(1,1)$ have coefficients r_1, r_2, r_3 satisfying

$$(b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2).$$

A degree-one class $\lambda = \lambda_1x_1 + \lambda_2x_2$ instead has restriction coefficients $(\lambda_1, \lambda_2, \lambda_1 + \lambda_2)$.

If 2^a is the exact power of two dividing N, with $a \geq 2$, then

$$H^*(BE[N];\mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2.$$

Its degree-two classes are $A_1y_1 + A_{12}x_1x_2 + A_2y_2$. The mixed class x_1x_2 is invisible on every cyclic subgroup. For N odd, all positive-degree mod-two cohomology vanishes. For $a = 1$, the first polynomial-ring description applies.

Proof. The classifying space of $\mathbb{Z}/2$ is $\mathbb{RP}^\infty$. Restriction to the three subgroups substitutes $(x,0)$, $(0,x)$ and (x,x), giving the displayed formulas.

For $\mathbb{Z}/2^a$ with $a \geq 2$, the periodic resolution gives one mod-two cohomology class in each degree, with a periodicity generator y in degree two. Its degree-one generator x lifts to a $\mathbb{Z}/4$ character. Its Bockstein, hence x^2, is zero. The cyclic ring is therefore $\Lambda(x) \otimes \mathbb{F}_2[y]$, and Künneth gives the rank-two formula.

On a cyclic subgroup whose order is divisible by four, the two restricted degree-one classes are multiples of one exterior generator, so their product vanishes. If its order has two-primary part exactly two, both restrictions vanish: an element of order two in $(\mathbb{Z}/2^a)^2$ has even coordinates. A cyclic subgroup of odd order has no positive mod-two cohomology. Finally, transfer annihilates positive cohomology for an odd-order group, because its order is invertible in $\mathbb{F}_2$. □

If a finite class is the restriction of $\alpha \in H^2(BE;\mathbb{F}_2)$, the inclusion $E[2] \hookrightarrow E$ sends u_i to x_i^2. Such a restriction has $b_{11} = 0$. A stabilizer class obtained from an equivariant gerbe on a moduli space must be computed from that gerbe. It is not an arbitrary class of BE, and it is not assumed independent of its global restrictions.

Remark 8.50 *(Square roots and their equivariant choices).* For a specified square-root gerbe on a stabilizer classifying space, a trivialization exists only when its degree-two obstruction vanishes. When it does, isomorphism classes of trivializations form a torsor under $H^1(BH;\mathbb{F}_2)$. The degree-two obstruction and the degree-one choice have distinct roles. An invariant Picard class does not choose either an equivariant cocycle or a trivialization.

An equivariant line descends to the quotient *stack* with its stabilizer representation. Triviality of that representation is an additional requirement for descent to a coarse quotient, or for a specified invariant frame. Thus a demand that a character vanish must name its descent target. These cohomology calculations give criteria once the actual orientation class is supplied. They do not compute that class for the compact Hall moduli.

Remark 8.51 *(Characters of odd order).* A sign character detects only two-primary information. A line linearization has a character in $\mathrm{Hom}(H,\mathbb{C}^\times)$, which can be nontrivial for H of odd order. For example, a generator of $\mathbb{Z}/3$ can act by a primitive cube root of unity, while $H^1(B(\mathbb{Z}/3);\mathbb{F}_2) = 0$. Descent of that line requires its actual character, not only its sign reduction.

8.12 Cusp polarization and chamber transport

Definition 8.52 *(Cusp order).* The standard cusp positivity convention orders $(m,n,-\ell)$ lexicographically. Inside the positive semigroup Γ_{eff}, retain the Jacobi-active indices Γ_{act} defined in §6. A region where every active monomial has absolute value less than one is $\mathrm{Im}\tau = \mathrm{Im}\sigma = t$, $\mathrm{Im}z = -a$, with $0 < a < t$. Taking t sufficiently large gives the usual cusp region.

For active indices with $n + m > 0$, the inequality $\ell^2 \le 4nm + 1$ and integrality give $|\ell| \le n + m$. Thus $t(n + m) - a\ell > 0$. The remaining index $(0,-1,0)$ has imaginary exponent $a > 0$. Hence each active monomial has absolute value less than one in this chamber. This is an order on Fourier indices, not a positivity theorem for microscopic charges.

Remark 8.53 *(Arithmetic chamber transport).* The index form

$$\langle \gamma, \gamma' \rangle_{\mathrm{ind}} = 2(nm' + n'm) - \ell\ell'$$

is $-1/2$ times the root pairing under α. An element of $\mathbf{Sp}_4(\mathbb{Z})$ transports the automorphic section and its expansion to a cusp with transported local coordinates. This arithmetic action does not specify an action on every sheaf charge or on a stability condition. Such a geometric action requires its own functor and charge comparison.

In the four-dimensional $N = 4$ dyon problem, the electric and magnetic vectors lie in a rank-28 Narain lattice [35, §1]. The three quadratic invariants enter a genus-two generating function. This does not identify its Siegel modular group with the full microscopic duality group or identify the rank-24 Mukai lattice with that Narain lattice.

Remark 8.54 *(Wrapped objects under a change of chamber).* Projection to the fixed curve E determines whether an object is wrapped. Changing an automorphic cusp alone does not change that projection. A comparison that transports the local and wrapped categories must specify the functors and prove the corresponding support statements.

Definition 8.55 *(Chamber-compatible Hall comparison).* Choose two stability conditions and cusp expansions. A chamber-compatible comparison consists of maps of their finite moduli and extension systems, with coherent coefficient and orientation maps. Require them to intertwine the charge maps in the chosen coordinates, preserve the indicated support categories, and commute with all transition maps. Wall-crossing identities of scalar invariants alone do not supply such a comparison.

The identity $\overline{\Pi}_X(\widehat{c} \star \widehat{c}') = \overline{\Pi}_X(\widehat{c}) + \overline{\Pi}_X(\widehat{c}')$ holds before any chamber is chosen. It remains a lattice identity until these categorical maps have been constructed.

8.13 Constructible coefficients for local operations

Deformation and obstruction complexes on stable-sheaf moduli underlie Donaldson–Thomas theory [110]. A Hall construction retains cohomological coefficients and extension operations before taking numerical invariants. Comparing it with an algebra of BPS states, as in [55], requires a map between the specified state spaces that preserves their products.

For an open $U \subset X$, select finite-type moduli $\mathfrak{M}_c(U)$ of the desired compactly supported objects. Assume they carry oriented critical charts and a constructible vanishing-cycle complex Φ_c with compatible descent. This requires a specified coefficient formalism for these stacks. On oriented d-critical schemes the construction is provided by [12, Theorem 6.9]. Applying it to a quotient requires compatible charts, orientations and equivariance.

Definition 8.56 *(Protected coefficient complex).* Suppose $R\Gamma_c(\mathfrak{M}_c(U), \Phi_c)$ is a bounded complex of finite-dimensional rational vector spaces. Fix the convention

$$C^{\mathrm{BM}}(\mathfrak{M}_c(U); \Phi_c) = R\,\mathrm{Hom}_{\mathbb{Q}}\big(R\Gamma_c(\mathfrak{M}_c(U), \Phi_c), \mathbb{Q}\big), \qquad \mathcal{A}_X(U) = \bigoplus_c C^{\mathrm{BM}}(\mathfrak{M}_c(U); \Phi_c).$$

Here the sum runs over the specified geometric charge support. An algebra structure on $\mathcal{A}_X(U)$ consists of homogeneous maps induced by the selected extension spans, with stated Gysin shifts, pushforward hypotheses and coherent coefficient comparisons.

For $\Phi_c = \mathbb{Q}$, the convention is the usual Borel–Moore complex $R\Gamma(\mathfrak{M}_c(U), \omega_{\mathfrak{M}_c(U)})$ by Verdier duality. The orientation is used to construct Φ_c. It is a coherent square-root line on the critical geometry, so it cannot be tensored directly with a rational constructible complex. A change of orientation can instead produce a specified sign local system in the constructible category. Orientation also enters the proposed vanishing-cycle definition of Gopakumar–Vafa invariants [89, §1.2].

For disjoint opens $U_1, \dots, U_k$, objects on their disjoint union decompose among the opens. If F and G have disjoint supports, the sheaf $R\mathcal{H}om(F, G)$ is zero, hence $R\,\mathrm{Hom}(F, G) = 0$.

This assertion does not require compact support. Compactness and finite type enter the coefficient functors and pushforwards. When the chosen moduli and orientations respect this product decomposition, Thom–Sebastiani and Künneth give maps

$$\mathcal{A}_X(U_1) \otimes \cdots \otimes \mathcal{A}_X(U_k) \longrightarrow \mathcal{A}_X(U_1 \sqcup \cdots \sqcup U_k)$$

[86, 106]. Extension to overlapping opens and collision strata requires further factorization data. A comparison with the D-module formalism of [7, §3.4] requires a realization on configuration spaces, compatible with their diagonals.

The Lie–commutative duality of [40] concerns chiral Lie algebras and commutative coalgebras in the chiral tensor category. An associative-to-associative comparison [81, 82] additionally requires the

associative products and compatible collision maps on both sides. These operations are not supplied by the disjoint-support calculation.

For the unreduced moduli of coherent sheaves and perfect complexes on a smooth projective Calabi–Yau threefold, orientation data do exist [64, Theorem 4.4]. The theorem applies to $S \times E$ with the chosen trivialization of its canonical bundle. It supplies isomorphism classes of square roots compatible with the indicated exact-sequence or triangle operations. Compatibility with a chosen scalar and trace reduction, translation quotient, boundary degeneration and higher flag maps must still be established for the system considered here.

Remark 8.57 *(Scope of the coefficient construction).* The shifted symplectic form, an oriented critical chart, its constructible complex, and a Hall product are separate structures. The product needs actual pull-push maps on extension stacks. Multiplicative orientation requires their compatibility on longer flags. Translation reduction requires the common-torsor spans and equivariant descent described above. None is constructed by the scalar superdimension of a coefficient complex.

8.14 Orientation and modular comparison

Four geometric requirements organize the orientation comparison. Condition $(D\mathrm{o})$ asks for a Do degeneration with compatible reduced complexes, semiregularity cosections, coefficient specialization, orientation and transition maps. Condition $(O\mathrm{1})$ asks for a square root of the reduced determinant line on the elliptic quotient, including its boundary behavior. Condition $(O\mathrm{1})^{+}$ asks for coherent Weyl lifts and a trivialization of their orientation cocycle. Condition $(O2_{\mathrm{atlas}})$ asks for compatible type-II wall charts with a normal skew complex $[V \xrightarrow{D_u} V^{\vee}]$ in degrees $[-1, 0]$, where $V = O^2$ and $\mathrm{pf}(D_u) = u$. The chart includes a compatible reflection lift on the root line and a reflection-invariant tangential unit. The corresponding conditional criteria are stated in Appendix 7.

These orientation requirements do not alone identify a geometric Pfaffian with an automorphic form. The finite geometric datum (P_{fin}) of Theorem 10.1 also specifies the reduced skew complexes, Pfaffian sections, Fourier coefficients, line maps, cusp frames and compatible limits. Under all those hypotheses, the line and section comparison gives

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5, \qquad \epsilon_o = \widetilde{\nu}_{\Delta_5} \quad \text{on } W^{(2)}(\Lambda_{II}^{2,1}).$$

Construction of the compact geometric hypotheses remains open.

The scalar normalization is a different statement. In the primitive K3 class regime and variable convention of the opening chapter,

$$Z_{\mathrm{OP}}^{X} = -\frac{1}{\chi_{10}^{\mathrm{OP}}} = -4096\Delta_5^{-2}, \qquad \chi_{10}^{\mathrm{OP}} = (\Delta_5/64)^2.$$

The reduced stable-pair and Gromov–Witten comparisons underlying this identity have the hypotheses stated there [99–101]. The reduced Donaldson–Thomas theory uses weighted Euler characteristics after quotienting the Hilbert scheme by elliptic translation [15].

The normalized scalar $Z_{\mathrm{BPS}}^{K3\times E} := -Z_{\mathrm{OP}}^{X}/4096$ is Δ_5^{-2}. Calling it a trace of an independently constructed geometric state space requires the state comparison. The inverse square loses the sign character of Δ_5. The formal set of twelve signs has $2^{12} = 64^2$ elements. A geometric orbit of that size requires its own bijection. The numerical equality supplies no map from an orientation line to an automorphic line.

Remark 8.58 *(Numerical invariants and comparison coefficients).* The tuple

$$(\chi(\mathcal{O}_X), \sum_q (-1)^q h^{0,q}(X), \operatorname{rk}\Gamma_{\mathrm{gram}}, \operatorname{wt}\Delta_5, \chi_{\mathrm{top}}(S)) = (0, 0, 3, 5, 24)$$

records five defined numerical invariants. Replacing its third entry by a genus-one chiral or Heisenberg coefficient requires a chosen complex, trace normalization and a comparison with the Gram rank [79]. A matching number does not supply this comparison. Nor does the scalar Igusa identity construct a gravitational path integral on the compact target. A proposed local holomorphic-topological model needs a global gluing map, with its obstruction class and coefficient comparison, before it can yield that interpretation [80].

8.15 Fixed notation and the remaining maps

The cohomological charge group is Γ_X^{form}, its algebraic subgroup is Γ_X^{alg}, and geometric support is selected inside it. The map Π_X is quadratic. The split extension $\widehat{\Gamma}_X$ carries the additive map $\overline{\Pi}_X$. The hybrid parameter space records local and wrapped families, while their operations require the extension spans and coefficients just specified.

At each height, the proposed Dirac–Igusa system consists of these geometric data together with its Hall and primitive maps, Koszul comparison, Pfaffian line and section. A pro-object exists from a coherent cofiltered system of such stages. The construction problem is to supply the compact stages, the comparison maps and their transition identities. The lattice extension, support lemma and finite regrading results solve their stated parts without replacing that problem.

9 THE $N = 1$ BOUNDARY COMPARISON

A compact Hall realization must compare its source operations with the denominator datum, in addition to reproducing a scalar series. Here the geometric base is a smooth complex K3 surface S times a smooth elliptic curve E. Singular K3 limits, nodal or cuspidal elliptic curves, imprimitive multiple covers, the compact Hall cusp and multi-component charge degenerations each require their own coefficient and specialization maps.

The formal automorphic target has the ordered support

$$\begin{aligned}\Gamma_{\mathrm{eff}} = {} & \{(n,l,m) : m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \\ & \cup \{(n,l,0) : n > 0,\ l \in \mathbb{Z}\} \cup \{(0,l,0) : l < 0\}.\end{aligned}$$

Only the degrees with $f(nm, l) \neq 0$ contribute factors. They generate the positive cone

$$C_+ = \{(n,l,m) : n, m, n+m-l \geq 0\}, \qquad h(n,l,m) = 2(n+m) - l.$$

Positive-height completion is

$$\widehat{\mathbb{Z}[C_+]} = \mathbb{Z}[[qr, sr, r^{-1}]].$$

The Lorentzian index map is

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}, \qquad (\alpha(\gamma), \alpha(\eta)) = -2\langle\gamma, \eta\rangle_{\mathrm{ind}},$$

so $\alpha(n,l,m)^2 = -2(4nm - l^2)$. The simple roots and Weyl vector are

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3, \qquad \rho = f_2 - \tfrac{1}{2}f_3 + f_{-2}, \quad (\rho, \delta_i) = -1.$$

These formulas fix the lattice, chamber and completion before the comparison is formulated.

The scalar and algebraic targets are

$$\mathrm{Borch}_{\theta}(\phi_{0,1}^{\mathrm{Weil}}) = \Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z),$$
$$\mathrm{wt}(\Delta_5) = 5, \qquad Z^X_{\mathrm{OP}} = Z^X_{\mathrm{DT}} = -4096\Delta_5^{-2}.$$

The last equality has the primitive nonzero-K3-class and variable conventions of Chapter 8. More generally a specified scalar branch may be written $Z^X_{\square} = C_{\square}\Delta_5^{-2}$; its normalization constant does not define the geometric coefficient theory.

Put $W = W^{(2)}(\Lambda^{2,1}_{II})$. The orientation comparison requires

$$\epsilon_o|_W = \widetilde{\nu}_{\Delta_5}|_W$$

for the homogeneous orthogonal character, including its line action. Weyl transport of a chosen root pair and the section-compatible wall charts supply the geometric side. Rescaling Δ_5 changes none of these character values.

There are three distinct algebraic recipients. The denominator algebra is the graded Lie superalgebra $\mathfrak{g}_{\Delta_5}$. Its ordinary enveloping algebra uses the usual Koszul tensor. The associative Ran object $\mathcal{A}_{\Delta,E}$ of Section 1.5 uses its specified chiral convolution; comparison with a Beilinson–Drinfeld current envelope requires the separate current construction stated there. A notation such as $U^{\mathrm{ch}}_E(\mathfrak{g} \otimes \mathcal{O}_E)$ does not identify these recipients or supply a geometric source.

For a Hall–Drinfeld double, one must construct a compatible opposite half and the full pairing and cross relations in their actual tensor category. Serre duality of deformation complexes does not alone construct that double. The invariant form on the primitive Lie algebra has a different role from a Hopf-adjoint pairing; imposing both identities on the same ordinary primitive form would contradict the real $\mathfrak{sl}_2$ sector.

The recognition criterion of Theorem 1.160 asks for actual geometric representatives, the Borcherds relations, generation and equality of the relation kernel. It preserves both parity dimensions and the invariant form, and then induces the ordinary filtered enveloping isomorphism. It does not infer these hypotheses from a product exponent. Even when all signed dimensions agree, additional even and odd root vectors could cancel in the scalar series. The fixed presentation and its ordinary denominator determine the target parities; a geometric map must still identify them.

Finite degree sets retain partial operations with their actual output spaces. They are not automatically Lie subalgebras, and a symmetric root cutoff need not be a quotient preserving brackets. Relation identities must retain every intermediate degree. Quotienting a free presentation by a collection of defects constructs that quotient; it does not prove the same defects vanish on an independently defined compact source. Faithfulness and the absence of additional source relations remain necessary.

The inverse-limit comparison also needs a common cofinal system, compatible maps and a specified completion. Ordinary cohomology commutes with that derived limit only under the corresponding exactness hypotheses. Centres, associators, supported field operations, states and protected traces require their own comparisons; they do not follow from the ordinary primitive recognition theorem.

The eight Gritsenko–Cléry input pairs and the cyclic CHL orders are different indexing systems, as explained in Chapter 4. In particular, the labels F_5, F_6, F_7, F_8 by themselves provide no map to a CHL denominator, no identified simple real roots or Weyl-sum side, and no compact geometric realization. An order-four multiplier can be realized on the corresponding character cover, but this operation alone constructs no nonabelian denominator algebra. Applying a separate CHL denominator theorem requires

matching its actual seed, level, multiplier, cusp, lattice and presentation. A row map $\Phi_j^{\mathrm{cmp}} \leftrightarrow F_i$ must be constructed; it is not part of the $N = 1$ theorem.

The $N = 1$ automorphic and denominator data thus specify a concrete target. The compact problem is to build the coefficient, orientation, primitive, field and state maps that reach it while preserving their stated operations.

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